

ព្រះរាជាណាចក្រកម្ពុជា
ជាតិ សាសនា ព្រះមហាក្សត្រ



មគ្គុទ្ទេសក៍ព្យាបាលគ្លីនិក
CLINICAL PRACTICE GUIDELINES

ផ្នែក

FOR

សាស្ត្រសាស្ត្រ
SURGERY

ភាគ ១

នាយកដ្ឋានសេវាសុខភាព

ឆ្នាំ ២០២៥

Kingdom of Cambodia
Nation Religion King



**CLINICAL PRACTICE
GUIDELINES**

FOR

SURGERY

Part 1

Department of Health Services

December 2025



Ministry of Health

CLINICAL PRACTICE GUIDELINES FOR SURGERY

Part
1

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ព្រះរាជាណាចក្រកម្ពុជា
ជាតិ សាសនា ព្រះមហាក្សត្រ

ក្រសួងសុខាភិបាល

លេខ ០៣៦ ៩៩០ ប.ស/៤ ស.ស.ប.ក

ប្រកាស
ស្តីពី

ការដាក់ឱ្យអនុវត្តមគ្គុទ្ទេសក៍ព្យាបាលគ្លីនិក ផ្នែកសង្ស័យសាស្ត្រ

រដ្ឋមន្ត្រីក្រសួងសុខាភិបាល

- បានឃើញរដ្ឋធម្មនុញ្ញនៃព្រះរាជាណាចក្រកម្ពុជា
- បានឃើញព្រះរាជក្រឹត្យលេខ នស/រកត/០៨២៣/១៩៨១ ចុះថ្ងៃទី២២ ខែសីហា ឆ្នាំ២០២៣ ស្តីពីការតែងតាំងរាជរដ្ឋាភិបាលនៃព្រះរាជាណាចក្រកម្ពុជា
- បានឃើញព្រះរាជក្រឹត្យលេខ នស/រកត/០២២៤/២០៥ ចុះថ្ងៃទី២១ ខែកុម្ភៈ ឆ្នាំ២០២៤ ស្តីពីការបំពេញបន្ថែមសមាសភាពរាជរដ្ឋាភិបាលនៃព្រះរាជាណាចក្រកម្ពុជា
- បានឃើញព្រះរាជក្រឹត្យលេខ នស/រកត/០៩២៤/១១៦៩ ចុះថ្ងៃទី២០ ខែកញ្ញា ឆ្នាំ២០២៤ ស្តីពីការកែសម្រួល និងតែងតាំងសមាសភាពរាជរដ្ឋាភិបាលនៃព្រះរាជាណាចក្រកម្ពុជា
- បានឃើញព្រះរាជក្រឹត្យលេខ នស/រកត/១១២៤/១៤៧៧ ចុះថ្ងៃទី២០ ខែវិច្ឆិកា ឆ្នាំ២០២៤ ស្តីពីការកែសម្រួល និងតែងតាំងសមាសភាពរាជរដ្ឋាភិបាលនៃព្រះរាជាណាចក្រកម្ពុជា
- បានឃើញព្រះរាជក្រមលេខ នស/រកម/០៦១៨/០១២ ចុះថ្ងៃទី២៨ ខែមិថុនា ឆ្នាំ២០១៨ ដែលប្រកាសឱ្យប្រើច្បាប់ ស្តីពីការរៀបចំ និងការប្រព្រឹត្តទៅនៃគណៈរដ្ឋមន្ត្រី
- បានឃើញព្រះរាជក្រមលេខ នស/រកម/០១៩៦/០៦ ចុះថ្ងៃទី២៤ ខែមករា ឆ្នាំ១៩៩៦ ដែលប្រកាសឱ្យប្រើច្បាប់ស្តីពីការបង្កើតក្រសួងសុខាភិបាល
- បានឃើញព្រះរាជក្រមលេខ នស/រកម/១១១៦/០១៤ ចុះថ្ងៃទី១៦ ខែវិច្ឆិកា ឆ្នាំ២០១៦ ដែលប្រកាសឱ្យប្រើច្បាប់ ស្តីពីការគ្រប់គ្រងអ្នកប្រកបវិជ្ជាជីវៈសុខាភិបាល
- បានឃើញអនុក្រឹត្យលេខ២៣៩ អនក្រ.បក ចុះថ្ងៃទី០៣ ខែធ្នូ ឆ្នាំ២០២៥ ស្តីពីការរៀបចំ និងការប្រព្រឹត្តទៅរបស់ក្រសួងសុខាភិបាល
- យោងតាមការចាំបាច់របស់ក្រសួងសុខាភិបាល



សម្រេច

ប្រការ១._

ត្រូវបានដាក់ឱ្យអនុវត្តមគ្គុទ្ទេសក៍ព្យាបាលគ្លីនិក ផ្នែក សសសសស (Clinical Practice Guidelines For Surgery) ដែលជាឧបសម្ព័ន្ធនៃប្រកាសនេះ។

ប្រការ២._

ប្រកាសនេះមានវិសាលភាពអនុវត្តចំពោះគ្រប់មូលដ្ឋានសុខាភិបាលសាធារណៈ និងឯកជនក្នុងព្រះរាជាណាចក្រកម្ពុជា។

ប្រការ៣._

បទប្បញ្ញត្តិទាំងឡាយណាដែលផ្ទុយនឹងប្រកាសនេះ ត្រូវទុកជានិរាករណ៍។

ប្រការ៤._

អគ្គលេខាធិការ អគ្គនាយកបច្ចេកទេសសុខាភិបាល អគ្គាធិការក្រសួងសុខាភិបាល គ្រប់ប្រធានអង្គភាពក្រោមឱវាទក្រសួងសុខាភិបាល ទាំងមូលដ្ឋានសុខាភិបាលសាធារណៈ និងឯកជន ត្រូវទទួលបន្ទុកអនុវត្តតាមប្រកាសនេះឱ្យមានប្រសិទ្ធភាព ចាប់ពីថ្ងៃចុះហត្ថលេខាតទៅ។

ថ្ងៃ ៣០ ខែ កុម្ភៈ ឆ្នាំម្សាញ់ សប្តស័ក ព.ស.២៥៦៩
ធ្វើនៅរាជធានីភ្នំពេញ ថ្ងៃទី ៣១ ខែ ៥ ឆ្នាំ២០២៥



រដ្ឋមន្ត្រីក្រសួងសុខាភិបាល

សាស្ត្រាចារ្យ ឈាន វ៉ាន

កន្លែងទទួល៖

- ទីស្តីការគណៈរដ្ឋមន្ត្រី
- ក្រសួងសេដ្ឋកិច្ចនិងហិរញ្ញវត្ថុ
- ឧទ្ធរណ៍យឯកឧត្តមសាស្ត្រាចារ្យរដ្ឋមន្ត្រីក្រសួងសុខាភិបាល
- ដូចប្រការ៤
- រាជកិច្ច
- ឯកសារ កាលប្បវត្តិ

អារម្ភកថា

ក្រោមកិច្ចដឹកនាំរបស់រាជរដ្ឋាភិបាលនីតិកាលទី៧ នៃរដ្ឋសភា នៃ ព្រះរាជាណាចក្រកម្ពុជា ក្រសួងសុខាភិបាល យកចិត្តទុកដាក់ខ្ពស់ដល់សុខភាព និងសុខុមាលភាពប្រជាជនគ្រប់រូប និង ប្តេជ្ញាចិត្ត យ៉ាងម៉ឺងម៉ាត់លើកកម្ពស់ការផ្តល់សេវាសុខាភិបាលប្រកបដោយ គុណភាព សុវត្ថិភាព និង សមធម៌ ដោយផ្ដោតលើអ្នកជំងឺ ជូនដល់ប្រជាជនកម្ពុជាគ្រប់រូប ស្របតាម ផែនការយុទ្ធសាស្ត្រ សុខាភិបាល ២០២៥-២០៣៤ ក៏ដូចជា ផែនទីបង្ហាញផ្លូវឆ្ពោះទៅការគ្របដណ្តប់សុខភាពជាសកល នៅកម្ពុជា ឆ្នាំ ២០២៤-២០៣៥ របស់រាជរដ្ឋាភិបាល។

មគ្គុទ្ទេសក៍ព្យាបាលគ្លីនិកផ្នែកសល្យសាស្ត្រ នេះជាមាតិកាណែនាំថ្នាក់ជាតិពាក់ព័ន្ធនឹងពិធីសារ នៃការគ្រប់គ្រងបញ្ហាសុខភាពផ្នែកសល្យសាស្ត្រ ជាអាទិភាព ដោយផ្ដោតសំខាន់លើបែបបទនៃការធ្វើ រោគវិនិច្ឆ័យ និងការថែទាំ អន្តរាគមន៍កាត់ ការព្យាបាល និងការបង្ការជំងឺ រាប់ទាំងការតាមដានបញ្ហា សុខភាព ឬជំងឺ ។

មគ្គុទ្ទេសក៍ព្យាបាលគ្លីនិកផ្នែកសល្យសាស្ត្រនេះជាសៀវភៅដែលបានចងក្រងលើកដំបូង មាន ពីរភាគ៖ ភាគទី១ និងភាគទី២ នៅកម្ពុជា ដោយក្រុមការងារបច្ចេកទេសជំនាញ នៃក្រសួងសុខាភិបាល និងមានការចូលរួមពីដៃគូសំខាន់ៗ អ្នកជំនាញ ឯកទេសជំនាញពីផ្នែកពាក់ព័ន្ធទាំងឡាយ រួមទាំង គ្រូពេទ្យព្យាបាល និងគ្រូពេទ្យវះកាត់ បម្រើការងារនៅតាមមន្ទីរពេទ្យគ្រប់លំដាប់ថ្នាក់ មន្ត្រីបំពេញ ការងារនៅការិយាល័យសុខាភិបាលស្រុកប្រតិបត្តិ និងមន្ទីរសុខាភិបាល នៃរដ្ឋបាលរាជធានី ខេត្ត កម្មវិធីជាតិ សមាគមវិជ្ជាជីវៈសុខាភិបាលផ្សេងៗ ក៏ដូចជាអង្គការដៃគូពាក់ព័ន្ធ ។

មគ្គុទ្ទេសក៍ព្យាបាលគ្លីនិកនេះ ត្រូវបានរៀបចំឡើងសម្រាប់អ្នកវិជ្ជាជីវៈសុខាភិបាល និងអ្នក វិជ្ជាជីវៈពាក់ព័ន្ធ ជាពិសេសអ្នកផ្តល់សេវាសុខភាព រាប់បញ្ចូលទាំង វេជ្ជបណ្ឌិត និងអ្នកប្រកបវិជ្ជាជីវៈ សុខាភិបាលផ្សេងទៀត នៅមូលដ្ឋានសុខាភិបាលសាធារណៈ និងឯកជន ក៏ដូចជាសិក្ខាកាម អ្នក រៀបចំគោលនយោបាយ អ្នកពាក់ព័ន្ធនានា គណៈវិជ្ជាជីវៈ និងសមាគមវិជ្ជាជីវៈផ្សេងៗ។

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រដ្ឋមន្ត្រីក្រសួងសុខាភិបាល

សាស្ត្រាចារ្យ លាង វ៉ាន់

FOREWORD

Under the leadership of the Royal Government of the 7th Legislature of the National Assembly of the Kingdom of Cambodia, the Ministry of Health pays high attention to the health and well-being of the entire population. The Ministry of Health is strongly committed to ensuring the provision of equitable, safe and quality patient-centered health services for the Cambodian population in line with the Health Strategic Plan 2025-2034 and the Royal Government Roadmap Towards Universal Health Coverage in Cambodia 2024-2035.

This Clinical Practice Guidelines (CPG) for Surgery provides national guidance on the management protocols of priority health problems related to surgery, with a focus on diagnosis and care, including surgical interventions, curative and preventive care, as well as follow-up of health problems.

The Guidelines, first ever published in Cambodia in two parts, Part 1 and Part 2; were developed by the Technical Working Group of the Ministry of Health with participation from various stakeholders and individuals, including experts from relevant fields ranging from clinicians working in hospitals at all levels to district health officers, Provincial Health Departments, national programs, health professional associations, as well as partner organizations.

The Guidelines is intended for use by health professionals and other relevant professionals, especially health care providers, including doctors and other health practitioners in both public and private health facilities as well as trainees, policymakers, relevant individuals, professional councils and professional associations.

The Ministry of Health guides all health professionals and relevant individuals to use this Guidelines in their professional practice with the aim of improving effectiveness, efficiency and quality in patient care and safety, as well as contributing to human resource capacity building.

Phnom Penh, 31 December, 2025
Minister of Health

Prof. CHHEANG RA

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We would also like to express our deep gratitude to all partners involved, including the World Health Organization and other partners, for their technical and financial contributions to the formulation of the guidelines.

This Clinical Practice Guidelines for Surgery represent another significant and commendable achievement of the Ministry of Health of the Kingdom of Cambodia.

Phnom Penh, 31 December, 2025

**Chair of the Steering Committee for National
Medical Care and Therapy**



H.E. Prof. Yit Sunnara

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ABBREVIATION

Cleaned List of Abbreviations – Surgery CPGs

ABG	– Arterial Blood Gas
ANA	– Antinuclear Antibodies
ATLS	– Advanced Trauma Life Support
CBC	– Complete Blood Count
COM	– Chronic Otitis Media
CRP	– C-Reactive Protein
CSF	– Cerebrospinal Fluid
CT	– Computed Tomography
CTS	– Carpal Tunnel Syndrome
DR	– Diabetic Retinopathy
EBV	– Epstein-Barr Virus
EMG	– Electromyography
ENT	– Ear, Nose, and Throat
ESR	– Erythrocyte Sedimentation Rate
FNA	– Fine Needle Aspiration
FT3	– Free Triiodothyronine
FT4	– Free Thyroxine
GCS	– Glasgow Coma Scale
HIV	– Human Immunodeficiency Virus
HPV	– Human Papillomavirus
ICP	– Intracranial Pressure
IOL	– Intraocular Lens
IOP	– Intraocular Pressure
IV	– Intravenous
LOC	– Loss of Consciousness
MESS	– Mangled Extremity Severity Score
MRI	– Magnetic Resonance Imaging
NSAIDs	– Non-Steroidal Anti-Inflammatory Drugs
ORIF	– Open Reduction and Internal Fixation
PET	– Positron Emission Tomography
PT	– Prothrombin Time
PVD	– Posterior Vitreous Detachment
RB	– Retinoblastoma
RF	– Rheumatoid Factor
ROM	– Range of Motion
RRD	– Rhegmatogenous Retinal Detachment
TBI	– Traumatic Brain Injury
TRD	– Tractional Retinal Detachment
TSH	– Thyroid Stimulating Hormone
VA	– Visual Acuity

INTRODUCTION

I. BACKGROUND

The Ministry of Health of Kingdom of Cambodia is committed to improving the health and well-being of all Cambodian population through the provision of quality, safe, acceptable, equitable and affordable healthcare services. Implementing evidence-based, appropriate, and cost-effective health interventions within care packages at various levels of the health system, guided by contextually relevant best practices in clinical practice guidelines (CPGs), can significantly enhance population health.

Clinical Practice Guidelines (CPGs) are essential diagnostic and care tools in achieving UHC by ensuring that healthcare services are provided timely, effectively, safely, continuously and with patient-centeredness. The guidelines provide evidence-based and or up-to-health services at different levels of the health system. They ensure that health problems are appropriately diagnosed and clinical decisions are based on the best available evidence and experiences, leading to improved patient diagnosis and care outcomes as well as more efficient use of resources. By adhering to CPGs, healthcare providers can offer consistent, high-quality care that meets national and international standards.

The 1st Clinical Practice Guidelines (CPGs) for Surgery and Pediatrics were published in 2013. They are the only existing CPGs in Cambodia. Since then, many new developments have taken place, especially in the area of laboratory tests, medical imaging, diagnosis tools and treatment options. Regularly reviewing and updating the CPGs offer up-to-date and more evidence-based recommendations for health services and interventions tailored to Cambodia's current and future needs. Updating the clinical practice guidelines aims to enhance both quality and coverage by providing a solid evidence-based foundation for changes in services. This includes the diagnosing, treatments, and support necessary to deliver these services as defined in the essential packages of care under the CPA guidelines.

II. DEVELOPMENT PROCESS

This Clinical Practice Guidelines was formulated by “Technical Working Group of Clinical Practice Guidelines for Surgery” of the Ministry of Health through face-to-face and online meeting of the working group and small group meeting and discussions with guidance and endorsements from the Steering Committee for National Medical Care and Therapy of the Ministry of Health mainly through its meeting. The development process of the guidelines was actively participated by various stake holders and individuals, including experts from related fields ranging from clinicians working in hospitals at all levels, district health officers, Provincial Health Department, national institutes, national programs, health professional associations as well as experts and officials partner organizations. Each health section follows standard format except in few cases. These guidelines contain three parts-Part and Part II with each separate volume presented in its hard copy division and publication. The guideline covers 174 topics/diseases/conditions within 11 specialized chapters. The guideline do not cover the national programs

topics/diseases/conditions such as HIV/AIDS, TB, Malaria, Dengue, Schistosomiasis and vaccine preventable diseases. The list of the protocols/guidelines of these national programs are included in the annex B of the CPGs.

III. PURPOSE AND USERS

The main aim of these guidelines is to assist health care providers and practitioners make informed decisions on clinical diagnosing and treatment based on recommended protocols based on systematic review of evidence and an assessment of the benefits and harms of alternative care options. The guidelines is used for health professionals and relevant stakeholders in both public and private sectors, including doctors, related health professionals, trainees and medical students, policy makers, professional societies, in both public and private health facilities and institutions, including Provincial and district hospitals, private clinic, training and research institutions.

The main end users of the CPG are clinicians, managers and other health professionals practicing in patient care settings in both private and public sectors, particularly at referral hospitals and private clinics to support the delivery of consistently safe, quality and cost effective and equitable health care. They should be harmonized with the CPA guideline, IPC guideline, AMR guideline, Cambodian Hospital Accreditation Standards (CHAS) and Quality Enhancement Monitoring Tool (QEMT) and other MoH guidelines to motivate and facilitate their relevance and use. CPG can also be used by other relevant bodies and people such as in training and research institutions as well as by other organization or institutions as per the relevancy of their mandates.

CHAPTER I

ABDOMINAL SURGERY

1. Anal fissure
2. Fistula-in-Ano
3. Biliary peritonitis
4. Colorectal cancer
5. Cholecystitis
6. Biliary tree CBD obstruction
7. Appendicitis
8. Breast mass
9. Thyroid diseases
10. Gastric cancer
11. Hemorrhoid
12. Inguinal hernia
13. Intestinal obstruction
- 14&15. Liver abscess & Liver abscess ruptured peritonitis
16. Gastroesophageal reflux disease (GERD)
17. Pneumothorax
18. Splenic abscess
19. Pancreatic pseudocyst
20. Perianal abscess
21. Gastric perforated peritonitis
22. Large bowel perforated peritonitis
23. Small bowel perforated peritonitis
24. Rectal tumor
25. Tracheotomy and Tracheostomy
26. Traumatic diaphragmatic injury
27. Hemothorax
28. Pleural effusion
29. Abdominal traum

MANAGEMENT OF ABDOMINAL TRAUMA

Tuo Sothunea, Duong Chhay, Lay Nhork

I. CASE DEFINITION

Abdominal trauma refers to any injury that affects the abdominal organs, which may be due to blunt or penetrating force. This includes injuries to the stomach, intestines, liver, spleen, pancreas, kidneys, bladder and blood vessels within the abdomen.

II. ETIOLOGY

A. Blunt Abdominal Trauma

Blunt abdominal trauma refers to an injury to the abdomen that occurs without an object penetrating the skin. It is caused by a forceful impact or collision, often seen in situations like car accidents, falls, or physical assaults. The trauma can damage internal organs, blood vessels, and structures within the abdominal cavity, such as the liver, spleen, kidneys, intestines, and pancreas, without external wounds or breaks in the skin. The severity of the injury can range from minor bruises to life-threatening internal bleeding and organ damage.

- Motor Vehicle Accidents: Most common cause; includes collisions and rollovers.
- Falls: High-energy falls, especially from significant heights.
- Assaults: Physical violence involving kicks, blows, or blunt objects.
- Sports injuries: High-impact sports can cause abdominal injuries.

B. Penetrating Abdominal Trauma

Penetrating abdominal trauma refers to an injury to the abdomen that occurs when an object pierces the skin and enters the abdominal cavity, potentially damaging internal organs, blood vessels, and other structures. This type of trauma is commonly caused by sharp objects such as knives or bullets and can result in severe internal bleeding, organ damage, and infection. The severity of the injury depends on the trajectory and the organs affected.

- Motor vehicle Accidents: Most common cause; includes collisions and rollovers.
- Stab wounds: Often result from interpersonal violence.
- Gunshot wounds: Common in areas with high rates of violence or conflict.
- Industrial accidents: Includes injuries from sharp or heavy equipment.

III. DIAGNOSTIC PROCEDURE

A. Initial Assessment: Primary Survey

The primary survey is a rapid assessment focusing on life-threatening conditions. Remember that *this is not just about abdominal injury*. It follows the ABCDE approach:

A = Airway:

Assessment: Ensure the airway is open and unobstructed. Check for signs of obstruction (e.g., choking, stridor).

Intervention: If the airway is compromised, perform immediate interventions such as suctioning, airway positioning (e.g., head tilt-chin lift), or intubation if necessary.

B = Breathing:

Assessment: Evaluate breathing rate, depth, and effort. Look for signs of respiratory distress, such as cyanosis or use of accessory muscles.

Intervention: Provide supplemental oxygen if needed, and address any visible chest injuries or tension pneumothorax.

C = Circulation:

Assessment: Check pulse rate, blood pressure, and capillary refill time. Look for signs of shock, such as pallor, cool skin, or altered mental status.

Intervention: Establish IV access and start fluid resuscitation with crystalloids or blood products if there is evidence of significant blood loss or shock. Control any external bleeding with direct pressure or tourniquets. In case of internal bleeding, surgical intervention should be indicated.

D = Disability (Neurological Status):

Assessment: Use the Glasgow Coma Scale (GCS) to assess level of consciousness and responsiveness. Evaluate pupil size and reaction to light.

Intervention: Address any immediate neurological deficits. Stabilize the cervical spine if there is a suspected spinal injury.

E = Exposure/Environment:

Assessment: Fully expose the patient to examine for hidden injuries while maintaining normothermia.

Intervention: Prevent hypothermia with blankets or warm fluids as needed.

B. Clinical Presentation

The clinical presentation of abdominal trauma can vary depending on the severity and type of injury. Key aspects include:

i. History of Trauma:

Mechanism of Injury: Understanding how the injury occurred (e.g., motor vehicle accident, fall, assault) can provide insights into possible internal damage. High-energy impacts or penetrating injuries are more likely to cause severe damage.

Onset of Symptoms: Determine when symptoms began, as immediate symptoms might indicate severe injuries, whereas delayed symptoms could suggest less obvious damage.

ii. Symptoms and Physical Findings:

- Pain: Location and severity of abdominal pain can indicate the affected organs. For example:
 - Right Upper Quadrant Pain: Often associated with liver injuries.
 - Left Upper Quadrant Pain: Common in splenic injuries.
 - Lower Abdominal Pain: May suggest injuries to the intestines or pelvic organs.
- Tenderness: Localized or diffuse tenderness upon palpation. Guarding and rebound tenderness may indicate peritoneal irritation.
- Distension: Abdominal distension or swelling can be a sign of internal bleeding or fluid accumulation.
- Ecchymosis: Bruising around the abdomen, flanks (Grey Turner's sign), or around the umbilicus (Cullen's sign) can indicate severe internal bleeding or trauma.
- Bowel Sounds: Altered or absent bowel sounds may suggest bowel injury or ileus (intestinal obstruction).

iii. Signs of Shock:

Vital Signs: Hypotension (low blood pressure), tachycardia (rapid heart rate), and altered mental status can indicate significant blood loss or shock.

Capillary Refill: Delayed capillary refill time might suggest decreased perfusion due to shock.

iv. Specific Signs:

Peritoneal Signs: Positive signs include rebound tenderness or guarding, which may indicate peritoneal irritation from internal bleeding or inflammation.

Pelvic Examination: For female patients, a pelvic examination may reveal signs of injury to the reproductive organs or evidence of internal bleeding.

v. Diagnostic Imaging

- Ultrasound (FAST - Focused Assessment with Sonography for Trauma): Used to quickly identify free fluid in the abdomen.
- Chest X-Ray: May be used to identify free air under the diaphragm (indicating perforation) or to assess for rib fractures and other intrathoracic injuries.
- CTScan (performed only in stable condition, without signs of shock): Provides detailed images to assess the extent of injuries to abdominal organs and identify internal bleeding.

vi. Laboratory Test

- Complete Blood Count (CBC): Assesses for anemia and signs of infection.
- Serum Electrolytes and Renal Function Tests: Evaluates the impact on

systemic function and fluid balance.

- Type and Crossmatch: Prepares for potential blood transfusions based on the extent of blood loss.

C. Secondary Survey

The secondary survey should be conducted after the primary survey has been completed and life-threatening conditions have been addressed. These are some principals of the secondary survey:

- Completion of Primary Survey: Ensure that the patient's airway, breathing, circulation, disability, and exposure/environment have been thoroughly assessed and stabilized.
- Stabilization of Critical Conditions: Address any immediate threats to life, such as severe bleeding, compromised airway, or shock, before proceeding.
- Transition to Secondary Survey: Once the primary survey is complete and the patient is stabilized (or ongoing critical interventions are in place), proceed with the secondary survey to perform a detailed and systematic evaluation of all injuries and conditions.
- Purpose: The secondary survey allows for a comprehensive assessment of less obvious injuries and conditions that may not be immediately life-threatening but still require attention.

During this moment, head-to-toe examination is important to find other non-life-threatening injuries. The imaging depends on the suspicious injuries during this examination.

- Head and Face:
 - Inspection: Look for signs of trauma such as lacerations, contusions, or deformities. Check for swelling or asymmetry.
 - Palpation: Gently palpate the scalp, skull, and facial bones to detect tenderness, crepitus (grating sound), or deformities. Examine the orbits and check for periorbital ecchymosis ("raccoon eyes").
 - Eyes: Assess pupil size, shape, and reaction to light. Look for signs of ocular injury or abnormal eye movements.
 - Nose and Ears: Check for bleeding (rhinorrhea or otorrhea), deformities, or bruising. Inspect for Battle's sign (bruising behind the ear), which can indicate a skull fracture.
 - Mouth: Inspect the oral cavity for bleeding, lacerations, or missing teeth. Check for airway patency.
- Neck:
 - Inspection: Look for bruising, swelling, or deformities. Assess for signs of tracheal deviation or jugular venous distension.
 - Palpation: Gently palpate the cervical spine and soft tissues for tenderness or crepitus. Be cautious of potential spinal injuries and avoid excessive movement.

- Assessment: Evaluate for signs of subcutaneous emphysema, which can indicate injury to the trachea or esophagus.
- Chest:
 - Inspection: Look for asymmetry, bruising, or abnormal movements. Check for signs of chest wall injury or respiratory distress.
 - Palpation: Assess for tenderness, rib fractures, and crepitus. Palpate for subcutaneous emphysema.
 - Percussion: Check for dullness or hyper-resonance, which may indicate fluid or pneumothorax.
 - Auscultation: Listen for normal breath sounds and note any abnormal findings like decreased or absent sounds, which may suggest pneumothorax or hemothorax.
- Abdomen:
 - Inspection: Observe for distension, bruising, or asymmetry.
 - Palpation: Gently palpate all quadrants to assess for tenderness, rigidity, or masses. Note any rebound tenderness or guarding, which may indicate peritoneal irritation.
 - Percussion: Assess for tympany or dullness, which can indicate fluid or gas accumulation.
 - Auscultation: Listen for bowel sounds and check for any abnormal sounds.
- Pelvic:
 - Inspection: Look for bruising, deformity, or instability.
 - Palpation: Gently palpate the pelvis to assess for tenderness, stability, and any signs of fracture. Perform a gentle compression test to check for pelvic stability.
 - Assessment: In females, perform a pelvic examination, if necessary, to check for internal bleeding or trauma to reproductive organs.
- Extremities:
 - Inspection: Check for deformities, swelling, bruising, or open wounds. Assess for limb symmetry and compare bilaterally.
 - Palpation: Palpate each limb for tenderness, swelling, and crepitus. Check joints for stability and range of motion.
 - Neurovascular Assessment: Check distal pulses, capillary refill time, and sensation. Assess motor function and compare with the uninjured side.
- Back and spines:
 - Inspection: Look for signs of injury or bruising along the spine.
 - Palpation: Gently palpate along the vertebral column to identify tenderness or deformities. Be cautious of possible spinal injuries and avoid excessive

movement.

- Assessment: Evaluate for signs of spinal cord injury, such as changes in sensation or motor function.

IV. DIFFERENTIAL DIAGNOSIS

- Gastrointestinal Perforation: Can cause acute peritonitis and requires urgent intervention.
- Splenic Rupture: Often associated with left upper quadrant pain and hypovolemic shock.
- Hepatic Injury: Can cause right upper quadrant pain, jaundice, and coagulopathy.
- Kidney Injury: Presents with flank pain and hematuria.
- Pancreatic Injury: Can lead to abdominal pain, nausea, and potentially delayed onset of symptoms.

V. THERAPEUTICAL APPROACH

A. General Principles

- Stabilization: Ensure airway, breathing, and circulation are stabilized.
- Pain Management: Provide adequate analgesia.
- Fluid Resuscitation: Administer IV fluids and blood products as needed.
- Antibiotics: Administer prophylactic antibiotics if there is a risk of contamination or infection.

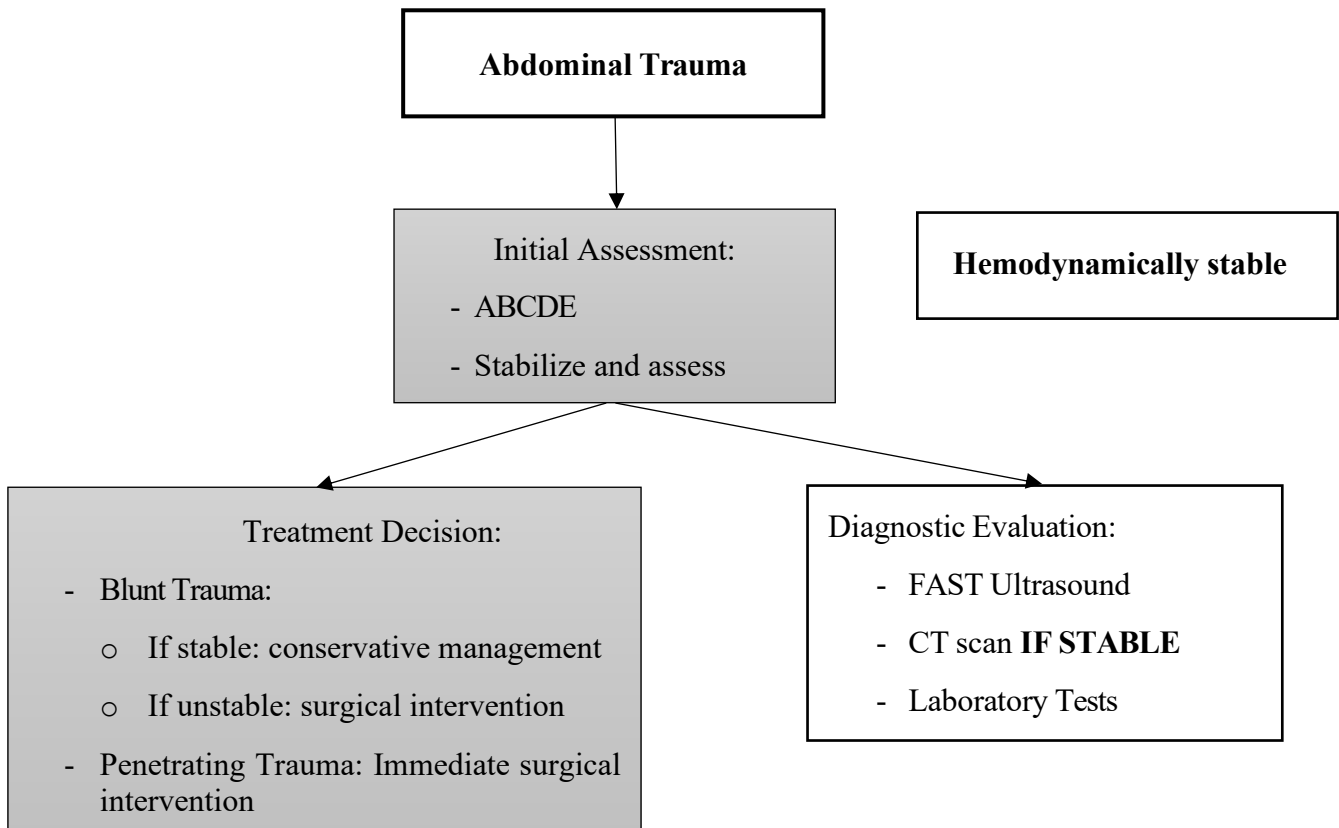
B. Specific Management

- Blunt Trauma: Management may be conservative or surgical depending on the extent of the injury.
 - Observation: For stable patients with minor injuries.
 - Surgical Intervention: Required for significant injuries such as massive hemorrhage or organ perforation.
- Penetrating Trauma: Generally, requires *immediate surgical intervention*.
 - Exploratory Laparotomy: For identification and repair of damaged organs and vessels.
- Exploratory Laparotomy:
 - Role: This surgical procedure is performed if there is suspected severe internal bleeding or organ injury that cannot be managed through conservative methods or imaging alone. It is not part of the primary assessment but is indicated based on the findings from the assessment. If the primary assessment reveals signs of hemodynamic instability or significant abdominal trauma, exploratory laparotomy is considered to identify and control internal bleeding or organ damage.

C. Postoperative Care

- Monitoring: Regular assessment of vital signs, urine output, and abdominal examination.
- Pain Management: Adequate analgesia to ensure comfort.
- Rehabilitation: Ensure proper wound care and gradual return to normal activity.

VI. ALGORITHM



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ACUTE CHOLECYSTITIS

BUN Sela, PLOK Vuthy

I. CASE DEFINITION

Acute cholecystitis refers to inflammation of the gallbladder. Mostly, it occurs as a complication of gallstones but sometimes it may develop also without gallstone called acalculous cholecystitis (1).

II. ETIOLOGY

The etiology of acute cholecystitis is defined by the blockage of the cystic duct which leads to inflammation. About 95% of patient with acute cholecystitis are due to gallstones, only 5-10% of acute cholecystitis are alithiasis (2).

Theses stones are formed when homeostasis is disrupted, due to:

- ☐ Bile stasis
- ☐ Supersaturation from the liver of cholesterol and lipids
- ☐ Disruption in the concentration process
- ☐ Cholesterol crystal nucleation.

The prevalence of gallstones is related to many factors:

- ☐ Age
- ☐ Gender
- ☐ Obesity, diet, physical activity, and weight loss
- ☐ Genetics and ethnic background
- ☐ Other factors: drugs, infections, parasites, surgery

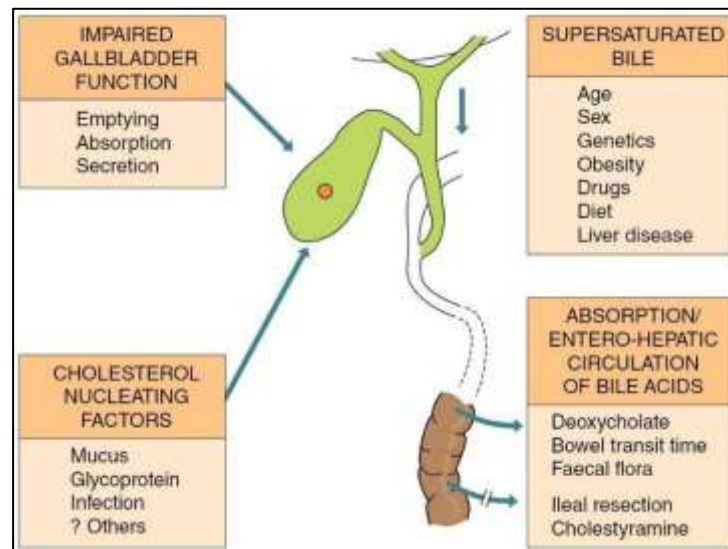


Figure 1: Etiology of gallbladder stone (3)

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

History: Fever + Abdominal pain

- ☐ Localization: right upper quadrant or epigastrium
- ☐ Duration: Steady, severe and is typically prolonged (greater than four to six hours)
- ☐ Radiation: Right shoulder or back.
- ☐ Associated complaints may include fever, nausea, vomiting, and anorexia.

Physical examination:

- ☐ **Murphy sign:** clinician's fingers are wrapped under the right costal margin lateral to the rectus abdominis muscles, and the patient is instructed to take a deep breath. If this maneuver elicits pain or abrupt cessation in inspiration, the test is **positive** and raises concern for cholecystitis (3).
- ☐ Note that abdominal tenderness on palpation and jaundice may also be present during the general abdominal examination.

ii. Technical procedure

a. Laboratory test

Patient might present with:

- ☐ Left shift leukocytosis with an increased number of band forms
- ☐ Elevation of CRP
- ☐ However, mild elevations in serum aminotransferases and amylase, along with hyperbilirubinemia and jaundice, have been reported even in the absence of common bile duct stone complications and may be due to the passage of sludge, or pus (4).

b. Imaging studies

Ultrasound: sensitivity and specificity = 88% and 80% respectively (5). Sonographic features may include:

Gallbladder wall thickening (greater than 4 to 5 mm)

Pericholecystic fluid, or edema (double wall sign).

A "sonographic Murphy's sign": is similar to the Murphy's sign elicited during abdominal palpation, except that the positive response is observed during palpation with the ultrasound transducer. This is more accurate than hand palpation because it can confirm that it is indeed the gallbladder that is being pressed by the imaging transducer when the patient catches his or her breath.

CT scan: sensitivity and specificity = 94% and 59% respectively (6). Abdominal scan is performed to rule out complications of acute cholecystitis in patients with sepsis (gangrene), generalized peritonitis (perforation), abdominal crepitus (emphysematous cholecystitis), or bowel obstruction (gallstone ileus). Its sensitivity and specificity are 94% and 59% respectively.

Abdominal CT is not routinely required to diagnose acute cholecystitis but is often performed in the initial evaluation of patients with abdominal pain to rule out other etiologies.

CT findings of acute cholecystitis include gallbladder wall edema pericholecystic stranding and fluid, and high-attenuation bile (7).

MRI: is not the first intention test for acute cholecystitis. It is requested mostly in cases of suspected choledocholithiasis (8).

IV. DIFFERENTIAL DIAGNOSIS

It is important to know, one can have pain due to temporary obstruction by gallstones, and that is called biliary colic. The diagnosis of biliary colic is upgraded to acute calculous cholecystitis if the pain does not resolve in six hours. If no stone is identified, it is called acute acalculous cholecystitis (9).

Biliary colic – As with acute cholecystitis, biliary colic causes pain in the right upper quadrant. However, the pain of biliary colic typically lasts at least 30 minutes, plateauing within an hour. The pain then starts to subside, with an entire attack usually lasting less than six hours. An episode of right upper quadrant pain lasting for more than six hours should raise suspicion for acute cholecystitis. Unlike acute cholecystitis, constitutional

symptoms such as malaise or fever and peritoneal signs are absent in patients with biliary colic and laboratory studies are normal (10).

Acute cholangitis – Acute cholangitis may present a right upper quadrant pain, fever, and leukocytosis. In contrast to patients with acute cholecystitis, patients have evidence of cholestasis on laboratory testing and biliary dilation on imaging. It is important to note that patients can have both acute cholangitis and acute cholecystitis (10).

Fitz-Hugh Curtis syndrome – In women with acute pelvic inflammatory disease, perihepatitis can cause right upper quadrant abdominal pain. Pericholecystic fluid on imaging studies may be confused with acute cholecystitis, but a hepatic iminodiacetic acid scan is negative (10).

Other conditions that can give rise to pain in the epigastrium or right upper quadrant abdomen (eg, acute pancreatitis, peptic ulcer disease) can usually be differentiated from acute cholecystitis by the clinical setting in which they occur and by obtaining the appropriate diagnostic studies.

V. THERAPEUTIC APPROACH

Patients diagnosed with acute calculous cholecystitis (ACC) should be admitted to the hospital and provided with supportive care including (11):

- ☐ Intravenous hydration.
- ☐ Correction of any electrolyte abnormalities.
- ☐ Pain control.
- ☐ Intravenous antibiotics.
- ☐ Patients should be kept fasting, and although uncommonly needed, those who are vomiting should have placement of a nasogastric tube.

Antibiotics:

In complicated acute calculous cholecystitis (gangrene/necrosis, rupture, or emphysematous cholecystitis), broad-spectrum antibiotics are required. For those with uncomplicated form, antibiotics administration are also suggested mostly in diabetic and immunocompromise patient (12,13).

Once started, antibiotic therapy should continue until either the gallbladder is removed or the cholecystitis clinically resolves. In general, antibiotic should be continued a day after cholecystectomy in uncomplicated form as supported by many randomized trials (14,15). For complicated form, it is discussed in case by case.

Antibiotic can be considered either Piperacillin/Tazobactam alone or a combination (Cephalosporine + Metronidazole) as recommended by Tokyo guideline 2018 on a table below (16).

Severity	Community-acquired biliary infections			Health-care-associated biliary infections*
	Grade I	Grade II	Grade III*	
Antimicrobial agents	Cholangitis and cholecystitis	Cholangitis and cholecystitis	Cholangitis and cholecystitis	Health-care-associated cholangitis and cholecystitis
Penicillin-based therapy	Ampicillin/sulbactam [§] is not recommended if >20% resistance rate	Piperacillin/tazobactam	Piperacillin/tazobactam	Piperacillin/tazobactam
Cephalosporin-based therapy	Cefazolin, ^Δ or cefotiam, ^Δ or cefuroxime, ^Δ or ceftriaxone, or cefotaxime ± metronidazole [◊]	Ceftriaxone, or cefotaxime, or cefepime, or ceftazidime ± metronidazole [◊]	Cefepime, or ceftazidime, or ceftazidime ± metronidazole [◊]	Cefepime, or ceftazidime, or ceftazidime ± metronidazole [◊]
	Cefmetazole, ^Δ cefoxitin, ^Δ flomoxef, ^Δ cefoperazone/sulbactam	Cefoperazone/sulbactam		
Carbapenem-based therapy	Ertapenem	Ertapenem	Imipenem/cilastatin, meropenem, doripenem, ertapenem	Imipenem/cilastatin, meropenem, doripenem, ertapenem
Monobactam-based therapy	–	–	Aztreonam ± metronidazole [◊]	Aztreonam ± metronidazole [◊]
Fluoroquinolone-based therapy [§]	Ciprofloxacin, levofloxacin, pazufloxacin ± metronidazole [◊]	Ciprofloxacin, levofloxacin, pazufloxacin ± metronidazole [◊]	–	–
	Moxifloxacin	Moxifloxacin		

Antimicrobial recommendation for acute biliary infection (16)

Indication of emergency cholecystectomy:

- Complicated acute cholecystitis (gangrene, emphysematous or perforated gallbladder)
- Disease progression: high fever, hemodynamic instability, or intractable pain in spite of best supportive care (including antibiotics and gallbladder drainage) suggested possible gangrene cholecystitis. Surgery is indicated to prevent perforation and sepsis.

Technique: Laparoscopic cholecystectomy is the **gold-standard** treatment for patients with acute calculous cholecystitis (17).

Risk stratification: In good surgical candidate, cholecystectomy is recommended during the same hospitalization. However, for poor surgical candidates (septic shock or cardiopulmonary contraindications to anesthesia) but without indications for emergency cholecystectomy, nonoperative management (antibiotics + gallbladder drainage) is preferred.

Timing: Regard to timing to cholecystectomy, although the 2013 Tokyo Guidelines suggested that surgery be performed within 72 hours of symptom onset, it is not always practical, as some patients present after more than three days of symptoms (18). The updated 2018 Tokyo Guidelines concluded that in patients for whom more than 72 hours has passed since symptom onset, there are still benefits to performing cholecystectomy early, and recommended early cholecystectomy in low-risk patients with acute calculous cholecystitis regardless of how much time has passed since symptom onset (11).

The 2020 World Society of Emergency Surgery (WSES) guidelines recommended early laparoscopic cholecystectomy for ACC as long as it can be completed within 7 days of admission and 10 days of symptom onset (19). Patients who have had symptoms for longer than 10 days should not undergo early cholecystectomy unless their disease progresses; delaying cholecystectomy beyond six weeks is a safer option for those patients to allow inflammation to subside.

VI. COMPLICATION

Left untreated, symptoms of cholecystitis may abate within 7 to 10 days. However, as complications are frequent and have the potential of being severe such as:

Gangrenous cholecystitis—the most common complication of cholecystitis (up to 20 %), particularly in older patients, patients with diabetes, or those who delay seeking therapy (20). The presence of a sepsis-like picture in addition to the other symptoms of cholecystitis suggests the diagnosis, but gangrene may not be suspected preoperatively.

Perforation—approximately 10 percent of cases and usually occurs in patients with a delay in diagnosis or failure to respond to initial therapy. The perforation is often localized and occurs at the fundus of the gallbladder after the development of gangrene (21). The resulting pericholecystic abscess may be palpable and can be visualized on abdominal computed tomography. Less commonly, there is free perforation into the peritoneum, leading to generalized peritonitis and associated with a high mortality.

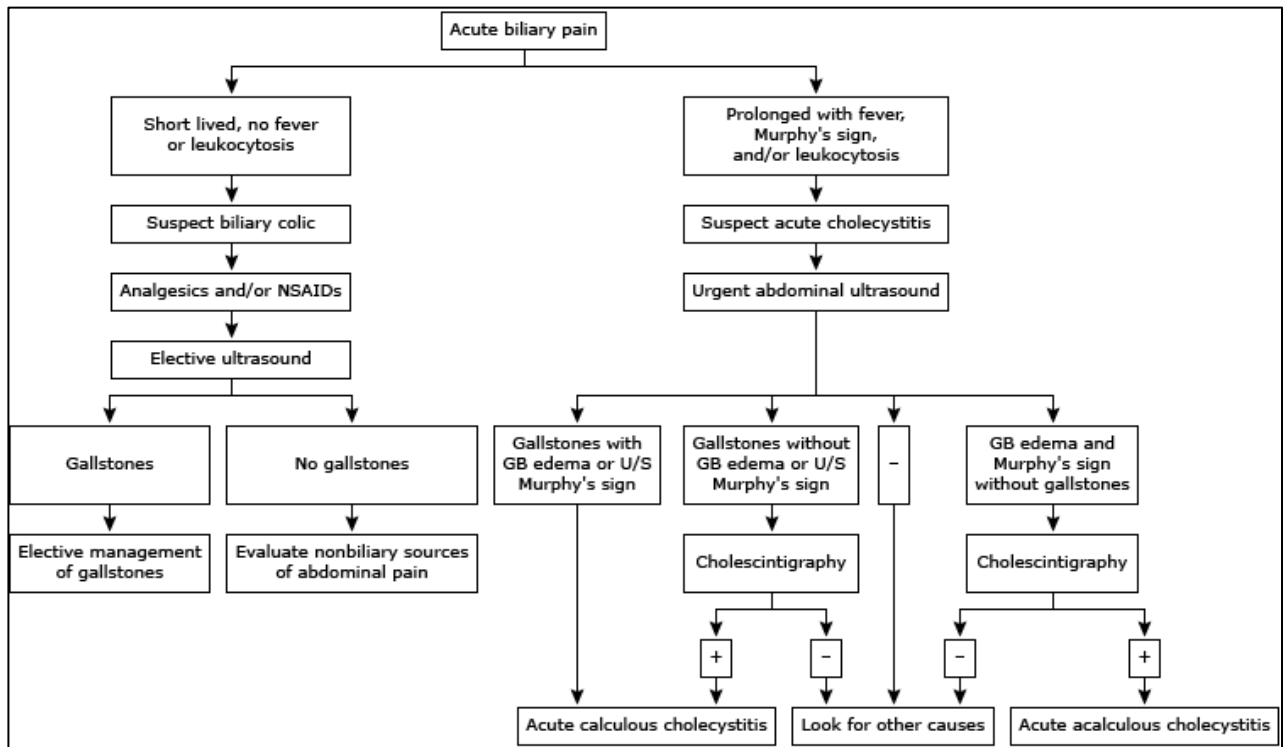
Emphysematous cholecystitis—Emphysematous cholecystitis is caused by secondary infection of the gallbladder wall with gas-forming organisms (such as *Clostridium welchii*) (22,23). Like other patients with acute cholecystitis, patients with emphysematous cholecystitis usually present with right upper quadrant pain, nausea, vomiting, and low-grade fever. Peritoneal signs are usually absent, but crepitus in the abdominal wall adjacent to the gall bladder may rarely be detected. When such crepitus is present, it is an important clue to the diagnosis.

Cholecystoenteric fistula—A cholecystoenteric fistula may result from perforation of the gallbladder directly into the intestinal lumen. Fistula formation is more often due to long-standing pressure necrosis from stones than due to acute cholecystitis. It complicates 2 to 3 percent of all cases of gallbladder stone disease but it can rarely occur as a result of acute calculous cholecystitis. While the majority of fistulas are cholecystoduodenal fistulas,

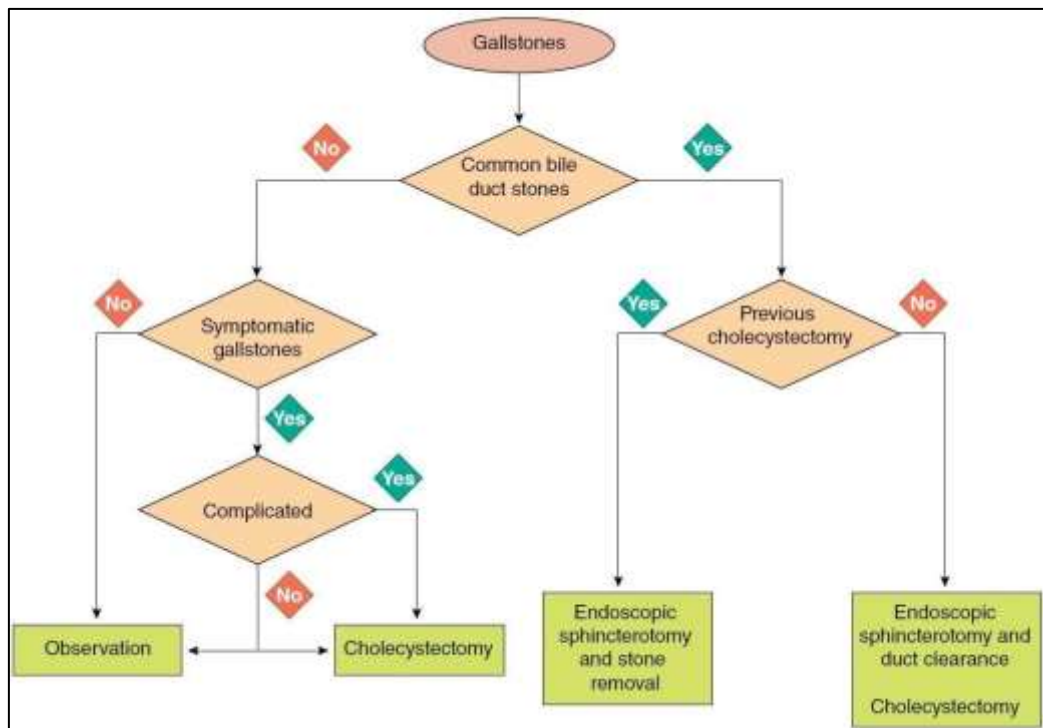
approximately 15 percent are cholecystocolonic, most commonly to the hepatic flexure (24). Symptoms of a cholecystocolonic fistula include bile acid diarrhea and rarely intestinal obstruction due to gallstone ileus.

Gallstone ileus — Passage of a gallstone, usually larger than 2.5 cm, through a cholecystoenteric fistula may lead to the development of mechanical bowel obstruction, usually in the narrowest part of the terminal ileum which is approximately two feet proximal to the ileocecal valve (25).

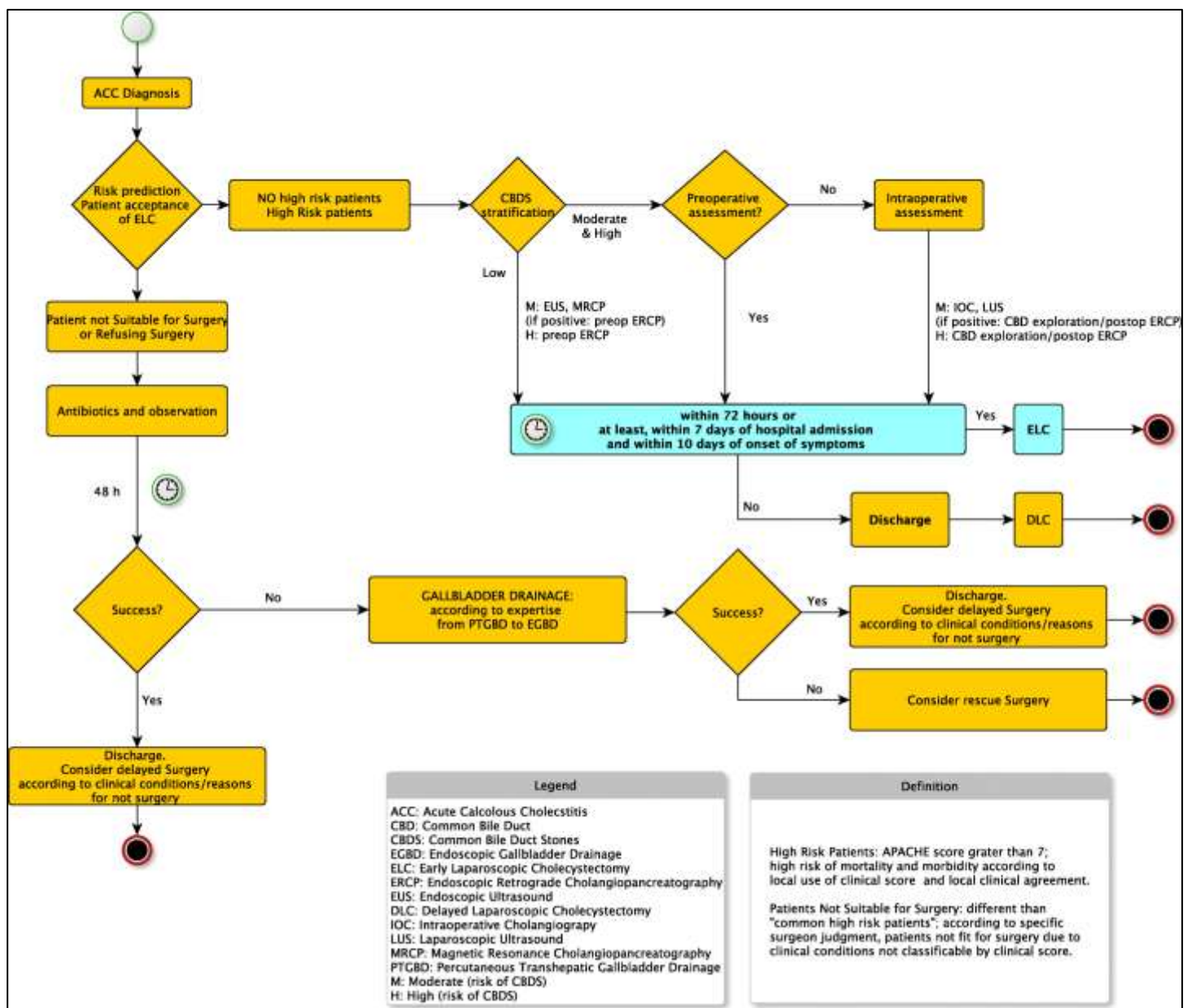
VII. ALGORITHM



Algorithm for the diagnosis of acute cholecystitis (10)



Management of gallstone (3)



Management acute calculous cholecystitis (19)

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ANAL FISSURE

MO Satdin, EM Sokhom

I. DEFINITION

A linear tear in the anal canal that may stretch from the mucocutaneous junction to the dentate line is known as an anal fissure. Anal fissures are usually the result of constipation and diarrhea. Despite being a painful condition, its etiopathogenesis remains obscure.

II. ETIOLOGY

The etiology of the fissure depends upon whether the fissure is primary or secondary.

The causes of a primary fissure include:

- Diarrhea
- Passage of hard stools
- Injury due to trauma or anal sex
- Vaginal delivery [2]

A secondary fissure may be due to:

- Inflammatory bowel diseases like Crohn's [2]
- Granulomatous disease like tuberculosis, sarcoidosis
- Surgical procedure in the anorectal region
- Malignancy
- Infection like HIV and syphilis [2]

Other theories which are related to the formation of anal fissures are:

- Tearing of anal valves due to passage of hard stool
- Anal infection theory related to the crypto-glandular origin
- Loss of elasticity due to infection and fibrosis
- Straining during parturition and chronic constipation
- Iatrogenic: as a complication of any inappropriate anal surgery [3]. Anal stenosis after hemorrhoidectomy can lead to a chronic anal fissure

III. DIAGNOSTIC PROCEDURES

i. Clinical argument

Anal fissures cause moderate to severe pain in the anal area. The pain starts during defecation and continues for hours. It is sharp cutting in nature. Due to the trauma caused by hard stool, painful bleeding is usually present. The blood is bright red and in the form of a streak. The chronic anal fissures are not always associated with bleeding.

A longitudinal superficial tear is seen extending up to the dentate line which may be tenderness and associated with bleeding.

The tear in a chronic fissure can be deep enough to expose the internal anal sphincter's muscular. The sentinel piles may be seen at the distal end of the fissure

During
palpation:

- Presence of sentinel tags
- Induration and fibrosis in chronic wounds
- Hypertrophic anal papillae
- Visible internal sphincter fibers Digital rectal examination
- In acute fissure, the anal tone is increased due to spasm
- It is mandatory to perform a thorough rectal examination to assess the exact etiology

ii. Proctoscopy

Proctoscopy is a must to rule out other associated anorectal disorders. It may not be possible to do it initially as the patient has severe pain.

iii. Role of anal manometry

Anal manometry is employed to measure the resting anal pressure. Anal fissure patients have a higher resting pressure [6,7].

- Normal resting anal pressure 40-60mmHg
- Increased anal tone > 80mmHg
- Decreased anal pressure < 40mmHg

iv. Various characteristic of anal fissures

a. Classification of anal fissure based on morphology [4,5]

Superficial anal fissure

- Severe pain and bleeding
- Anoderm separation on the surface with sharp edges
- Internal anal sphincter is not breached at the base of the fissure
- Heals spontaneously with days or weeks after conservative treatment

Deep anal fissure

- Internal anal sphincter fibers are often visible
- Wide pear-shaped ulcer
- A triad of indurated ulcer edges, sentinel pile and proximal hypertrophic anal papillae

b. Grading of anal fissures

- Grade 1: superficial linear tear
- Grade 2: deep linear tear with a small sentinel pile
- Grade 3: deep linear tear with fibrosed margin and sentinel pile
- Grade 4: wide tear with visible fibers of the internal anal sphincter and sentinel pile
- Grade 5: wide pear-shaped tear with visible fibers of the internal sphincter with hypertrophic anal papillae and sentinel pile

c. 2 types of anal fissures [4]

- Acute fissure: is formed as a linear tear that heals within 6 weeks and has fresh mucosal edges
- Chronic fissure: heals after 6 weeks and may be associated with sentinel pile, hypertrophic anal papillae and internal sphincter fibrosis at the fissure site

IV. DIFFERENTIAL DIAGNOSIS

- Thrombosed external hemorrhoids
- Perianal abscess
- Perianal ulceration in inflammatory bowel disease, tuberculosis
- Anal sexually transmitted diseases

V. THERAPEUTIC APPROACH

v. Dietary modification

The patient is advised to increase dietary fiber and fluid intake to keep the stool soft. The recommended dose of fiber intake is about 38 g in men and 28 g in women. The water intake should be 2 to 3 L/day. Strong recommendations are based on moderate-quality evidence [8].

vi. Sitz bath

A Sitz bath reduces the sphincter spasm and increases the blood flow, leading to pain relief. This is more effective than topical anesthetics and topical hydrocortisone [8].

vii. Medical treatment

Medical management includes the use of laxatives and topical agents. The osmotic laxatives are used as they soften the stools. The most typical used laxatives include lactulose and polyethylene glycol. The common topical ointments used are nitroglycerine (GTN) and calcium channel blockers (diltiazem, nifedipine). Botulinum toxin can also be used in the management of fissures.

Calcium Channel Blockers

They act by decreasing the intracellular availability of calcium [9]. The topical nifedipine reduces anal sphincter tone, which helps in promoting

blood flow and faster healing. Nifedipine combined with 2% lidocaine jelly produces a dual effect as lidocaine causes numbness of the area but does not heal the fissure. Diltiazem promotes muscular sphincter relaxation by inhibiting calcium channels in smooth muscles [1].

According to studies, topical nifedipine has a greater healing rate than diltiazem [10].

Botulinum Toxin (Botox)

Clostridium botulinum generates botulinum toxin, which works by preventing acetylcholine from being released at the neuromuscular junction [11]. It blocks sympathetic nerve function and the myogenic tone of the internal anal sphincter. Hence, it helps in reducing the sphincter's increased tone and increasing the local tissue perfusion leading to healing of chronic anal fissure. It can cause local paralysis, lasting for months [12]. Botox has a similar action as that of lateral sphincterotomy [13].

viii. Surgical procedures

Any fissure that fails to respond to conservative management is an indication of surgical intervention. Surgical management aims to reduce anal spasms and hence increases the blood flow.

4 techniques of surgical management:

- Anal dilatation
- Fissurectomy
- Lateral internal sphincterotomy
- Advancement flap (anoplasty)

Indications

- Chronic anal fissure
- Lack of response to medical management
- An increase in anal tone associated with fissure
- Fissure associated with underlying fistula
- Presence of persistent pain and bleeding
- Relapsing of refractory fissure

Anal Dilatation

- Stretching of the anal canal was the first treatment proposed for anal fissures. However, this procedure is obsolete because of a high incontinence rate [14].

Fissurectomy

- Fissurectomy is another technique practiced by surgeons to treat chronic anal fissures.

- The fissure is excised using a scalpel or scissors. Hence, an acute wound is created without scar tissue and is left to heal by secondary intention [15].

Lateral internal sphincterotomy

- It is deemed the gold standard surgical treatment for chronic anal fissure.
- The sphincter tone is minimized by dividing the internal anal sphincter up to the dentate line, the resting pressure without affecting the physiology of the anal canal.
- A lateral incision through the anoderm is made across the inter-sphincteric groove at 3 o'clock.

Advancement Flap (Anoplasty) [16]

- Anoplasty is a procedure that involves curettage of the fissure followed by placement of a flap.
- Anoplasty is performed by elevating a 1 cm thick subcutaneous cellular tissue and local skin flap. The flap is sutured to the rectal mucosa covering the fissure.

Laser lateral internal sphincterotomy

- Lateral internal sphincterotomy with laser can be performed either by open or closed technique. The principle and the basic technique remains the same as in open or closed lateral internal sphincterotomy. The only difference is that instead of scissors or cautery, a laser fiber is used for sphincter division. The cutting by laser is precise and immediate hemostasis is achieved.

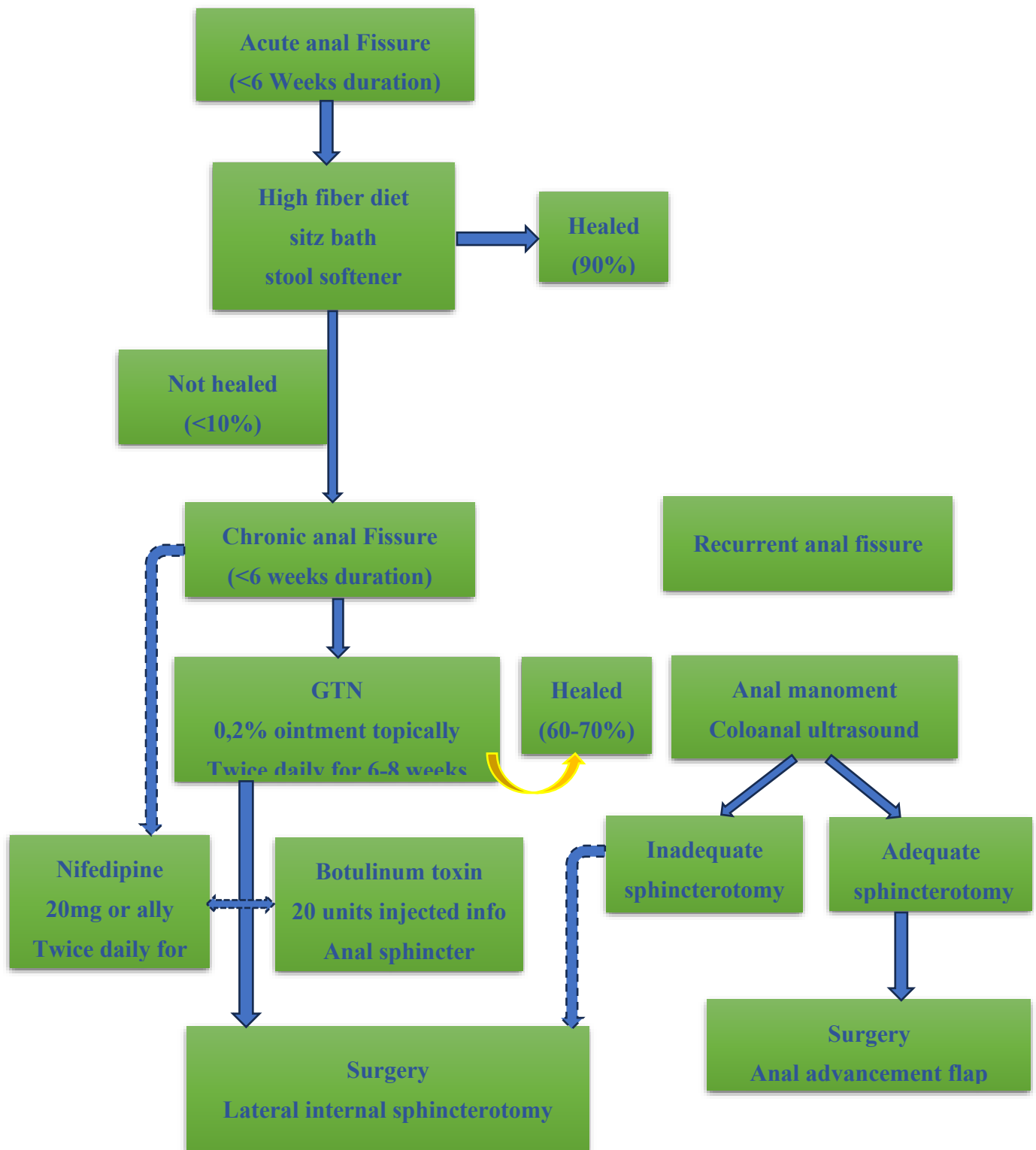
VI. COMPLICATION

If left untreated, the fissure may cause:

- Chronic painful
- Infection
- Bleeding
- Fistula formation

VII. ALGORITHM

Algorithm 1. Referral management of Anal fissure



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APPENDICITIS

Preap Ley, Norng Sanareth, Koeu Kimsath, Cheav Darareak, Ong Visal

I. DEFINITION

Appendicitis is defined as inflammation of the vermiform appendix and worldwide is the most common reason for emergency abdominal surgery. Globally, the annual incidence is 96.5 to 100 cases per 100 000 adult population (USA).

Many kinds of appendicitis condition were classified, but the most in practice is:

1. Uncomplicated Appendicitis - includes the acutely mildly or severe inflamed, phlegmonous or suppurative appendix.
2. Complicated Appendicitis - includes gangrenous appendicitis, perforated appendicitis localized purulent collection at operation (localized peritonitis), generalized peritonitis and periappendiceal abscess.
3. Equivocal Appendicitis - a patient with right lower quadrant abdominal pain who presents with an atypical history and physical examination and the surgeon cannot decide whether to discharge or to operate on the patient.

Appendiceal position is:

1. Retrocaecal (75%)
2. Pelvic (21%)
3. Preileal (anterior) (1%)
4. Postileal
5. Paracaecal
6. Subcaecal (2%)
7. Subhepatic

II. ETIOLOGY

- a. Development of appendicitis is related to swelling of the lymphoid tissue in the submucosa in response to viral or bacterial infection. This may then proceed to inflammation. (Acute nonobstructive appendicitis may lead into resolution, fibrosis, recurrent appendicitis or eventual obstructive appendicitis. In severe acute appendicitis → localization can occur by omentum and dilated ileum without pus inside → forming appendicular mass or periappendicular phlegmon).
- b. The presence of gangrene or perforation is associated with the presence of faecolith or stricture, adhesions and kinking, foreign body, round worm or threadworm obstructed. Approximately 50% of cases of gangrenous or perforated appendicitis are associated with a faecolith. (acute obstructive appendicitis → perforation at the tip or at the base → peritonitis. In contrast, after perforation → localization by greater omentum and dilated ileum occurs → with suppuration and pus inside forming appendicular abscess).

- c. Diet may be involved (reduced fiber content) as the disease is very common in the West but rare in Africa. Consumption of green vegetables and tomatoes may be protective.
- d. Family history may be relevant in 30% of appendicitis in children with appendicitis occurring in first degree relatives.
- e. In elderly patients there is some evidence that chronic intake of non-steroidal anti-inflammatory drugs may increase the risk.
- f. It is common in May and August—seasonal variation—often called as epidemic appendicitis.

III. DIAGNOSTIC PROCEDURE

- g. Clinical argument (Acute appendicitis)
 - i. HISTORY OF PRESENTING ILLNESS: History and physical examination are the initial steps in diagnosing acute appendicitis.
 - 1. Classically, abdominal pain is the first symptom of appendicitis and typically presents in the periumbilical region and then 50% to 60% of patients, the pain migrates to the right lower quadrant within 24 hours (Symptom duration more than 24 hours is a risk factor for perforation).
 - 2. 80% to 85% of patients report anorexia after the onset of abdominal pain and
 - 3. 40% to 60% report nausea with or without emesis.
 - ii. PHYSICAL EXAMINATION: Common signs include:
 - 1. Fever with or without chill
 - 2. McBurney sign (tenderness at the point one-third of the distance from the right anterior superior iliac spine to the umbilicus),
 - 3. Rovsing sign (pain in the right lower quadrant with palpation of the left lower quadrant)
 - 4. Psoas sign (right lower quadrant abdominal pain with passive right hip extension- Retrocecal Appendix),
 - 5. Obturator sign (right lower quadrant pain with flexion of the right hip and knee, followed by internal rotation of the right hip- Pelvic Appendix).
- h. Technical procedure
 - i. Laboratory test
 - Individually, the white blood cell (WBC) count and inflammatory biomarkers (procalcitonin, calprotectin, CRP, and the APPY1 biomarker) lack accuracy for the diagnosis of acute appendicitis. However, laboratory tests are helpful when combined with signs and symptoms, or in combination with imaging studies as part of a structured evaluation.
 - Laboratory evaluation of patients with suspected appendicitis should include a complete blood cell count with differential. A mild leukocytosis (white blood cell count >10 000/μL) is present in 67% to 90% of patients with acute

appendicitis, and approximately 80% have a left shift in the differential.

- alternative diagnoses can include urine analysis (to rule out urinary tract infection) and pelvic examination with urine β -human chorionic gonadotropin level (to rule out uterine or ectopic pregnancy).

ii. Imaging study

- The most common modalities for imaging are abdominal ultrasound. If Ultrasound is non-diagnostic consider a CT abdomen. If CT abdomen is still non-diagnostic and available should consider MRI.
- In low resource setting and imagery modalities are not available or non-diagnostic:
 - High clinical suspicion = Operate
 - Intermediate/ Low clinical suspicion = Observe

Ultrasound is the first modality recommended for patients with suspected appendicitis, although it is acknowledged that a negative ultrasound or non-visualization of the appendix does not exclude acute appendicitis. Some key signs in ultrasound:

- Elongated, blind-ending structure that is fixed and non-compressible with a diameter > 6mm
- Increased echogenicity of fat suggest inflammation
- Irregular contour and peri-appendiceal fluid suggest perforation CT with intravenous contrast administration is recommended because it improves examination accuracy, particularly in patients with appendiceal perforation or abscess formation, and facilitates detection of alternative disease processes. Non-contrast CT is an acceptable alternative, however, if intravenous contrast is contraindicated because of the relatively high diagnostic accuracy of CT without contrast. The routine administration of oral contrast is not necessary in the setting of suspected acute appendicitis. Some key signs in CT:

- Diameter > 7 mm, wall thickness > 2 mm and fluid filled appendix are signs of appendicitis
- Edema and inflammation are seen as T2 hyperintensity around or within the appendix
- In pregnant patients it can be difficult to delineate blind ending tubular due to displacement by the uterus MRI would be the next diagnostic test of choice, if It is available. It is no radiation, useful in pregnancy, can identify other intra-abdominal pathology, but takes time to organize and complete. Some key signs in MRI:

- Enlarged, blind-ending tubular structure associated with inflammatory stranding in surround fat
- Diameter > 6mm, thickening of wall > 1mm and abnormal wall enhancement

- Appendicolith CT scan and Ultrasound is recommended for equivocal appendicitis

IV. DIFFERENTIAL DIAGNOSIS

- i. Gastrointestinal
 - i. Acute cholecystitis
 - ii. Mesenteric adenitis
 - iii. Terminal ileitis
 - iv. Diverticulitis
 - v. Carcinoid tumour
 - vi. Meckel's diverticulum
- j. Gynaecological
 - i. Ectopic pregnancy
 - ii. Torted ovarian cyst
 - iii. Pelvic inflammatory disease
 - iv. Mittelschmerz pain
- k. Genito-urological
 - i. Testicular torsion
 - ii. Ureteric colic
 - iii. Pyelonephritis
- l. Others
 - i. Diabetic ketoacidosis
 - ii. Herpes zoster
 - iii. Porphyria

V. THERAPEUTIC APPROACH

- m. Stabilize patient
 - 1. When the decision is made to perform an appendectomy for acute appendicitis, the patient should keep NPO and proceed to the operating room with little delay to minimize the chance of progression to perforation.
 - 2. Patients with appendicitis may be dehydrated from fever and poor oral intake, so intravenous fluids should be begun, and pulse, blood pressure, and urine output should be closely monitored.
 - 3. Markedly dehydrated patients may require a Foley catheter to ensure adequate urine output.

4. Severe electrolyte abnormalities may be significant in cases of perforation. Any electrolyte deficiencies should be corrected prior to the induction of general anesthesia.
 5. Intravenous broad-spectrum antibiotics if there is evidence of perforation. No standardized antibiotic regimen exists. Acceptable options include a second-generation cephalosporin or a combination of antibiotics directed at gram negatives and anaerobes.
 6. There are many meta-analyses of randomized controlled trials showed that the use of opioids or other painkillers after appendicitis diagnosis is a paramount.
- n. Empirical treatment
- 1) Appendectomy, via open laparotomy or via laparoscopy, is the standard treatment for acute appendicitis. Even emerging evidence suggests that Antibiotic, IV fluids and Analgesics therapy may be considered a first-line and possibly sole therapy in selected patients with uncomplicated appendicitis.
 - 2) Patient should receive a prophylaxis dose of IV antibiotics within 30mn to 1 hour prior to skin incision.
 - 3) Post-operative antibiotics
 - A. Macroscopically normal appendix: No post-operative antibiotics
 - B. Simple appendicitis: No post-operative antibiotics
 - C. Complicated appendicitis:
 - a. Antibiotics to be administered for a minimum of 72 hours post-operatively
 - b. Consider conversion to oral antibiotics earlier than 72 hours when
 - i. The patient has been afebrile (Temp <37.5°C) for over 24 hours
 - ii. The patient is tolerating oral intake well
 - iii. There is no antibiotic resistance to the selected antibiotic in culture
 - c. Consider stopping antibiotics between 3 to 5 days after appendectomy when
 - i. The patient has been afebrile (Temp <37.5°C) for over 24 hours
 - ii. The white cell count is below 10
 - d. If a patient continues to be febrile on D5, continue antibiotics and consider investigation for intra-abdominal collection if no improvement by D7.
- o. Directed treatment
- Consent patient appropriately
 - Surgical Steps of open appendectomy for uncomplicated appendicitis:
 1. Incision at McBurney's Point [In open technique, a small oblique or

transverse incision centered on McBurney's point is sufficient. If the diagnosis is uncertain, either a vertical (midline or right paramedian) or transverse incisions are used. These incisions may be extended to address other conditions should the diagnosis of acute appendicitis prove wrong] or point of maximum tenderness (during pregnancy).

2. Bluntly divide external, internal and transverses muscle fibers parallel to their directional course
 3. Enter peritoneum
 4. Pull cecum into wound with sponge or Babcock (it may be necessary to divide peritoneal attachments to lateral sidewall to accomplish this)
 5. Identify appendix
 6. Make window at base of appendix
 7. Crush-clamp appendiceal base, suture ligate with absorbable suture; divide appendix above (keep clamp on to avoid spillage)
 8. Suture ligate mesoappendix en masse or dissect out and clip appendiceal artery
 9. Remove appendix
 10. Optional: invaginate stump with purse-string suture
 11. Assess for hemostasis; suction out any purulent fluid
 12. Close wound in layers (The incision may be closed primarily in patients with complicated appendicitis)
 13. Drains are not recommended in acute appendicitis cases
 14. The appendix should always be sent for histopathological examination
 15. A standardized operation note is recommended
- For gangrenous appendicitis, the recommended form of management is to treat in the same manner as uncomplicated appendicitis.
 - Most patients with perforated appendicitis can be treated with routine appendectomy. The base of the appendix proximal to the obstruction is often reasonably normal, and secure closure is possible. An attempt should be made to find and remove the fecalith. Rarely is the perforation right at the cecum. The cecal edges may be so indurated that the perforation cannot be closed. The surgeon has two alternatives: Either place a cecostomy tube and drain the RLQ or perform a limited right hemicolectomy.
 - For localized peritonitis, perform appendectomy and no necrotic tissue or purulent material should be left behind as much as possible. General peritoneal lavage is not recommended. Intraperitoneal drains, while most useful in patients with a well-established and localized abscess cavity, should be selectively utilized.
 - Patients occasionally present with appendicular mass or abscess. These are the sequelae of missed diagnosis and late presentation.

- i. The treatment is usually conservative, with intravenous fluid, antibiotics and analgesics for the appendiceal mass. Usually patient need the interval appendectomy 6 to 8 weeks later. If ineffective, surgery may be required, but it will cause faecal fistula postoperatively, should be warned.
- ii. A peri-appendiceal abscess should undergo surgery as soon as the diagnosis is made with appendectomy with base secure and put a drain to cavity and remove the necrotic tissue or purulent materials. But some patients, abscess may be drained with a catheter placed percutaneously under ultrasound or CT guidance. The patient is placed under broad-spectrum antibiotics and carefully followed clinically. If the patient fails to improve or deteriorates, surgical intervention is indicated. Often the patient improves, and the purulent catheter drainage subsides slowly. Such a patient may be discharged from hospital and readmitted for interval appendectomy 6 to 8 weeks later.

p. Monitoring

i. Clinical monitoring

- ☐ A collaborative approach between surgeon, physician, nurses, patient and family is advocated for all patients.
- ☐ Each patient should be reviewed a minimum of once every 24 hours by a senior decision maker until discharge.
- ☐ Intravenous fluid managements use with body weight calculation and electrolyte requirements. Assess the fluid status clinically. Measure fluid input and output hourly and calculate fluid balance with a subtotal every 12 hours and total every 24 hours. Control bloodglucose according to diabetic requirement. Calculate fluid requirement as per NICE guidance. Use isotonic solution to replace ongoing losses (e.g.NG tube). IVF should be stop as soon as patient can drink and eat.
- ☐ Foley catheters can be removed when patient can mobilize out of bed and passing urine by his/her self to reduce urinary tract infection.
- ☐ IV lines should be removed promptly when patients are no longer required to reduce the risk of line-associated infection.
- ☐ Early mobilization out of bed and go to bathroom by his/her self should be encouraged and supported.
- ☐ Chest exercise with deep breathing or other additional physiotherapy may be required for patient with low saturations, oxygen requirement or signs of worsening atelectasis and ensure that the pain is controlled enough to allow he/she to do this.
- ☐ NG tubes are not required for the majority of patients after appendicectomy. Patients with significant peritoneal contamination or bowel dilatation pre-operatively are at increased risk of post-operative ileus and may need it. It should be removed after bowel working.
- ☐ Analgesia: Patient's pain should be controlled enough to allow them to sleep comfortably, to cough and to take deep breaths exercise and mobilize.

Regular assessment using the tools described in the ‘analgesia’ section should be utilized. Step-down through the analgesic ladder as a patient recovers from their operation should occur.

ii. Laboratory monitoring

- There are no lab tests needed after appendectomy without postoperative complications.
- For postoperative complication cases, laboratory monitoring may need occasionally.

VI. COMPLICATIONS

q. Intra-abdominal collection (IAC) after appendectomy: Patient who remain or become febrile (temperature > 38C) on D7 or beyond after appendectomy, or who have ongoing abdominal pain, loose stools or other potential symptoms of IAC should be investigated for a focus of infection. Investigations should be governed by clinical symptoms and signs and may include:

i. Full blood count, and CRP

ii. Chest X-Ray with AFB sputum smear

iii. Blood, Urine and Stool culture

iv. Ultrasound or CT abdomen and pelvis

1. Intravenous antibiotics are recommended as a first-line treatment according to the results of intra-operative culture and sensitivity.
2. Large intra-abdominal collections which are accessible percutaneously or rectally should be considered for drainage, but only by specialist in standard center.
3. Re-look laparoscopy or laparotomy is rarely required with an IAC.

r. Wound infection following an appendectomy for a perforated appendicitis is another cause of fever. It can be prevented by delaying closure of the skin. When a wound abscess is detected, open the overlying skin for drainage and continue clean with wet to dry dressing change.

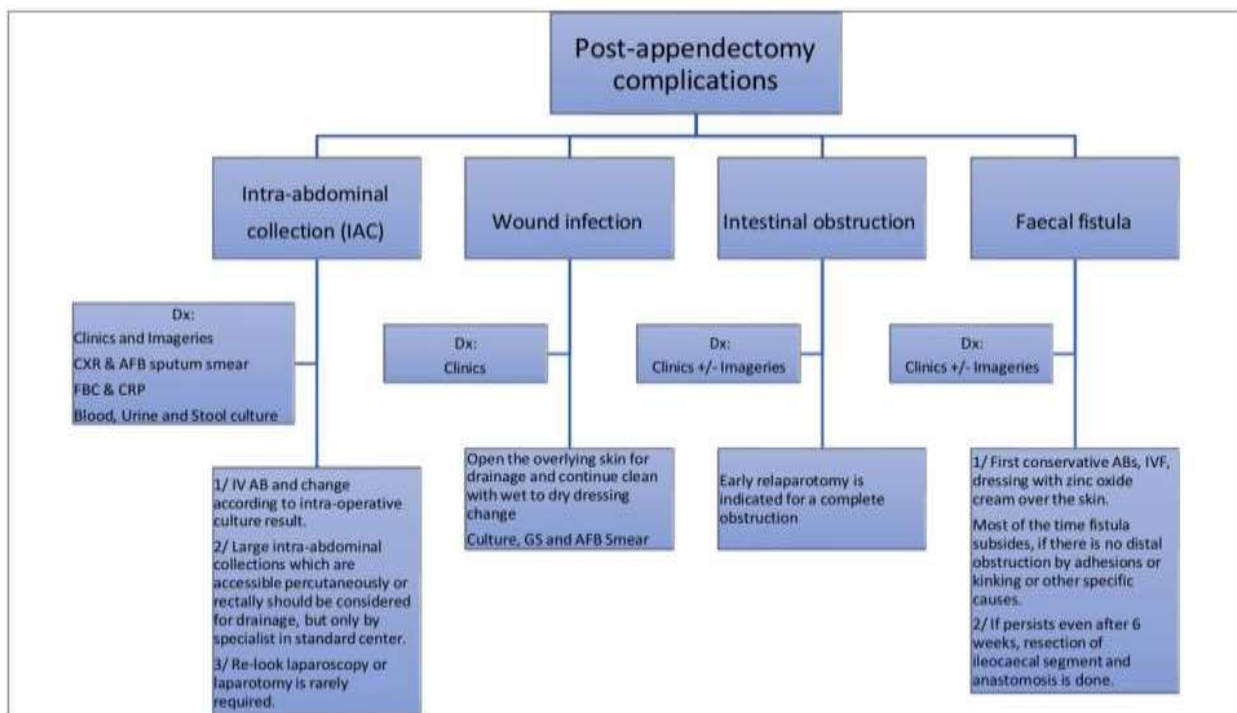
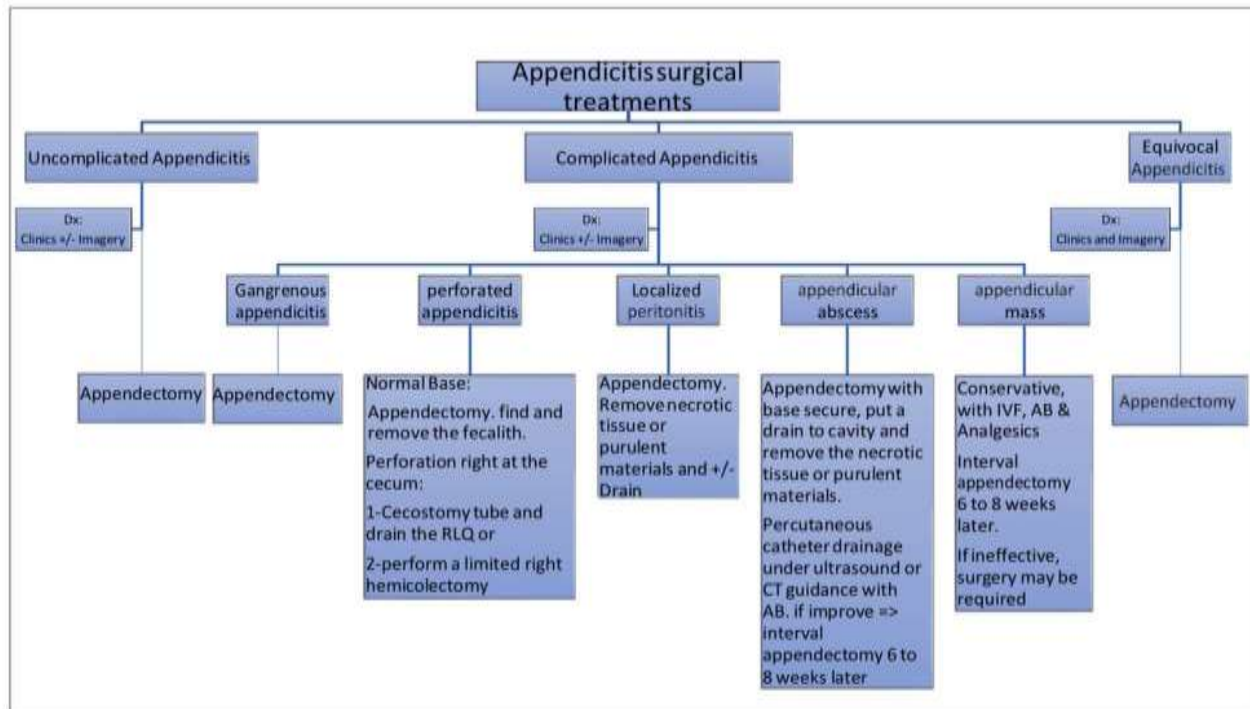
s. Intestinal obstruction due to adhesions occasionally occurs during the postoperative period, especially when there is some degree of peritonitis. Early relaparotomy is indicated for a complete obstruction.

t. Faecal fistula after appendectomy.

The causes can occur when appendectomy is done in gangrenous, perforated, friable base appendix. It also can occur after drainage of appendicular abscess or appendectomy is done/attempted in appendicular mass, underlying additional pathology like Crohn ‘s disease, carcinoma, ileocaecal tuberculosis and actinomycosis. Its features are feculent, foul smelling discharge from either main wound or drain site. It need work up with abdominal CT to find out the other pathology, CT fistulogram to delineate the track and other relevant investigations, Hb%, albumin level, etc. The treatments are first conservative antibiotics, IV fluids, dressing with zinc oxide cream over the skin. Most of the time fistula subsides

provided there is no distal obstruction by adhesions or kinking or specific causes like carcinoma or tuberculosis. If persists even after 6 weeks, resection of ileocaecal segment and anastomosis is done.

VII. ALGORITHM



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BILIARY PERITONITIS

MO Satdin, EM Sokhom

I. DEFINITION

The inflammatory response to generalized or localized leakage of bile into the peritoneal cavity. Bile in the peritoneal cavity results in severe chemical peritonitis and requires immediate surgical intervention. More than 35 years after the first Laparoscopic Cholecystectomy (LC), Bile Duct Injuries (BDI) remains one of the most worrisome complications of bile peritonitis [1].

II. ETIOLOGY

- Perforated gall bladder secondary to inflammation or obstruction [2] (especially empyema)
- Spontaneous rupture of the intrahepatic bile ducts [3]
- Post-cholecystectomy:
 - Cystic duct stump leakage
 - Leakage from an accessory duct in the gall-bladder bed
 - Bile duct injury [4]
- T-tube drain dislodgement [5] (or tract rupture on removal)
- Following other operations/procedures
 - Duodenal injury.
 - Leaking duodenal stump post gastrectomy.
 - Leaking biliary-enteric anastomosis.
 - Leakage around percutaneously placed biliary drains [6].
- Blunt or penetrating hepatobiliary or duodenal trauma.

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

Abdominal pain and a systemic inflammatory response, including fever, mild jaundice, tachycardia, and tachypnoea. Abdominal rigidity and guarding suggests the presence of peritonitis. Hypotension and hypoperfusion signs such as lactic acidosis, oliguria, and acute alteration of mental status are indicative of ongoing sepsis [7].

ii. Technical procedure

- a. Laboratory test
- b. Elevation of the white blood cell count (WBC) or leucopenia
- c. Elevation of C reactive protein
- d. Elevation of procalcitonin

- e. Elevation of serum lactate > 2mmol/l
- f. Creatinine, urea and electrolytes: acute kidney injury related to sepsis.
- g. Liver function, arterial blood gas (ABG)
- h. Blood culture, urine culture

iii. Imaging study

Patients who present with biliary peritonitis often receive extensive imaging, including computed tomography [9], ultrasonography, and radiography, as well as magnetic resonance cholangiopancreatography (MRCP)[10]. However, imaging studies should be chosen judiciously to avoid delaying definitive management, moving the acutely ill patient out of the resuscitation room, exposing the patient to ionizing radiation, and introducing potentially spurious data to the decision-making process. Here we discuss the role of imaging in the investigation of patients with biliary peritonitis and review the diagnostic value of specific radiographic finding.

Patients with generalized peritonitis or localized peritonitis with hemodynamic instability do not need imaging, as this would not alter the need for laparotomy.

IV. DIFFERENTIAL DIAGNOSIS

- ☐ Peptic ulcer perforated
- ☐ Acute pancreatitis
- ☐ Acute appendicitis (subhepatic)
- ☐ Spontaneous bacterial peritonitis
- ☐ Liver tumor or cyst rupture
- ☐ Small or large bowel rupture

V. THERAPEUTIC APPROACH

i. Stabilize patient

Stabilize severe sepsis or septic shock

- ☐ Oxygen therapy
- ☐ 2 large peripheral IV
- ☐ Fluid therapy
 - Crystalloids: 30 ml /kg (usually 500 to 1000ml)
 - Red blood cells if Hb < 8g/dl
 - blood lactate levels greater than 4 mmol/dL
 - Norepinephrine: goal MAP > 65 mmHg
- ☐ Correction of metabolic imbalances

- Nasogastric tube (empty stomach)
- Foley catheter (urine output)

ii. Empirical treatment

- Blood culture prior to antibiotic prescription
- Antibacterial therapy: Factors affecting the efficacy of antibacterial therapy include the activity of the agent against the common biliary tract pathogens and pharmacokinetic properties such as tissue distribution and the ratio of concentration in both bile and serum to the minimum inhibitory concentration for the expected micro-organism. Antimicrobial therapy is usually empirical. Initial therapy should cover the Enterobacteriaceae, in particular *Escherichia coli*. In severely ill patients with septicaemia, an antibacterial combination is preferable.

Table 1. The antibiotics are usually used to treat biliary tract infections and their biliary penetration ability (indicated as the ratio of bile to serum concentrations).

Good penetration efficiency (≥ 1)		Poor penetration efficiency (< 1)	
Antibiotics	Bile/serum	Antibiotics	Bile/serum
Tazocin	60	Cefotaxime	0.75
Tigecycline	38	Meropenem	0.75
Augmentin	30	Ceftazidime	0.5
Ciprofloxacin	30	Vancomycin	0.5
Unasyn	9	Amikacin	0.3
Ceftriaxone	5	Gentamicin	0.3
Levofloxacin	5	Cefepime	0.1
Penicillin G	5	Imipenem	0.01
Cefazolin	3		
Clindamycin	3		
Doripenem	1.17		
Cefuroxime	1		
Metronidazole	1		

Augmentin = amoxicillin–clavulanate; Bile/serum = bile concentration/serum concentration; Tazocin = piperacillin–tazobactam; Unasyn = ampicillin–sulbactam.

iii. Directed treatment

The treatment of etiology following:

- Perforated gall bladder secondary to inflammation or obstruction: laparotomy cholecystectomy + lavage + drainage.
- Bile leaks from the cystic duct or from a subvesical bile duct of Luschka (Strasberg type A lesion). These are accessory ducts that originate in the right hepatic lobe and have variable drainage into the bile duct (most frequently in the right hepatic or CBD). Postoperatively, depending on the patient's condition, treatment generally requires only drainage, either laparoscopically or percutaneously. They occasionally require an ERCP and sphincterotomy to decompress the biliary tree [11].
- Injury to the right posterior hepatic duct (RPHD) If the RPHD was clipped or ligated (Strasberg type B lesion), the patient will usually run an asymptomatic course producing atrophy of the affected segments. If the patient develops repeated cholangitis, with concomitant parenchymal atrophy there may be an indication for a hepatic lobectomy [12]. Otherwise, restoring the biliary tree continuity through a Roux-en-Y Hepaticojejunostomy (HJ) might be considered. If the posterior RPHD was divided (Strasberg type C lesion) there will be a biliary fistula, with the previously described manifestations. In these cases, primary suturing of the duct may be attempted, but in most cases an HJ over this compromised duct is definitive [13]. If a stricture of the duct develops, percutaneous balloon angioplasty dilatation may be attempted and if unsuccessful, definitive treatment may require a liver resection [14].
- **Lateral common bile duct injury** (Strasberg type D): This type of injury can be repaired by performing ERCP with stent insertion or percutaneous dilatation if the stenosis is short (such as in the case of an endobiliary instrumentation injury) or, if not effective, reconstruction through an HJ would be definitive [15].

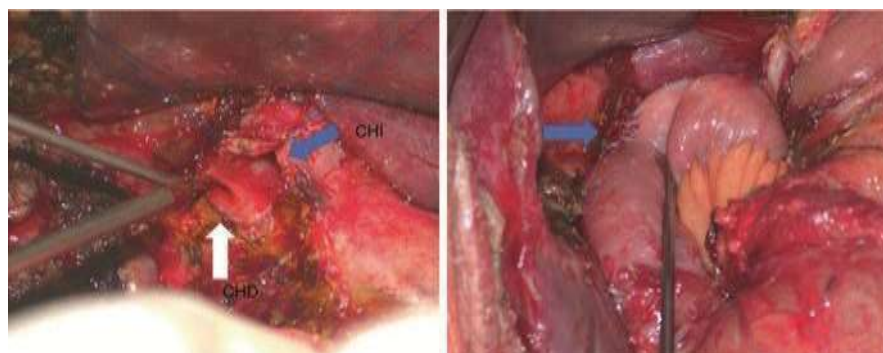


Fig.1 Patient with complex BDI. (a): RHD is marked with the white arrow, and LHC with a blue arrow, being prepared for an anastomosis. (b) Hepaticojejunostomy (blue arrow)

- **In cases of complex lesions injuries such as *Strasberg type E*:**

The definitive treatment consists of performing an HJ. In cases of high

stenosis (Bismuth type III and IV), exposure of the left hepatic duct (Hepp Couinaud operation), separate anastomosis of the hepatic ducts or of the sectoral collectors, or resection of hepatic segment IV B to expose the confluence of the hepatic ducts will be necessary [16]. To obtain good long-term results, it is necessary to perform the HJ with excellent technique. Technical failures are the main factor conditioning anastomotic strictures that manifest within a few months following the reconstruction and would require further therapies, like percutaneous balloon dilatations or eventually a re-do HJ [17]. In cases with a poor general condition, evidence of SBC and/or of portal hypertension, and very high stenosis, we recommend percutaneous treatment as first-line therapy (balloon dilatation without prosthesis placement). Otherwise, we prefer reoperation and anastomosis as described above. Despite refined techniques, and related to the complexity of these injuries, the future of these patients is uncertain. Some will require several treatments and yet will eventually progress to CBS, where the only valid treatment would be a LT [18].

□ **When Is the Best Time to Perform the Definitive Surgical Repair?**

The best time to perform an HJ after a BID is a matter of continuous debate and depends on several factors. Academically, the timing is divided into three different opportunities within *1 week (early)*; *1–6 weeks (intermediate)*; *6 weeks–6 months (late)*. However, there are several aspects to consider, other than time [19]. Immediate reconstruction is more likely feasible when the lesion is detected intraoperatively during cholecystectomy or in the very early postoperative period, since an early repair, whenever possible, allows for adequate anastomosis without associated complications and a related systemic inflammatory response [20]. When the injury is detected after the primary surgery, the timing of repair will depend on several factors such as sepsis, the patient's general condition, and the surgeon's training. Adequate sepsis control is associated with fewer postoperative complications and a higher likelihood of successful repair; therefore it should be first and foremost guaranteed. On the other hand, late repairs (after 6 weeks from the primary procedure) offer a less inflamed site, with more defined and adequately vascularized bile ducts and with control of the infection. Repairs between 8 days and 6 weeks it has been shown to increase the overall post-operative complications and failure rates of the reconstruction. Therefore, decisions should be made multidisciplinary on a case-by-case basis, carefully weighing the risk and benefits of the different therapeutic options.

iv. Monitoring

a. Clinical monitoring

I. General appearance

II. HEENT

III. Lung

IV. Heart

V. Abdomen (Peristalsis)

VI. Extremities

- I. Pain score**
- II. Vital signs**
- III. Maintenance fluids and nutrition**
- IV. Urine output**
- b. Laboratory monitoring
 - I. Complete blood count (CBC)**
 - II. Blood sugar (BS)**
 - III. Complete metabolic panel, including renal and liver function tests**

VI. COMPLICATION

i. Immediate

- ☐ Bleeding (from the wound or internally).
- ☐ Lung blockage or collapse.
- ☐ Shock.
- ☐ Heart problems.
- ☐ Pulmonary embolism
- ☐ Severe infection (septicemia).
- ☐ Acute kidney injury.

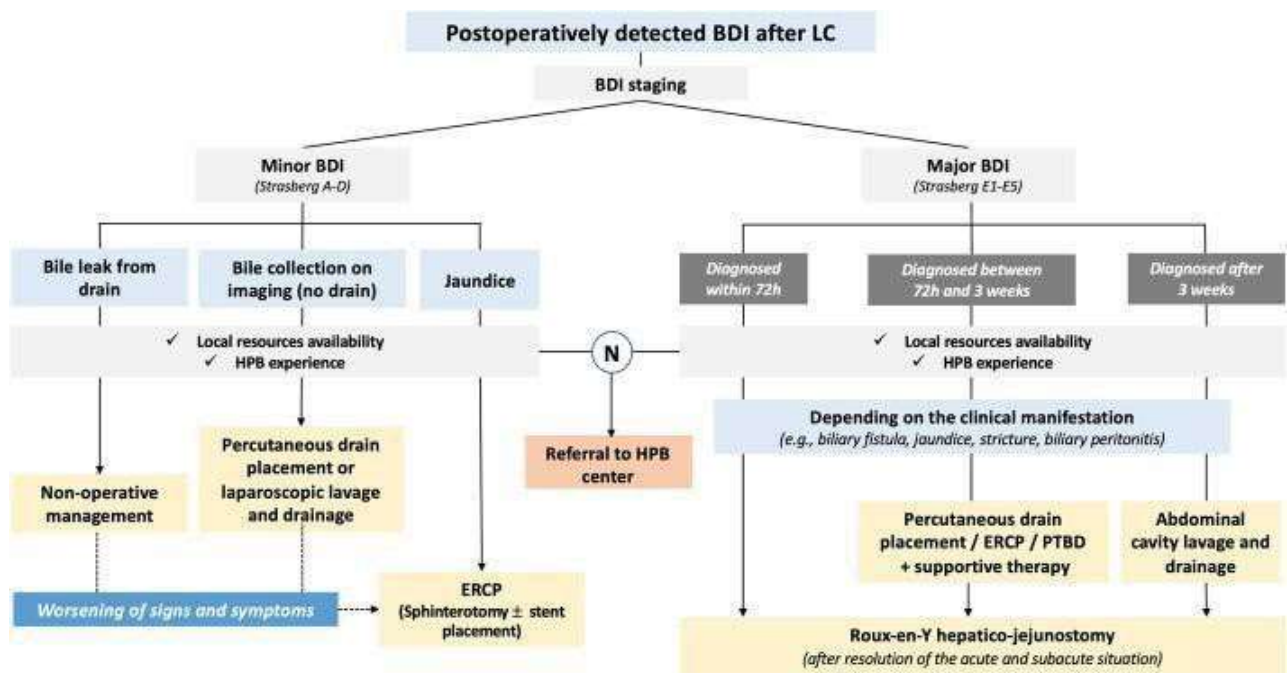
ii. Early

- ☐ Pain.
- ☐ Nausea, Vomiting, Confusion
- ☐ High temperature (fever).
- ☐ Bleeding.
- ☐ Wound breakdown.
- ☐ Deep vein thrombosis (DVT)
- ☐ Acute urinary retention
- ☐ Infections: Pneumonia, wound infection, urinary tract infection.
- ☐ Pressure sores
- ☐ Bowel problems (Constipation)

iii. Late

- ☐ Bowel blockage due to scarring inside the tummy
- ☐ Incisional hernia
- ☐ Persistent sinus.
- ☐ Thickening or tightening of scar.

VII. ALGORITHM



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BREAST MASS AND SURGERY

Preap Ley, Norng Sanareth, Koeu Kimsath, Cheav Darareak, Ong Visal

I. DEFINITION

The breast mass is the most common symptom of women presenting to the doctors among all breast diseases. Although most such lesions are benign, the presence of a mass can cause considerable anxiety because of the concern that cancer may be present. The most important task of the physician evaluating a breast mass is to provide accurate diagnosis of benign or malignancy, and then its treatment plans. Evaluation of a palpable breast mass requires obtaining a careful **clinical feature** (History and physical examination), **radiographic examination** and **biopsy results**.

The breast problems most commonly seen in a medical practice not specialized in breast diseases are

A. Benign tumors

a. Fibroadenoma. Fibroadenoma Variants:

- i. Giant Fibroadenoma
- ii. Juvenile Fibroadenoma
- iii. Lipoma
- iv. Hamartoma
- v. Phyllodes Tumor

b. Fibrocystic Changes

c. Cysts

d. Fat necrosis or Infection (Abscess)

B. Malignant tumors

a. Breast Cancer

b. Breast Cancers with Unusual Clinical Presentation

- i. Paget's Disease of the Nipple
- ii. Inflammatory Carcinoma

II. ETIOLOGY

According to breast diseases

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

When evaluating a patient who presents with a breast mass, the clinician needs to carefully on:

A. History:

- a) Patient age
- b) Precise location of the mass
- c) Length of time it has been present
- d) Any change in size (particularly with menstrual cycle)
- e) Associated symptoms (pain, skin changes, nipple discharge or retraction, arm swelling)
- f) History of other breast masses or fibrocystic disease
- g) History of previous breast biopsies and pathology result
- h) Menstrual history (age of menarche, age at first pregnancy, number of pregnancies, age at menopause)
- i) Personal history of cancer and how was it treated
- j) Hodgkin's disease and chest wall radiation
- k) Use of hormone replacement therapy: How long, what kind (estrogen only or estrogen and progestin)?
- l) Family history of all cancers, mother's and father's side

B. Physical Examination: The breast examination includes **inspection** and **palpation** of both the breast and the regional lymph nodes (axillary, supra-clavicular, and cervical). Always keep in mind that this is an awkward examination for most women, and every attempt should be made to keep the patient comfortable and relaxed, maintain her privacy, and perform the examination in a professional manner. The examination should begin with

- a) Patient sitting up, facing the examiner, arms to the side
 - ✓ Inspection for symmetry, contour, scars, skin lesions, erythema, nipple inversion
 - ✓ Palpation of the cervical and supraclavicular basins
 - ✓ Palpation of the mass in the upright position
- b) Patient sitting up, arms above head, touching
 - ✓ Inspection for dimpling, retraction, protruding mass
- c) Patient sitting up, arms on hips, pressing inward
 - ✓ Inspection for dimpling, retraction, protruding mass
- d) Patient sitting up, physician supporting weight of the arm
 - ✓ Palpation of bilateral axilla
- e) Patient in the supine position, arm raised over the head
 - ✓ Inspection for contour, scars, skin lesions & color, protruding masses, dimpling, or retraction
 - ✓ Examination of the nipple including gentle attempt to express discharge (color & smell)
 - ✓ Palpation of the mass in the supine position. Note size, consistency, borders, fixation, and location (including clock position and distance from areola)
 - ✓ Palpation of the entire breast parenchyma including inframammary fold, axillary tail
 - ✓ Note locations of any additional masses including size and location

- f) Examine the opposite breast in same way, noting symmetry between any areas of concern
- ii. Technical procedure
 - a. Laboratory test

These include the following:

 - ✓ Complete blood count, electrolytes, renal and hepatic profile with other panel depend on her/his comorbidities.
 - ✓ Breast tumor markers are not routine needed.
 - ✓ Bleeding panel.
 - ✓ Genetic testing is unavailable in our country.
 - b. Imaging study
 - ✓ Prompt diagnostic imaging is essential for any woman with a palpable dominant breast mass, regardless of her age. The system called the Breast Imaging Reporting and Data System or BI-RADS is use to classify the results into categories numbered 0 through 6.
 - ✓ Ultrasound is the initial test of choice for women under the age of 30.
 - ✓ In women between the ages of 30 and 39, ultrasound is the preferred initial imaging modality, although diagnostic mammography.
 - ✓ In women over the age of 40, diagnostic mammography with or without ultrasonography are the preferred imaging modalities. Chest and spinal cord X-ray $\pm$ computed tomography imaging (CT) if possible, if suspicion of breast cancer with metastasis to the lung or spinal cord.
 - ✓ Abdominal ultrasound $\pm$ CT of abdomen if possible, if suspicion of breast cancer with metastasis to abdominal cavity (Liver).
 - ✓ Magnetic resonance imaging (MRI) is limited to centers with a specialization of the radiologist in high-resource settings.

IV. DIFFERENTIAL DIAGNOSIS & PLANS

- Mass found as a collection pus=Breast Abscess
- Mass found as a collection fluid= Breast Cyst
- All masses found to be solid (Benign or Malignancy), especially on imaging require triple-test evaluation, which refers to physical examination, radiologic examination, and pathology result from FNA or Trucut biopsy or Incision/Excision biopsy. Remember that only histology may rule out breast cancer.

V. THERAPEUTIC APPROACH

- iii. Directed treatment
 - A. BREAST CYST: Aspiration of the cyst to dryness is the treatment of choice. Normally, the cyst fluid is discarded. If the fluid is uniformly blood stained, it should be sent for cytology. Following aspiration of a cyst, the patient should be reexamined, and should confirm that the previously palpable lump has disappeared.

Any residual areas of nodularity should be investigated similarly to any other discrete lump (FNA or Trucut biopsy or Incision/Excision biopsy).

- B. **BREAST ABSCESS:** Management consists of antibiotics and Drainage (I&D). However, many of these can be successfully treated by needle aspiration rather than incision and drainage.

Surgical Technique:

- a. Ultrasound guidance for the aspiration is preferred to ensure complete aspiration of the abscess.
- b. When needle aspiration is not possible, or not effective, incision and drainage may be necessary.
- c. The operation should be performed in the operating room with sedation because local anesthesia for an abscess is difficult to achieve.
- d. The incision is placed directly over the abscess, where the skin is thinnest.
- e. Pus cultures should be obtained.
- f. All loculations should be disrupted and the wound irrigated with saline until clean.
- g. A small sample of the abscess wall should be sent for pathology study to rule out cancer, even though this is extremely unlikely, especially among lactating women.
- h. The wound is packed with gauze, clean and change and allowed to heal by secondary intent.
- i. Antibiotics need only be continued until the local inflammation resolves and the patient has remained afebrile for 3 days.

- C. **EXCISION OF BENIGN TUMOR:** Fibroadenomas are well encapsulated lesions.

They can be excised by enucleating them from the surrounding breast tissue, especially in young patients. In older patients, especially as they approach the age at which the incidence of breast cancer peaks, the excision should include a rim of normal-appearing breast tissue surrounding the capsule.

Surgical Technique: The procedure is typically done in an outpatient facility. Local anesthesia and monitored intravenous sedation is used.

- a. Preparation and Anesthesia
 - The patient is positioned eccentrically on the operating table, with the side being operated on near the edge of the table.
 - The local anesthetic agent is injected into two separate planes: the subcutaneous tissue layer and the breast parenchyma. If the lesion is located deep within the breast, the local anesthetic is also injected deep to the lesion. The needle is introduced below and behind the lesion, just above the breast crease and parallel to the chest wall. The anesthetic agent is injected as the needle is withdrawn.
- b. Incision

- Except in the case of lesions located near or behind the areola (for which a curvilinear circumareolar incision is the incision of choice), a radial incision should be made over the area where the tumor is located. The incision should be large enough to facilitate good exposure. To avoid or minimize a hypertrophic scar, radial incisions should follow the Kraissl's skin lines.
- The skin incision is deepened through the subcutaneous tissue to the breast tissue layer. That layer is easy to recognize because of its whitish color
- Hemostasis is achieved using electrocautery or tie.
- Two skin flaps (including skin and subcutaneous tissue) are raised by sharp dissection and extended far enough to provide good exposure.

c. Excision

- Fibroadenomas are separated from the surrounding breast tissue by a layer of fibrous connective tissue. They can be excised by sharp dissection in that cleavage-tissue plane. Dissecting scissors can be used.
- Dissection of the lesion continues around its circumference until only the vascular pedicle is keeping the lesion attached to the capsule. The pedicle carries the blood supply. It should be clamped and tied before transection.

d. Closure

- To obtain a good cosmetic result and to minimize a hypertrophic scar or keloid, the defect in the breast should be carefully approximated in its anatomical layers (breast tissue, subcutaneous tissue, dermis, and skin).
- The defect in the breast tissue is approximated with 2/0 absorbable sutures. Depending on the depth, one or more layers of stitches may be necessary.
- If the rent at the breast tissue layer is large, approximation may be difficult. To prevent a breast deformity in such cases, the breast tissue layer should not be closed, and the resulting dead space drained. The defect will eventually fill with fibrous tissue that will maintain breast contour.
- Hemostasis should be thorough. Before closure of the breast tissue layer, the wound cavity is irrigated with sterile water to remove any blood clots that may be obliterating open veins or capillaries.
- The subcutaneous tissue layer is closed with interrupted sutures using 3/0 absorbable suture.
- Some surgeons also approximate the dermis, using a third layer of stitches. This approach reduces tension at the skin.
- The skin is approximated with a continuous suture placed in the subcuticular layer. An absorbable monofilament suture material is used.

D. **MODIFIED RADICAL MASTECTOMY (MRM):** Surgery is the most effective treatment option ever employed against breast cancer. Even there are few kinds of surgical procedures, but the most effective surgery is MRM used in this developing world regardless cancer stage (BHGI for LMIC) having to remove all breast tissue and lymphatic-bearing tissue in the axilla. It can also be termed as a total mastectomy with an axillary lymph node dissection. MRM of Auchincloss's procedure is the most preferred for breast cancer surgeon with similar overall survival rate and good quality of life compares to the others (Patey, Scalton).

Surgical Technique:

- a. Preparation and Anesthesia
 - Always need general anesthesia
 - Patients should be in a supine position with the side of the MRM close to the edge of the operating table.
 - Arms should not be abducted greater than 90 degrees to prevent injury to the brachial plexus.
 - Surgical preparation should include the breast and axilla and extend across the midline, inferiorly onto the abdomen, superiorly to the neck and the upper arm, and laterally to the level of the operating table.
 - Draping should ensure preservation of a sterile field with exposure of the whole breast and axilla.
- b. Skin incision and Creation of flaps
 - The incision should include the skin overlying the tumor and a 1 to 2cm margin around the tumor and include the previous biopsy site.
 - The classic Stewart incision is a transverse elliptical incision, provides access to the axilla, make sure there will be adequate skin for a tension-free closure without excess skin for redundancy.
- c. Dissection from the pectoral muscle
 - A small edge of tissue is raised below the dermal layer to develop the flap. The superior flap should be extended up to the inferior border of the clavicle, and the medial flap should be dissected to the sternal border. The lateral flap should be extended to the latissimus dorsi and the inferior flap should be dissected to the inframammary fold.
 - When the flaps are completed, the breast tissue is dissected off the pectoralis muscle taking the pectoral fascia with the breast tissue using the electrocautery. The dissection should be started superiorly at the level of the clavicle. The axillary tail of the breast should remain attached in order to keep the breast and axillary contents as one pathologic specimen and also to guide the axillary dissection.
- d. Axillary dissection
 - The clavipectoral fascia should be incised along the edge of the pectoralis major muscle, which will free up the pectoralis major and minor muscle from the interpectoralis nodal tissue. Close attention should be paid to identify the medial pectoral neurovascular bundle.

- The inferior border of the dissection should extend to the 4th or 5th rib to ensure that level I nodes are included in the specimen.
- For ALND describes as lateral to medial approach visualizing the latissimus dorsi laterally is needed.
- The next step is to identify the axillary vein.
 - Once the vein is identified, the dissection should continue inferiorly to it.
 - Level II nodes are located underneath the pectoralis muscle, and level III nodes are located medial to the muscle. Level III nodes are rarely involved in most breast cancer metastasis and are not included in the traditional axillary lymph node dissection.
 - The dissection should continue to extend medially along the axillary vein, and the lymphatic tissue on the chest wall should be retracted inferolaterally. This tissue should be dissected off the chest wall with ties or clips to prevent lymphatic leaks.
 - The long thoracic nerve lies along the chest wall and is superficial to the investing fascia of the serratus anterior muscle. Care should be taken to not retract the nerve off the chest wall with the specimen. Dissection of the lymphatic tissue close to the nerve should be done with sharp dissection and not electrocautery.
 - The thoracodorsal bundle is found lateral to the long thoracic nerve. The lymphatic tissue between the two nerves should be dissected and removed. The branches of the thoracodorsal vein and artery entering the chest medially should be preserved where possible.
 - Attempts should also be made to try to preserve the intercostobrachial nerves, but if the path of the nerve is directly through the specimen, it can be ligated that ligating the nerve will cause loss of sensation to their upper inner arm.
 - The breast and attached nodal tissue should be excised en bloc by dividing the remaining lateral pedicle.
 - The specimen should be sent to pathology with sutures placed for orientation.

e. Drain placement

- One or Two closed suction drains should be placed to prevent seroma formation. The drain should be placed in the axillary space. The drain should be sutured in place and dressed appropriately based on surgical preference.

f. Closure

- A cosmetically pleasing closure is important for psychologic reasons.
- The dermal layer should be closed with an inverted interrupted closure using absorbable suture.
- The skin should be approximated with an absorbable 4/0 monofilament suture in a running subcuticular closure.

2. Monitoring

i. Clinical & Laboratory monitoring

- After breast abscess I&D, the wound cleaning and the antibiotic using according to pus culture result are important until healing.
- Breast cyst may be single or multiple. Cysts may refill, particularly if not completely aspirated, and require repeated aspiration or, rarely, if large and recurring after repeated aspiration. A cyst that recurs within 4-6 weeks after aspiration can either be due to a benign lesion (large intraductal papilloma) or a malignancy (intracystic or partially cystic carcinoma). Irrespective of this, these cases warrant surgical excision of the cyst and send to pathologist. No specific laboratory tests needed.
- The management of a solid mass depends on the degree of clinical suspicion and the age of the patient. If a benign lesion is diagnosed after a triple assessment, the options include surgical excision or follow-up of the lesion. It is not necessary to excise all benign solid breast masses, and a selective policy is recommended based on the nature of the lesion and patient preference. In the event of a conservative approach being preferred or after excision of mass, there must be a defined follow-up plan to facilitate the early detection of a missed cancer. The patient is examined every 3-6 months for one year to ensure stability or recurrent of the mass. The conservative mass is measured at each visit and compared with the size at initial presentation. No specific laboratory tests needed.
- After surgical MRM wound healing, patients on continuing active treatment with Oncologist should be followed up until such treatment has been completed. The surgeon also continues their clinical follow up, the present practice of seeing women at 3-6-month intervals for a number of years or high-risk patients may be followed up more closely to early identify loco-regional recurrence or another breast health condition. In addition, surgical follow up could manage treatment related problems and provide other supports. Surgeon can check her breast or abdomen with ultrasound, can do chest or spinal cord X-Ray in every 6-12 months according to patient's complaints. Some tumor marker like CA15-3 is benefit for breast cancer recurrent checkup, but not routine perform yet in our practice.

VI. COMPLICATION

i. Breast abscess I&D:

- Once the acute infection resolves, the patient is often left with a small indurated area with an overlying scar. Because these are susceptible to repeated cycles of inflammation and infection, they should be excised electively.

ii. Complications from excision of benign breast tumor surgery should be minimal or nonexistent.

- Good hemostasis throughout the procedure and a meticulous dissection and closure will prevent skin ecchymosis, disruption of the suture line, seromas, and hematomas.
- Occasionally, the patient may develop thrombophlebitis of a vein near the incision, distal to the wound. Thrombophlebitis of a superficial vein of the breast

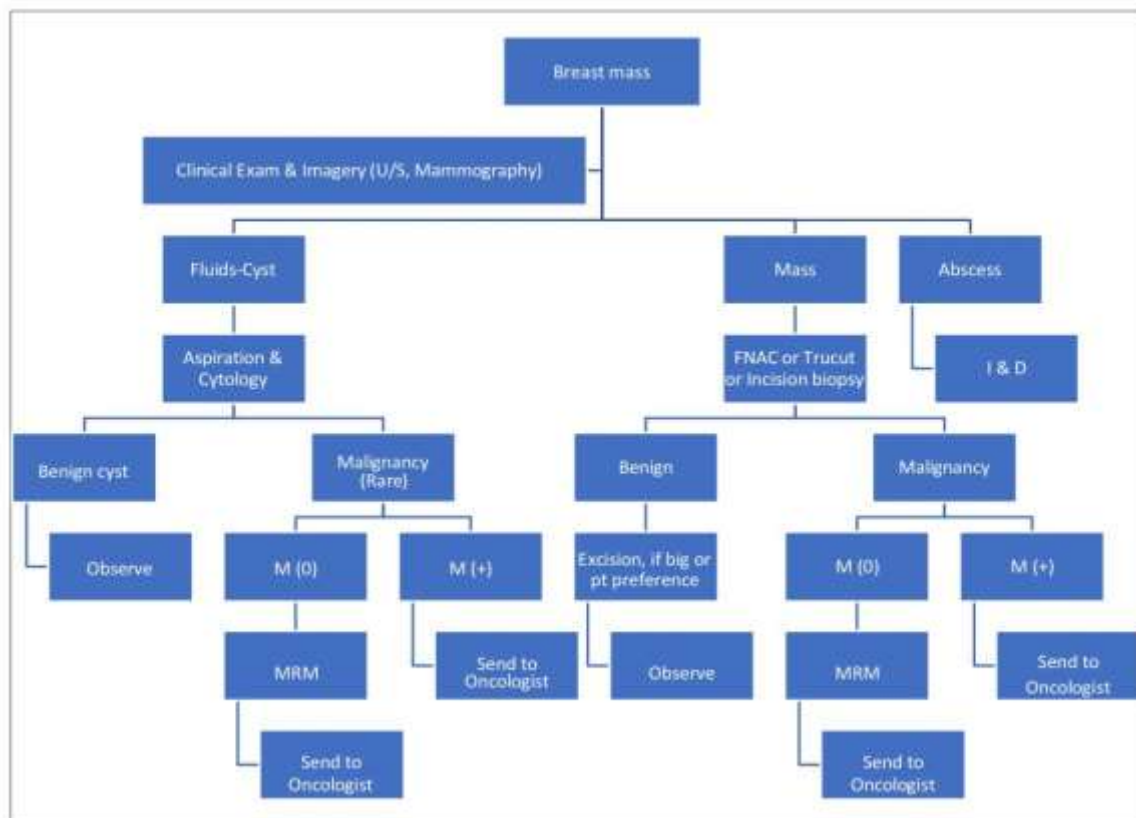
is called Mondor's disease. The condition may last for several weeks, but it usually resolves spontaneously.

iii. Complications of MRM:

- Postoperative loss of sensation on the anterior chest wall and the inner surface of the ipsilateral arm is a common occurrence with a modified radical mastectomy. The severing of the sensory nerve branches of the lateral and anterior cutaneous nerves of the second through the sixth intercostal nerves and of the intercostobrachial nerves produce this loss of sensation, which is not considered a complication.
- Complications that may follow a total mastectomy or modified radical mastectomy are
 1. wound infection,
 2. bleeding,
 3. seroma,
 4. skin slough,
 5. nerve injury, and
 6. lymphedema of the arm.
 - Postoperative wound infection and bleeding can be prevented by careful surgical technique and by proper hemostasis during dissection of the skin flaps.
 - Seroma is one of the most common complications following a modified radical mastectomy. It occurs more frequently in obese patients. To prevent this complication, the drains should remain in place for a longer period. Small seromas are treated using aspiration. Large, recurrent seromas should be incised and drained.
 - A truly dreadful complication of modified radical mastectomy is sloughing (necrosis) of the skin flaps. Viable skin flaps can always be obtained if dissected evenly and in the proper tissue plane, with a thin layer of subcutaneous tissue left attached. Careful dissection of the flaps and tensionless closure can prevent this complication. Unless the area of the slough is extensive, the complication can be resolved with local debridement and frequent skin cleansing.
 - The adverse effects of the injury to the nerves (pectoral, thoracodorsal, long thoracic):
 1. injury to the medial pectoral nerve causes atrophy in the clavicular portion of the pectoralis major muscle.
 2. Injury to the lateral pectoral nerve causes atrophy in the sternal and abdominal portions of the same muscle.
 3. Injury to the thoracodorsal nerve results in impairment to external rotation of the ipsilateral arm.
 4. Injury to the long thoracic nerve causes a "winged" scapula, which is the most disabling and cosmetically unacceptable of the nerve complications.

- Transient swelling of the arm is not uncommon following a modified radical mastectomy. Lymphedema of the arm is preventable. The complication is related not to the number of nodes removed, but to the axillary vein being subjected to trauma during the axillary dissection. To avoid or to minimize the complication, dissection of the axillary vein should be gentle and should leave the adventitia attached to the vein.
- Occasionally, lymphedema of the arm may be complicated by cellulitis of the skin. The arm becomes hard in a clinical presentation called "brawny lymphedema". In the acute phase, brawny lymphedema should be treated with antibiotics and elevation of the arm. When the cellulitis subsides, the patient should wear a pressure gradient elastic sleeve. Lymphedema can appear years after the surgery. It is routine to advise patients to avoid arm swelling and infection by taking special care not to suffer cuts and scrapes on the operated side, to avoid injections, blood-pressure monitoring, and blood draws, to avoid constricting clothing and jewelry, etc.

VII. ALGORITHM



BIRADS®

Class	Description	Recommendation
Category 1	Normal	Annual follow-up
Category 2	Benign	Annual follow-up
Category 3	Probably benign	Short interval (6 month) follow-up
Category 4	Suspicious	Biopsy recommended
Category 5	Highly suggestive of malignancy	Biopsy mandatory
Category 6	Known cancer (biopsy completed)	Treatment required

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COLORECTAL CANCER

Chea Theanlong, Preap Ley

I. CASE DEFINITION

- a. Colorectal cancer (CRC) is the fourth most commonly diagnosed cancer in the United States.
- b. Risk factors include a positive family history, hereditary syndromes, diet, and a number of conditions, such as inflammatory bowel disease.
- c. Most colorectal cancers (95%) are adenocarcinomas.
- d. Clinical signs are often nonspecific and may include a change in bowel habits, lower GI bleeding, and weight loss.
- e. These features as well as iron deficiency anaemia in men older than 50 years of age and postmenopausal women are red flags for CRC.

f.

Risk factors:

- Age: older age (> 40 years)
- Hereditary syndromes: Family history individuals with colorectal cancer (CRC)
- Familial adenomatous polyposis
- Hereditary nonpolyposis colorectal cancer (HNPCC)
- Associated conditions:
 - Colorectal adenomas and serrated polyps
 - Inflammatory bowel disease: chronic inflammation → hyperplasia → non-polypoid dysplasia → neoplasia
 - Ulcerative colitis
 - Crohn disease

Epidemiology:

- 46.000 new cases diagnose in 2015 in France
- 16.000 dies per year (second cause of dead by cancer in France).
- Average age: 70 years old (rare before 50 years old: 6 %)
- Peak incidence: between 65 and 74 years of age
- Mortality: third leading cause of cancer-related deaths in the US overall

II. AETIOLOG

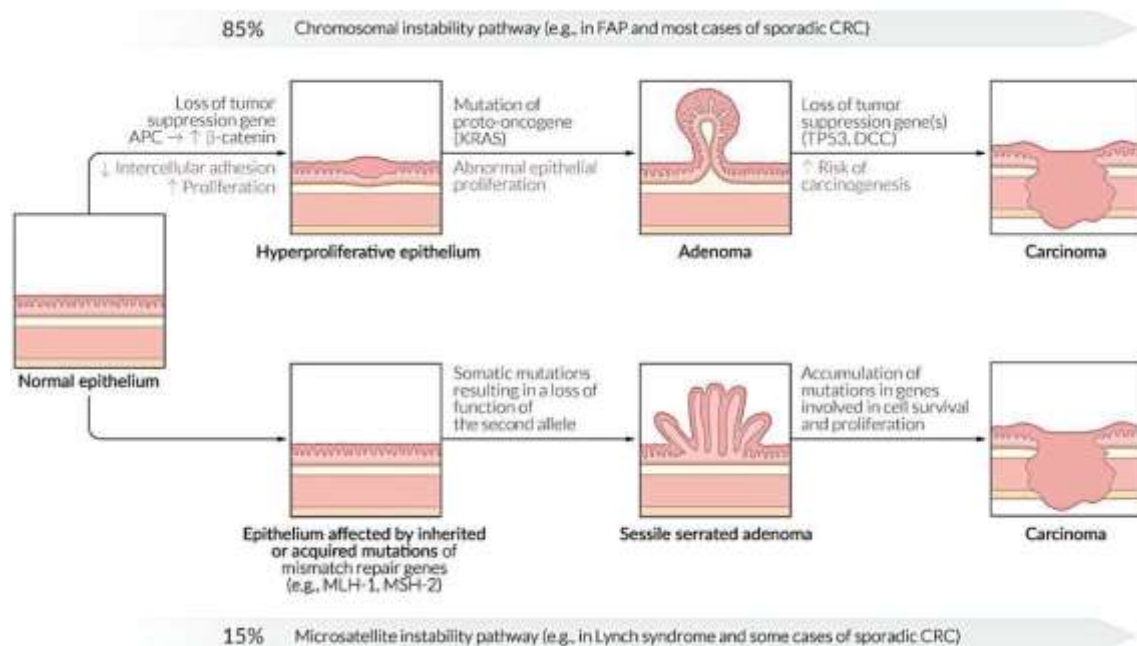
f. Chromosomal instability pathway in colon cancer:

- ☞ The adenoma-carcinoma sequence: mutations in oncogenes and tumor suppressor genes → slow transformation of adenomas into carcinomas
- ☞ APC gene mutation (loss of cellular adhesion and increased cellularproliferation) → KRAS gene mutation (unregulated cellular signaling and cellular proliferation) → TP53 and DCC gene mutation

g. Microsatellite instability pathway in colon cancer:

- ☞ Hypermethylation phenotype pathway in colon cancer

- h. COX-2 overexpression:
- ☞ Associated with colorectal cancer
 - ☞ Possible protective effect of long-term use of aspirin and other NSAIDs



III. DIAGNOSTIC PROCEDURE

i. Clinical argument

- ☞ Colorectal cancer can be asymptomatic, particularly during the early stages.
- ☞ **Constitutional symptoms:**
 - ☐ Weight loss
 - ☐ Fever
 - ☐ Night sweats
 - ☐ Fatigue
 - ☐ Abdominal discomfort (symptoms similar to diverticulitis, especially in carcinoma of the rectosigmoid or descending colon)
- ☞ **Right-sided colon carcinomas:**
 - ☐ Definition: large bowel malignancies arising from the cecum, ascending colon, or transverse colon
 - ☐ Clinical features
 - Occult bleeding or melena
 - Manifestations of iron deficiency anemia (due to chronic bleeding)
 - Diarrhea
 - Iron deficiency anemia in men > 50 years of age and postmenopausal

women should raise suspicion for colorectal cancer.

☞ **Left-sided colon carcinomas:**

- Definition: large bowel malignancies arising from the splenic flexure, descending colon, sigmoid colon, or the rectosigmoid junction.
- Clinical features
 - Changes in bowel habits (size, consistency, frequency) Blood-streaked stools
 - Colicky abdominal pain (due to obstruction)
 - Bowel obstruction occurs earlier in left-sided colon carcinomas because the distal colon has a smaller lumen than the proximal colon and contains solid faecal matter.

☞ **Rectal carcinomas:**

- Definition: large bowel malignancies located ≤ 15 cm from the anal verge
- Clinical features
 - Hematochezia
 - ↓ Stool caliber (pencil-shaped stool)
 - Rectal pain
 - Tenesmus
 - Flatulence
 - Fecal incontinence
 - Cancers located ≤ 15 cm proximal to the anal verge are considered rectal carcinomas
 - Consider colorectal cancer in every patient with rectal bleeding, even if there is a history of hemorrhoids or diverticular disease.

☞ **Metastatic disease:**

- CRC can metastasize through hematogenous, lymphatic, transperitoneal, and contiguous routes. Symptomatic metastases may be the first manifestation of CRC.
- Liver metastases (most common site of metastasis; 40–50%)
 - Ascites, abdominal distention
 - Hepatomegaly, RUQ pain
 - Jaundice
 - Anorexia, early satiety
- Lung metastases
 - Dyspnea
 - Cough
 - Hemoptysis
 - Pleural effusion
- Peritoneal metastases
 - Ascites, abdominal distension, diffuse abdominal pain

- Bowel obstruction
- Evidence of distant lymphatic spread: Virchow node (rare)
- Typically, cancers of the colon and upper rectum initially metastasize to the liver via the portal vein, and cancers of the lower rectum initially metastasize to the lung via the inferior vena cava)

j. Technical procedure

- All patients with suspected CRC should undergo a complete colonoscopy with biopsy of suspicious lesions.
- Initial workup: → Digital rectal examination
- Indication: all patients with lower gastrointestinal bleeding (LGIB) or other red flags for CRC
 - + Findings:
 - Distal rectal cancers may be palpable
 - Evidence of blood on DRE may indicate CRC
 - Endoscopic evaluation is essential if there is any suspicion for CRC, such as:
 - Inconclusive DRE in a patient with LGIB
 - Any risk factor for CRC
 - Any red flag for CRC
 - Assess sphincter tone during DRE to plan optimal surgical resection for rectal cancer.

k. Laboratory studies

- ☞ Carcinoembryonic antigen (CEA): Obtain baseline levels in all patients before initiating treatment.
- ☞ Monitor CEA levels during treatment and follow-up to assess response to treatment and evaluate recurrence.
- ☞ Should not be used for screening
- ☞ CBC: may show microcytic anaemia (iron deficiency anaemia)
- ☞ Liver chemistries and coagulation: may be abnormal in patients with multiple hepatic metastases
- ☞ Counselling and genetic testing: for patients < 50 years of age with CRC or those with a family history of CRC at a young age.

Colorectal cancer staging			
AJCC staging 8 th edition (simplified) [31]	TNM stage	Corresponding Dukes classification stage	Description
0	• Tis, N0, M0	• A	• Tumor invasion: none (tumor in situ) • Lymph node involvement: none (N0) • Distant metastases: none (M0)
I	• Up to T2, N0, M0		• Tumor invasion ◦ T1: submucosa ◦ T2: muscularis propria • Lymph node involvement: none (N0) • Distant metastases: none (M0)
II	• Up to T4, N0, M0	• B	• Tumor invasion ◦ T1: submucosa ◦ T2: muscularis propria ◦ T3: subserosa ◦ T4a: visceral peritoneum ◦ T4b: adjacent tissue or organs • N0: no lymph node involvement • M0: no distant metastases
III	• Any T, N1/N2, M0	• C	• Tumor invasion: any structure • Lymph node involvement ◦ N1: 1–3 regional lymph nodes ◦ N2a: 4–6 regional lymph nodes ◦ N2b: ≥ 7 regional lymph nodes • Distant metastases: none (M0)
IV	• Any T, any N, M1	• D	• Tumor invasion: any structure • Lymph node involvement: any number • Distant metastases ◦ M1a: affecting only one organ or site ◦ M1b: affecting more than one organ or site ◦ M1c: peritoneal metastases (regardless of metastasis to other organs) [31][32]

1. Imaging study

☞ Flexible sigmoidoscopy with or without anoscopy:

- Indication: consider in patients with scanty intermittent hematochezia and all of the following features
 - + Findings:
 - Age < 40 years
 - No other red flags for CRC
 - No risk factor for CRC
 - + Important considerations:
 - Patients who do not fulfil any of these criteria require a complete colonoscopy
 - Inconclusive sigmoidoscopy or lesion that raises concern for malignancy: Perform a complete colonoscopy.
 - Benign pathology confirmed: Manage accordingly (see, e.g., “Haemorrhoids” and “Anal fissures”).

☞ **Complete colonoscopy:**

- Colonoscopy is the gold standard test for CRC as it allows for direct visualization complete colonoscopy after neoadjuvant chemotherapy if there is evident biopsy of polyps and suspicious lesions.
- Indication: all patients with suspected CRC
- Typical findings:
 - Ulcero-proliferative friable mass
 - A biopsy is required to confirm the diagnosis
- Important considerations:
 - Consider the following to identify synchronous tumours if colonoscopy cannot be completed (patients with occlusive CRC)
 - Pre-treatment CT colonography or capsule endoscopy
 - Intra or postoperative colonoscopy

☞ **Double-contrast barium enema (uncommonly performed):**

- Indication: an alternative to CT colonography in patients who decline/cannot undergo a complete colonoscopy at presentation
- Typical findings:
 - Endoluminal filling defect typically with irregular margins
 - Apple core lesion (napkin ring sign): sharply defined circumferential narrowing of the bowel caused by astenosing CRC
- Important considerations:
 - Double-contrast barium enema is not routinely recommended because of its low diagnostic yield and the widespread availability of CT colonography
 - Findings that raise suspicion for CRC on barium enema require biopsy and histopathological confirmation



☞ **Preoperative staging**

- Assess for local and distant spread (T stage, N stage, and M stage)
- CT abdomen, pelvis, and chest (with IV and oral contrast)
- Consider CTA or MRI of the liver if hepatic metastases are suspected. Evaluate for specific sites of metastasis based on clinical suspicion (e.g., diagnostic laparoscopy and peritoneal cytology for suspected peritoneal metastasis).
 - Hepatic metastases: multiple hypodense lesions during the portal venous phase of CTA ; may show peripheral washout
 - Pulmonary metastases: multiple, peripheral nodules of varying sizes
- Additional staging workup in rectal cancer
 - Pelvic MRI with and without IV contrast: to determine T stage and N stage
 - Rectal endoscopic ultrasound: Consider in early tumor stages or if MRI is contraindicated.
 - Rigid proctosigmoidoscopy: to assess the distance of the tumor from the anus.

IV. DIFFERENTIAL DIAGNOSIS:

m. Differential diagnoses based on clinical presentation:

- Lower GI bleeding
- Diarrhoea
- Constipation
- Large bowel obstruction
- Metastatic liver disease

n. **Small bowel neoplasms:**

- Less than 5% of all gastrointestinal cancers
- Mainly benign lesions (> 95%)
- Malignant small bowel cancers (1–3%)
- 40% adenocarcinoma
- 30% neuroendocrine tumours
- 20% lymphoma
- 10% gastrointestinal stromal tumors (GIST), leiomyosarcoma, liposarcoma, metastatic disease
- Diagnostics
 - . Endoscopy, Ultrasound, Hydro-MRI, Capsule endoscopy
- Treatment
 - . Resection of the small bowel
 - . R1 resection requires adjuvant chemotherapy
- Complications

- . Bleeding
- . Ileus
- . Metastatic spread
- . Carcinoid syndrome in the case of neuroendocrine cancers of the small bowel



Tumor in the small intestine



MALT lymphoma of the small intestine

V. THERAPEUTIC APPROACH

Stage I	• Curative surgery
Stage II	• Curative surgery • Consider adjuvant chemotherapy in patients with high-risk stage II disease
Stage III	• Curative surgery • Adjuvant chemotherapy
Stage IV	• All patients ◦ Individualized, multidisciplinary approach ◦ Consider neoadjuvant treatment if downstaging may enable curative surgical resection. • Resectable metastatic disease ◦ Curative surgery ◦ Adjuvant chemotherapy • Unresectable metastatic disease or severe comorbidities preventing curative surgical treatment ◦ Consider palliative surgery. ◦ Consider palliative chemotherapy.

o. General principles

i) Stabilize patient

- ☞ A multidisciplinary approach is recommended.
- ☞ Treatment depends primarily on the location of the tumor and the TNM stage.
- ☞ Curative treatment is primarily surgical and may involve neoadjuvant or adjuvant chemotherapy and/or radiation therapy.
- ☞ Palliative treatment may involve chemotherapy, radiation therapy, and, in some cases, surgery.

ii) Empirical treatment

☞ Surgery of primary tumor

- ☐ Indicated in any resectable primary tumor with no metastasis or resectable distant metastases
- ☐ Conventional or laparoscopic approach is possible
- ☐ Complete resection with clear margins (R0 resection) is associated with the best prognosis.

☞ Regional lymph node dissection: performed routinely alongside resection of the primary tumor

- ☐ Resection of mesenteric lymph nodes along the path of lymphatic drainage
- ☐ Histological analysis of 12 removed lymph nodes to accurately define the lymph node status is required for accurate staging

☞ Resection of metastases

- ☐ Indicated in patients with resectable metastases (e.g., liver and/or lung metastasis)

- May significantly improve the survival of patients with limited metastatic disease

p. **Palliative surgery**

☞ **Surgery of primary tumor**

Consider in patients with **nonresectable distant metastases** to prevent or treat complications of colorectal cancer.

- **Intestinal bypass** (e.g., ileocolonic anastomosis, colostomy) or enteral stenting for obstructing/occlusive CRC
- **Tumor resection** to manage immediately life-threatening complications, such as complete bowel obstruction, persistent GI bleeding, or perforation.

q. **Directed treatment**

☞ **Chemotherapy regimens:**

- FOLFOX: folinic acid (leucovorin) PLUS 5-fluorouracil (5-FU) PLUS oxaliplatin
- FOLFIRI: folinic acid (leucovorin) PLUS 5-fluorouracil (5-FU) PLUS irinotecan
- CAPOX: capecitabine PLUS oxaliplatin
- Anti-VEGF antibodies (e.g., bevacizumab)
- EGFR antibodies (e.g., cetuximab)

☞ **Radiation therapy:**

- Rectal cancer: standard treatment modality in most stages of rectal cancer
- Colon cancer:
 - Not a standard modality because of adverse effects on the small intestine leading to enteritis and strictures
 - May be used in palliative care to treat bone and brain metastases

☞ **Surgery:**

A. Principle of surgery for colon cancer

- Colectomy with lymph node dissection is indicated in all resectable tumors
- The extent of the resection depends on the location of the tumor as well as the blood supply and lymphatic drainage of the affected region
- En bloc resection of infiltrated adjacent tissue is recommended to obtain an R0 resection
- In patients with peritoneal metastases, consider cytoreductive surgery in combination with intraperitoneal chemotherapy (HIPEC or hyperthermic intraperitoneal chemotherapy).
- Bowel continuity should be restored via anastomosis when feasible; a stoma (temporary/permanent) may be needed.

Typical surgeries for colon cancer			
Type of resection		Description	Indication
Hemicolectomy	Right hemicolectomy	• Resection of part of the distal ileum, the ileocecal valve, cecum, ascending colon, hepatic flexure, and proximal third of the transverse colon	• Tumor in the cecum and ascending colon
	Extended right hemicolectomy	• Right hemicolectomy and resection of the transverse colon	• Tumor near the hepatic flexure or in the proximal or middle transverse colon
	Left hemicolectomy	• Resection of the distal third of the transverse colon, the splenic flexure, descending colon, and sigmoid colon	• Tumor in the descending colon
Sigmoid colectomy		• Resection of the sigmoid colon	• Tumor in the sigmoid colon
Subtotal or total abdominal colectomy		• Resection of most of or the entire colon	• Multifocal carcinomas • Underlying colonic disease
Subtotal or total abdominal colectomy		• Resection of most of or the entire colon	• Multifocal carcinomas • Underlying colonic disease
Less commonly used techniques		• Extended left hemicolectomy • Transverse colectomy [42] • Anterior resection	

Treatment of rectal cancer by stage	
AJCC Stage	Treatment approach
Stage I	• Curative surgery
Stage II	• Neoadjuvant chemotherapy and/or radiotherapy • Curative surgery • Adjuvant chemotherapy [39] • Consider adjuvant radiotherapy in patients with high-risk stage II disease. [39]
Stage III	• Neoadjuvant chemotherapy and/or radiotherapy • Curative surgery • Adjuvant chemotherapy • Consider adjuvant radiotherapy.
Stage IV	• All patients ◦ Individualized treatment with a multidisciplinary approach ◦ Consider neoadjuvant treatment in potentially resectable metastatic disease. [39] • Resectable metastatic disease ◦ Curative surgery ◦ Individualized chemotherapy and/or radiotherapy with curative intent • Unresectable metastatic disease or severe comorbidities preventing surgical treatment ◦ Palliative chemotherapy ◦ Consider palliative surgery.

B. Surgery for rectal cancer

- The extent of the resection depends on the location of the tumor and the TNM stage.
- The sphincter tone and the distance of the tumor from the anal verge (e.g., via rigid proctosigmoidoscopy) should be assessed preoperatively to plan appropriate surgical resection.
- Consider gynecology and/or urology consult if imaging shows a regional spread past the rectum.
- Histopathology

- Most common: adenocarcinoma (95%)
- Mucinous adenocarcinoma
- Signet ring cell carcinoma
- Small cell carcinoma
- Adenosquamous carcinoma (rare)

Typical surgeries for rectal cancer			
	Description	Indication	Tumor stage
Transanal excision	Minimally invasive excision of small superficial tumors	Small (< 3 cm) T1 tumors with no risk factors	Early, localized disease (stage I)
Low anterior resection (LAR) resection (LAR)	- Sphincter-preserving resection of the rectum and sigmoid Total mesorectal excision (TME) - En bloc excision of the rectum, mesorectum, regional lymph nodes, and vasculature - Indicated in tumors of the vasculature - Indicated in tumors of the middle and lower third of the rectum - Immediate reconstruction (e.g., side-to-side anastomosis) - Optional diverting ostomy	- Tumor location allowing for sphincter preservation in patients with good preoperative sphincter function - Appropriate distal resection margins depend on the location of the tumor: 5 cm for tumors of the upper third of the rectum - For tumors closer to the anal sphincter, smaller 5 cm for tumors of the upper third of the rectum - For tumors closer to the anal sphincter, smaller resection margins may be tolerated. - Shorter margins may be acceptable when neoadjuvant therapy has been successful.	- Stage I tumors ineligible for transanal excision - Locally excision - Locally advanced disease (stages II–III) - Resectable metastatic disease (stage IV)
Abdominoperineal resection (APR)	- Resection of the rectum, sigmoid, and anus - TME - Permanent colostomy	Tumor too close to the sphincter to achieve an adequate distal margin without compromising the sphincter or cancer that has infiltrated the sphincter	

r. Monitoring

☞ Clinical and Laboratory

- All patients with CRC should be followed up closely after curative treatment to ensure early identification and management of recurrence. (American Society of Colon and Rectal Surgeons' 2015 guidelines).
- Patient history, physical examination, CEA level
 - Every 3–6 months for 2 years
 - Every 6 months for an additional 3 years
- CT chest/abdomen/pelvis: annually for 5 years
- Colonoscopy
 - 1 year after preoperative colonoscopy

- Every 3–5 years in the further follow-up, depending on findings
- Proctoscopy/sigmoidoscopy
 - Recommended additional follow-up modality after the treatment of rectal cancer
 - Every 6–12 months for 3–5 years
 - May be supplemented with endorectal ultrasound
- Recommended duration of close follow-up: 5 years following the completion of curative treatment

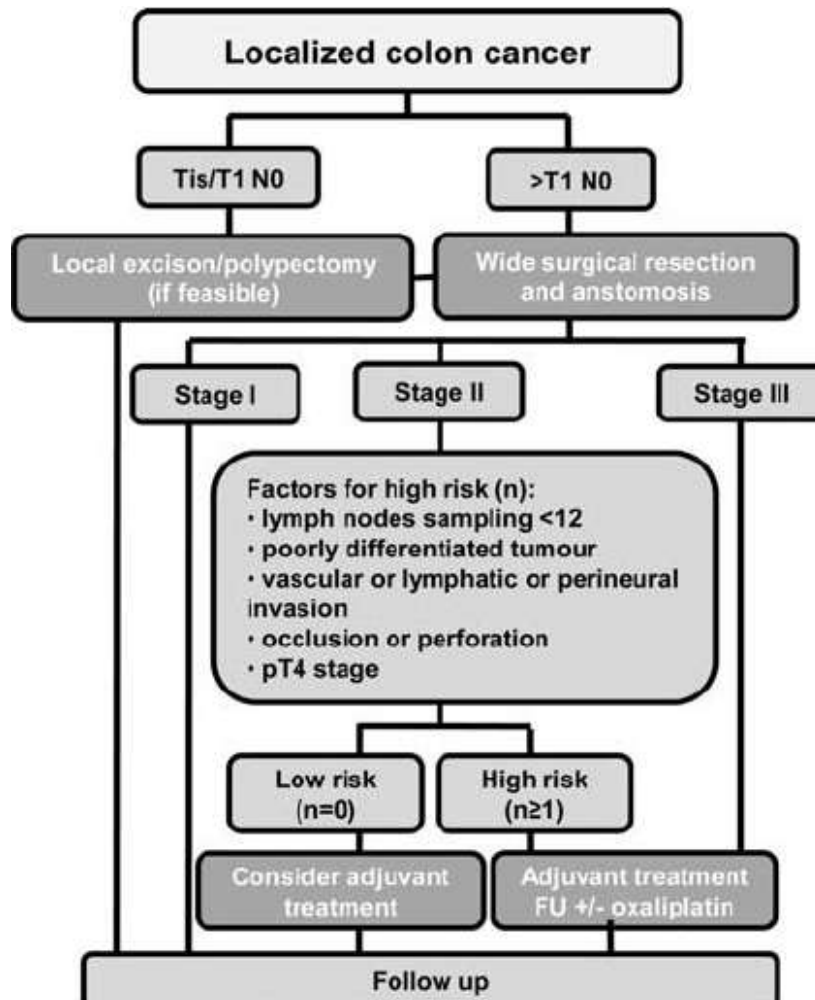
VI. COMPLICATION

s. Peritoneal carcinomatosis

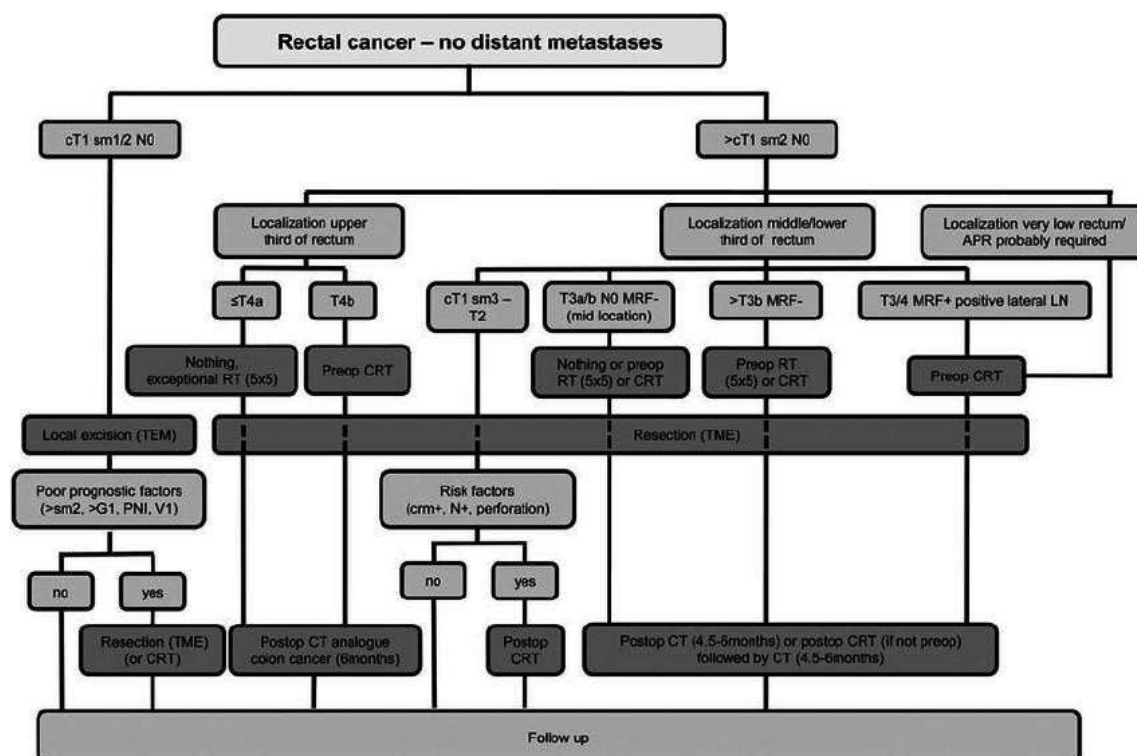
- ☞ Definition: a terminal feature of abdominal cancers (most commonly of the ovary, appendix, or colon) characterized by seeding of the tumor to the peritoneum
- ☞ Epidemiology peritoneum: depends on underlying malignancy; peritoneal carcinomatosis develops in ~ 10% of individuals with colorectal cancer
Etiology: primary tumor metastasizes to the peritoneal surface
Clinical features
 - Ascites
 - Fistulas
 - Fatigue
 - Abdominal pain, nausea, cachexia
- ☞ Diagnostics: CT is preferred; typically shows metastatic studding (tumor nodules studding the surface of the peritoneum) and omental caking
- ☞ Treatment: a terminal feature of abdominal cancers (most commonly of the ovary, appendix, or colon) characterized by seeding of the tumor to the peritoneum
 - Operable disease: involves cytoreductive surgery (CRS) followed by hyperthermic intraperitoneal chemotherapy (HIPEC), systemic chemotherapy, targeted therapy, and/or immune checkpoint inhibitors
 - Inoperable disease: experimental pressurized intraperitoneal aerosolized chemotherapy or repeated intraperitoneal chemotherapy infusion
- ☞ Complications: bowel, biliary, and/or ureteral obstruction
- ☞ Prognosis: poor; 5-year survival rates after CRS and HIPEC are approx. 40%.

VII. ALGORITHM

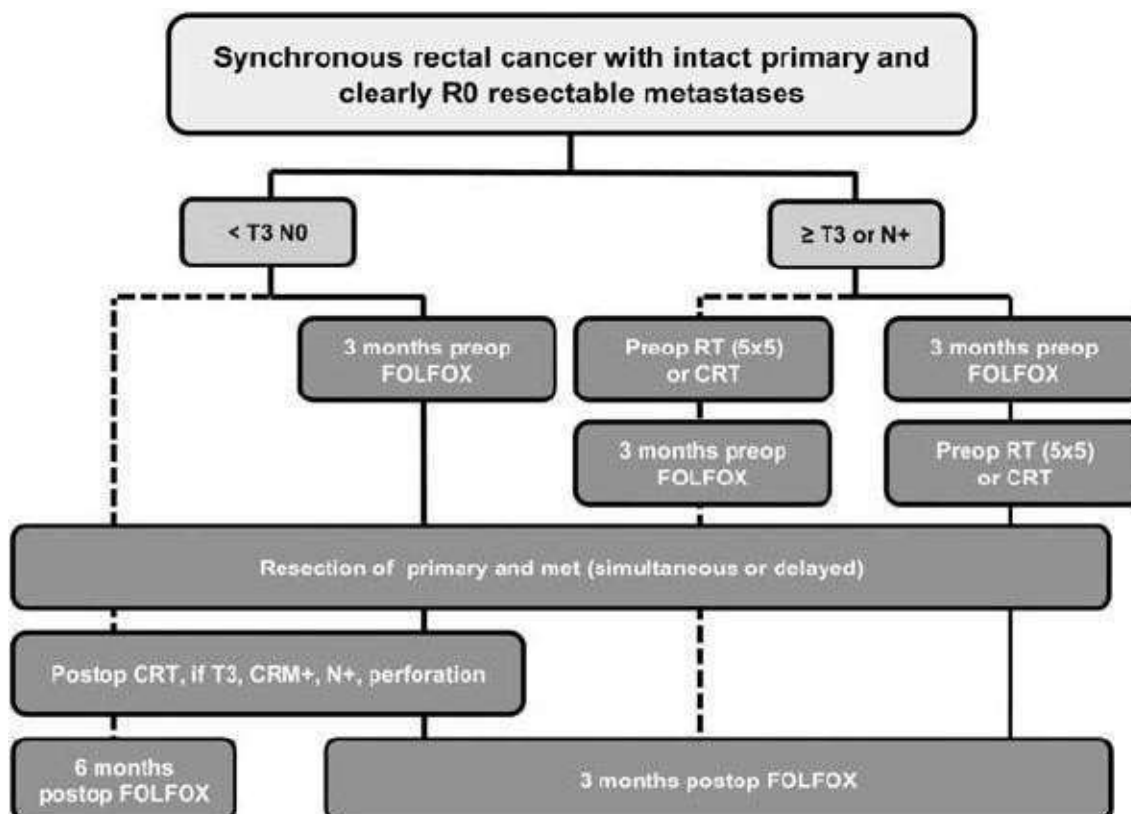
Algorithm for localized early colon cancer



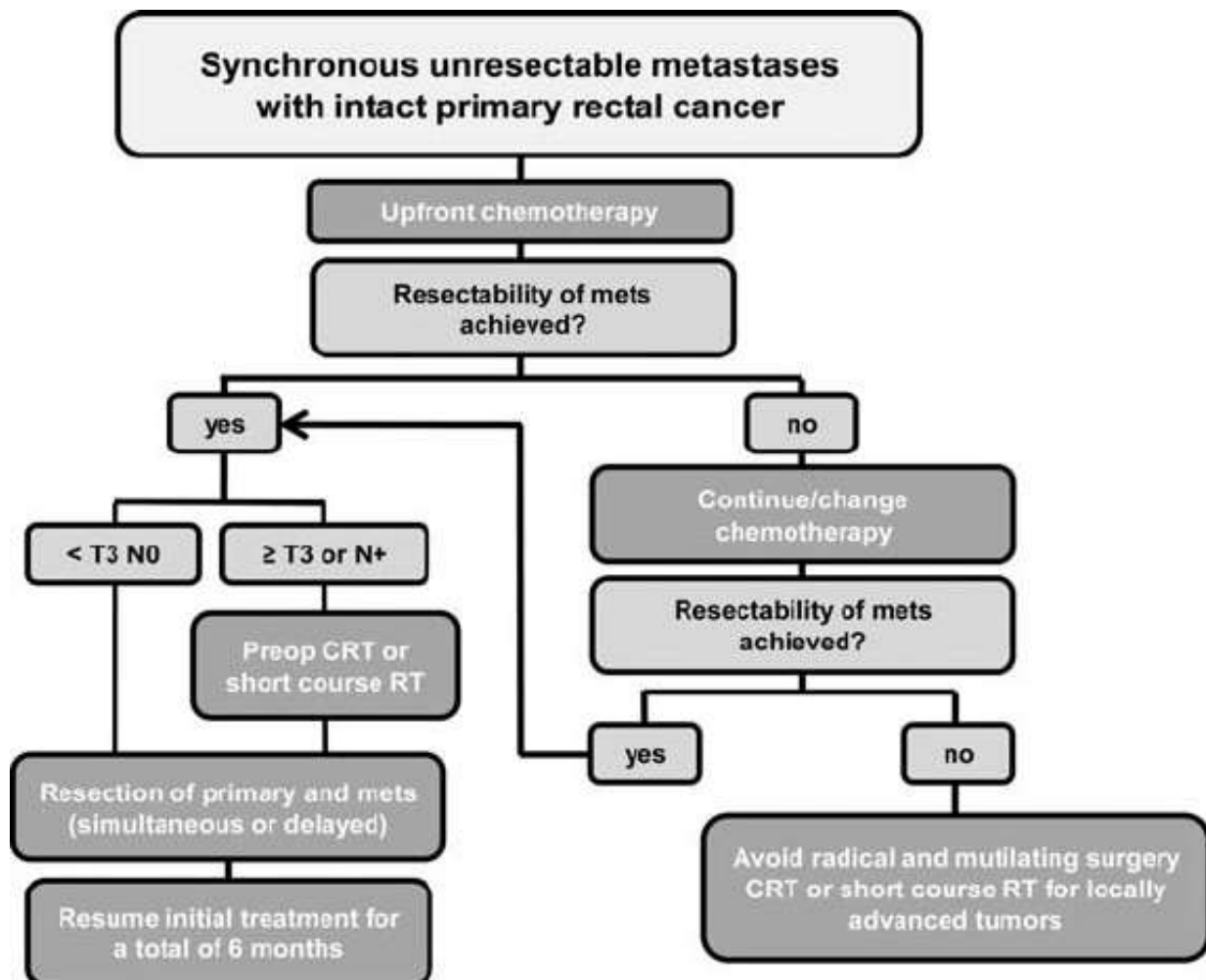
Algorithm for localized rectal cancer



Algorithm for resectable synchronous metastatic rectal cancer



Algorithm for resectable synchronous metastatic rectal cancer



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COMMON BILE DUCT OBSTRUCTION (TUMOR AND STONE)

BUN Sela, PLOK Vuthy

I. CASE DEFINITION

Common bile duct obstruction commonly refers to blockage of the common bile duct system leading to impaired bile flow from the liver into the intestinal tract (1).

II. ETIOLOGY

The common bile duct obstruction can be caused by either benign or malignant etiologies (2).

For benign pathologies, these include:

- Choledocholithiasis (gall stones in the common bile duct), (**most common cause**)
- Choledochal cysts (dilation/cysts of bile ducts),
- Mirizzi syndrome (gall stones in cystic duct pressing on bile duct),
- Benign structuring diseases like primary sclerosing cholangitis (psc),
- Fibrotic strictures from gall stone passage or iatrogenic strictures from bile duct cannulation.
Neoplastic causes can include:
 - Cholangiocarcinoma (bile duct cancer)
 - Pancreatic head cancer leading to distal common bile duct stricture
 - Ampullary carcinoma or adenoma.Other etiologies of biliary obstruction can include
 - Infectious diseases like parasitic cholangiopathy (Clonorchis sinensis, Ascaris lumbricoides) causing intra- or extrahepatic obstruction to bile flow,
 - Inflammatory and autoimmune diseases like AIDS cholangiopathy, and autoimmune cholangiopathy.

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

History:

- ☐ Jaundice with clay-colored urine
- ☐ Pruritis
- ☐ Depending on underlying etiology, patient can have right upper quadrant abdominal pain, fever, nausea and vomiting, and weight loss.

Physical examination:

- A thorough physical examination for signs of cholangitis (fever +/- rapid heart rate) is mandatory since it can be fatal if not treated promptly (3).
- General physical exam to assess for any distress, pallor, skin jaundice, scleral icterus, palmer erythema, and signs of malnutrition are essential.
- Access for possible Murphy Sign, hepatomegaly and splenomegaly, ascites or any palpable mass.
- Assessment in left supraclavicular lymphadenopathy, pleural effusion may be helpful in context of neoplastic causes.

ii. **Technical procedure**

c. Laboratory test

Patient might present with:

- Left shift leukocytosis with an increased number of band forms (mostly in case of cholangitis)
- Elevation in serum alkaline phosphatase (ALP) and gamma-glutamyl transpeptidase (GGT, GGTP).
- In jaundice patient, total and conjugated bilirubin are increased.
- In malignant bile duct strictures with complete obstruction, the level of total serum bilirubin is generally much higher than that observed in benign strictures, and a bilirubin level of greater than 20 mg/dL is highly suggestive of malignant obstruction. Again, in malignant strictures causing only partial obstruction (eg, Klatskin tumor), a rise in the alkaline phosphatase level may not be accompanied by a rise in the bilirubin level (4).

d. Imaging studie

Ultrasound: 1st line test to find out (5) :

- Gallstones, tumor
- Dilations of extrahepatic bile ducts
- Assess for patency of blood vessels using doppler.

However, it is not accurate in differentiating between benign and malignant case with also lower sensitivity (25-58%) in detecting choledocholithiasis (6) and tumor. Based on the ultrasound results, an abdominal **CT scan** can be the next step. This can help assess for pancreatic head lesions and liver lesions and rule out other intra- abdominal processes.

MRCP (magnetic resonance cholangiopancreatography) is a very sensitive test to assess further intra- and extrahepatic bile duct abnormalities like cholangiocarcinoma, strictures, or bile duct stones.

Endoscopic tests like **EUS** (endoscopic ultrasound) with needle aspiration and **ERCP** with cytology and biopsies can be further considered based on noninvasive testing results. ERCP can be both diagnostic and therapeutic in cases of biliary obstruction caused by gallstones (5).

IV. DIFFERENTIAL DIAGNOSIS

The differential diagnosis for biliary obstruction is very broad, ranging from benign disease to malignant diseases, such as (7):

Benign Diseases:

- Acute cholecystitis and biliary colic from gallstones (cholelithiasis)
- Acute or chronic pancreatitis
- Alcoholic hepatitis
- Cirrhosis from various etiologies
- Viral hepatitis, including chronic hepatitis B and

hepatitis C Malignant Diseases:

- Gallbladder cancer
- Pancreatic cancer
- Hepatocellular carcinoma
- Enlarged malignant lymph nodes

V. THERAPEUTIC APPROACH

General Principles:

- Initial management for acute cholangitis requires admission to the hospital (medical- surgical urgency) for intravenous fluid resuscitation and electrolyte replacement, pain control, and antibiotics with activity against enteric streptococci, coliforms, and anaerobes are generally needed.
- Low-risk community-acquired infections can be treated with ertapenem or piperacillin- tazobactam or combination therapy like ceftriaxone, ciprofloxacin, or levofloxacin with metronidazole.
- High-risk community-acquired infections can be treated with imipenem-cilastatin, meropenem, or piperacillin-tazobactam or combination therapy such as cefepime or ceftazidime with metronidazole.
- Treatment for biliary obstruction depends on the cause but likely will be either surgical or non-surgical, including endoscopic.
- Surgical may include cholecystectomy to remove the gallbladder if the cause is gallstones, along with ERCP with sphincterotomy to remove any gallstones from the biliary tract causing an obstruction.

- In case of tumor of common bile duct, surgical intervention procedure will be applied such as tumor removal with biliary by pass technique. The chemotherapy needs to be discussed depend on the stage of the tumor and the result of anatomopathological finding.

Drainage Procedures: Most patients with cholangitis will eventually require one of the following:

- Decompression procedure with ERCP (endoscopic retrograde cholangiopancreatography)
- Endoscopic removal of biliary stones with drainage of bile, often with sphincterotomy, PTC (percutaneous transhepatic cholangiography)
- Insertion of a needle into the biliary tract percutaneously for drainage, removal of stones, and possible placement of the biliary stent or surgical drainage
- Open or laparoscopic surgical intervention with biliary decompression, removal of the obstruction (tumor or stone), and cholecystectomy if warranted and the patient is stable (8)

VI. COMPLICATION

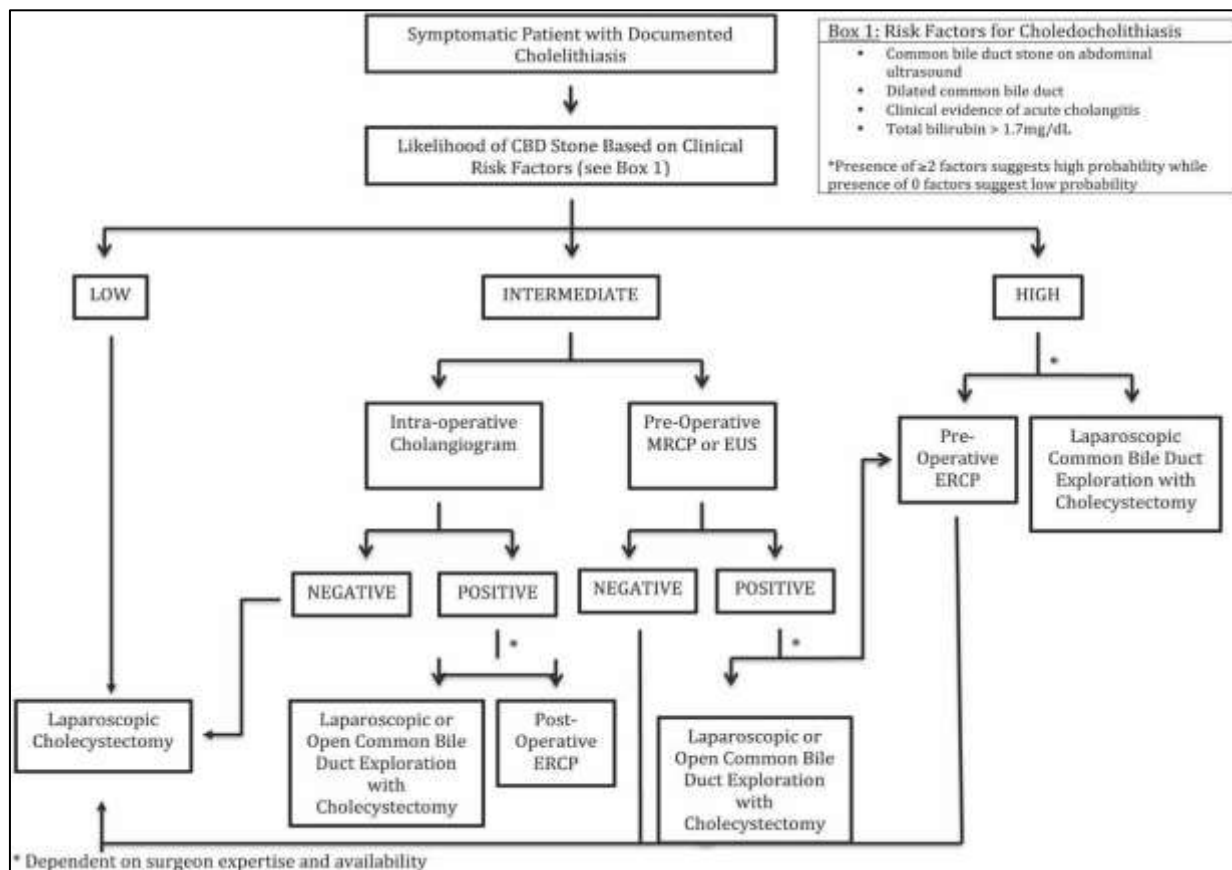
Acute ascending cholangitis: the most serious complication of common bile duct obstruction which is an infection that can quickly lead to systemic sepsis and resultant multi-organ failure.

Acute cholangitis can present as Charcot's triad (right upper quadrant pain, fever, and jaundice) or Reynold's pentad (Charcot's triad with mental status change and hemodynamic shock).

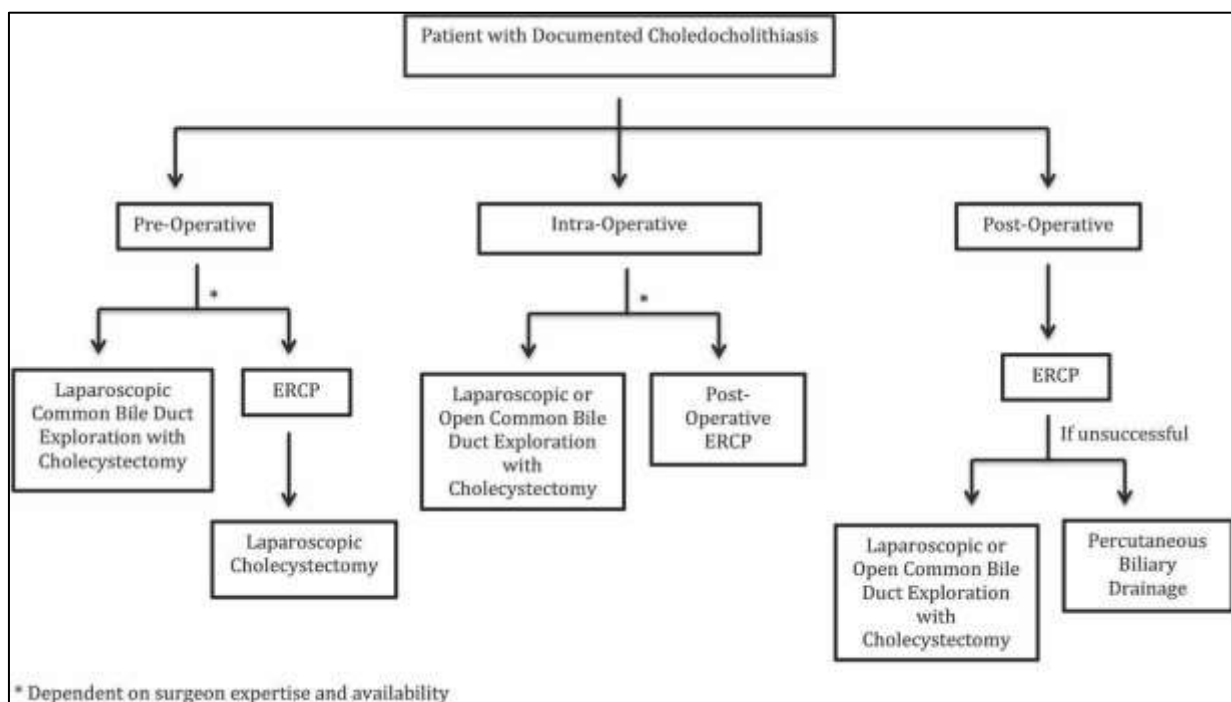
Severe cases of cholangitis present with (9):

- Hypotension requiring any vasopressor
- Mental status change
- PaO₂/FiO₂ less than 300
- Serum creatinine greater than 2.0 mg/dL
- INR greater than 1.5
- Platelet count less than 100000/mm³

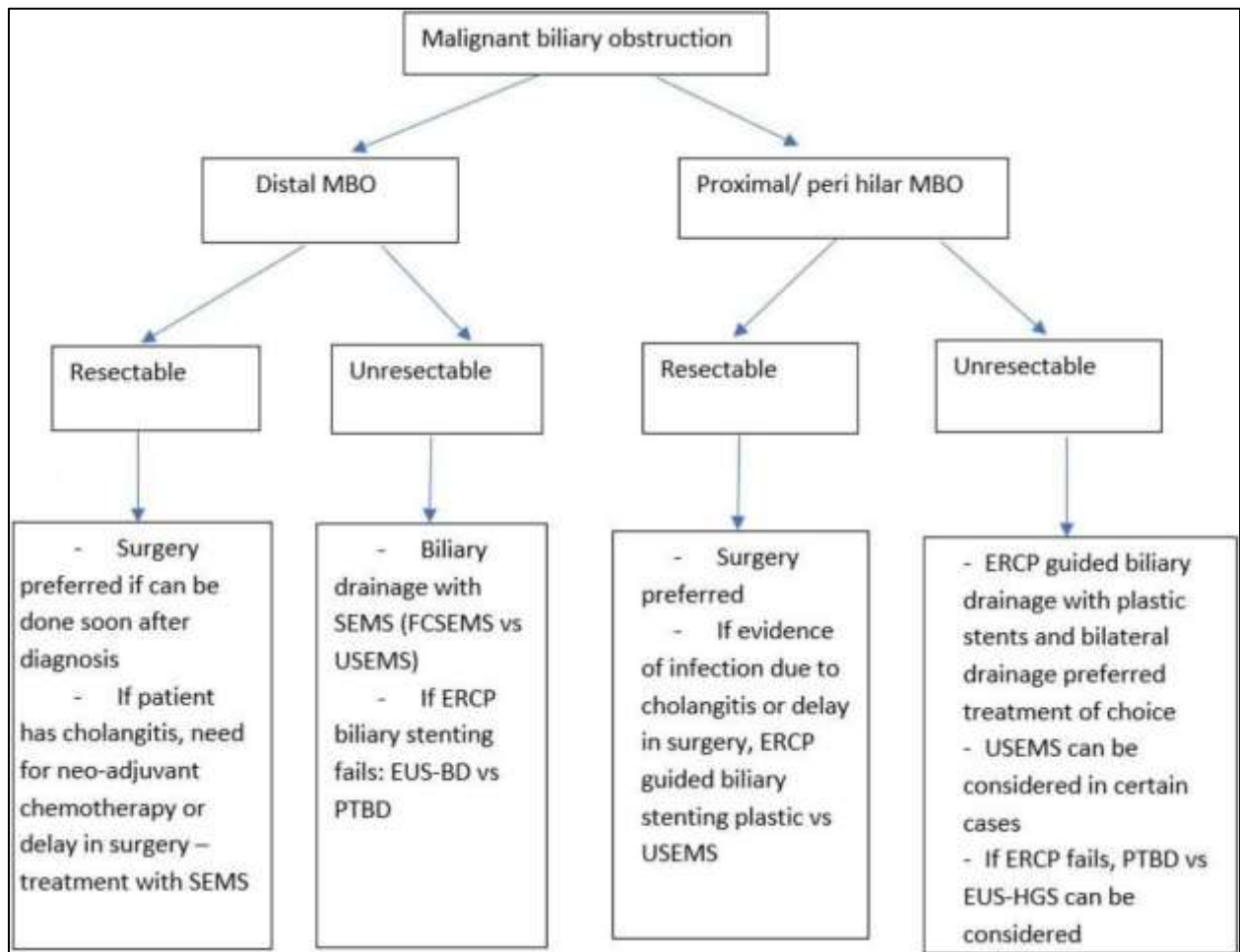
VII. ALGORITHM



Management algorithm for patients based on probability of choledocholithiasis (10)



Treatment algorithm for patients with documented choledocholithiasis based on time of diagnosis (10)



Treatment algorithm for management of malignant biliary obstruction (11)

MBO: Malignant biliary obstruction; SEMS: self-expanding metal stents; FCSEMS: fully covered self-expanding metal stents; USEMS: uncovered self-expanding metal stents; ERCP: endoscopic retrograde cholangiopancreatography; EUS: endoscopic ultrasound; BD: biliary drainage; PTBD: percutaneous transhepatic biliary drainage; HGS: hepaticogastrostomy.

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ANAL FISTULA

Dr. MO Satdin, Dr. EM Sokhom

I. DEFINITION

Anal fistula is an epithelial-lined tract connecting the anal canal to the perianal skin. It may be associated with external or internal openings, pus discharge in the perianal area, or at the time of defecation.

II. ETIOLOGY

According to cryptoglandular theory, obstruction in the ducts causes stasis in the anal glands, leading to infection, abscess, or fistula formation. Once the anal glands become infected, they fail to drain through the anal ducts at the dentate line. Inadequate draining of these glands causes abscess formation, which extends along a path of minimal resistance, usually into the intersphincteric, ischiorectal, or supralelevator regions [23, 25]. Anal fistulas and anorectal abscesses are trajectory phases of the same pathogenic process. An anorectal abscess is an acute inflammatory phase, while the anal fistulas are the chronic phase of the same disorder [26]. Other causes include [24]:

Patho-etiology	Common organisms
▪ Tuberculosis Actinomycosis Crohn's disease ▪ Malignancy and radiation therapy Anal fissure infecting the anal gland ▪ Pelvic infections resulting from appendicitis, diverticulitis, and gynecologic sepsis [24] ▪ As a postoperative complication of episiotomy, hemorrhoidectomy, and closed internal sphincterotomy ▪ Immunocompromised patients ▪ Penetrating injuries like gunshots, stab wounds, sexual trauma caused by anal sex, and accidental injuries ▪ ▪	▪ Escherichia coli ▪ Bacteroides fragilis ▪ Klebsiella pneumonia ▪ Prevotella ▪ Peptostreptococcus ▪ Porphyromonas ▪ Clostridium species ▪ Fusobacterium ▪ Streptococcus ▪ Staphylococcus aureus

III. DIAGNOSIS PROCEDURE

i. Clinical argument

- Itching around the anus due to pus discharge.
- Continuous throbbing pain that worsens during sitting. This is more frequent in an anal fistula associated with an abscess.
- Smelly discharge from or near the anus.
- Presence of an external opening with bloodstained pus discharge.

- Soiling of the clothes with purulent discharge.
- Fever in the presence of an abscess.

The clinical assessment, must be determined a detail history followed by physical examination is essential.

- Episodes of previous perianal abscess. If present, the frequency of the episodes and treatment received should be noted.
- History of obstetrics surgical procedure.
- History of Crohn's disease, radiation history to the pelvis and detail sexual history is vital as lymphogranuloma venereum can cause a perianal fistula [1].
- The procedural or surgical history in the anorectal area, since a fistula may occur after any surgical procedure.
- A history of tuberculosis disease should always be taken.
- A detailed history in case of recurrent fistulas, as it appears at the same anatomic location after healing of an abscess.

ii. Clinical examination

The left lateral posture is the preferred position for examination. Assess the patient by inspecting the perianal area. The purpose of clinical assessment is to recognize:

- External opening
- Internal opening
- Fistula tracts
- Underlying abscess,

Careful palpation will help to identify the fibrosed fistula tract and its course. The entire perineum should be palpated with gentle hands; the external opening may appear as an open sinus or elevated granulation tissue with pus or a blood-stained discharge. It realizes whether the external opening is posterior or anterior to a hypothetical line passing transversely over the middle of the anus. The internal opening may appear like an induration, pit, or groove and pus oozing at the anal crypt [2].

Digital rectal examination, is the key to examining the internal opening in the case of a fistula and an abscess, it makes pain, tenderness, or growth. With the tip of the finger, one can feel the tract as a hard, fibrous cord.

iii. Evaluating Incontinence

Wexner's score includes the frequency of soiling and allows for examining the three components of fecal incontinence (liquid, solid, and flatus). The difficulty in controlling the transit of the solid part of the stool is referred to as fecal incontinence. Anal incontinence, on the other hand, is the failure to manage the flatus and the liquid part of the stool [27].

iv. Technical procedures

a. Laboratory test

No any specific laboratory test, just notified of high white blood cells, CRP or Procalcitonin, which is a high specificity of bacterial infection.

b. Imaging study

Proctoscopy becomes challenging to perform if a fistula is associated with an abscess. But it will harm a severe painful for the patient.

Sigmoidoscopy or colonoscopy should be done if there is a suspicion of any disease like Crohn's disease, ulcerative colitis, or neoplastic lesion.

X-ray Fistulography and Computerized Tomography aren't used because of the suboptimal visualization of fistulas with a contrast medium and lack of accurate markers [6].

Anal endosonography may see the rectal wall, intersphincteric, and transsphincteric fistulas and their association with anal sphincters [6]. On the other hand, is inconvenient for assessing extrasphincteric and suprasphincteric tracts or secondary extensions due to the limited field view. Secondly, they are operator-dependent.

Magnetic Resonance Imaging, it's a great extent of soft-tissue distinction that helps evaluate fistula tracts and their specific course and underlying anatomy to plan a surgical procedure [7]. Complex and recurrent fistulas can be correctly identified up to 90% and 70%, respectively, using MRI [8].

c. Classification

Goodsall's rule, suggested by David Henry Goodsall, is the most widely used approach for evaluating fistula clinically [3]. It also describes the typical path followed by the fistula, which aids in locating the internal opening precisely [4]. An imaginary transverse line formed in the middle of the anal orifice divides it into anterior and posterior fistula.

Classification of fistulas determines the type of tract, spread, and the extent of the sphincters involved, making its implication important both anatomically and therapeutically. An anal fistula can be blind (only internal opening), complete (external and internal openings), or incomplete (external opening but no internal opening).

Park classified the fistula-in-ano into four primary types based on the anatomy and its relationship to the sphincter muscle [5]:

- Intersphincteric: The fistula extends up to the intersphincteric plane.
- Trans-sphincteric: The fistula extends beyond the external sphincter into the ischiorectal space. It may be low or high, depending on the extent of the external anal sphincter involved.

- Suprasphincteric: The fistula penetrates through the intersphincteric plane above the puborectalis and moves downwards to the ischiorectal fossa into the skin via a levator plate.
 - Extrasphincteric: The fistula tract passes outside the sphincter complex to enter the perianal skin from the rectum, passing through the levator muscle and ischiorectal fat.

Another classification subdivides fistulas into simple and complex [9,10]:

- Simple fistula defined when the tract is intersphincteric or low trans-sphincteric. A fistulotomy is the best surgical option since it affects only 30 percent of the external sphincter.
 - Complex fistula will follow these components:
 - Any fistula that involves over 30% of the external sphincter
 - Fistulas with multiple openings
 - All recurrent fistulas or secondary to radiation
 - Anterior fistulas in females
 - Fistulas with pre-existing incontinence
 - Inflammatory bowel and Crohn's related fistula

IV. DIFFERENTIAL DIAGNOSIS

- Infected Bartholin's cyst
- Subcutaneous abscess
- Infected anal fissure
- Hidradenitis suppurativa
- Rectorectal vestigial cyst with fistula formation
- Foreign body affected in the rectal wall
- Crohn's disorder
- Infected pilonidal cyst
- Infections (actinomycosis, tuberculosis, gonococcal infection, lymphogranuloma venereum)
- Progressive septic granulomatosis
- Anal colloid adenocarcinoma
- Diverge pathologies (prostatic abscess, osteomyelitis, etc.)
- Lymphoma

V. THERAPEUTIC APPROACH

v. Medical management

Antibiotics are often unnecessary in the setting of an uncomplicated and drained rectal abscess. Instead, antibiotics are reserved for immunocompromised patients, those with systemic signs of illness and the presence of cellulitis at the abscess. HIV patients may benefit from wound cultures and antibiotics.

vi. Minimal invasive procedures

a. Fibrin glue

The glue can be applied in surgical procedures to achieve a water-tight seal and hemostasis [12]. Occlusion of the fistula tract with sealant helps prevent tract contamination with mucus, stool, pus, and blood and helps in promoting wound healing and tissue growth. The advantages include minimum discomfort to the patient, no or low risk of damaging the sphincter complex, and ease of application.

b. Fistula plugs

The fistula plugs are made of acellular organic components of varied forms and sizes. They act as a framework for fibroblasts and native tissues to grow and close the fistula tract [13]. The porcine collagen anal fistula plug induces tissue remodeling, resulting in complete closure of the fistula tract.

c. Ligation of intersphincteric fistula tract (LIFT)

LIFT is an approach that Dr. Arun Rojanasakul designed in 2007 as a sphincter-sparing procedure [16]. Fistula tract ligation in an intersphincteric space close to an internal opening with separation of the distal tract. Suture ligations on the distal and proximal end of the tract are done using 3-0 polyglactin. After the procedure, complications were noted in nearly 14% of patients which are dehiscence and infection [17].

d. Video-Assisted Anal Fistula Treatment (VAAFT)

The VAAFT, first designed since 2006 approach is based on three principles [18]:

- Identification of internal opening
- Destruction of a tract from inside
- Preserving the function of the anal sphincter

The VAAFT technique helps detect and close the internal opening and destroy the distal tract for proper healing. With the VAAFT procedure, an internal opening is detected in almost 82.6% of the cases. Finally, the internal opening is closed with a stapler, an advancement flap, or a suture.

e. Fistula Laser Closure (FiLaC)

FiLaC is a sphincter-saving technique that “ablates” the fistula tract with a radial laser fiber and destroys the unhealthy granulation tissue. The principle of laser fistula surgery is to ablate the entire fistula tract with a diode laser source, followed by the internal opening closure [21]. The fiber used is a 360 degree corona radial fiber.

vii. Surgical fistula management

a. Fistulotomy

Fistulotomy, is a surgical intervention that removes the entire fistulous tract by cut open and allowed to heal secondary intention. It is still a gold standard for simple fistula tracts.

There are two ways to tackle it:

- **Park’s Technique:** The internal opening is removed by excising the mucosa, submucosa, surrounding tissue, and fibers of the internal anal sphincter reaching the deep intersphincteric area. An oval incision is given starting 5 mm above the internal opening and carried downwards up to the anal verge [20].
- **Eisenhammer’s Technique:** The alternative is to open the tract, curette, and scoop the primary infected source in the intersphincteric space [21].

b. Fistulectomy

A fistulectomy is a surgical technique in which a fistulous tract is entirely removed. The tract is laid open and curetted in fistulotomy, but fibrous tissue and glandular epithelium may be left behind. By Kronborg [14], the fistulectomy patients took longer to heal as more tissue was excised than fistulotomy, leaving an open wound to recover by secondary intention.

c. Seton placement procedure

A seton is a surgical thread inserted into the fistula tract and left for several weeks to keep it patent [14]. This facilitates the fistula to drain without requiring the sphincter to be severed. It is draining the pus and sepsis control before starting definitive treatment stimulating fibrosis. The cutting seton results in slow transection of an external sphincter muscle due to pressure necrosis with the cut ends being minimally separated [15].

d. Endorectal advancement flap

Elting, in 1912, was the first to apply this principle of anorectal advancement flap for fistula surgery with a desire to preserve the normal function of the sphincters.

The essential principle for curing the anal fistula is comprised of [11]:

- Severance of communication between the fistulous tract and bowel
- Removing the affected part of the bowel and the fistulous opening
- Mucosal advancement flap aims at covering the internal fistulous opening

e. Submucosal Ligation of Fistula Tract (SLOFT)

The procedure, is described in 2014 [19], performed to obliterate the internal opening and separate the distal tract by ligating and disconnecting it near the internal opening. The ligation is superficial and at the internal opening, thus leaving behind no stump.

f. Hybrid procedure

A hybrid procedure is a synergistic combination of two or more surgical techniques to yield a better outcome. They are not just procedures but concepts of thinking out of the box to treat complex fistulas. Hybrid procedures are an opportunity to combine the various surgical techniques, which are genuinely novel, with less morbidity, faster recovery, and least recurrence.

The different procedures are:

- Distal laser and proximal SLOFT (DLPS)
- Distal laser and proximal SLOFT with distal tract coring (DCPS)
- Distal laser with proximal LIFT
- VAAFT with LIFT with laser ablation of the distal tract (VA-LIFT)
- Distal coring with proximal fistulotomy with laser ablation (DCPF)

viii. Postoperative management of anorectal wounds

The most common postanorectal surgery complaints are pain and itching in the wound area. There are two patterns of pain reported following anorectal surgery: discomfort during resting and pain experienced during defecation. The resting pain lasts for around 24 to 48 h after surgery and gradually decreases. Postsurgery, the internal anal sphincter spasm causes pain during defecation, which takes a few days to go away. General principles to follow for healing the wounds after anorectal surgery include:

Ice pack: an ice finger is introduced into the anal canal to decrease the edema and counter the thermal effect. This ice finger is removed after 8 to 10 min. The cold compressions can be given in the initial 24 h.

Sitz bath: helps to promote blood flow, eliminate wound soiling, and relaxes the internal anal sphincter to expedite wound healing. After anorectal surgery, a hot sitz bath significantly reduces postoperative pain. The pelvic area and perineum are immersed for 15 to 20 min that can be repeated two to three times a day.

Topical ointments: These ointments relieve pain and swelling of the postoperative anorectal wounds. The patient is advised to apply the ointment 10 min before and after defecation.

Laxatives: Dietary modifications and toilet training remain the mainstay of management. Dietary changes include the use of a high-fiber diet. The laxatives (bulk-forming and osmotic agents) are used to keep the stools soft and decrease the patient's discomfort and fear of pain during the defecation.

VI. COMPLICATIONS

Infection

Any type of surgery carries a risk of infection. If the fistula is not completely removed, an infection in the tract (channel) can sometimes spread to other parts of the body.

Fecal incontinence

In rare cases, surgery can damage the anal sphincter muscles (the ring of muscles that open and close the anus). If you had some bowel incontinence before surgery, this may get worse. Rates of incontinence vary, although most studies report incontinence in 3-7% of people. The incontinence rate is 17% for seton techniques and around 6-8% after an advancement flap procedure.

Recurrence of anal fistula

In some cases, the fistula can recur despite surgery. After having a fistulotomy, the recurrence rate can vary from 7-21%, depending on factors such as whether it is simple or complex. After an advancement flap procedure, the recurrence rate may be as high as 36%.

VII. ALGORITHM

Table 1. Wexner's score, evaluation anal continence

Incontinence type	Always	Usually	Sometime	Never	Rarely	Frequency
Liquid	4	3	2	1	0	Never: 0 Rarely: <1/month Usually: <1/day >1/week Sometime: >1/month; <1/week Always: >1/day 20 pt: complete incontinence Total score: 0 perfect
solid	4	3	2	1	0	
Lifestyle alteration	4	3	2	1	0	
Wears pad	4	3	2	1	0	
gas	4	3	2	1	0	

Table 2. Intersphincteric fistula management

Type of intersphincteric fistula	Sphincter involvement and amount of sphincter to be cut	Outcome and alternatives
A1 —Simple intersphincteric fistula	Only the lower half of IAS	No risk of incontinence
A2 —Intersphincteric fistula with high blind tract	The whole IAS can be divided to lay open the tract	No risk of incontinence A hybrid procedure is a better approach
A3 —Intersphincteric fistula with an opening in the lower rectum	The whole IAS can be divided to lay open the tract	No risk of incontinence A hybrid procedure is a better approach
A4 —High intersphincteric fistula without perineal opening	The tract can be laid open by dividing IAS	No risk of incontinence Transanal opening of intersphincteric space with fistulotomy at 6 o'clock is done A hybrid procedure is a better approach
A5 —High intersphincteric fistula with pelvic extension	Such fistulas should be drained into the rectum A suprasphincteric fistula may occur if the ischiorectal fossa is drained	No risk of incontinence Transanal opening of intersphincteric space with fistulotomy at 6 o'clock is done A hybrid procedure is a better approach
A6 —Intersphincteric fistula extending from pelvic disease	Manage pelvic disease	The fistula will heal after eradicating the pelvic disease

Table 3. Trans-sphincteric fistula management

Type of trans-sphincteric fistula	Sphincter involvement and amount of sphincter cut	Outcome and alternatives
B1 —Uncomplicated	The superficial and subcutaneous parts of EAS and the lower part of IAS are divided	Little disturbance in continence
B2 —Trans-sphincteric fistula with a high blind tract	Fistulotomy up to the superficial part of EAS. The upper extension will heal on its own. Always prefer a transanal approach	Primary sphincter repair can be carried out The laser procedure is a better option

Table 4. Supra-sphincteric fistula management

Type of fistula	Sphincter involvement and amount of sphincter cut	Outcome and alternatives
Suprasphincteric fistula	Avoid sphincter cutting procedures	The laser procedure or seton is a better option

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GASTRIC CANCER AND SURGICAL MANAGERMENTS

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I. CASE DEFINITION

A. Gastric cancer is one of the leading causes of cancer death worldwide, second only to lung cancer. Despite the decreasing incidence of gastric cancer due to changes in diet and food preparation, prognosis is still poor with overall 5-year survival rates ranging from 5 to 15 % in the USA and other Western countries. In Japan where gastric cancer is endemic, the introduction of mass screening in the 1970s has led to early diagnosis with dramatic improvement in the 5-year survival rate. There is not clear data in Cambodia, but as a surgeon, we meet gastric cancer cases more and more in our daily practice. Malignant gastric neoplasms are comprised of several histologic subtypes including

- adenocarcinoma,
- lymphoma,
- carcinoid tumors,
- gastrointestinal stromal tumors (GIST), and
- leiomyosarcoma.

Approximately 95 % of all malignant gastric tumors are adenocarcinomas.

B. According to pathological analysis, there are two gastric cancer classifications as bellow:

a. WHO classifies gastric cancer into four types:

- i. papillary,
- ii. tubular,
- iii. mucinous, and
- iv. signet ring cell.

b. The Lauren classification divides gastric carcinoma into

- i. the intestinal and
- ii. diffuse subtypes.

C. These variants differ in pathologic epidemiologic, etiologic, and prognostic features.

- The intestinal subtype is usually found in the distal stomach, more common in elderly and found in conjunction with atrophic gastritis.
- Diffuse adenocarcinoma is mostly found in the cardia (but can arise in any part of the stomach), occurs more frequently in young patients without any predisposing condition, and prognosis is worse.

II. ETIOLOGY

The exact causes of stomach cancer are currently not known. However, factors which increase the risk of stomach cancer include:

- Age: Older people have a greater risk of developing stomach cancer.
- Gender: Stomach cancer is more than twice as common in men (2:1).
- Family History-10%: People with a family history of stomach cancer have a greater risk.
- Race: It is more common in Asia than in Europe and the United States.
- Foods: Pickled, salted and smoked foods may increase the risk of stomach cancer. Fruits and vegetables help reduce the risk.
- Infection with *Helicobacter pylori*: This bacterium causes inflammation and peptic ulcers, which may increase the risk of stomach cancer. It increases 6-fold in incidence. It causes intestinal type of gastric cancer.
- Chronic gastritis (atrophic, autoimmune), intestinal metaplasia.
- Gastric dysplasia: Chronic gastritis → gastric atrophy → intestinal metaplasia → dysplasia → carcinoma in situ → carcinoma—this cycle occurs in body and distal stomach and is often called as Correa cycle.
- Occupation: Exposure to certain chemicals and dust (Rubber workers, coal workers. Zinc, lead, talc, asbestos all can cause carcinoma stomach) is positively associated with stomach cancer.
- Lifestyle: Alcohol, tobacco, and a diet low in fruits and vegetables increase the risk of stomach cancer.

III. DIAGNOSTIC PROCEDURE

a. Clinical argument

All patients with a known or suspected gastric mass should undergo a history, physical examination carefully. There are usually no noticeable symptoms during the **early stage of stomach cancer**. However, the symptoms in the advanced stage are similar to those caused by other conditions, such as peptic ulcer and viral gastroenteritis. But in general, patient experience with

- Recent onset of loss of appetite and weight, early satiety, fatigue.
- Microcytic, hypochromic anaemia (iron deficiency) is common (40%). Anaemia, cachexia.
- Haematemesis (15%), melaena.
- Upper abdominal pain.
- Nausea and vomiting with features of gastric outlet obstruction, i.e. [+ve ausculto-percussion test, +ve succussion splash (to be checked with 4-6 hours empty

stomach)].

- Mass abdomen: Mass in pylorus lies above the umbilicus, nodular, hard, with impaired resonance, mobile, moves with respiration, all border well made out.
- Dysphagia with mass in upper epigastrium.
- When it arises from the body of stomach, it may present as only mass abdomen.
- In the late stage with peritoneal metastasis, patient can have ascites, Krukenberg tumours, Sister Joseph's nodules, rectovesical secondaries (Blumer shelf), Troisier's and Trousseau signs.
- Occasionally carcinoma stomach can present as perforation to begin with (4%).

b. Technical procedure

i. Laboratory test

Need lab tests done including CBC, Crea, Electrolytes, LFT, RFT, PT & INR, and some patient may need tumor markers such as CAE, CA 19-9 and CA 12-5. The lab tests also need to check according to her current condition and his/her co-morbidity.

ii. Imaging study

Chest and abdominal X-Ray with ECG need to evaluate his/her lung and heart

Gastroscopy with biopsy is the first choice for all patient with gastric disorders

Endosonography: EUS is useful to detect the involvement of layers of the stomach, nodal status and to define whether tumor is early or advanced.

US abdomen to see liver secondaries, ascites, nodes, ovaries. It is not as sensitive as EUS and CT scan.

CT scan abdomen and CT thorax with IV contrast to see size, extent, infiltration, L/Ns status, secondaries and operability.

Laparoscopy to stage disease. Laparoscopy will pick up peritoneal secondaries; detect occult metastases, organ invasion. Guided biopsy of peritoneum and nodes, peritoneal lavage for cytology. It will avoid unnecessary laparotomy. Signs of inoperability will be better identified like—peritoneal deposits (CT may miss), fixity, liver secondaries, fixed coeliac nodes, paraaortic nodes, ascitic fluid positivity on analysis. It is more accurate than CT.

IV. DIFFERENTIAL DIAGNOSIS

- Acid peptic disease; pyloric stenosis with gastric outlet obstruction.
- Gastritis.
- Pancreatic mass—carcinoma.

- Transverse colon mass—carcinoma.
- Advanced fixed stomach mass may mimic retroperitoneal or nodal mass.

V. THERAPEUTIC APPROACH

c. Stabilize patient

- Correction of anaemia, nutrition, fluid and electrolyte
- Cardiac, respiratory and renal status assessments
- Stomach wash using normal saline, but in some facility, they irrigate stomach with a Coke
- Need chest physiotherapy with incentive spirometer to prevent post-op lung collapse

d. Empirical treatment

- Prophylactic antibiotic as achlorhydria in gastric lumen allows colonisation of *Streptococcus faecalis*, *E. coli*, *bacteroides*, *Staphylococcus aureus*
- Blood/FFP may be needed preoperatively and for surgery

e. Directed treatment

Surgery is the treatment of choice for carcinoma stomach. Ideally the surgery is depending on tumor location and stage. For operable case:

- A. If it is an early growth, as in case of pylorus, distal partial gastrectomy with proximal 5 cm clearance, along with removal of greater omentum, lesser omentum, all lymph nodes, spleen (to clear splenic lymph nodes), tail of pancreas (when required); and later Billroth II or Roux-en-Y anastomosis.
 - B. In case of growth in the OG junction or upper part of the stomach, proximal partial gastrectomy is done along with removal of spleen, both omentum, lymph nodes; and later oesophagogastric anastomosis. Total gastrectomy is ideal here.
 - C. In case of growth in the body or linitis plastica, total gastrectomy (radical) with oesophagojejunal Roux-en-Y anastomosis is done.
- D. Some surgical mandates:
- Distal margin clearance towards duodenum is at least 1 cm from tumor end; proximal clearance is 5 cm.
 - D1 gastrectomy is clearance of perigastric nodes; D2 gastrectomy is clearance of nodes along major vessel trunks—along coeliac and splenic arteries. D2 is always better than D1.
 - D2 gastrectomy and adjuvant chemoradiation is ideal present concept therapy in operable carcinoma stomach.
 - D2 gastrectomy includes—removal of stomach with growth, omental bursa, greater and lesser omentum, anterior layer of mesocolon, anterior pancreatic capsule, D2 lymphadenectomy.

- Prophylactic distal pancreatectomy and splenectomy should not be routinely performed; a splenectomy should only be done when the hilum is involved.
- A meta-analysis of 12 randomized trials comparing D1 and D2 demonstrated no significant difference in OS, but a subgroup analysis of patients who underwent a D2 dissection without a splenectomy and/or pancreatectomy demonstrated a trend towards OS benefit.
- Presently newer staging for dissection is practiced in Japan—D1, D2, D3, D4 depending on levels of lymph nodes cleared ('D' for dissection). D1 or D2 dissections are commonly done.
 - However, D2 should only be performed by experienced and trained surgeons.
 - D1 is done when nodes are N0. It is removal of nodes along the lesser and greater curves and pylorus (Group I: stations 3-6).
 - D2 is done when nodes are N1. Here nodes like left gastric/ common hepatic/splenic/retropancreatic are removed. Splenic nodes can be cleared with or without splenectomy along with removal of tail of pancreas. Here node stations 1-11 are removed. Stations 12-18 are not removed. Average 27 nodes are removed.
 - D3 is done when nodes are N2. Nodes removed for clearance are hepatoduodenal, nodes along mesentery, middle colic (1-16 stations). Average 43 nodes are removed.
 - D4 is not commonly advocated. It is removal of stations 1-18.
- f. Monitoring
 - i. Clinical & Laboratory monitoring
- General management:
 - Similar to other intraabdominal operation, patients with gastric surgery should undergo a regimen of aggressive pulmonary hygiene, early ambulation, and physical therapy.
 - Careful attention should be paid to the volume status, electrolytes, and input/output balance.
 - DVT prophylaxis: We start patients on DVT prophylaxis within 24 hours after completion of the procedure unless contraindicated.
- NG tube: We keep the NG tube until 2 to 3 days postoperatively. If the tube gets dislodged accidentally, we do not replace it. If replacement of an NG tube is necessary, it should be performed under fluoroscopy to avoid accidental disruption of the anastomosis by a blindly placed NG tube.
- Diet:
 - A clear liquid diet is initiated on postoperative day 3 or 4 and advanced as tolerated.
 - We consult a dietitian to educate patients on issues related to postgastrectomy diet.
- Seventy-four percent of recurrences occurred within 2 years after surgery, and less than 4% occurred after 5 years following surgery.

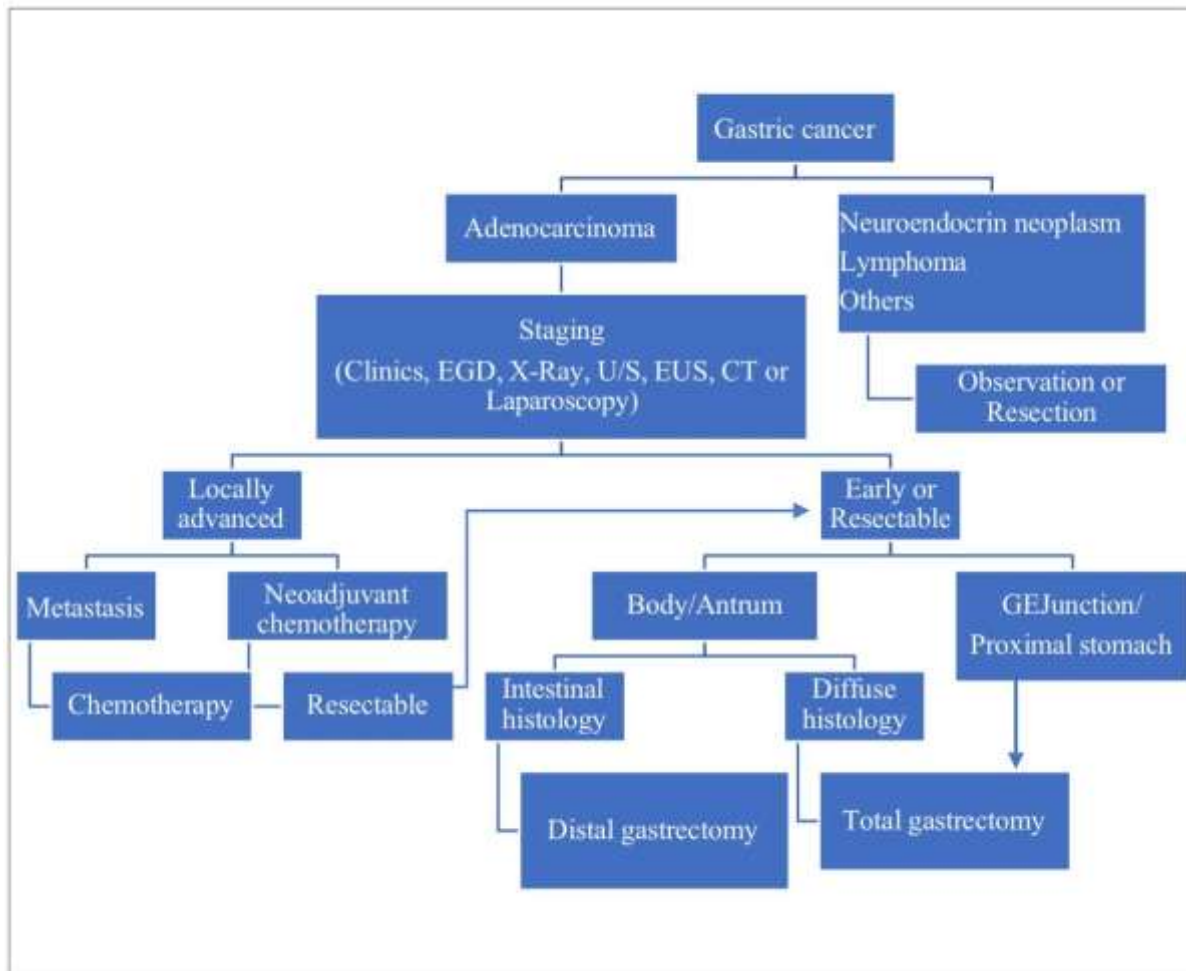
- Follow-up includes a complete history and physical examination. every 3–6 months for 1–2 years, every 6–12 months for 3–5 years, and then annually thereafter.
- Routine imaging and laboratory values should be avoided unless patients have signs and symptoms that necessitate a workup.
- Vitamin B12 and iron deficiency should be monitored and treated for patients who had a gastrectomy.

VI. COMPLICATIONSZ

- a. Postoperative bleeding:
 - Intraluminal bleeding can generally be managed by supportive measures in the form of correction of coagulopathy and volume resuscitation. Bleeding that causes hemodynamic compromise or does not respond to supportive measures requires careful endoscopic therapeutic measures. Reoperation for bleeding not controlled with endoscopy is rare.
 - Intraabdominal bleeding requires volume resuscitation and correction of coagulopathy. Patients with hemodynamic compromise or who do not immediately respond to supportive measures (suggesting ongoing bleeding) should return to the operating room for definitive control of the bleeding source.
- b. Anastomotic leak:
 - When anastomotic leak or intraabdominal sepsis is suspected, a CT scan of the abdomen and pelvis with oral and IV contrast is performed.
 - CT scan may suggest the presence of anastomotic leak and would demonstrate the presence of any drainable abdominal collection.
 - The anastomotic leak can further be confirmed and characterized with a Gastrografin (diatrizoic acid) swallow study.
 - Most leaks can be managed by making the patient NPO, parenteral nutrition, and drainage of any intraabdominal collection.
 - Weekly Gastrografin swallow studies can document when the leak is healed.
 - Rarely is surgical intervention required.
 - Small leaks can be closed primarily and buttressed with an omental patch.
 - Larger leaks may require a revision of anastomosis with wide drainage and jejunostomy tube for enteral feeding.
- c. Duodenal stump blowout:
 - A CT scan will demonstrate an abscess in the right upper quadrant and will not contain oral contrast.
 - A percutaneous drain with a contrast study through the drain will confirm connection with duodenum.
 - Management is largely through supportive measures including NPO, parenteral or enteral nutrition, and replacement of fluid losses.

- A percutaneous transhepatic duodenal diversion can assist with closure of the leak by diverting the bile and pancreatic juices away from the fistula.
 - Operative management might be required if the fistula does not close.
 - If the duodenal blowout presents as acute abdomen, then exploratory laparotomy, placement of a tube duodenostomy, and wide drainage of right upper quadrant is performed.
- Delayed complications
- a. Delayed gastric emptying
 - Treatment is largely supportive and includes gastric decompression with NG and trial of promotility agents such as metoclopramide (10mg three times daily) and/or erythromycin (up to 250mg four times a day).
 - Alkaline reflux gastritis
 - b. Alkaline reflux gastritis does not respond well to medical therapy.
 - However, a trial of medical therapy with cholestyramine, sucralfate, or antacid is still warranted.
 - Surgical treatment involves diverting duodenal contents away from the stomach by converting loop gastrojejunostomy to a Roux-en-Y gastrojejunostomy with a 45- to 60-cm Roux limb.
 - A vagotomy should be performed at the time of reoperation to reduce the risk of marginal ulcer.
 - c. Afferent and efferent loop syndrome
 - Acute afferent loop syndrome is an emergency and requires immediate operative exploration.
 - In case of chronic afferent loop syndrome, conversion of a loop gastrojejunostomy to Roux-en-Y anastomosis addresses the problem.
 - Surgery is usually required for efferent loop obstruction and involves correction of the underlying cause.
 - d. Nutritional consequences: Iron, Vit B12 and other Vit & minerals need
 - e. Dumping syndrome
 - Dumping syndrome exists in either a late (2 to 3 hours after a meal) or an early form (within 20 to 30 minutes following ingestion of a meal). Dumping syndrome is treated by adjusting the diet and pattern of eating, for example, eating smaller, more frequent meals and limiting liquids.

VII. ALGORITHM



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GASTROESOPHAGEAL REFLUX (GERD)

Dara vithiea, San corine, Meas Vutha

I. INTRODUCTION

Gastroesophageal reflux disease (GERD) is notable for its high prevalence, variety of clinical presentations, under-recognized morbidity, and substantial economic consequences.

II. TERMINOLOGY

Some degree of reflux is physiologic [1]. Physiologic reflux episodes typically occur postprandially, are short-lived, asymptomatic, and rarely occur during sleep.

Pathologic reflux is associated with symptoms or mucosal injury and often occurs nocturnally.

- Gastroesophageal reflux disease (GERD) is a condition that develops when the reflux of stomach contents causes troublesome symptoms and/or complications [2].

GERD is classified based on the appearance of the esophageal mucosa on upper endoscopy into the following:

- **Erosive esophagitis** Erosive esophagitis is characterized by endoscopically visible breaks in the distal esophageal mucosa with or without troublesome symptoms of GERD.
- **Nonerosive reflux disease** Nonerosive reflux disease or endoscopy negative reflux disease is characterized by the presence of troublesome symptoms of GERD without visible esophageal mucosal injury.

III. EPIDEMIOLOGY

In a systematic review of 15 epidemiological studies, the prevalence of gastroesophageal reflux disease (GERD) was found to be 10 to 20 percent in the Western world and less than 5 percent in Asia [3]. The incidence in the Western world was approximately five per 1000 person-years or 0.5 percent per year. In a subsequent population-based survey in the United States, 22 percent of respondents reported that they had heartburn or regurgitation within the last month while 16 percent reported regurgitation [4]. Heartburn or regurgitation was clinically significant ($\geq$ twice weekly) in 6 and 3 percent, respectively.

There are limitations in the epidemiologic estimates of the prevalence of GERD, as they are based upon the assumption that heartburn and/or regurgitation are the only indicators of the disease [3,4]. However, patients with objective evidence

of GERD (eg, esophagitis or Barrett's esophagus) do not always have heartburn and heartburn is not always sufficiently severe to be indicative of GERD [5].

IV. CLINICAL FEATURES

Clinical manifestations Classic symptoms of gastroesophageal reflux disease (GERD) are heartburn (pyrosis) and regurgitation.

- Heartburn is typically described as a burning sensation in the retrosternal area, most commonly experienced in the postprandial period [2]. Heartburn is considered troublesome if symptoms occur two or more days a week [2].
- Regurgitation is defined as the perception of flow of refluxed gastric content into the mouth or hypopharynx [2]. Patients typically regurgitate acidic material mixed with small amounts of undigested food.

Other symptoms of GERD include dysphagia, chest pain, water brash, globus sensation, odynophagia, extraesophageal symptoms (eg, chronic cough, hoarseness, wheezing), and, infrequently, nausea.

- Dysphagia is common in the setting of longstanding heartburn and is often attributable to reflux esophagitis but can be indicative of an esophageal stricture [6]. Odynophagia is an unusual symptom of GERD but, when present, usually indicates an esophageal ulcer.
- GERD-related chest pain can mimic angina pectoris, and is typically described as squeezing or burning, located substernal and radiating to the back, neck, jaw, or arms. The pain can last anywhere from minutes to hours and resolve either spontaneously or with antacids. It usually occurs after meals, awakens patients from sleep, and may be exacerbated by emotional stress [1]. Patients with reflux-induced chest pain may also have typical reflux symptoms.

However, heartburn is a poor predictor of whether patients with chest pain have evidence of GERD by objective reflux testing (eg, esophageal pH monitoring) [7].

- Water brash or hypersalivation is a relatively unusual symptom in which patients can foam at the mouth, secreting as much as 10 mL of saliva per minute in response to reflux.
- Globus sensation is the almost constant perception of a lump in the throat (irrespective of swallowing), which has been related to GERD in some studies. However, the role of esophageal reflux in globus is uncertain [8].
- Nausea is infrequently reported with GERD, but a diagnosis of GERD should be considered in patients with otherwise unexplained nausea [9].

1. **Radiographic findings** Double contrast **barium** swallow examination is of limited diagnostic utility in patients with GERD. ~~The~~ demonstration of reflux of barium during the study is of dubious significance since it can be provoked in 25 to 71 percent of symptomatic patients and 20 percent of normal controls [10]. In one study,

the diagnostic accuracy of barium radiography in patients with mild, moderate, and severe esophagitis on upper endoscopy is 25, 82 percent, and 99 percent, respectively [11].

Mucosal changes associated with esophagitis and peptic strictures seen on double contrast [Barium](#) radiography include the following:

- In early stages of reflux esophagitis, the mucosa of the distal third of the esophagus appears granular or nodular with numerous ill-defined, 1 to 3 mm lucencies [12].
- In patients with erosive esophagitis, shallow ulcers and erosions may be visualized as tiny collections of [barium](#) in the distal esophagus near the gastroesophageal junction, sometimes surrounded by a radiolucent halo of edematous mucosa.
- A benign peptic stricture due to GERD has the appearance of a smooth, tapered area of concentric narrowing in the distal esophagus that is usually 1 to 4 cm in length and 0.2 to 2.0 cm in diameter. However, many peptic strictures have an asymmetric appearance with puckering of one wall of the stricture due to eccentric scarring.

Complications Complications from GERD can arise even in patients who lack typical esophageal symptoms. These complications may be esophageal (eg, Barrett's esophagus, esophageal stricture, esophageal adenocarcinoma) or extra-esophageal (eg, chronic laryngitis, exacerbation of asthma).

V. DIAGNOSIS

Patients with classic symptoms The diagnosis of gastroesophageal reflux disease (GERD) can often be based on clinical symptoms alone in patients with classic symptoms such as heartburn and/or regurgitation [2,13]. However, patients may require additional evaluation if they have alarm features, risk factors for Barrett's esophagus, or abnormal gastrointestinal imaging performed for evaluation of their symptoms.

Although 40 to 90 percent of patients with symptoms suggestive of GERD have a symptomatic response to proton pump inhibitors (PPIs), a response to antisecretory therapy is not a diagnostic criterion for GERD [14,15]. A meta-analysis of diagnostic test characteristics found that a response to PPIs did not correlate well with objective measures of GERD such as ambulatory pH monitoring [15]. Pooled sensitivity and specificity were 78 and 54 percent, respectively. Thus, a response to PPIs does not correspond to a GERD diagnosis based on reflux testing.

Patients without classic symptoms Other symptoms (eg, chest pain, globus sensation, chronic cough, hoarseness, wheezing, and nausea) may be seen in the setting of GERD, but are not sufficient to make a clinical diagnosis of GERD in the absence of classic symptoms of heartburn and regurgitation. Other disorders need to be excluded before attributing the symptoms to GERD. As an example, unexplained chest pain should be evaluated with an electrocardiogram and exercise stress test prior to a gastrointestinal evaluation.

VI. EVALUATION IN SELECTED PATIENTS

Additional evaluation is required in selected patients with suspected gastroesophageal reflux disease (GERD) to rule out alternative etiologies, confirm the diagnosis of GERD, and assess for complications (eg, Barrett's esophagus).

a) Upper gastrointestinal endoscopy

Indications Upper endoscopy is indicated in patients with suspected GERD to evaluate alarm features or abnormal imaging if not performed within the last three months. Upper endoscopy should also be performed to screen for Barrett's esophagus in patients with risk factors. On upper endoscopy, biopsies should target any areas of suspected metaplasia, dysplasia, or, in the absence of visual abnormalities, normal mucosa to evaluate for eosinophilic esophagitis [16].

Upper endoscopy is not required to make a diagnosis of GERD. However, upper endoscopy can detect esophageal manifestations of GERD (eg, Barrett's metaplasia, erosive esophagitis) and can rule out an upper gastrointestinal tract malignancy. Upper endoscopy can also rule out other etiologies in patients with GERD symptoms that are refractory to a trial of proton pump inhibitor (PPI) therapy.

Alarm features Alarm features that are suggestive of a gastrointestinal malignancy include:

- New onset of dyspepsia in patient ≥ 60 years
- Evidence of gastrointestinal bleeding (hematemesis, melena, hematochezia, occult blood in stool)
- Iron deficiency anemia
- Anorexia
- Unexplained weight loss
- Dysphagia
- Odynophagia
- Persistent vomiting
- Gastrointestinal cancer in a first-degree relative

Risk factors for Barrett's esophagus Screening for Barrett's esophagus is typically recommended for patients with multiple risk factors (one of which must be duration of GERD of at least 5 to 10 years). Risk factors for Barrett's esophagus include:

- Duration of GERD of at least 5 to 10 years
- Age 50 years or older
- Male sex
- White individuals
- Hiatal hernia
- Obesity
- Nocturnal reflux
- Tobacco use (past or current)
- First-degree relative with Barrett's esophagus and/or adenocarcinoma

Multiple societies have issued guidelines regarding screening patients for Barrett's esophagus with varying recommendations.

Abnormal upper gastrointestinal tract imaging — Abdominal imaging is not required to establish the diagnosis of GERD but may have been performed for evaluation of concurrent symptoms. In such cases, luminal imaging abnormalities in the upper gastrointestinal tract may warrant diagnostic evaluation with upper endoscopy.

Endoscopic findings — Upper endoscopy may be normal in patients with GERD, or there may be evidence of esophagitis of varying degrees [17]. Among untreated GERD patients, approximately 30 percent will have endoscopic esophagitis. The severity and duration of symptoms correlate poorly with the severity of esophagitis.

In contrast to infectious and medication-induced esophagitis, which tend to be in the mid-esophagus, the ulcerations seen in peptic esophagitis are usually irregularly shaped or linear, multiple, and are in the very distal esophagus.

Other endoscopic findings in patients with longstanding GERD include peptic strictures, Barrett's metaplasia, and esophageal adenocarcinoma.

Grading the severity of esophagitis – Erosive esophagitis is graded according to its severity to guide management. Several endoscopic grading schemes have been devised to decrease inter-operator variability of endoscopy in assessing the severity of peptic esophagitis. Of these, the Los Angeles classification is the most thoroughly evaluated classification for esophagitis and is the most widely used [18].

- Los Angeles classification – The Los Angeles classification grades esophagitis severity by the extent of mucosal abnormality, with complications recorded separately. In this grading scheme, a mucosal break refers to an area of slough adjacent to more normal mucosa in the squamous epithelium with or without overlying exudate.

- Grade A – One or more mucosal breaks each ≤ 5 mm in length
- Grade B – At least one mucosal break > 5 mm long, but not continuous between the tops of adjacent mucosal folds
- Grade C – At least one mucosal break that is continuous between the tops of adjacent mucosal folds, but which is not circumferential
- Grade D – Mucosal break that involves at least three-fourths of the luminal circumference

Histology — Approximately two-thirds of patients with symptoms of GERD and no visible endoscopic findings (ie, nonerosive reflux disease) have histologic evidence of esophageal injury that responds to acid suppression [19]. However, histology is not specific for GERD and similar findings may be seen in patients with eosinophilic esophagitis. The most consistently observed histologic finding of GERD is the dilation of the intercellular spaces seen on transmission electron microscopy. Other histologic features include the presence of neutrophils and eosinophils, dilated vascular channels in papillae of the lamina propria, thickening of the basal cell layer with pale, distended squamous ("balloon") cells, and elongation of the papillae of the epithelium.

Cytokine triggered inflammation stimulates histopathological events in the development of esophagitis. Lymphocytic inflammation and dilated intercellular spaces occur deep in the epithelium, not at the luminal surface, and regenerative changes (basal cell hyperplasia, papillary elongation) are initiated prior to the development of surface necrosis.

Esophageal manometry — In patients with suspected GERD with chest pain and/or dysphagia and a normal upper endoscopy, an esophageal manometry should be performed to exclude an esophageal motility disorder. Manometry is useful in ensuring that ambulatory pH probes are placed correctly but cannot diagnose GERD. It is also used to evaluate peristaltic function before antireflux surgery for GERD [20].

Ambulatory esophageal pH monitoring — Ambulatory pH monitoring is also used to confirm the diagnosis of GERD in those with persistent symptoms (whether typical or atypical, particularly if a trial of twice-daily PPI has failed) or to monitor the adequacy of treatment in those with continued symptoms [21,22]. The approach to management of patients with refractory GERD is discussed in detail separately.

Ambulatory pH monitoring can be performed with either a trans nasally placed catheter or a wireless, capsule-shaped device that is affixed to the distal esophageal mucosa [23,24]. Esophageal pH monitoring with impedance is an alternative to wired or wireless pH studies with the advantage of detecting weakly acid reflux in addition to acid reflux, making it more useful in ascertaining symptom-reflux correlation.

In each case, the pH sensor is coupled with compact, portable data recorders, and computerized data analysis. The catheter type pH electrode is positioned 5 cm above the manometrically defined upper limit of the lower esophageal sphincter. In the case of the

wireless device, the pH capsule is attached 6 cm proximal to the endoscopically defined squamocolumnar junction. Tests are traditionally conducted for a 24-hour period with patients advised to consume an unrestricted diet. Studies with the wireless device are conducted for two to four days, potentially varying the dietary and/or therapeutic circumstances among days [25]. Increasing the monitoring period in patients undergoing evaluation using a wireless device increases the yield of the study for detecting reflux episodes and correlating those events with symptoms [24,26].

VII. DIFFERENTIAL DIAGNOSIS

The differential diagnosis of gastroesophageal reflux disease (GERD) includes infectious esophagitis, pill esophagitis, and eosinophilic esophagitis. Other causes of dysphagia include esophageal rings/webs, and impaired peristalsis due to an esophageal motility disorder. Slowly progressive dysphagia for solids with episodic esophageal obstruction is suggestive of a stricture or an esophageal cancer.

Odynophagia may be due to infectious or medication-induced esophagitis. GERD can be distinguished from these conditions by esophageal manometry and upper endoscopy with biopsies of the esophagus. The differential diagnosis and evaluation of patients with dysphagia is discussed in detail separately.

Frequent heartburn may also be due to reflux hypersensitivity or functional heartburn. GERD can be distinguished from these conditions by pH or pH-impedance testing. Patients with reflux hypersensitivity have normal acid exposure but a positive symptom association with acid or weakly acid reflux. Patients with functional heartburn have normal acid exposure and a negative symptom reflux association.

Patients with functional dyspepsia have heartburn in one third of cases, but early satiety and postprandial fullness are the predominant symptoms.

VIII. TREATMENT APPROACH BY PRESENTING SYMPTOMS

Definite indications for treatment of GERD are:

- Heartburn (distinct and recurrent symptoms)
- Esophagitis (documented on endoscopy)

By contrast, extraesophageal disease (asthma or pneumonia) usually has causes other than GERD. Treatment for GERD is appropriate only for selected patients

The treatment approach for each of these presenting symptoms is outlined in the following sections, followed by a more detailed discussion of each type of intervention.

Heartburn — Patients with both heartburn and alarm symptoms (dysphagia, weight loss, odynophagia, hematemesis, or recurrent forceful vomiting) should be referred to a gastroenterologist for further evaluation, which may include upper endoscopy.

For patients with typical esophageal symptoms (eg, heartburn, increased salivation, retrosternal or epigastric discomfort, and no alarm symptoms), we recommend a "step-up" approach, starting with conservative options and progressing to more intensive therapy if needed.

- **Mild** – For children and adolescents with mild or infrequent heartburn (averaging once a week or less), we suggest a trial of lifestyle changes (weight management for overweight patients; cessation of smoking and vaping and avoidance of tobacco smoke; head of bed elevation; and avoidance of chocolate, caffeine, spicy foods, alcohol, and other foods that exacerbate symptoms). Occasional use of antacids or histamine type 2 receptor antagonists (H2RAs) is acceptable for short-term relief of symptoms, but frequent use of these drugs (more than once/week) calls for advancement of therapy [2].
- **Moderate or severe** – If the heartburn symptoms are moderate or severe, or fail to respond to lifestyle changes, we recommend a trial of acid-suppression therapy. For this trial, we typically use a proton pump inhibitor (PPI). An H2RA is a reasonable alternative but may be limited by the possibility of tachyphylaxis that can occur during a typical trial duration of four to eight weeks [1,3]. If symptoms do not improve, or recur after stopping treatment, the patient should be evaluated by a pediatric gastroenterologist. If an H2RA is used initially and there is no improvement or partial improvement in two to four weeks, or if long-term treatment is required, therapy should be advanced to a PPI.

For neurologically impaired children with symptoms consistent with GERD, empiric treatment with acid-suppressing medications, usually a PPI, is appropriate because GERD is common in this population [4]. However, if long-term pharmacotherapy is planned, endoscopy should be performed to evaluate for esophagitis.

Children with autism spectrum disorder who are nonverbal often present with nonspecific, atypical signs and symptoms, including food refusal, chest tapping, pica, chewing on objects, and/or excessive salivation.

b) Esophagitis

- **Treatment strategy** – For patients with endoscopically documented esophagitis, we employ a "step down" approach, in which we treat with a PPI for 8 to 12 weeks, then withdraw treatment as tolerated, or extend treatment if necessary. We suggest using a PPI because this class of medication [is](#) more effective than H2RAs for initial healing of esophagitis [1,2]. Dosing and adverse effects are outlined in the table.\
- **Follow up** – Further evaluation and treatment of patients with esophagitis depends on the severity of the esophagitis at baseline and on the degree of symptom relief with treatment. The histologic findings and differential diagnosis of esophagitis, and grading of erosive esophagitis are reviewed separately.
- Patients with only mild esophagitis (eg, histologic findings consistent with GERD but no grossly visible findings) can be treated for four to eight weeks and reassessed

based on the degree of symptom relief.

- Patients with peptic erosive esophagitis at baseline should undergo a repeat endoscopy after three to six months of treatment; the repeat endoscopy is important to demonstrate healing and also to identify the possibility of stricture.

Distinguishing peptic esophagitis (ie, esophagitis caused by acid reflux) from eosinophilic esophagitis (EoE) may be difficult because both disorders are characterized by esophageal eosinophilia, and they may have similar symptoms. However, in peptic esophagitis the inflammation is usually greatest in the distal esophagus, and in EoE the number of eosinophils often exceeds 15 per high-power field.

Six to eight biopsies from different areas of the esophagus are recommended [5,6]. Esophagitis and symptoms that persist despite several months of effective acid suppression also should raise suspicion for EoE.

When continuous therapy is necessary to control symptoms, we suggest treatment with a PPI; however, H2RAs are a reasonable choice for brief or intermittent treatment [1]. Long-term treatment with PPIs is sometimes necessary but should be undertaken for clear indications and with caution due to potential long-term side effects. If the symptoms are well controlled, a trial of discontinuing treatment should be performed every three to six months, unless the patient has either a history of severe and recurrent esophagitis, or the endoscopy demonstrates Barrett esophagus (replacement of the normal squamous epithelium with metaplastic columnar epithelium). Long-term treatment with acid-suppressing medications is appropriate if symptoms or endoscopy indicate chronic or recurrent esophagitis, without evidence for EoE; such patients should be managed by a pediatric gastroenterologist, if possible.

Dysphagia or odynophagia — For patients with dysphagia (difficulty swallowing) or odynophagia (pain with swallowing), empiric treatment for reflux (lifestyle changes or pharmacotherapy) is **not** recommended unless other causes of the symptoms have been excluded. Instead, these patients should have a careful diagnostic evaluation including upper endoscopy.

Dysphagia may be caused by GERD but also may be due to other causes, such as EoE, pill-induced esophagitis (direct mucosal injury from swallowed medications, such as [tetracycline](#) that remain in the esophagus causing ulceration), esophageal dysmotility (eg, achalasia), or foreign body. Pill esophagitis is also an important cause of odynophagia; other causes include infectious esophagitis caused by cytomegalovirus, herpes, or *Candida*.

Rumination — Children who do not respond to conventional treatment for GERD should be evaluated for rumination. A small study of children with refractory GERD reported that 44 percent met criteria for rumination, but interpretation of these findings is limited because the study used novel high-resolution impedance indices to diagnose rumination rather than the standard antroduodenal manometry. Reflux events in patients with rumination were more likely to occur in the proximal esophagus, were often nonacidic, occurred more often

postprandially, and were highly associated with symptoms [7].

c) Extraesophageal manifestations of GERD Asthma with gastroesophageal reflux

- **With GERD symptoms** – For patients with persistent moderate to severe asthma (particularly, nocturnal wheezing) **and clear symptoms** suggesting GERD, including frequent heartburn or regurgitation, we suggest a trial of effective acid suppression for four to eight weeks [1]. Management after the trial depends on the response.

This empiric approach to treatment is appropriate because although gastroesophageal reflux (GER) is an established trigger for asthma in many patients, it is difficult to predict which patients will respond to treatment [8]. In approximately 60 percent of children with asthma and clinical or laboratory evidence of GER, asthma control will improve with effective treatment for reflux [9]. In adults with GERD symptoms and moderate to severe asthma (or nocturnal asthma), randomized controlled trials of PPIs demonstrated small but significant improvements in numbers of asthma exacerbations.

- **Without GERD symptoms** – A trial of acid-suppressing medication is generally **not** recommended for children or adults with asthma in the absence of GERD symptoms [1]. However, in some patients with asthma, GERD may be silent and diagnosed only with esophageal pH-impedance monitoring (pH-MII). Thus, pH-MII or an empiric trial of antireflux treatment is reasonable in selected patients with asthma who are difficult to control (steroid-dependent) and/or have nocturnal onset of asthma, even in the absence of apparent symptoms of GERD. Other causes of wheezing should be carefully excluded, including foreign body aspiration.

Antireflux surgery (fundoplication) is generally **not** recommended as a treatment for asthma that is thought to be related to GERD. Evidence to support the efficacy, patient selection, and role of antireflux surgery for patients with asthma and GERD is inadequate [10]. Moreover, outcomes following fundoplication in children with asthma have a low rate of success, and as many as 75 percent of children continue to need acid-suppressing medications following surgery [1].

Recurrent pneumonia — When aspiration and recurrent pneumonia are strongly suspected to be related to GERD, aggressive treatment is appropriate to prevent chronic progression of pulmonary disease [1,11]. If pulmonary disease is mild, medical treatment with acid suppression can be used; pulmonary function should be carefully monitored in follow-up.

There are no definitive tests to determine whether GERD is the cause of recurrent pneumonia in a given patient. Clinical decisions must be based on an individual patient's clinical risk and a combination of tests for GER, aspiration, and immune

deficiency. The possibility of antegrade aspiration (aspiration during swallowing) should be fully investigated because this is a more common cause of recurrent pneumonia than retrograde aspiration (GERD). If the pneumonia is consistently in the same region of the lung, the patient should be evaluated for a rare H-type tracheoesophageal fistula. If pulmonary disease is severe or progressive, transpyloric feeding or surgical treatment with fundoplication should be considered, but the success rate of fundoplication for treating pulmonary disease is poor [1]. (See 'Surgery' below.)

Other extraesophageal symptoms — GERD is not a common cause of isolated chronic cough in children, except in those with neurologic abnormalities predisposing to aspiration; however, whether GERD causes cough remains unknown [12]. Stridor, hoarseness, sinusitis, and otitis media have been associated with GERD, mostly in case reports and case series in children. Neither the association with GER nor the benefits of antisecretory therapy have been established [1].

2. MEDICAL TREATMENT

Lifestyle changes — We suggest lifestyle changes as an initial approach to treatment for children and adolescents with mild or infrequent symptoms of gastroesophageal reflux (GER), such as occasional heartburn or painless regurgitation. Lifestyle changes also may be helpful as an adjunct to pharmacologic treatment in patients with moderate or severe symptoms suggestive of GERD, or with documented esophagitis.

The type of lifestyle changes that may be beneficial depend on the patient's age and symptom characteristics. An empiric approach is appropriate since there is little evidence that one of these approaches is more effective than others. Because few studies have examined the efficacy of lifestyle modifications in children and adolescents with GERD [1], recommendations are mostly extrapolated from studies in adults. Even in adults, there is limited evidence supporting efficacy for these interventions.

The following approaches have some benefit in adult patients with reflux, but have not been rigorously evaluated in children [1,13]:

- Weight loss or weight management for individuals who are overweight [1].
- Head of bed elevation and left lateral positioning – This is important for individuals with nocturnal or laryngeal symptoms, but its value for other situations is unclear. A trial of positional therapy is reasonable for children with symptoms of GERD but is not recommended for sleeping infants [1].

The following lifestyle approaches also are used frequently. The efficacy of these measures for managing symptoms is inconsistent, but there is some evidence that they improve laboratory measures of reflux (such as lower esophageal sphincter pressure) [1,13]:

- Dietary modification – Sensitivity to dietary triggers varies among individuals.

Therefore, dietary restriction should be limited to those foods that are associated with symptoms in a specific patient, as determined by trials of individual foods. Options

include:

- Avoid foods that tend to induce reflux by reducing lower esophageal sphincter pressure, including chocolate, peppermint, and caffeinated beverages.
- Avoid acidic foods, including colas, orange juice, and tomato sauce. Although their contribution to gastric acidity is probably minimal, ingestion of these foods seems to exacerbate symptoms in some patients.
- Avoid high-fat foods in selected patients. This strategy is recommended in adults with GERD because fatty foods tend to slow gastric emptying and thereby promote reflux; however, the benefit of reducing fat intake in children is unproven. Decisions about whether to restrict dietary fat depend on the infant or child's overall nutritional status and the clinical response.
- Positioning – Avoid the supine position soon after eating.
- Salivation – Promote salivation by either chewing gum or using oral lozenges.

Salivation neutralizes refluxed acid, thereby increasing the rate of esophageal acid clearance. Alcohol, tobacco, and vaping – Avoid alcohol and tobacco (including passive exposure to tobacco smoke). Substances that contain nicotine reduce lower esophageal sphincter pressure, and smoking also diminishes salivation [14].

Lifestyle modifications also are used to manage GER in infants, including changes in feeding, positioning, and avoidance of tobacco smoke and nicotine exposure. These approaches are discussed in detail in a separate topic review.

Pharmacotherapy — Drugs that are used for treatment of GERD can be grouped into the following categories:

- Proton pump inhibitors (PPIs)
- Histamine type 2 receptor antagonists (H2RAs)
- Antacids
- Surface agents
- Prokinetics

3. SURGERY

Gastrostomy placement — Placement of a percutaneous endoscopic gastrostomy (PEG) is often considered in neurologically impaired children with feeding problems, poor weight

gain, and GERD. The PEG usually should be performed without fundoplication, followed by medical antireflux therapy if clinically indicated [4]. This is because fundoplication increases the risk of complications and does not improve reflux-related outcomes. This was shown in a retrospective multiinstitutional study of infants with neurologic impairment who underwent PEG placement with or without fundoplication, which showed that fundoplication did not reduce hospital admissions for reflux-related problems during one-year follow-up. For the minority (5 to 29 percent) of patients who fail medical therapy, fundoplication can be performed as a second step.

Another alternative to antireflux surgery is transpyloric or jejunal feeding (via nasojejunal or gastrojejunal feeding tubes, or surgical placement of a jejunal feeding ostomy), which reduce reflux by bypassing the stomach [1]. The efficacy of this approach is variable, and its utility is limited by tube dislodgement and other practical considerations [4].

d) Fundoplication

Indications and presurgical evaluation — Antireflux surgery is sometimes indicated in patients with debilitating GERD that is refractory to medical management, after other causes of the symptoms have been excluded [1]. Many candidates for antireflux surgery have underlying neurologic impairment such as cerebral palsy. Surgery should be approached with caution because indications for surgery have not been definitively established, estimates of efficacy and failure rates vary widely, and complications are common, as discussed below.

Candidates for surgery should be carefully evaluated to exclude other causes of the symptoms and signs. The diagnostic tests and surgical considerations depend on the patient's clinical presentation:

- **Intractable esophagitis** – Children with chronic esophagitis unresponsive to acid suppression might be considered for antireflux surgery. They should first undergo a full evaluation to exclude other causes of esophageal inflammation, especially eosinophilic esophagitis (EoE). At a minimum, the evaluation should include upper endoscopy with biopsies and esophageal pH monitoring while on proton pump inhibitor (PPI) therapy, to confirm whether gastric acid was adequately suppressed.
- **Pulmonary disease that is thought to be due to aspiration** – In children with neurologic impairment and suspected aspiration, the possibility of anti-reflux surgery should be approached with caution. This is because it may be difficult to discriminate retrograde aspiration (due to GERD) from antegrade aspiration (due to swallowing dysfunction). Moreover, surgery has low success rates for preventing progression of pulmonary disease, as outlined below. Although the fundoplication can reduce gastroesophageal reflux (GER), it can also impede esophageal clearance and increase the likelihood of aspiration. For this group of patients, alternative approaches to reducing aspiration should be considered, such as transpyloric feeding or gastrostomy.

The evaluation of patients suspected to have pulmonary disease related to GERD often includes review of all radiographic studies, swallowing studies, esophageal pH and impedance monitoring, polysomnography, and bronchoalveolar lavage. The combined results of all these studies should be interpreted by clinicians with experience in swallowing dysfunction because no single test can determine the role that GERD is playing in pulmonary disease.

Relative contraindications — Fundoplication is relatively contraindicated for individuals with:

- Gastrointestinal dysmotility, recurrent vomiting or retching (rather than regurgitation/effortless reflux) – These individuals are at high risk for feeding intolerance and intractable retching after fundoplication.
- Neurologic disability – These individuals tend to have poor outcomes after fundoplication, including retching, failure of the surgery (due to breakdown of the wrap), and other complications that may be life-threatening. Because they have impaired central inhibition, gastrointestinal dysmotility (impaired gastric accommodation) and visceral hypersensitivity may occur after fundoplication, activating the emetic reflex and causing breakdown of the wrap. Percutaneous gastrostomy feedings are an alternative approach to managing chronic GERD and poor feeding in these patients.

Although postoperative retching is particularly common in the above groups of patients, it can also occur in patients without these characteristics.

Technique — The most commonly performed operation is a Nissen fundoplication. This procedure originally involved passage of the gastric fundus behind the esophagus to encircle up to 6 cm of the distal esophagus. However, many variations and modifications have been described, and a Nissen fundoplication may be performed differently by different surgeons. Variables include approach (transthoracic or abdominal); portion of stomach wall used (anterior and posterior or anterior only); combination with other procedures (eg, vagotomy or gastroplasty); as well as the tightness, the length, and the completeness (270° to 360°) of the wrap.

The Nissen fundoplication is now performed laparoscopically at many centers. In experienced centers, short-term complication rates are similar for laparoscopic and the open techniques. Some studies suggest that laparoscopic surgery has fewer complications, such as retching, while others report somewhat higher rates of reoperation and recurrence of GERD compared with open fundoplication.

Efficacy — Symptom improvement has been described in approximately 60 to 90 percent of children undergoing fundoplication [1], but "failure rates" (variously defined) vary widely among reports, ranging from 2 to 50 percent. As an example, in one study, nearly two-thirds of children had ongoing reflux requiring treatment two months following fundoplication; a majority of the subjects had neurodevelopmental delay. In studies that

have reported longer-term outcomes, most demonstrate higher rates of failure within one to three years. Several studies suggest higher failure rates in children with neurologic impairment, chronic respiratory conditions, repaired esophageal atresia, and in infants. In particular, antireflux surgery has low success rates as an intervention for aspiration pneumonia. Similar unsatisfactory outcomes were seen in a report of children with cystic fibrosis and GERD. Nearly one-half of the children had ongoing symptoms of GERD after fundoplication, and overall there was little benefit to pulmonary function. Twelve percent of patients had complications that required a subsequent surgical procedure.

The wide variation in efficacy estimates is likely due to the use of different inclusion criteria, with short follow-up periods and reported variable endpoints; many of these studies focused on neurologically impaired children. These limitations and the absence of controlled trials are critical impediments to defining the role of antireflux surgery in the long-term management of GERD in children.

Experience in adults has demonstrated that patients who do not respond to adequate doses of medical therapy (particularly those with typical features of GERD, such as heartburn) also may have a poor response to surgery, questioning the role of antireflux surgery in patients with symptoms refractory to medical therapy. Furthermore, long-term follow-up of patients who underwent standard antireflux surgery reveals that even after surgery, many continue to depend on medical therapy to control symptoms.

Complications — Complications of antireflux surgery range from 2 to 45 percent. The most common complications included breakdown of the wrap (1 to 13 percent), small bowel obstruction (1 to 11 percent), gas bloat syndrome (2 to 8 percent), infection (1 to 9 percent), atelectasis or pneumonia (4 to 13 percent), perforation (2 to 4 percent), persistent esophageal stricture (1 to 9 percent), and esophageal obstruction (1 to 9 percent). Complications that occurred less frequently included an incisional hernia, gastroparesis, and dumping syndrome. Reoperation was required in 3 to 19 percent of patients, and operative mortality ranged from 0 to 5 percent. The presenting symptoms of dumping syndrome are often irritability after feeds, diaphoresis, or diarrhea; management is described separately.

Retching (dry heaves) is common in children who have undergone fundoplication, particularly children who are neurologically impaired. The presence of preoperative forceful vomiting (versus effortless regurgitation) is a significant predictor of the development of postoperative retching. Treatment for this complication is difficult, and, if the retching is not controlled, it may disrupt the surgical repair. The optimal treatment varies among patients, so a trial-and-error approach is appropriate. Options include modification of feeding content and route (including jejunal feeding), use of antiemetics and motility agents ([ciproheptadine](#), [ondansetron](#), [aprepitant](#), alimemazine), neuromodulation, and other novel therapies (eg, tricyclic antidepressants, gabapentinoids, cannabinoids, acupuncture or vagal nerve/gastric electrical stimulation). Most of these interventions are supported by only low-quality evidence (case series or expert opinion). Symptoms tend to improve after several months.

Patients with neurologic impairment, obesity, or those requiring a second antireflux operation appear to have a somewhat higher frequency of complications. Some surgeons will not perform funduplications on patients with a significantly elevated body mass index. However, children without underlying disorders also may have poor outcomes.

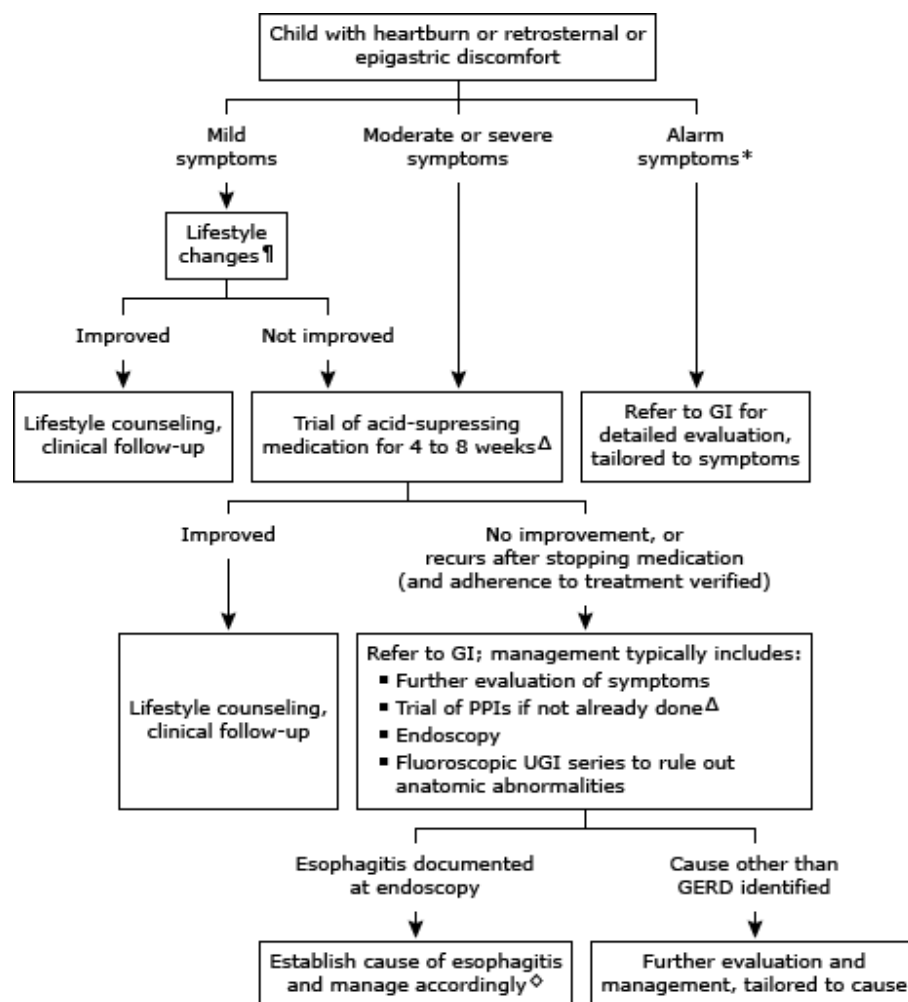
Other surgical interventions — Other techniques such as endoscopic radiofrequency ablation of the lower esophageal sphincter, endoscopic full-thickness fundoplication, and total esophagogastric disconnection have been described, primarily in adults. None of these techniques is recommended at this time for primary therapy of GERD in children [1]. Total esophagogastric disconnection may be appropriate as a "rescue" procedure for selected children with neurologic impairment who have failed fundoplication

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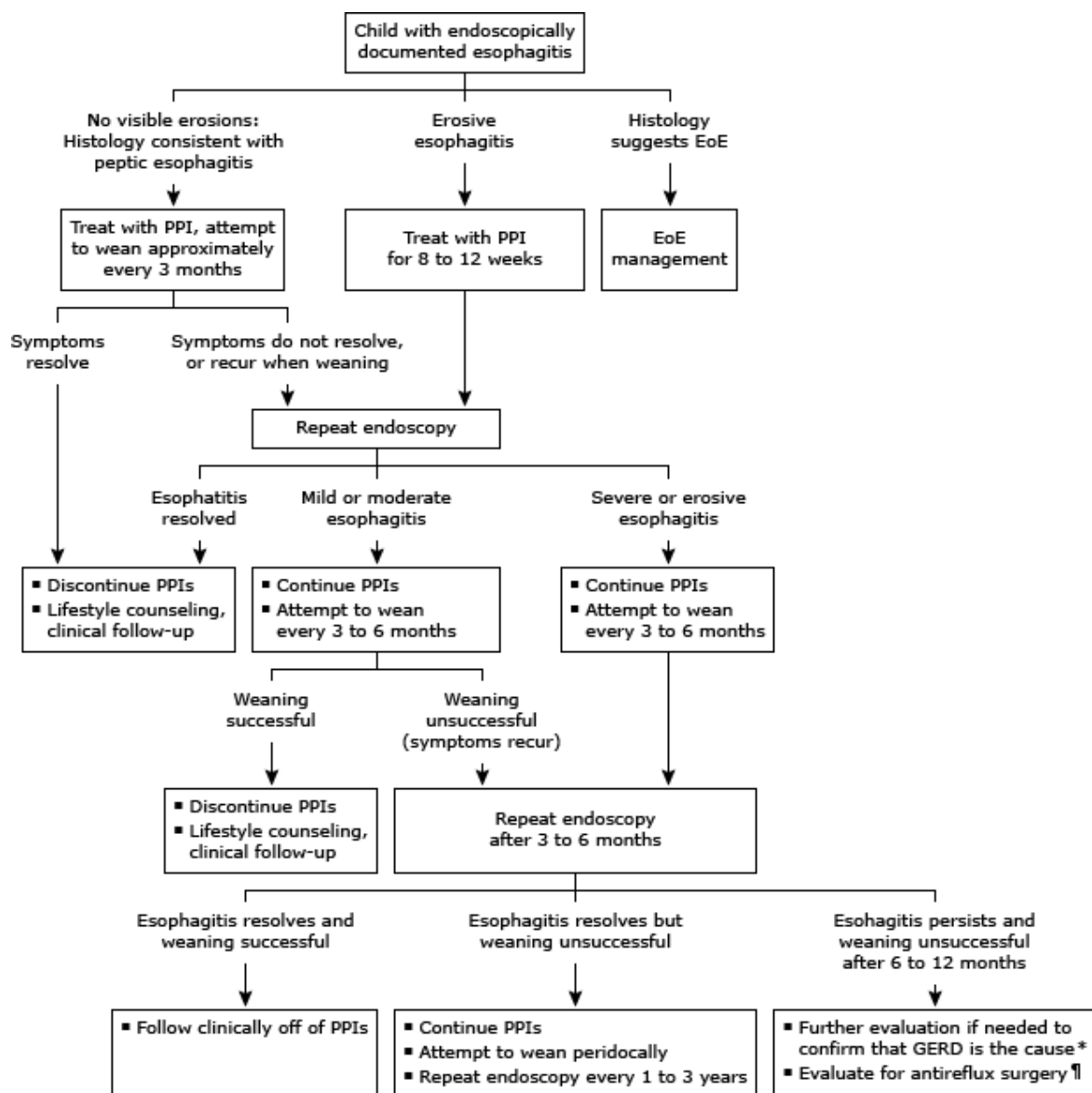
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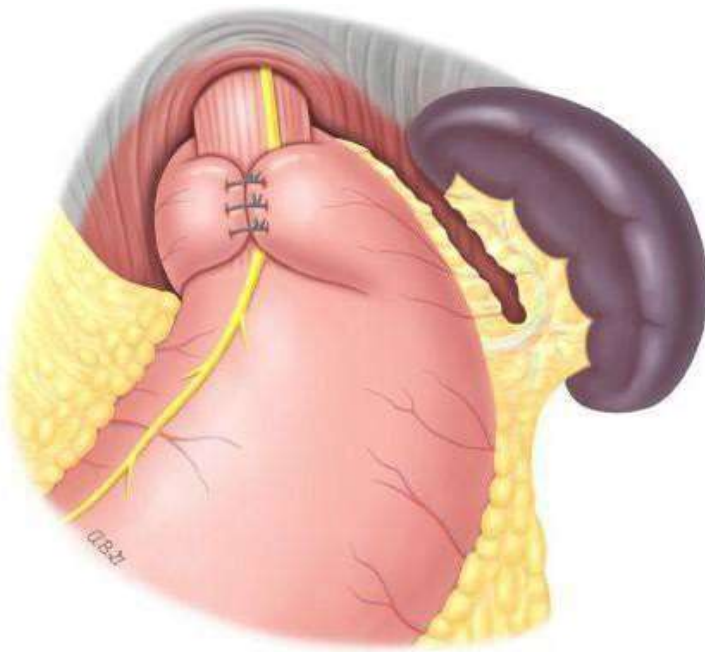
X. MANAGEMENT OF SUSPECTED GASTROESOPHAGEAL REFLUX IN CHILDREN



XI. MANAGEMENT OF ESOPHAGITIS IN CHILDREN



XII. NISSEN FUNDOPLICATION



The Nissen fundoplication provides a 360° "wrap" or collar around the esophagus. Mobilization of the fundus is generally accomplished by dividing the short gastric vessels to the spleen.

HEMORRHOID

MO Satdin

I. DEFINITION

Hemorrhoids are vascular cushions that underlie the distal rectal mucosa and contribute approximately 15–20% of the resting anal pressure, ensuring complete closure of the anal canal. Hemorrhoids became symptomatic when enlarged, inflamed, thrombosed, or prolapsed, and the most common symptom is painless rectal bleeding, which is why hemorrhoids are one of the most frequent causes of severe acute lower gastrointestinal bleeding. In the United States, it is estimated that more than 50% of the population over 50 years of age has experienced hemorrhoid problems. Furthermore, Hippocrates (460 BC) was the first to use the term “hemorrhoid” (from the ancient Greek word “hema,” blood, and “rhoos,” flow) meaning flow of blood.

II. ETIOLOGY

Although the exact pathophysiology of hemorrhoids remains unknown, it is considered that hemorrhoids primarily result from abnormally congested and descended anal cushions. It was assumed that hemorrhoids were a variety of anorectal varices. However, there is clear evidence that hemorrhoids and anorectal varices are distinct entities. Since the development of hemorrhoids are multifactorial, several risk factors have been reported to be associated with the formation of hemorrhoids – such as pregnancy, elderly, constipation, chronic diarrhea, and internal rectal prolapse [16][19]. The concepts of hemorrhoid formation could be categorized into four groups: sliding anal cushions (loss of fixation network), vascular abnormality, rectal redundancy, and an increased pressure on anorectal vascular plexus [15].

The concept of sliding anal cushions or sliding anal canal lining is widely accepted [3]. It develops when the supporting tissues of the anal cushion disintegrate or deteriorate. The shearing force of fecal material, particular hard and bulky stools, causes muscle fibers and connective tissue within anal cushions to tear which is resulting in the downward displacement of anal cushions.

The vascular abnormality and dysregulation of blood supply to, from, and within anal cushions might be associated with the formation of hemorrhoids [15]. Imbalance between vasoconstrictor and vasodilator substances causes the dysregulation of vascular tone. Venous hypertension may be a result of insufficient venous drainage (e.g., increased intraabdominal pressure and pregnancy) and venous reflux. Besides the high arterial blood flow and venous hypertension of anorectal vascular plexus, vascular hyperplasia and neovascularization were evident in patients with hemorrhoids [15].

The rectal redundancy might interfere the proper fixation of supporting tissue within anal cushions to the rectal wall.

Intrarectal pressure is markedly high in patients with functional constipation or those with obstructed defecation disorder [13]. An abnormally high intraabdominal pressure and intrarectal pressure affect the venous drainage of anorectal vascular plexus thus resulting in venous engorgement of anal cushion and the formation of hemorrhoids.

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

The assessment and classification of hemorrhoidal disease will typically take place in an outpatient or ambulatory setting and should be performed by an appropriately qualified clinician subspecializing (i.e., colorectal surgeon), or with an accredited specialist interest, in coloproctology. The most common symptoms described by patients with hemorrhoids are:

- Rectal bleeding
- Perianal pain/discomfort
- Pruritus
- Prolapse

Other symptoms that may be associated with symptomatic hemorrhoids include perianal swelling, soiling, mucus discharge, or tenesmus [14]. Hemorrhoids most typically cause bleeding as their primary symptom. The blood is normally bright red in color (often described as “fresh”), and may appear on the paper while wiping, or can drip into the pan.

Internal hemorrhoids typically cause painless bleeding, unless they prolapse and result in thrombosis. External hemorrhoids can cause pain again if thrombosis occurs and can cause bleeding when ulceration happens as a result of pressure necrosis [14].

The general physical examination, it is important to perform at rectal examination in an outpatient setting by the best performed with the patient in the left lateral decubitus position, with their knees drawn up to their chest. By gently spreading the buttocks, a full inspection of the perianal area can be completed. Digital rectal examination can identify any distal rectal masses, can elicit tenderness in the presence of abscesses or fissures, and can demonstrate mucus discharge or blood. Internal hemorrhoids are not normally palpable unless thrombosed.

ii. Technical procedure

a. Imaging study

Anoscopy or proctoscopy should complete the ambulatory examination when investigating for hemorrhoids. From this point, it may be possible to identify hemorrhoids, fissures, the internal openings of fistula tracts, polyps, and on request

of the Valsalva maneuver, may replicate symptoms of prolapse. 99% of anal lesions were identified on anoscopy whereas 78% and 54% were identified on colonoscopy.

Flexible sigmoidoscopy or colonoscopy should be considered in order to rule out other more serious pathology that may present in a similar way as hemorrhoids. The more suspicious symptoms of altered blood or clots or change in bowel habit would raise the concern for an alternative diagnosis, and such patients should undergo more expensive investigation of the colon, through the means of flexible sigmoidoscopy or colonoscopy.

b. Classification of hemorrhoids

Hemorrhoids may be classified by their relation to the dentate line. These may be defined as of internal, external, or mixed origin. The dentate line is described as lying 2 cm from the anal verge and is the area of demarcation between the upper anal canal (lined with columnar epithelium) and the lower anal canal (lined with sensitive squamous epithelium) [10]. Internal hemorrhoids have been classified in 4 degree of prolapse as suggested by Banov et al. [5] :

- Grade I : Hemorrhoids without prolapse
- Grade II : Hemorrhoids with bleeding and protrusion, with spontaneous reduction
- Grade III : Hemorrhoids with bleeding and protrusion that require manual reduction
- Grade IV : Irreducible hemorrhoids

An alternative staging system from Lunniss et al. (Table1) aims to further differentiate internal hemorrhoids, with an aim to aid decision making in the management of such hemorrhoids.

Kraemer et al. describe the proctological symptom scale, which measures the four main proctological symptoms on visual analogue scales (1–10), these being:

- Pain
- Itching
- Discharge and bleeding

Another severity score that has been suggested is the Sodergren hemorrhoid symptom severity scoring system described by Pucher et al. [19] which again takes into consideration the effect on the quality of patient life. A cutoff score of 5 or greater, should be considered for operative (non ambulatory) treatment. Again, this score requires further assessment, and no assessment of score improvement following successful therapy has yet been shown.

IV. DIFFERENTIAL DIGNOSIS

Hemorrhoids should be differentiated from other diseases that cause anal discomfort and pain with defecation such as rectal cancer, anal fissure, anal abscess, and anal fistula etc.

Diagnosis		Clinical features	Findings
Anal cancer		Pain in the anal area and, in severe instances, loss of weight	Ulcerating lesion of the anus
Anal condylomas		Anal intercourse history anal mass without bleeding	Cauliflower like lesions
Colorectal cancer		Blood in stool, weight loss, changing bowel habits, abdominal pain, family history	Abdominal mass or tenderness
Anal fissure		Tearing pain and bleeding with bowel movement	Examining the anal canal in a fiss is painful
Perianal abscess	Pain onset gradually		In contrast to the rectal mucosa, t mass is covered with sensitive sk
Inflammatory bowel disease		Constitutional signs, abdominal pain	Abdominal mass or tenderness
Rectal polyp		Changing stool color, rectal bleeding, excess mucus, pain, iron deficiency anemia, changing bowel habits (i.e., frequency), abdominal pain	Mushroom-like growth in the ana canal
Skin tags	No bleeding		Around the anus, tags are visuali
Rectal prolapse		Pain, blood and mucus, constipa- tion, incomplete evacuation, difficulty in passing motion, protrusion of the rectum through the anus	Concentrically arranged mucosal folds should easily distinguish complete rectal prolapse

V. THERAPEUTIC APPROACH

Treatment choices can be broadly separated into lifestyle, medical, ambulatory (outpatient), and surgical.

iii. Life style change

Lifestyle measures should be recommended in all patients both as a primary treatment and to reduce the risk of recurrence in those who require further intervention. Patients should be advised to include more fiber in their diet and increase oral fluid intake with the aim to reduce constipation, straining, and poor toileting habits. Advice about toilet habits, such as the use of sitz baths and the avoidance of straining may also be useful, although there is limited scientific evidence to support this.

iv. Medical therapy

Medical therapy may include the use of usually fiber-based laxatives to maintain soft stools and avoid straining. While numerous topical treat- ments are available on the market in the forms of creams, ointments, and suppositories, there is no convincing evidence of their efficacy. These include various combinations of local anesthetics, corticosteroids, vasoconstrictors, antiseptics, and astringents. Phlebotonics (such as oral flavonoids) may have some role in

the control of acute symptoms and may be useful in reducing symptoms after surgical treatment

v. Ambulatory procedures

Nonsurgical treatments include injection sclerotherapy, rubber band ligation (RBL), infrared coagulation, bipolar diathermy, and cryotherapy. These may be carried out in an outpatient setting without the need for anesthetic.

Injection sclerotherapy can be carried out using various different substances (most commonly 5% oily phenol) which are injected into the submucosa at the base of the hemorrhoid (not into the hemorrhoid itself) and cause sclerosis and local thrombosis of the hemorrhoidal vessels.

Rubber band ligation (RBL) can be carried out using either a forceps ligator or disposable suction applicator which allows placement of a band over the base of the hemorrhoidal pedicle. Rubber band ligation appears to be the most effective of these treatments [6] although injection sclerotherapy may still have a role in patients with bleeding tendencies.

Infrared coagulator (IRC) produces infrared radiation which coagulates tissue and evaporizes water in the cell, causing shrinkage of the hemorrhoid mass. A probe is applied to the base of the hemorrhoid through the anoscope and the recommended contact time is between 1.0-1.5s, depending on the intensity and wavelength of the coagulator [21].

Cryotherapy ablates the hemorrhoidal tissue with a freezing cryoprobe. It has been claimed to cause less pain because sensory nerve endings are destroyed at very low temperature.

Radiofrequency ablation (RFA) is a relatively new modality of hemorrhoidal treatment. A ball electrode connected to a radiofrequency generator is placed on the hemorrhoidal tissue and causes the contacting tissue to be coagulated and evaporized [22]. By this method, vascular components of hemorrhoids are reduced and hemorrhoidal mass will be fixed to the underlying tissue by subsequent fibrosis. RFA can be performed on an outpatient basis and *via* an anoscope similar to sclerotherapy.

vi. Surgical procedures

Surgical treatments fall into three main categories: formal excisional surgery, stapled hemorrhoidopexy, and hemorrhoidal artery ligation. Surgical hemorrhoidectomy is considered the gold standard and the most effective and definitive treatment for grades 3 and 4 hemorrhoids. Milligan-Morgan's and Ferguson's procedures are the most widely used hemorrhoidectomy techniques throughout the world.

Excisional surgery involves excising the external component of hemorrhoid along with its vascular pedicle. This is most commonly done using an open (Milligan procedure [1]) or closed (Ferguson procedure [2]) technique where the proximal part of the vascular pedicle is suture ligated.

Milligan Morgan's opened hemorrhoidectomy

In 1937, Milligan and Morgan modified Miles' technique, popularly known as Milligan Morgan's hemorrhoidectomy, which is considered the gold standard. The incisions are so marked that sufficient bridges of intervening anoderm between the excised hemorrhoidal tissue are preserved. Dissection is started from the largest hemorrhoid, or if of similar size, then the one on the left lateral position. A V-shaped incision starting at the distalmost external component is extended proximally in an elliptical manner. The hemorrhoidal tissue from the submucosal plane is raised and separated from the internal sphincter by blunt and sharp dissection, converging at the pedicle. The pedicle is securely ligated at the apex of the hemorrhoidal mass by an absorbable 2-0 polyglactin suture, and finally, the hemorrhoid tissue is excised [23].

Ferguson's Closed Hemorrhoidectomy

Ferguson described this technique in 1959, where excision is continued superiorly up to the anorectal ring. An interrupted fine resorbable stitch was used to close the wound from the anorectal ring to the perianal skin [24], leaving a small margin for draining the collection.

Submucosal Hemorrhoidectomy (Park's Procedure)

This another alternative of Milligan hemorrhoidectomy, developed since 1950 to reduce stenosis and post-operative pain. Because of its complexity and length, considerable blood loss, and danger of incontinence, the submucosal reconstructive hemorrhoidectomy procedure [25] has never been popular.

Whitehead hemorrhoidectomy, removing the entire pile-bearing mucous membrane ring with circular incision. The whole section of dilated hemorrhoidal masses and the surrounding mucosa are excised, and the proximal end is sutured to the skin below [27]. It is almost abandoned right now.

Stapled haemorrhoidectomy according to Longo for treating reducible anal and mucosal prolapse appears to be very simple technically. The device to resect a ring of redundant mucosa from above the hemorrhoidal bundle [7].

Hemorrhoidal artery ligation (HAL), is a relatively new procedure which disrupts the blood supply to the hemorrhoidal cushions by suture ligating the distal hemorrhoidal arterial branches using Doppler guidance. The proctoscope is inserted through the anal canal reaching the low rectum, about 6–7 cm from the anal verge. The surgeon can decide to start the operation at any point of the rectal circumference and proceed in a clockwise or anticlockwise direction. Doppler signal corresponding to all six main trunks of the hemorrhoidal arteries, which are usually located at 1, 3, 5, 7, 9, and 11 o'clock of the lower rectal circumference, is sought by slowly rotating and/or tilting the proctoscope. Mucopexy is carried out with a continuous suture, including the redundant and prolapsing mucosa and submucosa, in a proximal-to-distal direction, along a longitudinal axis.

Modern hemorrhoidectomy techniques are a safe and fast technique that fulfills the requirements of low complication rate, fast wound healing and quick return to work. It will be reduced in postoperative pain, and hospitalization. The LigaSureTM vessel sealing system and Harmonic Scalpel have been recently introduced as an instrument conceived to upgrade the conventional treatment of hemorrhoids. It reduces anal spasm and allows to perform a bloodless hemorrhoidectomy with reduced postoperative pain and fast healing. [8][26][28]

Laser hemorrhoidoplasty (LHP) has an upper edge since it is hemostatic, less painful, bactericidal, leads to faster healing, is associated with lesser complications, and maintains the physiology of the anal canal by preserving the anal cushions [29]. Lasers can be used in two ways to treat hemorrhoids: First, by excising the hemorrhoidal mass, as with the CO₂ laser, and second, by delivering the laser energy into the hemorrhoidal mass and causing fibrosis by initiating protein denaturation. [30][31]

vii. Monitoring

a. Preoperative setting

Prophylactic antibiotic by using of metronidazole and ceftriaxone, reduce rates of surgical site infection (SSI) after hemorrhoidectomy have been reported to be as low as 1.4% [18].

Preemptive analgesia is provided to minimize postoperative pain by preventing sensitization at the central nervous system level before surgery commences. The aim is to prevent postoperative hyperesthesia from developing [13].

There is ample evidence that for the vast majority of patients, there is no need for general anesthesia and endotracheal intubation for hemorrhoid procedures. These surgeries can be safely done with total sedation (propofol) and local anesthetic [15]. Neuraxial techniques of regional anesthesia, such as spinal and epidural anesthesia, are well recognized as advantageous for lower abdominal, pelvic, perineal, and lower extremity surgeries.

A pudendal nerve block is a very simple, quick, and effective technique that should be employed by every surgeon that performs anorectal surgery. This easy to perform peripheral nerve block has been shown to significantly decrease postoperative pain, decrease the intake of analgesics, and decrease pain during the first bowel movement. To perform the block, the 24G needle is introduced on both right and left lateral perianal areas at a location halfway between the external sphincter and the ischial spine.

b. Postoperative setting

The goal after hemorrhoid surgery is to provide adequate pain relief while allowing the patient to be as comfortable as possible, all while keeping side effects to a minimum. It is clear that a multimodal analgesia approach is the best option for pain management of the surgical patient and even more important in the ambulatory setting. Nonsteroid anti-inflammatory drugs (NSAIDs) are commonly used, which are an effective adjunct in post-hemorrhoidectomy pain control, and well tolerated and inexpensive. The use

of gabapentin in post-hemorrhoidectomy pain control was able to demonstrate an improvement in postoperative pain on days 1, 7, and 14 in the patients who received.

Topical anesthetics have been used to aid in pain control after hemorrhoid surgery. Sucralfate has angiogenic, antibacterial, and wound healing properties and other compounds like cholestyramine 15% or even vitamin E have also been proven advantageous in decreasing postoperative pain after hemorrhoidectomy [20].

Local injection of lidocaine and bupivacaine decreases postoperative pain, but unfortunately these effects are short lived. A formulation of liposomal bupivacaine (Exparel®, Pacira Pharmaceuticals, Parsippany, NJ) is now available, which can provide analgesia for up to 72 h when injected locally.

Multiple studies have sought to prove the efficacy of oral metronidazole as an adjunct in the pain management of the patient who has undergone a hemorrhoidectomy. One such study compared the administration of metronidazole 500 mg every 8 h against placebo administration over 7 days following surgery.

The bowel regimens has been prevented of constipation, straining, and fecal impaction is of paramount importance care of these patients. This becomes especially important in the case of patients receiving opioids. Usually, a postoperative regimen should include stool softeners around the clock and stronger “rescue” laxatives as needed.

c. Follow up

The follow-up includes four different time; At the first visit, 7–10 days after the procedure, a digital anorectal examination is never carried out, but only an external inspection to avoid the risk of pulling on the stitches. Usually, bleeding is no longer present.

The second follow-up visit is made after 1 month. The patient’s anorectum is digitally explored and assessed. Rectal pain, discomfort, and tenesmus should no longer be present.

At the 3month follow-up visit, the patient is also evaluated with anoscopy. At that time, when the procedure is successful, all symptoms are resolved. A long-term annual follow-up may be established. If any symptom related to a possible recurrence of hemorrhoidal disease is reported, the patient undergoes digital examination and anoscopy.

VI. COMPLICATIONS

ii. Complications of hemorrhoid

Bleeding through rectum is one of the most familiar complaints of anorectal patients requiring medical or surgical intervention.

Continuous pain which is not associated with defecation is indicative of anorectal abscess or thrombosed hemorrhoids. Pain associated with defecation is suggestive

of anal fissure. Deep-seated throbbing pain with difficulty in sitting indicates a perianal abscess.

Mucus discharge from the anal area is one of the complaints of anorectal diseases.

Prolapse hemorrhoid is a common complication developed during non-providing any treatment.

iii. Complications of hemorrhoid surgery

A stricture is thought to occur when the surgeon excises too much hemorrhoidal and anal canal tissue. With healing, scar formation further reduces the anal canal aperture and results in a symptomatic stricture. Randomized control trials have reported strictures in up to 5% of patients having conventional or stapled hemorrhoidectomy and 2% with energy device hemorrhoidectomy.

Urinary retention is a well-known complication of hemorrhoidectomy and tends to occur more frequently in males, particularly in the elderly and very young.

Fecal incontinence caused by damaging the anal sphincter muscles.

Ectropion can occur rarely after hemorrhoidectomy. Symptoms of pruritus, soilage, and mucosal weeping, with a physical examination showing rectal mucosa outside the anal canal is the usual presentation.

Recurrent symptoms can occur following all treatment options for hemorrhoids.

VII. ALGORITHM

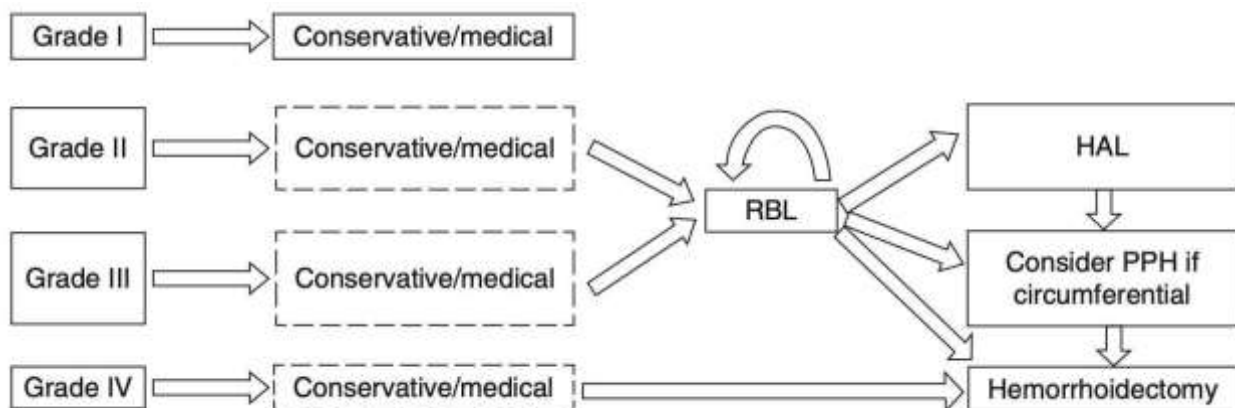
Table 5. Internal hemorrhoid staging and management [12]

Stage	Morphology	Principal presentation	Possible additional features	Visual confirmation related to size	Presentations	Common conventional management
<i>Nonprolapsing</i>						
0	Anal cushions	Bleed very rarely No prolapse	None	No definite increase	Present from birth	Dietary/ lifestyle advice; medications; consider sclerotherapy or IRC for nonresponders
1	Small hemorrhoids	Bleed intermittently No prolapse	None	Minor definite increase seen on proctoscopy	Adolescence onwards, but usually before middle age	Sclerotherapy/ IRC Consider banding or stapling for nonresponders
<i>Prolapsing</i>						
2	Intermediate hemorrhoids	Prolapse but return spontaneously Bleed frequently and sometimes profusely	Pruritus Skin tags rare	Moderate increased size of individual pile masses which prolapse on straining	Adults (30+ years)	Banding/ stapling Consider surgery for nonresponders
3	Large hemorrhoids	Prolapse, and need aid to reduce Bleed frequently and often profusely	Pruritus Discomfort Some skin tags common	Major increase in size, including circumferentially Prolapse easily and require replacement	Middle age onwards (40+ years)	Operation by usual techniques
4	Very large hemorrhoids +/- additional features (inter-external hemorrhoids)	Prolapse which is permanent and irreducible Bleed profusely which can soil underwear	Pain Pruritus Soiling Many skin tags usual Complications, e.g., Thrombosis	Extreme increase in size with hemorrhoids also in secondary positions and skin tags	Middle age and older	Operation, frequently by modified technique

Table 6. Sodergren hemorrhoid symptom severity scoring system [19]

Symptom		Points scored
How severe are your symptoms of itching or irritation? (circle number from 1 to 5)	0	No symptoms
	1	Mild/do not really bother me
	2	
	3	Moderately bothersome
	4	
	5	Severe
How severe are your symptoms of pain or discomfort at rest? (circle number from 1 to 5)	0	No symptoms
	1	Mild/do not really bother me
	2	
	3	Moderately bothersome
	4	
	5	Severe
How severe are your symptoms of pain or discomfort on opening your bowels? (circle number from 1 to 5)	0	No symptoms
	1	Mild/do not really bother me
	2	
	3	Moderately bothersome
	4	
	5	Severe
How often to you feel that you might have a lump at your anus (prolapse)?	0	Never
	1	Less than once a month
	2	More than once a month
	3	More than once a week
	4	Every day
		Final score (0–14)

Algorithm 2. Management of hemorrhoids by grade. (RBL rubber band ligation, HAL hemorrhoidal artery ligation, PPH procedure for prolapsing hemorrhoids)



Algorithm 3. Current management of hemorrhoid by grade

Treatments	Grade I	Grade II	Grade III	Grade IV	Acute thrombosis or strangulation
Dietary and lifestyle modification	×	×	×	×	×
Medical treatment	×	×	×		
Non-operative treatment					
Sclerotherapy	×	×			
Infrared coagulation	×	×			
Radiofrequency ablation	×	×			
Rubber band ligation	×	×	×		
Operative treatment					
Plication		×	×		
DGHAL		×	×		
Hemorrhoidectomy		×	×	×	×
Stapled hemorrhoidopexy			×	×	

Algorithm 4. Analyses result post-hemorrhoidecto

Comparison of Outcomes Between Different Surgical Procedures for Treatment of Hemorrhoids					
Procedure	Resolution of symptoms	Reduction of prolapsing tissue (mucopexy)	Likelihood of recurrence	Amount of post-surgical pain	Longer recovery time
Banding (i.e., rubber band ligation)	++	+	++	++	+
Infrared photocoagulation	+	Not applicable	+++	+	+
Open hemorrhoidectomy	+++	++	+	+++	+++
Closed hemorrhoidectomy	+++	++	+	+++	+++
Stapled hemorrhoidopexy	++	+++	++	++	++
Hemorrhoidal artery ligation (without mucopexy)	++	Not applicable	++	+	+
Hemorrhoidal artery ligation (with mucopexy)	++	++	++	++	+
+ = Outcome less likely. ++ = Outcome relatively neutral in comparison with other surgical procedures. +++ = Outcome more likely. Information from references 7, 21, and 23 through 28.					

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MANAGEMENT OF HEMOTHORAX

Tuo Sothunea, Duong Chhay, Lay Nhork

I. CASE DEFINITION

Hemothorax has been consistently defined as a collection of blood in the pleural space or a pleural fluid hematocrit of greater than 50%. A hematocrit value of pleural fluid below 50% does not necessarily rule out hemothorax. It has been reported that a hemothorax may become diluted with pleural fluid in as little as 3-4 days, mimicking a hemorrhagic effusion with a hematocrit between 25-50%.

II. ETIOLOGY

The source of blood may be the chest wall, intercostal vasculature, internal mammary arteries, great vessels, mediastinum, myocardium, lung parenchyma, diaphragm, or abdomen. Although some authors have stated that a hematocrit value of at least 50% is necessary to differentiate a hemothorax from a bloody pleural effusion, most have not agreed on any specific distinction.

There are three major etiologic categories of hemothorax: spontaneous, iatrogenic, and traumatic. Thoracic trauma is the most common cause.

A. Spontaneous hemothorax

The most common cause of spontaneous hemothorax is spontaneous pneumothorax. Besides this, there are four more categories of spontaneous hemothorax such as coagulopathic, vascular, neoplastic, and miscellaneous.

- Coagulopathic etiologies: congenital diseases (hemophilia or thrombocytopenia) and anticoagulation treatment.
- Vascular etiologies: rupture or dissection of descending aorta or genetic diseases (Ehlers-Danlos Syndrome and Rendu-Osler-Weber Syndrome).
- Neoplastic etiologies: this can be result of spontaneous rupture of a tumor or direct invasion of vasculature in or near the pleural cavity.

B. Iatrogenic hemothorax

Iatrogenic hemothorax occurs during surgery or procedures when the lung parenchyma, surrounding vessels, or organs are inadvertently damaged.

Thoracentesis, central venous catheterization (especially subclavian), tube thoracostomy, and thoracotomy may all result in iatrogenic hemothorax.

C. Traumatic hemothorax

Traumatic hemothorax can be due to either blunt or penetrating injury. Hemothorax, hemopneumothorax, and pneumothorax are the most common complications of either penetrating or blunt thoracic trauma.

Blunt thoracic trauma typically occurs far more frequently, mortality is significantly higher in penetrating chest trauma. The mortality of penetrating chest trauma can also depend upon the means of injury.

III. DIAGNOSTIC PROCEDURE

D. Clinical Argument

Clinical findings of hemothorax are broad and may overlap with pneumothorax. These include respiratory distress, tachypnea, decreased or absent breath sounds, dullness to percussion, chest wall asymmetry, tracheal deviation, hypoxia, narrow pulse pressure, and hypotension. There are other physical findings that should be performed in case of chest trauma such as:

- Distended neck veins: pericardial tamponade, tension pneumothorax, cardiogenic failure, air embolism
- "Seat belt sign": deceleration or vascular injury; chest wall contusion/abrasion
- Paradoxical chest wall movement: flail chest
- Facial/neck swelling or cyanosis: superior mediastinum injury with occlusion or compression of superior vena cava (SVC)
- Subcutaneous emphysema: torn bronchus or lung parenchyma laceration
- Scaphoid abdomen: diaphragmatic injury with herniation of abdominal content into the chest
- Excessive abdominal movement with breathing: chest wall injury

E. Technical Procedure

a. Imaging Study

Imaging plays a very critical role in the management of traumatic injuries. First described by Rozycki et al., the Focused Assessment with Sonography in Trauma (FAST) exam evaluates the pericardium, perihepatic, perisplenic, and pelvis for pathologic fluid or air collection.

i. Chest radiography

Chest radiography has traditionally been the initial tool in the emergency room for rapid evaluation of thoracic injury. Upright chest radiography is preferred, because in the supine position the blood will be distributed along the entire posterior aspect of the affected pleural space rather than the diaphragmatic surface.

In upright radiography, *blunting of the costophrenic angle* is the most common sign of hemothorax and pleural effusions. This technique has well known limitations. 300–500 mL of blood is required to blunt the costophrenic angle. In case which the patient is in the supine position, the blood up to 1000 mL can be overlooked. Since most severely injured patients arrive in the supine position, and upright chest radiography cannot be safely performed, the amount of blood in the chest is often *underestimated, and occasionally missed*

completely. Upright chest radiography cannot be used to accurately quantify the volume of hemothorax.

ii. Ultrasound and FAST/eFAST

The Focused Assessment with Sonography in Trauma (FAST) is a bedside test developed in the mid 1990's for use in acute trauma patients to rapidly assess for intra-abdominal hemorrhage and to rule out clinically significant pericardial tamponade. The Extended Focused Assessment with Sonography in Trauma (eFAST) adds additional views of the hemithoraces to look for signs of pneumothorax and hemothorax.

These include the right and left pleural spaces (anterior axillary line between 6th and 9th intercostal spaces) and left and right anterior pleural spaces (midclavicular line between 2nd and 3rd intercostal spaces). Ultrasound can quantify volume, detecting 100 ml of pleural fluid with 100% accuracy, and as little as 20 ml. In case of hemothorax, ultrasound has an overall sensitivity of 67% and specificity of 99%.

Additional benefits to the use of ultrasound include the ability to be used in a pre-hospital setting, portability, reproducibility, non-invasiveness, and employs no contrast or radiation.

There are some limitations of ultrasound. The sensitivity of ultrasound is influenced by the experience of the operator and the frequency of the transducer. While ultrasound has a higher sensitivity than chest radiography for detecting hemothoraces, *it is not as sensitive as computed tomography*.

iii. Computed tomography

Chest computed tomography is commonly used to detect additional injuries following chest radiograph and bedside sonography.

The patient may initially be studied using chest radiography and possibly ultrasound to quickly evaluate the extent of injuries, especially in cases of *hemodynamic instability*. When *the patient becomes stable*, a chest CT, preferably *with intravenous contrast*, can be performed to evaluate further pathology in detail. In cases of hemothorax, CT can reveal a collection of blood missed by the chest radiograph.

IV. DIFFERENTIAL DIAGNOSIS

- Chylothorax
- Pleural effusion
- Pneumothorax
- Hydrothorax
- Cardiac tamponade

V. THERAPEUTICAL APPROACH

The management of hemothorax depends on situation of the hemothorax. This can be acute hemothorax which needs an early management (pleural drainage or thoracotomy) and delayed hemothorax which needs to be diagnosed properly. The delayed hemothorax is defined as a hemothorax that occurs 24h after a negative initial imaging.

There are different options of management of hemothorax. The conservative management (non-interventional) is indicated in case which the patient is stable and the hemothorax is less than 300mL. Patients with hemothorax volumes less than 260–300cc were 72%–92% likely to resolve without significant complication and with no intervention. During the conservative management, the vital signs, especially the respiratory status, need to be observed. The painkillers, oxygen and physiotherapy will be prescribed as supportive treatment.

Some other interventions can be performed to treat the hemothorax.

D. Needle thoracocentesis

Needle thoracocentesis (or thoracentesis) is used to evacuate the hemothorax and to identify the nature of pleural effusion (infection or malignancy). This technique is still a common treatment of pleural effusion, but it is considered obsolete in management of hemothorax.

Equipment required for pneumothorax aspiration (needle thoracocentesis):

- Sterile field
- Sterile dressing pack and gloves
- Antiseptic solution
- Analgesia
 - 4mls of 1% or 2% Lidocaine
 - Orange (25G) needle (x1)
 - Green (19G) needle (x1)
 - 5ml Syringe (x1)
 - 16–18-gauge cannula
 - 3-way tap
 - 50 ml syringe
 - Ultrasound machine

Pre-procedure:

- Gain consent from the patient – ideally consent should be written
- Consent for pain, failure of procedure, bleeding, infection, damage to surrounding structures
- Familiarise yourself with the landmarks
- 2nd intercostal space in the midclavicular line, midaxillary line (supine position) or posterior mid-scapular line by inserting above rib to avoid neurovascular bundle on posterior aspect of second rib.
- Mark area if necessary
- Set up a sterile trolley with equipment

Procedure of needle thoracocentesis:

- Don sterile gloves and clean area with Chlorhexidine/Betadine.
- Apply sterile field.
- Bedside ultrasound can be performed to identify the location of the hemothorax.
- Insert Lidocaine 5-10ml initially under the skin and then into subcutaneous tissues and pleural space.
- During insertion of catheter, apply negative pressure until a loss of resistance is felt and a steady flow of fluid is obtained.
- Remove the needle.
- Attach three-way tap to cannula.
- Connect the cannula to the drainage bag.
- Apply dressing.
- Send the liquid for cyto-bacteriologic analysis.

Post-procedure care

- Analgesia
- Chest X-ray
- Physiotherapy

E. Tube thoracostomy or chest drainage

Tube thoracostomy is the primary intervention of hemothorax. According to international practice management guidelines, all hemothoraces regardless of size should be considered for tube thoracostomy.

If treated early with adequate drainage, the need for additional surgery for retained or recurrent hemothoraces should be relatively low. Typically, bloody effusions that remain after tube thoracostomy are reabsorbed in 4–5 weeks without complication.

The drainage with large bore tubes (>24Fr) is more recommended because it can evacuate better quantify blood loss and it is less obstructed by blood clot. But it depends on availability and surgeon preference.

After the drainage, the tube should be connected to suction with at least 20 to 40 cm of water negative pressure. Recent studies suggest that initially placing the chest tube to suction rather than water seal may be associated with improved outcome regarding duration of chest tube therapy, length of hospital stay, persistent air leak, and risk of empyema.

Drainage from the hemithorax should in general be less than 100 ml in 24 hours before removal of the chest tube. A repeat chest radiograph should be performed prior and after removal to ensure complete lung re-expansion with no residual or recurrent pneumothorax.

Equipment required for an intercostal drain (chest drain/pleural drain):

- Sterile ultrasound sheath
- Sterile field
- Sterile dressing pack and gloves
- Antiseptic solution
- Analgesia
 - 4mls of 1% or 2% Lidocaine.
 - Orange (25G) needle (x1)
 - Green (19G) needle (x1)
 - 5ml Syringe (x1)
 - Chest tube
 - Drainage suction system
 - Sterile water/saline
 - Suture kit
 - Sterile dressing

Pre-procedure set-up:

- Set up sterile trolley
- Prepare drain
- Pour sterile water/saline into chest drain bottle up to the prime line
- Attach chest drain tubing ensuring the end stays within the package and sterile
- Review imaging and examine patient to confirm side of insertion
- It is advised to have a nurse and a helper to assist during the procedure
- Position the patient with leaning forward with arms outstretched or sat at 90 degrees with arm lifted and hand resting behind their head. In elderly or frail patients, the nurse may be required to help support this position.

Procedure for tube insertion

- Wash hands and don sterile gown and gloves
- Clean insertion site: either the site identified by ultrasound or – for pneumothorax – insert drain in the “safe triangle”
- Lower border of axilla to the 5th intercostal space; the lateral edge of pectoralis major and the lateral edge of latissimus dorsi
- Apply sterile field
- Inject local anesthetic into the skin, subcutaneous tissue, and periosteum of the rib below the insertion site.
- Fluid or air should be able to be aspirated with the green needle
- Skin incision for a size of the tube, over the rib below the chosen intercostal space.
- Blunt dissection by using hemostats to bluntly dissect through the subcutaneous tissue and intercostal muscles, reaching the pleura.
- Remember the pathway which have been created.
- Insertion of the Joly tube by holding the external tip with dominant hand. With non-dominant hand, holding 4-5cm from the internal tip to prevent accidental intrathoracic injury while pushing the tube.
- Apply steady pressure to advance the trocar and tube into the pleural space.
- Remove the trocar, leaving the chest tube in place.
- Ensure that the tube is positioned correctly (all side holes should be inside the pleural cavity)
- Connecting drain to chest drain tubing.
- Fixation of the tube to the skin and dressing

Post-procedure

- Analgesia
- Post procedure chest x-ray
- Document procedure clearly and document length of drain inserted
- Advise patient to always hold drain bottle below level of insertion
- Respiratory review and advise on onward management

F. Video-Assisted Thoracoscopy Surgery (VATS)

This technique is indicated in stable patient. If the patient remains hemodynamically unstable, it means that the patient is still actively bleeding

The timing of VATS is also very important when evacuating a hemothorax. Within 48–72 hours, VATS is usually successful for hemothorax evacuation and allows for early re-expansion of the lung with reduction in short and long-term morbidity and

mortality. Performing VATS on the sixth post-trauma day or later is associated with a 15.8% possibility of conversion to an open thoracotomy related to dense adhesion and pleural thickening.

The VATS provides better visualization, and a higher yield of detecting small injuries with less post-operative pain, decreased hospital length of stay, and avoidance of excess bleeding compared to thoracotomy. But this technique needs experience surgeon and anesthesiologist and some special instruments.

G. Thoracotomy

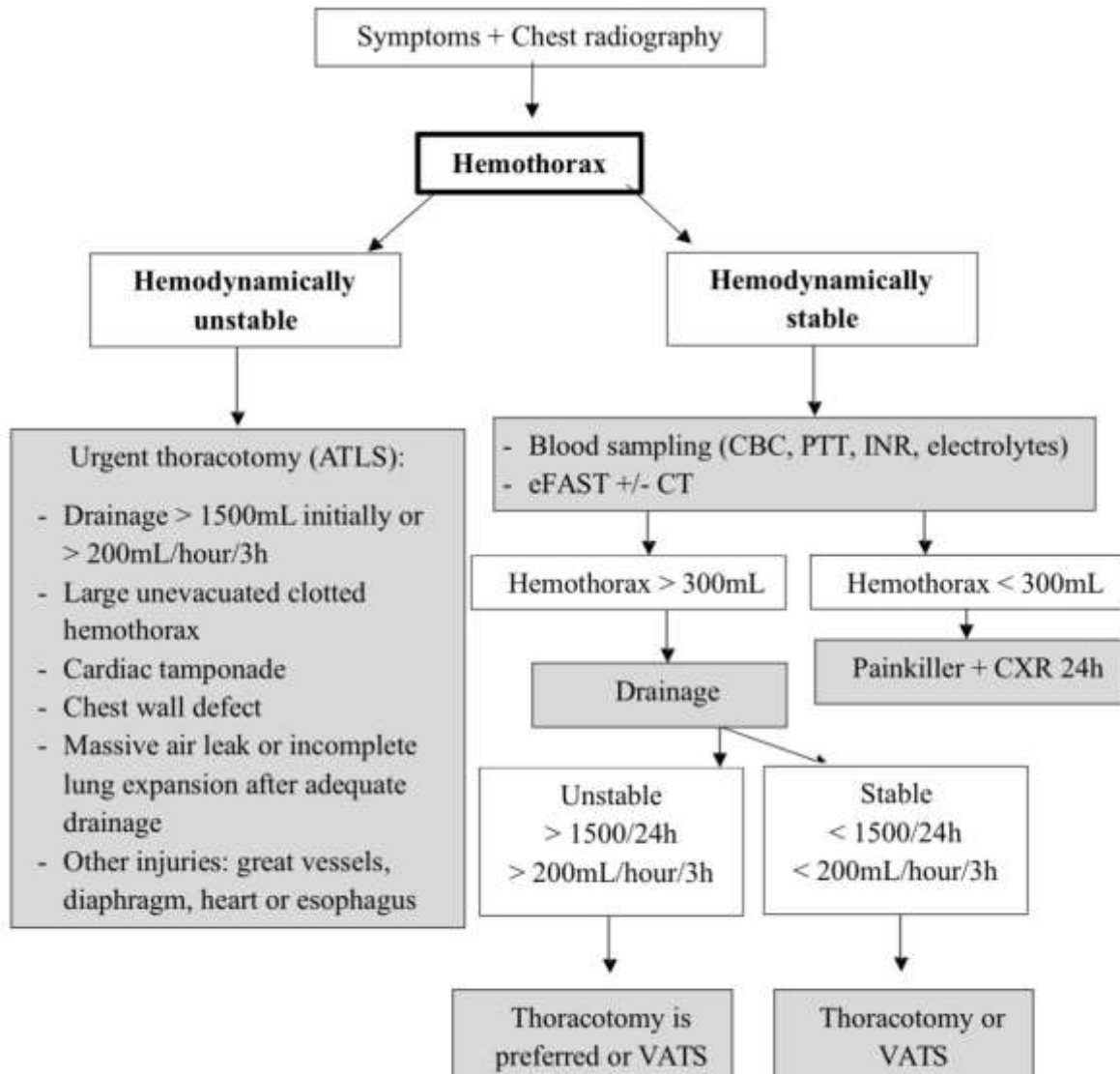
If the patient is actively bleeding into the thoracic cavity and remains hemodynamically unstable despite initial attempts at resuscitation, thoracotomy is the procedure of choice.

This can be lateral or anterior thoracotomy. The indications of immediate thoracotomy depend on the output from chest drainage:

- 1000–1500 mL total in 24 hours *in stable patient* or more than 1500 mL of blood immediately after chest tube in *unstable patient*.
- Or 200–500 mL/hour for 2 to 3 consecutive hours. (>200 mL/hour for 3 hours)
- Specific internal injuries including esophageal injury, diaphragmatic injury, cardiac injury, massive hemothorax and great vessel injury.

Therefore, it has been recommended that emergent thoracotomy not be performed in patients with no signs of life upon arrival to the hospital.

VI. ALGORITHM



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INGUINAL HERNIA

Sokhean Manaty, Preap Ley

I. CASE DEFINITION

A protrusion, bulge, or projection of an organ or a part of an organ through the body wall that normally contains it. Collectively, inguinal and femoral hernias are known as groin hernias.

a) Inguinal hernia:

- Direct inguinal hernia: The hernia passes directly through the muscles through the area of weakness of the fascia transversalis medial to the inferior epigastric vessels. It is always an acquired hernia.
- Indirect inguinal hernia: The hernia passes through the deep inguinal opening outside the inferior epigastric vessels and is found in contact with the spermatic cord or the round ligament in women. It is also acquired in adults, but can be congenital in children

b) Crural/Femoral hernia:

- The hernia passes through the area of weakness of the fascia transversalis. More common in women, they are almost always located inside the external iliac vessels.

II. ETIOLOGY

- Increased intra-abdominal pressure, recorded in obesity, chronic cough, ascites, pregnancy, heavy lifting and straining due to constipation or dysuria, is also thought to play a role in the development of inguinal hernia.
- Studies have also shown that certain diseases such as chronic obstructive pulmonary disease (COPD), Ehlers-Danlos syndrome and Marfan syndrome contribute to an increased incidence of inguinal hernia.
- There is evidence to suggest that genetics also play a role. Patients with known family history of hernia are at least 4 times more likely to have an inguinal hernia than patients with unknown family history.

III. DIAGNOSTIC PROCEDURE

g. Clinical argument

- a. Clinicals: Inguinal swelling or mass, impulsive and expansive with exertion and coughing, reducible to the finger traveling its route to the collar, most often asymptomatic in the absence of complications. Sometimes, moderate local pain, particularly with exercise, or a feeling of heaviness.
- b. Physical examination:
 - i. Inspection: Inguinal swelling
 - Above the Malgaigne line: Inguinal hernia

- Below the Malgaigne line: Crural hernia
- ii. Palpation: Palpation in both the supine and standing positions. A Valsalva maneuver or cough can also help in the detection of a hernia. Usually there are 3 tests needed for inguinal hernia:
 - Finger invagination test
 - Deep ring occlusion test
 - Ziemen test (Three finger test)
- ** Malgaigne line: the line between the anterior superior iliac spine and the pubis.
- h. Technical procedure: The diagnosis is therefore clinicals → no additional examination.
- i. Laboratory test: CBC, Crea, Ionogram, PT and according to his/her comorbidities.
- ii. Imaging study: Chest x-ray. In case of doubt about an occlusion, may need abdominal x-ray or U/S or CT.

IV. DIFFERENTIAL DIAGNOSIS

Hydroceles and Groin lymph nodes can be differentiated from inguinal hernias.

V. THERAPEUTIC APPROACH

- i. Stabilize patient
 - All hernias in adults should be repaired unless the patient's condition would result in an unacceptably high mortality. Support with trusses or surgical belts may be appropriate conservative management in the very high-risk patient.
 - Irreducible or sliding inguinal hernia may be due to adhesions or its volume.
 - Strangulated hernia is an urgent case to prepare for surgery:
 - Need placement of a venous line, rehydration, fluid-electrolyte rebalancing, analgesics.
 - Keep in NPO or need NGT in case of vomiting.
- j. Empirical treatment
 - In case of strangulated hernia and need urgent surgery, it must start antibiotics as soon as possible and continue postoperatively according to hernia organs condition.
 - Most elective cases do not need antibiotics. But they need a shot of prophylaxis for hernia repair with mesh only.

k. Directed treatment

- Surgical hernia repair or Herniorrhaphy is only treatment option for inguinal hernia. There are Open surgical repair (Ex: Bassini, McVay, Shoudice, Liechtenstein) and Laparoscopic surgical repair (Ex: TEP and TAPP). Surgical repair goal 3 steps are:
 - Find hernia sac and reduction of herniated healthy contents back into abdominal cavity
 - Excision of hernia sac at its neck (Herniotomy)
 - Closure of the inguinal canal with various surgical techniques.
- Gentle and progressive manual reduction can be attempted in patients with hernial encroachment (non-reducible but painless hernia and without signs of occlusion). Manual reduction is absolute contraindicated in cases of local or general signs of inflammation which should lead to surgical exploration.
- Example of two Basic surgical procedures:

Bassini's procedure

- Under general anesthesia with endotracheal intubation, the patient was placed in the supine position and the entire abdomen was prepared with povidone-iodine. It was draped in the usual manner.
- A transverse skin incision in RLQ just 2cm beneath Bikini line about 5 cm long was made, and it was carried down to the subcutaneous tissue.
- All bleeders were controlled with Bovie electrocoagulator.
- The external oblique aponeurosis was incised along its fibers.
- The incised external oblique aponeurosis was retracted downward and upward for the good operative field. The ilioinguinal and iliohypogastric nerves were identified and preserved.
- The cremasteric muscle fibers were divided and blunt dissection was done in order to identify to the spermatic cord.
- After the identification of the spermatic cord, the index finger was introduced beneath the cord from medial side just above the pubic tubercle in order to assist in the blunt dissection and feeling of the cord from the underlying inguinal introduction. And then the umbilical tape was drawn through the cord for traction.
- The hernial sac was identified on the anteromedial aspect of the cord structure and hernial sac itself was seen as a definitive white membrane and wall of sac was lifted up and opened, (If there were any hernia organs, they must be check out for its healthy and then put it back into abdominal cavity).
- And then the sac was divided into proximal and distal part, with index finger in the proximal hernial sac, the vas deference and accompanying vessels were dissected freely by sharp and blunt dissection to the preperitoneal fat line.
- Distal part of hernial sac was removed by blunt or sharp dissection.
- Transfixing suture with 3/0 Vicryl on the base of sac was performed doubly.

- The posterior wall was reinforced by approximation of Paupart's ligament and conjoined tendon with interrupted suture using by 0/0 Prolene.
- The operative field was reexamined and complete hemostasis was confirmed.
- The operative wound was closed in layers; using running 0/0 Vicryl sutures for the external oblique aponeurosis, and using 3/0 Vicryl suture for the subcutaneous tissue, using 3/0 Monocryl for subcuticular suture.
- After the dressing, the patient was delivered to the recovery room in the good general condition.

Lichtenstein's

- Under general anesthesia with endotracheal intubation, the patient was placed in the supine position and the entire abdomen was prepared with povidone-iodine. It was draped in the usual manner.
- A transverse skin incision in RLQ just 2cm beneath Bikini line about 5 cm long was made, and it was carried down to the subcutaneous tissue.
- All bleeders were controlled with Bovie electrocoagulator.
- The external oblique aponeurosis was incised along its fibers.
- The incised external oblique aponeurosis was retracted downward and upward for the good operative field.
- The Ilioinguinal and iliohypogastric nerves were identified and preserved,
- The cremasteric muscle fibers were divided and bluntly dissected with gauze in order to identify to the spermatic cord.
- After the identification of the spermatic cord, the index finger was introduced beneath the cord from medial side just above the pubic tubercle in order to assist in the blunt dissection and feeling of the cord from the underlying inguinal introduction. And then the umbilical tape was drawn through the cord for traction.
- The hernial sac was identified on the anteromedial aspect of the cord structure and hernial sac itself was seen as definitive white membrane and wall of sac was lifted up and opened. (If there were any hernia organs, they must be check out for its healthy and then put it back into abdominal cavity). And then the sac was divided into proximal and distal part. With index finger in the proximal hernial sac, the vas deference and accompanying vessels were dissected freely by sharp and blunt dissection to the preperitoneal fat line.
- Distal part of hernial sac was removed.
- Transfixing suture with 3/0 Vicryl on the base of sac was performed doubly.
- A Prolene mesh was laid on space between external oblique aponeurosis and transverse abdominis muscle and transfixing sutures were done with Prolene 2/0 to few sites of mesh edge.
- The operative field was reexamined and complete hemostasis was confirmed.

- The operative wound was closed in layers; using running 0/0 Vicryl sutures for the external oblique aponeurosis, and using 3/0 Vicryl suture for the subcutaneous tissue, using 3/0 Monocryl for subcuticular suture.
- After the dressing, the patient was delivered to the recovery room in the good general condition.

1. Monitoring

i. Clinical monitoring

- ☐ Patient should stay in bed for a few hours, if he/she was under SA and start on free diet and normal mobilization. Painkillers are requested according to wound pain scale.
- ☐ Patient not allow to lift up the weight more than 5Kg in the first 2 months postoperatively.

ii. Laboratory monitoring

Not specific, but according to his/her co-morbidities.

VI. COMPLICATION

a) Intraoperative complication:

- ☐ Allergy to anesthesia drugs
- ☐ Hemorrhage due to damage to any the epigastric or femoral vessels)
- ☐ Harmful to vas deferens or testicular vessels
- ☐ Nerves damage

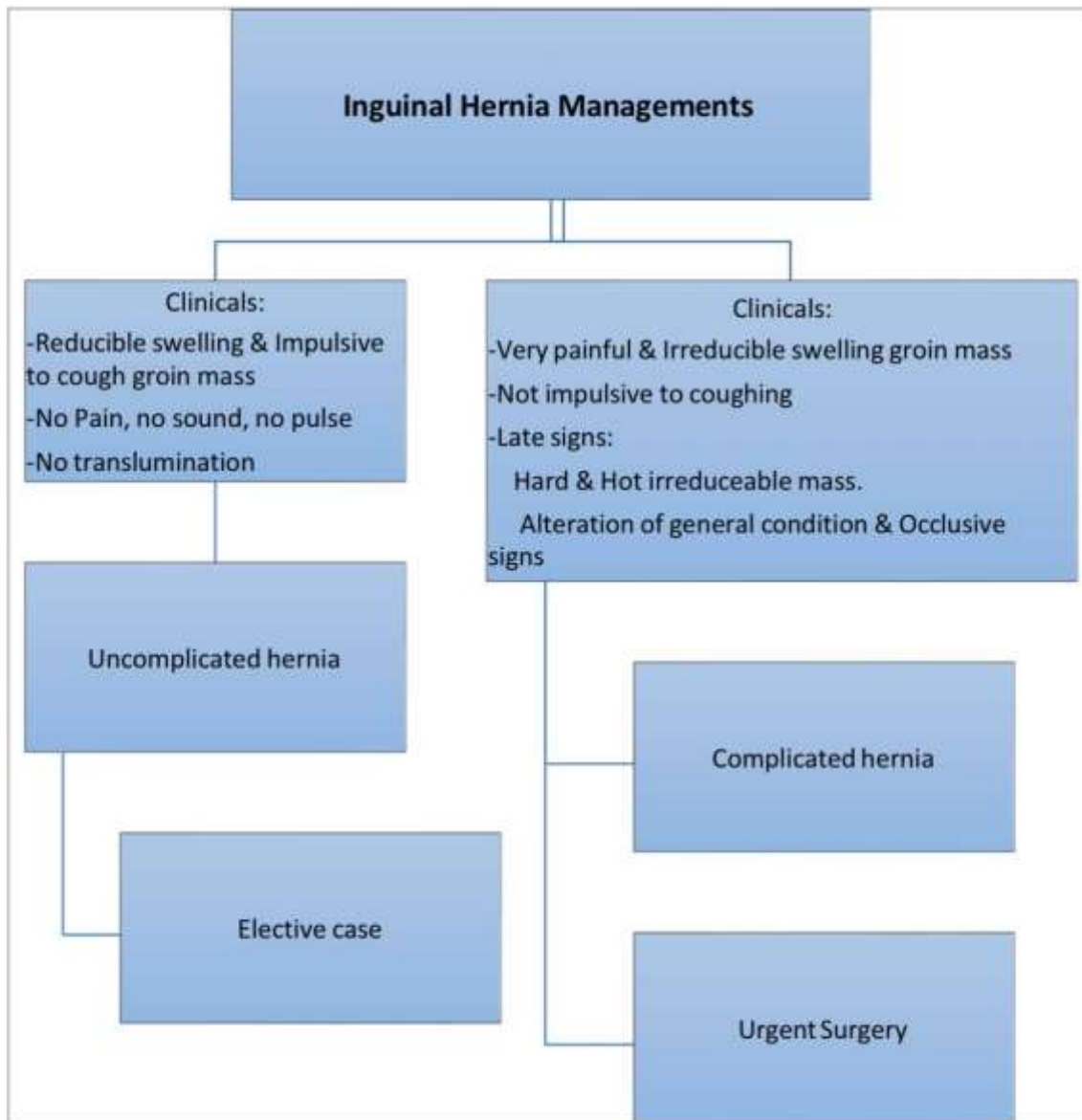
b) Early post-operative complications:

- ☐ Hematoma: Most common complication. It exposes the risk of secondary infection and must be evacuated in the operating room, if it is large.
- ☐ Infection: Rare but very serious for prosthesis implantation case. It requires open wound and drainage. Need wound care until secondary closure with appropriate antibiotics or have to remove implants.
- ☐ Seroma or hydrocele: Collection of citrine color occurring during the first post-operative days. If not infected, no treatment is necessary and the seroma resolves spontaneously. Puncture and aspiration of the seroma helps relieve pain but exposes to the risk of infection.

c) Late post-operative complications

- ☐ Residual pain: 6% of patients (nerve damage or muscle tension suture)
- ☐ Hydrocele, testicular atrophy (< 1% after the first treatment but 4% in case of repeat)
- ☐ Recurrence of the hernia: Less than 5% regardless of the technique used (lower after use of a prosthesis). They are often early (50% during 5 years).

VII. ALGORITHM



VIII. REFERENCES

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INTESTINAL OBSTRUCTION

Chhay Raksmeay

I. CASE DEFINITION:

Intestinal obstruction is significant **mechanical impairment or complete arrest** of the passage of contents through the intestine due to pathology that causes blockage of the bowel. **Adhesions are becoming the more common cause** of intestinal obstruction than hernia nowadays.

Common causes for adhesions are gynaecologic surgeries, **appendicectomy, colonic and pelvic surgeries**. **Adhesions and hernias** are most common causes of intestinal obstruction.

Depending on degree of block intestinal obstruction can be **complete, partial or partial circumferential**.

Depending on the vascular compromise it can be – **simple obstruction** with intact blood supply or **strangulated with compromised blood supply**.

II. ETIOLOGY

The etiology of intestinal obstruction are two types, dynamic or mechanic and adynamic.

Dynamic type is **mechanical blockage of normal propulsion** and passage of intestinal contents.

Obstruction may be **external/internal, partial (incomplete or subacute)/complete, acute/chronic or chronic, simple/closed loop/strangulation, congenital/acquired; proximal/distal**.

Adhesions commonly cause small bowel obstruction than large bowel.

Classification depends on mechanism of intestinal obstruction

	Dynamic		Adynamic
<i>Outside the wall</i>	<i>In the wall</i>	<i>In the lumen</i>	
Hernia—25%	Tuberculous stricture	Gallstones	Cessation of peristalsis
Adhesions—40%	Crohn's disease	Roundworm	Postoperative period
Volvulus	Malignancy	Inspissated faeces	Electrolyte imbalance
Intussusception		Meconium ileus	Spinal injuries
			Uraemia
			Diabetes mellitus
			Retroperitoneal—haematomas and surgeries
			Renal surgeries
			Mesenteric ischaemia
			Pseudo-obstruction

Classification depends on site of intestinal obstruction

	<i>Proximal small bowel</i>	<i>Distal small bowel</i>	<i>Large bowel</i>
Clinical features:	Severe vomiting, dehydration, no or less distension, colicky pain	Central distension, vomiting, dehydration Central abdominal pain	Constipation, distension— early; Late vomiting less pain
Special features:			
Plain X-ray	Valvulae conniventes	Characterless Central fluid level	Dilatation and haustration

Depend on site of intestinal obstruction

	<i>Proximal small bowel</i>	<i>Distal small bowel</i>	<i>Large bowel</i>
Site of obstruction	Duodenum and jejunum	Ileum	Anywhere in large intestine
Causes:	Congenital Lipomas Leiomyomas Malignancy Bands and adhesions	Tuberculosis strictures Malignancy Crohn's Gallstones Hernias—common cause Roundworm Congenital	Malignancy Tuberculosis stricture Anorectal malformation Volvulus Congenital megacolon Bands

Causes of Intestinal Obstruction

Location	Cause
Colon	<u>Tumors</u> (usually in left colon), <u>diverticulitis</u> (usually in sigmoid), volvulus of sigmoid or cecum, fecal impaction <u>Hirschsprung disease</u> , <u>Crohn disease</u>
Duodenum	
• Adults	Cancer of the duodenum or <u>cancer of the head of pancreas</u> , <u>ulcer disease</u>
• Neonates	Atresia, <u>volvulus</u> , bands, <u>annular pancreas</u>
Jejunum and ileum	
• Adults	<u>Hernias</u> , adhesions (common), tumors, <u>Meckel diverticulum</u> , Crohn disease (uncommon), <u>Ascaris infestation</u> , midgut volvulus, intussusception by tumor (rare), <u>foreign body</u> , <u>gallstones</u> (rare)
• Neonates	<u>Meconium ileus</u> , volvulus of a malrotated gut, atresia, <u>intussusception</u>

III. IAGNOSTIC PROCEDURE 1-

Clinical arguments

a. Abdominal pain:

Colicky and intermittent later continuous and severe Pain is the first symptom to develop which is sudden and severe. Initial colicky pain suggests obstruction and eventual diffuse persistent pain suggests strangulation.

Pain begins usually around umbilicus in small bowel obstruction. In small bowel obstruction, it is crampy, recurrent paroxysms occurring as short crescendo/decrecendo episodes (of 30 seconds).

In large bowel obstruction, it is of longer episodes of minutes (In paralytic/adynamic ileus, pain is diffuse and mild).

b. Vomiting:

- In jejunal obstruction, it is early and persistent.
- In ileal obstruction, it is recurrent occurring at an interval; initially bilious later faeculent.
- In large bowel obstruction, vomiting is a late feature.

c. Distension:

- It is absent or minimal in case of jejunal obstruction
- Obvious with visible intestinal peristalsis (VIP) and borborygmi sounds in case of ileal obstruction—Step ladder peristalsis.
- It is enormous in case of large bowel obstruction.

d. Absence gas and stool:

- It is absolute, i.e. neither faeces nor flatus is passed.

e. Dehydration:

- Leads to oliguria → renal failure.
- Features of toxaemia and septicaemia (tachycardia, tachypnoea, fever, sunken eyes, cold periphery)

f. Abdominal tenderness:

- It is initially localised but later becomes diffuse—is a feature of intestinal obstruction. Rebound tenderness and guarding will not be present in simple obstructions which are features of strangulation.

g. Features of strangulation:

- Continuous severe pain, shock, tenderness, rebound tenderness (Blumberg's sign).
- Guarding and rigidity, absence of bowel sounds.
- In case of strangulated hernia, a swelling which is tense, tender, rigid, irreducible, no expansile impulse on coughing and history of recent increase in size is seen.

h. Temperature:

- Fever signifies inflammation in the bowel wall/ ischaemia/perforation.
- Hypothermia can occur when septicaemia develops due to lack of pyrogenic response. It suggests poor prognosis.

i. Bowel sounds:

- They are increased—high-pitched metallic (rushes and groans) sounds followed by metallic tinkling sounds of dilated bowel. Eventually once fatigue occurs or gangrene develops, bowel sounds are not heard—silent abdomen of peritonitis

develops (in paralytic ileus, there are only continuous metallic sounds of dilated bowel).

j. Per-rectal examination:

Shows empty, dilated rectum, often with tenderness. If rectal growth is the cause for obstruction, it may be palpable

2- Imaging study

2.1. Plain X-ray abdomen:

- + Initially supine abdominal X-ray is taken, later if needed, X-ray in erect posture is taken if perforation is suspected)
 - Multiple air-fluid levels.
 - Proximal the obstruction → Lesser the air fluid level. Distal the obstruction → More the air fluid level.



Fig. 23.10: Plain X-ray showing valvulae conniventes with intestinal obstruction.



Fig. 23.8: Plain X-ray abdomen in erect posture showing dilated bowel and colon—a feature of intestinal obstruction.

- + Normally, three fluid levels can be seen in plain X-ray film—at fundus of stomach, at duodenum and often at caecum.
 - Jejunum shows concertina effect due to **valvulae conniventes** (Herring bone pattern)—by the valves of Kerckring.
 - Ileum is smooth and characterless (by Wangenstein).
 - **Large bowel shows haustration.**
 - **Pneumobilia** (gas in biliary tree) maybe due to gall-stone ileus.
 - Distended caecum is shown as round gas shadow in the right iliac fossa.
 - Dilated caecum signifies large bowel obstruction.
 - Small bowel >3 cm diameter; proximal large bowel >9 cm; transverse colon

>5.5 cm; sigmoid colon >5 cm are suggestive of intestinal dilatation. But this increased diameter need not suggest intestinal obstruction every time.

2.2 CT scan:

- + Small bowel loop >2.5 cm suggests dilatation. It can show dilated loop, transition zone and collapsed part which are definitive features of intestinal obstruction. It can also give idea of changes in the bowel wall, ischaemia, strangulation, mesenteric oedema and thickening. It also shows bowel wall gas, portal venous gas and mass lesion.



Fig. 23.11: CT abdomen showing dilated bowel loop; CT is ideal investigation for intestinal obstruction.

2.3. Barium (micro bar solution) enema or gastrografin contrast enema X-ray:

- + Barium meal is usually contraindicated in acute intestinal obstruction. However dilute (micro bar) barium meal/gastrografin meal follow through X-ray may be done with caution in suspected subacute/partial intestinal obstruction under fluoroscopy, otherwise it may precipitate complete obstruction or may cause perforation and barium peritonitis which is very dangerous.

2.4. US abdomen:

- + Useful to see dilated bowel and fluid in the peritoneal cavity.
- + It is better than X-ray but not as good as CT scan. It has got 95% sensitivity; 80% specificity; 80% accuracy.
- + Doppler US is useful in detecting strangulation.

3- Laboratory study

- a. **Haematocrit, blood urea and serum creatinine; arterial blood gas analysis** (acidosis is common), **LFT, platelet count** (in severe sepsis, there will be **altered LFT with thrombocytopenia**).
- b. **Serum electrolytes** estimation. Hypokalaemia is common.
- c. **Total white blood count is increased. But can be significantly low in severe stage of sepsis.**
- d. Estimation of **serum D lactate, CPK-BB isoenzyme, intestinal fatty acid binding proteins** are different investigations may be useful to predict bowel ischaemia/gangrene.

IV. DIFFERENTIAL DIAGNOSTIC

- 1- Paralytic obstruction
- 2- Pseudo obstruction
- 3- Ascites

V. THERAPEUTIC APPROACH FOR INTESTINAL OBSTRUCTION

1) Stabilized patient

- a. Nasogastric aspiration: to reduce toxic effects, to reduce bowel distension which indirectly improves pulmonary ventilation and to reduce possibility of aspiration pneumonia
- b. IV route for replacement of fluid and electrolytes
- c. Urinary catheter for monitoring urine and kidney function
- d. Oxygen

2) Empirical treatment

- a. Analgesic or antispasmodique for strangulate occlusion
- b. Antibiotics: Ampicillin, gentamicin, metronidazole, cephalosporins
- c. Blood transfusion: FFP or platelet transfusions are often needed in critical patient.
- d. ICU critical care: Systemic management of complications like ARDS, DIC, SIRS are important. If there is hypotension, dopamine/dobutamine are also needed.
- e. CVP for fluid and monitoring: PCWP (pulmonary capillary wedge pressure) monitoring are often needed in haemodynamically unstable patient.

3) Surgical treatment

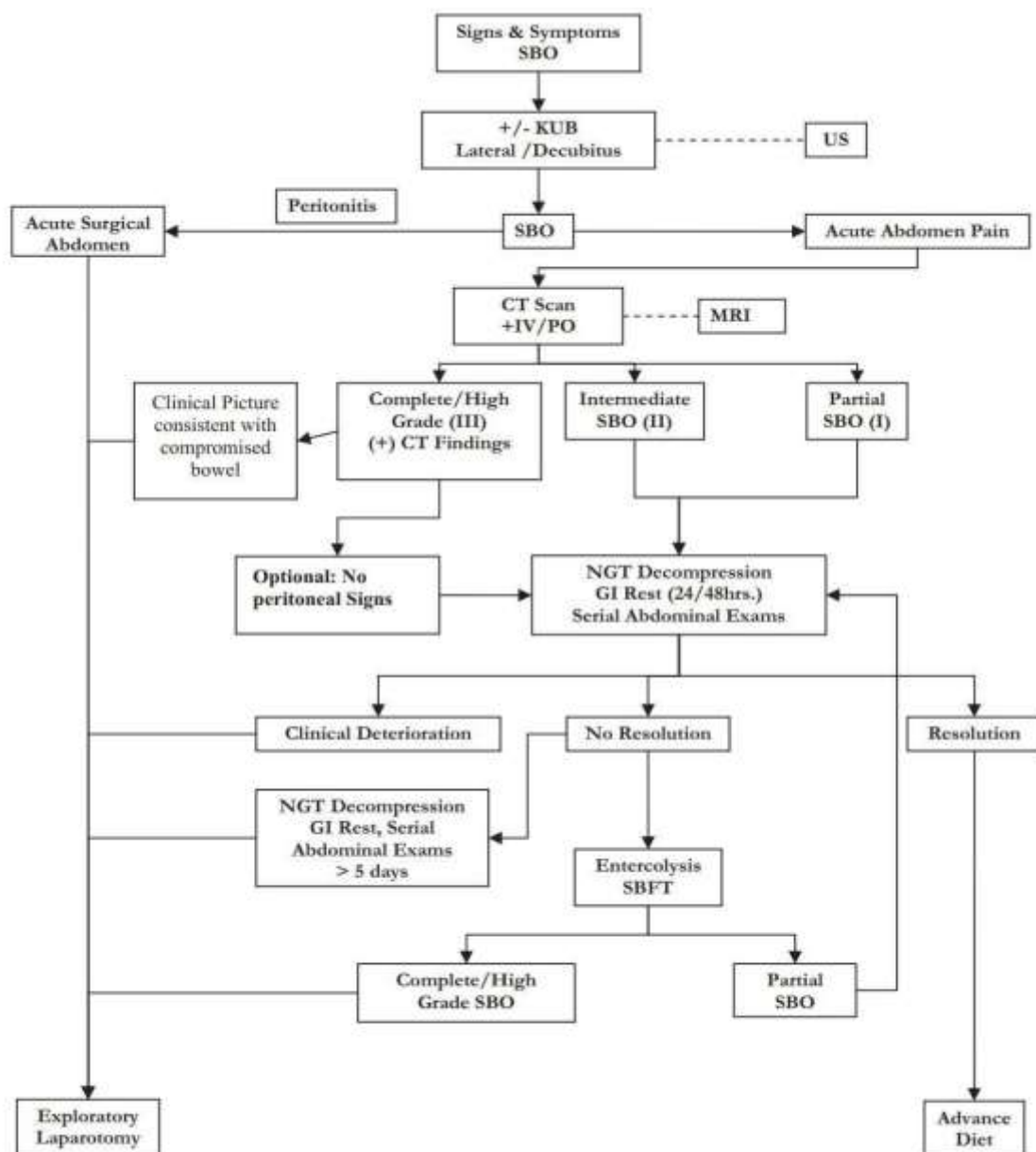
Goal of surgical treatment is to reestablish the continuity of intestinal route to guarantee the intestine can pass the content of intestine till terminal.

It depends on etiology, mechanism and severity of pathology.

It can be laparotomy or laparoscopic, depend on surgeon and instruments disponible.

- a. For intestinal obstruction by adhesion, procedure can be adhesiolysis, resection of intestine, primary anastomosis or stoma
- b. For right colon tumor, it can be right hemicolectomy and anastomosis or ileostomy depend on severity
- c. For left colon tumor, it can be left colectomy and frequently Hartmann procedure
- d. For small bowel obstruction, it can be resection some segment and primary anastomosis or stoma
- e. For colon volvulus, it can be resection and colostomy or pexy

VI. ALGORITHM



1. Clinical flow diagram for SBO.

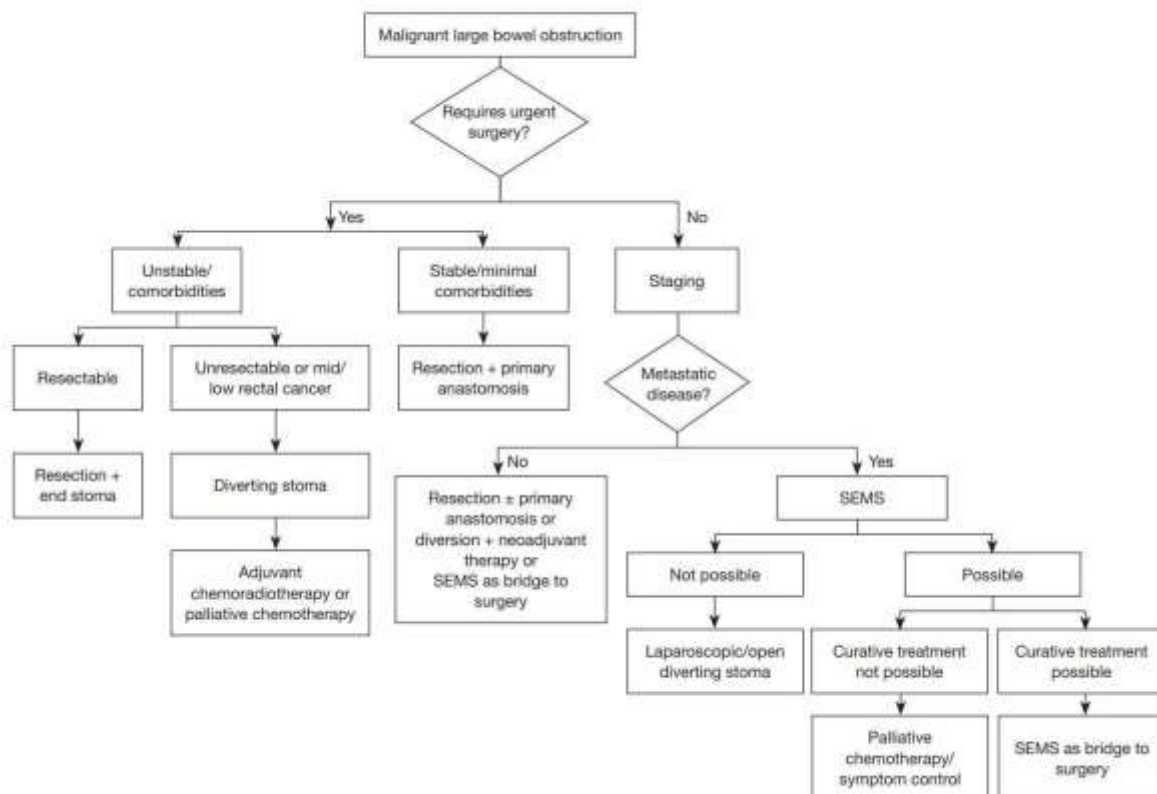


Figure 1 Suggested decision-making tree for malignant LBO (26). SEMS, self-expanding metal stent; LBO, large bowel obstruction.

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LIVER ABSCESS

Dara Vithiea, San corine, Meas Vutha

I. CASE DEFINITION:

A liver abscess is defined as a pus-filled mass in the liver that can develop from injury to the liver or an intraabdominal infection disseminated from the portal circulation.[1] The majority of these abscesses are categorized into pyogenic or amoebic, although a minority is caused by parasites and fungi. Most amoebic infections are caused by *Entamoeba histolytica*. *The pyogenic abscesses are usually polymicrobial, but some organisms are seen more commonly in them, such as E.coli, Klebsiella, Streptococcus, Staphylococcus, and anaerobes.* While the incidence is low, it is essential to understand the severity of these abscesses because of the high mortality risk in untreated patients.[2]

The usual pattern of abscess formation is that there is leakage from the bowel in the abdomen that travels to the liver through the portal vein. Many cases have an infected biliary tract that causes an abscess via direct contact.

Liver abscesses can be classified in a variety of ways: One is by location in the liver. 50% of solitary liver abscesses occur in the right lobe of the liver (a more significant part with more blood supply), less commonly in the left liver lobe or caudate lobe. Another method is by considering the source: If the cause is infectious, the majority of liver abscesses can be classified into bacterial (including amebic) and parasitic sources (including hydatiform cyst).

II. ETIOLOGY:

Appendicitis used to be the main reason why people develop a liver abscess, but that has decreased to less than 10% since better diagnosis and management of the disease has become available. Nowadays, biliary tract disease (biliary stone, strictures, malignancy, and congenital anomalies) are the major causes of pyogenic liver abscesses. (Figure 1)

Half of the bacterial cases are developed by cholangitis. Less often, causes are hepatic artery bacteremia, portal vein bacteremia, diverticulitis, cholecystitis, or penetrating trauma.[3] Some may be from a cryptogenic origin. The most common organisms include *E.coli*, *Klebsiella*, *Streptococcus*, *Staphylococcus*, and anaerobic organisms but are generally polymicrobial. If strep or staph is isolated solely, the focus should be on finding another source of infection (endocarditis) that has hematogenously spread to the liver. *Klebsiella pneumoniae* is a prominent etiology in Southeast Asia and is thought to be related to or associated with colorectal cancer there as well.[4][5] It is usually present with no hepatobiliary system disease and is exclusively monomicrobial. It happens in a background of diabetes and is more severe than other forms of bacterial abscesses, possibly because of enhanced virulence factors of the bacteria.[6]

If the source is an anaerobe, the most common organism is *Entamoeba histolytica*. It affects the liver by first causing amebic colitis, then seeding the portal system and

migrating to the liver and causing an amebic liver abscess. Although rare in the United States, it can still be found in immigrants or travelers from other countries.

Another rare but important parasitic organism is *Echinococcus granulosus*, which causes a hydatid cyst of the liver. Infection is caused by the metacestode stage of the tapeworm *Echinococcus* which is part of the Taeniidae family. Patients typically present with symptoms that include abdominal pain, diarrhea, and hepatomegaly. Hydatid cysts are acquired from dogs and can take a while to cause symptoms in the host. Most cases are discovered in the late stages and incidentally.[7]

III. DIAGNOSTIC PROCEDURE

1. Clinical argument

It is vital to obtain a thorough history and carry out a detailed physical examination before choosing any diagnostic measures. This includes but is not limited to, gathering the patient's personal history, occupation, travel, place of origin, recent infections, or treatments. Certain risk factors promote the development of liver abscesses, such as diabetes, cirrhosis, male gender, elderly, immunocompromised states, and people with proton pump inhibitor usage.

After gathering a history, the review of systems and physical examination can provide a lot of additional information. On analysis of systems, patients may complain of the following symptoms: fever, chills, night sweats, malaise, nausea or vomiting, right shoulder pain (due to phrenic nerve irritation), right upper quadrant pain, cough, dyspnea, anorexia, or recent unexplained weight loss. Fever is present in 90% and abdominal pain in about 50-75% of patients.[17] Dark urine is present, much like it is present in other forms of hepatitis.

On physical examination, a patient can have hepatomegaly with an enlarged mass and jaundice. Although Charcot triad (right upper quadrant pain, jaundice, and fever) is a sign of cholangitis, the physician will need to consider liver abscess as a differential.[18] A small number of patients with hepatic abscesses may present in distress or even overt shock (septic shock or anaphylactic shock in the case of the hydatiform mole rupture).

In the case of *Klebsiella* liver abscesses, in addition to the symptoms mentioned above, it also sends septic emboli to the eye, meninges, and brain. Therefore symptoms of these organ systems can be present, and it can last after the liver abscess is drained.

In the case of *Echinococcus* infection, there is an initial asymptomatic phase in a child. Years later, some of these patients will show clinical symptoms from reactivation of the infection.[19] The clinical manifestations depend on the type, size, and site of the cysts present. Small cysts in non-vital organs can go undetected, but large ones in critical locations can present with signs of compression or rupture.[20] The usual rate of cyst progression is 1 to 5 centimeters in a year. The liver is affected in two-thirds of cases of *Echinococcus* infection. Symptoms of compression usually start when the diameter is 10 cm and include biliary colic, cholangitis, obstructive jaundice, portal and venous obstruction, Budd-Chiari syndrome, bronchial fistula, If it ruptures, overt peritonitis or anaphylaxis will be present.

2. Technical procedure

a. Laboratory test

After the history and physical examination, the next step is to obtain laboratory and diagnostic evidence to determine the cause of the patient's chief complaint and confirm or rule out a liver abscess. Laboratory tests include a complete blood count with differential, tests for hepatocellular injury (liver enzymes that are usually elevated in half the cases), liver synthetic function tests (Prealbumin and INR), alkaline phosphatase (elevated in around 90% of patients), C-reactive protein, erythrocyte sedimentation rate, and blood cultures to rule out bacteremia.[21]

If an amebic abscess is suspected (such as in residents or travelers from Southeast Asia, Africa, etc.), stool test or serology for *Entamoeba histolytica* should be performed. For the hydatid cyst, serology for *Echinococcus* is needed. Enzyme-linked immunosorbent assay (ELISA) seems to be the most sensitive and specific for *Echinococcus*. After initial screening with ELISA, confirmatory tests with immunoelectrophoresis and immunoblotting are needed. Serology positivity is dependant on the size and site of the cysts. Liver and bone cysts produce positive serology, whereas lung, brain, eye, splenic, or calcified cysts do not. Calcification is usually a sign of non-viable material.

b. Imaging study

Many radiographical tests can be performed, and certain tests should take precedence. It is important to limit a patient's exposure to unnecessary radiation and tests. There can be findings in the chest x-ray, pointing to this diagnosis, such as elevated hemidiaphragm on the right and pleural effusion over the liver. However, the initial test of choice is an abdominal (Table1) ultrasound (US), which shows hyper or hypoechoic lesions with occasional debris or septation. Computed tomography (CT) with contrast is the next step, and slightly more sensitive. Rim enhancement and edema are not typical but very specific for infection. The US or CT is followed by needle aspiration under guidance to identify the exact causative organism, which is essential for diagnostic as well as therapeutic purposes (small cysts).[22] Technetium scan is another test with an 80% sensitivity (less than CT), which is 50-80% for gallium and 90% for indium.[23][24] If inner cyst walls are in-folded (separation of hydatid membrane from the wall of the cyst) during the US, hydatid disease is more likely.

Stains and cultures should be obtained from the direct aspirate. Drains that are in place will get contaminated with skin flora and are not accurate for cultural purposes. Stains and cultures should be done for aerobes and anaerobes (special handling might be needed), but occasionally more specific cultures need to be done, such as fungi, *Mycobacterium*, *Entamoeba histolytica*, and parasites.

Diagnosis is confirmed when there are cystic or solid areas in the liver that, upon aspiration, will yield fluid with positive cultures. It is important to obtain these tests quickly and start treatment because of the high complication rate if left untreated.

IV. DIFFERENTIAL DIGNOSTIC

Liver abscess manifests with right upper quadrant pain, fever, and hepatitis. Therefore, many liver and non-liver ailments are in its differential diagnoses, such as:

- Viral hepatitis
- Cholecystitis
- Cholangitis
- Right lower lobe pneumonia
- Appendicitis
- Necrotic liver masses

Other causes of hepatitis such as autoimmune, drug-induced and acetaminophen toxicity can be mistaken although they are usually not painful.

V. THERAPEUTIC APPROACH

1. Empirical treatment

Antibiotics should be started promptly, as soon as blood cultures have been drawn and before drainage procedures, to limit the systemic effects of septicemia [8], [9]. Small HA, less than 3–5 cm, especially when multiple, can be treated by antibiotics alone without drainage, although no general consensus exists on this point. In 2008, Hope et al. reported a 100% success rate with antibiotic therapy alone for unilocular HA < 3 cm in their series of 107 patients [10]. Similarly, in a literature review of 465 medically-treated abscesses, 176 of which were located in the liver, the 5 cm cut-off was the main factor associated with success of medical treatment alone, and the success rate for HA in this review was over 80% [11].

Empirical antibiotic choice should be directed at the microbes typically responsible, i.e. with activity against aerobic Gram-negative bacilli and Gram-positive cocci, [piperacillins, tazobactam, amoxicillin-clavulanic acid, 3rd generation cephalosporins, (cefotaxime, ceftriaxone)], in combination with an aminoglycoside (gentamicin) [9], [10], [11]. An anaerobic drug such as metronidazole can be used in combination if the chosen antibiotic is not active against anaerobes (or when an amebic abscess is a possibility) [9], [10], [11]. Antibiotic selection should be based on culture and sensitivities. The duration of antibiotic therapy is not clearly established, but is generally between 2 and 6 weeks [9], [10], [11].

2. Directed treatment

a) Aspiration and drainage

Percutaneous drainage of HA was initially described in Hong Kong in 1953 by McFadzean et al. [12]. Sonographic or CT-guided needle puncture with aspirative drainage is currently the first line treatment of HA (Figure 2). Needle aspiration also allows identification of the causative germ and may incidentally reveal evidence of biliary tract communication. After drain insertion, complete evacuation of the collection can proceed.

In 1998, Rajak et al. published a randomized study comparing simple needle aspiration versus catheter drainage of HA [13]. In this study of 50 patients, with either bacterial or amebic abscess, catheter drainage was found to be more effective than needle aspiration alone (success rate 100% vs. 60%) [13], [14]. Another randomized study, published in 2004 by Yu et al., compared repeated needle aspiration versus catheter aspiration in 64 patients with bacterial HA larger than 3 cm in diameter [15]. The authors concluded that the two techniques were equivalent in terms of success (positive needle aspiration rate = 97%), morbidity and mortality, and duration of hospital stay [15]. Of note, however, 59% of patients required more than one procedure [15].

In countries where interventional radiology is readily available, surgical drainage is rarely indicated. However, for some authors, there is a role for surgical drainage if percutaneous treatment for HA fails, for large abscesses > 5 cm, and/or multilocular HA [10], [16], [17], or when surgical treatment of the underlying cause of HA is necessary. Nonetheless, experienced radiologists can usually manage complex situations by “re-drainage” and/or improvement of existing drainage systems (larger drains, irrigation/lavage systems...), or even by innovative endoscopic wall collapsing techniques [18]. If surgery is planned, superficial HA can be treated with simple fenestration and drainage while deep abscesses can be treated either by intra-operative echo-guided drainage or limited hepatic resection [10], [16], [17].

3. Monitoring

a. Clinical monitoring

It is vitally important to follow up with patients after treatment to prevent and recognize some of these complications. Immediate treatment for intraabdominal infection is the best preventive measure for liver abscesses. Conditions like liver abscesses need long-term monitoring to avoid further complications in the future. This monitoring involves checking the treated liver after removing pus or abscess. Through monitoring, you can proactively check the status of your liver condition and prevent the spread of pus.

Long-term monitoring of liver abscess involves:

- Weekly monitoring of the liver through Ultrasound Examination and Computed Tomography (CT) to check the drainage of the abscess cavity in the liver.
- Monitoring fever curves.
- More aggressive drainage is needed when you face continuous fever, even after two weeks.
- Try to maintain drains until the output is less than 10ML/day.

Radiographic abnormalities will take longer than clinical or lab abnormalities to go away, as is with many other medical problems. So enough time should elapse before the graphs go back to normal.

b. Laboratory monitoring

Postoperative prevention includes the continuation of antibiotics, checking blood count, renal function, bilirubin, and aminotransferases.

VI. COMPLICATION

If untreated, hepatic abscesses can rupture, cause peritonitis and shock. At times the area will get walled off, and chronic pain and discomfort in the right upper quadrant with occasional nighttime fever can follow. Complications are also possible after drainage and include liver or kidney failure, intraabdominal lesions, infections, or recurring liver abscesses.[31] Other complications include subphrenic abscess, fistula to organs nearby (such as pleuropulmonary and hypobranhial fistula), acute pancreatitis, abdominal or hepatic venous thrombosis, and liver pseudoaneurysm. Infectious metastatic complications include endophthalmitis or central nervous system septic emboli.

It is vitally important to follow up with patients after treatment to prevent and recognize some of these complications. Immediate treatment for intraabdominal infection is the best preventive measure for liver abscesses. Four to six weeks of antibiotic therapy post-drainage can prevent almost any complications. Antibiotic prophylaxis during chemoembolization or endoscopic retrograde cholangiography help in the prevention of future abscess formation.

Radiographic abnormalities will take longer than clinical or lab abnormalities to go away, as is with many other medical problems. So enough time should elapse before the graphs go back to normal.

VII. ALGORITHM

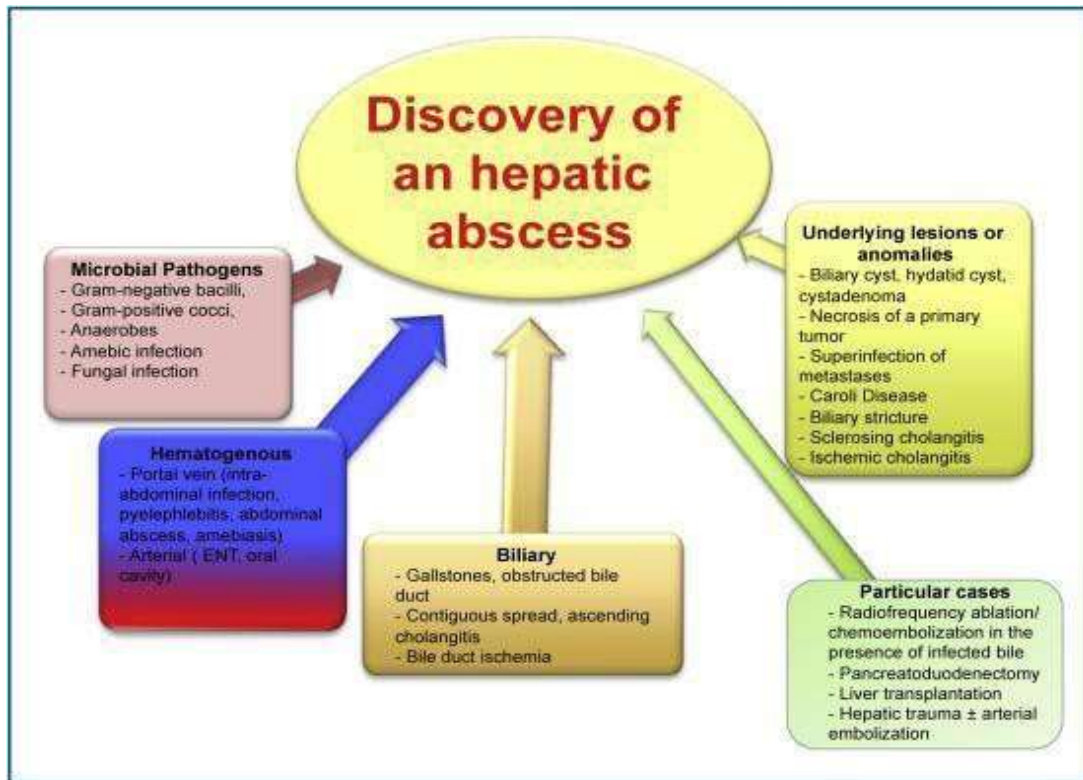


Figure 1: The different etiologies that must be considered when a hepatic abscess is discovered.
 Table 7: Diagnostic radiologic techniques

Study	Sensitivity	Findings	Benefits	Limitations
Ultrasound	80-90%	Hypoechoic areas ≥1cm	Bedside test for diagnosis. Therapeutic drainage guidance.	Operator dependent
CT scan W/IV contrast	95-100%	Sharply demarcated mass, hypodense to surrounding parenchyma +/- enhanced rim sign. Double target sign on dynamic study. Gas detected in 20% of lesions	Noninvasive, allows evaluation of concurrent pathology. Assists with therapeutic drainage.	
Hepatic scans				Not useful when bilirubin >3 Long delay until diagnosis Cannot distinguish between abscess, tumor, or cyst. Must confirm Dx
Technecium	80%	Localized area of decreased uptake, rim sign	Detects lesions < 2cm in diameter	
Gallium	50-80%	Localized area of increased uptake		
Indium	90%	Localized area of increased uptake	Tagged cells localize to abscess. Requires the least time of all hepatic scans	

Table7: Diagnostic radiologic techniques

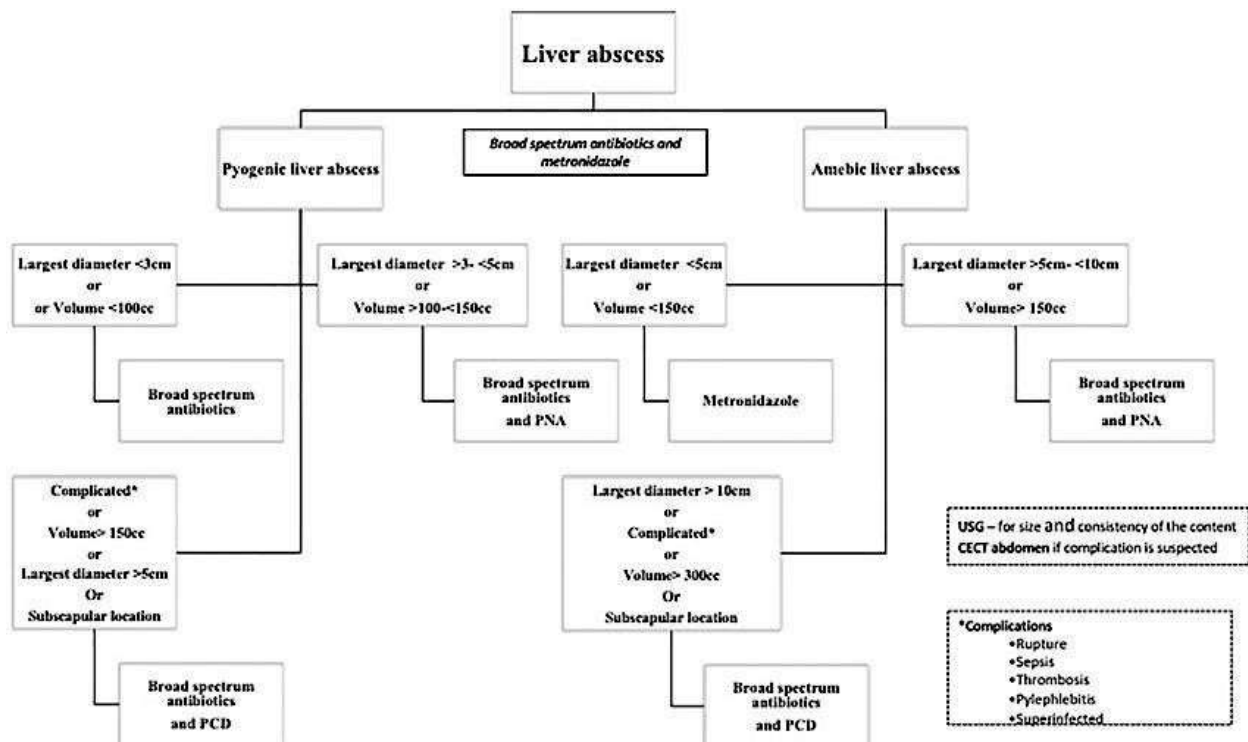


Figure 2: Management of liver abscess

(PNA : Percutaneous Needle Aspiration, PCD: Percutaneous Catheter Aspiration, SD: Surgical Drainage.

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PANCREATIC PSEUDOCYST

Kaing Kimsan

I. CASE DEFINITION

Pancreatic pseudocyst is a collection of pancreatic juice surrounded by a wall of fibrous or granulation tissue.[1]

Pancreatic pseudocysts are common sequelae of acute pancreatitis or chronic pancreatitis and the most common cystic lesion of the pancreas.[2]

II. ETIOLOGY

Pseudocyst are formed when disruption of the main pancreatic duct or its branches, either from inflammation or direct injury, causes extravasation of pancreatic enzymes into the parenchyma and eventually forms a distinct collection. Pseudocysts arise as a complication of pancreatitis. Therefore, it follows that the causes of pseudocyst formation are the same the cause of pancreatitis.[3]

Pseudocyst can occur after pancreatitis in any age group. The incidence of pseudocysts is higher in males as it follows the incidence of pancreatitis, which is slightly male predominant.[3]

Walt et al reported data from Wayne State University Hospital in Detroit USA. The causative factors in the 357 admissions for pancreatic pseudocysts included alcohol use in 251 case (70%), biliary tract disease in 28 (8%), blunt trauma in 17 (5%), penetrating trauma in 4 (1%), operative trauma in 1 (0,3%), and idiopathic in 56 (16%). Most of the patients in the idiopathic group were thought to have been alcohol-related, but no definite evidence was recorded. [4][5]

III. DIAGNOSTIC PROCEDURE

1.1.Clinical argument [10]

- Patients with small pseudocysts are usually asymptomatic.
- Patients may become symptomatic if the pancreatic pseudocyst:
 - . Expands, get infected, digests the wall of the adjacent vessels,
 - . occludes the adjacent veins, or ruptures into the peritoneal cavity, pleural space, or pericardium.
- Expansion of the pseudocyst may cause:
 - . abdominal pain
 - . vomiting, early satiety (duodenal or gastric obstruction)

- . Jaundice (biliary obstruction)
- . Upper GI bleeding (vascular occlusion/thrombosis of the portal vein, splenic vein or both).
- Infection of the pseudocyst may cause:
 - . Fever and elevation of the white blood cell count (WBC)
 - . sepsis
- Digestion of adjacent vessel by the pseudocyst may cause:
 - . Arterial pseudoaneurysm:
 - >> Sudden painful expansion of the pseudocyst (bleeding pseudoaneurysm)
 - >> Gastrointestinal bleeding, pseudoaneurysm rupture into the pancreatic duct (hemorrhage)
 - >> Unexplained drop of the hematocrit (pseudoaneurysm leakage)
- Venous compression may cause:
 - . Thrombosis of the splenic, superior mesenteric, or portal vein
 - . Upper GI bleeding
 - . Splenomegaly
- Disruption of the pancreatic duct may cause:
 - . Abdominal distention (pancreatic ascites)
 - . Pleural effusion (shortness of breath)
 - . Pericardial effusion (cardiac failure)

1.2. Technical procedure

a. Laboratory test

- Amylase and lipase levels are often elevated. [5]
- Carcinoembryonic antigen (CEA) and CA-125 (low in pseudocysts and elevated in tumor).[6]
- Bilirubin and liver chemistries may be elevated if the bile duct is obstructed from stone, extrinsic compression from the pseudocyst or from underlying liver disorder (eg. Alcoholic hepatitis).[5]
- Some laboratory tests may provide clues to the underlying etiology of pancreatitis (eg. Elevated triglycerides or calcium level). [5]

b. Imaging study

- Transabdominal Ultrasound

Pancreatic pseudocyst appears as an echoic structure associated with distal acoustic enhancement on ultrasound examination. They are well defined and round or oval, and they are contained within a smooth wall.

During the early phase of their development, pseudocysts can appear more complex, with varying degrees of internal echoes.

Usually, this appearance results from the presence of necrotizing pancreatitis than in chronic pancreatitis related pseudocyst. The debris is cleared over time in most cases.

The pseudocyst can appear more complex in 2 other instances: when hemorrhage occurs into the cyst or when infection of the cyst complicates the clinical course. Color Doppler or duplex scanning should always be performed in cystic lesion to ensure that in the lesion in the question is not a giant pseudoaneurysm.

Sensitivity rates for ultrasound in the detection of pancreatic pseudocysts are 75% to 90%. Therefore, ultrasound is inferior to CT, which has a sensitivity of 90% to 100%. Ultrasound has several limitations, as compared with CT, in the initial diagnosis of a pseudocyst: the presence of overlying bowel gas decreases the sensitivity of ultrasound, ultrasound examinations are highly operator dependent. [5][7]

- CT Scan

The identification of a thick-walled, rounded, fluid-filled mass adjacent to the pancreas on an abdominal CT Scan in a patient with a history of acute or chronic pancreatitis is virtually pathognomonic for pancreatic pseudocyst. Positive CT scan findings in this clinical situation do not require confirmation with another diagnostic modality. In the acute setting, a CT scan is the better choice because significant amounts of bowel gas resulting from ileus or obstruction decrease the sensitivity of ultrasound, CT scan provides more detailed information regarding the surrounding anatomy and can demonstrate additional pathology, including pancreatic duct dilatation and calcification, common bile duct dilatation, and extension of the pseudocyst out-side the lesser sac. The major weakness of CT scanning is the relative inability to differentiate pseudocyst from cystic neoplasm, especially mucinous cystadenomas and intraductal papillary mucinous neoplasm (IPMN). [5][8]

- Magnetic resonance imaging (MRI)

MRI and Magnetic resonance cholangiopancreatography (MRCP) are sensitive diagnostic modalities for pancreatic pseudocysts. They are generally not routinely used because CT scanning typically offers all the diagnostic information that is required. However, the increased contrast provides for better characterization of fluid collections. MRI or MRCP is superior to CT in depicting debris within fluid collection and pseudocyst. On T2-weighted images, a fluid-filled cystic mass produces high signal intensity and appears bright. The pancreatic duct and biliary

systems are easily visualized in detail, although interpreting the status of pancreatic duct integrity may be difficult. [5][9]

IV. DIFFERENTIAL DIAGNOSIS [10]

- Oligocystic serous cystadenoma
- Mucinous cystic neoplasm
- Intraductal papillary mucinous neoplasm (IPMN)
- Cystic neuroendocrine tumor
- Solid pseudopapillary tumor of pancreas
- Lymphoepithelial cyst
- Duplication cyst (GI)
- Necrotic GIST tumor

V. PSEUDOCYST MANAGEMENT [10]

- Currently, the initial management of an asymptomatic pseudocyst regardless of the size is conservative.
- Close surveillance with Ultrasound, CT, or MRI is recommended.
- Indication for intervention:
 - . Persistent pain
 - . Infection
 - . Increasing size
 - . Obstruction of the biliary system
 - . Obstruction of the stomach or duodenum
 - . Location: mediastinum
- Treatment Options:
 - . Percutaneous drainage
 - . Endoscopic drainage
 - . Internal drainage (surgery)
 - . Surgical resection

1.3. Percutaneous Drainage under Ultrasound or CT

- As effective as surgery in the treatment of both sterile and infected pseudocyst.
- Does not require 6 weeks delay to allow the pseudocyst wall to thicken and mature.
- The pseudocyst can be drained under ultrasound or CT guidance.
- Catheters used: pigtail APD catheters (8-10-12-14F gauge).

- Approach: direct, transgastric, transhepatic.
- Catheter drainage is continued until the flow rate falls to 5-10ml/day.
- If reduction in flow does not occur, subcutaneous delivery of 50-200 ug octreotide every hour is recommended.
- Mean duration of drainage: 3-8 weeks.

1.4.Endoscopic Drainage

- With this method a double pigtail stent is placed into the pseudocyst with endoscopic guidance.
- Stent placement options:
 - Endoscopic cystogastrostomy (pseudocyst adherent to the stomach)
 - Endoscopic cystroduodenostomy (pseudocyst adherent to the duodenum)
 - Insertion of a stent through the ampulla directly into pancreatic duct and then into the pseudocyst itself (this procedure is associated with the development of acute pancreatitis)
- Complication:
 - . Bleeding, retroperitoneal perforation, infection.

1.5 Surgical Drainage

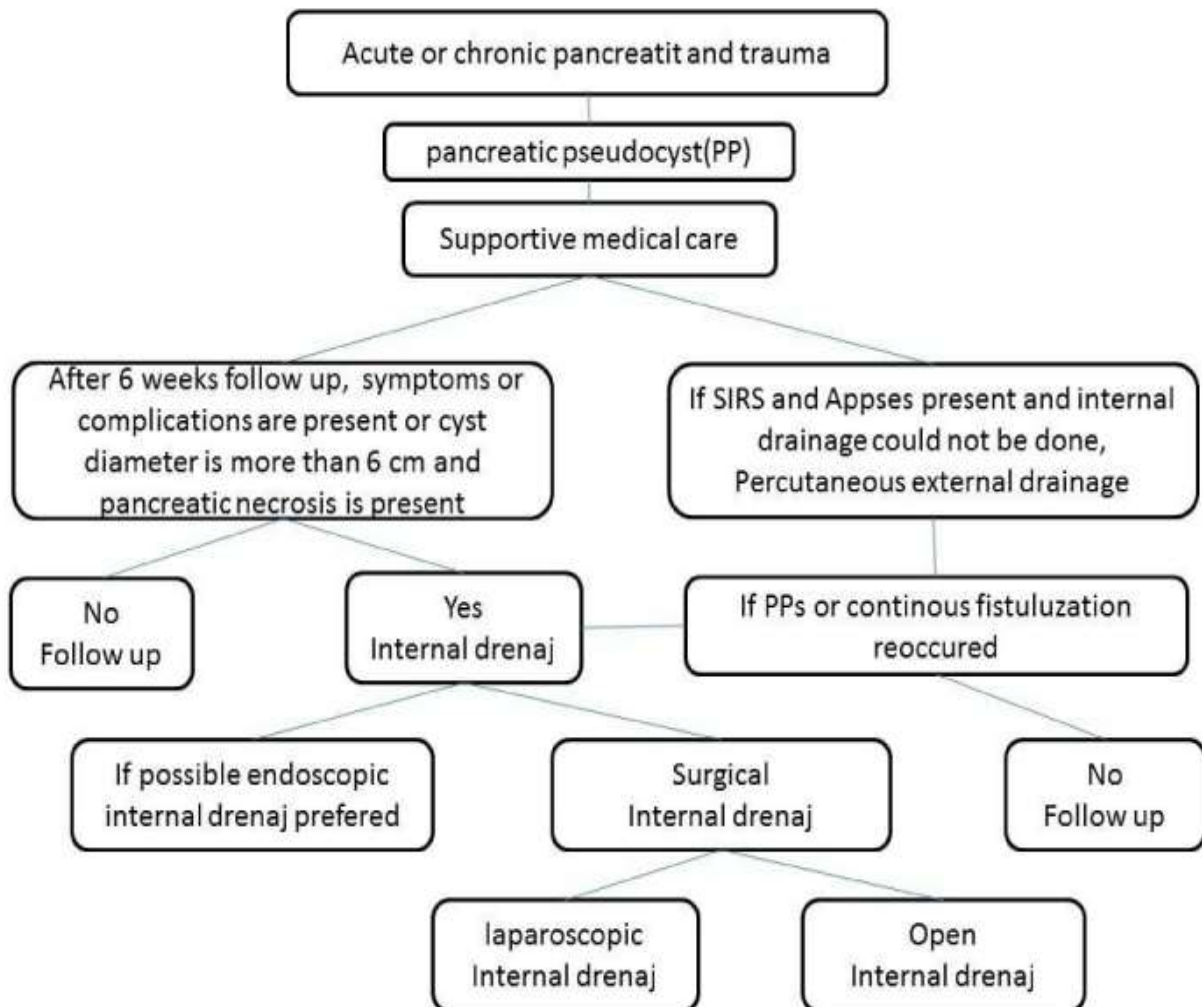
- Failures or technical impossibilities of radiological and endoscopic treatments constitute the majority of indication for surgical treatments.
- Surgical options:
 - . Cystogastrostomy
 - . Cystoenterostomy
 - , Pancreaticoduodenectomy (if the pseudocyst is associated with common bile obstruction or difficult to differentiate from necrotic malignant tumor).
 - . Subtotal distal pancreatectomy if the pseudocyst is associated with splenic vein thrombosis
 - . Distal pancreatectomy (pseudocyst tail of the pancreas).
- These approaches are associated with substantial morbidity (25%) and mortality (5%).

2. Management of the Pseudocyst Complications: [10]

- Pseudocyst infection: percutaneous catheter drainage, antibiotics
- Pseudoaneurysm: selective arterial embolization (intravascular coils)
- Rupture into the peritoneal cavity: percutaneous drainage of peritoneal pancreatic ascites and endoscopic bridging intraductal stent placement.
- Rupture into the pleural cavity: percutaneous drainage of pleural pancreatic fluid and

endoscopic bridging intraductal stent placement.

VI. ALGORITHM [11]



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PERIANAL ABSCESS

MO Satdin, EM Sokhom

I. DEFINITION

Perianal abscess is the most typical form of an anorectal abscess. It presents as a tender superficial swelling outside the anal verge. The incidence reported is 40–60%. Patients with a perianal abscess may complain of painful swelling that may increase in intensity after defecation or sitting. 50% of anorectal abscess will develop into the anal fistula.

II. ETIOLOGY

Anal sepsis can occur at any age though it is more common in third or fourth decade of life. In children, it is more common in babies under 12 months of age. Males are affected more frequently than females, with a male-to-female ratio of 2–3:1 (Isbister 1987). Approximately 30 % of patients with anorectal abscesses have a previous history of similar abscesses (Ramunjam et al. 1984).

Many factors have been associated with development of anal sepsis like malignancy, trauma, tuberculosis, foreign bodies, inflammatory bowel disease, simple skin infection, and actinomycosis. However, most anal sepsis originates from an infected anal gland. Obstruction of these glands leads to stasis, bacterial overgrowth, and ultimately abscess formation.

III. DIAGNOSTIC PROCEDURES

i. Clinical argument

Erythema, induration, or fluctuance are present on local examination. On digital examination, the patient may experience pain and tenderness. Although proctoscopy becomes challenging to perform due to the presence of pain, if possible, it may demonstrate pus oozing out from the base of the crypt, indicating an internal opening. Digital rectal examination may reveal boggiess in the anal region.

Classification of anorectal abscess:

- Perianal abscess (simple type), occur immediately adjacent to the anal verge.
- Ischiorectal abscess, penetrates through the external anal sphincter into the ischiorectal space.
- Intersphincteric abscess, locate between internal and external sphincter.
- Supralelevator abscess, that travel superiorly within the intersphincteric plane to the supralelevator space.

ii. Laboratory test

Usually laboratory investigations yield little information; however, frequently patients will have an elevated white blood cell count.

iii. Imaging study

Ultrasound is usually not tolerated for patient with acute abscess collection.

CT scan with IV or rectal contrast can differentiate perianal cellulitis from abscess or when deep anorectal abscess is suspected, but can't be palpated on digital.

MRI is the most useful imaging tool especially when evaluating potential fistula-in-ano cases, suspected perianal Crohn disease, or recurrent complex abscesses.

IV. DIFFERENTIAL DIAGNOSIS

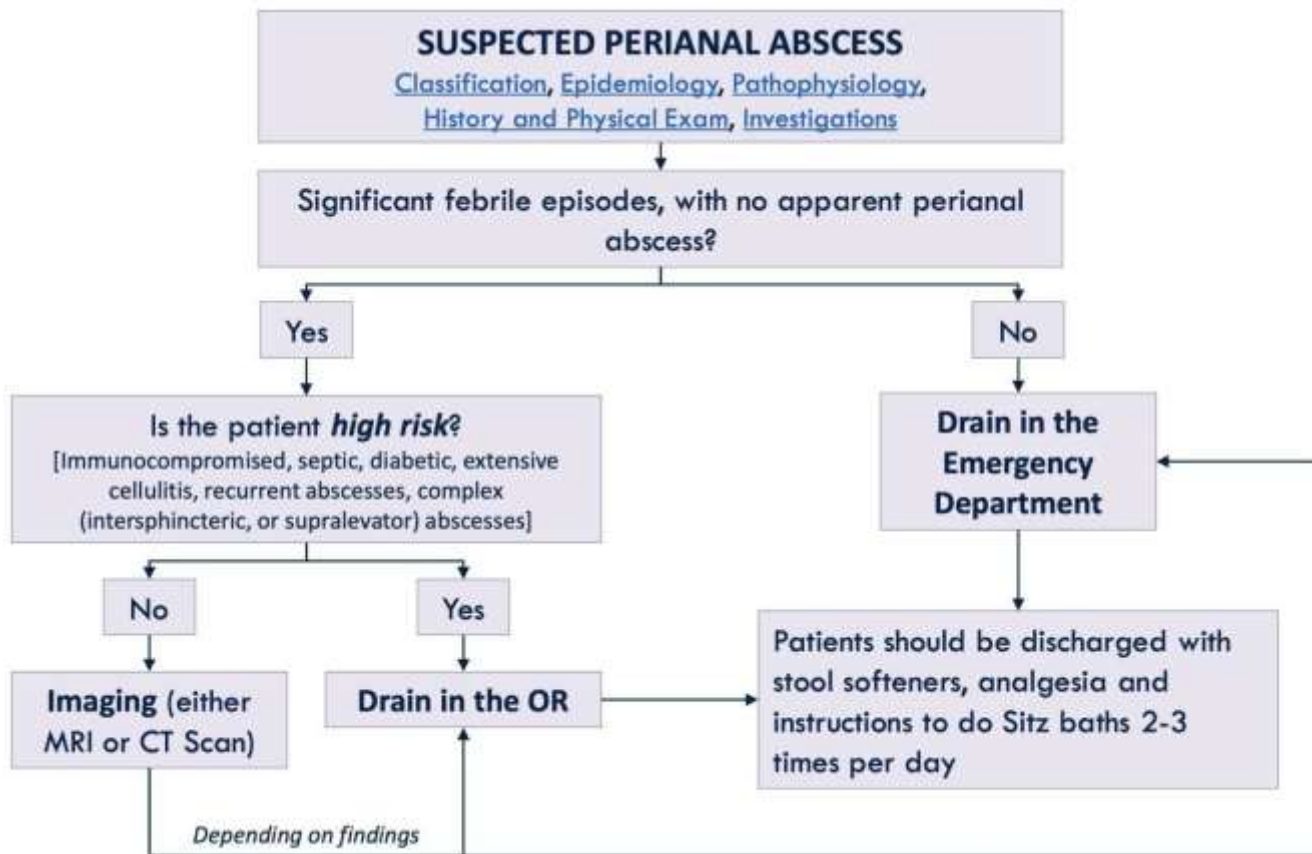
- Hidradenitis suppurativa
- Thrombosed
- Skin furuncles
- Herpes
- HIV
- Tuberculosis
- Actinomycosis
- Ulcerative colitis
- Bartholin cyst

V. ALGORITHM

Algorithm 5. Etiology and pathogenesis of perianal abscess

1	Infection originates from an obstructed anal crypt gland. Anal crypt glands are arranged within the anal canal at the level of the dentate line. The glands penetrate the internal sphincter and end in the intersphincteric plane.	2	When the gland becomes obstructed with inspissated debris, bacterial growth occurs, with resultant abscess formation.
3	Perianal abscesses occur immediately adjacent to the anal verge. Ischiorectal abscesses are larger, more complex and usually present as a tender buttock mass.	4	Supralelevator and intersphincteric abscesses are rare and there may be no external manifestations. A tender mass on digital rectal exam above the anatomic anal canal, or within the canal is clinically suggestive.
5	Undrained anorectal abscesses can expand into adjacent spaces and progress to generalized systemic infection	6	50% of anorectal abscess will develop in to anorectal fistula

Algorithm 6. Strategy of management of perianal abscess



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PERITONITIS DUE TO GASTRIC PERFORATION

Oum Sethykum, Preap Ley

I. CASE DEFINITION

Is inflammation of the peritoneum caused by full thickness injury of the gastric wall. perforation of the gastric wall creates a communication between the gastric lumen and peritoneal cavity. Gastric content freely enters the general peritoneal cavity causing inflammatory reaction and infection.

II. ETIOLOGY

- Peptic ulcer disease (most common cause of stomach perforation)
- Spontaneous gastric perforation (uncommon in the neonatal period)
- Trauma (penetrating abdominal injury, blunt abdominal trauma...)
- Malignancy-Related perforation (perforation by direct penetration and necrosis or by producing obstruction and following chemotherapy, radiation treatment)
- Iatrogenic (endoscopy, lesser ...)

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

a. Symptoms:

Sudden onset of abdominal distension and pain or chest pain, less common presenting symptoms are ileus, respiratory distress, fever, emesis, hematemesis or hematochezia. Patient on immunosuppressive or anti-inflammatory agents may have impaired inflammatory response and some may have less pain and tenderness.

b. Signs:

General sign: severe sepsis, septic shock, oliguria, multi-organ failure
Infection sign: fever or hypothermia, tachycardia, tachypnea

Digestive sign: general abdominal tenderness and distension, hyperresonance, bowel sound may be absent and rebound and guarding are likely to be present.

ii. Technical procedure

a. Laboratory:

- CBC: hyperleukocytosis or leukopenia, CRP elevation, neutrophilia,
- Creatinine and urea: common acute renal failure linked to sepsis.
- Transaminase, blood gas
- Blood culture and urinary analysis
- Coagulation panel
- Electrolyte

b. Imaging:

The sensitivity of plain films for detecting extraluminal free air ranges from 50%-70%. Ultrasound has also been studied and show excellent potential for identifying pneumoperitoneum. The most useful imaging modality is an X-ray and CT scan that is highly sensitive and specific for free air

IV. DIFFERENTIAL DIGNOSIS

The differential diagnosis for the sudden onset abdominal pain seen with a gastric perforation is broad and includes but not limited to:

- Peptic ulcer
- Duodenal ulcer
- Biliary disease
- Embolic mesenteric insufficiency
- Splenic infection
- Gastritis
- Esophageal perforation
- Abdominal aortic aneurysm rupture

V. TREATMENT / MANAGEMENT

i. Stabilize patient

Hospitalization surgery or intensive care unit for the patient with organ failure and comorbidities.

- Nothing by mouth or (NPO)
- Peripheral intravenous line
- Correction shock or hypovolemia condition
- Correction fluid and electrolyte disorders
- Oxygen therapy
- Intravenous broad-spectrum Antibiotics
- Intravenous analgesia and PPIs should be given as necessary
- Nasogastric tube should also be placed
- Urinary catheters enable close monitoring of urine output

ii. Empiric approach

Medical management is mainly symptomatic:

- IV fluids
- Electrolyte repletion
- Analgesics and antipyretics as needed
- Start antibiotic therapy with Intravenous broad-spectrum
- Preoperative assessment and anesthesia consultation

iii. Directed management

Most patients with gastric perforation should be managed with urgent explorative laparotomy

- procedure: laparotomy exploration and complete exploration of the peritoneal cavity and multiple samples of different effusion for bacteriology and cytology examination.

Verify perforated area and causative factor and repair technique:

- *Primary repair*: The defect is primarily closed with suture, this is appropriate for most traumatic perforation
- *Graham patch repair*: The defect is simply plugged with a well-vascularized omental pedicle

- *Modified Graham patch repair*: Primary closure of the defect then application of flap
- *Wedge resection*: the perforated area may be resected from healthy tissue, particularly if it is on the greater curvature and distend from the gastroesophageal junction or the pylorus

Options for reconstruction:

- *Billroth I*: Gastroduodenostomy, anastomosis between the gastric remnant and the duodenum
- *Billroth II*: Gastrojejunostomy, side to side anastomosis between gastric remnant and loop of jejunum with the closure of the duodenal stump
- *Roux-en- Y gastrojejunostomy*: The creation of the jejuno-jejunostomy forming Y shape figure of the small bowel.
- Restoring continuity is most contraindicated because of the significant risk of anastomotic fistula. However in certain cases, anastomosis is possible (localized peritonitis, patient in good condition.
- Abundant washing of the peritoneal cavity with between 15-20 liters
- Drainage and closure

iv. Monitoring

- a. Clinical monitoring
 - Monitoring vital signs, airway patency, and neurological status
 - Monitoring for possible complication (hemorrhage, infection, fistula ...)
 - Monitoring and managing pain
 - Monitoring fluid and electrolyte
 - Monitoring drainage and urine output
- b. Laboratory monitoring
 - CBC
 - Electrolyte
 - CRP
 - Kidney function
 - Liver function

- Albumine
- Blood glucose

VI. COMPLICATIONS

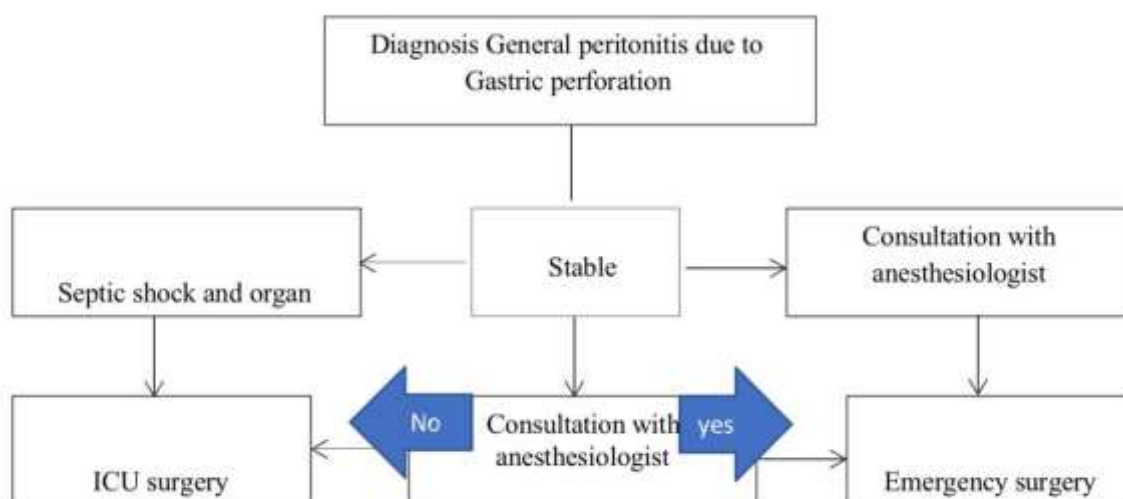
Complications following gastric perforation include the following:

- Wound infection
- Sepsis
- Malnutrition
- Multiorgan failure
- Adhesion and bowel obstruction
- Delirium

The risk of complications is increased in the presence of:

- Advanced age
- Dementia
- Sepsis
- Electrolyte and metabolic abnormality
- Hypoxia
- Intraoperative complication

VII. ALGORITHMS



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PERITONITIS DUE TO LARGE BOWEL PERFORATION

Oum Sethykum, Preap Ley

I. DEFINITION

Peritonitis is an inflammation of the peritoneum, the serous membrane that lines part of the abdominal cavity. Peritonitis may be localized or generalized, and may result from infection and from a non-infectious process.

Large bowel perforation is a full-thickness loss of large bowel wall integrity caused by air, digestive contents and fecal material into peritoneal cavity that results in perforation peritonitis. Most common perforation of large bowel is appendicitis and diverticulitis.

II. ETIOLOGY

- Malignancy Related perforation (perforation by direct penetration and necrosis or by producing obstruction and following chemotherapy, radiation treatment)
- Bowel ischemia
 - Bowel obstruction (i.e., adhesions, volvulus, malignancy)
 - Acute mesenteric ischemia
- Infections
 - Diverticulitis (colonic diverticula)
 - Acute appendicitis
 - Toxic megacolon
- Inflammatory bowel disease: Ulcerative colitis, Crohn disease
- Trauma
 - Penetrating abdominal trauma (e.g., stab injury)
 - Blunt abdominal trauma
 - Iatrogenic perforations (e.g., during colonoscopy, laparoscopic surgery)
- Foreign body ingestion
- Drug-induced (e.g., NSAIDs, glucocorticoids, cocaine).

III. PATHOPHYSIOLOGY

In normal conditions, the peritoneum appears to greyish and glistening. It becomes dull 2-4 hours after the onset of peritonitis, initially with serous or slightly turbid fluid.

Bowel perforation results from a violation of the mucosal layers of the intestinal tract resulting in the spilling of air and digestive contents into the peritoneal cavity. These contents can range from highly acidic gastric contents in more proximal bowel perforation, to fecal material from a more distal area of perforation. Areas of partial erosion can become full-thickness tears over time, or a specific insult can cause spontaneous rupture. Bacterial proliferation occurs, edema of the tissues results and exudation of fluid develop in a short time.

Fluid in the peritoneal cavity becomes turbid with increasing amounts of protein, white blood cells, cellular debris, and blood. The immediate response of the intestinal tract is hypermotility, followed by paralytic ileus with an accumulation of air and fluid in the bowel. Later on, exudate becomes creamy and suppurative. It may spread to the whole peritoneum.

IV. DIAGNOSTIC PROCEDURE

i. Clinical Argument

- Abdominal pain may be gradual or sudden, but it usually progresses severity
 - Abdominal distention
 - Nausea, vomiting, obstipation
 - Fever, tachycardia, tachypnea, hypotension
 - Signs of peritonitis or shock
 - Decreased or absent bowel sounds due to paralytic ileus
 - Tenderness – initially localized later becomes diffused
 - Rebound tenderness (Blumberg sign)
 - Guarding and rigidity, dull flanks on percussion representing peritonitis may develop
 - Tenderness on per rectal examination.
- History suggestive of specific locations
- Perforated diverticulitis: constipation, previous LLQ pain
 - Perforated appendicitis: progressively worsening RLQ pain, migratory pain
 - Perforated malignancy or IBD: anorexia, weight loss, melena, change in bowel habits.

- Localization of pain
 - Diffuse: in patients with free intraperitoneal perforation
 - Localized RLQ pain: contained perforated appendicitis
 - Localized LLQ pain: contained perforated diverticulitis.

ii. Technical Procedure

a) Laboratory Tests

- CBC: hyperleukocytosis or leukopenia, neutrophilia
- Inflammatory marker such as C-reactive protein or procalcitonin
- Coagulation Panel (PT,aPTT, INR)
- Electrolyte, Liver Function test, Amylase, Lipase, Glycemia
- Urea and Creatinine: common acute renal failure linked to sepsis.

b) Imaging Study

Plain radiography or X-ray abdomen (50% to 70% sensitive)

- Indications: Consider only for patients too unstable to safely undergo CT scanning.
- Findings: free intraperitoneal air (pneumoperitoneum) under the diaphragm and/or between liver and lateral abdominal wall suggestive of bowel perforation. The likelihood of diagnosing free air is higher if the patient has been sitting upright for at least 15 minutes prior to the film. Lateral decubitus films can be used if the patient is unable to sit upright.

Ultrasound Abdomen

- Indication: preferred in patients with contraindications to radiation exposure (e.g., pregnancy)
- Findings: pneumoperitoneum, localized fluid collection, localized thickening of a bowel segment. However, this modality is especially dependent on the experience level of the operator.

Computer tomography scan (CT scan)

- CT scan abdomen and pelvis with IV contrast (most sensitive) is the modality of choice for not only diagnosing free air but also for localization of the site of perforation. A CT scan can also help to determine if the area has spontaneously walled off if there has been a progression to abscess formation or if there is an inflammatory involvement of surrounding structures. Signs of perforated bowel is loss of bowel wall continuity or localized mesenteric fat stranding.

V. DIFFERENTIAL DIAGNOSES

- Acute Cholecystitis and Biliary colic
- Acute gastritis

- ☐ Acute pancreatitis
- ☐ Appendicitis
- ☐ Constipation
- ☐ Crohn's disease
- ☐ Endometriosis
- ☐ Fallopian tube disorders
- ☐ Gastric perforation
- ☐ Inflammatory bowel disease
- ☐ Meckel diverticulum surgery
- ☐ Ovarian (Adnexal) torsion
- ☐ Small bowel perforation
- ☐ Pelvic inflammatory disease
- ☐ Peptic ulcer disease
- ☐ Typhoid Fever.

VI. THERAPEUTIC APPROACH

i. Stabilize Patient

It must not delay surgical treatment which is an emergency. Hospitalization surgery or intensive care unit for the patient with organ failure and comorbidities.

- Bowel rest (NPO)
- IV access with two large-bore peripheral intravenous line (Intravenous broad-spectrum Antibiotics)
- Correction shock or hypovolemia condition
- Correction fluid and electrolyte disorders and Oxygen therapy
- Intravenous analgesia (Opioid or NSAIDs)
- Nasogastric tube should also be placed
- Parenteral antiemetics and PPIs should be given as necessary
- Urinary catheters enable close monitoring of urine output.

ii. Empirical Treatment

Medical management is mainly symptomatic:

- IV fluids
- Electrolyte repletion
- Analgesics and antipyretic as needed

- Start antibiotic therapy with Intravenous broad-spectrum
- Reevaluate frequently (serial abdominal examination, vitalsigns), as the patient's condition may rapidly deteriorate
- Preoperative assessment and anesthesia consultation if patient must be for operation.

iii. Directed Treatment

Most patients with large bowel perforation should be managed with urgent explorative laparotomy.

- Indications: Signs of generalized peritonitis or Signs of sepsis.
- Surgery the objective:
 - Exploratory laparotomy with midline incision is usually preferred.
 - Establish the cause of peritonitis
 - Control the origin of sepsis by removal of the inflamed or ischemic organ
 - Perform effective peritoneal toilet/lavage
 - Complete exploration of the abdominal cavity
 - Suture closure of the perforated hole
 - Exteriorize the segment of the digestive tract
 - Avoid anastomosis in septic environment
 - Directed fistulization
 - Not excess debridement
 - Peritoneal lavage with greater volume
 - Drainage.
- Specific Procedure:
 - Perforation of Appendice: Appendectomy lavage and drainage.
 - Perforation of Cecum or Ascending colon: Right hemicolectomy and anastomosis ileus with transverse colon with or without loop ileostomy protective or End Ileostomy.
 - Perforation of Transverse colon: Transverse colectomy and anastomosis ascending colon with descending colon with or without loop ileostomy protective or End Colostomy.
 - Perforation of Descending colon: Left hemicolectomy and anastomosis transverse colon with rectum with or without loop ileostomy or Hartmann's procedure.
 - Perforation of Sigmoid colon: Sigmoidectomy and anastomosis descending colon with rectum with or without loop ileostomy protective or Hartmann's procedure.

- Perforation of Rectum: Anterior resection or Hartmann's procedure.

iv. Monitoring

a) Clinical Monitoring

Postoperative care should include a continued focus on hemodynamic resuscitation with monitoring of expected urine output, airway patency, neurological status, electrolyte repletion, infection monitoring and control measures, pain management, and nutritional considerations.

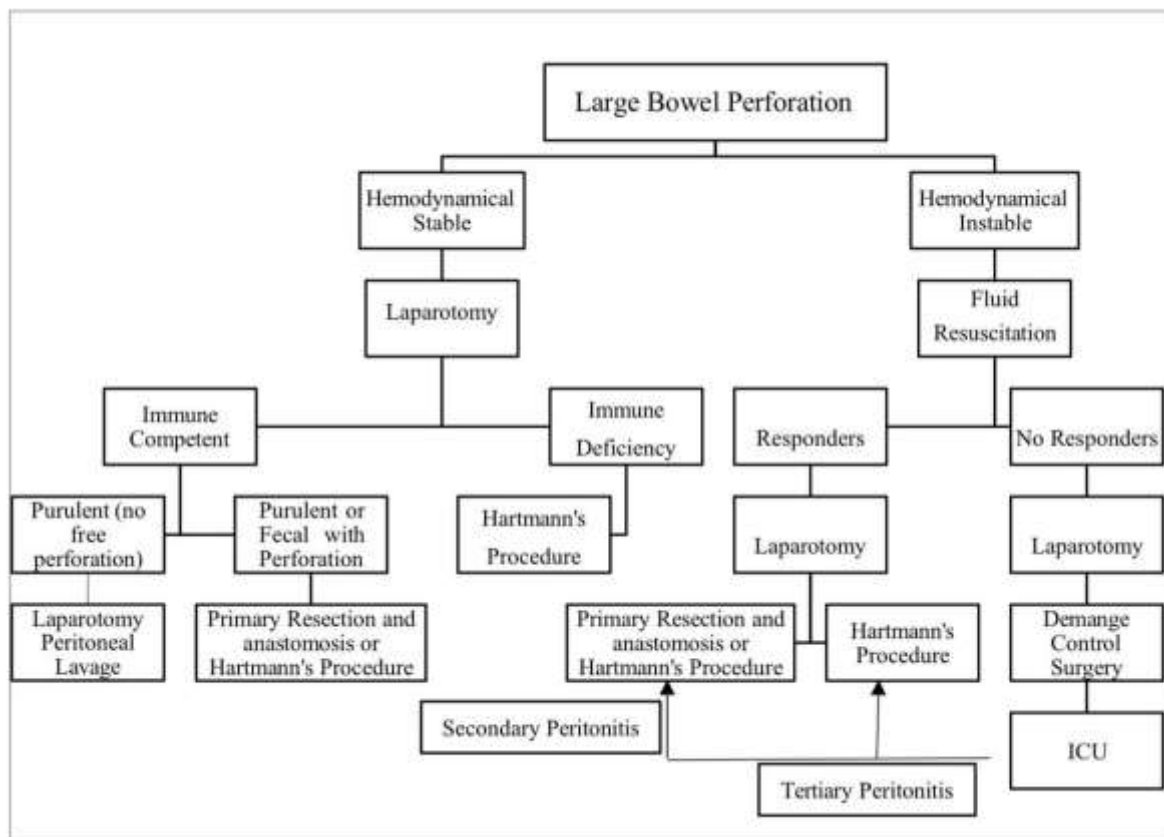
b) Laboratory Monitoring

- CBC
- CRP
- Electrolyte
- Urea and Creatinine
- Liver Function Test
- Albumin
- Glycemia.

VII.COMPLICATIONS

- ☐ Peritonitis
- ☐ Bacteremia
- ☐ Hemodynamic instability leading to hypoperfusion, shock, sepsis
- ☐ Multiorgan dysfunction
- ☐ Local abscess formation: o Intraabdominal abscess o Subhepatic abscess
- ☐ Pyogenic liver abscess
- o Pelvic abscess
 - ☐ Delayed wound healing
 - ☐ Postoperative complications: intraabdominal adhesions leading to bowel obstruction, fistula formation, and hernias.

VIII. ALGORITHM



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PERITONITIS DUE TO SMALL BOWEL PERFORATION

Oum Sethykum, Preap Ley

I. DEFINITION

Peritonitis is defined as inflammation of the peritoneal cavity and may be classified according to the underlying cause (primary or secondary), extent (localized or generalized), or the presence of infectious agents (septic or nonseptic).

Small bowel (SB) perforation is an acute pathological condition resulting from a discontinuity of the small bowel wall secondary to different etiologies with subsequent leakage of intestinal gas and contents into the peritoneal cavity.

II. ETIOLOGY

There are ultimately four mechanisms that can lead to a perforation of the intestinal tract.

1. Ischemia (bowel obstruction, necrosis)
2. Infection (appendicitis, diverticulitis)
3. Erosion (malignancy, ulcerative disease)
4. Physical disruption (trauma, iatrogenic injury)

III. DIAGNOSTIC

i. Clinical presentation

- Abdominal pain
- Abdominal distension
- Nausea
- Vomiting
- Fever
- Altered Bowel Habit

ii. Physical examination

- Abdominal tenderness
- Abdominal guarding
- Diminished or absent bowel sound
- Shock
- Tachycardia
- Dehydration
- Obliteration of liver dullness

iii. Investigations

a. Laboratory tests

Labs may include a complete blood count, basic metabolic panel, liver function tests, lipase, amylase, and inflammatory markers such as C-reactive protein.

b. Imaging study

Plain radiograph

- free gas under the diaphragm is a classic sign of pneumoperitoneum on the erect chest and abdomen radiographs and it is suspicious for bowel perforation
- other signs of pneumoperitoneum, including Rigler sign

Ultrasound

- although not a primary modality for evaluating pneumoperitoneum, free gas may be detected on ultrasound when gas shadowing is present along the peritoneum
- free intraperitoneal fluid collection

CT

CT is the imaging modality of choice in the diagnosis of small bowel perforations especially in the early stages.

IV. DIFFERENTIAL DIAGNOSIS

- Acute Cholecystitis and Biliary Colic
- Acute Gastritis
- Acute Pancreatitis
- Appendicitis
- Constipation
- Crohn Disease
- Endometriosis
- Fallopian Tube Disorders
- Inflammatory Bowel Disease
- Meckel Diverticulum Surgery
- Ovarian (Adnexal) Torsion
- Pelvic Inflammatory Disease
- Peptic Ulcer Disease
- Small Intestinal Diverticulosis
- Typhoid Fever

- Ulcerative Colitis

V. MANAGEMENTS

Treatment of a patient with suspected small bowel perforation involves establishing intravenous (IV) access and initial hemodynamic management, especially if the patient presents with any signs or symptoms concerning sepsis or shock. Initiation of broad-spectrum antibiotics aimed at gram-negative and anaerobic organisms is essential and should be initiated early. If the patient is hemodynamically stable and there is no concern for peritonitis, such as in the case of a spontaneously contained perforation, the option for non-surgical management with antibiotics and observation may be pursued in consultation with the surgical team.

Most cases will progress to require direct investigation via laparoscopic or open (laparotomy) exploration. This allows for primary repair as well as infection control measures. Minimally invasive exploration and repair are often successfully performed. However, it should be noted that exploratory laparotomy is the surgical intervention of choice for any signs of clinical deterioration or hemodynamic instability.

VI. POSTOPERATIVE AND REHABILITATION CARE

Postoperative care should include a continued focus on hemodynamic resuscitation with monitoring of expected urine output, electrolyte repletion, infection monitoring and control measures, pain management, and nutritional considerations.

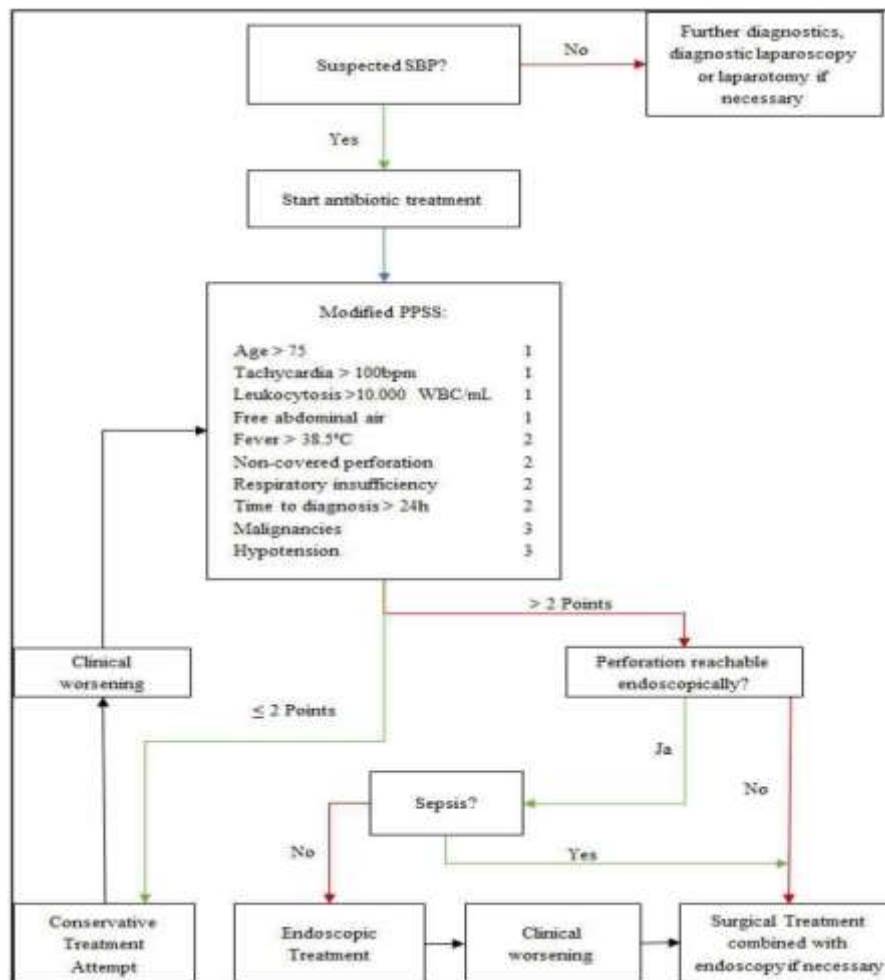
VII.COMPLICATIONS

Complications caused by an intestinal perforation can be related to the disease process that caused it or the treatments that are enlisted. Perforation and subsequent leakage of intestinal contents can lead to peritonitis and eventually sepsis if left untreated. Adequate resuscitation and antibiotic initiation early on are important to decrease the physiologic detriment of the infection.

If necessary for control of leakage, surgery can be undertaken, and it is important to discuss potential complications of surgery with patients and their families. Risks of infection, bleeding, potential anastomotic leakage, hernia formation exist. Other risks of pulmonary complications, thromboembolic events, cardiovascular events, and possible prolonged need for ventilator support must also be discussed as well as expected recovery and possible level of function.

VIII. ALGORITHM

A clinical scoring system used to stratify patients to possible therapeutic options could be advantageous. In the algorithm for diagnostic and therapeutic steps, a patient's clinical presentation is measured with the Pittsburgh Perforation Severity Score (PPSS) . This score was created to stratify patients with esophageal perforations for surgical or non-surgical treatment concepts. We believe that this score is transferable to patients with SBP. We present this algorithm for clinical use in cases of suspected SBP



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MANAGEMENT OF PLEURAL EFFUSION

Tuo Sothunea, Duong Chhay, Lay Nhork

I. DEFINITION

Pleural effusion refers to the accumulation of fluid within the pleural space, which can be caused by a variety of underlying conditions, impairing lung function and often causing respiratory symptoms.

II. ETIOLOGY

Pleural effusion can result from a variety of conditions affecting the pleural space. The underlying cause often determines the type of pleural effusion (transudative vs. exudative) and influences management strategies.

Transudative Pleural Effusions

Transudative effusions occur due to systemic factors that alter the balance of hydrostatic and oncotic pressures. These effusions are typically associated with conditions that affect fluid balance without direct inflammation of the pleura.

i. Cardiac Causes

- a) Congestive Heart Failure: Elevated hydrostatic pressure in the pulmonary circulation due to left-sided heart failure leads to fluid leakage into the pleural space. In this case, it is often bilateral; typically, more prominent on the right side in severe cases.
- b) Cor pulmonale: Right-sided heart failure secondary to chronic lung disease, causing fluid accumulation.

ii. Hepatic Causes

- a) Cirrhosis: Decreased oncotic pressure due to hypoalbuminemia and portal hypertension leading to ascites and pleural effusion.
- b) Clinical Context: Often associated with ascites; right-sided pleural effusion is common.
- c) Hepatic Hydrothorax: Fluid from the abdominal cavity (ascites) translocates into the pleural space through diaphragmatic defects.

iii. Renal Causes

- a) Nephrotic Syndrome: significant protein loss in the urine leading to decreased oncotic pressure, resulting in fluid accumulation in the pleural space. It is often bilateral; associated with edema and ascites.
- b) Acute Kidney Injury/Chronic Kidney Disease: Fluid overload and/or hypoalbuminemia leading to transudative effusions.

iv. Other

- a) Hypoalbuminemia

- b) Peritoneal Dialysis...

Exudative Pleural Effusions

Exudative effusions are typically caused by local inflammation or malignancy that increases vascular permeability. They are often associated with disease processes that directly involve the pleura or nearby structures.

- i. Infectious Causes
 - a) Pneumonia (Parapneumonic Effusion): Infection of the lung parenchyma can spread to the pleural space, leading to inflammation and effusion.
 - b) Tuberculosis (TB): Mycobacterial infection causing granulomatous inflammation in the pleura.
 - c) Viral Infections: Mechanism: Viral infections (e.g., adenovirus, coxsackievirus) can cause pleuritis leading to effusion.
- ii. Malignant Causes
 - a) Lung Cancer: Tumor cells invade the pleura or lymphatic system, leading to fluid accumulation.
 - b) Metastatic Disease: Cancer spread from other organs to the pleura.
 - c) Lymphoma: Lymphoid tissue involvement in the pleura or mediastinum causing effusion.
- iii. Autoimmune and Inflammatory Causes
 - a) Rheumatoid Arthritis: Autoimmune inflammation affecting the pleura.
 - b) Systemic Lupus Erythematosus (SLE)
 - c) Asbestosis: Chronic exposure to asbestos leading to pleural fibrosis and effusion.
- iv. Post traumatic or post operative

III. DIAGNOSTIC PROCEDURE

F. Clinical Argument

Symptoms:

- Dyspnea: The most common, often progressive, depending on volume of fluid.
- Pleuritic Chest Pain: Sharp, stabbing pain that worsens with deep inspiration or coughing
- Cough: Non-productive cough, usually mild, resulting from lung irritation and reduced lung volume.
- Orthopnea: Difficulty breathing while lying flat, more common in transudative effusions related to heart failure.
- Fatigue: May result from the effort required to breathe and decreased oxygenation.

- Decreased Exercise Tolerance

Symptoms of complicated pleural effusion (e.g. Empyema...):

- Fever and Chills: Common in empyema or parapneumonic effusions.
- Weight Loss: Seen in chronic conditions such as tuberculosis or malignancy.
- Night Sweats: Often associated with infections like tuberculosis or malignancy.

Clinical Examination:

- Inspection:
 - Reduced chest wall movement on the affected side.
 - Tachypnea (rapid breathing).
- Palpation: Decreased Tactile Fremitus: Less vibration felt on the chest wall during speech due to fluid acting as a sound barrier.
- Percussion: Dullness to Percussion: Over areas of fluid collection.
- Auscultation:
 - Diminished or Absent Breath Sounds: Over areas with pleural fluid.
 - Pleural Rub: If there is inflammation of the pleura.

G. Technical Procedure

b. Imaging Study

iv. Chest radiography

- Posterior-Anterior (PA) View: May show blunting of the costophrenic angle with effusions > 200mL.
- Lateral Decubitus View: Confirms fluid mobility and quantifies smaller effusions.

v. Thoracic Ultrasound

- Advantages: Preferred method for detecting small effusions, guiding thoracentesis, and identifying septations in loculated effusions.
- Can differentiate solid masses from fluid and guide drainage procedures.
- Can differentiate solid masses from fluid and guide drainage procedures.

vi. Computed Tomography

- Indications: Suspected malignancy, loculated effusions, or complex pleural disease.
- Findings: Visualizes lung parenchyma, pleura, and mediastinal structures in detail.

- vii. Thoracentesis (Pleural Fluid Aspiration):
 - Indications: Diagnostic thoracentesis is performed to analyze pleural fluid when the etiology is unknown.
 - Procedure: Under ultrasound guidance, a needle is inserted into the pleural space to aspirate fluid.
- c. Pleural Fluid Analysis:
 - Biochemical Studies: Protein, Lactate Dehydrogenase (LDH), Glucose: Used to differentiate transudative from exudative effusions (Light's Criteria).
 - Microbiological Studies: Gram Stain and Culture: To identify infectious pathogens.
 - Cytological Examination: Essential for detecting malignant cells in cases of suspected cancer.
 - Special Tests:
 - Adenosine Deaminase (ADA): Elevated in tuberculous pleural effusions.
 - pH Measurement: Low pH (<7.2) in parapneumonic effusion suggests the need for chest tube drainage.

IV. DIFFERENTIAL DIAGNOSIS

- Chylothorax
- Pneumothorax
- Hemothorax
- Cardiac tamponade

V. THERAPEUTICAL APPROACH

H. Treatment of Underlying Cause

- a. Transudative Effusions:
 - Heart Failure: Diuretics, ACE inhibitors, or beta-blockers.
 - Cirrhosis: Sodium restriction, diuretics, and therapeutic paracentesis if necessary.
- b. Exudative Effusions:
 - Parapneumonic Effusion: Antibiotics and drainage if complicated.
 - Empyema: Urgent drainage with chest tube and appropriate antibiotic therapy.
 - Malignant Effusion: Therapeutic thoracentesis, pleurodesis, or indwelling pleural catheter.
 - Tuberculosis Effusion: Anti-tuberculous therapy

I. Therapeutic Intervention:

- Thoracentesis: For symptom relief in large effusions, especially in patients with dyspnea.
- Chest Tube Insertion: Indicated in cases of empyema, complicated parapneumonic effusions, or large symptomatic effusions.
- Pleurodesis: Indicated for recurrent malignant pleural effusions. Chemical agents such as talc are commonly used.
- Indwelling Pleural Catheters: Consider for chronic or recurrent malignant effusions in palliative care settings.

J. Intrapleural Therapy

Intrapleural Fibrinolytic Therapy (IPFT): Used for complicated parapneumonic effusions and empyema, especially if chest tube drainage is insufficient. Fibrinolytics, such as tissue plasminogen activator (tPA), help break down fibrinous adhesions, while deoxyribonuclease (DNase) reduces the viscosity of pus. This combination has been shown to improve drainage and outcomes in patients with loculated pleural effusions.

- Indications: Patients with loculated effusions, failure of simple chest tube drainage, and complex parapneumonic effusions.
- Dosing: Common regimen involves intrapleural administration of 10 mg of tPA and 5 mg of DNase twice daily for 3 days.
- Monitoring: Regular imaging (e.g., ultrasound) to assess resolution of the effusion.

K. Surgical Management

a. Indication

Surgical interventions are typically reserved for cases where medical management is insufficient, such as in empyema, loculated effusions, or recurrent pleural effusions, particularly in malignant cases.

- Empyema (Infected Pleural Effusion): Indicated when antibiotics and chest tube drainage fail.
- Loculated Effusions: Surgical drainage is necessary when fibrinolytic therapy is ineffective.
- Malignant Pleural Effusion: For recurrent symptomatic effusions not responsive to therapeutic thoracentesis.

b. Types of Surgical Procedures

- Video-Assisted Thoracoscopic Surgery
- Thoracotomy
- Pleurodesis

- Indwelling Pleural Catheter (IPC): Placement of a catheter that allows repeated drainage of pleural fluid at home.
- Decortication
- Rib resection: Empyema and Complex Parapneumonic Effusion or TB.

L. Monitoring and Follow-Up

- Radiological Monitoring: Repeat chest X-ray or ultrasound to assess response to treatment.
- Clinical Follow-up: Evaluate symptom resolution, need for further intervention, and underlying disease management.

VI. ALGORITHM

Step	Action	Details/options
1. Initial Assessment	Clinical Evaluation	Assess symptoms, history, and physical examination.
2. Imaging Studies	Obtain Chest X-ray	Look for signs of pleural effusion; use PA and lateral decubitus views.
	Perform Thoracic Ultrasound	Guide thoracentesis; assess effusion size and characteristics.
	Consider CT scan if indicated	For complex cases, loculated effusions, or malignancy.
3. Diagnostic Thoracentesis	Perform Thoracentesis	Sample pleural fluid for analysis; guide with ultrasound if needed.
4. Pleural Fluid Analysis	Analyze Fluid	Determine if transudative or exudative using Light's Criteria.
	Transudative Effusion	Investigate systemic causes (e.g., heart failure, cirrhosis).
	Exudative Effusion	Investigate local causes (e.g., infection, malignancy).
5. Treatment Based on Etiology	Transudative	Treat underlying condition (e.g., heart failure management, diuretics).
	Exudative	Address underlying cause (e.g., antibiotics for infection, chemotherapy for malignancy).
6. Recurrent or Complex Cases	Consider Additional Procedures	If symptomatic despite treatment, or if fluid is loculated or complicated.
	Pleurodesis	For recurrent malignant or chronic effusions.
	Indwelling Pleural Catheter (IPC)	For recurrent malignant effusions, especially in palliative care.
	<u>Surgical Intervention</u>	
	- VATS	For drainage, debridement, or pleurodesis.
	- Thoracotomy with Decortication	For thickened pleura or trapped lung.
	- Rib Resection (if indicated)	Consider in cases of trapped lung or complex empyema.
7. Follow-Up	Monitor and evaluate	Regular follow-up to assess response to treatment and detect recurrence.

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MANAGEMENT OF PNEUMOTHORAX

Tuo Sothunea, Duong Chhay, Lay Nhork

I. CASE DEFINITION

Pneumothorax is defined as presence of air in pleural cavity (between the lung at chest wall) [1]. It is an urgent situation that must be treated immediately upon diagnosis.

II. ETIOLOGY

On each sized of chest cavities, there are two membranes: parietal and visceral pleura. Between the two layers, there is a small amount of lubricating serous fluid which is created in a virtual space. The negative pressure inside the pleural cavity helps the lungs to be fully inflated.

Pneumothorax can develop in different conditions. It can appear after damage to the chest wall or damage to the lung itself. Sometimes, it can be caused by production of gas by microorganisms in the pleural space.

There are 2 types of pneumothoraces [2] [3]. The pneumothorax can be spontaneous or traumatic.

- Spontaneous pneumothorax:
 - Primary spontaneous pneumothorax (PSP) is considered the one that occurs without an apparent cause and in the absence of significant lung disease. Until now the cause of primary spontaneous pneumothorax has not been identified, however; several risk factors have been identified such as smoking, male sex, and a family history of pneumothorax.
 - Secondary spontaneous pneumothorax (SSP) occurs in the presence of existing lung pathology such as chronic obstructive pulmonary disease, tuberculosis, necrotizing pneumonia, pneumocystis carinii, lung cancer, sarcoma, sarcoidosis, endometriosis, cystic fibrosis, acute severe asthma, idiopathic pulmonary fibrosis, Rheumatoid arthritis, ankylosing spondylitis, polymyositis and dermatomyositis, systemic sclerosis, Marfan's syndrome and Ehlers-Danlos syndrome, histiocytosis X and lymphangioleiomyomatosis (LAM).
- Traumatic pneumothorax occurs when the chest wall is pierced, such as when a stab wound, or gunshot wound allows air to enter the pleural space. Sometimes pneumothorax can be caused accidentally during a procedure which is called an iatrogenic pneumothorax [4]

III. DIAGNOSTIC PROCEDURE

1. Clinical Argument

The symptoms of PSP such as chest pain and dyspnea may be minimal or absent. But in SSP, symptoms are greater even if the size of pneumothorax is relatively small

In PSP, the symptoms can be delayed several days after the onset. The longer of this period, the greater is the risk of re-expansion pulmonary oedema.

In SSP, most patients could experience breathlessness which is out of proportion of the size of pneumothorax.

When severe symptoms accompanied by signs of cardiorespiratory distress, tension pneumothorax must be considered.

Physical examination can find reduced lung expansion, hyper-resonance and diminished breath sounds on the side of pneumothorax. In case of tension pneumothorax, there can be some critical signs such as cyanosis, sweating, severe tachypnea, tachycardia, and hypotension.

2. Technical Procedure

A. Imaging Study

Standard erect chest x-rays in inspiration are recommended in initial diagnosis. Besides this, there are several modalities which are used for diagnosis and management of pneumothorax such as lateral x-ray, supine and lateral decubitus x-ray, and chest ultrasound [1]. Sometimes, in case of uncertain or complex pneumothorax, CT scanning is preferred to estimate the size and cause of pneumothorax.

- Standard erect PA chest x-ray is used to find displacement of pleural line and sometimes air-fluid level at costophrenic angle (50% of cases). The presence of bullous lung disease can lead to misdiagnose the pneumothorax. In this case, the CT scanning should be indicated.
- Lateral x-ray is used to detect suspected pneumothorax which cannot be found in PA chest film. But this is less used now because of the less reliability.
- Supine and lateral decubitus is less sensitive than erect PA x-ray. This is mostly employed for trauma patient who cannot be safely moved.
- Chest ultrasound is commonly used in the management of supine trauma patients. This technique is performed during Focused Assessment with Sonography in Trauma (FAST).
- CT scanning is the gold standard in the detection of small size pneumothorax and its cause.

During imaging study, there should be a precision of pneumothorax size. In erect PA x-ray, 2 cm of depth is considered as a large pneumothorax. There are different guidelines to measure this depth. According to British Guidelines, it is measured from the chest wall to the outer edge of lung at the hilum level.

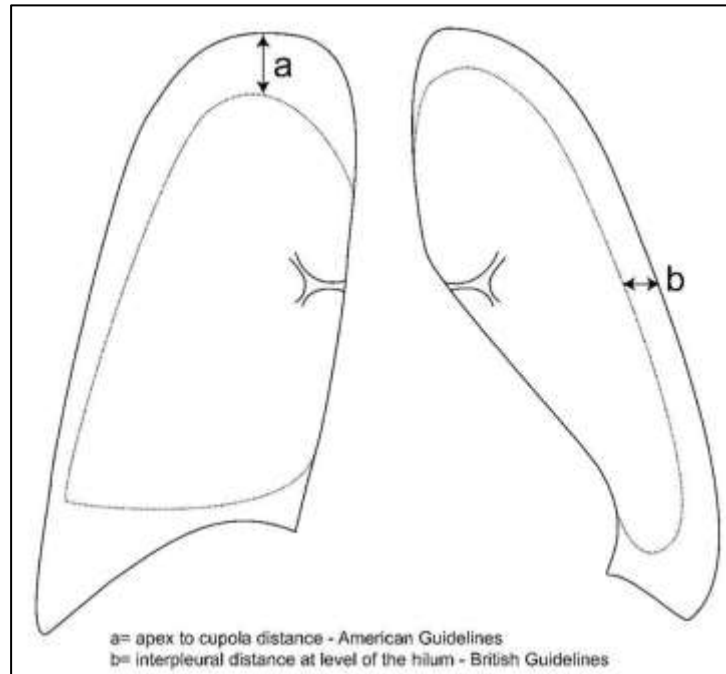


Figure 1: Depth of pneumothorax

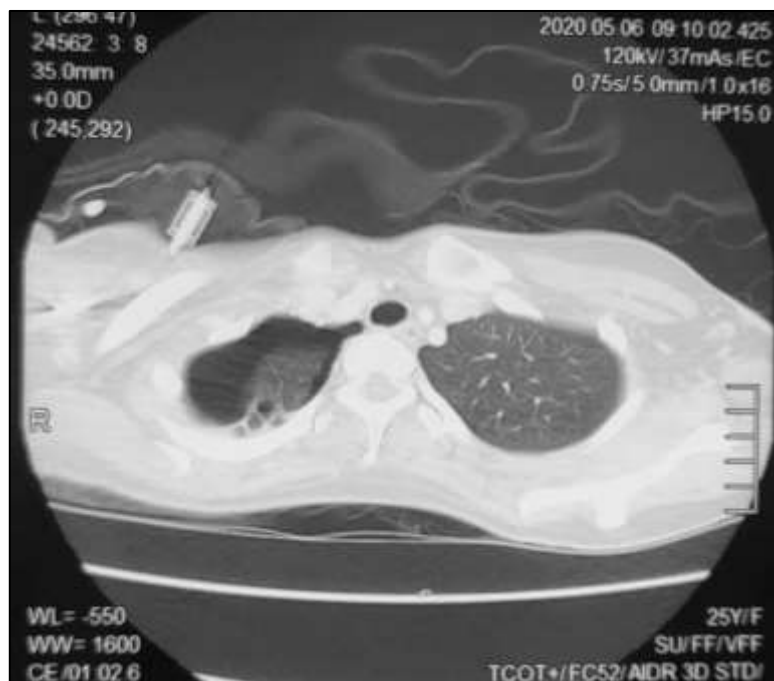


Figure 2: Bullous of the right apical lung

B. Other Test

Arterial blood gas measurements are frequently abnormal in patients with pneumothorax, with the arterial oxygen tension (PaO₂) being <10.9 kPa in 75% of patients, 31 but are not required if the oxygen saturations are adequate (>92%) on breathing room air.

Pulmonary function tests cannot predict the presence or size of pneumothorax and are generally avoided in this situation.

IV. DIFFERENTIAL DIAGNOSIS

With the respiratory symptoms there are a lot of differential diagnosis of a pneumothorax such as pneumonia, acute asthma exacerbation, bronchitis, pulmonary embolism, aortic dissection, costochondritis, acute coronary syndrome, anxiety or panic attack, diaphragmatic injuries, GERD, esophageal spasm, Mallory-Weiss tear, Boerhaave's syndrome, mediastinitis, myocarditis, pericarditis, pleurodynia, tuberculosis, pulmonary empyema, lung abscess [5].

V. THERAPEUTICAL APPROACH

For an appropriate management of pneumothorax, there should a distinction between PSP and SSP since the patients with pre-existing disease have less tolerance to pneumothorax.

Breathlessness indicates the need for active intervention as well as supportive treatment (including oxygen). The size of the pneumothorax determines the rate of resolution and is a relative indication for active intervention [1].

There are several treatment options of pneumothorax:

- Observation
- Supportive treatment
- Needle aspiration (14 – 16 G)
- Chest drainage
- Surgery

During drainage, continued bubbling of air through a chest drain after 48h is defined as persistent air leak. The persistent air leak should be discussed with a thoracic surgeon at 48h.

It has been suggested that optimal suction should entail pressures of -10 to -20 cm H₂O (compared with normal intrapleural pressures of between -3.4 and -8 cm H₂O, according to the respiratory cycle). High-pressure high-volume suction may lead to air stealing, hypoxemia, or the perpetuation of air leaks.

Complex drain management is best affected in areas where specialist medical and nursing expertise are available.

i. Observation

In case of pneumothorax depth is less than 2 cm or asymptomatic PSP, observation is the treatment of choice.

ii. Supportive treatment

Oxygen, painkillers, and physiotherapy are included in supportive management of pneumothorax.

iii. Needle aspiration

There are some indications of needle aspiration [6]:

- Primary spontaneous pneumothorax greater than 2cm (at level of hilum) +/- breathlessness
- Primary spontaneous pneumothorax under 2cm but symptomatic
 - BTS guidelines advise 2 attempts at aspiration can occur
 - If the primary pneumothorax is less than 2cm and the patient is asymptomatic re-image in 2-3 weeks
- Tension pneumothorax

Equipment required for pneumothorax aspiration (needle thoracocentesis):

- Sterile field
- Sterile dressing pack and gloves
- Antiseptic solution
- Analgesia
 - 4mls of 1% or 2% Lidocaine
 - Orange (25G) needle (x1)
 - Green (19G) needle (x1)
 - 5ml Syringe (x1)
- 16-18 gauge cannula
- 3-way tap
- 50 ml syringe

Pre-procedure:

- Gain consent from the patient – ideally consent should be written
- Consent for pain, failure of procedure, bleeding, infection, damage to surrounding structures
- Familiarise yourself with the landmarks
- 2nd intercostal space in the midclavicular line (above rib to avoid neurovascular bundle on posterior aspect of second rib)
- Mark area if necessary
- Set up a sterile trolley with equipment

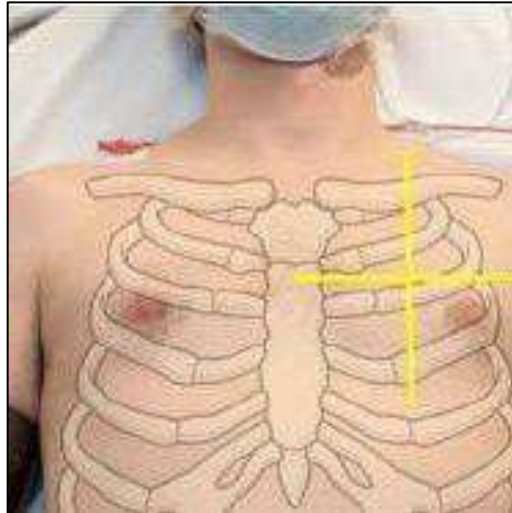


Figure 3: Site of needle aspiration

Procedure for pneumothorax aspiration (needle thoracocentesis):

- Don sterile gloves and clean area with chlorhexidine
- Apply sterile field
- Insert lignocaine 5-10ml initially under the skin and then into subcutaneous tissues and pleural space.
- Air should be aspirated with green needle and local anaesthetic
- Take cannula and insert at 90 degrees in second intercostal space, midclavicular line
- Begin to remove needle once at the depth air was aspirated with the green needle but advance plastic sheath to the hilt. Be cautious not to kink the cannula
- Attach three way tap to cannula and then syringe to three way tap
- Aspirate air via three way tap, ensuring tap is closed whenever the syringe is removed so no air is allowed into the chest.
- Aim to aspirate until resistance o Usually this is less than 2.5L o Remove cannula
- Apply dressing Post-procedure care
- Analgesia
- Chest X-ray
- If pneumothorax remains greater than 2cm, proceed to a further attempt at aspiration.
- If second attempt fails to resolve pneumothorax, seek senior respiratory review with a view to chest drain insertion.
- If pneumothorax remains but is less than 2 cm and the patient's symptoms have improved they can be discharged with future follow up arranged.
- Patients who have suffered a pneumothorax must be advised that they can never

do diving (due to pressure changes) and that they cannot fly for at least two weeks after full resolution of the pneumothorax.

- The diving rule can only be overturned if the patient undergoes chemical pleurodesis.
- Advise patients who smoke to stop.
- Smokers have an increased risk of recurrence.

iv. Chest drainage

The intercostal drain insertion or chest drainage is usually indicated after needle aspiration. These are other indications of chest drainage in pneumothorax [7]:

- Unresolved primary pneumothorax greater than 2 cm after 2 attempts at aspiration.
- Secondary pneumothorax greater than 2 cm.
- Tension pneumothorax after needle decompression.
- To facilitate pleurodesis

Equipment required for an intercostal drain (chest drain/pleural drain):

- Sterile ultrasound sheath
- Sterile field
- Sterile dressing pack and gloves
- Antiseptic solution
- Analgesia
 - 4mls of 1% or 2% Lidocaine
 - Orange (25G) needle (x1)
 - Green (19G) needle (x1)
 - 5ml Syringe (x1)
- Chest drain tubing and bottle.
- Sterile water/saline
- Suture kit
- Sterile dressing

Pre-procedure set-up:

- Set up sterile trolley
- Prepare drain
- Pour sterile water/saline into chest drain bottle up to the prime line
- Attach chest drain tubing ensuring the end stays within the package and sterile

- Review imaging and examine patient to confirm side of insertion
- It is advised to have a nurse and a helper to assist during the procedure
- Position the patient with leaning forward with arms outstretched or sat at 90 degrees with arm lifted and hand resting behind their head. In elderly or frail patients, the to help

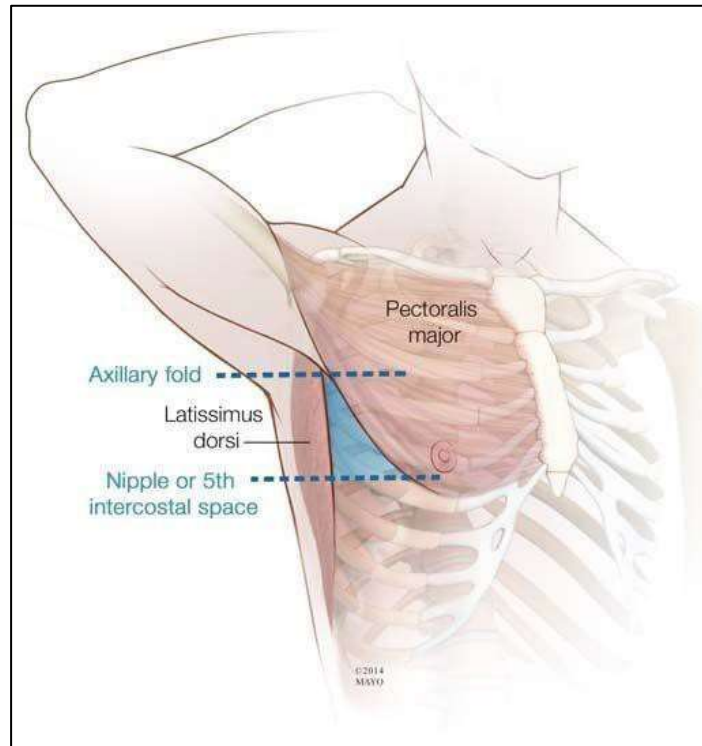


Figure 4: Safe triangle for pleural drainage

Procedure for intercostal drain insertion (using of Seldinger chest drainage kit):

- Wash hands and don sterile gown and gloves
- Clean insertion site: either the site identified by ultrasound or – for pneumothorax – insert drain in the “safe triangle”
- Lower border of axilla to the 5th intercostal space; the lateral edge of pectoralis major and the lateral edge of latissimus dorsi
- Apply sterile field
- Insert Lidocaine cutaneously, subcutaneously and then into the pleural space.
- Fluid or air should be able to be aspirated with the green needle
- Take the Seldinger needle and attach this to the 10ml syringe provided
- Insert needle in the same plane as the lignocaine, aspirating as you advance. Insert needle to the same distance as air was aspirated with the
- green needle. Once air is aspirated inset 0.5cm further and confirm ongoing air aspiration
- Remove the 10ml syringe ensuring you place your thumb over the open needle

- Take the Seldinger wire and insert through the needle. Ensure you hold the wire and needle at all times
- Remove Seldinger needle over the wire
- Take scalpel and make a 0.5cm incision in the skin
- Scalpel sharp edge should always be facing away from the wire
- Take the Seldinger dilator and pass it over the wire, gently but firmly insert the dilator over the wire through the skin and intercostal muscles
- Warn the patient they will feel some pushing
- Do not be too forceful as this will kink the wire
- If the dilator is not advancing it may indicate you are pushing in the wrong plane or against bone
- Once dilated remove the dilator and pass the chest drain over the wire
- Ensure that you have a hold of the wire out the end of the drain before advancing
- Insert the drain over the wire and remove the wire
- Attach three way tap to the drain and ensure it is closed
- Then confirm air or fluid aspiration with a syringe via the 3-way tap
- Close 3-way tap once position confirmed and suture drain in place
- This needs to be firm but not pinch the skin or occlude the drain
- Dress the drain so the insertion sight is visible
- Attach drain to chest drain tubing

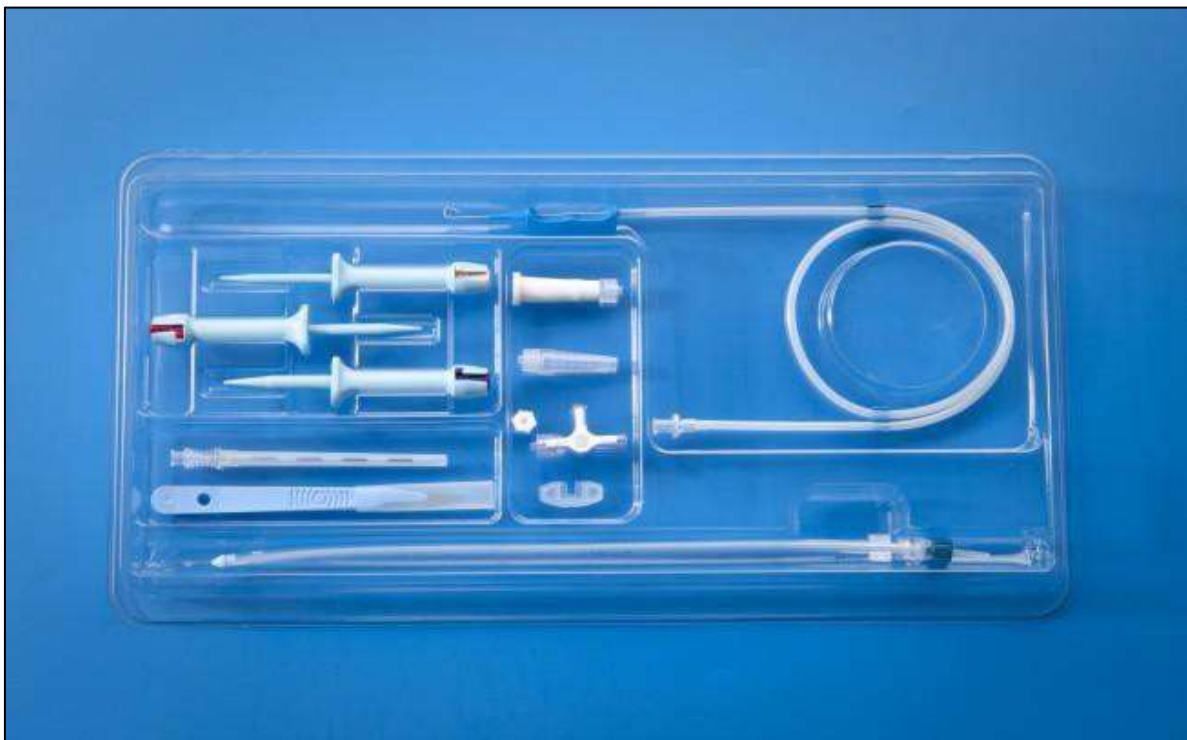


Figure 5: Seldinger chest drainage kit

Procedure for Joly tube insertion:

- Wash hands and don sterile gown and gloves
- Clean insertion site: either the site identified by ultrasound or – for pneumothorax – insert drain in the “safe triangle”
- Lower border of axilla to the 5th intercostal space; the lateral edge of pectoralis major and the lateral edge of latissimus dorsi
- Apply sterile field
- Insert Lidocaine cutaneously, subcutaneously and then into the pleural space.
- Fluid or air should be able to be aspirated with the green needle
- Skin incision for a size of the tube.
- Fat and muscles dissection with Kelly just above the rib
- Entering the pleural cavity by piercing of parietal pleura. Try to avoid not to force inside because this can cause lung injury.
- Remember the pathway which have been created.
- Insertion of the Joly tube by hold the external tip with dominant hand.
- With non-dominant hand, holding 4-5cm from the internal tip to prevent accidental intrathoracic injury while pushing the tube.
- After position 10-15 cm of the tube inside, connecting drain to chest drain tubing.
- Fixation of the tube to the skin
- Dressing

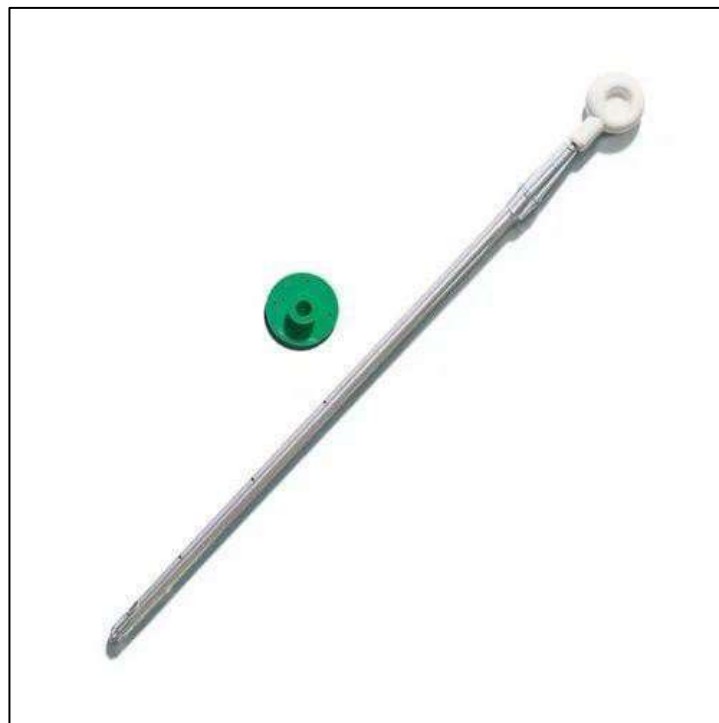


Figure 6: Joly chest tube Post-

procedure

- Analgesia
 - Post procedure chest x-ray
 - Document procedure clearly and document length of drain inserted
 - Advise patient to always hold drain bottle below level of insertion
 - Respiratory review and advise on onward management
- v. Surgery: Open thoracotomy or video-assisted thoracic surgery (VATS) Surgical indication of pneumothorax [1]:
- Second ipsilateral pneumothorax.
 - First contralateral pneumothorax.
 - Synchronous bilateral spontaneous pneumothorax.
 - Persistent air leak (despite 5-7 days of chest tube drainage) or failure of lung re-expansion.
 - Spontaneous haemothorax.
 - Professions at risk (e.g. pilots, divers).
 - Pregnancy.

Objectives of surgery in pneumothorax:

- To treat the pneumothorax (e.g. remove the bullous lung)
- To prevent the recurrent (pleurodesis: chemical or mechanical)

The types of intervention:

- Pleurodesis
- Pleural abrasion
- Pleurectomy
- Bullectomy



Figure 7: Bullous lung

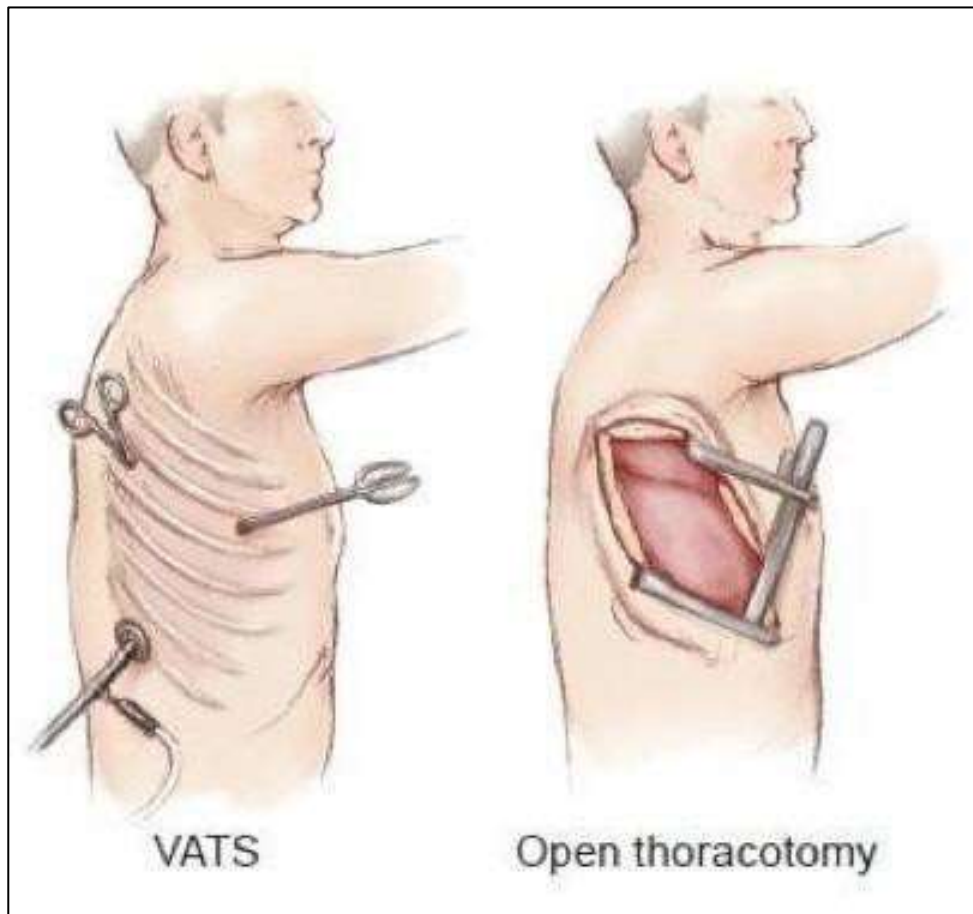


Figure 8: VATS and open thoracotomy

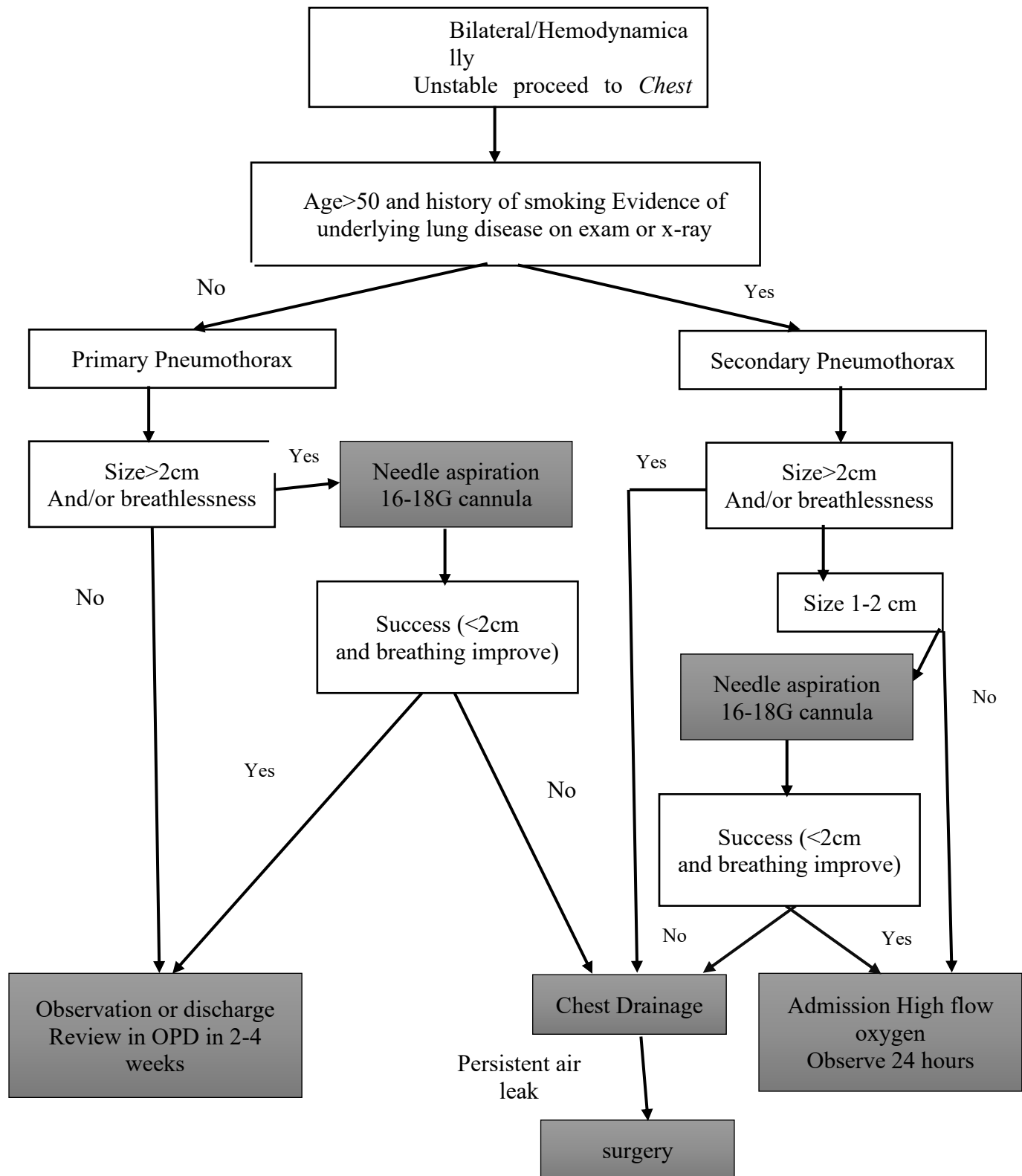
vi. Management of subcutaneous emphysema

Subcutaneous emphysema is defined as presence of gas under the skin. It is result of imbalance between a large air leak and a relatively small-bore chest drain. This can be caused by a malpositioned, kinked, blocked, or clamped chest drain. It will subside spontaneously after a few days. It can lead to airway obstruction or chest compression which cause respiratory compromise. In this condition, the patient may need tracheostomy, skin incision decompression and insertion of large-bore chest drains have all been used.

vii. Monitoring

- Monitoring
- Degree of dyspnea and chest pain
- Chest x-ray
- Bubbling of air through drain
- Quantity of drained liquid

VI. ALGORITHM



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RECTAL TUMOR

Mam Keovorchira

I. CASE DEFINITION

Rectal cancer is one of the frequent human malignant neoplasms and the second most common cancer in large intestine.

Colorectal cancers (CRCs) are the second most common cancers in human and major public health problems worldwide [1].

II. ETIOLOGY

Factors that may increase the risk of rectal cancer are the same as those that increase the risk of colon cancer. Colorectal cancer risk factors include:

- A personal history of colorectal cancer or polyps. Your risk of colorectal cancer is higher if you've already had rectal cancer, colon cancer or adenomatous polyps.
- Black race. Black people in the United States have a greater risk of colorectal cancer than do people of other races.
- Diabetes. People with type 2 diabetes may have an increased risk of colorectal cancer.
- Drinking alcohol. Heavy drinking increases the risk of colorectal cancer.
- A diet low in vegetables. Colorectal cancer may be related to a diet low in vegetables and high in red meat.
- Family history of colorectal cancer. You're more likely to develop colorectal cancer if you have a parent, sibling or child with colon or rectal cancer.
- Inflammatory bowel disease. Chronic inflammatory diseases of the colon and rectum, such as ulcerative colitis and Crohn's disease increase your risk of colorectal cancer.
- Inherited syndromes that increase colorectal cancer risk. In some families, DNA changes passed from parents to children may increase the risk of colorectal cancer. These changes are involved in only a small percentage of rectal cancers. Inherited syndromes may include familial adenomatous polyposis, also known as FAP, and Lynch syndrome.
- Genetic testing can detect these and other, rarer inherited colorectal cancer syndromes.
- Obesity. People who are obese have an increased risk of colorectal cancer compared with people considered to be at a healthy weight.
- Older age. Colorectal cancer can be diagnosed at any age, but most people with this type of cancer are older than 50. The rates of colorectal cancer in people younger

than 50 have been increasing, but healthcare professionals aren't sure why.

- Radiation therapy for previous cancer. Radiation therapy directed at the abdomen to treat previous cancers may increase the risk of colorectal cancer.
- Smoking. People who smoke may have an increased risk of colorectal cancer.
- Too little exercise. If you're inactive, you're more likely to develop colorectal cancer.
- Getting regular physical activity may reduce your risk of cancer [2].

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

Although a large number of asymptomatic cases in early stages are diagnosed as a result of current screening programs worldwide, a significant number of cases are diagnosed after the onset of symptoms. Extension of a rectal tumor into adjacent organs or into the lumen of gastrointestinal tract leads to symptomatic presentation. Therefore, symptoms usually reflect at least a locally advanced cancer.

Rectal bleeding is the most common presentations of rectal cancer. In later stages of the disease, other symptoms such as tenesmus, incomplete stool evacuation, diminished caliber of stools cramping, pelvic and rectal pain or obstructive symptoms might present [3]. Hematochezia and change in bowel habit are more common in rectal cancers and left-sided CRCs.

Abdominal pain can occur in both left- and right-sided tumors. It can be a symptom of partial obstruction, peritoneal dissemination of the tumor, intestinal perforation or even peritonitis. Patients suffering from metastatic rectal cancer may present with clinical symptoms referable to their metastatic site. Based on the venous drainage of the upper rectum via the portal system, most common site of hematogenous metastasis is liver, followed by the lungs and bone; however, distal rectum drains into the inferior rectal vein (and then into the inferior vena cava) and it may metastasize initially to the lungs [4-5].

In rare situations, rectal tumors can also present emergently with intestinal obstruction, acute gastrointestinal bleeding or peritonitis following its perforation into the peritoneal cavity. Fistula formation into adjacent organs (such as bladder), fever of unknown origin, abscesses (due to a localized perforated cancer), bacteremia or sepsis (due to *Streptococcus bovis* or *Clostridium septicum*) has also been reported as other rare presentations [6-7].

ii. Technical procedure

a. Laboratory test

Laboratory tests include:

- **Complete blood count.** Also known as CBC, this test reports the numbers of different types of cells in the blood. A CBC shows whether your red blood cell count is low, called anemia. Anemia suggests that the cancer is causing blood loss. A high level of white blood cells is a sign of infection. Infection is a risk if a rectal cancer grows through the wall of the rectum.

- Blood tests to measure organ function. A chemistry panel is a blood test to measure levels of different chemicals in the blood. Worrying levels of some of these chemicals may suggest that cancer has spread to the liver. High levels of other chemicals could mean problems with other organs, such as the kidneys.
- Carcinoembryonic antigen(CEA) and carbohydrate antigen 19-9 (CA 19-9). Cancers sometimes produce substances called tumor markers. These tumor markers can be detected in blood. Due to their low sensitivity
- for early- stage disease and possibility to increase in non-cancer medical conditions, they cannot be used as a screening or diagnostic test for CRCs [8][9].

b. Imaging study

Colonoscopy:

Colonoscopy is a test to look at the colon and rectum. It uses a long, flexible tube with a camera at the end, called a colonoscope, to show the colon and rectum.

Biopsy:

A biopsy is a procedure to remove a sample of tissue for testing in a lab. To get the tissue sample, a healthcare professional passes special cutting tools through a colonoscope. The health professional uses the tools to remove a very small sample of tissue from inside the rectum. The tissue sample is sent to a lab to look for cancer cells. Other special tests give more details about the cancer cells. Your healthcare team uses this information to make a treatment plan.

Computed tomographic colonography (CTC) or virtual colonoscopy

CTC is non-invasive and safe diagnostic tool. Not only CTC provides endoluminal visualization of the colon and rectum, but also it enables the examination of extra colonic organs [10].

CT scan of the chest, abdomen and pelvis

CT scan is a helpful staging test for identifying local and distant metastasis and evaluation of tumor-related complications (such as obstruction, perforation or fistula formation) [11].

MRI of the pelvis

An MRI provides a detailed image of the muscles, organs and other tissues surrounding cancer in the rectum. An MRI also shows the lymph nodes near the rectum and different layers of tissue in the rectal wall more clearly than a CT does [2]. According to a meta-analysis of 21 studies, published by Al-Sukhni et al, [12] MRI had 87% and 77% sensitivity for evaluation of the tumor size and nodal involvement, respectively. Its specificity was 75% for the

size and 71% for the nodal status. Based on another meta-analysis, done by Nickel et al, [13] MRI has also been recommended as the preferred first-line imaging study for evaluating CRC liver metastases in patients who have not previously undergone therapy.

Staging

TNM (Tumor, Node and Metastasis) definition of colorectal cancers

Primary tumor (T)	
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	Carcinoma in situ: intraepithelial or invasion of lamina propria
T1	Tumor invades submucosa
T2	Tumor invades muscularis propria
T3	Tumor invades through the muscularis propria into pericorectal tissues
T4a	Tumor penetrates to the surface of the visceral peritoneum
T4b	Tumor directly invades or is adherent to other organs or structures
Regional lymph nodes (N)	
NX	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastasis
N1	Metastasis in 1-3 regional lymph nodes
N1a	Metastasis in one regional lymph node
N1b	Metastasis in 2-3 regional lymph nodes
N1c	Tumor deposit(s) in the subserosa, mesentery, or nonperitonealized pericolic or perirectal tissues without regional nodal metastasis
N2	Metastasis in four or more regional lymph nodes
N2a	Metastasis in 4-6 regional lymph nodes
N2b	Metastasis in seven or more regional lymph nodes
Distant metastasis (M)	
M0	No distant metastasis
M1	Distant metastasis
M1a	Metastasis confined to one organ or site (eg. Liver, lung, ovary, nonregional node)
M1b	Metastases in more than one organ/site or the peritoneum

Staging of colorectal cancers

Stage	T	N	M
0	Tis	N0	M0
I	T1	N0	M0
	T2	N0	M0
IIA	T3	N0	M0
IIB	T4a	N0	M0
IIC	T4b	N0	M0
IIIA	T1-2	N1/N1c	M0
	T1	N2a	M0
IIIB	T3-T4a	N1/N1c	M0
	T2-T3	N2a	M0
	T1-T2	N2b	M0
IIIC	T4a	N2a	M0
	T3-T4a	N2b	M0
	T4b	N1-N2	M0
IVA	Any T	Any N	M1a
IVB	Any T	Any N	M1b

IV. DIFFERENTIAL DIAGNOSIS

Colorectal cancer must be differentiated from other diseases that cause lower abdominal pain and fever like appendicitis, diverticulitis, inflammatory bowel disease, cystitis, and endometritis [14][15][16][17][18].

V. THERAPEUTIC APPROACH

Different types of treatment modalities have been proposed for patients with rectal cancer. Surgery is the mainstay of treatment for cases with resectable rectal cancers. According to the location of the tumor and the stage, surgical resection can be performed as the sole treatment modality or in combination with other neoadjuvant and/or adjuvant therapies [19].

Procedures used for rectal cancer may include:

- **Transanal local excision, removing very small cancers from the inside of the rectum.**

Very small rectal cancers may be removed using a colonoscope or another specialized type of scope inserted through the anus. Surgical tools can be passed through the scope to cut away the cancer and some of the healthy tissue around it.

This procedure might be an option if cancer is small and not likely to spread to nearby lymph nodes. If a lab exam of cancer cells shows that they are aggressive or more likely to spread to the lymph nodes, additional surgery may be needed.

- **Low anterior resections (sphincter- sparing procedures):**

A low anterior resection colon and rectum is resected to a level where the distal

margin is free of tumor followed by a primary anastomosis between the descending colon and the distal rectum. For cancers located in the distal rectum without invasion to the anal sphincter, a very low anterior resection (VLAR) or ultra-low anterior resection (ULAR) with coloanal anastomosis have been recommended [20].

In order to protect the anastomosis, a temporary diverting ileostomy has been recommended if the anastomosis is low, under tension, presence of an air leak on proctoscopic testing, preoperative chemoradiation, or history of immunosuppressive medication[21][22].

➤ **Abdominoperineal resection (APR):**

is an accepted surgical approach for low rectal tumors not indicated for sphincter-sparing procedures. It includes the resection of sigmoid colon, rectum, and anus followed by a permanent colostomy. It is indicated when achieving a negative distal margin is not possible with the sphincter-sparing procedures or as a salvage procedure for local recurrence or locally advanced rectal tumors

Monitoring

c. Clinical monitoring

Postoperative follow-up and surveillance for colorectal cancer are:

- A visit to the clinician for clinical examination every 3 to 6 months for the first 3 years, and every 6 months in the 4th and 5th year.
- Computed tomography (CT) of the chest, abdomen, and pelvis yearly for at least 3 years.
- Colonoscopy at 1 year and subsequently at 3 to 5 years intervals (provided a full length colonoscopy was done prior to surgery, if not, then a colonoscopy at 3 months post operatively).
- Flexible proctosigmoidoscopy every 6 months for 3 to 5 years in patients with rectal cancers and low anterior resection who have not received pelvic irradiation.

d. Laboratory monitoring

CEA testing can be helpful in monitoring your response to treatment [2].

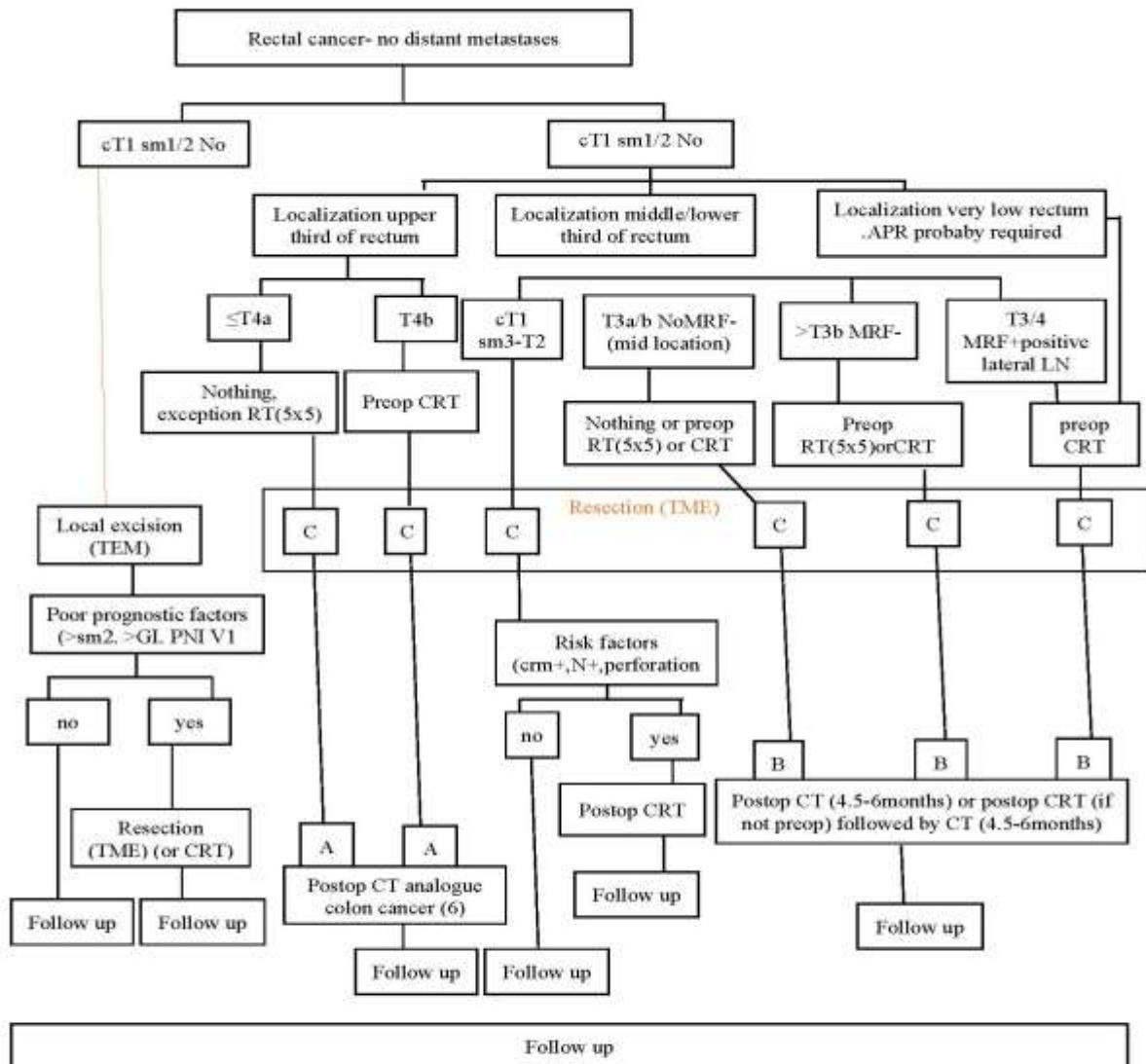
VI. COMPLICATION

Rectal cancer can lead to complications, including [2]:

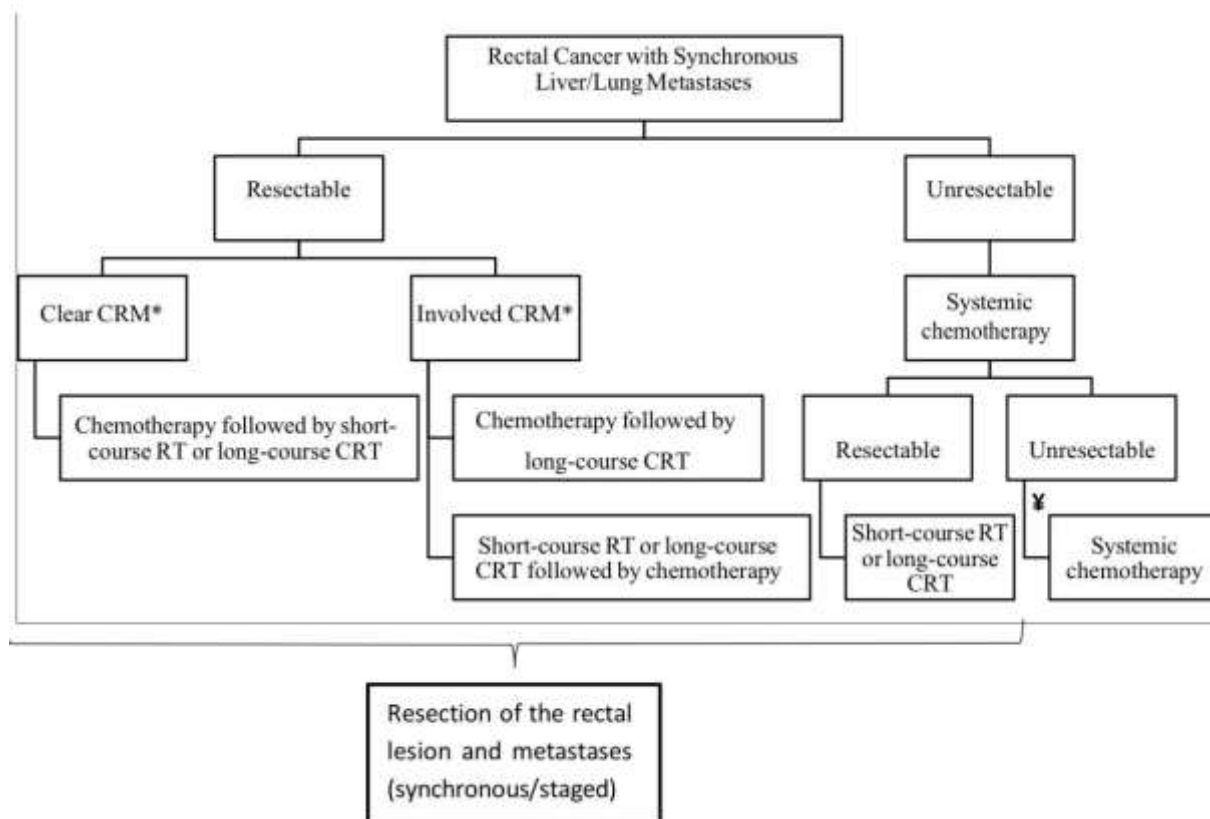
- Bleeding in the rectum. Rectal cancer often causes bleeding in the rectum. Sometimes the amount of blood is worrying, and treatment might be needed to stop it right away.
- Intestinal obstruction. Rectal cancer can grow to block the intestines. This prevents stool from leaving the body. Surgery to remove the cancer often relieves an obstruction. If you can't have surgery right away, you might need other treatments to ease the blockage.
- Intestinal perforation. Rectal cancer may cause a tear in the intestines. Perforation typically requires surgery

VII. ALGORITHM

Treatment algorithm for localized rectal cancer [23]



Treatment Algorithm for Stage IV Rectal Cancer with Synchronous Liver/Lung Metastases [24].



CRM: Circumferential Resection Margin, CRT: Chemoradiotherapy, RT: Radiotherapy

*CRM is the closest distance of the tumor to mesorectal fascia. Involved CRM: Tumor deposits within 1 mm of mesorectal fascia or levator muscles (for lower rectal tumors), invasion into or beyond intersphincteric plane.

¥ If there is progression of the primary tumor, palliative RT or long course CRT can be considered.

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SPLEEN ABSCESS & SURGICAL MANAGERMENTS

Kaing Kimsan

I. CASE DEFINITION

Spleen abscess is a collection of infectious origin secondary to a hematogenous or contiguity infection. It's a rare condition potentially serious due to local complication, and the risk of infection caused by functional asplenia.

II. ETIOLOGY

- Hematogenous dissemination: the spleen's infected during sepsis, which can represent up to 75% of cases.
- The intrinsic theory: infection following an alteration of splenic structure/function (infarction, hematoma or hemoglobinopathy).
- The extrinsic theory: contiguity infection.

III. DIAGNOSTIC PROCEDURE

a) Clinical Examination

- General signs: Fever (37.5-38.5°C), Weight loss
- Functional sings: The average time between diagnosis and the start of symptoms is two weeks (ranging from 2days to 4weeks).
 - Trial of Sarr and Zuidema: fever, pain in the left hypochondrium and tender mass.
 - Vomiting
 - Nausea
- Physical examination:
 - Palpation: Mass in left hypochondrium (Splenomegaly)

b) Investigation

- Biology:
 - CBC: hyperleukocytosis > 10000/ml in 60 to 100% of cases.
 - CRP: high, it has good sensitivity and specificity for diagnosis.

o Bacterial ecology

The most frequently encountered microorganisms are Gram-positive cocci (GPC) (staphylococci and streptococci), enterobacteria, particularly salmonella, and anaerobes (mainly Bacteroides and Peptostreptococcus spp).

In addition, other germs can be involved such as *Pseudomonas* spp., *Brucella* spp., *Bartonella* spp., including in immunocompetent subjects, or even *Candida* spp. As well as mycobacteria particularly in the immunocompromised host. Other intracellular germs such as *Coxiella burnetii* can be incriminated in an exceptional way, apart from any endocarditis.

Finally, amoebic abscesses can also affect the spleen, in addition to the liver, in the consequences of a digestive infestation.

- Microbiology diagnosis:

The diagnosis is based on taking blood cultures which are positive in 24 to 80% of cases, in the case of multiple abscesses (70%) than a single lesion (14%).

Culture of the abscess after drainage has a better sensitivity, estimated at 50 to 80%. but requires puncture with a major risk of hemorrhagic complication. However, it has been shown that when the puncture is carried out by trained teams, under ultrasound control, it can prove profitable and low risk, for etiological diagnosis with a sensitivity of 75%

the often polymicrobial nature of these abscesses, observed in 28 to 55% of cases, the concordance of blood cultures with the culture of abscess pus would be 38%.

- Images

Ultrasound is abandoned in favor of CT scanner, whose sensitivity and specificity for the diagnosis of splenic abscess are higher, estimated at 95 and 92%.

The abscess appears as a hypodense image with discreet peripheral contrast enhancement. Sometimes this contrast enhancement can be missing, and it is the clinical context which allows us to suggest the diagnosis of abscess.

The scanner also makes it possible to carry out a guided puncture, at the cost of morbidity of 5% and mortality of less than 1% for trained teams.

The risks of surgical puncture are higher, with morbidity varying from 11 to 28% and mortality from 6 to 14%.

Possible complications following these procedures are pleural effusion (gas or liquid), hemorrhage, empyema and fistula

Chest X-ray shows nonspecific signs such as left pleural effusion or elevation of the left hemidiaphragm.

IV. DIFFERENTIAL DIAGNOSIS

- Splenic infarction
- Primary spleen hematoma or lymphoma
- Splenic peliosis
- Hepatosplenic candidiasis

V. THERAPEUTIC APPROACH

A lot of strategies are possible and sometimes complementary in the management of splenic abscesses. Most often, surgical or percutaneous drainage is associated with medical treatment with broad-spectrum antibiotics active on BGN, gram-positives and anaerobes, for 3 to 4 weeks according to the authors.

This treatment strategy remains non-consensual; it is based solely on clinical observations and not on data with a high level of evidence. The duration of treatment as well as the recommended antibiotics are similar to what is proposed for the management of liver abscesses.

In the literature, the main advantage of percutaneous drainage is to reduce the length of hospitalization and prevent the risk of intraperitoneal rupture, with an effectiveness depending on the series of 70 to 100%. Furthermore, according to some authors, antibiotic treatment alone is associated with mortality of 30 to 80%.

The advantage of percutaneous drainage with CT or ultrasound identification is to avoid the need for splenectomy, the immediate and longer-term risks of which are known (in particular the risk of serious infection linked to asplenia). This technique is, however, only recommended in cases of solitary abscess with a homogeneous, easily accessible wall and contents.

In the absence of a microbiological diagnosis, drainage should be favored when possible, but should be the subject of medical-radio-surgical consultation depending on the size of the abscess (particularly if it proves to be greater than 5 cm, by analogy with liver abscesses), of the patient's terrain, as well as the available technical platform.

The advantage of the surgical approach is that it can be completed immediately by a splenectomy.

The role of splenectomy remains debated and, according to some authors, is the only effective way to prevent the recurrence of splenic abscesses. However, more recent data increasingly report therapeutic success in the absence of splenectomy, which avoids the increased risk of death from post-splenectomy infection.

In addition, conservation of the spleen is particularly interesting in immunocompromised patients and children with moderate and incomplete functional asplenia. There is no "gold standard" in the treatment of splenic abscesses, particularly with regard to splenectomy.

In the event of splenectomy, vaccination to prevent infections with encapsulated germs should be offered, particularly against pneumococcus (50-80% of infections), *Haemophilus influenzae* (5-15%) and meningococcus.

Beta-lactams and fluoroquinolones are usually prescribed as monotherapy in intra-abdominal infections but there is currently no pharmacokinetic data concerning the diffusion of these antibiotics in the spleen. The few data supporting the use of these two classes of

antibiotics are based on studies conducted mainly in patients seen in surgery.

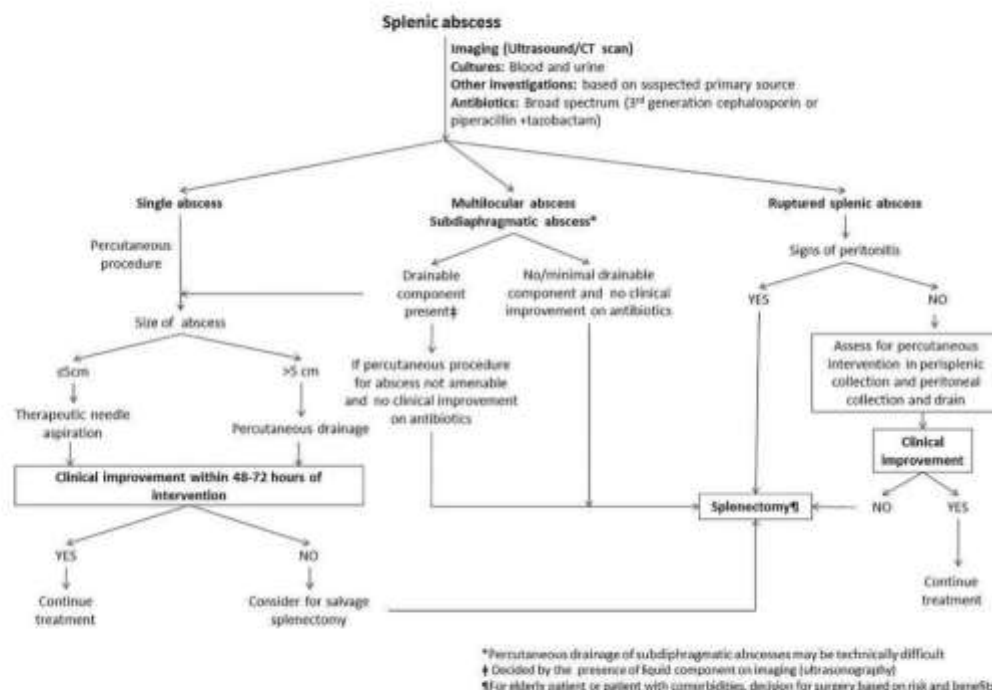
The combination of amoxicillin, clavulanic acid, cephalosporins and carba penemes have shown their effectiveness in the conservative treatment of uncomplicated acute appendicitis. The effectiveness of fluoroquinolones, including ciprofloxacin, has been established for the treatment of digestive infections. especially traveler's diarrhea.

In the event of community infection, in order to target the germs most frequently observed in these digestive infections, it seems legitimate to first propose a beta-lactam combination (injectable 3rd generation cephalosporin type, ex. cefotaxime) + metronidazole, or even amoxicillin + clavulanic acid. In case of allergy to beta lactams, or as an alternative treatment, monotherapy with levofloxacin seems just as appropriate.

VI. COMPLICATION

- Pneumothorax
- Atelectasis
- Left pleurisy
- Left subdiaphragmatic abscess
- Gastric, intestinal or colon perforation
- Pancreatic fistula
- Post splenectomy thrombocytosis
- Pneumonia
- Life-threatening hemorrhage
- Post splenectomy sepsis

VII.ALGORITHM



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THYROID DISEASES AND SURGERY

Preap Ley, Norng Sanareth, Koeu Kimsath, Cheav Darareak, Ong Visal

I. DEFINITION

Thyroid disease is common and results in many different conditions that can be treated by drugs and/or surgery-Thyroidectomy. Depending on the extent of thyroid disease, few types of surgery can be performed. Example:

- Hemithyroidectomy or Thyroid lobectomy (half of the thyroid gland is removed)
- Subtotal thyroidectomy (almost all the thyroid gland is removed)
- Total thyroidectomy (the entire thyroid gland is removed)

II. ETIOLOGY

Thyroid surgery (thyroidectomy) is performed for the following conditions:

- m. **Benign thyroid goiter:** An enlarged thyroid gland is known as a goiter and it may be diffuse or nodular. Its function may be normal (euthyroid or nontoxic goiter), overactive (hyperthyroidism or toxic goiter), or underactive (hypothyroid goiter or Hashimoto's disease). If the goiter is small, it is unlikely that surgery is required. In most cases, medication is ineffective in shrinking benign goiters. If the goiter becomes large and causes compression signs such as discomfort, vein compression, difficulty to breath or swallowing secondary to compresses the trachea or esophagus, surgery is necessary. In case of retrosternal goiter, even in the absence of obstruction, patient need thyroidectomy. Goiter classifications:

WHO Classification:

- Stage I: Slightly enlarged, detected then patient close to examining doctor
- Stage II: Clearly enlarged, detected easily then patient enters room
- Stage III: enormous goiter

Goiter (Pan American Health Organization)

- Lateral lobe volume > terminal phalanges of the thumb
- n. **Hyperthyroidism, Thyroidtoxicosis and Graves' Disease:** These conditions result from an overproduction of thyroid hormones by an overactive gland. These diseases are initially treated with medication. However, for patients who develop hyperthyroidism again after stopping medication, treatments are through should undergo radioactive iodine treatment or surgery.

- o. **Thyroid Cancer:** Malignant tumors of the thyroid gland are uncommon. An accurate method to assess a lump and determine whether it is benign or malignant is by a Fine Needle Aspiration Cytology (FNAC). The recommended treatment is by surgery to remove the entire thyroid gland (total thyroidectomy) and any enlarged lymph nodes (Neck dissection). If necessary, radioactive iodine is given four to six weeks after surgery to destroy any remaining thyroid cells. After total thyroidectomy, patients have to be on lifelong thyroxine medication.
- p. **Cosmetic problems:** should not be an indication!!!

III. DIAGNOSTIC PROCEDURE

- q. **Clinical argument**

Benign thyroid goiter:

SYMPTOMS & SIGNS

- ☐ Swelling in the neck
- ☐ Breathing and swallowing difficulty, if large
- ☐ Irregular or diffuse thyroid swelling

Hyperthyroidism, Thyroidtoxicosis and Graves' Disease:

SYMPTOMS & SIGNS

- ☐ Weight loss despite increased appetite
- ☐ Excessive sweating
- ☐ Moist palms
- ☐ Heat intolerance
- ☐ Nervousness and irritability
- ☐ Menstrual irregularity and sub-fertility
- ☐ Tremors
- ☐ Staring or protruding eyes (Exophthalmosis)
- ☐ lid lag
- ☐ Pretibial myxedema
- ☐ Acropachy (clubbing of the fingers and toes)
- ☐ Rapid pulse rate which may be irregular (Tachycardia)
- ☐ Heart failure
- ☐ Goiter often present but not always
 - **Smooth and diffuse goiter in Grave's disease**
 - **Irregular goiter in toxic multi-nodular goiter.**

Hypothyroidism:

- Lethargy
- Weight gain
- Cold intolerance
- Constipation
- Hair loss
- Dry skin
- Depression
- Bradycardia
- Memory impairment
- Menorrhagia
- Goiter often present but not always

Thyroid Cancer:

- Thyroid cancer is far less common than benign thyroid nodules are. Cancer of the thyroid often appears as an enlargement on one side of the thyroid and usually develops slowly and initially does not cause any symptoms. But some cases (Undifferentiated thyroid carcinoma) are fast growing thyroid nodule.
- The most common first sign is a small lump in the neck, which is painless.
- Other symptoms which may develop as the cancer grows include: Hoarseness or difficulty in speaking in a normal voice, swollen lymph glands in the neck, difficulty swallowing or breathing as the cancer presses on the esophagus or trachea. In a few cases can have pain in the throat or neck.

r. Technical procedure

i. Laboratory test

- Beside the **basic lab tests & ECG** for standard preoperative care in other surgical diseases, initial screening should include **TSH** given the sensitive assays in the absence of symptoms of hyper or hypothyroidism, further testing is not required. An assessment of **T4** and **T3** would be the next step in the evaluation (Table).
- Further laboratory testing is based on presentation and results of screening studies and may include thyroid antibodies (antithyroid peroxidase formerly the antimicrosomal antibodies /**TPOAb** in Hashimoto's thyroiditis or Antibodies to the TSH receptor/**TRAb** in Graves' disease and antithyroglobulin antibody/**Anti-TG** in autoimmune thyroiditis & Hashimoto's thyroiditis), thyroglobulin, sedimentation rate and **Calcitonin** in an individual at high risk for medullary carcinoma of the thyroid.

Laboratory results in different metabolic situations

Diagnosis	TSH	T3, fT4
1. Euthyroid goiter:	Normal	Normal
2. Hyperthyroid goiter:	Low	high
3. T3 Hyperthyroidism:	Low	fT4 norm, T3↑
4. Hypothyroid goiter:	High	Low
5. Subclinical hyperthyroidism:	Low	Normal
6. Subclinical hypothyroidism:	High	Normal
7. Hypothyroidism caused by low function of pituitary gland:	Low	Low

ii. **Imaging** study

- Ultrasound
 - Establish and follow goiter size, consistency, presence of calcification and nodularity.
 - Regional lymph nodes.
 - Localize nodules for ultrasound-guided biopsy.
- X-Ray
 - Assess extent of a goiter and presence of calcification.
 - Visualize the position of trachea.
- In some facilities, they do laryngoscopy to determine preoperative vocal cord function.
- Computed tomography scan
 - CT scan is more precise than x-rays.
 - CT scan can be used to delineate size and goiter extent.
 - CT scan does a much better job of determining the effect of the thyroid gland on nearby structures.
 - May be useful in the follow-up of patients with thyroid cancer that shows evidence of recurrence.
 - CT scan can be used to guide biopsy of the thyroid.
- Magnetic resonance imaging has the same indications as CT scan.
- Radionuclide uptake and radionuclide scan
 - Assess thyroid function and anatomy in hyperthyroidism.
 - Useful in the patient with neck or superior mediastinal masses.

- Allows determination of the function of a nodule. Function of a thyroid nodule has value both diagnostically and therapeutically.
- Barium swallow is used to assess esophageal obstruction.
- Spirometry: The flow-volume loop is useful in determining the functional significance of compressive goiters.

iii. **Procedures**

- Fine-needle aspiration cytology (FNAC) is used for cytologic diagnosis. Fine-needle aspiration of the thyroid is used to determine the cause of an enlarged gland, especially for cancer diagnosis.
- Negative FNA: NOT 100 % certainty, that no malignancy. Oncocytic cells in FNA: 20 - 30 % chance of a thyroid cancer.

IV. DIFFERENTIAL DIAGNOSIS

- Branchial cleft cyst
- Thyroglossal duct cyst
- Cystic hygroma
- Pseudogoiter
- Lymphadenopathy
- Carotid artery aneurysm
- Parathyroid cyst
- Parathyroid adenoma
- Sarcoma
- Fibroma
- Thyroid abscess
- Granulomatous disease of the thyroid
- Infectious thyroiditis

V. THERAPEUTIC APPROACH

s. Stabilize patient

- For thyroid surgery, the surgeon must
 - have a thorough understanding of the pathophysiology of thyroid disorders,
 - be versed in the pre- and postoperative care of patients,
 - have a clear knowledge of the anatomy of the neck region, and
 - finally, use an unhurried, careful, meticulous operative technique.

- Preoperative preparation
 - Euthyroidism: Most patients with euthyroid goiter are not needing of any
 - specific preoperative preparations other than the usual examinations and
 - basic work up to determine their risk for a general anesthetic from other medical problems.
 - **Hypothyroidism:** If a patient is hypothyroid, it is appropriate to use L-Thyroxine treatment preoperatively and restore a euthyroid or at least
 - subclinical state. Although mild hypothyroidism probably has little
 - deleterious effect, severe hypothyroidism increases both the anesthetic and surgical risks.
 - **Hyperthyroidism:**
 - Patients with Graves' disease or other thyrotoxic states should be treated preoperatively with the antithyroid drugs, Ex: Carbimazole, in order to restore a euthyroid or at least sub clinical state in 2 months to prevent thyroid storm, a severe accentuation of the symptoms and signs of hyperthyroidism that can occur during or after operation. Thyroid storm results in tachycardia, or cardiac arrhythmias, fever, disorientation, coma, and even death.
 - In the mild cases of thyrotoxicosis due to Graves' disease, iodine (SSKI) therapy can be used for preoperative preparation. Lugol's solution is added to the regimen for 10 days to 2 weeks before operation. It decreases the vascularity and increases the firmness of the gland. It is our practice to administer 0.3cc of 5%, three times daily. This medication is taken in milk or orange juice to make it palatable.
 - The beta-adrenergic blocker Propranolol has increased the safety of thyroidectomy for Graves' disease. We use it with antithyroid drugs to block β adrenergic receptors and ameliorate the major signs of Graves' disease by decreasing the patient's pulse rate and eliminating tremor to make the operation easier since the thyroid gland is smaller and less friable. We continue the propranolol for 6 days postoperatively, for they are still thyrotoxic immediately after operation, although the peripheral manifestations of their disease have been blocked (T4 half-life is around 6 days).
- t. Empirical treatment & Directed treatment (Basic thyroidectomy Procedure)

Step 1. Position:

- a. Put the patient in semi-Fowler position.
- b. Patient's neck should be hyperextended.
- c. Place small pillow at the area of the upper thoracic spine, beneath the shoulders.
- d. Place a doughnut support under the head.

Step 2. Preparation of skin:

- e. Use Betadine or any other solution of the surgeon's choice.
- f. Be sure the chin and long axis of the body are aligned at the midline.
- g. With 2-0 silk, mark the location of the incision, two fingerbreadths above the sternal notch.

Step 3. Incise the low collar symmetrical mark of the skin, carrying out the incision **through** the superficial fascia (subcutaneous fat and platysma). Establish good hemostasis by electrocoagulation or ligation using silk.

Step 4. Formation of flaps: By blunt dissection, elevate the upper flap to the notch of the thyroid cartilage and the lower flap to the jugular (sternal) notch. Use Mahorner's, Murphy's, or other self-retaining retractors.

Step 5. Opening of the deep fascia: The opening is accomplished by a longitudinal midline incision along the raphe of the strap muscles, which is actually the deep fascia.

Step 6. Elevation of the strap muscles: The sternohyoid muscles are easily elevated, but the thyrohyoid and sternothyroid muscles are attached to the false thyroid capsule and should be separated carefully to avoid injuring the gland and causing bleeding. In extremely rare cases, when the thyroid gland is huge, section of the strap muscles becomes necessary. Divide them at the proximal (upper) one-third to avoid paralysis due to injury of the ansa hypoglossi (C1, C2, C3, and XII).

Step 7. Exposure and mobilization of the gland: With all strap muscles elevated and retracted, the index finger of the surgeon is gently inserted between the thyroid and the muscles. A lateral elevation is also taking place, occasionally using all the fingers except the thumb. It not only breaks the remaining muscular or pathological attachments, but enables the surgeon to appreciate the gross pathology of the gland in toto. Occasionally the strap muscles should be divided. The anatomy of the normal and the abnormal must be studied carefully regarding size, extension, consistency, and fixation of the gland. Is a pyramidal lobe present? How thick is the isthmus? Are a Delphian node and other lymph nodes present? If so, excision of the Delphian node and perhaps one or two of the other palpable lymph nodes is in order. Frozen section should follow, if available in your facility.

Step 8. Total lobectomy:

- a. Retract the lobe medially and anteriorly by special clamps or deep sutures outside the lesion.
- b. Ligate the middle thyroid vein.
- c. Identify the recurrent laryngeal nerve by blunt dissection into the tracheoesophageal groove.
- d. Identify and protect the parathyroids.
- e. Ligate the inferior thyroid artery.
- f. Ligate the lower pole vessels.
- g. Carefully ligate the upper pole. Perform en masse ligation.
- h. Dissect the lobe from the trachea.

- i. If the pyramidal lobe is present, ligate its most distal part and remove it together with the lobe.
- j. Ligate the isthmus, if present.
- k. Send the specimen to the lab for frozen section, if available in your facility.
- l. Obtain meticulous hemostasis.

Step 9. Subtotal thyroidectomy:

- a. Apply multiple hemostats at the thyroid parenchyma and partially transect the gland.
- b. Use 4-0 silk suture ligature for suture ligation of the thyroid parenchyma and surface veins.
- c. If possible, approximate the segment to the trachea.
- d. Both upper pole remnants or the tracheoesophageal remnant should weigh approximately 5–6 g.
- u. Monitoring
 - i. Clinical and Laboratory monitoring

- o **Benign thyroid surgery- Postoperative treatment**

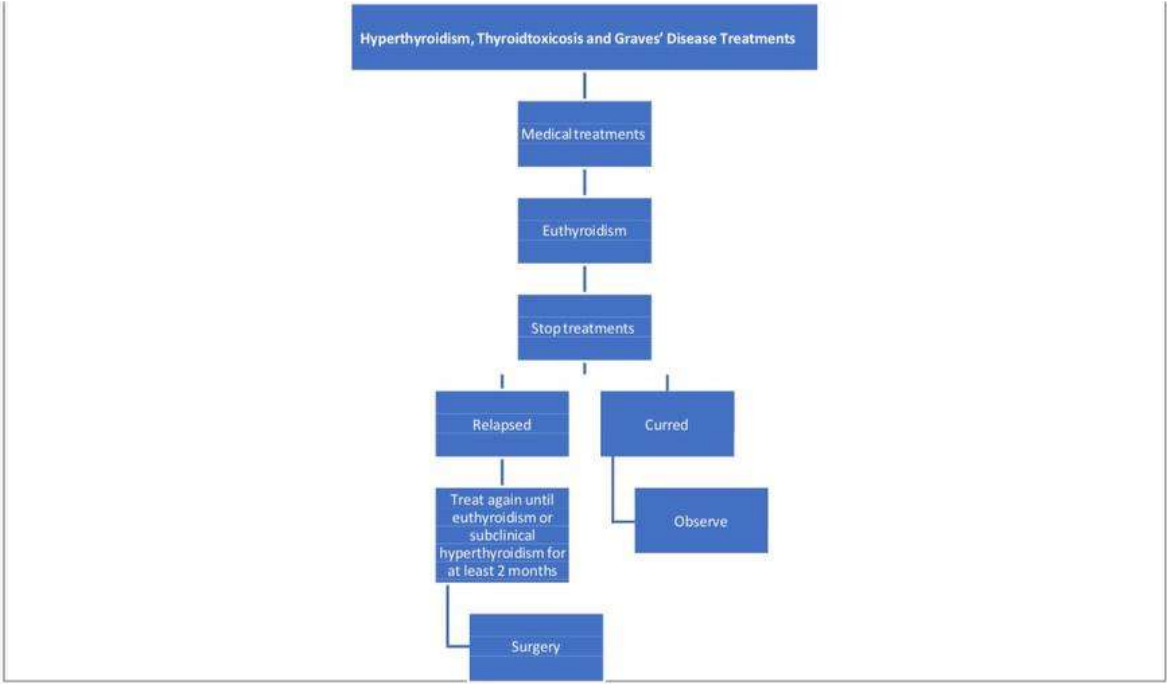
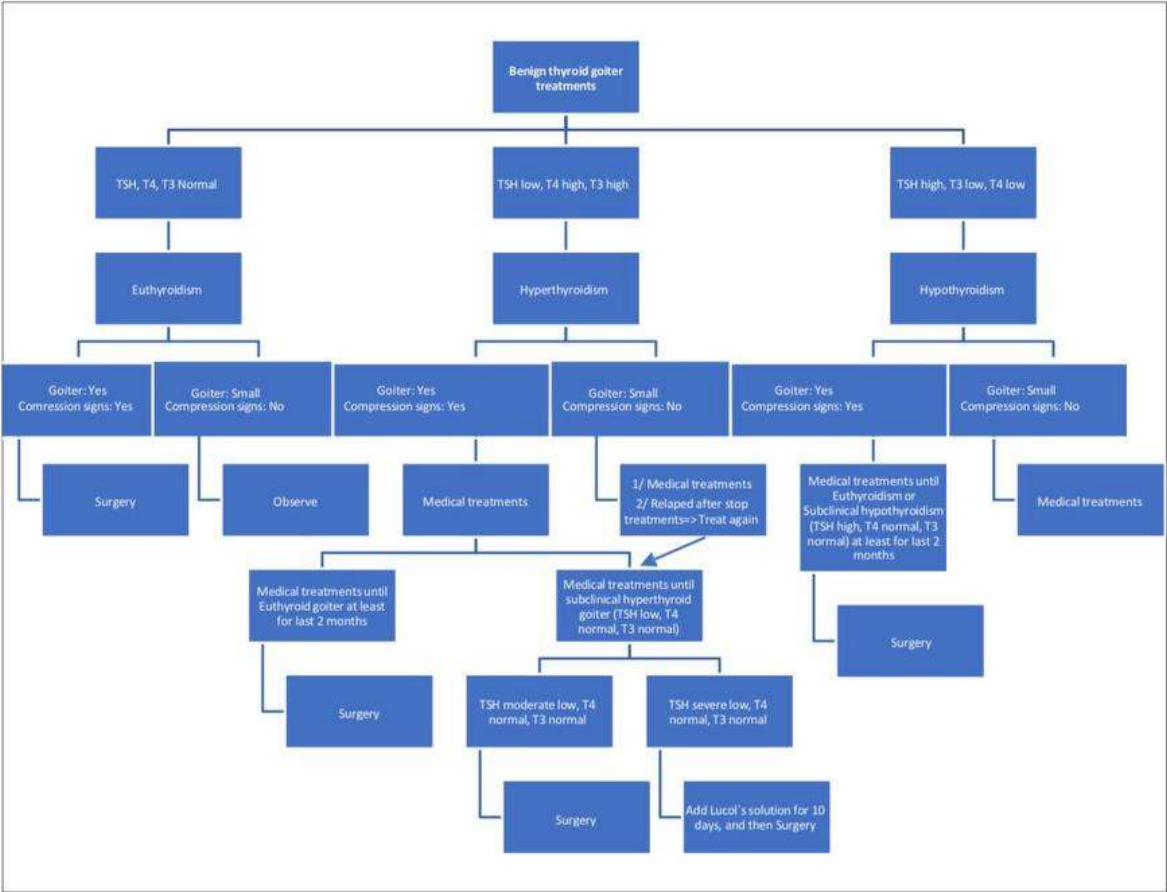
- Following total or subtotal thyroidectomy patients must take thyroid hormone replacement for life to prevent hypothyroidism. Most patients require L-thyroxine in doses between 75 and 100 µg given once daily in the first 6 weeks postoperatively to keep their circulating TSH in normal range. This dose will be changed depend on TSH result checked in first 6 weeks and in every 3 months follow up.
- Hypocalcaemia, even, it is permanent and serious complication that requires lifetime therapy due to dissection in the region of the parathyroid glands. When postoperative hypocalcaemia is mild, no treatment is needed but careful, watchful waiting should be employed. Serum calcium levels should be determined every 12 hours if the patient remains hospitalized. If the patient becomes symptomatic, 1g calcium gluconate can be slowly given intravenously; then 2 g of this preparation should be added to the intravenous bottle and dripped continuously every 8 hours. However, more calcium in the infusion may be needed to alleviate most symptoms and get a normal blood calcium level. Step by step, we replace that IV calcium to oral calcium is frequently used in a dose of 1.5 to 2.0 g calcium ion per day. Usually, when significant hypocalcaemia is present, the fastest way to attain a normal serum calcium value postoperatively is to add a vitamin D to the oral calcium regimen. This is frequently done in order to be able to safely discharge the patient as early as possible. The same as TSH, calcium serum have to check regularly in every follow up and the doses of calcium and vitamin D have to change to get its normal range.
- If there is wound infection, the treatment requires antibiotics for cellulites and drainage for an infected seroma or hematoma.

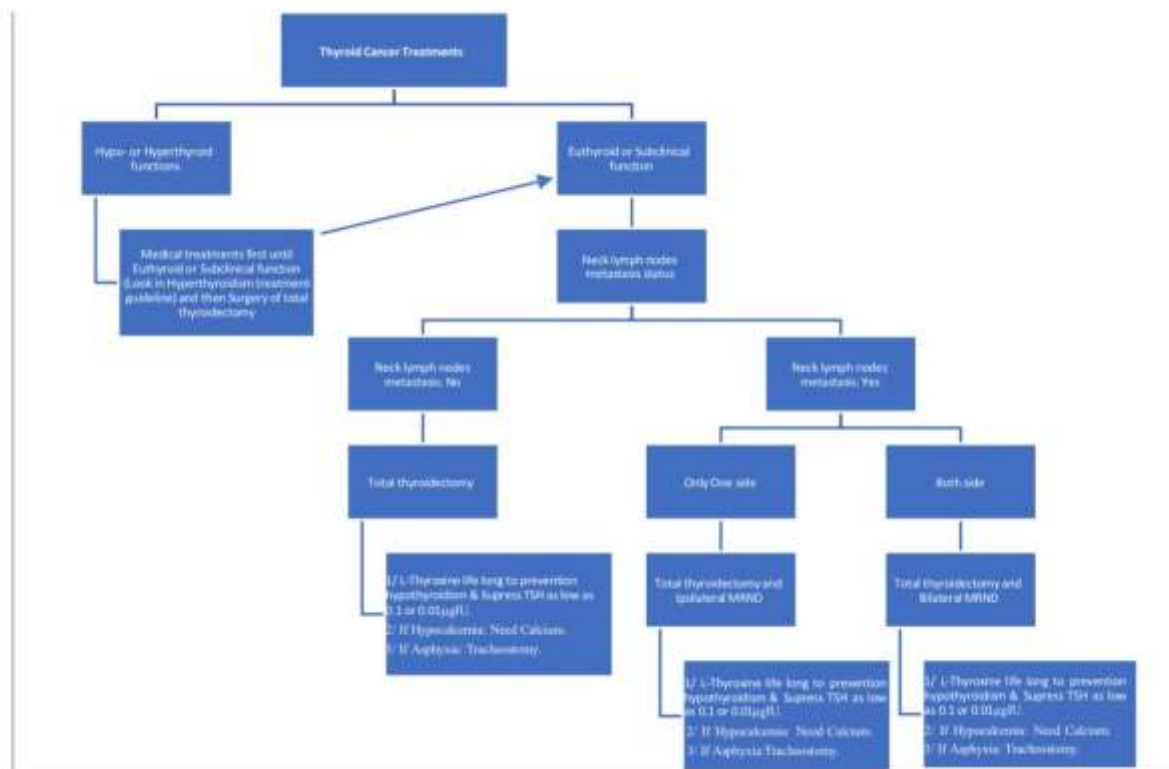
- Damage to the recurrent laryngeal nerve can be unilateral or bilateral and temporary or permanent. No treatment available in our country for this complication. It faces to tracheostomy, if patient gets asphyxia.
- **Thyroid cancer- Postoperative treatment**
 - For cancer patients after total thyroidectomy, patients have to take
 - L-thyroxine as much as possible not only for treatment of hypothyroidism, but also to prevent its recurrence by suppression of the circulating TSH levels to be less than 0.1 or 0.01 μ IU.
 - Radioiodine: Normal thyroid tissue remaining after surgery needs to be destroyed first by using radioiodine.
 - External radiation: Limited role, because well-differentiated cancer is radio-resistant. Trial in undifferentiated thyroid carcinoma.
 - Chemotherapy: Uncertain results, no cure, only improving of survival
 - Follow up every 3 months to check: Local situation, Lymph nodes, TSH must be as low as our goal, Chest X-ray every year (pulmonary metastasis?).

VI. COMPLICATION

- a. Bleeding: Can be so severe that the wound needs to be opened in the ward to remove the clots, because otherwise the trachea might be compressed.
- b. Recurrent laryngeal nerve injury: If unilateral, resulting in hoarseness. If bilateral, stridor, needing tracheotomy.
- c. Hypoparathyroidism: All four parathyroid glands removed or ischemia of the parathyroid glands. Result: Hypocalcemia increased neuromuscular excitability. Symptoms of hypocalcemia:
 - Mild: Tingling around mouth, fingers, toe, anxiety, depression, confusion.
 - Severe: Tetany, carpopedal spasm, convulsion, laryngeal stridor
 - i. Clinical signs: Chvostek's sign, Trousseau's sign
 - ii. Treatment: Calcium Po/IV
- d. Hypothyroidism: Depends on the extent of surgery and the nature of the disease.

VII. ALGORITHM





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TRACHEOTOMY

Chek Sokchea

I. INTRODUCTION

- Tracheotomy is a surgical procedure in which a trachea-cutaneous airway is created in the anterior neck [1].
- Tracheotomy is a commonly performed procedure.
- There is still debate on optimal timing and technique of tracheotomy.
- Management of tracheotomy is important to prevent complications [2].

II. ANATOMY

- The trachea is a structure composed of incomplete cartilaginous rings, beginning at the subglottic larynx and terminating at the carina and mainstem bronchi.
- The cervical portion of trachea is related to [3]:
 - Anterior: thyroid gland (isthmus, especially), sternohyoid and sternothyroid muscles.
 - Immediately lateral: recurrent laryngeal nerves and some peritracheal lympho-fatty tissue.
 - Lateral: common carotid arteries.
 - Posterior: esophagus
 - Inferior-anterior: thymus and innominate artery.
- It is essential that tracheotomy is carried out at least one to two rings beyond the cricoid.
- Usually, the trachea is entered between the second and the third or between the third and the fourth cartilage rings.
 - When the tracheotomy is too high (close to cricoid), there is a risk of a subglottic stenosis, which is difficult to treat.
 - A too low tracheotomy may result in bleeding from the brachiocephalic trunk [2].

Anatomic Landmarks for Tracheostomy [3]

- Thyroid notch - a palpable landmark to identify the superior aspect of the larynx in the midline.
- Cricothyroid membrane - a palpable depression between the cricoid and thyroid cartilages. This is the location for an emergent cricothyrotomy.
- Cricoid cartilage - a palpable landmark to identify the junction of the larynx and trachea.

- The skin incision is typically placed 1 to 2 cm inferior to the cricoid.
- Sternal notch - a palpable landmark to identify the thoracic inlet. It is important to palpate here to detect the possibility of a high-riding innominate artery that may be encountered during tracheostomy

III. INDICATION

hh. Emergency tracheotomy

- Acute upper airway obstruction with failed endotracheal intubation (foreign body, angioedema, infection, anaphylaxis, etc.)
- Post-cricothyrotomy (if a cricothyrotomy has been placed, it should be immediately formalized into a tracheostomy once an airway has been secured)
- Penetrating laryngeal trauma
- Le Fort III fracture ¹

Emergent tracheostomy is most commonly carried out in the setting of acute airway obstruction, such as aspiration of a foreign body into the upper airway, Ludwig's angina, or penetrating trauma to the airway that is not amenable to endotracheal intubation. Emergent tracheostomy may also be necessary for the setting of severe facial or cervical trauma, particularly in pan-facial fractures where craniofacial dislocation presents a contraindication to nasal intubation [4].

ii. Elective tracheotomy

- Prolonged ventilator dependence
- Prophylactic tracheostomy prior to head and neck cancer treatment
- Obstructive sleep apnea refractory to other treatments
- Chronic aspiration
- Neuromuscular disease
- Subglottic stenosis

The timing of elective tracheostomy for prolonged intubation (failure to wean from mechanical ventilation) has been a subject of much debate. Classic teaching dictates tracheostomy be carried out 5 to 7 days after endotracheal intubation in order to minimize the risk of complications associated with long-term intubation, most notably subglottic stenosis. The development of low-pressure cuffs on endotracheal tubes (with a maximum pressure of 20 cmH₂O) may allow this time to be extended if the likelihood of extubation exists. Alternatively, early tracheostomy has been advocated in order to enhance patient comfort, decrease sedation, and potentially decrease ICU/ventilator days [5][6].

2. Contra-indication

- Absolute contraindication for tracheotomy is skin infection and prior major neck surgery which completely obscures the anatomy [2].

¹ Le Fort type III fractures are caused by impact to the nasal bridge and upper part of the

maxilla.

- End of life issues should be discussed in the terminally ill patient and goals of continued care established before proceeding with tracheostomy or any invasive procedure [3].

IV. SURGICAL PROCEDURE (OPEN TRACHEOTOMY) [7]

- In the routine, anesthetized, elective tracheotomy, a patient is positioned supine.
- A shoulder roll should provide moderate extension of the neck. Overextension of the neck can lead to narrowing of the airway or unintentional low placement of the tracheotomy.
- A 2 to 3 cm horizontal incision in a natural, relaxed skin tension line midway between the cricoid cartilage and the sternal notch.
- Electrocautery is used to dissect through the subcutaneous fat and platysma muscle in the midline of the neck.
- Dissection should proceed in a vertical fashion through the raphe.
- Retractors are used bilaterally to retract strap musculature laterally, and a finger should be placed in the right inferior aspect of the wound bed to identify and prevent potential injury to a high riding innominate artery.
- Midline dissection continues until the thyroid gland is encountered. The thyroid gland can be retracted superiorly, however most often the isthmus must be divided.
- After division of the thyroid gland, the pre-tracheal fascia is encountered. Loose fascia is bluntly removed from the anterior trachea and the cricoid cartilage should be palpated superiorly.
- Exposure of the trachea can be dramatically improved by placing a single sharp hook into the cricoid cartilage and pulling in an anterosuperior direction.
- At this point in the operation, the anesthesia team is alerted to remove tape or ties from the endotracheal tube and secure it manually. The inspired fraction of oxygen should be reduced if possible.
- The cuff of the endotracheal tube is deflated to limit the chance of balloon rupture during the tracheotomy in the event the endotracheal tube must be utilized to ventilate the patient again during the procedure.
- Tracheotomy is then planned between the second and third tracheal rings. Finger palpation of the cricoid cartilage should be used to verify the location of each tracheal ring.
- After tracheotomy, the endotracheal tube is pulled back superiorly until the surgeon sees the tip of the tube just above the level of the tracheotomy.
- A tracheotomy tube with its obturator is then inserted into the trachea, the obturator is removed, the inner cannula is inserted, the cuff on the tracheotomy tube is inflated, and the respiratory circuit is connected to the tracheotomy tube.

- All retractors and the endotracheal tube are left in place while the anesthesia team assures that the patient can be adequately ventilated via the tracheotomy tube.
- This is done by bilateral chest auscultation and confirmed return of carbon dioxide from the respiratory circuit.
- After affirmation of accurate placement of the tracheotomy tube, retractors can be removed and the tracheotomy tube secured by suture to the skin of the anterior neck in the four corners of the tracheotomy tube flanges.
- The tracheotomy tube is also secured by means of a tracheotomy tie around the neck.
- The endotracheal tube may be removed after securing the tracheotomy tube. The tube is maintained until this point to allow for readvancement of the tube if the airway is lost during the procedure.

3. Choix of tracheotomy tube

- Tubes, varying in shape, are available in graded lengths and diameters [8].
- Tracheotomy tubes can be metal or plastic.
- We suggest to use plastic cuffed tracheotomy tubes with an inner cannula, for example the Bjork-Shiley tubes or Portex tubes.

V. COMPLICATIONS

- Some complications are related to the procedure and others to the cannula.
- Late complications of tracheotomy are difficult to categorize and to quantify, as many patients are critically ill and lost to follow-up secondary to discharge before decannulation [9] [10][11][12].

a. Early complications

- **Haemorrhage:**
 - Bleeding is reported in +/- 5% of tracheostomies.
 - That can be minor and can be controlled by packing and by ensuring that the cuff of the tracheotomy tube is inflated.
 - Major bleeding may require reoperation.
- **Wound infection:**
 - A tracheotomy is considered a clean contaminated wound.
 - Prophylactic antibiotics are usually not warranted.
 - True infection is rare and requires only local therapy.
 - In case of necrotizing tracheal infection, conversion to oral tracheal intubation is necessarily followed by wide debridement of involved tissues.
- **Subcutaneous emphysema:**

- Subcutaneous emphysema can be caused by positive pressure ventilation or coughing against a tightly sutured or packed wound.
- It can be prevented by not suturing the wound around the tube.
- The emphysema will resolve spontaneously within a few days.
- A chest radiograph should be obtained to rule out a pneumothorax.

□ **Tube obstruction:**

- The tube can be obstructed by mucus, blood clots, displacement into surrounding soft tissues or abutment of the tube's open tip against the tracheal wall.
- Failure to re-establish adequate ventilation by suctioning through the tube requires immediate replacement of the inner cannula or the entire tube.

□ **Fausse route** during the procedure or early tube displacement.

- Fausse route or early displacement (within 5 days after placement of tracheotomy) of the tube creates an airway emergency.
- Orotracheal intubation should be performed when the tract cannot be re-established immediately.

b. Late complications

□ **Swallowing problems.** Factors that contribute to disturbed swallowing are as follows:

- Decreased laryngeal elevation,
- Esophageal compression and
- Obstruction from the tracheotomy tube cuff.

□ **Tracheal stenosis.**

- Tracheal stenosis occurs in approximately 1—2% and usually results from ischemia, devascularization and chemical erosion.
- Causes include the use of high-pressure cuffs (now eliminated after the introduction of large-volume, low-pressure cuffs), forced angulation of
- a stiff tube or hyperinflation of the cuff which result in tracheal damage.
- Stenosis may occur at the stoma, the cuff side or the tip of the tracheotomy tube.

□ **Tracheo-innominate artery fistula.**

- This complication is rare (<0.7%). Most fistulas appear to result from direct pressure of the cannula against the innominate artery.
- Risk factors include low placement of tracheotomy, high-pressure cuffs and excessive head or tracheotomy tube movement. Overall survival is only 25%.

- Preventive measures include correct placement of the tracheotomy tube at the level of the second or third cartilaginous ring, avoiding prolonged or excessive hyperextension of the neck, using a soft, re-adjustable tracheotomy tube and lightweight tubing to avoid dragging on the tracheotomy tube [13].

□ **Tracheoesophageal fistula.**

- Tracheoesophageal fistula is a rare complication occurring in less than 1% of patients.
- Cause is mostly iatrogenic due to erosion by the tracheotomy cuff but also a right angle of the tube can cause undue pressure against the posterior tracheal wall. It is more commonly seen when a nasogastric tube is in place as well.
- Fistula repair is performed by a cervical incision with interposition of viable tissue [14][15].
- Postintubation tracheoesophageal fistula is usually best treated with tracheal or laryngotracheal resection and anastomosis and primary esophageal closure even in the absence of tracheal damage [16].

□ **Granuloma formation.**

- Granulomas may result from a foreign body reaction to the tracheotomy tube or specific parts of it.
- They are more common with fenestrated tracheotomy tubes. These granulomas can be treated with the YAG laser. Especially worrisome are granulomas at the lower end of the tracheotomy tube where bronchoscopic removal only provides temporary relief.

□ **Persistent stoma.**

- This situation usually results when the tube has been left in position for a prolonged period, permitting epithelialisation between the skin and the tracheal mucosa.
- Although the opening might become very small due to wound contracture, it may be troublesome for the patients.
- Surgical closure is proposed.

VI. TRACHEOTOMY MANAGEMENT

Recommendations on tracheotomy management have been made by the panel on the basis of personal experience and after careful reading of several published guidelines and protocols [17][18][19][20][21].

c. Nursing care

Some general principles are advocated for immediate post-tracheotomy care:

- The tracheotomy cannula is secured in place and the tracheotomy is left to heal for 5—7 days.
- Angulation of the tracheostomy tube should be avoided since this may lead to stenosis.
- The tracheotomy wound has to be kept clean and dry to prevent post-incisional wound infection.
- When a dual cannula is used, there is usually no need to change the outer cannula.
- The inner cannula is changed daily or more frequently if necessary.
- Changing the outer cannula within 5—7 days.
- The only indication of outer cannula changing is when the cuff has been damaged or when a tracheotomy tube of different size or shape is found to be necessary.
- The early change of the outer tracheotomy tube should be performed by a practitioner who is trained in the technique.

d. Tube cuff pressure

- Tracheotomy tube cuffs require monitoring to maintain pressures in a range of 20—25 mmHg [18].
- High cuff pressures above 25—35 mmHg exceed capillary perfusion pressure and can result in compression of mucosal capillaries, which promotes mucosal ischaemia and tracheal stenosis.
- Low cuff pressures below 18 mmHg may cause the cuff to develop longitudinal folds, promote microaspiration of secretions collected above the cuff and increase the risk for nosocomial pneumonia.
- Cuff pressures should be monitored with calibrated devices and recorded at least once every nursing shift and after manipulation of the tracheotomy tubes.
- It has been suggested to deflate tracheotomy cuffs on a regular schedule to allow mucosal perfusion at the site of the cuff and to improve clearance of secretions accumulated around the cuff, but too little evidence is available to advocate this practice.
- To minimize damage to the tracheal wall by the distal end of the tracheotomy tube, the tube has to be maintained in a central position, avoiding angling and contact between tracheal mucosa and tube.
- Traction as well as unnecessary movement of the tube is to be avoided.
- The curvature and length of the tube has to be chosen to ensure collinearity with the trachea.

e. Humidification of inspired air

- Ensuring free airway by means of tracheotomy bypasses upper airway functions to filter, heat and humidify inspired gas.
- In tracheotomised patients, inspired air may have a significant humidity deficit which may lead to pathological deep airway changes such as mucosal damage, loss
- of mucociliary transport and thickening of airway secretions.
- These changes may in turn increase the risk for lower respiratory tract infection and provoke airway obstruction by endoluminal mucous impaction. These physiopathological arguments make it mandatory for inspired gas delivered through tracheotomy to be humidified and heated. To ensure humidification and heating, various techniques are used.
- During mechanical ventilation, the most frequently applied devices are heated humidifiers and passive humidifiers, of which heat-and-moisture exchangers are most commonly used. Heated humidifiers, in which gas flow is directed through a heated water bath prior to inspiration, are the most efficient, but also the most expensive.
- Passive humidifiers in the shape of heat-and-moisture exchangers (so called ‘artificial noses’) consist of a filter placed between ventilator and tube (to which a bacterial filter is often added), collecting heat and moisture from the patient’s expired air and delivering part of it at the following inspiration.
- Because of lesser cost and a probably decreased risk of ventilator-associated pneumonia associated with its use, heat-and-moisture exchangers have been recommended in mechanically ventilated patients.

f. Clearance of secretions

- Because suctioning is uncomfortable and is associated with ventilatory complications such as airway collapse and alveolar derecruitment, it should be performed only when indicated and not at a fixed frequency.
- Frequency of airway suctioning should hence be determined on a patient level, taking into account factors such as viscosity and quantity of mucus, neurological and muscular performance and presence of active cough reflexes and efforts.
- The upper airway should also be suctioned periodically to remove oral secretions and to minimize stasis of pooled secretions above the tracheotomy cuff with subsequent potential aspiration into the lower airways.
- Shallow suctioning, involving the insertion of the aspiration catheter to a premeasured depth not beyond the distal end of the tracheotomy tube, is preferred to deep suctioning, in which the aspiration tube is inserted beyond the tracheotomy tube until resistance is met.
-
- Deep suctioning should be avoided since it is associated with airway mucosal damage and inflammation and may induce bronchial bleeding with subsequent risk of airway occlusion.

g. Speech

- Ventilator-dependent patients who require low minute ventilations may accomplish whispered speech during periods of partial deflation of the tracheotomy tube cuff, provided a good swallowing function is present and stasis of secretions above the cuff is limited.
- After removal from mechanical ventilation, the inner cannula is removed in awake and adequate patients, allowing expiratory airflow through the larynx when the external end of the tracheotomy tube is occluded transiently.
- Application of a one-way valve permits inspiratory airflow through the tracheotomy tube during inspiration but closes during expiration, promoting airflow around a deflated cuff.
- Again, when loosening the airway seal, one should take into account the patient's risk of aspiration.
- Some centres will use a fenestrated tube promoting airflow through the tube fenestrations during expiration.

h. Nutrition

- The presence of a tracheotomy not only provides opportunities for oral nutrition but also complicates feeding because of tube interference with normal swallowing and airway control.
- Between 20 and 70% of patients with a chronic tracheotomy experience at least one episode of aspiration every 48 h.
- Patients with tracheotomy are kept with their heads elevated to 45° during periods of tube feeding, to decrease risks for nosocomial pneumonia caused by aspiration.
- Oral feeding should be supervised by a caregiver and carefully assessed for aspiration or regurgitation.
- In patients with prolonged critical illness or anticipated impaired swallowing dysfunction, a first trial of oral feeding should preferably be preceded by a fiber optic evaluation of the swallowing function and guided by swallowing reeducation if dysfunctional.

4. Timing of tracheotomy

- There are not enough well-designed studies to establish clear guidelines regarding the timing of tracheotomy.
- Factors other than absolute time may play a role in the decision to perform a tracheotomy: improved patient comfort and communication as well as enhanced nursing care.
- A systematic review and meta-analysis comparing early tracheotomy versus late tracheotomy in trauma patients found no difference in days on mechanical ventilation, length of ICU stay, frequency of pneumonia except for patients with severe brain injury [22].

- An international study found that a tracheotomy was performed after a median of 11 days [23].
- **Recommendation:** In critically ill adult patients requiring prolonged mechanical ventilation, tracheotomy performed at an early stage (within the first week) may shorten the duration of artificial ventilation and length of stay in intensive care.

5. Removal of tracheotomy decannulation

- Tracheotomy removal after acute upper airway obstruction will only be performed if a patent airway is reestablished.
- Tracheotomy removal in patients who were weaned from prolonged mechanical ventilation can be considered if they are clinically stable, do not have psychiatric disorders, have adequate swallowing and are able to expectorate

a. Criteria for tracheotomy decannulation [24]

- Stable arterial blood gases.
- Absence of distress.
- Haemodynamic stability.
- Absence of fever or active infection.
- $\text{PaCO}_2 < 60 \text{ mmHg}$.
- Absence of delirium or psychiatric disorder.
- Normal endoscopic examination or revealing stenotic lesion occupying $<30\%$ of the airway.
- Adequate swallowing.
- Able to expectorate.

b. Technique of decannulation

- Before removal, one can try the deflated-cuff tracheotomy occlusion procedure.
- After deflation of the tracheotomy cuff, a gloved finger occludes the opening of the tube and one observes the patient for objective signs of respiratory distress.
- In case of problems, one promptly returns the patient to breathing through the tracheotomy tube and performs an endoscopic examination to check for upper airway obstruction. If no lesions are present, one should consider whether the tube is not too large and try again after changing the tube.

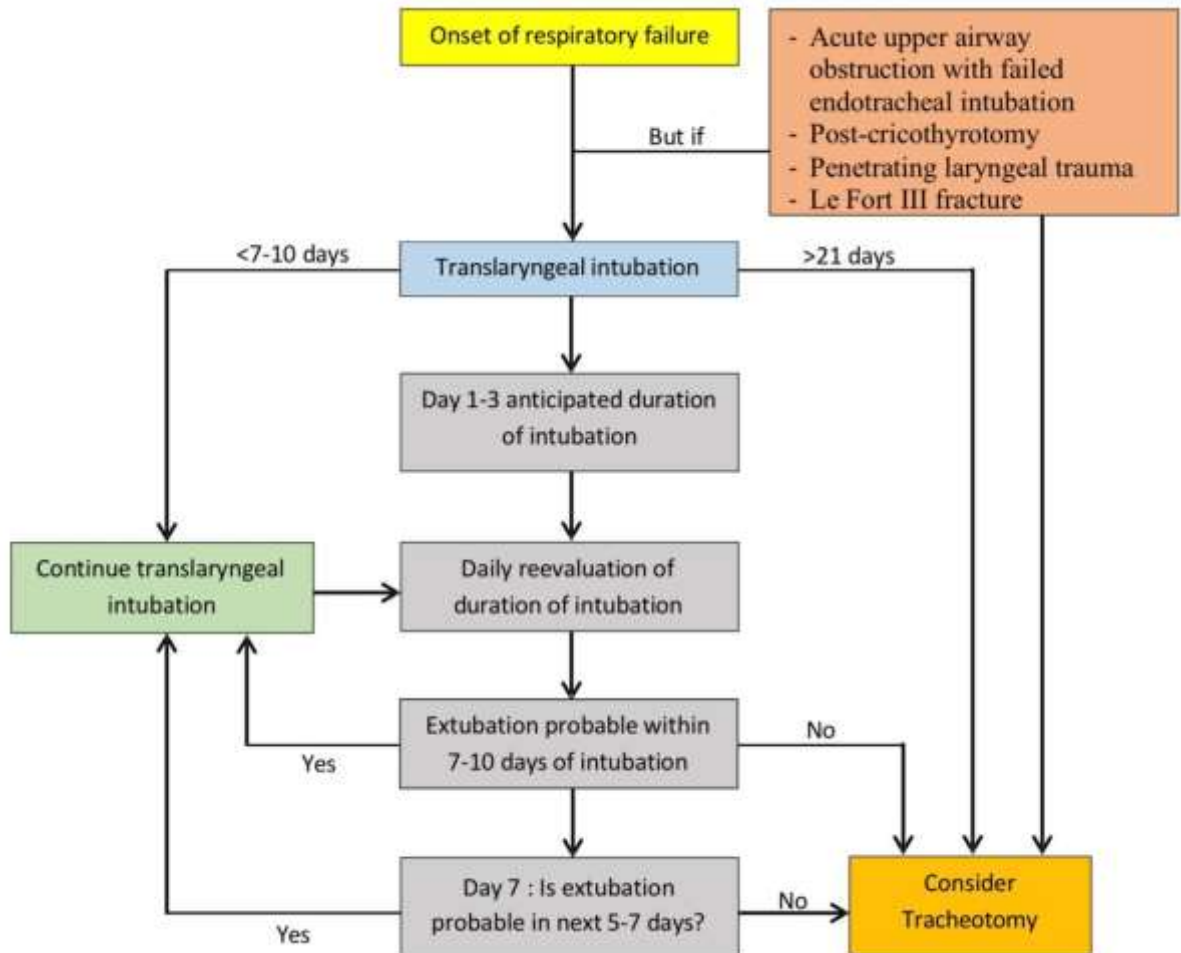
- The tube can be removed and the opening is covered with sterile dressings.
- The wound spontaneously heals in +/- 10 days in most cases. After a long period of breathing through a tracheotomy tube, it is common to take an intermediate step before removal by using a tracheotomy button or a speech valve.
- In patients with copious sputa or expectoration problems, the tube can be downsized.
- Some patients benefit from logopedic support.

c. Recommendation

- Tracheotomy removal in patients who were weaned from prolonged mechanical ventilation can be considered if they are clinically stable, do not have psychiatric disorders, have adequate swallowing and are able to expectorate.
- Different techniques are used and can be tailored according to the patient's need.

VII.ALGORITHM

Algohythm of Tracheotomy



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TRAUMATIC DIAPHRAGMATIC INJURY (TDI)

Sok Pheng, Preap Ley

I. CASE DEFINITION

- Diaphragmatic rupture refers to a tear or hole in the diaphragm. This condition can be caused by trauma, such as blunt or penetrating injuries to the chest or abdomen.
- Although the true incidence remains unknown in our country, autopsy series in westerners of patients sustaining blunt abdominal injury suggest an incidence of approximately 5%. The injury is found more frequently in patients who sustain penetrating trauma, and in some subsets, such as anterior wounds below the nipple, the occurrence may be as high as 30%.
- In blunt trauma diaphragmatic tear is significantly larger than in penetrating injuries.
- Diaphragmatic injury following penetrating trauma may be difficult to identify because of the resulting small defects in the hemidiaphragm. These may not produce symptoms, clinical signs, or radiographic abnormalities when the patient is first seen. As a result of negative pleural pressures, intra-abdominal positive pressures and diaphragmatic motion during respiration, these defects can enlarge over time and lead to late complications including hollow viscera obstruction, strangulation, and perforation.
- This potential for serious and possibly lethal complications makes a thorough and aggressive search for this injury imperative after any thoracoabdominal injury.

According to Moore EE, Malangoni MA and Cogbill TH, et al in 1994, TDI is graded and Scaled as bellow:

TDI Scale		
Grade	Description of injury	AIS-90
I	Contusion	2
II	Laceration <2cm	3
III	Laceration 2-10cm	3
IV	Laceration >10cm with tissue loss $\leq 25\text{cm}^2$	3
V	Laceration >25cm ²	3

** Advanced one grade for bilateral injuries up to grade III

**Organ injury scaling IV: Thoracic vascular, lung, cardiac & diaphragm.

II. ETIOLOGY

- Blunt mechanisms include motor vehicle accidents, falls from a significant height, or pedestrian strikes. In blunt trauma, an abrupt increase in the pleuroperitoneal gradient causes an increase in kinetic energy transfer to the diaphragm, which results in the disruption. This happens most commonly on the left due to a congenital

weakness. The right hemidiaphragm is less vulnerable to injury due to the presence of the liver directly beneath it.

- Penetrating injuries can include gunshot wounds, stab wounds, or any foreign object. In penetrating trauma, the injury follows the tract of the foreign body and is typically small, with over 80% being smaller than 2 cm.
- Associated Injuries:
 - Diaphragmatic ruptures are often associated with other injuries, such as rib fractures, pulmonary contusions, abdominal organ injuries, or spinal injuries, depending on the mechanism and severity of trauma.
 - Classification systems for diaphragmatic ruptures may vary depending on the context, such as trauma severity scoring systems used in trauma centers or specific criteria used in surgical literature for planning and performing repairs.

III. DIAGNOSTIC PROCEDURE

d. Clinical argument

- A clinical history involving high-velocity motor vehicle crashes, falls from great heights, a direct blow to the abdomen, thoracoabdominal stabbing, gunshot wounds, and abdominal crush injuries should heighten awareness of the possibility of a TDI with or without herniation.
- Physical examination may reveal tenderness over the chest or abdomen, particularly on the side of the diaphragmatic injury. The other clinical signs include a difficulty breathing, gastrointestinal symptoms, subcutaneous emphysema, respiratory distress and cardiovascular instability.
- For those patients who present in a delayed fashion, their diaphragmatic injury often includes herniation of abdominal viscera. Symptoms here depend on the size of the diaphragmatic defect, which tends to increase over time. The physical findings can vary based on the anatomical point of obstruction and the severity of the herniation. Incarcerated viscera may present with partial obstructing symptoms, whereas strangulated viscera present with a graver clinical picture, leading to ischemia and impending

perforation with septic shock.

e. Technical procedure

i. Laboratory test

- There are no specific laboratory finding which directly diagnoses this entity. Basic chemistries, hepatic panel, and pancreatic enzymes can help to rule out injuries to the liver or pancreas. Unfortunately, these numbers are not directly helpful nor reliable in acute trauma patients.

ii. Imaging study

- In a stable patient without an indication for emergent operative intervention, imaging studies are obtained secondary to the physical exam. Typically, a chest x-ray is the initial preferred study.

- It may be immediately diagnostic in the case of significant herniation where bowel content is visualized in the chest cavity (Air-fluid levels are present above the level of the diaphragm, a nasogastric tube apex terminating back into the chest).
- In the acute phase, if the defect is small, organ displacement into the chest may not be present.
- Subtle but non-specific findings precluding TDIs may include atelectasis, pneumothorax, hemothorax, and/or pulmonary contusion.
- Although chest x-ray is the most common initial test in the workup for traumatic diaphragmatic hernias, computerized tomography (CT) is the gold standard.
- Multiple radiographic signs have been studied and are used in the practice of reading and detecting diaphragmatic injuries. They can be broken down into direct and indirect signs.
 - Direct signs include diaphragmatic discontinuity, intrathoracic visceral herniation, collar sign, dependent viscera sign, diaphragm thickening, and dangling sign.
- Although ultrasound and magnetic resonance imaging (MRI) may help diagnose diaphragm injuries, they are often not clinically feasible in the poly-traumatized patient.
- Because delayed diagnosis of TDI can result in significant morbidity and at times mortality, a high index of suspicion must be maintained and an aggressive diagnostic approach pursued, which may include invasive diagnostic techniques include diagnostic peritoneal lavage (DPL), laparoscopy, laparotomy, thoracotomy, and video-assisted thoracoscopic surgery (VATS).

IV. DIFFERENTIAL DIAGNOSIS

- The differential diagnosis is much more extensive in blunt trauma. This is due to the magnitude of associated injuries that produce nonspecific signs seen on initial imaging, such as pneumothorax, pulmonary contusion, atelectasis, and hemothorax. Findings suggestive of diaphragm injury may suggest a phrenic nerve paralysis or injury.
- In the presence of a defect, not all defects are secondary to trauma. These include congenital or acquired hernias such as Bochdalek, Morgagni, paraesophageal hernias, or a diaphragmatic eventration.

V. THERAPEUTIC APPROACH

iii. Stabilize patient

- Early management of TDI is the same as the other trauma patient and consists of a concise and rapid assessment to identify and address life-threatening injuries. Those patients presenting in hemorrhagic shock or with

systolic blood pressure less than 90 mm Hg are considered unstable and should receive immediate resuscitation as per Advanced Trauma Life Support (ATLS) guidelines.

- Surgical management of traumatic diaphragm injuries varies based on patient presentation, mechanism of injury, associated injuries, and the need for immediate surgical intervention.
 - Enough painkillers, antibiotics and adjunctive procedures such as NGT, intubation or chest tube placement should be performed as indicated during the primary and secondary surveys.

iv. Empirical & Directed treatment

- Amid instability, regardless of mechanism, TDI patients should proceed immediately to the operating room, where an exploratory surgical laparotomy is indicated also to combat the other life-threatening injuries. These injuries should be repaired immediately when possible as they do not heal but tend to worsen over time.
- The gold steps are
 - If present, intraperitoneal blood and clots are rapidly evacuated and packs are placed to control ongoing hemorrhage.
 - The abdomen is explored in a systematic and thorough fashion.
 - Blood loss is controlled and gross soil age is limited by closing any hollow-organ perforation.
 - The diaphragm is then carefully evaluated by visual inspection and manual palpation to find a tearing.
 - If there is a herniated viscera in a tear, are reduced from the thoracic cavity by gentle traction.
 - The pleural space is thoroughly irrigated with warm saline to remove retained blood and clot. Several liters of irrigation are required if the injury is associated with a perforated viscus.
 - And then the tear is needed to repair right away with sutures or mesh patches.
- Delayed diagnoses in missed TDIs:
 - First, these patients have a progressive clinical course owing to pain, intermittent obstruction, strangulation, and in some cases, perforation. Addressing the potential contamination that accompanies these hernias is the same: debridement, irrigation, contamination control, and wide drainage.
 - Second, the diaphragm tends to regress laterally over time. A missed injury is subject to an inability to reapproximate the tissue, primarily when repaired at a later date. These require non-absorbable prosthetic mesh placement to bridge the defect.

v. **Monitoring**

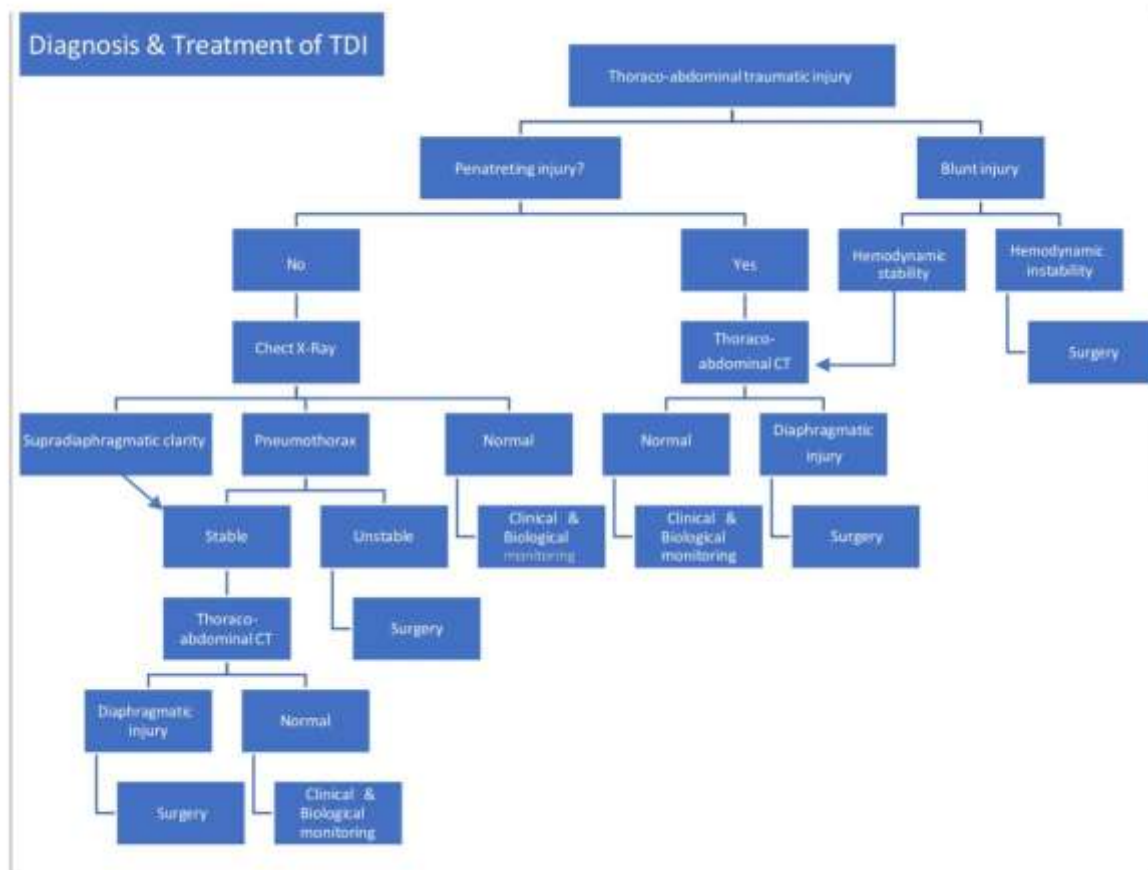
- Clinical and Laboratory monitoring
- After surgical repair, patients require close monitoring in the intensive care unit and follow all ICU guideline with pain management and respiratory therapy.
- Patients also may require long-term follow-up to monitor for recurrence of herniation.

VI. COMPLICATION

TDI complications can be subdivided into morbidity as a direct result of the diaphragm repair and secondary to the trauma.

- Complications associated with the repair include iatrogenic phrenic nerve paralysis and suture dehiscence. Both complications can lead to the need for additional invasive procedures and potentially long-term complications.
- Morbidity associated with traumatic injury varies depending on the presence of associated injuries but could include empyema, subphrenic abscess, and resultant ventilator-dependent respiratory failure.
- When patients present in a delayed fashion due to a missed injury (few weeks to years), the complications such as atelectasis, respiratory failure, pneumonia, bowel obstruction, strangulation, and perforation have all been reported.

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CHAPTER II

ENT

SURGERY

1. Cholesteatoma
2. Chronic suppurative otitis media (COMC)
3. Chronic tonsillar hypertrophy
4. Deviated septum & Chronic rhinosinusitis
5. Epistaxis
6. Nasal cavity tumor
7. Nasal polyps
8. Nodular goiter
9. Parotid tumor
10. Submandibular gland tumor
11. Tongue tumor

CHOLESTEATOMA

Po Ratha, Loek Sothearith, Meng sour, Theng Sakphea, Heng Sambo, Nen sophat,

Yos Sophorn

I. CASE DEFINITION

Cholesteatoma is defined as the presence of keratinising squamous epithelium within the middle ear, or in other pneumatized areas of the temporal bone. This keratinising epithelium exhibits independent growth, leading to expansion and to resorption of underlying bone. Focal erosion of external canal bone with accumulation of keratin is called external canal cholesteatoma.

II. AETIOLOGY

Cholesteatoma may be considered acquired or congenital.

- **Acquired cholesteatoma** occurs in several ways. In many cases, it is due to retraction of an area of the pars flaccida with or without associated atrophy of the pars tensa. The epithelium becomes trapped and infected, and proliferates to form a cholesteatoma. Alternatively, squamous epithelium may migrate through a defect in the tympanic membrane, or the cholesteatoma may form due to implantation of viable keratinocytes into the middle ear cleft following otological surgery or after a traumatic blast injury. This increased risk is secondary to poor Eustachian tube function.
- **Congenital cholesteatoma** is considered to be present if there is no history of previous ear surgery and no perforation or retraction of the tympanic membrane. It is believed to arise from developmental epidermoid rests present before birth that persist in the middle ear space. Alternative theories include: invagination of squamous epithelium from the developing ear canal; seeding of the middle ear by squamous cells in the amniotic fluid; epithelial ingrowth from the surface of the tympanic membrane after infection; and micro-perforation.

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

- Hearing loss (common)
- Ear discharge resistant to antibiotic therapy (common)

Recurrent or chronic purulent aural discharge that may be unresponsive to antibiotic therapy is common in acquired cholesteatoma. Discharge is malodorous and may be scant.

- Attic crust in retraction pocket (common): Otoscopy typically shows crust or keratin in the attic (upper part of the middle ear), the pars flaccida, or the pars tensa (usually posterior superior aspect), with or without a perforation of the tympanic membrane. white mass behind intact tympanic membrane: occurs in congenital cholesteatoma.
- Other diagnostic factors such as: tinnitus, otalgia, altered taste, dizziness and facial nerve weakness

ii. Technical procedure

a. Laboratory test

- Ear cultures are obtained mainly in patients who present with an aural discharge unresponsive to antimicrobial therapies. Swabs from the ear demonstrate bacterial infection, commonly *Pseudomona aeruginosa* , and may demonstrate other pathogens such as *Staphylococcus aureus* and anaerobic bacteria.

b. Imaging study

- **Audiometry** should be performed in all patients to determine the patient's hearing status.
- **High-resolution CT scan** of the petrous temporal bones is recommended as part of the initial work-up of patients with middle ear cholesteatoma. It can provide confirmation of disease in patients with an atypical presentation, and can be used to assess the ear for mastoid pathology and complications such as cochlear, semicircular canal, or intracranial involvement. In patients with cholesteatoma, CT shows opacification of the middle ear or mastoid; with or without erosion of the scutum, ossicular chain, labyrinth, facial canal, tegmen, or bony capsule of the sigmoid sinus. CT of the temporal bones without intravenous contrast medium is usually sufficient in the investigation of suspected cholesteatoma and in surgical planning. However, administration of intravenous contrast medium is helpful in demonstrating extra-osseous soft- tissue extension, if there is clinical concern.
- MRI is used to characterise soft-tissue pathology in cholesteatoma, complementing CT imaging, which better defines bony involvement. MRI of the head with thin sections across the petrous temporal bone (to image the internal auditory canal), with and without intravenous contrast, is useful in defining the extent of a middle ear cholesteatoma; and in assessing for intracranial extension, prior to surgical intervention. MRI is the preferred imaging modality for certain intracranial complications of cholesteatoma, including meningoencephalitis; temporal lobe abscess formation; and venous sinus thrombosis.

IV. DIFFERENTIAL DIAGNOSIS

- **Chronic suppurative otitis media:** on otoscopy is perforation of the pars tensa but no evidence of cholesteatoma.
- **Otitis externa:** on otoscopy there is swelling of the external canal and a scant discharge. The tympanic membrane is not visible or if visible appears inflamed but no evidence of cholesteatoma.
- **Benign necrotizing otitis externa:** the patient report severe otalgia, a history of diabetes or other immunosuppression. On otoscopy there are granulations in the ear canal but no evidence of cholesteatoma.
- **Myringitis:** on otoscopy there is inflammation of the tympanic membrane with or without granulation but no evidence of cholesteatoma.

V. THERAPEUTIC APPROACH

i. Stabilize patient

ii. Empirical treatment

On initial presentation, prior to definitive surgical treatment, it may be appropriate to treat the aural discharge with topical antibiotics. Agents containing quinolone (e.g., ciprofloxacin and ofloxacin) are effective, in adults and in children, either alone or in combination with a topical corticosteroid. Aural cleaning may also reduce symptomatic discharge.

iii. Directed treatment

The definitive treatment of cholesteatoma is surgery. The main goal is to remove the disease; provide a dry, safe ear that does not discharge; and prevent potential complications. Treatment is also aimed at improving the hearing threshold. This may not always be possible at initial surgery but may be possible following a second procedure. There are different surgical approaches: the canal wall up mastoidectomy or the canal wall down mastoidectomy. The use of endoscopy has been shown to positively impact the management of cholesteatoma, and its use is recommended during surgery.

Canal wall up mastoidectomy (combined approach tympanoplasty)

This procedure allows removal of cholesteatoma but leaves the canal wall intact. The procedure involves removal of the mastoid air cells lateral to the facial nerve and otic capsule, leaving the posterior and superior parts of the external canal wall intact. It is often preferred for children, as it avoids long-term complications of a mastoid cavity. However, it necessitates a second-look procedure after 9 to 12 months to examine for residual/recurrent disease. Patients will require continued follow up.

Canal wall down mastoidectomy

This procedure aims to remove the disease by drilling from the attic wall posteriorly. The size of the resultant cavity will depend on the extent of the cholesteatoma. A less invasive procedure resulting in a minimal cavity is called an atticotomy or atticotomy; a more invasive procedure resulting in a larger cavity is called a modified radical mastoidectomy. A canal wall down procedure may allow direct examination of the cavity for recurrence, but if the attic has been reconstructed a second-look procedure or a non-echo-planar diffusion-weighted MRI may be necessary to examine the middle ear for recurrent disease.

iv. Monitoring

a. Clinical monitoring

Patients are followed up in the ear, nose, and throat clinic to monitor for symptoms of cholesteatoma recurrence. During otoscopy, the cavity is examined for signs of a keratinous ear discharge. The mastoid cavity is often dry and may initially require wax removal in an aural care clinic. A canal wall up mastoidectomy necessitates a second-look procedure at 9 to 12 months to examine for residual/recurrent disease. Alternatively, a non-echo-planar imaging diffusion-weighted MRI may be used in some patients. There continues to be debate regarding which type of MRI best examine for recurrent cholesteatoma. Some authors advocate routine MRI scans (such as non-echo-planar, fast spin echo, or diffusion-weighted sequences) for follow-up, but caution that a negative scan may not be completely accurate as residual or recurrent disease may not yet be detectable. Non-echo-planar imaging was found to be more reliable compared with echo-planar imaging in identifying residual/recurrent cholesteatoma in one systematic review. Patients will require continued follow-up. A canal wall down procedure may allow examination of the cavity for recurrence, but if the attic has been reconstructed a second-look procedure or a non-echo-planar imaging diffusion-weighted MRI may be necessary to examine the middle ear for recurrent disease.

b. Laboratory monitoring

VI. COMPLICATION

a. recurrence of cholesteatoma

Cholesteatoma has a tendency to recur despite treatment, and may occur long after the initial surgical excision. The risk of recurrence after surgery in children has been shown to be higher if the disease involves the posterior mesotympanum, and if the ossicular chain is interrupted after excision, or both.

Recurrent disease may be detected on follow-up otoscopy or present as a persistent aural discharge. Treatment is by further surgery.

b. Hearing loss

Patients with cholesteatoma commonly present with a conductive hearing loss. sis).

c. labyrinthine fistula

One of the most common complications of cholesteatoma (occurs in 7% of patients). The horizontal semicircular canal is most often involved (90%). Symptoms include vertigo, and patients may have a positive fistula test. Diagnosis can be made preoperatively with a CT scan of the petrous temporal bone (60% sensitivity). Treatment is closure of the fistula at the time of mastoidectomy, but surgical manipulation may result in sensorineural hearing loss.

d. Facial palsy

Cholesteatoma may cause facial paralysis via involvement of a dehiscence facial nerve or via bone erosion of the facial canal. Facial paralysis is often slowly progressive and has a worse prognosis. Diagnosis is clinical. A CT scan of the petrous temporal bone can be useful in surgical planning as well as demonstrating other potential complications such as a labyrinthine fistula. Treatment involves mastoidectomy for removal of the cholesteatoma in contact with the facial nerve.

e. Meningitis

Meningitis occurs in approximately 0.1% of patients. Common symptoms are fever, persistent headache, nausea and vomiting, lethargy, and irritability. Clinical signs include nuchal rigidity, new-onset seizures, ataxia, and decreased mental status. If confirmed on cranial CT or MRI brain and lumbar puncture, treatment includes intravenous antibiotics and corticosteroids.

f. Brain abscess

This often occurs in the temporal lobe of the brain or cerebellum. *Proteus* species is the most common causative organism, but many abscesses are sterile on culture.

g. Sigmoid sinus thrombosis

Involvement of the internal jugular vein may cause septic pulmonary emboli. Classical presenting symptom is spiking temperatures (picket fence pattern) with headache and lethargy. Diagnosis is made with a cranial contrast CT or MRI. Treatment includes intravenous antibiotics and mastoidectomy.

h. Mastoiditis

This is an acute infection of the temporal bone that occurs more commonly in acute otitis media but may occur in chronic otitis media with cholesteatoma. Patients present with post-auricular erythema, swelling, tenderness, fever, and an inferiorly and laterally displaced pinna.

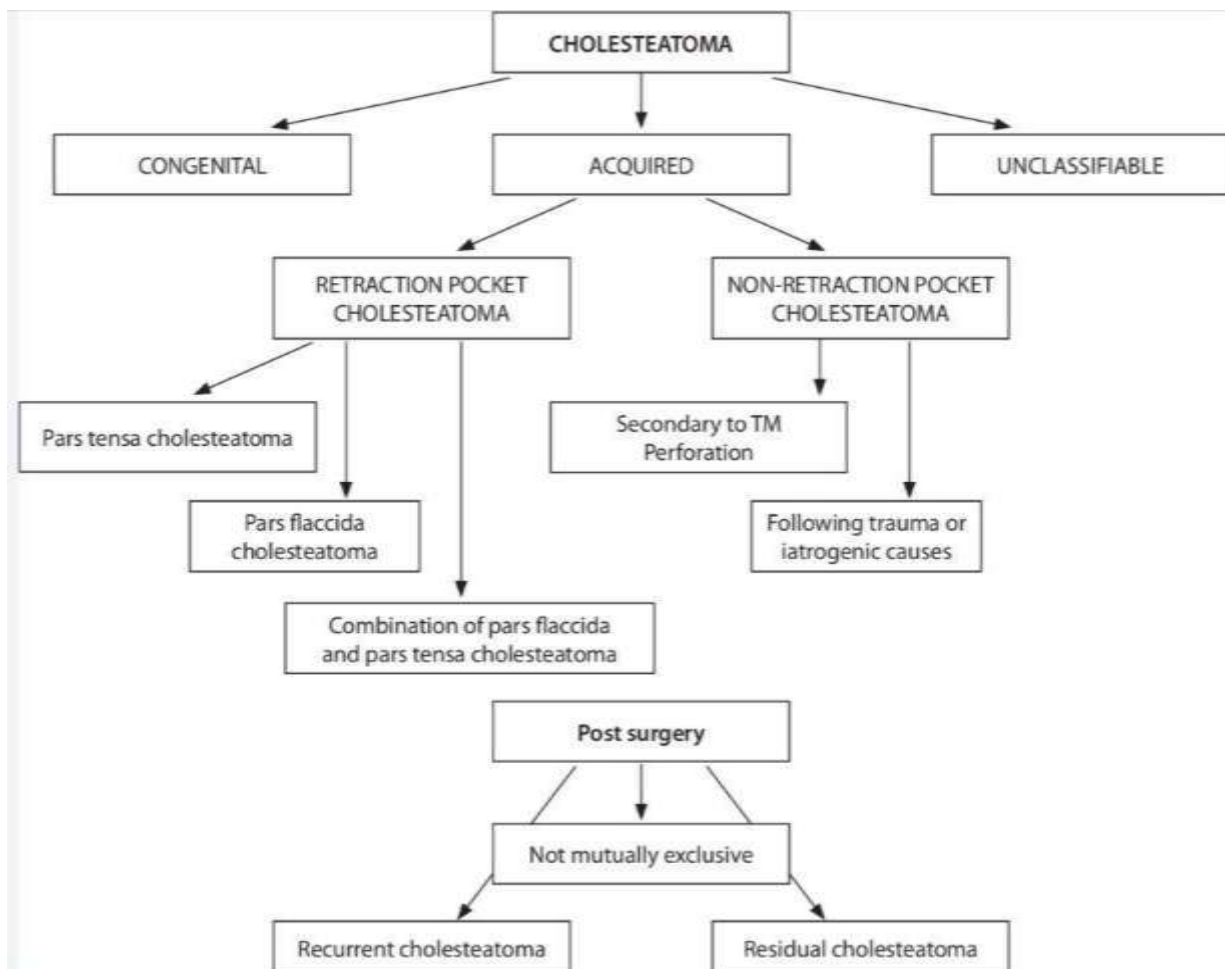
i. Petrous apicitis

Inflammation may spread to the petrous apex of temporal bone, resulting in localised osteomyelitis and reactive meningitis. Classic symptoms include deep retro-orbital pain, ear discharge, and sixth nerve palsy (Gradenigo's syndrome). Diagnosis is made via a CT scan of the petrous temporal bone.

j. Epidural abscess

Clinical signs may be subtle and include increased otalgia or headache. Diagnosis is made via a CT scan with contrast or an MRI scan. Treatment is surgical and the abscess may be drained via mastoidectomy if found intra-operatively.

VII. ALGORITHM



VIII. REFERENCES

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CHRONIC SUPPURATIVE OTITIS MEDIA

Po Ratha, Lock Sothearith, Meng sour, Theng Sakphea, Heng Sambo, Nen sopath, Yos Sophorn

I. DEFINITION

It is defined as chronic inflammation of mucoperiosteal lining of the middle ear cleft.

Classification (Adapted from Scott-Brown Otolaryngology, 6th edn.)

- a. Tubotympanic type/safe type mucosal type of otitis media (mucosa! type of otitis media)
 - Inactive mucosal COM (permanent perforation with no discharge)
 - Active mucosal COM (permanent perforation with discharge)
 - Healed COM (healed perforation/tympanosclerosis)
- b. Atticoantral type/dangerous type (squamosal type of otitis media)
 - Inactive squamous COM (retraction pocket with potential of being active with cholesteatoma)
 - Active squamous (retraction pocket with cholesteatoma)

AETIOLOGY OF TUBOTYMPANIC TYPE

a. Predisposing factors

- Inadequate/improper treatment of ASOM.
- Infection from surrounding areas like nose, nasopharynx and oropharynx.
- Some diseases like tuberculosis are chronic from the beginning.
- Pneumatization of mastoid-sclerotic mastoids is more prone for CSOM.

b. Exciting factors

- Gram-negative organisms like Pseudomonas, Proteus, E. coli
- Streptococcus.
- Staphylococcus.

II. DIAGNOSTIC PROCEDURE

i. Clinical argument

- **Discharge:** Profuse, intermittent, predominantly mucoid occasionally mucopurulent, non-foul smelling, whitish/yellowish and tenacious.
- **Deafness:** Mild conductive

- **Earache:** If associated with otitis externa.
- **Otoscopy:** Tympanic membrane shows central perforation occurs in the pars tensa without any involvement of the margin of the drum. Size of the perforation will range from small to very large perforation in the pars tensa.

ii. Technical procedure

- a. **Laboratory test:** Culture and sensitivity of the discharge in case of active stage.
- b. Imaging study
 - Examination under microscope
 - Pure tone audiogram: Mild conductive loss between 20 and 30 dB.
 - X-ray of mastoids: To rule out mastoiditis
 - CT-scan of temporal bone

III. DIFFERENTIAL DIAGNOSIS

IV. THERAPEUTIC APPROACH

i. Stabilize patient

Aural toilet by (a) dry mopping, (b) wet mopping or (c) suction cleaning. Wet mopping is discouraged nowadays. Suction cleaning is the best method but may not be possible in children. Antibiotic ear drops after culture and sensitivity report.

ii. Empirical treatment

- a. **Active stage:** Conservative treatment with medical management only and treatment of focus of infection especially rhinogenic.
- b. **Quiescent stage:** Treatment of focus of infection. If infective focus is limited to mastoid, a myringoplasty. Tympanoplasty can be done along with cortical mastoidectomy.
- c. **Inactive stage:** Treatment of focus of infection if any, Myringoplasty / tympanoplasty.

iii. Directed treatment

- **Myringoplasty** for persistent perforation if hearing loss is less than 40 dB.
- **tympanoplasty**, if above 40 dB.
- If X-ray mastoid shows mastoiditis, do cortical mastoidectomy.

iv. Monitoring

- c. Clinical monitoring

- d. Laboratory monitoring

V. COMPLICATION

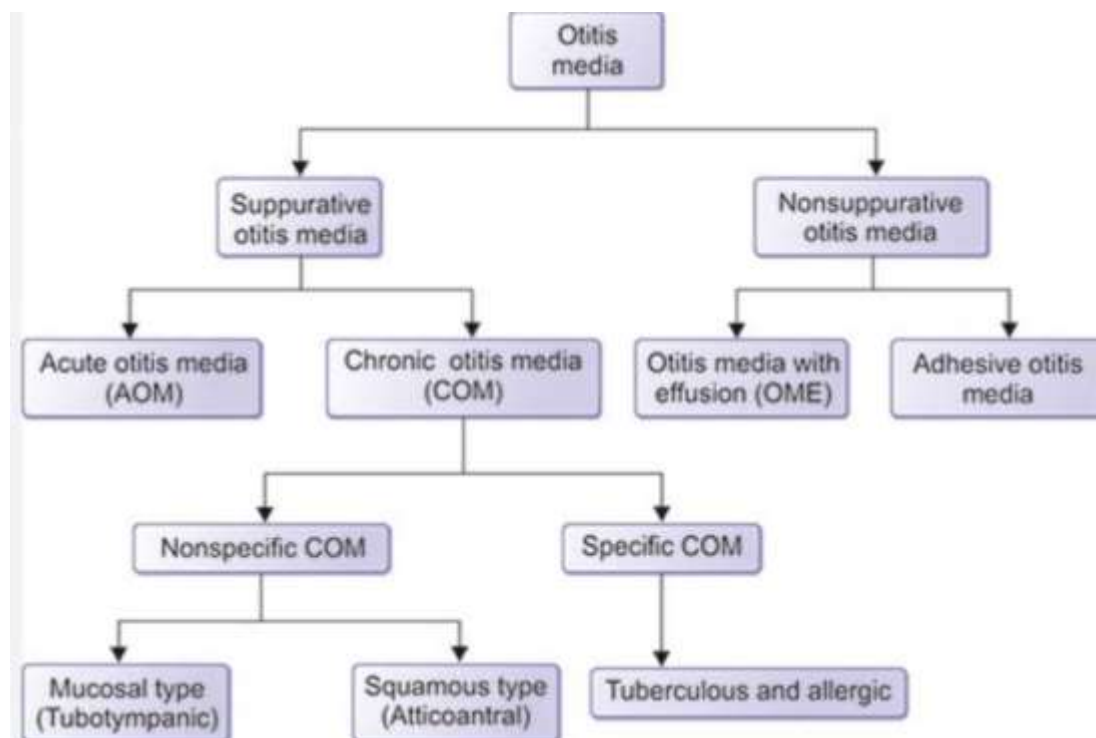
a. Extracranial

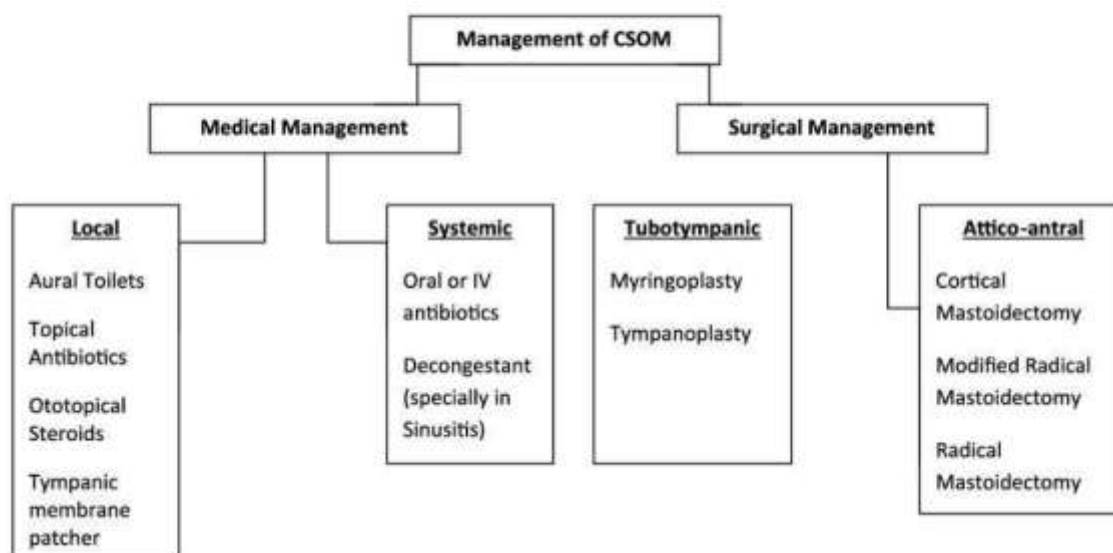
- Mastoiditis and mastoid abscess
- Facial nerve paralysis
- Labyrinthitis
- Petrositis
- Otogenic tetanus

b. Intracranial

- Extradural abscess
- Subdural abscess
- Brain abscess
- Lateral sinus thrombosis
- Otitic hydrocephalus
- Meningitis

VI. ALGORITHM





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CHRONIC TONSILLAR HYPERTROPHY

Po Ratha, Lock Sothearith, Meng sour, Theng Sakphea, Heng Sambo,
Nen sophat, Yos Sophorn

I. DEFINITION

Chronic tonsillar hypertrophy is the medical term for persistently enlarged tonsils. The tonsils are two small glands located on either side of the back of the throat. They're part of your immune system and help to fight off infections that can be caused by bacteria, viruses, and other germs that enter your body through your nose and mouth.

Enlarged tonsils can be a sign of infection or irritation from things like smoke or polluted air. Some people just naturally have larger tonsils. In other cases, there's no known cause.

Chronic tonsillar hypertrophy is particularly common in children, though it can also affect adults. Children commonly get enlarged tonsils, but the condition can affect adults too. Children's tonsils are generally larger than adult's because their bodies are busy fighting off the frequent colds and other viruses of childhood. Large tonsils often get smaller on their own as children age.

II. ETIOLOGY

Chronic tonsillar hypertrophy tends to affect children, but experts aren't sure why. Some children are simply born with larger tonsils. There might also be a genetic link, as tonsillar hypertrophy often runs in families.

In both children and adults, enlarged tonsils may also be a sign of an underlying bacterial or viral infection, such as:

- strep throat
- cold
- mononucleosis
- flu

These infections all share some common symptoms, including:

- sore throat
- fever
- fatigue
- swollen glands in the neck

Other things that can make your tonsils swell and look larger include:

- allergies
- exposure to irritants such as secondhand smoke and pollution in the air
- gastroesophageal reflux

III. DIAGNOSTIC PROCEDURE

a. Clinical argument

Enlarged tonsils don't always cause symptoms. However, if they're very large, they can partially block your throat, affecting your breathing.

Other possible signs and symptoms of enlarged tonsils include:

- difficulty breathing through the nose
- mouth breathing
- noisy breathing
- loud snoring
- obstructive sleep apnea
- restless sleep
- daytime sleepiness
- constant runny nose
- repeated ear or sinus infections
- trouble eating in young children
- bad breath

b. Technical procedure

Depending on your symptoms, they may also do a throat culture. This involves swabbing the back of the throat and testing the tissue for signs of a bacterial infection. You may also need an X-ray to give your doctor a better view of the soft tissues in your neck.

If you've been having symptoms such as trouble sleeping or loud snoring, your doctor might also suggest doing a sleep study to check for sleep apnea caused by tonsillar hypertrophy. To do this, you'll typically need to spend the night in a specially designed laboratory while a doctor monitors your breathing and other vital signs.

IV. DIFFERENTIAL DIAGNOSIS

- Membranous tonsillitis
- Diphtheria tonsillitis
- Infectious mononucleosis

- Candidiasis
- Vincent's angina
- Tonsillar neoplasm
- Agranulocytosis
- Traumatic ulcer
- Keratosis pharyngitis

V. THERAPEUTIC APPROACH

Tonsillar hypertrophy usually only requires treatment if it's interfering with your ability to sleep, eat, or breathe. However, if it's caused by an underlying infection, you may need antibiotics. If it's due to allergies, your doctor might recommend using a nasal corticosteroid spray or taking antihistamines to help with your symptoms.

Surgery

If your enlarged tonsils interfere with your breathing and aren't due to any underlying condition, you may need to have them surgically removed. This can help improve the symptoms of obstructive sleep apnea in both adults and children. Surgery to remove the tonsils is called a tonsillectomy.

During a tonsillectomy, your doctor might also remove your adenoids, which are two glands located in the back of your nose near the roof of your mouth.

A tonsillectomy is a straightforward procedure done under general anesthesia. Most people go home the same day as their surgery and make a full recovery within 7 to 10 days.

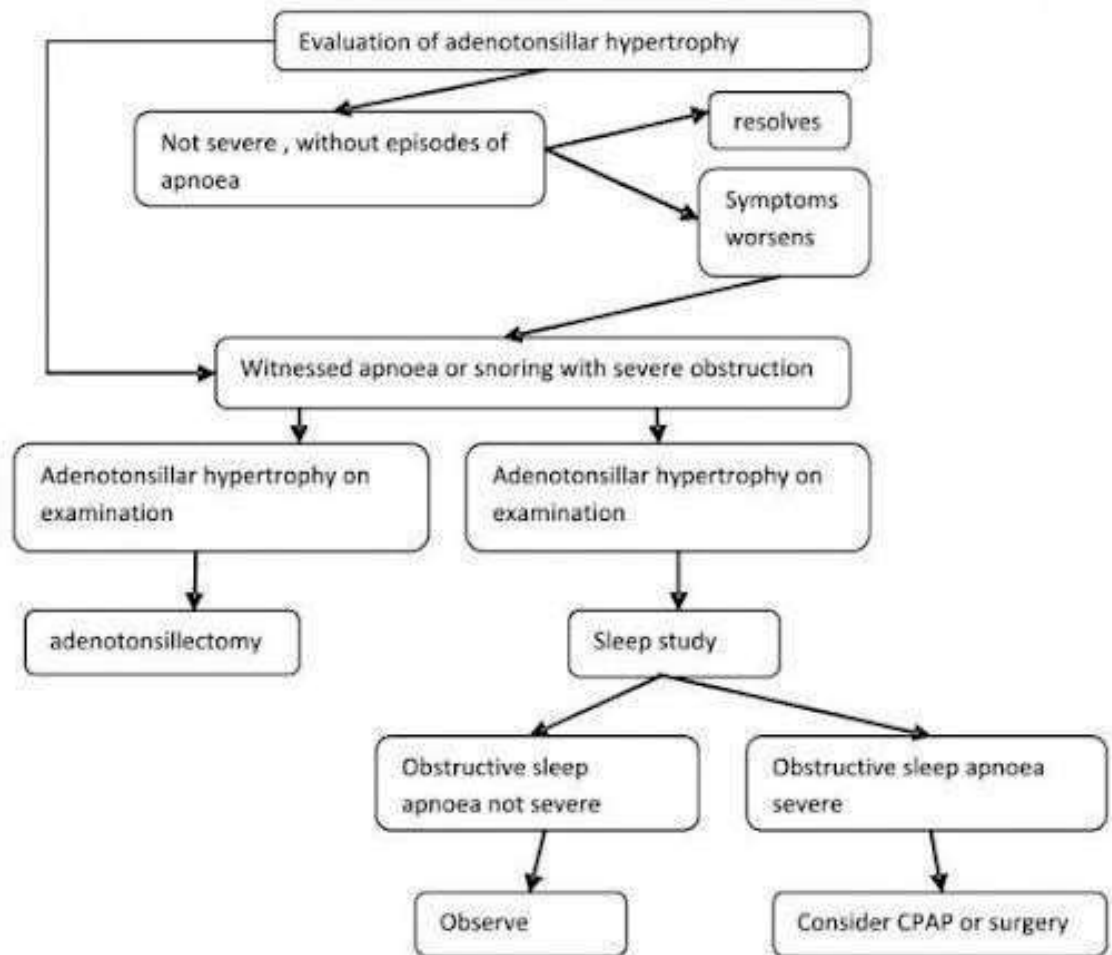
VI. COMPLICATION

When tonsillar hypertrophy leads to sleep apnea and trouble sleeping, it can cause a range of complications if left untreated, especially in children.

These include:

- heart and lung conditions, such as pulmonary hypertension or heart enlargement
- trouble in school
- behavioral problems
- frequent illness

VII. ALGORYTHM



VIII. REFERENCES

<https://www.healthline.com/health/tonsillectomy-scabs#timeline>.

DEVIATED SEPTUM & CHRONIC RHINOSINUSITIS

Po Ratha, Loek Sothearith, Meng sour, Theng Sakphea, Heng Sambo,

Nen sophat, Yos Sophorn

I. CASE DEFINITION

A deviated septum occurs when the nasal septum, the bone, and cartilage that divide the nasal cavity, is significantly off center or crooked, potentially causing breathing difficulties or other complications.

ETIOLOGY

- ❑ **Congenital:** Present at birth due to fetal development.
- ❑ **Traumatic:** Resulting from injury or trauma to the nose.
- ❑ **Developmental:** Changes occurring during growth that cause deviation.

II. DIAGNOSTIC PROCEDURE

Clinical Argument:

- ❑ History and physical examination focusing on nasal obstruction, difficulty breathing through the nose, recurrent sinus infections, nosebleeds, and facial pain.

Technical Procedure:

- **Laboratory Test:**
 - Generally, not required for diagnosing a deviated septum.
- **Imaging Study:**
 - Nasal endoscopy to visualize the internal nasal structure.
 - Standard X-rays of the sinus cavities to detect sinusitis, fractures, and other abnormalities.
 - CT scan of the sinuses for a detailed view if surgical intervention is considered or if chronic sinusitis is suspected.

III. DIFFERENTIAL DIAGNOSIS

- ❑ Nasal polyps
- ❑ Chronic rhinosinusitis
- ❑ Allergic rhinitis
- ❑ Turbinate hypertrophy

IV. THERAPEUTIC APPROACH

- Stabilize **Patient**:
 - Address acute nasal obstruction if present, especially in the case of trauma.
- Empirical Treatment:
 - Conservative management with nasal corticosteroids or decongestants to reduce inflammation and improve breathing.
 - Saline nasal irrigation for symptomatic relief.
- Directed **Treatment**:
 - Septoplasty: Surgical correction of the deviated septum, recommended for patients with significant symptoms not relieved by medical treatment.
- Monitoring:
 - Clinical **Monitoring**:
 - Follow-up visits to assess symptom improvement and surgical outcomes if performed.
 - Laboratory **Monitoring**:
 - Not typically necessary unless part of preoperative assessment or managing complications.

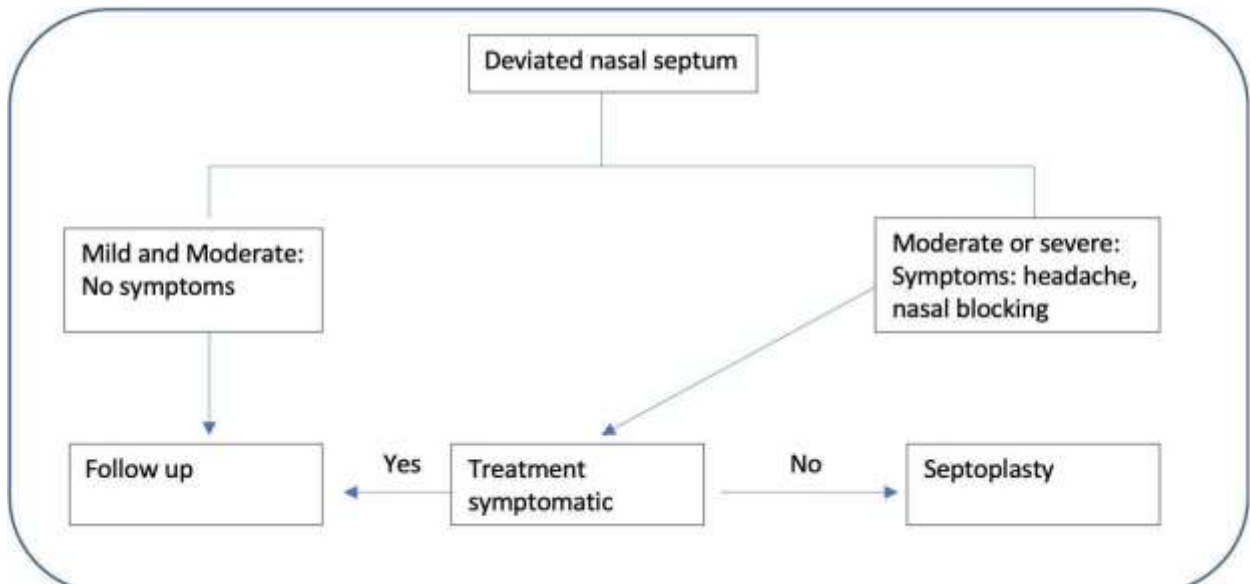
V. COMPLICATION

- Persistent symptoms post-surgery
- Nasal bleeding
- Infection
- Septal perforation
- Adverse reaction to anesthesia

VI. ALGORITHM

- a. Assess symptoms and clinical history.
- b. Perform physical examination and nasal endoscopy.
- c. Conduct imaging studies if needed.
- d. Initiate empirical treatment.
- e. Consider surgical intervention (septoplasty) if symptoms persist.

- f. Monitor patient response and adjust treatment as necessary.



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CHRONIC RHINOSINUSITIS

I. CASE DEFINITION

Chronic rhinosinusitis (CRS) is defined as inflammation of the nasal and paranasal sinus mucosa lasting for at least 12 weeks, despite medical management. Symptoms include nasal obstruction, nasal discharge, facial pain/pressure, and reduction/loss of smell.

II. ETIOLOGY

- **Infectious Causes:** Persistent or recurrent bacterial or fungal infections.
- **Non-infectious Causes:** Allergic reactions, nasal polyps, anatomical **variations**, immunodeficiency, environmental factors, and systemic diseases such as cystic fibrosis or aspirin-exacerbated respiratory disease (AERD).

III. DIAGNOSTIC PROCEDURE

- **Clinical Argument:**
 - Detailed history and physical examination focusing on symptom duration, severity, and impact on quality of life.
- **Technical Procedure:**
 - **Laboratory Test:**
 - Nasal swabs or cultures to identify pathogenic organisms.
 - Allergy testing if allergic rhinitis is suspected.
 - **Imaging Study:**
 - Standard X-rays of the sinus cavities to detect sinusitis, fractures, and other abnormalities.
 - CT scan of the sinuses to evaluate the extent of sinus involvement, anatomical abnormalities, and presence of polyps or other obstructions.
 - Nasal endoscopy to visualize the nasal passages and obtain cultures or biopsy if needed.

IV. DIFFERENTIAL DIAGNOSIS

- Allergic rhinitis
- Acute rhinosinusitis
- Non-allergic rhinitis
- Nasal polyps

- **Neoplastic lesions**

V. THERAPEUTIC APPROACH

□ Stabilize Patient:

- Manage **acute** exacerbations and ensure there are no immediate complications requiring urgent intervention.

□ Empirical Treatment:

- Nasal corticosteroids to reduce inflammation.
- Saline nasal irrigation to improve mucociliary clearance.
- Short courses of antibiotics for bacterial exacerbations.
- Oral corticosteroids for severe inflammation or polyps.

□ Directed Treatment:

- Long-term antibiotics for chronic bacterial infection.
- Antifungal treatment if fungal infection is identified.
- Surgery (e.g., functional endoscopic sinus surgery - FESS) for patients with persistent symptoms despite medical therapy, to improve sinus drainage and remove obstructions such as polyps.

□ Monitoring:

○ Clinical Monitoring:

- Regular follow-up visits to assess symptom control and response to treatment.
- Adjust treatment plan based on symptom persistence or recurrence.

○ Laboratory Monitoring:

- Repeat cultures or imaging if symptoms worsen or do not improve with standard therapy.

VI. COMPLICATION

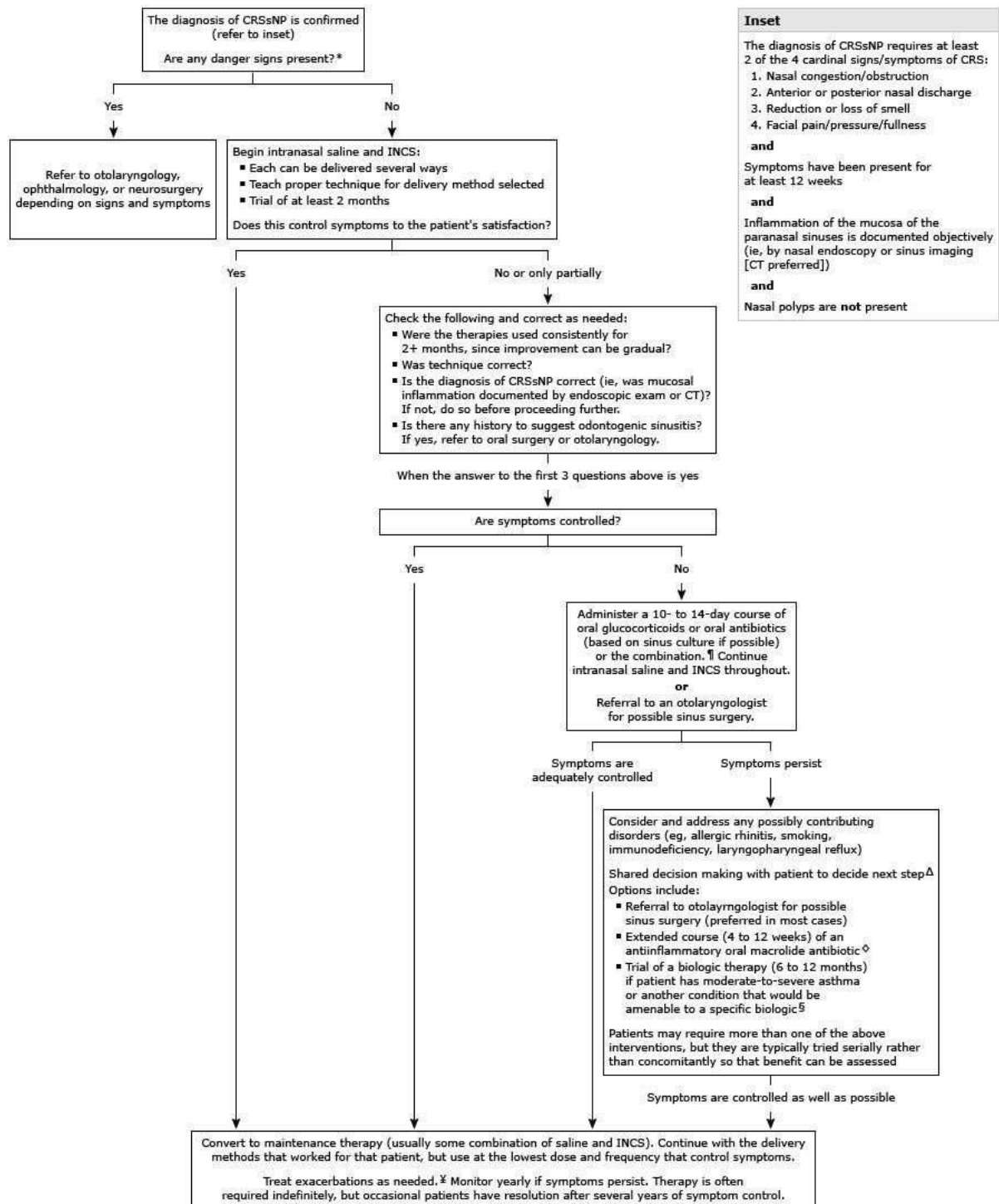
- Orbital cellulitis or abscess
- Intracranial complications such as meningitis or brain abscess
- Osteomyelitis of the frontal bone (Pott's puffy tumor)
- Chronic pain and impact on quality of life

VII. ALGORITHM

- Assess symptoms and clinical history.
- Perform physical examination and nasal endoscopy.

- c. Conduct imaging studies (CT scan).
- d. Initiate empirical treatment.
- e. Consider directed treatment based on clinical response.
- f. Evaluate need for surgical intervention if medical therapy fails.
- g. Monitor patient response and adjust treatment accordingly.

Management of chronic rhinosinusitis without nasal polyposis in adults



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EPISTAXIS

Po Ratha, Loek Sothearith, Meng sour, Theng Sakphea,
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I. DEFINITION

Epistaxis is bleeding from nose (nosebleed). Blood may flow anteriorly or go posteriorly.

Is one of the most common ear, nose, and throat (ENT) emergencies that present to the emergency department or the primary care clinic.

There are two types of nosebleeds: anterior (more common), and posterior (less common, but more likely to require medical attention).

The source of 90% of anterior nosebleeds is within Kiesselbach's plexus (also known as Little's area) on the anterior nasal septum.

There are five named vessels whose terminal branches supply the nasal cavity:

- 1) Anterior ethmoidal artery
- 2) Posterior ethmoidal artery
- 3) Sphenopalatine artery
- 4) Greater palatine artery
- 5) Superior labial artery

The watershed area of these five vessels is in the anterior nasal septum, comprising Kiesselbach's plexus. This lies at the entrance to the nasal cavity and so is subject to extremes of heat and cold, and of high and low moisture, and is easily traumatized. The mucosa over the septum in this area is especially thin, making this the site of the majority of epistaxis. More rarely, vessels in the posterior or superior nasal cavity will bleed, leading to the so-called "posterior" epistaxis. This is more common in patients on anticoagulants, patients who are hypertensive, and patients with underlying blood dyscrasia or vascular abnormalities. Management will depend on the severity of the bleeding and the patient's concomitant medical problems.

II. ETIOLOGY

There are multiple causes of epistaxis which can be divided into local, systemic, environmental, and medication-induced.

Local causes:

- Digital manipulation
- Deviated septum
- Trauma

- Chronic nasal cannula use

Systemic causes:

- Alcoholism
- Hypertension
- Vascular malformations
- Coagulopathies (von Willebrand disease, hemophilia)

Environmental factors:

- Allergies
- Environmental dryness (more common in winter months)

Medications:

- NSAIDs (ibuprofen, naproxen, aspirin)
- Anticoagulants (warfarin)
- Platelet aggregation inhibitors (clopidogrel)
- Topical nasal steroid sprays
- Supplement/alternative medications (vitamin E, ginkgo, ginseng)
- Illicit drugs (cocaine)

While epistaxis is a very common spontaneous problem, rarer etiologies such as neoplasms or vascular malformations must always be in the differential diagnosis, particularly if additional symptoms such as unilateral nasal obstruction, pain, or other cranial nerve deficits are noted.

III. DIAGNOSTIC PROCEDURE

A. Clinical argument

- Differentiating an anterior or posterior is key in management.
- Diagnosis of anterior bleeding is can be made by direct visualization using a nasal speculum and light source.
- A topical spray with anesthetic and epinephrine may be helpful for vasoconstriction to help control bleeding and to aid in the visualization of the source.
- Diagnosis of posterior bleeding is made after measures to control anterior bleeding have failed. Clinical features of posterior bleeding can include active bleeding into the posterior pharynx in the absence of an identified anterior source; high-flow posterior bleeds may cause blood to emanate from both nares.

B. Technical procedure

- Labs may be obtained if necessary, including a complete blood cell count (CBC), type and cross match, and coagulation studies, though should not delay treatment of an active bleed.
- Imaging such as x-ray or computed tomography have no role in the urgent or emergent management of active epistaxis.

IV. DIFFERENTIAL DIAGNOSIS

- ☐ Nasal tumor
- ☐ DIC
- ☐ Hemophilia
- ☐ Von Willebrand disease
- ☐ Rhinitis
- ☐ Foreign body in the nose
- ☐ Drug toxicity (Warfarin, NSAIDs)

V. THERAPEUTIC APPROACH

Start with a primary survey and address the airway, ensure the airway is patent.

Next, assess for hemodynamic compromise. Obtain large-bore intravenous access in patients with severe bleeding and obtain labs. Reverse blood clotting as necessary, if there is a concern with medication use.

All patients with moderate to severe nose bleeding should have two large-bore intravenous lines and infusion of crystalloid. The monitoring of oxygen and hemodynamic stability is vital.

Treatment for anterior bleeding can be started with direct pressure for at least 10 minutes. Have the patient apply constant direct pressure by pinching the nose over the cartilaginous tip (instead of over the bony areas) for a few minutes to try to control the bleed.

If that is ineffective, vasoconstrictors such as oxymetazoline or thrombogenic foams or gels can be employed. It is important to remove all clot with suction before any attempt at treatment is made. The reasons are two fold:

- 1) Clot will prevent any medication from reaching the vessel itself and
- 2) If packing becomes necessary, the clot can be pushed into the nasopharynx and aspirated.

If topical treatments are unsuccessful, proceed with nasal examination to identify and cauterize the vessel with silver nitrate.

If this too is unsuccessful, anterior nasal packing is necessary. This can be performed with absorbable packing material such as surgicel or fibrillar, or with devices such as anterior epistaxis balloons, or nasal tampons (Rapid Rhino). If silver nitrate is used to cauterize a septal blood vessel, only use it on one side of the septum to prevent septal perforation. Thermal coagulation is painful and should rarely be attempted in an emergent setting. Traditional petrolatum gauze can be used if one does not have access to balloons or tampons.

If none of this is successful, the bleeding may be from the posterior or superior nasal cavity. Symptoms can include active bleeding from both nostrils or active bleeding present in the posterior pharynx. Longer (7.5cm) nasal tampons are available that provide some more posterior pressure and can be employed in this situation. Formal posterior nasal packing should only be performed by experienced personnel as it requires admission and telemetry monitoring, and sometimes intubation. It is associated with higher rates of complications like pressure necrosis, infection, or hypoxia, and may trigger a nasal-cardiac reflex (sudden bradycardia after nasal packing - if this occurs, remove the pack immediately). Foley catheters can be used by experienced personnel to tamponade a posterior bleed. If a posterior pack is placed, a formal petrolatum gauze anterior pack must be placed as well to create a closed, tamponaded space in the nasopharynx.

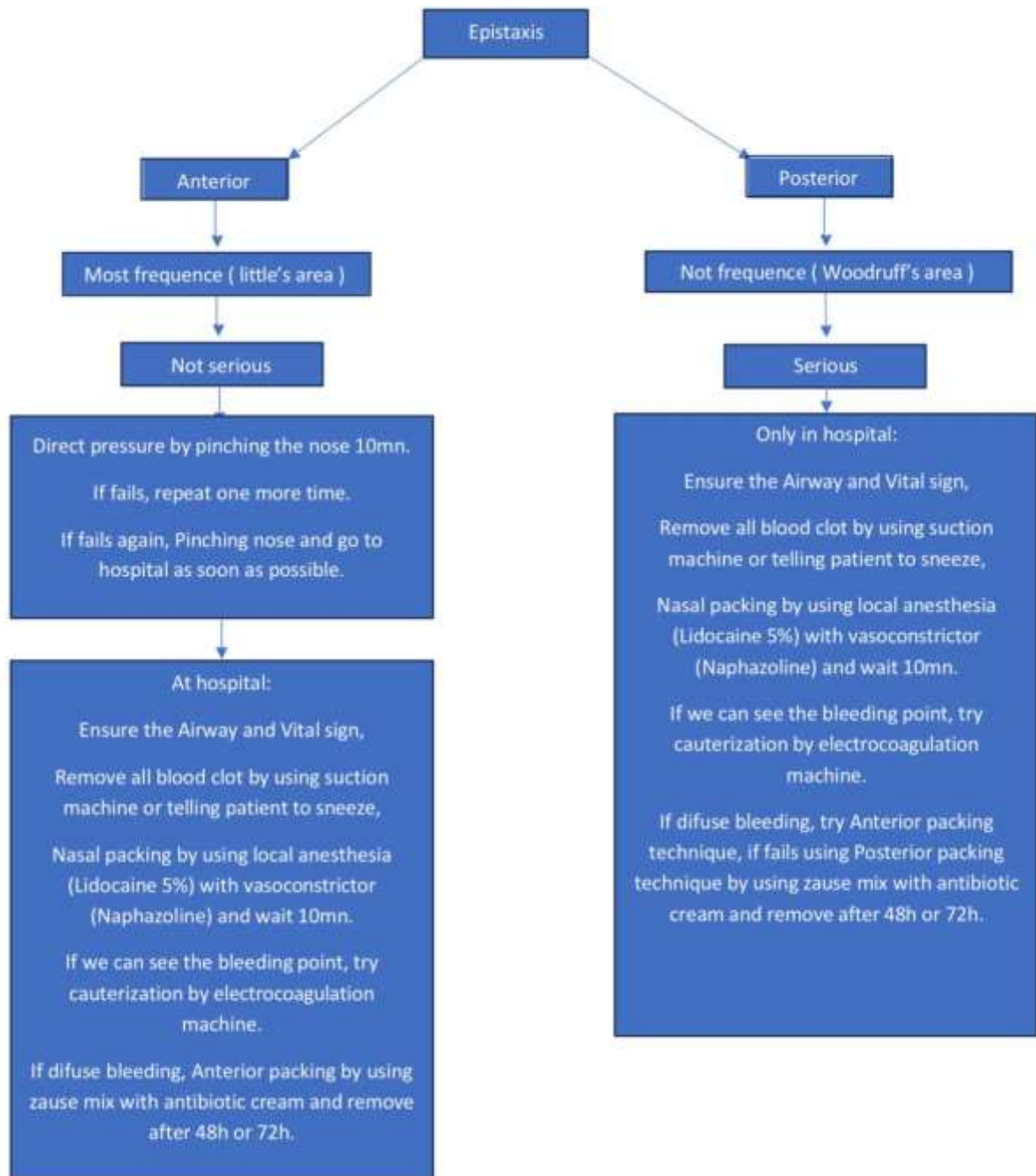
If all of these measures are unsuccessful, the patient should be intubated for airway protection and interventional radiology consulted emergently for embolization. If this service is unavailable, operative ligation of the sphenopalatine and ethmoid arteries can be performed in the operating room by an otolaryngologist.

VI. COMPLICATION

Complications of epistaxis may include the following:

- Sinusitis
- Septal hematoma/perforation
- External nasal deformity
- Mucosal pressure necrosis
- Vasovagal episode
- Balloon migration
- Aspiration

VII. ALGORYTHM



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NASAL POLYPS

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I. CASE DEFINITION

- Nasal polyps (NP) are benign swellings of the mucosal lining of the paranasal sinuses.
- They are invariably associated with chronic rhinosinusitis with the presentation correctly labelled as chronic rhinosinusitis with nasal polyps (CRSwNP).
- Polyp tissue comprises a mixture of loose connective tissue, oedema, inflammatory cells (mostly eosinophils), glandular cells, and capillaries.
- Polyps are covered with varying types of epitheliums, mostly respiratory pseudostratified epithelium with ciliated cells and goblet cells. [1]

II. ETIOLOGY

- The etiology of nasal polyps is not fully understood.
- They seem to arise due to chronic mucosal inflammation.
- Chronic rhinosinusitis with nasal polyps (CRSwNP) is closely associated with adult-onset, eosinophilic asthma.
- There is likely to be a shared etiology in such patients, as well as shared pathology, but the precipitating factors are unknown.
- There is a genetic component to CRSwNP, with a positive family history in 15% to 52% of patients, although specific risk genes are not well characterized.
- The role of environmental factors is unclear.[1] [2]

People with aspirin- and non-steroidal anti-inflammatory drug (NSAID)-exacerbated respiratory disease have a high incidence of nasal polyps (estimated at between 36% and 96%).

The combination of CRSwNP, asthma, and aspirin/NSAID hypersensitivity is commonly referred to as Samter's triad.

Allergic fungal rhinosinusitis (AFRS) features nasal polyps in 80% of patients. Here, it is assumed that an intense, IgE-driven, eosinophilic inflammatory response occurs to fungal hyphae that are able to colonise the nasal sinuses, such as *Aspergillus* species.

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

- nasal obstruction

- Occurs in 81% to 95% of patients with chronic rhinosinusitis.[3]
- often obstruction is bilateral in patients with nasal polyps.
- Most Nasal obstruction may be valvular, with obstruction in either inspiration or expiration, depending on the size and location of the polyps.
- Unilateral obstruction should be regarded as suspicious of a neoplasm until shown otherwise.
- nasal discharge
 - Anterior and/or posterior nasal drips are common in patients with nasal polyps. If there is associated infection, it may be mucopurulent.
- facial pain/pressure
 - Facial congestion-pressure-fullness occurs in 70% to 85% of patients with chronic rhinosinusitis.[3]
 - facial pain/pressure and nasal polyps are poorly correlated
 - facial pain/pressure is more common in patients with chronic rhinosinusitis without nasal polyps.
- direct visualization
 - NP must be seen on anterior rhinoscopy or nasal endoscopy, or indirectly on CT scan, for the diagnosis to be made.
- reduced sense of smell/anosmia
- cough

ii. Technical procedure

- a. Laboratory test
 - Biopsy
 - Nasal smear and culture
 - FBC with differential
 - Allergy testing [2]
 - Skin-prick tests and serum allergen-specific IgE tests
- b. Imaging study
 - Endoscopy image
 - Sinus X-rays
 - Paranasal CT scan

The endoscopic appearance of polyps is graded as follows:[1] Grade 0: absence of polyps

Grade 1: polyps in the middle meatus only

Grade 2: polyps beyond the middle meatus but not completely obstructing the nasal cavity

Grade 3: polyps completely obstructing the nasal cavity.

Test	Result
CT sinuses □ Undertaken if there are unilateral signs and symptoms or other features suggestive of neoplasia such as bleeding, crusting, cacosmia (perception of foul odour), periorbital oedema, displaced globe, impaired vision, ophthalmoplegia, severe frontal headaches, frontal swelling, signs of meningitis or focal neurology, or systemic signs of neoplasms such as weight loss. □ Also undertaken if surgical intervention is being considered as part of the pre-operative work-up.	polyps noted and extent defined
biopsy □ Indicated if there are concerns about alternative diagnoses, particularly neoplasia. Undertaken if there are unilateral signs and symptoms or concerning features such as bleeding, crusting, cacosmia (perception of foul odor), periorbital oedema, displaced globe, impaired vision, ophthalmoplegia, severe frontal headaches, frontal swelling, signs of meningitis or focal neurology, □ or systemic signs of neoplasms such as weight loss. Histology may change the diagnosis in 1% of surgically removed NP.[4]	confirms histology
nasal smear and culture □ A nasal smear demonstrating the presence of eosinophils is typical in chronic rhinosinusitis with nasal polyps, but may also be seen in allergic rhinitis and non-allergic	presence of eosinophilia and positive culture for organisms

rhinitis with eosinophilia syndrome. Nasal swabs/nasal mucus culture may be beneficial in revealing secondary bacterial infection, although Staphylococcus aureus carriage in the nose is very common.[2]	
FBC with differential An FBC is indicated in patients with rhinosinusitis. Patients with chronic rhinosinusitis with nasal polyps often have a peripheral blood eosinophil count at the upper end of the normal range or modestly elevated. Levels of 0.4×10^6 cells/L up to 1.5×10^6 cells/L may be seen, with higher levels in patients with asthma. Levels higher than this may raise the possibility of eosinophilic granulomatous polyangiitis (also known as Churg- Strauss syndrome).[2]	eosinophilia
skin-prick tests/serum allergen-specific IgE tests Used in rhinology clinics for the diagnosis of allergic rhinitis	positive in patients with allergic rhinitis; not a required feature of chronic rhinosinusitis with nasal polyps

IV. DIFFERENTIAL DIAGNOSIS

Condition	Differentiating sings /	Differentiation tests
Neoplasm	Features that may suggest neoplasm include unilateral symptoms, epistaxis/ bleeding, crusting, cacosmia (perception of foul odour),	CT head defines extent and characteristics of the lesion(s). MRI can be performed in

periorbital oedema, globe, double or red vision, ophthalmop severe frontal head frontal swelling, sig meningitis or focal neurology, and syst symptoms such as w loss.	displaced conjunction with CT uced for further evaluation. egia, • Biopsy confirmsches, histology, which is ns of typically an inverted papilloma. mic eight
Congenital lesions (e.g., encephalocele, concha bullosa)	Concha bullosa is often CT and MRI head defines asymptomatic, but may cause extent and characteristics of sinus obstruction. the lesion(s). Encephalocele symptoms are dependent on size and location.

Foreign body

- Usually sudden onset of unilateral symptoms. Epistaxis may feature.

- Nasal endoscopy may allow identification and possible retrieval of foreign body.
- CT may identify position, shape, and nature of foreign body.

V. THERAPEUTIC APPROACH

i. Stabilize patient

Treatment options for chronic rhinosinusitis with nasal polyps (CRSwNP)

- topical medications,
- intranasal corticosteroids,
- systemic medications and surgery.[5]

Nasal polyposis in children is rare (incidence of 0.1%), and should prompt additional investigations to exclude cystic fibrosis.[2] Nasal polyps may also be a feature of the nasal and sinus inflammation seen in primary ciliary dyskinesia.

Management of nasal polyposis in children is similar to that for adults; surgery is reserved for severe CRSwNP resistant to medical treatment.

ii. Empirical treatment

- For patients with mild to moderate symptoms (visual analogue scale [VAS] score 0-7),
 - the first-line treatment is a 3-month course of an intranasal corticosteroid spray,

such as fluticasone or mometasone.

- The patient is reviewed at 3 months and, if there has been satisfactory improvement, the patient may continue to use the intranasal corticosteroid spray, at the lowest effective dose, with review every 6 months.
- Intranasal corticosteroids improve symptoms, reduce polyp size, and prevent polyp recurrence after surgery.[6] [7]
- For patients with severe symptoms (VAS score >7-10),
 - the first-line treatment is an intranasal corticosteroid spray, plus consideration of a short course of an oral corticosteroid.[2]
 - Trial evidence for the use of oral corticosteroids is relatively weak and has not confirmed a sustained effect.[8] [9] However, in practice, patients typically see a clear improvement within a few days, particularly early on in the course of the disease. Following completion of the oral corticosteroid course, the patient then continues on the intranasal corticosteroid. The patient is reviewed at 1 month and, if there has been satisfactory improvement, the patient should continue on the intranasal corticosteroid with review at 3 months.
- Doxycycline may have a modest effect in addition to intranasal corticosteroids when used for 3-12 weeks.[1] [10] [11]
- Nasal saline irrigation can also be beneficial.[12]
- Leukotriene receptor antagonists (e.g., montelukast),

iii. Directed treatment

- If there is no or minimal improvement in symptoms at the initial 3-month follow-up, most clinicians suggest surgery for removal of polyps. This necessitates a preoperative computed tomography (CT) scan to examine the local anatomy and the extent and location of the polyps, in order to make an accurate surgical plan.
- Follow-up saline douching and intranasal corticosteroids.
- Postoperative doxycycline for 3 weeks or 12 weeks may have a beneficial effect.[10][11]
- Non-steroidal anti-inflammatory drugs (NSAIDs) should be used first-line for pain.[30]

iv. Monitoring

The management of nasal polyps is aimed at controlling the symptoms by limiting the extent of the disease.

There is no cure. Monitoring the patient's symptoms and adherence to medical management is essential.

Once symptoms are under control, patients are normally followed up at 6–12-month intervals or on an as-required basis if symptoms worsen

VI. COMPLICATION

- Periorbital and orbital cellulitis
- cavernous sinus thrombosis
- osteomyelitis
- intracranial infection
- lacrimal gland abscess
- nasal septal perforation
- septicemia
- topical corticosteroid-related nasal bleeding
- oral corticosteroid-related complications
- sinus surgery-related orbital complications
- sinus surgery-related intracranial complications
- sinus surgery-related bleeding
- sinus surgery-related complications: synechiae, hyposmia, osteitis, atrophic rhinitis, toxic shock syndrome, anosmia, and death.

VII. ALGORITHM

mild to moderate polyposis (VAS 0-7)		
	1st	intranasal corticosteroid Primary options » fluticasone propionate nasal: (50 micrograms/spray) 100 micrograms (2 sprays) in each nostril twice daily; (93 micrograms/spray) 93- 186 micrograms (1-2 sprays) in each nostril twice daily

		OR » mometasone nasal: (50 micrograms/spray) 100 micrograms (2 sprays) in each nostril once or twice daily » For patients with mild to moderate symptoms (VAS score 0-7), the initial treatment is a 3- month course of intranasal corticosteroid spray, such as fluticasone or mometasone.[2] There is insufficient evidence to suggest one type of corticosteroid over another. » The patient is reviewed at 3 months and, if there has been satisfactory improvement, the patient may continue to use the intranasal corticosteroid spray, at the lowest effective dose, with review every 6 months.
	adjunct	nasal saline irrigation
	adjunct	doxycycline » doxycycline: 200 mg orally on the first day, followed by 100 mg once daily » May have a modest effect in addition to intranasal corticosteroids when used for 3-12 weeks.[1] [10] [11]
	2nd	surgical polypectomy » If there is no or minimal improvement in symptoms after the initial 3 months of medical treatment, many clinicians suggest surgery to remove the polyps.
	plus	postoperative nasal saline irrigation (daily saline douching for 1 month after the surgery)
	plus	postoperative oral antibiotic

		» doxycycline: 200 mg orally on the first day, followed by 100 mg once daily for 3-12 weeks or » clarithromycin: 250 mg orally twice daily for 8- 12 weeks Or » azithromycin: 500 mg orally three times weekly for up to 12 weeks
		postoperative analgesia » ibuprofen: 400 mg every 4-6 hours when required, maximum 2400 mg/day [13]
		ongoing intranasal corticosteroid » fluticasone propionate nasal: (50 micrograms/spray) 100 micrograms (2 sprays) in each nostril twice daily; (93 micrograms/spray) 93- 186 micrograms (1-2 sprays) in each nostril twice daily Or » mometasone nasal: (50 micrograms/spray) 100 micrograms (2 sprays) in each nostril once or twice daily
plus		
severe polyposis (VAS >7)		
	1st	intranasal corticosteroid » fluticasone propionate nasal: (50 micrograms/spray) 100 micrograms (2 sprays) in each nostril twice daily; (93 micrograms/spray) 93- 186 micrograms (1-2 sprays) in each nostril twice daily

		Or » mometasone nasal: (50 micrograms/spray) 100 micrograms (2 sprays) in each nostril once or twice daily » For patients with severe symptoms (VAS score >7-10), the first-line treatment is an intranasal corticosteroid spray, plus consideration of a short course of an oral corticosteroid.[2]
	plus	consider short course of oral corticosteroid » prednisolone: 0.5 mg/kg orally once daily in the morning for 7-14 days, maximum 40 mg/ day » For patients with severe symptoms (VAS score >7-10), the first-line treatment is an intranasal corticosteroid, plus consideration of a short course of an oral corticosteroid.[2]
	adjunct	nasal saline irrigation
	adjunct	doxycycline » doxycycline: 200 mg orally on the first day, followed by 100 mg once daily » May have a modest effect in addition to intranasal corticosteroids when used for 3-12 weeks.[1] [10] [11]
	2nd	surgical polypectomy » If there is no or minimal improvement in symptoms after the initial 1 month of medical treatment, many clinicians suggest surgery to remove the polyps.

	plus	postoperative nasal saline irrigation (daily saline douching for 1 month after the surgery)
	plus	postoperative oral antibiotics » doxycycline: 200 mg orally on the first day, followed by 100 mg once daily for 3-12 weeks Or » clarithromycin: 250 mg orally twice daily for 8- 12 weeks Or » azithromycin: 500 mg orally three times weekly for up to 12 weeks
	plus	postoperative analgesia » ibuprofen: 400 mg every 4-6 hours when required, maximum 2400 mg/day [13]
	plus	ongoing intranasal corticosteroid » fluticasone propionate nasal: (50 micrograms/spray) 100 micrograms (2 sprays) in each nostril twice daily; (93 micrograms/spray) 93- 186 micrograms (1-2 sprays) in each nostril twice daily Or » mometasone nasal: (50 micrograms/spray) 100 micrograms (2 sprays) in each nostril once or twice daily

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NODULAR GOITER

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I. DEFINITION

A goiter or goitre (Latin gutteria, struma), is a swelling in the thyroid gland which can lead to a swelling of the neck or larynx (voice box). Goitre rarely occurs when the thyroid gland is functioning properly. Worldwide, over 90% cases of goitre are caused by iodine deficiency.

Classification: They can be classified in several ways by morphology, appearance, cause and other characteristics.

- Non-Toxic: Simple, Uninodular, Multinodular
- Toxic: Diffuse (Graves), Toxic multinodular, Toxic nodule
- Special: Cancer, Thyroiditis, Inflammatory
- Various causes: Chronic infection, Actinomycosis, Amyloidosis Other type of classification:
- Class I - palpation struma - in normal posture of the head, it cannot be seen; it is only found by palpation.
- Class II - the struma is palpative and can be easily seen.
- Class III - the struma is very large and is retrosternal; pressure results in compression marks.

II. ETIOLOGY

Worldwide, the most common cause for goiter is iodine deficiency. Selenium deficiency is also considered a contributing factor. In countries that use iodized salt, Hashimoto's thyroiditis is the most common cause.

Further causes include:

1. Thyroid autonomy
2. Autoimmune conditions of the thyroid (Hashimoto thyroiditis, Morbus Basedow)
3. Medications and substances such as lithium, antithyroid agents, thiocyanate
4. Inflammations (Thyroiditis)
5. Cysts
6. Benign and malignant neoplasms
7. Pituitary problems

8. Acromegaly
9. Thyroid hormone insensitivity
10. Sarcoidosis, Amyloidosis
11. Hydatiform mole

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

Physical examination involves performing a detailed assessment of the thyroid mass and any associated lymphadenopathy, evaluating whether the patient has signs of hypothyroidism or hyperthyroidism and evaluating whether the surrounding structures (trachea and oesophagus) are compressed.

Examination must include inspection and palpation of the anterior and lateral aspects of the neck to assess for thyroid enlargement, presence of nodules, and lymphadenopathy. Asking the patient to swallow during palpation, which elevates the thyroid, can improve the palpation of the thyroid gland and detection of nodules.

The signs of hyperthyroidism include tachycardia, arrhythmias, muscle wasting, tremor, brisk reflexes, and brittle hair.

The signs of hypothyroidism include bradycardia, thickened and puffy appearance of skin (myxoedema), and delayed relaxation phase of reflexes.

Visualisation of vocal fold movement is very important if the patient presents with dysphonia. A paralysed vocal cord ipsilateral to the thyroid mass is a concern for an invasive thyroid cancer.

ii. Technical procedure

a. Laboratory test

Patients presenting with thyroid nodules are usually euthyroid. This should be confirmed by a serum TSH level. If the TSH is below normal (suppressed), further assessment with free thyroxine (FT4) and calculated or free triiodothyronine (FT3) should be performed to confirm hyperthyroidism. Some practitioners prefer to order TSH and FT4 as a paired set.

Consider asking patients about their biotin (vitamin B7) intake, as a high consumption from dietary supplements may lead to falsely high or low thyroid function test results.

b. Imaging study

- Ultrasonography: Cervical ultrasonography is the initial imaging study of choice for thyroid nodules as well as many other anatomical lesions in the neck. Ultrasonography can detect and characterise nodules too small to be palpable, and provide exact dimensions to facilitate serial monitoring.

Malignancy cannot be definitively diagnosed or excluded with ultrasonography. However, the presence of the ultrasonographic features noted below should raise the level of suspicion about malignancy and the need

for FNA biopsy:

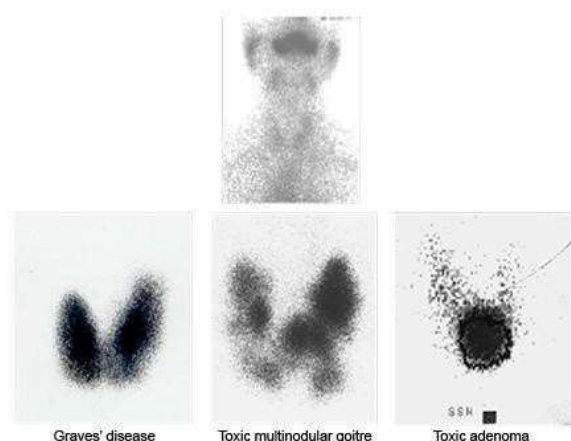
- Presence of a solid component
- Microcalcifications
- Irregular margins
- Taller than wide shape
- Hypoechoic sonographic appearance relative to thyroid parenchyma

Threshold size for FNA depends on the sonographic characteristics, but usually exceeds 1 cm. FNA biopsy of smaller nodules should be considered if there is a high-risk history or suspicious sonographic appearance.

Nodule sonographic or clinical features	Recommended nodule threshold size for FNA
High-risk radiographic features or history	>1 cm
Abnormal cervical lymph nodes	All
Solid nodule and iso- or hyperechoic	≥1 cm
Mixed cystic-solid nodule without suspicious ultrasound features	≥2 cm
Spongiform nodule	≥2 cm
Purely cystic nodule	FNA not indicated

Nodule sonographic characteristics and threshold size for fine needle aspiration (FNA).

- *Radioisotope imaging:* If TSH is suppressed, radioisotope imaging should be performed to identify autonomously hyperfunctioning ('hot') nodules and to distinguish between Graves' disease and thyroiditis. A thyroid nodule that exhibits focally increased radioisotope uptake with suppression of the surrounding thyroid parenchyma has a risk of malignancy of less than 2%. A radioisotope scan showing such a 'hot' nodule obviates the need to perform FNA if there are no other suspicious features or history.



Iodine uptake scans: typical appearances of absent uptake in thyroiditis (top panel); diffuse increased uptake in Grave's disease (lower left panel); areas of increased and

decreased uptake in toxic multinodular goitre (lower middle panel); single area of increased uptake in a toxic adenoma (lower right panel).

- *Computed tomography (CT) and magnetic resonance imaging (MRI)*: Neither CT nor MRI (cross-sectional imaging) has a primary role in the initial evaluation of a thyroid nodule; however, they can be used for accurate measurement and assessment of the size of large substernal goitres and they may identify lymph nodes that cannot be visualised by ultrasound. They are also helpful in cases of large goitre with posterior extension and diffuse bulky adenopathy. Administration of iodinated contrast for CT scanning should be carefully considered because it often prevents subsequent diagnostic and therapeutic use of radioactive iodine for 4 to 8 weeks or longer.
- *Fine needle aspiration (FNA) biopsy*: Following ultrasonography, ultrasound-guided

FNA is the procedure of choice for evaluation of suspicious thyroid nodules. Cytology and adjunctive molecular genetic testing may determine whether surgery is indicated. The procedure is well tolerated and carries a very low risk of complications. Local pain and haematoma are the most common complications, while serious adverse events are rare. Cytopathology from the FNA is reported as either unsatisfactory, benign, indeterminate, suspicious, or malignant. The Bethesda classification system includes recommended diagnostic categories, implied risk of malignancy, and recommended clinical management. The categories are designated as: non-diagnostic or unsatisfactory; benign; atypia of undetermined significance or follicular lesion of undetermined significance; follicular neoplasm or suspicious for a follicular neoplasm; suspicious for malignancy; and malignant. Actual management may depend on other factors (e.g., clinical, sonographic) besides the FNA interpretation. Unsatisfactory cytology results require repeat FNA. Cytologically benign, non-cystic lesions on FNA may still have up to a 5% risk of malignancy that may be even higher in nodules ≥ 4 cm (probably from sampling error). Such nodules therefore still require follow-up with annual neck examination and neck ultrasonography. Aspirated cyst contents often produce cytological results interpreted as inadequate or indeterminate because of the lack of cellular content and absence of colloid. Nodules with a Bethesda criteria cytological diagnosis of atypia of undetermined significance/follicular lesion of undetermined significance or equivalent may be considered for repeat FNA after 4 to 6 weeks to allow for healing of the previous FNA tract.

IV. DIFFERENTIAL DIAGNOSIS

The differential diagnosis of nodular goiter includes:

- Common:
 - Colloid nodule
 - Thyroid adenoma or hyperplastic nodule, single (solid or Complex)
 - Non-toxic multinodular goitre
 - Differentiated thyroid cancer (papillary, follicular)
- Uncommon
 - Toxic adenoma, single
 - Toxic multinodular goitre

- Medullary thyroid cancer
- Anaplastic thyroid cancer
- Lymphoma
- Simple epithelial-lined thyroid cyst
- Thyroglossal duct cyst
- Acute suppurative thyroiditis
- Subacute granulomatous thyroiditis
- Chronic lymphocytic (Hashimoto's) thyroiditis
- Painless lymphocytic thyroiditis
- Graves' disease
- Enlarged parathyroid gland(s): benign
- Parathyroid carcinoma
- Metastasis to the thyroid of non-thyroidal malignancies

V. THERAPEUTIC APPROACH

- i. **Stabilize patient:** Beta-blockers are generally recommended for older adults with symptoms or younger people with a heart rate >90 bpm. However, toxic MNGs generally cause milder symptoms than Graves' disease. Beta-blockers may be appropriate for patients with atrial fibrillation and rapid ventricular response. Older patients may have contra-indications to beta-blockers, such as bradycardia or heart block. If beta-blockers are contra-indicated, a calcium-channel blocker may be useful.
- ii. **Empirical treatment:** Antithyroid drugs are the preferred treatment during pregnancy. They should also be considered for patients who are not candidates for, or who refuse, definitive treatment. Antithyroid drugs may also be used to restore euthyroidism prior to definitive therapy. A course of antithyroid drugs may be used to normalise thyroid function prior to I-131 therapy in patients with severe symptoms, older age, or comorbidities such as heart disease.

Both thiamazole and propylthiouracil have been associated with birth defects when used during the first trimester of pregnancy, although the defects associated with thiamazole use tend to be more severe. Propylthiouracil does, however, have more frequent dosing compared with thiamazole, and its use may rarely cause fulminant hepatic failure. Thiamazole is, therefore, the preferred antithyroid drug outside the first trimester of pregnancy; when antithyroid drugs are needed in the first trimester propylthiouracil is recommended.

I-131 therapy is a preferred treatment for most non-pregnant and non-lactating patients. A nuclear medicine consultant referral is indicated. Full effect of therapy is achieved after several weeks to months. Persistent hyperthyroidism may require a second dose.

- iii. **Directed treatment:** Surgery is indicated for patients with large goitres that cause obstructive symptoms such as choking or dysphonia, or for patients who decline or are resistant to radioactive iodine, or who prefer surgery.

In pregnancy, surgery is occasionally indicated for uncontrolled hyperthyroidism due to adverse reaction or non-adherence to antithyroid drugs and is preferably performed in the second trimester. An experienced, high-volume surgeon is recommended to reduce the risk of complications.

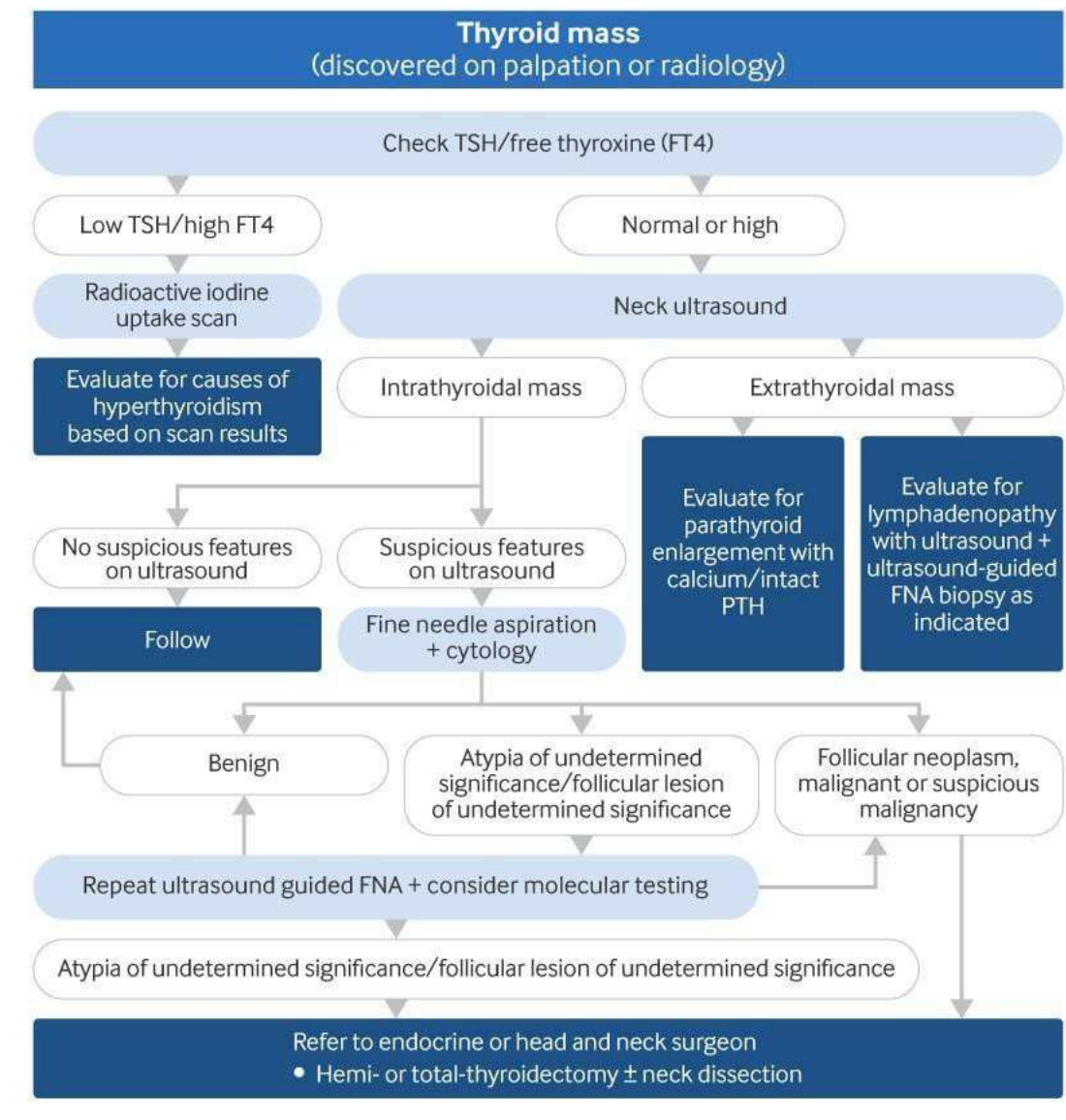
- iv. **Monitoring:** Thyroid function tests and clinical examination should be monitored after treatment. There are no specific guidelines for the monitoring schedule. One approach is to monitor at 6 weeks, 3 months, 6 months, and 12 months, and then yearly after I-131 therapy.

There is no evidence for efficacy of follow-up FBCs in predicting agranulocytosis. Baseline values are recommended. Immediate patient reporting of any symptoms of infection is mandatory.

VI. COMPLICATION

- Surgery-related recurrent laryngeal nerve damage
- Surgery-related hypoparathyroidism
- Bone mineral loss
- I-131-related hypothyroidism
- Atrial fibrillation
- Mass effect
- Thyroid storm
- Antithyroid drug-related agranulocytosis

VII. ALGORITHM



Algorithm for the evaluation of a thyroid mass

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PAROTID TUMOR

Po Ratha, Loek Sothearith, Meng sour, Theng Sakphea, Heng Sambo,
Nen sopath, Yos Sophorn

I. CASE DEFINITION

A discrete mass in the parotid gland is most commonly a salivary gland tumor, 80% of which are benign. It can also represent a lymph node within the parotid gland that can be inflammatory (eg Tuberculosis) or lymphoma or a metastasis most commonly from skin malignancies of the face or scalp.

i. Benign salivary gland tumors

Table 1. Benign salivary gland tumors and lesions

Pleomorphic adenoma
Myoepithelioma
Basal cell adenoma
Warthin tumor
Oncocytoma
Lymphadenoma
Cystadenoma
Sialadenoma
Ductal papillomas
Sebaceous adenoma
Canalicular adenoma and other ductal adenomas
Non- neoplastic epithelial lesions
Sclerosing polycystic adenosis
Nodular oncocytic hyperplasia
Lymphoepithelial sialadenitis
Intercalated duct hyperplasia
Benign soft tissue lesions
Hemangioma
Lipoma/sialolipoma
Nodular fasciitis
Hematolymphoid tumors
Extranodal marginal zone lymphoma of mucosa-associated lymphoid tissue (MALT lymphoma)

(Adapted from 2017 WHO Classification of Head and Neck Tumors).

ii. Malignant salivary gland tumors

Table 2. Malignant salivary gland tumors

Mucoepidermoid carcinoma
Adenoid cystic carcinoma
Acinic cell carcinoma
Polymorphous adenocarcinoma (formerly polymorphous low-grade adenocarcinoma)
Clear cell carcinoma
Basal cell carcinoma
Adenocarcinoma, not otherwise specified
Salivary duct carcinoma
Myoepithelial carcinoma
Epithelial-myoepithelial carcinoma
Carcinoma ex-pleomorphic adenoma
Secretory carcinoma (formerly mammary analogue secretory carcinoma)
Sebaceous adenocarcinoma
Carcinosarcoma
Poorly differentiated carcinoma
Undifferentiated carcinoma
Large cell neuroendocrine carcinoma
Small cell neuroendocrine carcinoma
Lymphoepithelial carcinoma
Squamous cell carcinoma
Oncocytic carcinoma
Salivary gland neoplasm of uncertain malignant potential (SUMP)
Sialoblastoma

(Adapted from 2017 WHO Classification of Head and Neck Tumors).

iii. Staging

The classification applies only to carcinomas of the major salivary glands. Tumors arising in minor salivary glands are not included in this classification. The staging of metastatic neck nodes for salivary gland cancer is similar to that for other metastatic diseases. T-stage, according to the 8th edition of the UICC/AJCC staging manual (2016) is as follows:

- *T1*: Tumor 2 cm or less in greatest dimension without extra-parenchymal extension*
- *T2*: Tumor more than 2 cm but not more than 4 cm in greatest dimension without extra-parenchymal extension*
- *T3*: Tumor more than 4 cm and/or tumor with extra-parenchymal extension*
- *T4a*: Tumor invades skin, mandible, ear canal, and/or facial nerve
- *T4b*: Tumor invades base of the skull, and/or pterygoid plates, and/or encases carotid artery

* Extraparenchymal extension is clinical or macroscopic evidence of invasion of soft tissues or nerve, except those listed under T4a and 4b. Microscopic evidence alone does not constitute extra-parenchymal extension for classification purposes.

II. ETIOLOGY

The causes of parotid tumors are generally unknown at this time.

Radiation exposure has been linked to parotid gland carcinomas 15 years after the event. Cigarette smoking and alcohol is associated with head and neck squamous cell carcinoma, skin malignancies in the head and neck have been known to metastases to the parotid glands. Some cases of a link between occupational exposure to silica dust and nitrosamines have been reported.

It is important to note that both alcohol and smoking are not linked to salivary gland tumors, except for Warthin tumors.

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

a. History

- ☐ Age and gender
- ☐ Location of disease (gland involved)
- ☐ Presence/absence of pain (higher suspicion of malignancy with pain)
- ☐ Temporal trend of growth
- ☐ Cranial nerve deficits (facial nerve/trigeminal nerve)
- ☐ Episodic vs progressive swelling
- ☐ Other symptoms
 - Xerostomia/xerophthalmia
 - Dental caries, trismus
 - Otalgia
 - Fever
 - Unintentional weight loss
 - Neck mass
- ☐ Social history: tobacco and alcohol exposure, travel, infectious exposure
- ☐ Family history: autoimmune disorders, head and neck malignancy lymphoma, genetic disorder/syndrome

b. Physical examination

- ☐ General
 - Facial symmetry
 - Cranial nerves (facial/trigeminal)

- Location, mobility, tenderness, and quality of mass
- Regional lymph node examination
- Signs of systemic disease
- Examination of eyes/lacrimal glands
- Parotid gland
 - Palpation of gland, overlying skin, bimanual palpation of buccal space, examination of Stenson's duct
 - Examination of oropharynx and/or nasopharynx for submucosal bulging from deep lobe parotid tumors in pre-styloid space
 - Trismus can occur with posteromedial extension of parotid tumors into infratemporal fossa
- Submandibular gland
 - Asymmetry between paired glands
 - Bimanual palpation of submandibular gland
 - Cranial nerves: hypoglossal, lingual, marginal mandibular branch of facial nerve
 - Intraoral examination of Wharton's duct
- Minor salivary glands
 - 500-1000 glands along upper aerodigestive tract
 - Most common site involved with salivary gland tumors is hard palate where majority of minor glands located

ii. Technical procedure

c. Laboratory test

- Consider lab work for diagnosis or exclusion of infectious, granulomatous, metabolic, autoimmune, hormonal or other systemic disorders
 - Complete blood count
 - Erythrocyte sediment rate (ESR)
 - Serum calcium and angiotensin-converting enzyme elevation (sarcoidosis)
 - Autoantibodies-rheumatoid factor (RF), antinuclear antibodies (ANA), anti-SSA and anti-SSB (Sjögren syndrome)
 - Thyroid stimulating hormone (thyroid dysfunction)
 - Complete metabolic panel (hepatic or renal dysfunction)
 - Human immunodeficiency virus test
 - Nutritional or vitamin deficiencies

d. Imaging studies

- MRI

- ☐ CT scan
- ☐ Ultrasound

e. e. Biopsy

- ☐ Fine-needle aspiration (FNA)
- ☐ Surgical excision

IV. DIFFERENTIAL DIAGNOSIS

Mucoepidermoid Carcinoma

- ☐ Necrotizing sialometaplasia
- ☐ Pleomorphic adenoma with squamous metaplasia
- ☐ Sclerosing polycystic adenosis

Adenoid Cystic Carcinoma

- ☐ Pleomorphic adenoma
- ☐ Polymorphous adenocarcinoma
- ☐ Epithelial-myoepithelial carcinoma
- ☐ Basal cell adenocarcinoma

Acinic Cell Carcinoma

- ☐ Secretory carcinoma

V. THERAPEUTIC APPROACH

iii. General principles

- a. Remove tumor with adequate margin to avoid local recurrence
- b. Extent of dissection is limited by size, location and histology of tumor

iv. Parotidectomy

- a. Modified Blair incision (incision in preauricular crease, extending around lobule and curving into natural skin crease in upper neck)
- b. May consider facelift approach to hide incision in hairline
- c. Types of parotidectomy:
 - ☐ Partial – removal of tumor with cuff of normal parotid tissue
 - ☐ Superficial – removal of entire superficial lobe
 - ☐ Total – removal of superficial and deep lobes
 - ☐ Extracapsular – removal of tumor with preservation of capsule only
- d. Great auricular nerve preservation can be considered though it can be associated with dysesthesia and minimal quality of life improvement

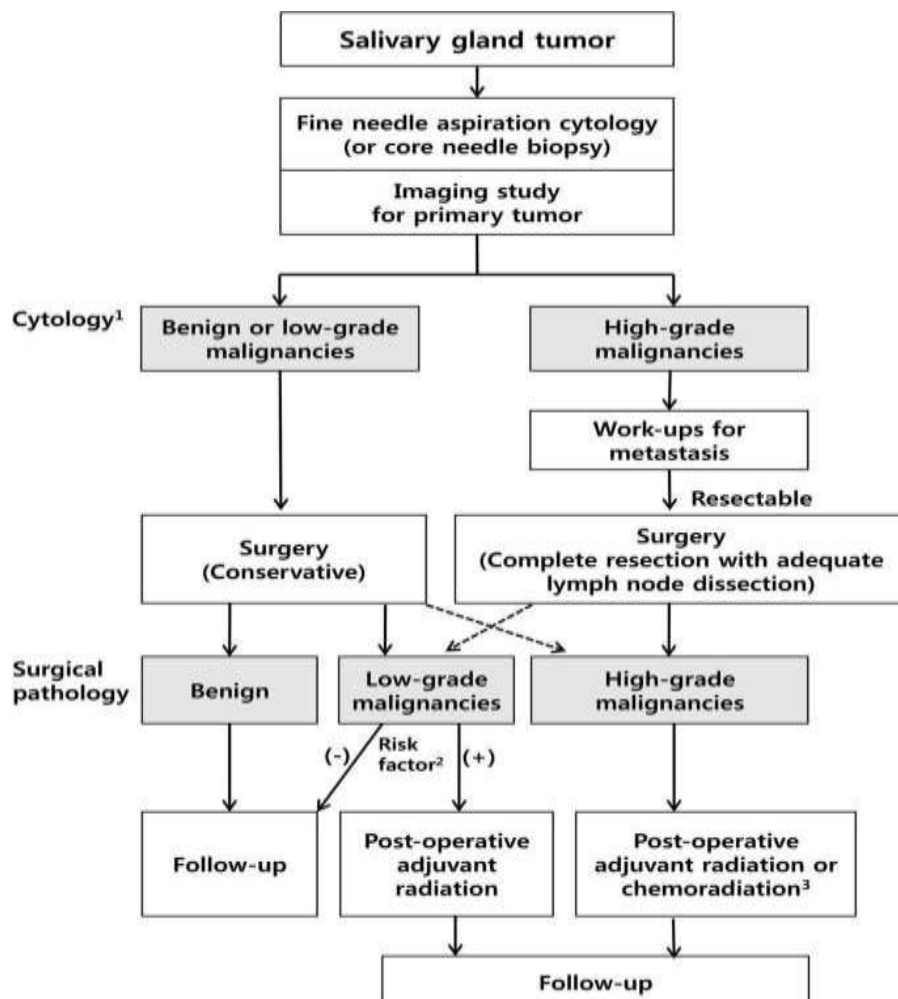
e. Anterograde facial nerve identification:

- Tragal pointer (main trunk 1-1.5cm deep and inferior)
- Tympanomastoid suture line (main trunk 6-8mm deep)
- Posterior belly of digastric define plane of nerve
- Stylomastoid foramen
- Retrograde facial nerve dissection may be helpful when tumor overlies main trunk of nerve

VI. COMPLICATION

- Facial nerve paresis/paralysis
- Ear lobe numbness
- Frey's syndrome
- Sialocele

VII. ALGORITHM



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SUBMANDIBULAR GLAND TUMOR

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I. DEFINITION

- Submandibular gland tumor is growth of abnormal cells that begin in the submandibular gland or in the tube (ducts) that drain the gland. Although most submandibular gland tumors are benign, some are malignant.

II. CAUSE

- The cause of submandibular gland tumor is unknown, but genetic research is being conducted to uncover it.
- Some factors have been implicated as potential causes:
 - o Radiation
 - o Smoking
- Viral infections may be associated with an increased risk of submandibular gland cancer:
 - o Epstein Barr virus (EBV)
 - o Human immunodeficiency virus (HIV)
 - o Human papillomavirus (HPV)
- Environmental factors and industrial exposure to factors such as rubber manufacturing, hair dressers, beauty shops, and nickel compounds.

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

Patients with a tumor of submandibular gland typically present with firm, usually painless mass or swelling. Slow swelling usually indicates a benign tumor, while a rapid swelling is more likely to indicate a malignant tumor or infection. Tumors with a high malignant potential are more likely to spread to regional lymph nodes and result in a palpable mass.

ii. Technical procedure

a. Laboratory test

- Submandibular gland biopsy or fine needle aspiration biopsy (FNAB) to determine whether the tumor is benign or malignant.
- FNAB is a widely used, safe and less traumatic. It serves to determine the nature of the lesion and in some cases, the specific diagnosis.

b. Imaging study

- Ultrasound is the first non-invasive option for evaluating submandibular gland tumors. It can help localize tumors; distinguish solid masses from cystic collections, and help guide fine-needle aspiration biopsy.
- CT scan can evaluate tumor extent, bony infiltration, and lymphadenopathy.
- Magnetic resonance imaging (MRI) is recommended to assess tumor extent, soft tissue invasion, and nerve involvement.
- Positron emission tomography (PET) is to detect locoregional and distant metastasis.

IV. DIFFERENTIAL DIAGNOSIS

The differential diagnosis of submandibular gland tumor includes:

- Salivary cysts
- Cysts of the first branchial cleft
- Salivary gland stones
- Sarcoid
- Sjögren disease
- Metastases from other tumors
- Lymphoepithelial cysts (particularly in an immunocompromised host)
- Chronic sclerosing sialadenitis (Küttner tumor)
- Regional lymphadenopathy from infectious, inflammatory, or malignant diseases.

V. THERAPEUTIC APPROACH

i. **Stabilize patient**

Stabilize the patient according to ABC (Airway, Breathing, Circulation)

ii. **Treatment**

- Benign tumors of submandibular gland are most commonly treated by simple excision of the gland.
- Malignant tumors: wide excision is appropriate for tumors confined to the gland combined with some form of neck dissection.
- In the N0 neck, patients should undergo clearance of nodes with a selective neck dissection (levels 1 and 2A). Clinically high-grade tumors with suspicious MRI appearances should have a selective neck dissection (levels 1–3).
- Adjuvant radiotherapy (RT) is recommended for all malignant tumors except in cases of small, low-grade tumors that have been completely excised
- Chemotherapy may be an option for people with advanced submandibular gland cancer. It's sometime used in combination with radiation therapy

iii. Monitoring

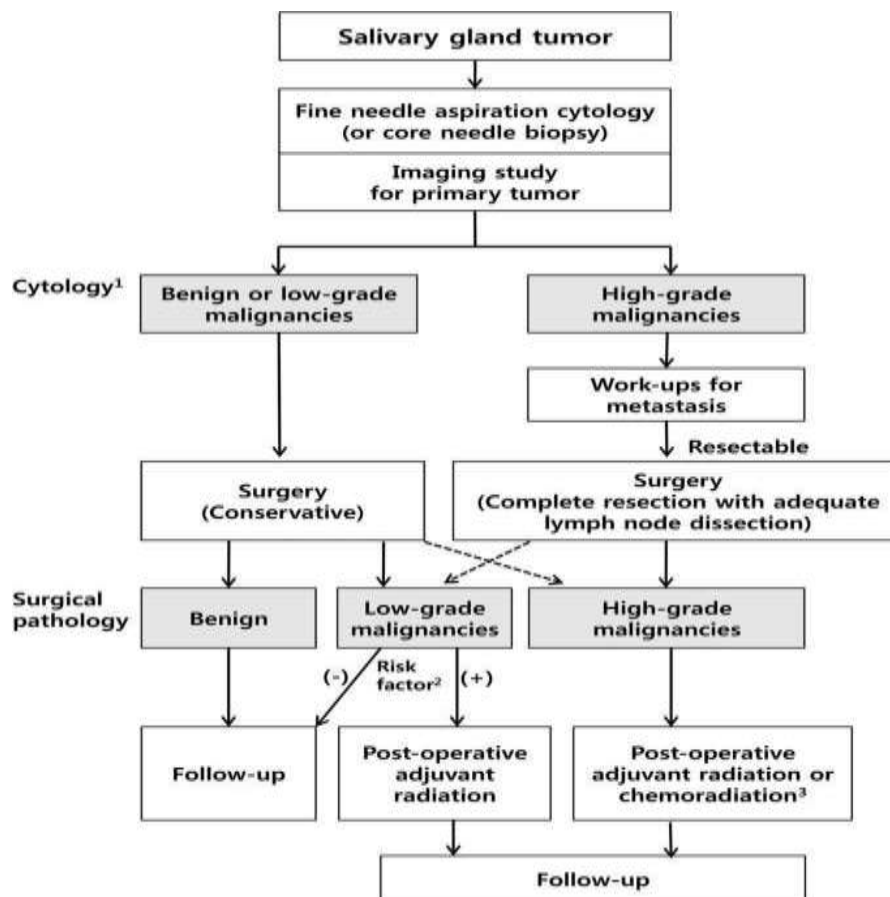
a. Clinical monitoring

Patients are followed up in the ear, nose, and throat clinic to monitor for symptoms of submandibular gland tumor recurrence or to monitor how well treatment is working starting 3 months after surgery. Follow up imaging should be done every 6 to 12 months for the first 2 years after treatment. For patients with malignant submandibular gland tumors, the intensity of follow-up is generally greatest in the first 2 to 4 years. However, continued follow-up is generally suggested as the risk of recurrence even beyond the first 5 years.

VI. COMPLICATION

- Hematoma
- Wound infection
- Salivary fistula
- Injury to the hypoglossal or lingual nerve
- Transient palsy of the marginal mandibular nerve

VII. ALGORITHM



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TONGUE TUMOR

Po Ratha, Loek Sothearith, Meng sour, Theng Sakphea,

Heng Sambo, Nen sopath, Yos Sophorn

I. DEFINITION

- Tongue tumor is growth of abnormal cells that begin in the tongue. It can be benign or malignant.

II. ETIOLOGY

The most common factors that can increase the risk of tongue tumor include:

- Tobacco use
- Consuming alcohol
- Being exposed to HPV
- Trouble maintaining oral hygiene
- Having a weak immune system

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

- The clinical history of any change in appearance or associated sensory changes (e.g. pain, burning, loss of taste) is important to elicit.
- Dysphagia, weight loss, dysarthria, and odynophagia are also common.
- The patient's ability to tolerate his usual diet should be addressed.
- There may be a history of longstanding leukoplakia or erythroplakia.
- Any history of recently noticed neck masses or nodules is also fundamental, as this may represent locoregional metastatic disease.

ii. Technical procedure

a. Laboratory test

- Complete blood count (CBC)
- Prothrombin time (PT)
- Activated partial thromboplastin time (aPTT)
- Erythrocyte sedimentation rate (ESR) and/or C-reactive protein (CRP)
- Biopsy of tongue tumor to determine whether the tumor is benign or malignant.

b. Imaging study

- Computerized tomography (CT). Conventional CT can evaluate tumor extent, bony infiltration, and lymphadenopathy.
- Magnetic resonance imaging (MRI) is recommended to assess tumor extent, soft tissue invasion, and nerve involvement.
- Positron emission tomography (PET). The role of PET is to detect locoregional and distant metastasis.

IV. DIFFERENTIAL DIAGNOSIS

The differential diagnosis of tongue tumor includes:

- Squamous cell carcinoma
- Leukoplakia
- Carcinoma in situ
- Rhabdomyosarcoma
- Lymphoma
- Rhabdomyoma
- Neurofibroma
- Pyogenic granuloma
- Papilloma
- Vascular or lymphatic malformation
- Lingual thyroid
- Dermoid cyst
- Epidermoid cyst

V. THERAPEUTIC APPROACH

i. **Stabilize patient**

During evaluation, rapidly identify and stabilize respiratory, circulatory, or neurologic compromise.

ii. **Treatment**

- Benign tumors are most commonly treated by simple excision of the tumor.
- The management of tongue carcinoma is primarily surgical with a selective neck dissection level I-III.
- Adjuvant radiation therapy (RT) with or without concurrent chemotherapy is the standard of care for patients with resected locoregionally advanced tongue

cancer.

iii. Monitoring

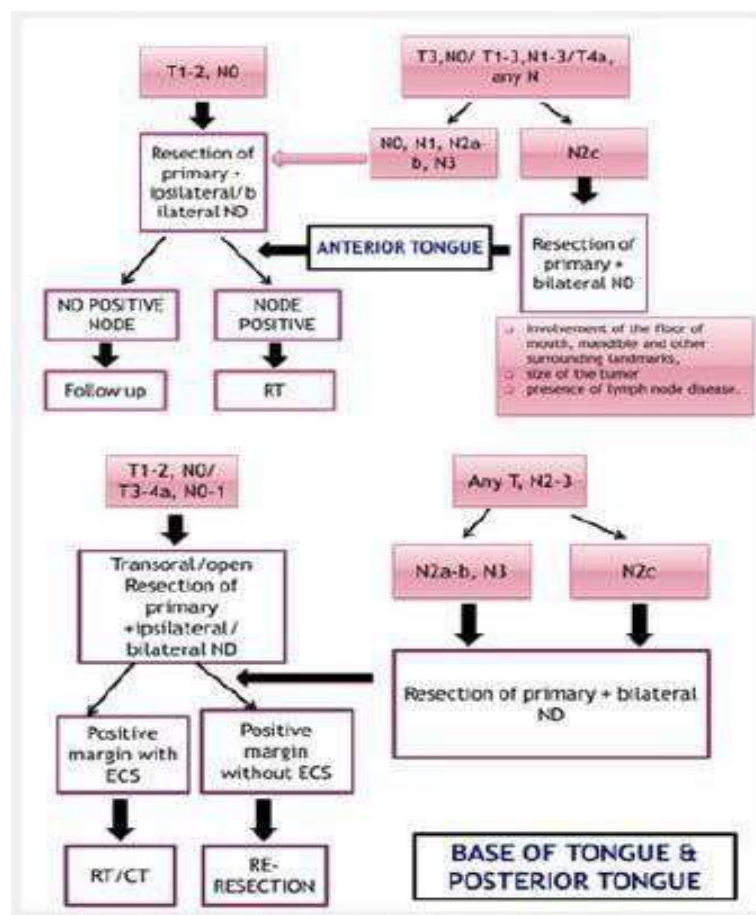
a. Clinical monitoring

Diagnostic examinations may be done to watch for potential recurrences or to monitor how well treatment is working starting 2 months after you finish treatment. A common follow-up schedule is every 2 months for the first year, every 4 months for the second year, every 6 months for the third year, and once a year after that.

VI. COMPLICATION

- Hemorrhage
- Wound infection
- Edema
- Pain

VII. ALGORITHM



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CHAPTER III

NEUROLOGICAL

SURGERY

1. Cervical spine disc herniation
2. Chronic subdural hematoma
3. Epidural hematoma
4. Thoraco-Lumbar spine fracture
5. Traumatic Brain Injury

CERVICAL SPINE DISC HERNIATION

CHOR Vannaroeth, SOK Chhay, HUOY Phearum, HENG Sopanha, SIM Sokchan

I. INTRODUCTION

Vertebrae, along with intervertebral discs, form the vertebral column, or spine. It extends from the base of the skull to the coccyx and includes the **cervical**, thoracic, lumbar, and sacral regions.

The spine has several significant roles in the body that include: protection of the spinal cord and branching spinal nerves, structural support, and allows for flexibility and mobility of the body.

The intervertebral discs are cartilaginous structures between adjacent vertebrae that support the spine by acting as shock-absorbing cushions to the axial loading of the body. [1,2]

The cervical spine has seven vertebral bodies, numbered C1 to C7, counting from the base of the skull to the thoracic spine. The structure of C1, C2, and C7 have distinctive properties that make them unique in comparison to the typical cervical vertebrae, C3 to C6.

The anatomy of C3 to C6 consists of a vertebral body, a vertebral arch, as well as seven processes. The vertebral arch is comprised of the pedicles, bony processes that project posteriorly from the vertebral body, and the lamina; these are the bone segments that form most of the arch. Together, the pedicles and the lamina form a ring around the spinal canal, which harbors the spinal cord. Completing the typical vertebra are seven processes, and they include two superior articular facets, two inferior articular facets, one spinous process, and two transverse processes that allow the passage of the vertebral vasculature.

There are three atypical vertebrae in the cervical region. C1 (atlas) articulates with the base of the skull and is unique in that it does not contain a body due to fusion with the C2 (axis) vertebrae, acting as a pivot point for the atlas to rotate. The most distinctive feature of the C2 vertebra is the presence of an odontoid process (dens) that rises from the superior aspect of its body and articulates with the posterior surface of the anterior arch of C1. C7 has two distinct features that make it unique to a typical cervical vertebra: first, the vertebral vasculature does not traverse through its transverse foramina, and second, it contains along the spinous process, making C7 to be commonly known as "vertebra prominens." [3]

While there are seven cervical vertebrae, there are eight pairs of cervical nerves, numbered C1 to C8. Each pair of cervical nerves emerge from the spinal cord superior to their corresponding vertebra, except for C8, which exits inferiorly to the C7 vertebra. [4] Cervical disc herniation is the result of the displacement of the nucleus pulposus of the intervertebral disc, which may result in impingement of these traversing nerves as they exit the neural foramen or directly compressing the spinal cord contained within the spinal canal.

II. ETIOLOGY

An intervertebral disc is a cartilaginous structure composed of three components: an inner nucleus pulposus, outer annulus fibrosus, and endplates that anchor the discs to adjacent vertebrae.

Disc herniations occur when part or all of the nucleus pulposus protrudes through the annulus fibrosus. This process can occur acutely or more chronically.

Chronic herniations occur when the intervertebral disc becomes degenerated and desiccated as part of the natural aging process; this typically results in symptoms of insidious or gradual onset that tend to be less severe.

In contrast, acute herniations generally are the result of trauma, resulting in the nucleus pulposus extruding through a defect in the annulus fibrosus. This injury will usually result in a sudden onset of more severe symptoms when compared to chronic herniations.[5]

III. EPIDEMIOLOGY

The prevalence of cervical disc herniation increases with age for both men and women and is most common in people in their third to fifth decades of life. It occurs more frequently in females, accounting for more than 60% of cases. For both sexes, the most frequently diagnosed patients were in the age group of 51 to 60.[6,7]

IV. DIAGNOSTIC PROCEDURE

1.1. History

Cervical disc herniations most commonly occur between C5-C6 and C6-C7 vertebral bodies. This, in turn, will cause symptoms at C6 and C7, respectively. History in these patients should include the chief complaint, onset of symptoms, alleviating and aggravating factors, radicular symptoms, and any past treatments.

The most common subjective complaints are axial neck pain and ipsilateral arm pain or paresthesia in the associated dermatomal distribution.

As part of the evaluation of neck pain, it is important to identify certain red flags that could be features of underlying inflammatory conditions, malignancy, or infection.[8] These include:

- Fever. Chills
- Night sweats
- Unexplained weight loss
- History of inflammatory arthritis, malignancy, system infection, tuberculosis, HIV, immunosuppression, or drug use
- Unrelenting pain
- Point tenderness over a vertebral body
- Cervical lymphadenopathy

1.2. Physical Examination

The clinician should assess the patient's range of motion (ROM), as this can indicate the severity of pain and degeneration. A thorough neurological examination is necessary to evaluate sensory disturbances, motor weakness, and deep tendon reflex abnormalities. Careful attention should also focus on any sign of spinal cord dysfunction.

Typical findings of solitary nerve lesions due to compression by a herniated disc in the cervical spine: **Table 1**

- C2 Nerve – eye or ear pain, headache. History of rheumatoid arthritis or atlantoaxial instability
- C3, C4 Nerve – vague neck, and trapezial tenderness, and muscle spasms
- C5 Nerve – neck, shoulder, and scapula pain. Lateral arm paresthesia. Primary motions affected include shoulder abduction and elbow flexion. May also observe weakness with shoulder flexion, external rotation, and forearm supination. Diminished biceps reflex.
- C6 Nerve – neck, shoulder, and scapula pain. Paresthesia of the lateral forearm, lateral hand, and lateral two digits. Primary motions affected include elbow flexion and wrist extension. May also observe weakness with shoulder abduction, external rotation, and forearm supination and pronation — diminished brachioradialis reflex.
- C7 Nerve – neck and shoulder pain. Paresthesia of the posterior forearm and third digit. Primary motions affected include elbow extension and wrist flexion. Diminished triceps reflex
- C8 Nerve – neck and shoulder pain. Paresthesia of the medial forearm, medial hand, and medial two digits. Weakness during finger flexion, handgrip, and thumb extension.
- T1 Nerve – Neck and shoulder pain. Paresthesia of the medial forearm. A weakness of finger abduction and adduction.[9]

Table 8: Anatomical levels and function according to cervical nerve distribution

Anatomical Level	Nerve root compromised	Motor function	Sensitive distribution	Reflex
C4-C5	C5	-Deltoid -Biceps	-Lateral arm (Axillary nerve)	-Biceps Reflex
C5-C6	C6	-Biceps -Wrist Extensors	-Lateral forearm (Musculocutaneous nerve)	-Brachioradialis Reflex
C6-C7	C7	-Triceps -Wrist Flexor -Finger Extensors	-Middle Finger	-Triceps Reflex
C7-T1	C8	-Interosseous -Finger Flexor	-Ring and Little finger -Distal half of the forearm's ulnar side	None

Provocative tests include the Spurling test, Hoffman test, and Lhermitte sign. Spurling test can help diagnose acute radiculopathy. This test is performed by maximally extending the neck and rotating towards the involved side while compressing the head to load the cervical spine axially. This will narrow the neuroforamen and may reproduce symptoms of radiculopathy.



Hoffman test and Lhermitte sign can be used to assess for the presence of spinal cord compression and myelopathy. The Hoffman test is performed by holding the long finger and flicking the distal tip downward. A positive sign results when there are flexion and adduction of the thumb. The Lhermitte sign is performed by flexing the patient's neck, which may result in an electrical sensation traveling down the spine and into the extremities. [10,11]

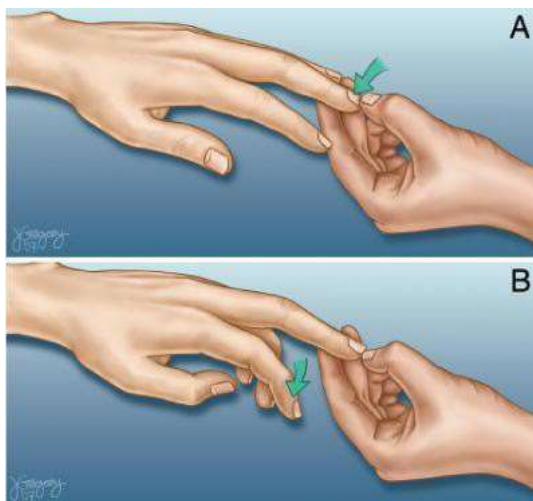


Figure 3: Hoffman sign



Figure 4: Lhermitte sign

1.3. Investigation

Most cases of acute spinal injury or herniation will resolve within the first four weeks, without any intervention. The use of imaging during this period is typically not recommended as management of these cases will usually not be altered. Imaging during this period is a recommendation when there is clinical suspicion of potentially serious pathology or in the presence of neurological compromise.

Additionally, patients that fail to respond to conservative treatment after a period of 4 to 6 weeks warrant further evaluation. [12] Also, patients that exhibit red flag symptoms listed above may warrant evaluation with lab markers. These can include:

Lab values:

- Erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP): These are inflammatory markers that should be obtained if a chronic inflammatory condition is suspected (rheumatoid arthritis, polymyalgia rheumatica, seronegative spondyloarthropathy). These can also be beneficial if an infectious etiology is suspected.
- Complete blood count (CBC) with differential: Useful to obtain in instances when infection or malignancy is suspected.

1.3.1. X-ray

The first test typically performed and one that is very accessible at most clinics and outpatient offices. Three views (AP, lateral, and oblique) views help assess the overall alignment of the spine as well as for the presence of any degenerative or spondylotic changes.

These can be further supplemented with lateral flexion and extension views to assess for the presence of instability. If imaging demonstrates an acute fracture, this requires additional investigation using a computed tomogram (CT) scan or magnetic resonance imaging (MRI). If there is a concern for atlantoaxial instability, the open mouth (odontoid) view may assist in diagnosis.

1.3.2. CT Scan

This imaging is the most sensitive test to examine the bony structures of the spine. It can also show calcified herniated discs or any insidious process that may result in bony loss or destruction. In patients that are unable to or are otherwise ineligible to undergo an MRI, CT myelography can be used as an alternative to visualize a herniated disc.

1.3.3. MRI

The preferred imaging modality and the most sensitive study to visualize a herniated disc, as it has the most significant ability to demonstrate soft-tissue structures and the nerve as it exits the foramen.

1.3.4. Electrodiagnostic testing

Electromyography and nerve conduction studies can be an option in patients that demonstrate equivocal symptoms or imaging findings as well as to rule out the presence of a peripheral mononeuropathy. The sensitivity of detecting cervical radiculopathy with electrodiagnostic testing ranges from 50% to 71%. [13]

V. DIFFERENTIAL DIAGNOSIS

1. Brachial plexus injury
2. Degenerative cervical spondylosis
3. Muscle strain
4. Parsonage-Turner syndrome
5. Peripheral nerve entrapment
6. Tendinopathies of the shoulder

VI. THERAPEUTIC APPROACH

1.4.Conservative Treatments:

Acute cervical radiculopathies secondary to a herniated disc are typically managed with non-surgical treatments as the majority of patients (75 to 90%) will improve. Modalities that can be used include [5,14]

1.4.1.Collar Immobilization:

In patients with acute neck pain, a short course (approximately one week) of collarimmobilization may be beneficial during the acute inflammatory period.

1.4.2.Traction:

May be beneficial in reducing the radicular symptoms associated with disc herniations. Theoretically, traction would widen the neuroforamen and relieve the stress placed on the affected nerve, which, in turn, would result in the improvement of symptoms. This therapy involves placing approximately 8 to 12 lbs of traction at an angle of approximately 24 degrees of neck flexion over a period of 15 to 20 minutes.

1.4.3.Pharmacotherapy:

There is no evidence to demonstrate the efficacy of non-steroidal anti-inflammatories (NSAIDs) in the treatment of cervical radiculopathy. However, they are commonly used and can be beneficial for some patients.

The use of COX-1 versus COX-2 inhibitors does not alter the analgesic effect, but there may be decreased gastrointestinal toxicity with the use of COX-2 inhibitors.

Clinicians can consider steroidal anti-inflammatories (typically in the form of prednisone) in severe acute pain for a short period. A typical regimen is prednisone 60 to 80 mg/day for five days, which can then be slowly tapered off over the following 5 m to 14 days. Another regimen involves a prepackaged tapered dose of Methylprednisolone that tapers from 24 mg to 0 mg over 7 days.

Opioid medications are typically avoided as there is no evidence to support their use, and they carry a higher side effect profile. If muscle spasms are prominent, the addition of a muscle relaxant may merit consideration for a short period. For example, cyclobenzaprine is an option at a dose of 5 mg taken orally three times daily.

Antidepressants (amitriptyline) and anticonvulsants (gabapentin and

pregabalin) have been used to treat neuropathic pain, and they can provide a moderate analgesic effect.

1.4.4. Physical Therapy:

Commonly prescribed after a short period of rest and immobilization. Modalities include range of motion exercises, strengthening exercises, ice, heat, ultrasound, and electrical stimulation therapy. Despite their frequent use, no evidence demonstrates their efficacy over placebo. However, there is no proven harm, and with a possible benefit, their use is recommended in the absence of myelopathy.

1.4.5. Cervical Manipulation:

There is limited evidence suggesting that cervical manipulation may provide short-term benefits for neck pain and cervicogenic headaches. Complications from manipulation are rare and can include worsening radiculopathy, myelopathy, spinal cord injury, and vertebral artery injury. These complications occur ranging from 5 to 10 per 10 million manipulations.

1.5. Interventional Treatments:

Spinal steroid injections are a common alternative to surgery. Perineural injections (translaminar and transforaminal epidurals, selective nerve root blocks) are an option with pathological confirmation by MRI. These procedures should take place under radiologic guidance. [14]

In the past few years, neuromodulation techniques have been used to a large extent to manage radicular pain secondary to disc herniations. [15] These neuromodulatory techniques consist mainly of a spinal cord stimulation device [16] and an intrathecal pain pump. [17] For patients who are not candidates for surgical intervention, these devices offer minimally invasive efficacious treatment options.

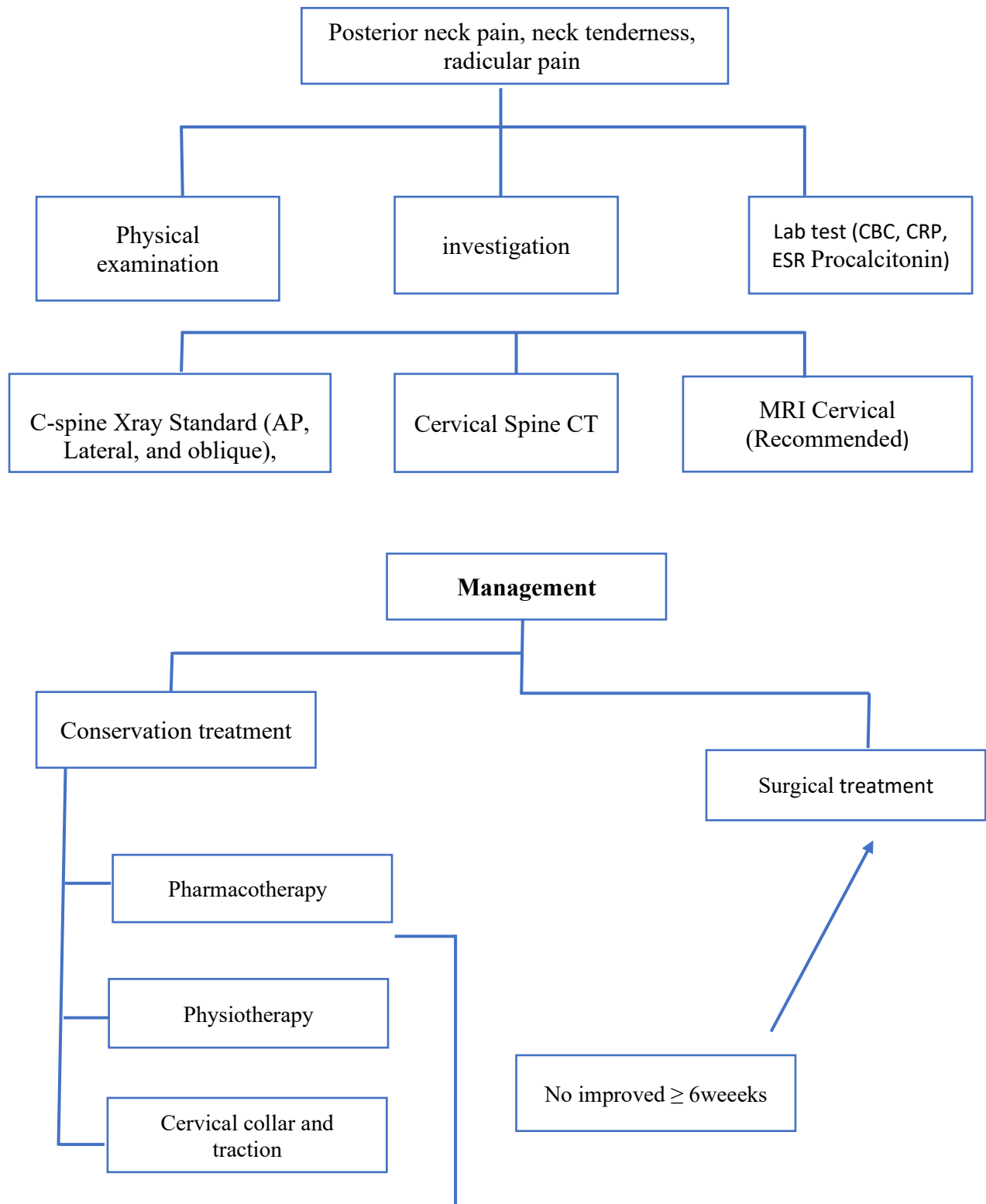
1.6. Surgical Treatments:

Indications for surgical intervention include severe or progressive neurological compromise and significant pain that is refractory to non-operative measures. Conservative therapy is recommended for 3-6 weeks for cervical radiculopathy without myelopathy. [18,19] There are several techniques described based on pathology.

The gold standard remains the anterior cervical discectomy with fusion, as it allows the removal of the pathology and prevention of recurrent neural compression by performing a fusion. A posterior laminoforaminotomy can be a consideration in patients with anterolateral herniations. Total disc replacement is an emerging treatment modality, where indications remain controversial.

Posterior cervical foraminotomy has increased in popularity as an alternative to anterior cervical discectomy and fusion, sparing the issues associated with fusion and surgical instrumentation, with equally good clinical results and the added advantage of motion preservation. [20] Cervical endoscopic decompression is becoming popular as a soft tissue-conserving minimally invasive technique for cervical foraminal decompression and discectomy for a prolapsed intervertebral disc and foraminal stenosis. [21]

VII. ALGORITHM



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CHRONIC SUBDURAL HEMATOMA

THO Vantha, MOB Seyha, SENG Samnang, SIM Sokchan

I. DEFINITION

Chronic Subdural Hematoma (CSDH): an abnormal collection of liquefied blood degradation underneath the dura matter that may result in brain tissue compression and subsequent neurological sequelae SDH. [\[1\]](#)

Depending on the length of time between the onset of symptoms and the inciting event, SDH can be classified into the following:

- Hyper acute SDH: within 24h.
- Acute SDH: Symptom onset within 3 days of the inciting event
- Subacute SDH: Symptom onset 4–20 days after the inciting event.
- Chronic SDH: symptom onset ≥ 21 days after the inciting event. [\[2\]](#)

II. ETIOLOGY AND RISK FACTOR

1. Head trauma is identified in 50–77% of cases.
2. Coagulopathies (including therapeutic anticoagulation)
3. Patients at risk for falls (e.g., with hemiplegia from previous stroke).
4. Seizures
5. CSF shunts (intra cranial hypotension)
6. Cerebral atrophy produce negative pressure within the cranium
7. Alcohol abuse
8. High altitude-induced cSDH have been reported in the literature, and it has been proposed that this may be partially due to increased venous pressure
9. Iatrogenic [\[3\]](#)

III. EPIDEMIOLOGY

Chronic subdural hematoma evacuation has been projected to become the most common cranial neurosurgical operation among adults by the year 2030 in the United States. Factors that drive these trends include the increased use of antithrombotics and the ageing population. [\[4\]](#) [\[5\]](#) [\[6\]](#)

The average age of onset reported from multiple different countries ranges between 60 in India to 81 years of age in North Wales.[\[7\]](#) [\[8\]](#)

IV. PHYSIOPATHOLOGY

It is generally well accepted that the initial hemorrhage in cSDH derives from bleeding of bridging veins. This can occur due to traumatic events, anatomical changes that result in cranio-cerebral disproportion (i.e. brain-shrinkage with age), intracranial hypotension, or through anatomical manipulation (cranial surgery). [9]

Many CSDHs probably start out as acute subdural. Blood within the subdural space evokes an inflammatory response. Within days, fibroblasts invade the clot and form neomembranes on the inner (cortical) and outer (dural) surface. This is followed by ingrowth of neocapillaries, enzymatic fibrinolysis, and liquefaction of blood clot. Fibrin degradation products are reincorporated into new clots and inhibit hemostasis. The course of CSDH is determined by the balance of plasma effusion and/or rebleeding from the neomembranes on the one hand and reabsorption of fluid on the other. [10] [11]

Another mechanism that has been proposed is that cSDH develops after formation of hygromas. This mechanism has been suggested due to the observation that cSDH often appears after subdural hygromas, and that both may manifest after cranial trauma. [12]

It is generally believed, however, that the CSF leakage into the dural cell layer creates a potential space and that the stretching of this layer results in the leaking of the aforementioned thin bridging veins. [13]

It has been shown that beta-trace protein is present in a relatively high degree compared with serum levels in both cSDH and hygromas, confirming not only that CSF is present in the fluid, but also that hygromas are composed of CSF and do not simply represent serum or effusions. [14]

Table 1

Histopathological description of the various neo-membranes as described by Nagahori T, Nishijima M, Takaku A: *Histological study of the outer membrane of chronic subdural hematoma: possible mechanism for expansion of hematoma cavity. No shinkei geka. Neurological surgery* 21:697- 701, 1993.

Original Description	
I, Non-inflammatory membrane.	Contains immature fibroblasts and collagen fibers, was associated with very slight or sparse cell infiltration and neo-capillaries.
II, Inflammatory membrane	Consisting of one layer of immature connective tissue, was associated with marked cell infiltration and vascularization throughout the entire thickness.
III, Hemorrhagic-inflammatory membrane.	Presents with structure of 2 or 3 layers, and was associated with capillaries with a large lumen on the side of the dura mater and marked cell infiltration and many thin new vessels on the side of the hematoma cavity. Some patients showed a layer consisting of only collagen fibers and fibroblasts between two such layers. In addition, hemorrhage into the membrane was often observed.
IV, Scar-inflammatory membrane.	Inflammatory cell infiltration, neovascularization and hemorrhage in the outer membrane of cicatricial tissue.

V. DIAGNOSIS

a. Clinical argument

The clinical presentation of cSDH can vary widely, ranging from asymptomatic to severely reduced Glasgow Coma Scale (GCS) scores.

1. Headache
2. Confusion
3. Language difficulties (e.g., wordfinding difficulties or speech arrest, usually with dominant hemisphere lesions)
4. TIA-like symptoms
5. Coma
6. Hemiplegia
7. Seizures (focal, or less often generalized)

Clinical Tests:

Pronator drift

(Barré test)

Mingazzini test

Romberg test

Assessment of orientation (time, space, self) Pupil exam (to assess intracranial pressure). [15]

❖ Technical procedure

Imaging study

1. Plain radiograph of the skull shows a dense calcified mass lesion that follows the curvature of the skull and crosses suture lines. [16]



Chronic classified sub-dural hematoma

- b. CT Scan:** Non-contrast CT remains one of the most useful tools in the evaluation of CSDH

cSDH typically presents on the cerebral convexities either unilaterally or bilaterally. Bilateral hematomas represent between 9 and 22% of cases. [17]

Additional CT findings which may be described include obliteration of sulci, herniation, collapse of ventricular spaces, midline shift, and “thickening” of the cortex (hematoma isointensity with cortex) [18]

The cSDHs were classified into four groups; hypodensity (<25 HU), homogeneous isodensity (25-35 HU), layered type, and mixed type on the basis of CT scans (Fig.2).

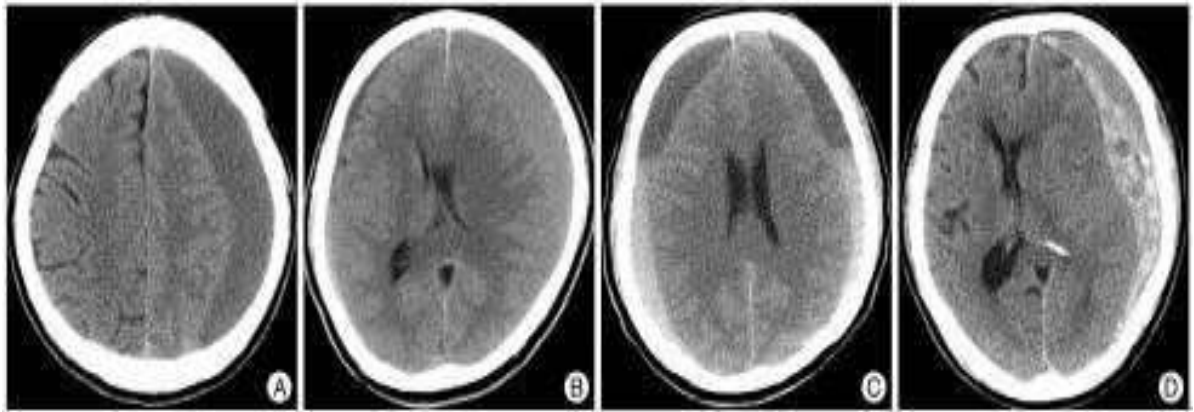


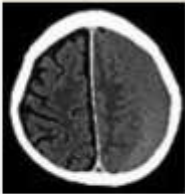
Fig. 2

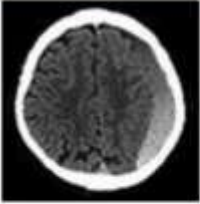
Chronic SDH classification according to CT density. A : Hypodensity (<25 HU). B : Homogeneous isodensity (25-35 HU). C : Layered type. D : Mixed type.

Other suggested by Nakaguchi et al., the type of CSDH with respect to the internal architecture is categorized as homogeneous, laminar, separated, or trabecular

TABLE 2

Type of chronic subdural hematoma on computed tomography by internal architecture

Architecture type	CT scan	Description
Homogeneous		Hematoma exhibiting complete homogeneous density (homogeneous hypo-, iso-, hyperdensity)
Laminar		Hematoma which had thin high-density layer along the inner hematoma membrane

Architecture type	CT scan	Description
Separated		Hematoma containing two component of different densities (lower density above higher density component) with a clear margin lying between them If the margin is indistinct, hematoma is the gradation type (subtype of separated type)
Trabecular		Hematoma with inhomogeneous density components and a high-density septum

- c. Magnetic resonance imaging (MRI) : internal structures of hematoma can be identified more accurately on MRI and it can show various stages of hemoglobin degradation in the hematoma.

Based on the MRI findings, Goto et al. classified CSDH into

- Hyperintensity:
- Iso or hypointensity,
- Mixed intensity,
- Layered intensity lesions on T1-weighted imaging (WI) and T2WI (Figure 3). [19].

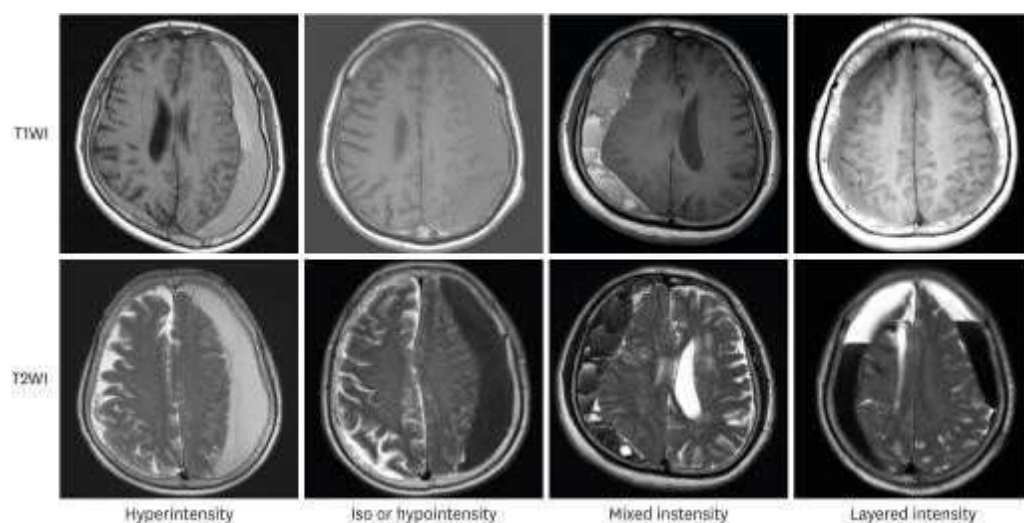


FIGURE 3

Type of chronic subdural hematoma on magnetic resonance imaging. Hematomas were classified as hyper-, iso-, or hypo-intensity, mixed intensity, and layered intensity lesions on T1WI and T2WI.

VI. DIFFERENTIAL DIAGNOSIS:

- a. Calcified [subdural empyema](#)
- b. Diffuse dural calcification
- c. Dural osteoma

VII. THERAPEUTIC APPROACH

- d. General management
 - a. Prepare for surgery if indications below are met.
 - b. Prompt surgery if signs of local mass effect, signs of herniation (increasing drowsiness, pupil changes, hemiparesis...) or cardiorespiratory abnormalities. Management includes: admit to ICU with q 1 hour neuro checks, NPO. Reverse coagulopathies.
- e. Medical treatment:

Non-surgical treatment of cSDH can only be considered if there are no signs of [intracranial hypertension](#) or cortical irritation, such as [epileptic seizures](#).

1. **Corticosteroids** remain a major target, as they are powerful anti-inflammatory substances that may inhibit formation of new blood vessels[20]

High dose dexamethasone treatment for 2 weeks: 8 mg twice daily on days 1 to 3

then 6 mg twice daily on days 4 to 6 then 4 mg twice daily on days 7 to 9 then 2 mg twice daily on days 10 to 12

and then 2 mg once daily on days 13 and 14

Dexamethasone as compared with placebo, would improve outcomes in patients with symptomatic chronic subdural hematoma by leading to clinical improvement that was adequate to enable nonoperative management and by reducing the risk of recurrence of the hematoma in patients who had undergone surgery [21]

2. [Atorvastatin](#) a low dose: 3 mg/kg/day was found to have anti-inflammatory and anti-angiogenic effects.

Results indicated that atorvastatin accelerated hematoma absorption, decreased recurrence risk, and surgical requirement. atorvastatin used as an adjunct to surgery could significantly decrease the risks of recurrence or prolong the time-to-recurrence hematoma. [22]

3. [Tranexamic](#) acid is a potential inhibitor of the fibrinolytic and inflammatory (kinin-kallikrein) systems. dosages of TXA varied from 750 mg daily (TRACS) to 500 mg, twice daily for eight weeks (an upper limit of 20 weeks of treatment)

Follow-up for each patient consisted of CT or MRI every 21 days from diagnosis to resolution of the CSDH. Administered it postoperatively with no reported cSDH recurrence. [23]

f. Surgical treatment

Surgical indications:

If symptomatic, or thicker than 10 mm or with more than 7 mm of midline shift, surgery is offered. [24]

Surgery remains the definitive treatment, while MMA (middle meningeal artery) embolization is potentially useful, but requires further investigation. Furthermore, MMA embolization is potentially useful for stopping further bleeding but will not provide direct help in reducing the mass effect of blood accumulation on the brain.

➤ **Surgical evacuation**

A wide variety of surgical techniques have been proposed and motivated several reviews and clinical trials: twist-drill craniostomy and spontaneous hematoma efflux, twist-drill craniostomy and catheter drainage, small craniotomy and endoscopic removal, subduro-peritoneal shunt as an alternative for infants and elderly patients, larger craniotomy, hematoma removal and membranectomy, or burr-hole evacuation without or with continuous closed-system drainage. This last option has been the most commonly applied all over the world, despite the fact that comparative outcomes are still debated. Regardless of several single-center reports and reviews promoting other surgical possibilities[25]

Single burr hole drainage with placement of a subdural drain, the current discussion seems to focus mostly on the use of drainage, while burr-hole evacuation has increasingly become the standard maintained for 24–48 hrs (removed when output becomes negligible), the current discussion seems to focus mostly on the use of drainage, while burr-hole evacuation has increasingly become the standard [26]

However, recent level I evidence shows lower rates of recurrence, drain misplacements, parenchymal injuries, as well as overall morbidity for subperiosteal/subgaleal drains but similar outcome and mortality when compared with subdural drains [27]

Burr-hole evacuation in the operating room can be performed under conscious sedation or general anesthesia as per surgeon preference or patient tolerance, compliance, and comorbidities. The technical aspects of this procedure are subject to several variants and preferences. In our hands, for a large convexity cSDH, two burr-holes are typically placed,

(1) anterior, just in front of the coronal suture, and (2) posterior, on the tuber parietalis, approximately 8 cm posterior to the first one. Both are placed cranial to the superior temporal line, as this normally coincides with the cranio-caudal center of the hematoma and avoids the risk of injuring eloquent brain areas (operculum or primary motor cortex) (Fig. 1a). If placed more medially and superiorly, the risk of injuring a bridging vein is higher and the gravity-induced passive hematoma efflux is lower. After opening the dura in a cruciate fashion, an outer membrane may be present and should be coagulated. However, the outer membrane should not

be cut beyond the edges of the burr-holes and evidence does not support opening the inner membrane. The vascularized membrane can bleed and such bleeding may be difficult to control if remote from the surgical site.

- **Placing** two burr holes, and irrigating through and through with tepid saline until the fluid runs clear,

Single “large” burr hole with irrigation and aspiration

- Twist drill craniostomy: note that small “twist drill” drainage without subdural drain has higher recurrence rate than e.g. burr holes

In our hands, for a large convexity cSDH, two burr-holes are typically placed, (1) anterior, just in front of the coronal suture, and (2) posterior, on the tuber parietalis, approximately 8 cm posterior to the first one. Both are placed cranial to the superior temporal line, as this normally coincides with the cranio-caudal center of the hematoma and avoids the risk of injuring eloquent brain areas (operculum or primary motor cortex) ([Fig. 1a](#)). If placed more medially and superiorly, the risk of injuring a bridging vein is higher and the gravity-induced passive hematoma efflux is lower. After opening the dura in a cruciate fashion, an outer membrane may be present and should be coagulated. However, the outer membrane should not be cut beyond the edges of the burr-holes and

evidence does not support opening the inner membrane

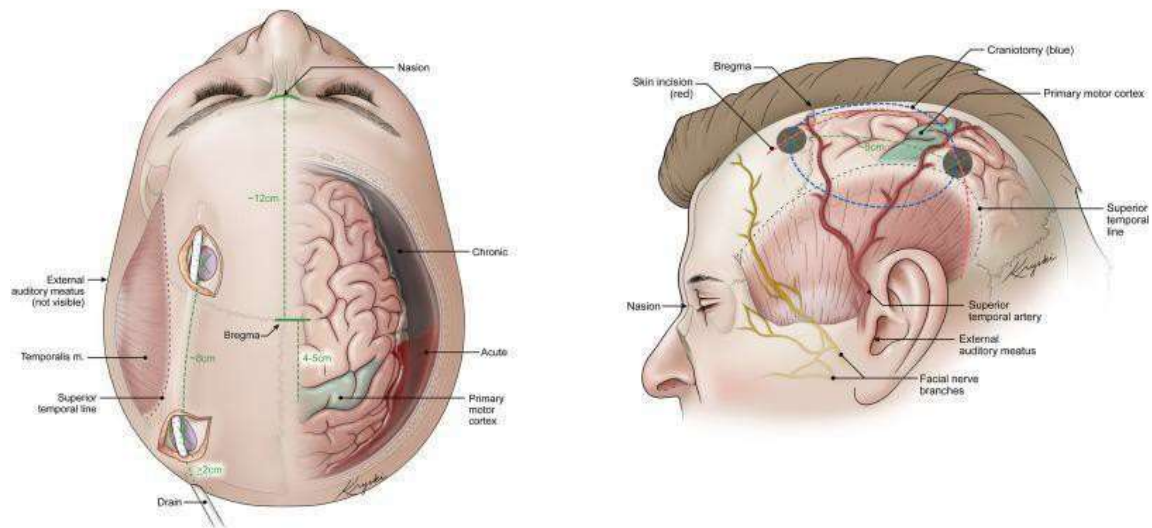


Fig. 1. Illustrations of a superior and lateral view of important anatomical landmarks to consider during surgical treatment of cSDH. **A)** The superior view highlights the placements of the burr-holes, with consideration of the [motor cortex](#). Shown here is the subcutaneous/subgaleal drain placement which is preferred by the authors. **B)** The lateral view demonstrated the same Burr-hole placements but also demonstrated the [skin incision](#) for a conversion to a [craniotomy](#). Anatomical landmarks are highlighted to demonstrate why this skin incision is preferred.

If hemorrhage cannot be controlled despite continued irrigation, consideration must be given to converting the burr-holes into a craniotomy (Fig. 1). Also, an upfront craniotomy for cSDH might be indicated for hematoma recurrence and presence of multiple cavities and membranes. When converting the two burr-holes into a craniotomy, the skin incision that we propose negotiates different concerns: (1) preservation of subcutaneous vascularization and superior temporal artery branches; (2) preservation of the frontal branch of the facial nerve; (3) reaching the upper part of the hematoma and cavity, between the superior temporal line and the superior sagittal sinus. Putting together these criteria, we propose a skin incision for such scenarios that we believe is safe and easy to replicate (Fig. 1b).

A last technical detail that frequently goes unnoticed is the long-term cosmetic outcome resulting from the scalp depression secondary to the burr-hole defect. Im et al. showed a good reconstruction with excellent cosmetic and functional outcomes and without significant complications after covering with titanium plates [28]

➤ Staging and head position during cSDH surgical evacuation

No clear evidence exists regarding a single-vs. two-staged procedure for bilateral cSDH. Whereas, a two-staged approach is done with the head horizontally on each side, a single-staged procedure is conducted with the head remaining in a neutral position thereby permitting simultaneous bilateral work.

➤ **Location of drainage and suction pressure**

Aside from placement of the draining catheter (e.g. subdural vs. subgaleal), application of suction to such a drain is also matter of debate. A partial suction with negative pressure would prevent air moving in whilst allowing the patient to elevate the head without the risk of a concomitant pneumocephalus. Early mobilization may decrease complications like thrombotic events or pulmonary inhalation and infection. On the other hand, bed rest is thought to reduce recurrence rates by facilitating brain re-expansion[29]

- **Bed-rest** restriction with the head of the bed flat (1 pillow is permitted) with mild overhydration for 24–48 hours post-op (or if a drain is used, until 24–48 hours after it is removed). May promote expansion of the brain and expulsion of residual subdural fluid. Allowing patients to sit up to 30–40° immediately post-op was associated with higher radiographic recurrence rate (2.3% for those kept flat, vs. 19% for those who sat up) but usually did not require reoperation.

❖ **Postoperative antithrombotic medication management**

An important factor that may explain this discrepancy is the timing of the re-initiation of antithrombotic medication. Recently, it has been shown that resumption of antithrombotics at day 3 postoperatively has no significant effect on recurrence rates. [30]

❖ **Surgical outcome**

There is clinical improvement when the subdural pressure is reduced to close to zero, which usually occurs after $\approx 20\%$ of the collection is removed. Patients who have high subdural fluid pressure tend to have more rapid brain expansion and clinical improvement than patients with low pressures. Residual subdural fluid collections after treatment are common, but clinical improvement does not require complete resolution of the fluid collection on CT.

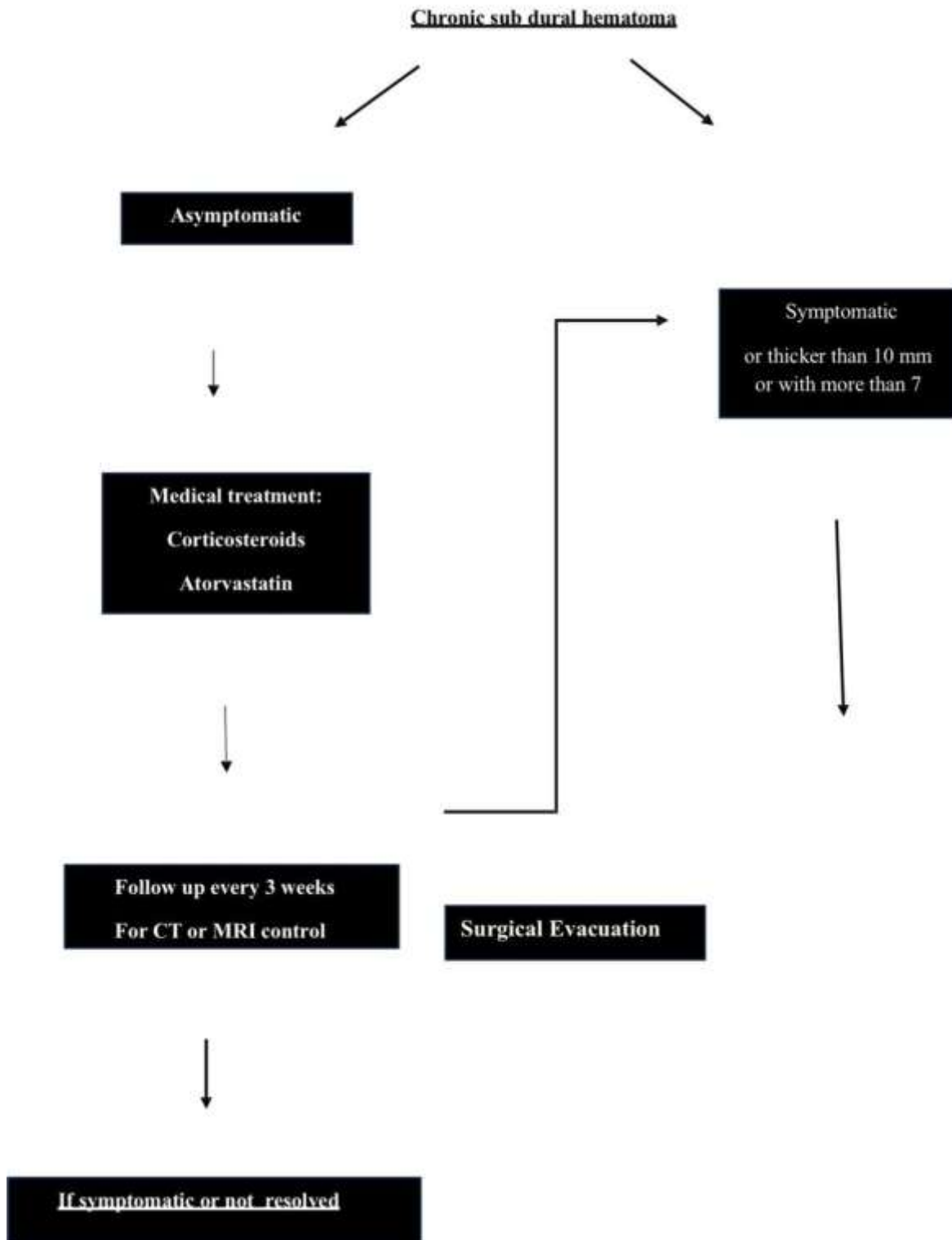
CTs showed persistent fluid in 78% of cases on post-op day 10, and in 15% after 40 days, and may take up to 6 months for complete resolution.

Recommendation: do not treat persistent fluid collections evident on CT (especially before ≈ 20 days post-op) unless it increases in size on CT or if the patient shows no recovery or deteriorates

❖ **Complications of surgical**

1. seizures (including intractable status epilepticus)
2. intracerebral hemorrhage (ICH): occurs in 0.7–5%. Very devastating in this setting: one-third of
3. these patients die and one third are severely disabled.
4. failure of the brain to re-expand and/or reaccumulation of the subdural fluid
5. tension pneumocephalus 5. subdural empyema: may also occur with untreated subdurals [31]

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EPIDURAL HEMATOMA

MOB Seyha, THO Vantha, SENG Samnang, SIM Sokchan

I. DEFINITION

An epidural hematoma (EDH) is an extra-axial collection of blood within the potential space between the outer layer of the dura mater and the inner table of the skull. It is confined by the lateral sutures (especially the coronal sutures) where the dura inserts. It is a life-threatening condition, which may require immediate intervention and can be associated with significant morbidity and mortality if left untreated. Rapid diagnosis

and evacuation are important for a good outcome.[\[1\]\[2\]\[3\]](#)

II. ETIOLOGY

It occurs in approximately 10% of traumatic brain injuries (TBI) requiring hospitalization. Both traumatic and non-traumatic mechanisms can cause an epidural hematoma.[\[4\]\[5\]](#)

The majority of cases related to a traumatic mechanism are a result of head injury due to motor vehicle collisions, physical assaults, or accidental falls.

Non-traumatic mechanisms include the following:

1. Infection/Abscess
2. Coagulopathy
3. Hemorrhagic Tumors
4. Vascular Malformations

III. EPIDEMIOLOGY

An epidural hematoma occurs in 2% of all head injuries and up to 15% of all fatal head traumas. Males are more often affected than are females. Furthermore, the incidence is higher among adolescents and young adults. The mean age of affected patients is 20 to 30 years, and it is rare after 50 to 60 years of age. As an individual's age advances, the dura mater becomes more adherent to the overlying bone. This decreases the chance that a hematoma can develop in the space between the cranium and dura.[\[6\]](#)

IV. PHYSIOPATHOLOGY

Arterial Injury

Most epidural hematomas result from arterial bleeding from a branch of the middle meningeal artery. The anterior meningeal artery or dural arteriovenous (AV) fistula at the vertex may be involved.[\[7\]\[8\]](#)

Venous Injury

Up to 10% of EDHs are due to venous bleeding following the laceration of a dural venous sinus.

In adults, up to 75% of EDHs occur in the temporal region. However, in children, they occur with similar frequency in the temporal, occipital, frontal, and posterior fossa regions.

A skull fracture is present in the majority of patients with EDH. These hematomas often present beneath a fracture of the squamous part of the temporal bone.

If this condition occurs within the spine, this entity is described as a spinal epidural hematoma.

Based on radiographic progression, it can be classified into one of the following

- i. Type I: Acute; occurs on day 1 and associated with a “swirl” of un-clotted blood
- ii. Type II: Subacute occurring between days 2 to 4 and usually solid
- iii. Type III: Chronic occurring between days 7 to 20; mixed or lucent appearance with contrast enhancement

V. DIAGNOSIS

b. Clinical argument

“Textbook” presentation (< 10%-27% have this classic presentation):

- brief posttraumatic loss of consciousness (LOC): from initial impact
- followed by a “lucid interval” for several hours
- then, obtundation, contralateral hemiparesis and ipsilateral pupillary dilatation as a result of mass effect from hematoma

Deterioration usually occurs over a few hours, but may take days and rarely, weeks (the longer intervals may be associated with venous bleeding).

Other presenting findings: H/A, vomiting, seizure (may be unilateral), hemihyperreflexia + unilateral Babinski sign, and elevated CSF pressure (LP is seldom used any longer). Bradycardia is usually a late finding. In peds, EDH should be suspected if there is a 10% drop in hematocrit after admission.

Contralateral hemiparesis is not uniformly seen, especially with EDH in locations other than laterally over the hemisphere. Shift of the brainstem away from the mass may produce compression of the opposite cerebral peduncle on tentorial notch which can produce ipsilateral hemiparesis (so called Kernohan's phenomenon or Kernohan's notch phenomenon)[9] a false localizing sign.

60% of patients with EDH have a dilated pupil, 85% of which are ipsilateral.

No initial loss of consciousness occurs in 60%. No lucid interval in 20%. NB: a lucid interval is not pathognomonic for EDH and may also be seen in other conditions (including subdural hematoma).

c. Technical procedure

i. Imaging study

1. Plain Skull X-Ray

Usually not helpful. No fracture is identified in 40% of EDH. In these cases the patient's age was almost always < 30 y

b. CT Scan

Classic CT appearance occurs in 84% of cases:

high density biconvex (lenticular) shape adjacent to the skull (Fig. 1). In 11% the side against the skull is convex and that along the brain is straight, and in 5% it is crescent shaped (resembling subdural hematoma). EDH may cross dural barriers (viz. falx or tentorium) (Fig 2) distinct from SDH, which is limited to one side, however EDH are usually limited by skull sutures which SDH are not. EDH usually has uniform density, sharply defined edges on multiple cuts, high attenuation (undiluted blood), is contiguous with inner table, and is usually confined to small segment of calvaria. Mass effect is frequent. Occasionally, an epidural may be isodense with brain and may not show up unless IV contrast is given,[10] although other clues are often present (e.g., inward displacement of the cortical ribbon). Mottled density may occur, and in some cases is associated with a hyperacute EDH [11]

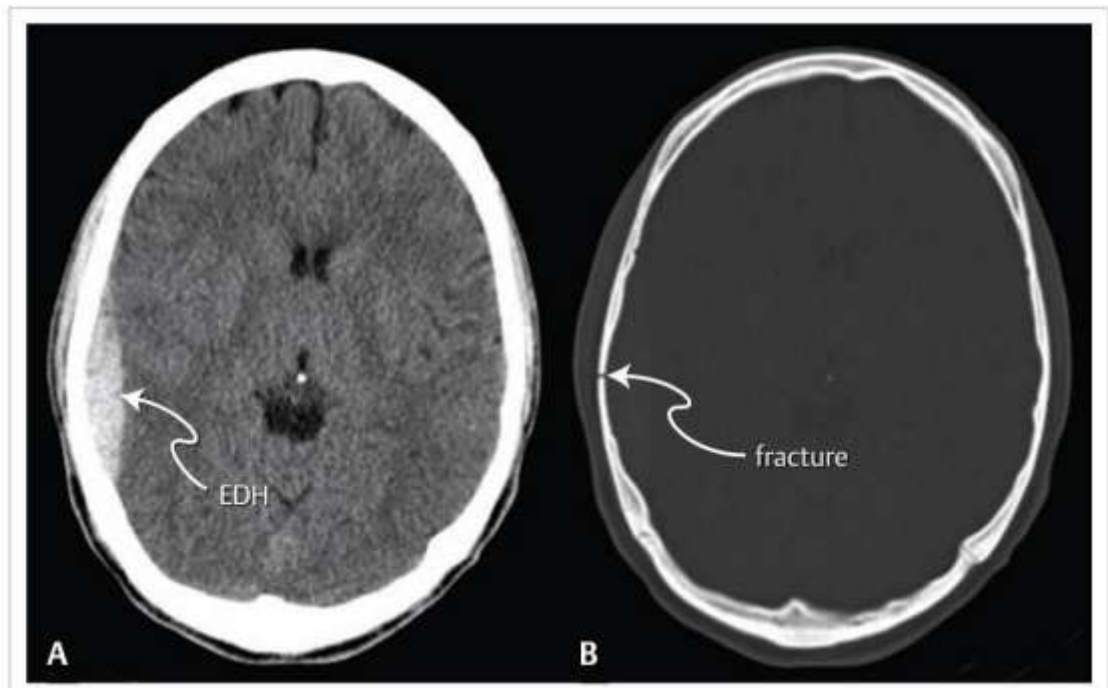


Fig.1 Epidural hematoma, right side.

Epidural hematoma (EDH) appears as biconvex (lenticular) shaped hyperdensity adjacent to inner table of skull. Note the underlying skull fracture seen on bone windows.

Image: axial CT scan. A: brain window. B: bone window.



Fig.2 Epidural hematoma, left side.

Image: coronal CT illustrating an EDH (yellow arrow- heads) crossing the tentorium (white arrow), which in addition to its biconvex shape, helps distinguish this from a subdural hematoma (SDH).

c. **Magnetic resonance imaging (MRI)**

Brain MRI is more sensitive than a CT scan, particularly when assessing for EDH in the vertex. It should be obtained when there is high clinical suspicion for EDH, accompanying a negative initial head CT scan.

In the situation of a suspected spinal EDH, a spinal MRI is the preferred imaging modality, as it affords higher resolution versus a spinal CT.

d. **Angiography**

When evaluating EDHs located in the vertex, the healthcare professional should evaluate for the presence of a dural arteriovenous (AV) fistula that may have arisen from the middle meningeal artery. Angiography may be required to evaluate the presence of such a lesion fully.

VI. DIFERRENCIAL DIAGGNOSIS

- d. subdural hematoma (see below for differentiating radiographic features)
- e. a posttraumatic disorder described by Denny-Brown consisting of a posttraumatic “lucid interval”
- f. followed by bradycardia, brief periods of restlessness and vomiting, without intracranial hypertension or mass. Children especially may have H/A and may become drowsy and confused. Theory: a form of vagal syncope. CT must be done to rule out EDH

VII. THERAPEUTIQUE APPROACH

g. General management

Prepare for surgery if indications below are met. Prompt surgery if signs of local mass effect, signs of herniation (increasing

drowsiness, pupil changes, hemiparesis...) or cardiorespiratory abnormalities. Management includes admit to ICU with q 1 hour neuro checks, NPO. Reverse coagulopathies.

h. Medical treatment

CT may detect small EDHs and can be used to follow them. However, in most cases, EDH is a surgical condition (see below).

Nonsurgical management may be attempted in patients not meeting surgical indications below, which generally limited to:

- a. *small (≤ 1 cm maximal thickness) subacute or chronic EDH[12]*
- b. minimal neurological signs/symptoms (e.g., slight lethargy, mild H/A) without focal deficit and without evidence of herniation
- c. **X**although medical management of p-fossa EDHs has been reported, these are more dangerous (see below) due to small volume of p-fossa and vulnerability of brainstem to compression by even modest hematoma expansion. Surgery is recommended .

Optional: for persistent headache a short course of steroids (1-3 days) may help, then taper. Follow-up CT: early after admission (e.g., 6 hours) and then any time the patient becomes symp- tomatic, or in 1-2 wks if clinically stable. Repeat in 1–3 mos (to document resolution).

In 50% of cases there will be a slight transient increase in size between days 5–16, and some patients required emergency craniotomy when signs of herniation occurred[13]

i. Surgical treatment

Surgical indications and timing

EDH in pediatric patients is riskier than adults since there is less room for clot. The threshold for surgery in pediatrics should be very low.

Indications for surgery

- i. EDH volume* $> 30 \text{ cm}^3$ should be evacuated regardless of GCS.
- ii. EDH with all of the following characteristics can be managed nonsurgically with serial CT scans and close neurological observation in a neurosurgical center:
 - a. volume* $< 30 \text{ cm}^3$
 - b. and thickness $< 15 \text{ mm}$
 - c. and midline shift (MLS) $< 5 \text{ mm}$
 - d. and GCS > 8
 - e. and no focal neurologic deficit

Timing of surgery

It is strongly recommended that patients with an acute EDH and GCS < 9 and anisocoria undergo surgical evacuation as soon as possible.

Surgical technical issues

Evacuation is performed in the OR unless the patient herniates in E/R and access to OR is not within acceptable timeframe (see Exploratory burr holes . Objectives:

- ◆ *clot removal: lowers ICP and eliminates focal mass effect. Blood is usually thick coagulum, thus exposure must large enough provide access to most of clot. Craniotomy permits more complete evacuation of hematoma than e.g., burr holes*[13]
- ◆ hemostasis: attempt to locate and neutralize the source of the hematoma (e.g., middle meningeal artery), coagulate bleeding soft tissue (dural veins & arteries). Requires an exposure larger than e.g., burr holes
- ◆ prevent reaccumulation: some bleeding or oozing typically occurs post op, and the dura is now detached from inner table facilitates recurrence of EDH. Strategies to prevent:
 - apply bone wax to intra-diploic bleeders
 - dural tack-up sutures: small sutures (e.g., 4:0 braided nylon) from the dura to small drill holes in the bone along the edges of the craniotomy and a central “tenting” suture through drill holes in the flap

- perforating the bone flap with a drill and placing a subgaleal drain which is maintained for several days or until output is scant

Special cases of epidural hematoma

- Delayed epidural hematoma (DEDH)

Definition: an EDH that is not present on the initial CT scan, but is found on subsequent CT. Comprise 9–10% of all EDHs in several series.[12,13]

Theoretical risk factors for DEDH include the following (NB: many of these riskfactors may be incurred after the patient is admitted following a negative initial CT):

- lowering ICP either medically (e.g., osmotic diuretics) and/or surgically (e.g., evacuating contralateral hematoma) which reduces tamponading effect
- rapidly correcting shock (hemodynamic “surge”) may cause DEDH[14]
- coagulopathies

DEDH tend to occur in patients with severe head injury and associated systemic injuries. However, DEDH have also been infrequently reported in mild head injury (GCS > 12) [15] Presence of a skull fracture has been identified as a common feature of DEDH[15]

Key to diagnosis: high index of suspicion. Avoid a false sense of security imparted by an initial “nonsurgical” CT. 6 of 7 patients in one series improved or remained unchanged neurologically despite enlarging EDH (most eventually deteriorate). 1 of 5 with an ICP monitor did not have a heralding increase in ICP. May develop once an intracranial lesion is surgically treated, as occurred in 5 of 7 patients within 24 hrs of evacuation of another EDH. 6 of 7 patients had known skull fractures in the region where the delayed EDH developed,[13] but none of 3 had a skull fracture in another report[14]

Posterior fossa epidural hematoma

Comprise ≈ 5% of EDH.[15,16] More common in 1st two decades of life. Although as many as 84% have occipital skull fractures, only ≈ 3% of children with occipital skull fractures develop p-fossa EDH. The source of bleeding is usually not found, but there is a high incidence of tears of the dural sinuses. Cerebellar signs are surprisingly lacking or subtle in most. See surgical indications. Overall mortality is ≈ 26% (mortality was higher in patients with an associated intracranial lesion).

□ Indications for surgery

Symptomatic posterior fossa mass lesions or those with mass effect on CT should be surgically removed. Note: mass effect on CT: defined as dislocation, compression or obliteration of the 4th ventricle; compression or loss of basal cisterns or the presence of obstructive hydrocephalus

asymptomatic lesions without mass effect on CT may be managed with close observation and serial imaging.

Timing of surgery

Posterior-fossa mass lesions meeting surgical criteria should be evacuated ASAP due to the potential for rapid deterioration

Surgical methods :

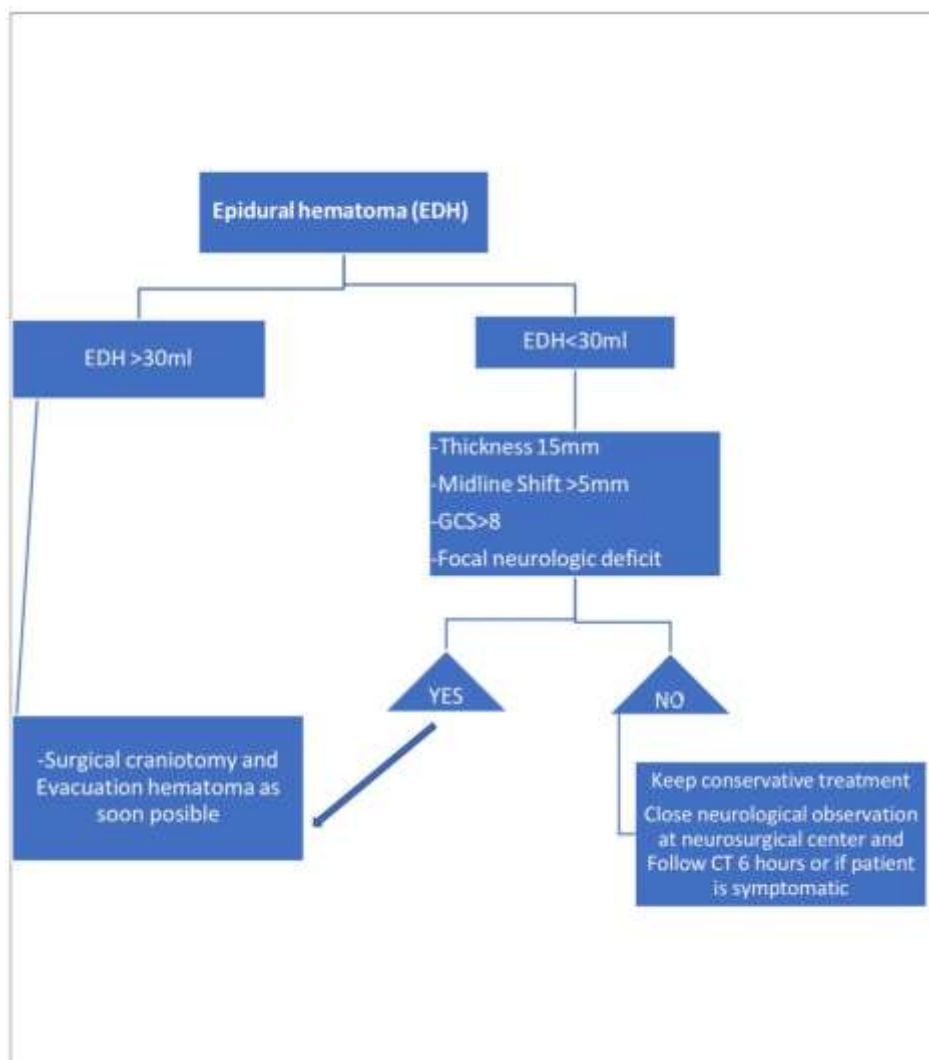
Suboccipital craniectomy (not replacing the bone) as opposed to craniotomy, is the recommended procedure because post-op bleeding or swelling is poorly tolerated in the small confines of the posterior fossa .

VIII. COMPLICATION

In general, patients with pure EDHs have an excellent prognosis of a functional outcome after the surgical evacuation, when it is rapidly detected and evacuated. A delay in diagnosis and treatment increases morbidity and mortality.

- Mass effect: compression of the brain if bleeding is significant
- Herniation
- Seizures

ALGORYTHM



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THORACO-LUMBAR SPINE FRACTURE

SENG Samnang, THO Vantha, MOB Seyha, SIM Sokchan

I. INTRODUCTION

Thoracolumbar fractures are breakages in the vertebrae of the spinal column in the thoracic and lumbar regions. They may be associated with disruption of the ligamentous complexes, and can result in instability or compression of neural structures.

Thoracolumbar fractures are the usual outcome of thoracolumbar trauma. Other outcomes include traumatic disk prolapse, ligamentous injury, and epidural hematoma causing pressure on the spinal cord or nerve roots; these occur very rarely without a fracture.

The thoracolumbar junction (T11–L2) is the most common site of traumatic spine injury in the thoracic and lumbar spine. Injury recognition is key in clinical decision-making and prevention of secondary neurologic injury.

Surgery is indicated for spinal cord injury, acute or delayed mechanical instability, and intractable pain.

II. ETIOLOGY

Injury to the thoracolumbar spine can occur from motor vehicle accidents, falls from height, recreational accidents, and occupational injuries. Because the thoracolumbar junction is more vulnerable than other segments of the spine, it is a very common area of injury due to its intrinsic biomechanics.

In the event of high-velocity spine injuries, there is an associated 25% risk for accompanying spinal cord injury (SCI), which can subsequently have devastating effects on the patient with loss of individual productive years and an associated high cost to society. As many as 25% of thoracolumbar fractures have another spine fracture elsewhere, most commonly the cervical spine.

III. EPIDEMIOLOGY

Thoracolumbar traumatic injury was most commonly reported in the 15-29 year age group before 2000, but now the median age is 35. Approximately 27% of the patients with thoracolumbar junction injuries had neurological deficits, which results in detrimental effects on society due to lifelong disability and loss of productive economic years. Approximately 70% of these fractures occur without immediate neurological injury, and eventually, 55% remain neurologically intact. Of the 45% who develop

neurological symptoms, 26% develop an incomplete injury, and 19% develop a complete injury. Up to 75% of spine fractures occur in the T10-L2 area.

According to the United States National Spinal Cord Injury Statistical Center, in approximately 54 cases per million people, 17,000 new spinal cord injury cases occur each year. Over 250,000 patients are currently living with permanent deficits. Blunt trauma is associated with an incidence of 1.9% thoracic fractures.

The incidence of thoracolumbar junction fractures after motor vehicle accidents is approximately 2.4%, but it has been increasing over the years. In blunt trauma patients, thoracolumbar junction fractures had an incidence of 6.9%. A review of patients with blunt trauma who underwent a chest/abdomen/pelvis computed tomographic (CT) scan showed that up to 25% had thoracolumbar fractures. Several fractures were undetected in the plain X-ray films. In adults, motor vehicle accidents were the most common mechanism in 36.70% of the cases, followed by falls from height for 31.70%. For people with paraplegia, the average lifetime costs are approximately 2.5 million dollars per patient.

A report from China found a thoracolumbar junction fracture incidence rate of 2.4 patients per million with a mean age of 49 years, with a male to female ratio of 1.4:1.

In children, thoracolumbar fractures are rare, with a reported incidence as low as 2% of all spine fractures. However, other series have reported an incidence of 5%-34%. The most common mechanism is falling from a height, with 25% presenting a complete deficit, and approximately 20% requiring surgery.

IV. DIAGNOSTIC PROCEDURE

1- Clinical argument

Initial triage is performed using the primary ATLS survey, ensuring that the airway is secured, breathing and ventilation are adequate, and there is proper cardiovascular support. After this initial assessment, the secondary survey focused on the spine will consist of a motor and sensory examination to determine the SCI level. The rectal exam for evaluation of tone, voluntary contraction, and sensation should never be overlooked. If the patient is critically ill and on mechanical ventilation, with associated head injuries such as traumatic brain injury (TBI), the rectal tone may be the only objective finding of the neurological physical exam in patients with SCI. The international standards for neurological classification of spinal cord injury, commonly referred to as the ASIA exam, was developed as a universal classification tool for SCI based on a standardized sensory and motor assessment.[\[18\]](#) It is a system used to define and describe the extent and severity of a patient's SCI and determine future rehabilitation and recovery needs.[\[19\]](#) It is ideally completed within 72 hours after the initial injury.[\[20\]](#) The patient's grade is based on how much sensation and movement are detected below the injury level.

TABLE 1. ASIA Impairment Scale grade according to International Standards for Neurological Classification of Spinal Cord Injury

ASIA grade	Type of injury	Definition of type of injury
Grade A	Complete	No motor or sensory function below the level of injury
Grade B	Incomplete	Sensory but not motor function is preserved below the level of injury
Grade C	Incomplete	Motor function is preserved, but more than 50% of key muscles below the neurological level have a muscle grade <3
Grade D	Incomplete	Motor function is preserved but the at least 50% of key muscles below the neurological level have a muscle grade ≥ 3
Grade E	Normal	Motor and sensory functions are normal.

a. Technical procedure

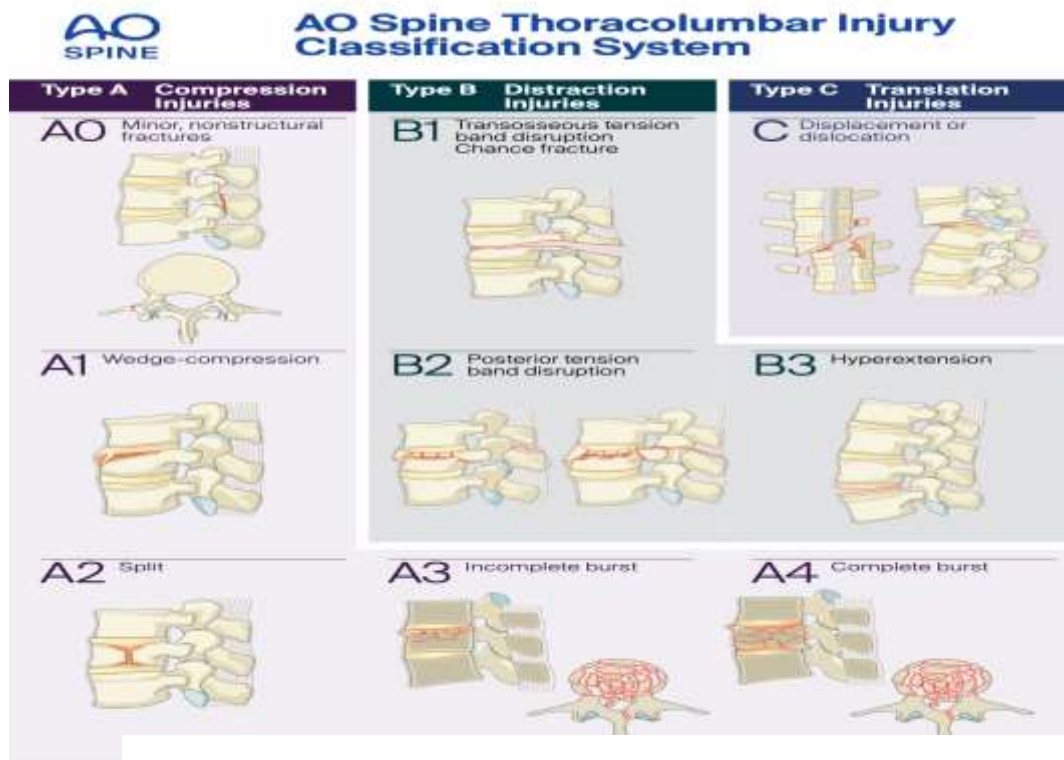
□ Imaging study

In the setting of a traumatic event suffered by the patient, especially a high-energy mechanism, it is imperative that full spinal imaging is performed as soon as the primary survey is completed. This should be performed once the patient is stabilized from acute life-threatening injuries (shock, TBI). Delaying time to image acquisition in an SCI setting with a neurological deficit may be detrimental to the patient by potentially decreasing the maximal benefit of early surgery. Delayed diagnosis of a thoracolumbar fracture or injury can occur in up to 5% of patients. In the case of unstable thoracolumbar fractures, it can result in neurologic deficits and pain

CT scan will give an adequate assessment of bony structures, including the vertebral bodies, pedicles, laminae, facets, and transverse and spinous processes. In contrast, MRI will provide a better assessment of the ligamentous and the neural structures, including the spinal cord, cauda equina, conus medullaris, and spinal nerve roots. Therefore, in the case of a patient presenting with neurological deficits, MRI will most likely be performed for a complete management algorithm.

TLICS 3 independent predictors

1	Morphology immediate stability	- Compression - Burst - Translation/rotation - Distraction	1 2 3 4	- Radiographs - CT
2	Integrity of PLC longterm stability	- Intact - Suspected - Injured	0 2 3	- MRI
3	Neurological status	- Intact - Nerve root - Complete cord - Incomplete cord - Cauda equina	0 2 2 3 3	- Physical examination
			0-3	- nonsurgical



Cont'd

V. DIFFERENTIAL DIAGNOSIS

- Anatomical variant
- Osteoporosis fracture
- Ankylosing spondylitis
- Stiff-spine diseases
- Spine tumor
- Spine metastatic disease
- Pseudoarthrosis from previous spine instrumentation

VI. THERAPEUTIC APPROACH

a. Stabilize

Spinal precautions should be implemented in all trauma patients until a spine fracture is ruled out. Limiting flexion, extension, rotation, and twisting of the spine can help avoid exacerbating an injury to the spinal cord. Precautions can be lifted once radiographic images and physical exams are reassuring for no spinal instability. Maintaining perfusion to the spinal cord with supportive care, including airway, breathing, and circulation management with a mean arterial pressure above 85 mmHg, is essential to enhance spinal cord perfusion. Based on the amount of instability or neurological involvement, the patient may require emergency surgery.

b. Bracing

High thoracic fractures – Stable fractures between T1 to approximately T5 may be treated with a cervicothoracic orthosis (CTO), for a TLSO does not extend far enough superiorly to provide adequate support. • Midthoracic to lumbosacral fractures – Stable fractures anywhere between the midthoracic spine to L5 may be treated with a hard TLSO. – In most cases of acute lumbosacral trauma, a soft lumbosacral orthosis (LSO) brace does not provide adequate stabilization. A TLSO remains the best option for external stabilization. • Always assess stability of fracture in brace with upright X-rays in brace.

c. Surgical intervention

For surgical intervention, approach alternatives include anterior, posterior, lateral, or combined approaches. The operative planning and stabilization of these fractures usually include a multifactorial decision-making process. The key factor for deciding an anterior versus a posterior approach is the location of the pathology to be addressed. Different techniques include posterior segmental instrumentation and fusion with or without posterior decompression, percutaneous pedicle screw instrumentation, lateral interbody fusion, and anterior column fusion. Spinal neuromonitoring is routinely used during the surgery. Postoperative bracing does not offer an advantage after surgical stabilization; however, it can be used for comfort.[\[36\]](#)

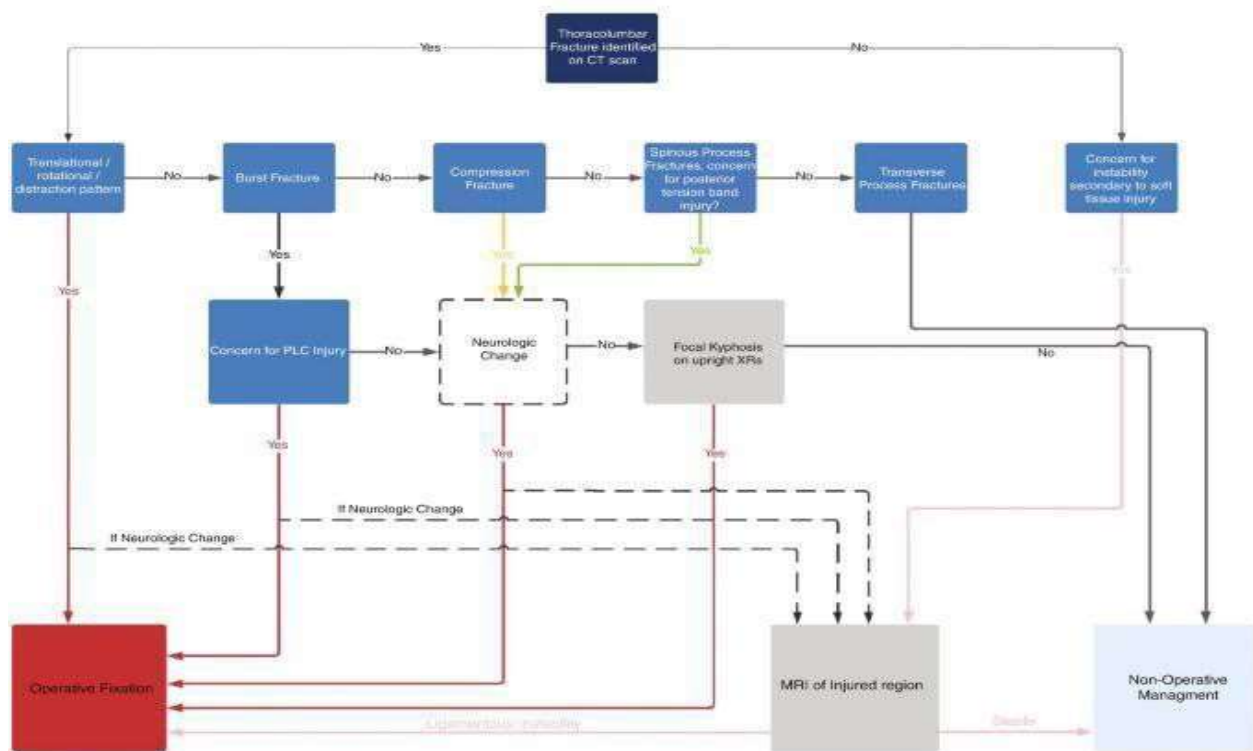
Burst fractures are stable if the posterior osteoligamentous complex and facet capsules are intact. Stable burst fractures can be treated with an orthosis. If unstable, posterior pedicle screw fixation of the adjacent levels is performed. If an anterior surgery is performed, the fractured bone is removed, and the defect filled with a solid bone graft or a metallic or synthetic cage. This technique restores the anterior column. Flexion-distraction injuries damage the posterior osteoligamentous complex. They are treated with posterior instrumentation and fusion over several levels to restore the sagittal alignment. Fracture-dislocations are very unstable and are treated with early decompression and posterior multilevel instrumentation and fusion

VII. COMPLICATION

- ☐ Sepsis
- ☐ Aspiration bronchopneumonia
- ☐ Neurological deficits
- ☐ Deep venous thrombosis
- ☐ Surgical wound infection
- ☐ Cerebrospinal fluid leakage
- ☐ Pseudoarthrosis
- ☐ Adjacent level disease
- ☐ Excessive blood loss during surgery
- ☐ Anesthesia-related complication

VIII. ALGORITHM

Algorithm for treatment of Thoracolumbar Injuries



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TRAUMATIC BRAIN INJURY

PUTH Chenda, SAM Sathya, SOK Chhay, CHOR Vannaroth, HUOY Phearum, SIM Sokchan

I. DEFINITION

The current definition of TBI is phenomenological. Often, there is confusion in the nosology of TBI especially in relation to mild TBI (mTBI), a term that implicitly refers to the TBI event consistent with acute concussion. TBI is categorized according to the clinical pillars of post-traumatic amnesia (PTA) and/or a disturbance of consciousness either alteration of consciousness (AOC) or loss of consciousness (LOC)]. Both clinical features, although correlated, allow for independent diagnosis of TBI severity. The overall TBI diagnosis is due to the severity of Primary Traumatic Brain Damage that is brain injury that results from mechanical forces producing tissue deformation at the moment of injury with direct damage to blood vessels, axons, neurons, and glia.

Brain injury from trauma results from two distinct processes:

- a. Primary brain injury occurs at time of trauma (cortical contusions, lacerations, bone fragmentation, diffuse axonal injury, and brainstem contusion) (fig.1)
- b. Secondary injury develops subsequent to the initial injury. Includes injuries from intracranial hematomas, edema, hypoxemia, ischemia (primarily due to elevated intracranial pressure (ICP) and/or shock), vasospasm. (fig.2)

II. ETIOLOGY

TBI is the result of head trauma. It can cause by some mechanism:

- a. Road traffic accidents
- b. Falls
- c. Sport accidental
- d. Violent
- e. Gun Shot or Blast injury(fig.3)

Based on the mechanism, TBI is classified as:

- a. Closed/blunt force injury caused by direct force to head, acceleration/deceleration, or rotational forces. Common causes include falls, assault, and motor vehicle collisions.
- b. Closed head injury
 - Blunt
 - Acceleration/ deceleration

- c. Penetrating
 - Sharp material
 - Bullet (fig 1)
 - Boom blast fragment

III. CLINICAL ARGUMENT

i. Clinical history

The history of bleeding from the ear, nose or throat should be sought. The duration and timing of loss of consciousness must be asked for. (fig.4)

The associated co-morbid conditions and any significant concurrent or past illness (e.g. diabetes, hypertension, ischemic heart disease, etc.) should be enquired for and taken note of.

ii. General examination

- a. **Mild TBI (MTBI):** GCS 13 to 15, the majority of patients with cranial trauma fall in this group. Patients are awake and may be confused but can communicate and follow commands.
- b. **Moderate TBI:** GCS 9 to 12, these patients are generally drowsy to obtunded but not comatose. They can open their eyes and localize painful stimuli. They are at high risk of clinical deterioration and must be monitored carefully.
- c. **Severe TBI:** GCS 3 to 8, these patients are obtunded to comatose, they do not follow commands and may exhibit decerebrate or decorticate posturing. They have significant structural and metabolic brain dysfunction and are at high risk of secondary brain injury and deterioration.

Table of Glasgow Coma Scale

Eye opening (E)	Score	Verbal response (V)	Score	Motor response (M)	Score
-		-		Obeys commands	6
-		Orientated	5	Localises to pain	5
Spontaneously	4	Confused	4	Flexion/withdrawal to pain	4
To speech	3	Inappropriate words	3	Abnormal flexion	3
To pain	2	Incomprehensible sounds	2	Extends	2
None	1	None	1	None	1

iii. General physical condition

- a. Basal skull fracture: (fig.5)
 - ☐ Raccoon's eyes (AKA Panda bear sign): periorbital ecchymoses
 - ☐ Battle's sign: postauricular ecchymoses (around mastoid air sinuses)
 - ☐ CSF rhinorrhea/otorrhea
 - ☐ Hemotympanum or laceration of external auditory canal
- b. Facial fractures
 - ☐ Le Fort fractures: palpate for instability of facial bones, including zygomatic arch
 - ☐ Orbital rim fracture: palpable step-off
 - ☐ Periorbital edema, proptosis

iv. Cranial nerve examination

- a. optic nerve function
 - ☐ if conscious: serial quantitation of vision in each eye is important.¹⁹ A Rosenbaum near vision card is ideal (see inside back cover), otherwise use any printed material. If patient cannot see this, check if they can count fingers. Failing this, check for hand motion vision and lastly light perception. Children may develop transient cortical blindness lasting 1–2 days, usually after a blow to the back of the head
 - ☐ if unconscious: check for afferent pupillary defect, best demonstrated with swinging flashlight test. Indicates possible optic nerve injury
 - ☐ funduscopic exam: check for papilledema, pre-retinal hemorrhages, retinal detachment, or retinal abnormalities suggestive of anterior optic nerve injury. If a detailed exam is required, pharmacologic dilatation with mydriatics (p.593) may be employed; however, this precludes pupillary exam for a variable period of time, and should be undertaken advisedly
- b. pupil: size in ambient light; reaction to light (direct & consensual)
- c. VII: check for peripheral VII palsy (p.1064) (facial asymmetry of unilateral upper and lower facial muscles)
- d. VI: abducens palsy following trauma may occur because of ↑ ICP or with clival fractures

v. Level of consciousness/mental status

- a. Glasgow coma scale for quantitating level of consciousness in poorly responsive patient

vi. Motor exam (assesses motor tracts from motor cortex through spinal cord)

- a. if patient is cooperative: check motor strength in all 4 extremities

- b. if uncooperative: check for appropriate movement of all 4 extremities to noxious stimulus (differentiate voluntary movement from posturing or stereotypical spinal cord reflex). This also assesses sensation in an unresponsive patient
- c. if any doubt about integrity of spinal cord: also check “resting” tone of anal sphincter on rectal exam, evaluate voluntary sphincter contraction if patient can cooperate, check anal wink with pinprick, and assess bulbocavernosus reflex

vii. Sensory exam

- a. cooperative patient:
 - ☐ Check pinprick on trunk and in all 4 extremities, touch on major dermatomes (C4, C6, C7, C8, T4, T6, T10, L2, L4, L5, S1, sacrococcygeal)
 - ☐ Check posterior column function: joint position sense of LEs
- b. uncooperative patient: check for central response to noxious stimulus (e.g., grimace, vocalization..., as opposed to flexion-withdrawal, which could be a spinal cord mediated reflex).

viii. Reflexes

- a. muscle stretch (“deep tendon”) reflexes if patient is not thrashing e.g., preserved reflex indicates that a flaccid limb is due to CNS injury and not nerve root injury (and vice versa)
- b. Check plantar reflex for upgoing toes (Babinski sign) in suspected spinal cord injury: the anal wink and bulbocavernosus reflex are checked on the
- c. Rectal exam (see above)

IV. IMAGING

General information

An unenhanced (i.e., non-contrast) CT scan of the head usually suffices for patients seen in the emergency department presenting after TBI or with a new neurologic deficit. Enhanced CT or MRI may be appropriate after the unenhanced CT in some circumstances but are not usually required emergently (exceptions include: suspected ischemic stroke, significant brain edema on non-contrast CT suggesting a neoplasm that cannot be demonstrated without contrast).

- i. Other tests: angiography may be needed acutely in certain circumstances, primarily in penetrating head trauma.
- ii. Skull X-rays are usually inadequate for primary evaluation, but may be helpful in certain situations, for example with some retained foreign bodies.
- iii. Structurally of the TBI in the image and the pathologies as a lesion post traumatic: (fig.6)
 - a. Epidural hematoma (EDH)
 - b. Subdural hematoma (SDH)

- c. Diffuse axonal injury (DAI) or traumatic axonal injury (TAI)
 - d. Traumatic subarachnoid hemorrhage (TSAH or SAH)
 - e. Intraventricular hemorrhage (IVH)
 - f. Brain Contusion: there is coup injury or contre coup injury
 - g. Skull fracture
- iv. Indication for initial brain CT

Noncontrast head CT for patients ☐☐ 16 years with GCS=14-15 following non TBI are indicated only if one or more of the following is present
Level I: For patients with LOC or posttraumatic amnesia
☐ Headache ☐ Vomiting ☐ Age > 60 years ☐ Intoxication (alcohol or drugs) ☐ Short-term memory deficits

☐ Physical signs of trauma above the clavicle ☐ Posttraumatic seizure ☐ GCS < 15 ☐ Focal neurological deficit Coagulopathy (including anticoagulant/antiplatelet drugs) *
Level II: For patients without LOC or posttraumatic amnesia
☐ Focal neurologic deficit ☐ Vomiting ☐ Severe headache ☐ Age ☐ ☐ 65 years ☐ Signs of basal skull fracture (includes hemotympanum, “raccoon’s eyes” CSF rhinorrhea or otorrhea, Battle’s sign) ☐ GCS < 15 ☐ Coagulopathy (Including anticoagulants/antiplatelet drugs)* Dangerous mechanism of injury: pedestrian hit by motor vehicle, ejection from automobile, fall from height>3 feet or 5 stairs
*data regarding the risk of adverse outcome in anticoagulated patients after mTBI is limited and of low quality.

- v. CT finding in head trauma
- a. Blood (hemorrhages or hematomas)
 - b. Hydrocephalus: enlarged ventricles may sometimes develop following trauma
 - c. Cerebral swelling: obliteration of basal cisterns, compression of ventricles and sulci...
 - d. Evidence of cerebral anoxia: loss of gray-white interface, signs of swelling
 - e. Skull fractures (best appreciated using bone windows)

- f. Ischemic infarction (stroke): findings are usually minimal or subtle if < 24 hrs since stroke (DWI MRI is the test of choice for this)
- g. Pneumocephalus: may indicate skull fracture (basal or open convexity)
- h. Shift of midline structures

V. MANAGEMENT

- i. Bed rest
- ii. Put the patient on collar and spinal brace
- iii. Airway and blood oxygenation requirements.
- iv. Intubation in case of coma (GCS< 8)
- v. Ventilation
- vi. Correction hypovolemia: Maintenance of systemic BP
- vii. Good intravenous line
- viii. Anticonvulsants
- ix. Brain protective agents
- x. Associated injuries should be identified

VI. COMPLICATION

- i. Post-traumatic amnesia
- ii. Post-concussion syndrome
- iii. Neurological deficit
- iv. Neuro-behavioral sequelae
- v. Post-traumatic epilepsy
- vi. Infections: Meningitis, osteomyelitis, abscess
- vii. Post-traumatic CSF fistulae and pneumocephalus
- viii. Normal pressure hydrocephalus
- ix. Metabolic abnormalities
- x. Vascular abnormalities like coagulopathies, carotid cavernous fistula, delayed intracerebral hematomas and others.

VII. ALGORIHM

The algorithm of the TBI

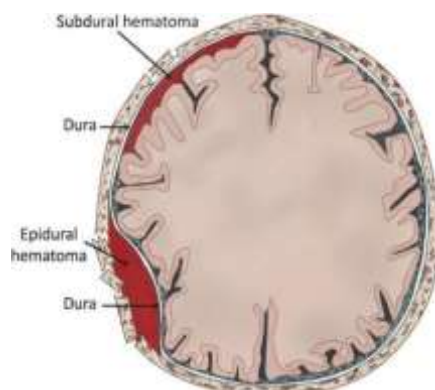
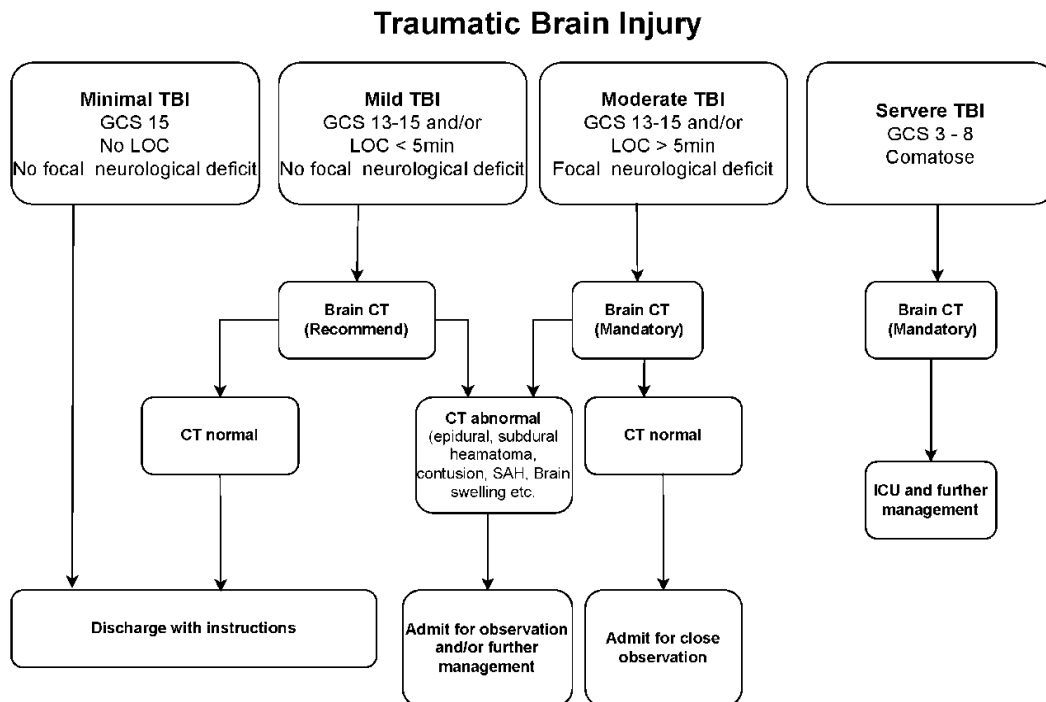


Fig.1: Primary brain injury

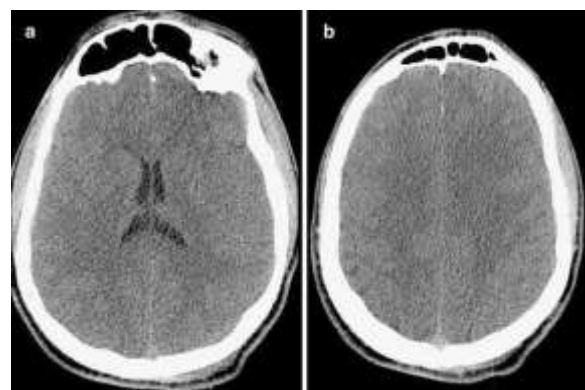


Fig.2: Cerebral edema

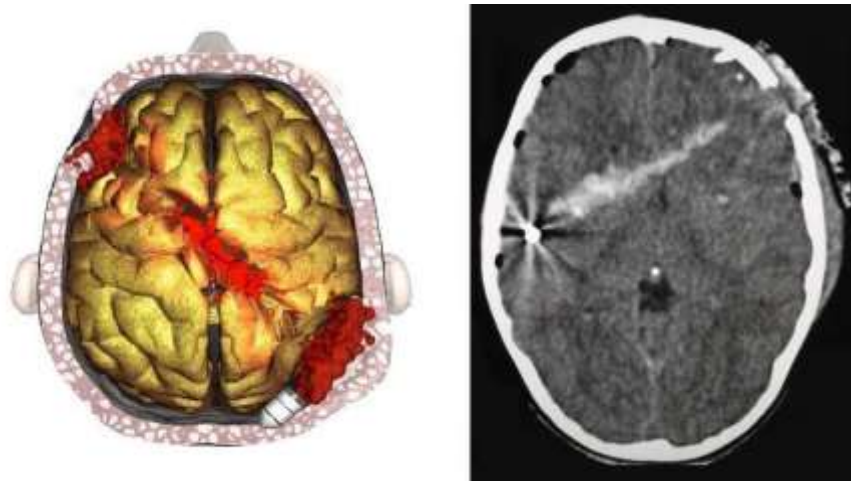


Fig.3: Brain injury by gun shot.



Fig.4: Bleeding from nose and ear (Rhinorrhagia and Otorrhagia)



Fig.5: Raccoon's eyes sign and Battle's sign

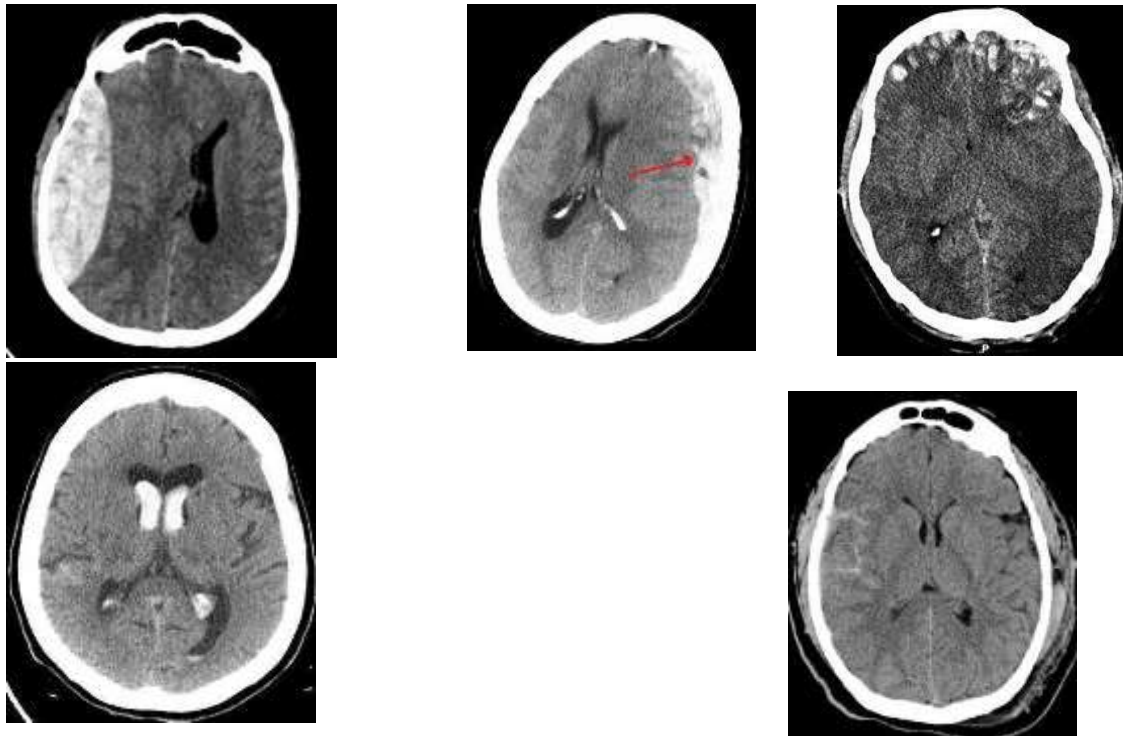


Fig.6: Epidural Hematoma; Acute Subdural Hematoma; Cerebral Contusion; Intraventricular Hemorrhage; Subarachnoid Hemorrhage.

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CHAPTER IV

OPHTHALMOLOGICAL

SURGERY

1. Blepharopstosis
2. Cataract
3. Chemical eye injury
4. Congenital Nasolacrimal Duct Obstruction
5. Congenital cataract
6. Corneal Ulcer
7. Corneal wound laceration
8. Diabetic retinopathy
9. Ectropion
10. Entropion
11. Esotropia
12. Exotropia
13. Glaucoma
14. Intraocular foreign bodies
15. Lens Removal with IOL Implantation and Trabeculectomy
16. Nasolacrimal duct obstruction
17. Primary closure angle glaucoma
18. Primary open angle glaucoma
19. Retinoblastoma
20. Rhegmatogenous Retinal Detachment
21. Tractional Retinal Detachment
22. Vitreous Hemorrhage
23. Zonular Dehiscence with Lens Subluxation or Dislocation Cataract

BLEPHAROPTOSIS

Krin Sreypeou, Pok Thon

I. CASE DEFINITION

Ptosis or blepharoptosis is abnormally low positioned upper eyelid, which can decrease or even occlude the vision completely. It may be congenital or acquired in origin.

II. ETIOLOGY

The classification of ptosis could be by onset, by caused and by severity. By onset, ptosis could be classified as congenital and acquired ptosis.

But any approach to the treatment of ptosis must be based on an accurate of its causes (by causes). Without such knowledge, even the most enthusiastic and perfectly executed surgical operation could be unsuccessful. The following mechanistic classification allow a more natural grouping of disorders responsible for ptosis and provides a more appropriate basis for its management. Clearly, in many cases of ptosis, good medical judgment will demand more complete clinical evaluation or modification in surgical correction.

1. Neurogenic ptosis
 - a. Third nerve palsy: dysfunction of oculomotor nerve from various of causes (microvascular, toxic, trauma, tumor...etc.)
 - b. Horner's syndrome (sympathetic causes)
 - c. Marcus Gunn Jaw Wink syndrome (synkinetic nerve)
2. Myogenic ptosis
 - a. Congenital developmental dystrophy of levator aponeurosis or levator muscles (Congenital ptosis)
 - b. Chronic progressive external ophthalmoplegia (CPEO), myotonic dystrophy, and oculopharyngeal muscular dystrophy.
3. Neuro-muscular ptosis: Ocular myasthenia gravis
4. Aponeurotic ptosis:
 - a. senile or involutional ptosis is the most common cause
 - b. Other: other causes of levator aponeurotic dehiscence: trauma...etc.
5. Mechanical ptosis
 - a. Periocular or eyelid mass or inflammation and infection

- b. Conjunctival scarring restricted lid movement

III. DIAGNOSTIC PROCEDURE

1. Clinical argument

Evaluation of the ptotic patient should be included an attempt to determine the precise etiology mention above. Once other treatable diseases have been ruled out, attention is directed at repair of the ptosis. Complete ophthalmic examination must be done to identify the causes, severity and emergency state of the condition, especially in acquired ptosis.

History

The age at onset of ptosis and its duration will usually distinguish congenital from acquired cases. If the history is not clear, old photographs may be helpful. It is also important to enquire about symptoms of possible underlying systemic disease, such as associated diplopia, variability of ptosis the day and excessive fatigue.

Clinical examination

The most important criterion in selecting a successful surgical procedure is levator muscle function. This must be measured with extreme care because in the poor function range, even a different of 1 or 2mm may result in the choice of an inappropriate operation. Levator muscle function is measured as maximum eyelid margin excursion from extreme downward gaze to extreme upward gaze positions. The frontalis muscle must be immobilized with the examiner's finger at the brow to eliminate its contribution to lid elevation. This especially important in children. By convention, more than 12mm of function is considered excellent, 8-11mm is good, 5 to 7mm is fair by 4mm is poor.

Other measuring beside levator function for ptosis evaluation:

- **MRD (Margin-reflex distance)** is the distance between the upper lid margin and the corneal reflection of pen torch held by the examiner, at which the patient is directly looking, normal is 4-4.5mm. Severity of ptosis is shown in the table below.
- **Palpebral fissure height** is the distance between the upper and lower lid margins, measured in the pupillary plane. The upper lid margin normally rests about 2mm below the upper limbus and 1mm lower above the lower limbus or at the limbus. Normal is around 9 to 12mm)
- **Upper lid crease** is the vertical distance between the lid margin and the lid crease in downgaze. In females it measures about 10mm and in males 8mm for Caucasian. It is smaller in Asians. Absence or low crease in patient with congenital ptosis is indirect evidence of poor levator function, whereas a high crease suggests an aponeurotic defect.

- **Bell's phenomenon** is the important evaluator pre-surgery, it helps surgeon to have clearer plan whether they should aim for full correction or under correction. The Bell's phenomenon, also called the palpebral-oculogyric reflex, refers to the movement of the eyeballs in an upward direction when the eyelids are forcefully closed. Poor Bell phenomenon is considered relative contraindication for ptosis surgery as it increases corneal exposure.
- **Corneal sensitivity** should be tested before operation. Reduce corneal sensitivity and dry eye can produce exposure keratopathy.
- **Marcus Gunn Jaw winking syndrome** also needs to be tested before the operation.

Severity Classification of Ptosis	
 MRD [4.5 to 5mm] A	Normal MRD: 4.5 to 5mm
 B	Mild ptosis
 C	Moderate ptosis
 D	Severe ptosis

2. Technical procedure

a. Laboratory

Laboratory test is normally not necessary unless the physician is aware of neurologic or neuromuscular causes. However, some blood test like hemogram, bleeding test and serology test are important for preparing surgery.

b. Imaging study

Not different from laboratory test, imaging is needed only when the neurological or neuromuscular or some form of muscular causes that could be associated with systemic involvement.

IV. DIFFERENTIAL DIAGNOSIS

The false impression of ptosis may be caused by the following:

I. Lack of orbital volume

This is poorly defined group of unrelated disorders resulting in ptosis and caused by defects in posterior eyelid support, changes in ocular position, or counter effects of the protractor muscles. Loss of orbital volume seen in microphthalmos, phthisis bulbi, enucleation and traumatic fat atrophy resulting in sagging of support to the levator muscle and Whitenall's ligament, with associated ptosis.

Correction may require orbital soft tissue volume augmentation or repositioning rather than standard ptosis repair.

II. Ipsilateral hypotropia

Hypotropia and ptosis are frequently related because of both anatomic and physiological mechanisms. Correction of the strabismus also corrects the ptosis in this condition.

III. Blepharospasm

Blepharospasm, either primary or secondary to ocular surface irritation or iritis, can increase protractor force against the levator muscle, resulting in pseudoptosis.

- ☐ Brow ptosis
- ☐ Contralateral eyelid retraction

V. THERAPEUTIC APPROACH

i. Stabilize patient

The most important part in management in ptotic patient is precise etiology mentioned above. Complete ophthalmic examination must be done to identify the causes, severity and emergency state of the condition, especially in acquired ptosis.

ii. Empirical treatment

Once other treatable diseases have been ruled out, attention is directed at repair of the ptosis.

iii. Directed treatment

The most important criterion in selecting a successful surgical procedure is levator muscle function. The indication for ptosis surgery could be for either functional or cosmetic concern or both.

Functional surgical indications currently in common use include impaired visual acuity (obscured visual axis or causing high astigmatism), decreased peripheral vision, a compensatory chin-up backward head tilt and difficulty wearing a prosthesis in an anophthalmia socket.

In most cases with fair to good levator muscle function, ptosis can be repaired by with aponeurotic advancement or resection.

Mild ptosis with good levator function may be corrected by Muller's muscle-conjunctival resection.

In patients with very poor or no levator function, the only operation that will achieve adequate eyelid elevation is frontalis suspension. In this procedure, the eyelid is fixed to the frontalis muscle at the brow. A variety of suspension materials have been advocated. Autogenous facial lata has proven to give good results with minimal complications but requires secondary surgery on the leg for harvesting of the fascia.

A silicone rod has become very popular and has the advantage of not scarring into position, so that it can adjust indefinitely.

Whenever possible, all ptosis surgery should be performed under local anesthesia to allow more precise adjustment of eyelid height and contour.

iv. Monitoring

All the cases of ptosis surgery require closed follow up for possible complication of lagophthalmos such as severe dry eye and exposure keratopathy. Extensive lubricant eyedrops and gel are given for postoperative ptotic patient to prevent such complication mentions above.

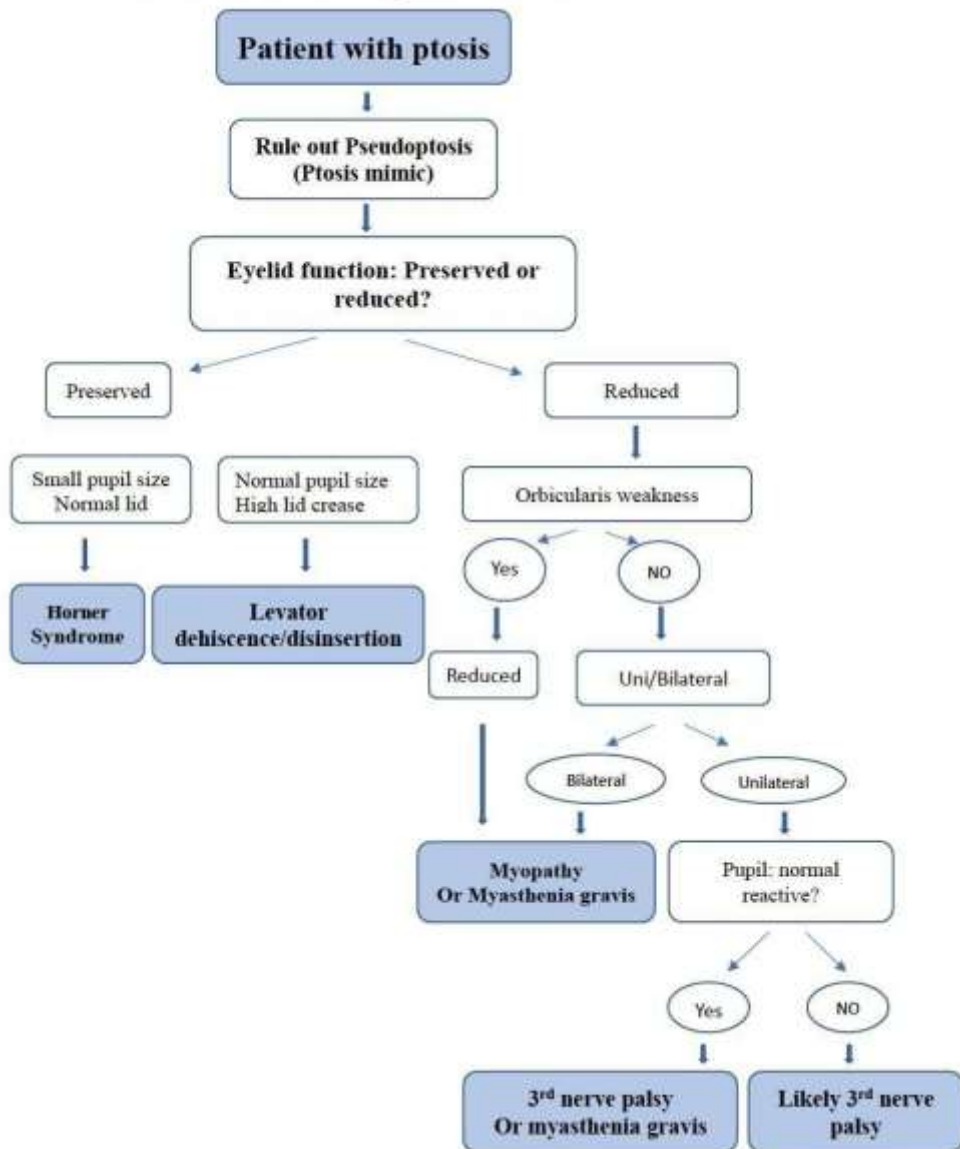
VI. COMPLICATION

Post operative complication of ptosis surgery included

- Under and overcorrection of ptosis
- Lagophthalmos and exposure keratopathy
- Asymmetry of eyelid crease
 - Poor eyelid contour
 - Wound infection

VII. ALGORITHM

Flowchart for the diagnosis of Ptosis



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CATARACT

Ou Vong Virak, Yin Vanlyda, Ny Povpronet, Sun Sarin

I. DEFINITION

A cataract is any opacity of the crystalline lens or its capsule leading to degradation of its optical quality, thereby causing decreased quality of vision.

II. ETIOLOGY

- Aging: the most common cause, where changes in the lens proteins and lens fiber cell structure.
- Ultraviolet radiation: exposure to UV light can accelerate the formation of cataracts.
- Diabetes: diabetic patients are at higher risk due to the osmotic stress from fluctuating blood glucose.
- Smoking: tobacco use significantly increases the risk of cataracts due to oxidative stress on the lens.
- Medications: certain medications, particularly corticosteroids and statins.
- Trauma: physical injury to the eye can lead to traumatic cataracts.
- Other health conditions: other systemic diseases such as hypertension and obesity.
- Genetic factors: hereditary genetic disorders can predispose individuals to congenital cataracts.

III. CLINICAL DIAGNOSIS

1- History

History taking is important for assessing the quality of the visual impairment due to this disability. Occupational and functional needs of the patients should be taken into consideration before planning for surgical intervention.

2- Ocular examination

- Visual acuity
 - With and without present corrections.
 - If spectacle is available, the power of the present correction should be recorded.
- External examination
 - Pressure over the sac to look for any regurgitation.

- If regurgitation is positive → syringing of naso-lacrimal duct.
- If duct is not free, with mucus or purulent discharge → dacryocystectomy or dacryocystorhinostomy is done and cataract surgery is done after one month.
- Eyelid examination
- Examination of ocular alignment and motility
- Assessment of pupillary shape, size and reaction
 - To determine the prognosis of surgical intervention.
- Slit-lamp biomicroscopy of the anterior segment
 - To examine the corneal endothelium for any guttata, for pupil pseudoexfoliation and the lens for any preoperative
- Dilated examination of the lens → reveal the extent of visual
 - To grade nuclear sclerosis and reveal any subluxation.
 - To measure amount of maximal dilation helps in planning the surgery.
- Dilated examination of the fundus
 - To exam the peripheral retina.
 - For the prognosis of the surgical intervention.

3- Investigations

- Measurement of intraocular pressure.
- Gonioscopy only if
 - Anterior chamber appears shallow
 - IOP is raised
 - Presence of any sign of secondary glaucoma (e.g. pseudoexfoliation)
 - Known cases of glaucoma
- Keratometry and A scan biometry
 - It should be performed in both eyes.
 - It should be repeated if needed for unusual powers of IOL.
 - SRK/T or appropriate formula should be used according to cases.
 - In case of scarred cornea and irregular surface of cornea, the K reading will not be possible → consider the other eye.
- Ultrasonogram of the posterior segment is media opacity.
- Blood pressure
- Screening for diabetes mellitus
- Physician fitness is mandatory for cardiac patients and those with advanced systemic problems.

- Patients who require general anesthesia need to undergo pre-anesthetic check-up.

IV. TREATMENT

4.1- Nonsurgical management

Patients with early cataract and all stages of nuclear cataracts that benefit by spectacles.

4.2- Indications for Surgery

The prime indication: An individual who is unable to carry out his/her desired activities due to dimness of vision for which cataract surgery is likely to restore the visual function.

The other reasons for cataract removal:

- Clinically significant anisometropia in the presence of a cataract to facilitate binocularity.
- Dense lens opacity is enough to evaluate and manage posterior segment conditions.
- Lens induced ocular inflammation
- Lens induced glaucoma

4.3- In-Patient

- Patients coming from a long distance
- Traumatic cataracts
- Pediatric patients requiring general anesthesia
- Mono ocular patients
- Cataracts with intraoperative complications

4.4- Referral criteria

- Cases needing specialized investigations before surgery (Specular microscopy)
- Cases not manageable by the facilities available
- Seeking a second opinion
- Complications needing specialized intervention (e.g. posterior vitrectomy)
- Detection of posterior segment disorder after cataract removal

4.5- Prognosis of surgery

It should be clearly explained to the patient in an understandable language.

4.6- Contra-indications for surgery

- Patients are not convinced about surgery.
- Patients have satisfactory functional vision either with or without visual aids.

- ☐ Cataract surgery is unlikely to improve vision.
- ☐ Patients are medically unfit.

4.7- Anesthesia

The planned mode of anesthesia should be decided after interaction between patient and the doctor. The process should be discussed with the patient so that he/she will know what to expect in terms of pain, discomfort, consciousness level, visual experiences, and complications.

- ☐ Retrobulbar or peribulbar anesthesia: for extracapsular cataract surgeries.
- ☐ Topical anesthesia: for clear corneal phacoemulsification.
- ☐ General anesthesia if needed for children and patients with medical, psychosocial, or surgical indications.

4.8- Infection Prophylaxis

- ☐ A thorough face washes before surgery is recommended.
- ☐ Using topical antibiotic eye drops before surgery.
- ☐ Preparing the periocular skin with 10% povidone iodine.
- ☐ Applying 5% povidone iodine to the conjunctival cul de sac.
- ☐ Careful sterile draping of the eyelid margins and eyelashes.
- ☐ A surgeon should stick to strict aseptic techniques of surgeries.
- ☐ Wound construction and closure should be meticulous.
- ☐ The surgery should end with a watertight incision.
- ☐ The surgeon should not hesitate to put sutures for security.
- ☐ Applying topical antibiotic eye drops after surgery.
- ☐ Taking care of personal hygiene of the patient.

4.9- Post Operative Medication

- ☐ Topical corticosteroids are in a tapered fashion for 4 weeks.
- ☐ Topical broad-spectrum antibiotics for 4 weeks.
- ☐ Cycloplegic drops, NSAID and anti-glaucoma medications according to ophthalmologist's clinical decision.

4.10- Surgical Techniques

Visual results can be improved by adhering to these common guidelines.

- ☐ A well-planned incision that minimizes surgically induced astigmatism
- ☐ Safe and secure incision
- ☐ Ensuring minimal loss of the corneal endothelial cells
- ☐ An appropriate posterior chamber IOL placed in the capsular bag

- Avoiding trauma to iris, and other ocular tissues

The preferred method to remove cataract

- Manual small incision cataract surgery (MSICS)
- Phacoemulsification (PE)

Steps of Manual Small-Incision Cataract Surgery

1. Superior rectus bridle suture
2. Conjunctival peritomy
3. Superior incision
 - About 1–1.5 mm behind the limbus.
 - The size is 5.5 to 7.5 mm of the incision depending on the size of the nucleus.
 - Configuration: straight, Chevron or Frown incision with side-port blade.
4. Sclerocorneal tunnel
 - A proper sclerocorneal tunnel is made with crescent blade
 - One third to half of the width of the thickness of sclera
 - Then an entry into the anterior chamber with keratome blade
5. Capsulotomy
 - Using a 26-gauge cystotome or Utrata's forceps
 - Trypan blue dye stains the anterior capsule for better visualization.
 - Three types
 - Continuous curvilinear capsulorhexis (CCC) for medium-grade nuclear sclerotic cataract
 - Can-opener capsulotomy for large nucleus as in brown cataract and cataract nigra
 - Envelope (linear) capsulotomy for Morgagnian, intumescent, or hypermature cataracts
6. Hydroprocedures
 - Using balanced salt solution (BSS) or ringer's lactate solution.
 - Hydrodissection: the dissection of the corticonuclear from the lens capsule.
 - Hydrodelineation: the separation of epinucleus from the nucleus.
7. Nucleus management
 - Nuclear prolapse out of the bag with the help of Sinsky hook.
 - Nucleus delivery out of the anterior chamber by hydroexpression, viscoexpression, Vectis-assisted delivery, sandwich technique, or the fish hook technique.
8. Irrigation and Aspiration (I/A)

- To remove the epinucleus and cortical matter from the capsular bag.
- 9. Intraocular lens (IOL) implantation: 6mm wide rigid IOL
- 10. Viscoelastic removal

Steps of Phacoemulsification

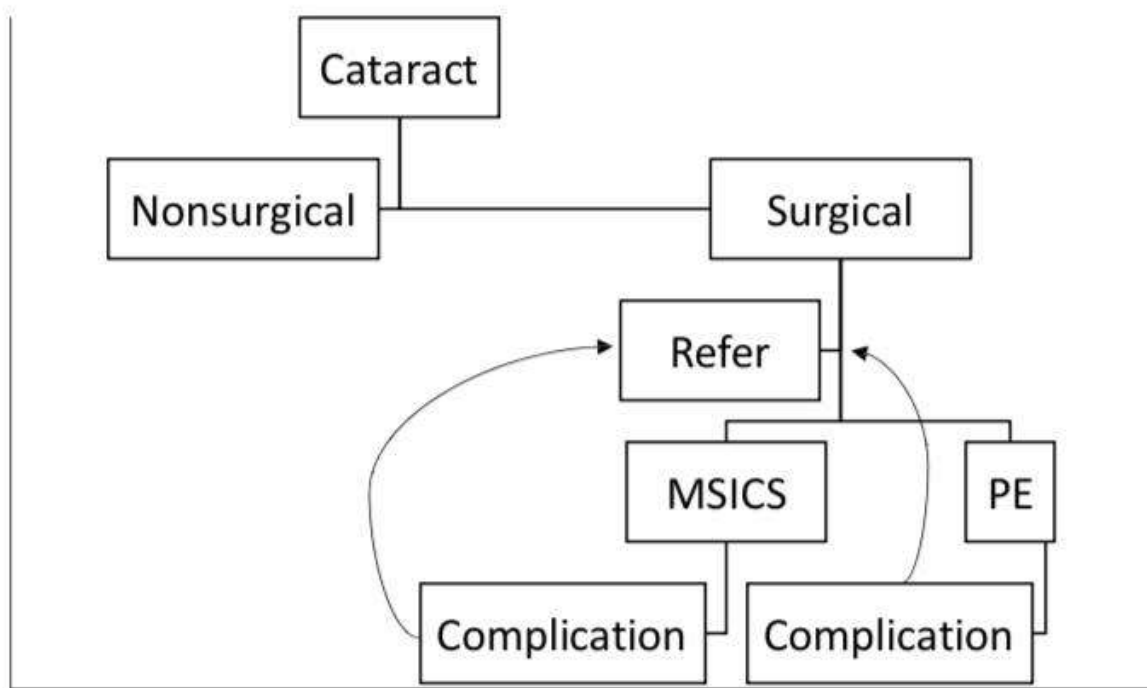
1. Side port incisions (Paracentesis)
 - Approximately 2 to 3 clock hours from the main precise corneal incision
2. Clear Corneal Incision
 - Temporally or superiorly approximately 2.2 to 3.2 mm wide
 - Large enough to accommodate the handpiece of phaco and IOL.
 - The incision can be uniplanar, biplanar, or tri-planar.
3. Continuous Curvilinear Capsulorrhexis (CCC)
 - The ideal size is 5 to 5.5 mm to accommodate a 6 mm optic of the IOL.
 - Performed with a bent 26 G needle the cystotome or rhexis forceps.
4. Hydrodissection
 - Using a 2 ml or 5 ml syringe with a 25-30 G cannula.
 - Separating the cortex from the capsule.
5. Hydrodelineation
 - To separate the endonucleus from the epinucleus and create a golden ring.
 - Indicated in posterior polar cataract.
6. Nuclear Rotation
 - It is easier to perform divide and conquer, stop and chop, and direct chop techniques of phaco.
 - If not proper, it will stress the zonules and result in zonular dialysis.
7. Techniques of Nucleus Break and Removal

The various steps: pre-phaco, sculpting, trenching, cracking, chopping, holding, and emulsification.
8. Irrigation and Aspiration (I/A)
 - To remove the epinucleus and cortical matter from the capsular bag.
 - For polishing the capsule.
9. IOL Implantation
 - The insertion of a foldable IOL with the help of IOL injectors.
10. Viscoelastic removal

V. COMPLICATION

- Posterior Capsule Rupture/Vitreous loss
- Cystoid Macular Edema
- Endophthalmitis
- Vitreous/Suprachoroidal Hemorrhage
- Retinal Tears/Detachment
- Lens Dislocation

VI. ALGORITHM



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CHEMICAL EYE INJURY

CHEA Piseth Dalin, KHOEM Namgech, SO Chamroeurn

I. CASE DEFINITION

Chemical Eye Injury (CEI) is an injury to the eye inducing chemical changes in the substance of contact that can cause mild to severe damage of the ocular annexa, ocular surface and anterior segment leading to visual threatening in one or both eyes. Chemical eye injury is an acute emergency which can threaten sight and life. These commonly occur at home or the workplace with the former being generally mild and the latter more severe and bilateral (Dua et al, 2020).

II. ETIOLOGY

The main causative agents are alkalis, acids and irritants or neutral agent like alcohols with alkalis being responsible for 60%.

Alkali

Alkali agents can penetrate tissues more quickly than acid given its lipophilic components. They can destruct the cornea stroma then secrete proteolytic enzymes that can further damage tissues (Madison et al., 2003; Breikreutz et al., 2009).

ALKALI	
Substance	Found in
Lime	Cement, mortar, plaster, whitewash
Ammonia	Cleaning agents, fertilizers, refrigerants
Lye	Drain cleaners, airbags
Magnesium Hydroxide	Firework sparklers, flares

Acids

Acids are generally less harmful than alkali substances. They coagulate, denature and precipitate proteins in the tissues they contact causing damage. Unlike alkali injuries this acts as a barrier to prevent further penetration (Breikreutz et al., 2009). However, hydrofluoric acid is an exception, where the fluoride ion rapidly penetrates the thickness of the cornea and causes significant anterior segment destruction (Greenhalgh et al., 2015).

ACID	
Substance	Found in
Citrus acid	Lemon or lime
Acetic acid	Vinegar, glacial acetic acid
Hydrofluoric acid	Glass polishing and mineral refining
Sulfurous acid	Bleach and refrigerant
Hydrochloric acid	Swimming pools
Sulfuric acid	Car Batteries

Other irritants such as alcohols and household detergents cause de-epithelialization of the ocular surface, which is the least severe of all chemical injuries and usually heals without visual consequences. Early recognition and treatment ensure the best possible outcome for visual potential and visual lost.

III. DIAGNOSTIC PROCEDURE

a. Clinical argument

Besides the nature of the agent, the duration of contact and surface area affected determine the depth of penetration and hence the severity of the damage. Depending on the degree of penetration, corneal epithelium, stroma, keratocytes and nerves can be affected with corresponding involvement of conjunctival epithelium, substantia propria, Tenon's capsule, episcleral and sclera. Further penetration can affect the iris, and lens. Damage to blood vessels at the limbus causes 'limbal ischemia' with loss of limbal stem cells; scleral and uveal ischemia contributes to serious damage. Damage to the ciliary body in the form of ischemia or distortion of the trabecular meshwork, due to shortening and shrinkage of the collagen fibrils, can lead to hypotony or elevation of the intraocular pressure.

b. Technical procedure

Classification of chemical burns of the eye

Dua et al. (2020) presented current two prognostic classifications used for grading CEI, namely Roper- Hall (modification of Balen) classification (Table 1) and Dua classification (Table 2).

Table 1 Roper-Hall classification of chemical eye injury.

Grade	Cornea	Limbus	Prognosis
I	Corneal epithelial damage	No limbal ischaemia	Good
II	Corneal haze, iris details visible	<1/3 limbal ischaemia	Good
III	Stromal haze, iris details obscured	1/3–1/2 limbal ischaemia	Guarded
IV	Opaque cornea, iris and pupil obscured	>1/2 limbal ischaemia	Poor

Table 2 Dua classification of chemical eye injury.

Grade	Limbal involvement (clock hours)	Conjunctival involvement (%)	Analogue scale	Prognosis
I	0	0	0/0%	Very good
II	≤3	≤30	0.1–3/1–30%	Good
III	>3–6	>30–50	3.1–6/30.1–50%	Good
IV	>6–9	>50–75	6.1–9/51–75%	Good - Guarded
V	>9 to <12	>75 to <100	9.1–11.9/75.1–99.9%	Guarded - Poor
VI	12 (total limbus)	100 (total conjunctiva)	12/100%	Very poor

According to McCulley's classification (McCulley, 1987), the clinical course of chemical eye injury (CEI) can be divided into three distinct stages:

- (1) Immediate phase (<7 days): direct result of the injury-causing tissue necrosis and sloughing
- (2) Intermediate phase (1–3 weeks): host healing and inflammatory responses (secondary to the release of a milieu of chemotactic and inflammatory cytokines) leading to a range of clinical changes such as corneal ulceration, melt, vessel recanalization and hemorrhages, conjunctivoplasties and pannus formation.
- (3) Late phase (>3 weeks): changes secondary to host repair and regeneration or lack thereof are manifest, including fibrovascular pannus, deep corneal vascularization, dry eye, neurotrophic keratopathy, persistent epithelial defect, and/or perforation.

IV. THERAPEUTIC APPROACH

a. Stabilize patient

Ocular pH measurement is essential in the evaluation of a chemical burn. Evaluate all patients with alkali injuries to the face for tracheal and esophageal burns, which are potentially life-threatening. Ocular burns can be associated with surrounding facial burns or more generalized burns which can be serious if the agent has been ingested or inhaled with gastrointestinal and respiratory symptoms, including laryngeal oedema, that can be life threatening. Cardiac arrhythmias precipitated by hypocalcemia and consequent hyperkalemia can be fatal. (Bizrah et al., 2019)

b. Laboratory test

No additional laboratory studies are warranted in cases of isolated ocular injury; additional studies should be considered if warranted by coexisting injuries. Hydrofluoric acid burns can cause significant hypocalcemia. Consider checking the calcium level for

burns that are not limited to the eye. Systemic toxicity can occur secondary to depletion of calcium and magnesium which are essential for cellular and enzymatic function.

Before irrigation, pH should be measured in both eyes even when one eye is affected as the unaffected fellow eye may serve as a “normal” control for the affected eye (Conner & Severn, 2009). The accuracy of the litmus paper strips has been questioned (Monaghan et al., 2019), but it has equally been argued that assessment of pH with the litmus paper is a crude but practical means of ascertaining whether the agent was an acid or alkali, but more so to know whether it is normal or not, and to serve as a baseline from which a change can be measured (Said & Dua, 2019).

c. Imaging study

Complete slit lamp examination and thorough assessment Once the eyes are thoroughly irrigated and pH is neutralized, a systematic and comprehensive assessment needs to be performed to fully evaluate the severity of CEI. This could be achieved through systematic examination in an anatomical order (e.g. from the outside towards the inside and from the front towards the back of the eye), starting from the face, ocular adnexa, lid margin, conjunctiva, limbus, cornea, intraocular pressure, iris, pupil, lens and retina. Fluorescein staining provides comprehensive view of ocular surface damage for grading and classification and must be performed in all cases.

In-Vivo Confocal Microscopic (IVCM) examination of conjunctival epithelium on cornea following chemical burns can show the central corneal epithelium has characteristic corneal epithelial phenotype appearing as dark polygonal spaces with a bright border while the limbus shows classical signs of stem cell deficiency in the form of hyper-reflective conjunctival cells, goblet cells, vessels and cystic spaces (Mastropasqua et al., 2020).

V. DIFFERENTIAL DIAGNOSIS

- [Corneal abrasion](#)
- Cornea ulcer
- [Fluoride Toxicity](#)
- [Ultraviolet Keratitis](#)

VI. THERAPEUTIC APPROACH

Treatment given and the time from injury to initiation of treatment influence the depth of penetration of the chemical agent, involvement of the extraocular and intraocular structures, the wound healing and repair and regeneration response.

iii. Pre-ophthalmic management of chemical burns:

To prevent further damage, profuse and prolong eye wash or irrigation is crucial to limit duration of chemical exposure, removing substance, restoring physiological pH (at least 30 minute); Diphoterine is amphoteric (can act as acid or alkali, i.e. neutralise

both to normal, irrigation should be stopped for ~5 min and the pH tested again. If the pH has changed, irrigation should be recommenced.

- Isotonic solution: Normal saline (NS), Lactated Ringer's, phosphate buffer solution, balance saline solution (BSS)
- Hypertonic Solution (Diphoterine): increase osmotic pressure to mobilize water and dissolved corrosives out of corneal tissues
- Precaution must be made with hypotonic solution (tap water): Increase water influx into the cornea, leading to further diffusion of corrosive material into the eye and increased corneal edema.

iv. Empirical treatment

Guided by this natural course of disease, the management can be broadly divided into early or acute and late stages (Baradaran-Rafii et al., 2017)

Management at early or acute stage (<4–6 weeks):

Control of acute inflammatory reaction Topical corticosteroids are the mainstay of treatment for controlling the acute inflammatory reaction, reduce risk of cornea opacity following CEI (Brodovsky et al., 2000; Noecker R, 2001; Davis et al., 1997). The general consensus is to taper the dose or frequency of topical steroids after 7–10 days. Anti- proteases play key roles in reducing and/or preventing corneal ulceration following CEI (Brodovsky et al., 2000; Bian F et al., 2016; Seedor et al., 1987; Pfister et al., 1981).

Potent corticosteroids: (7-10 days)

- Dexamethasone 0.1% (dose depending on grade)
- Prednisolone acetate 1%. (dose depending on grade) Anti-proteases:
- Oral tetracycline 250mg (QID)
- Topical tetracycline 1% suspension or 3% ointment (QID)
- Oral doxycycline 100mg (BD)
- Oral minocycline 100 mg BD
- Sodium citrate 10%
- Acetylcysteine
- EDTA

Management at late stage (>6 weeks):

Multidisciplinary approach to treatment of sequelae and complications that require reconstruction and restoration namely the lids and adnexa, the eye pressure, cataract, the ocular surface and the cornea (Dua et al., 2020; Mastropasqua et al., 2020).

- Lid repair and reconstruction if risk of exposure presents.
- Skin grafts and oral mucous membrane grafts for necrosis eyelid tissue lost.
- Tarsorrhaphy to promote surface healing and prevent exposure.

- Medical and surgical treatment of secondary glaucoma.
- A tube shunt in sever case subject to conjunctival health.
- Avoid Cyclo-destructive procedures (lead to increase inflammation)
- Urgent cataract extraction without IOL (in phacomorphic glaucoma) → Plan a secondary lens implant later.
- Defer planned cataract surgery until surface reconstruction provides a clear view.
- Ocular surface repair, restoration, regeneration and reconstruction include medical and surgical options, individually or usually in combination. In sever case, debridement, amniotic membrane transplantation, an osteo-odonto keratoprosthesis or a Boston type 2 keratoprosthesis are indicated. Dry eye disease
 - Conjunctival adhesions/symblepharon
 - Non-healing or persistent epithelial defects
 - Neurotrophic keratopathy (NK)
 - Limbal stem cell deficiency
 - Corneal scarring
 - Cornea decompensation
 - Cornea thinning, melting and perforation

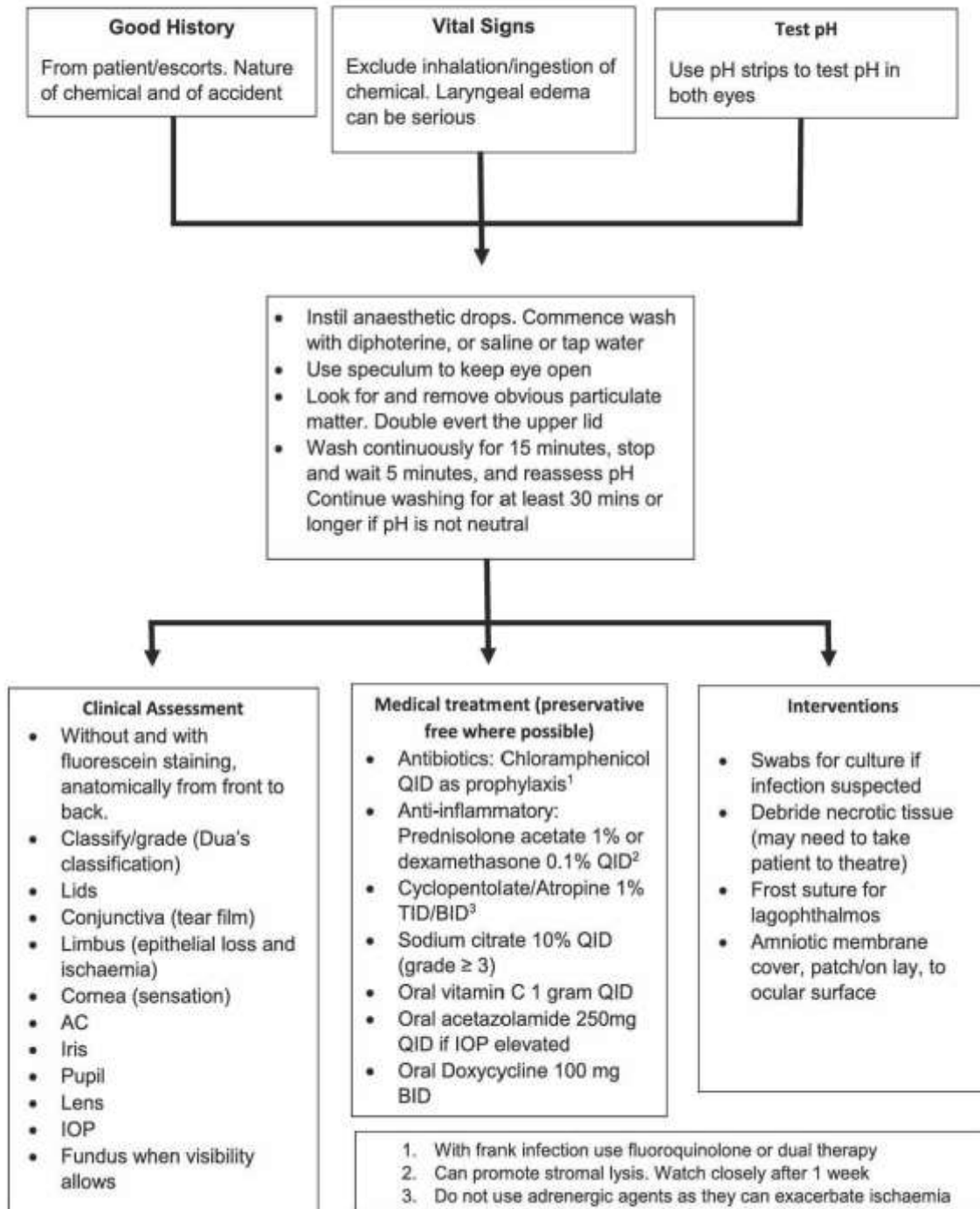
VII. ALGORITHM

Proposed paradigm for the initial treatment of chemical injury based on the Roper-Hall grad of injury (Colby, K. 2010).

Grade I	Grade II	Grade III	Grade IV
- Topical AB (QID) - Prednisolone acetate 1% (QID) - PF artificial tear as needed - Cycloplegic (TID) for pain (Cyclopentolate)	- Topical AB (QID) - Prednisolone acetate 1% (QID) - PF artificial tear as needed - Cycloplegic (TID) for pain (Atropine) - Oral Vit. C 2grams four times a day - Doxycycline 100mg twice daily (Avoid in children) - Sodium Ascorbate drops (10%) hourly while awake - Debridement of necrotizing tissues	- As for grade II - Consider AMT/Prokera placement (Ideally 1st week of injury) - Covering cornea, bulbar conjunctiva, palpebral conjunctiva suturing to lids	- As for grade III - Tenoplasty for significant necrosis to reestablish limbal vascularity - AMT for ocular surface damage

Dua et al. (2020) proposed a protocol for acute stage management of chemical eye injury in the eye casualty.

Emergency Management of Chemical Eye Injury in Eye Casualty



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CONGENITAL CATARACT

Morm Pheakdey, Reth Chongchiv, Teng Vannaroit, Sim Sreylin, Prof. Mar Amarin

I. DEFINITION

Congenital cataract is an opacity of crystalline lens, are of importance in children worldwide. The incidence varies, but in the UK adjusted cumulative incidence at age 1 year is 2.49/10000, increasing to 3.46/10000 by age 15 years. Bilateral cataract is more common than unilateral.

II. ETIOLOGY

i. Bilateral

- a. An etiology can be established about 70% of children, The most common etiology in Europe and the United state are mutation in genes for lens proteins.
- b. Less frequently cataract can be associated with aniridia, Trisomy 21, Lowe syndrome, Hallermann-Streiff-Francois syndrome
- c. Systemic associations include metabolic disorder such as galactosemia, Wilson disease, hypocalcemia and diabetes.
- d. Intrauterine infection including rubella, herpes simplex, toxoplasmosis, varicella and syphilis are another cause.
- e. Acquiring cataract may arise secondary to the used of corticosteroids, or external beam radiation therapy.

ii. Unilateral

- a. In contrast, most unilateral cataracts are not inherited or associated with a systemic disease and are unknown etiology although they do not rule out the possibility of an associated systemic disease.
- b. They are usually the result of local dysgenesis and may be associated with other ocular dysgenesis such as persistent fetal vasculature (PFV), posterior lenticonus or lentiglobus.
- c. Trauma is the most common cause of an acquired unilateral cataract in child. On occasion, there is no known history of trauma, but iris sphincter tear or corneal scar can help to establish a traumatic etiology.

III. DIAGNOSIS

i. History

- a. A family history of childhood cataracts, exposure to corticosteroids or radiation therapy, and ocular trauma.
- b. Comorbidities may suggest certain systemic diseases such as diarrhea with cerbrotendinous xanthomatosis.
- c. The age the child was first noted to squint or to have leukocoria should be noted.

- d. In children with bilateral cataracts, the parents should be asked when they first noted abnormal visual behavior or nystagmus.
- e. Reviewing photographs of the child is particularly helpful in children with unilateral cataract because the abnormal red reflex can be compared to the photographs are taken indoors with a flash and child looking slightly eccentrically.
- ii. Ocular examination
 - a. Visual acuity:
 - 1. The visual acuity of older children should be assessed using optotypes.
 - 2. In young children, the ability to fix and follow with each eye should be assessed.
 - 3. Preferential looking techniques can help to assess visual acuity in pre-verbal children.
 - b. Slit-lamp examination:
 - 1. A slit-lamp examination should be performed to evaluate the cornea, iris, and morphology of the cataract.
 - 2. In young children, a hand-held slit-lamp is often better tolerated.
 - 3. Slit-lamp examination of other family members can be helpful in establishing whether there is hereditary basis for the cataracts.
- iii. Laboratory work up
 - a. A systemic investigation is not usually indicated for
 - 1. children with unilateral cataracts
 - 2. If child has family history of isolated ocular problem.
 - b. But it can be helpful in selected children with bilateral cataracts.
 - 1. Infectious diseases: TORCH and varicella titer & VDRL
 - 2. Metabolic diseases:
 - a. Disorder of galactose metabolism: Urine for reducing substances, galactose-1-phosphate uridylyltransferase, galactokinase
 - b. Urine amino acids test: Lowe syndrome
 - c. Serum calcium: ↓ in hypoparathyroidism

IV. MANAGEMENT

iv. Non-surgical

- a. Observe: incomplete bilateral cataracts should only be removed after a careful assessment of the child's visual behavior. The density rather than the size of the lenticular opacity correlates best with the visual prognosis for the children with bilaterally incomplete cataracts. Nuclear cataract, although smaller in size than

lamellar cataracts, have a worse visual prognosis.

- b. Patching: If the incomplete cataract(s) is unilateral or asymmetric, part time patching therapy of the normal/better eye may be beneficial to improve or maintain vision in the most affected eye.
- c. Dilatation: Chronic dilatation of the pupil may be helpful in some children with bilateral incomplete cataracts, particularly if the lens opacities are central and child has miotics pupils. If a cycloplegic drop is used to achieve long-term dilation, bifocals should be used to correct any hypertropia and to facilitate near vision.

v. Surgical

- a. Total cataract
- b. Unilateral cataract
- c. Asymmetric cataract
- d. Secondary effect: Nystagmus, Strabismus
- e. Symptoms: VA worse than 6/18, photophobia
- f. The More anterior the lens opacity is, the less vision threatening.
- g. The more posterior and central the more vision threatening.

vi. Intraocular lens implantation: Controversial

- a. Single-piece acrylic foldable IOL or PMMA
- b. Unilateral: > 2years
- c. Bilateral: Leave aphakia until the child is older (7-8years old)
- d. Target refraction: 2 approaches
 - 1. Emmetropia: especially in unilateral → Myopia shift → require secondary procedure
 - 2. Undercorrection → require hyperopic spectacles/contact lens

vii. Surgical techniques

- a. A clear cornea, limbal or scleral incision should be created superiorly to introduce a 23- or 25 -gauge vitrector and a lateral incision should be created at the limbus for inserting an infusion cannula into the anterior chamber.
- b. The vitrectorhexis is initiated by making an opening in the center of the anterior capsule with the port of the vitrector facing downwards.
- c. The capsulorhexis should be performed with anterior chamber filled with an ophthalmic viscoelastic device.
- d. The vitrectorhexis is then gradually enlarges to 5-6mm by moving the vitrector away from the center of the lens.
- e. . If cataracts extraction with intraocular lens (IOL) implantation is planned, a continuous curvilinear capsulorhexis is preferable since the capsular edge created using this technique is less likely to tear during IOL implantation.
- f. In almost all cases, the lens nucleus and cortex of a pediatric cataract can then

aspirate using a vitrector.

- g. A foldable IOL is preferred in children to minimize the width of the entry site. A smaller entry site reduces the risk of iris prolapse intraoperatively and wound dehiscence postoperatively.
- h. When implanting an IOL in the eye of the young child, primary posterior capsulotomy and vitrectomy should be performed.
- i. The posterior capsulotomy can be performed using limbal or a pars plana approach.
 - 1. The limbal approach has the advantage that the same incision site used for cataract surgery can be used, but it makes it more difficult to implant the IOL in the bag. After completing the cataract surgery, the limbal incision site should be closed with 9-0 or 10-0 absorbable sutures
 - 2. A primary capsulotomy can also be performed through the pars plana after implanting the IOL. This simplifies IOL implantation into the capsular bag, but necessitates a second entry site in the pars plana. The pars plana site should be closed with 8-0 absorbable suture.

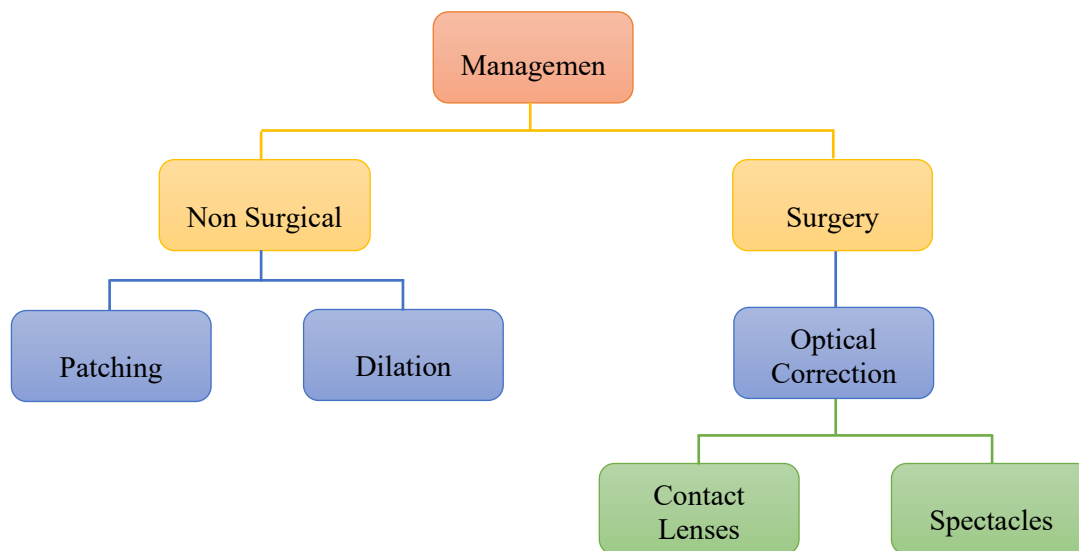
viii. Post operative care:

- a. Subconjunctival triamcinolone injection in selected cases.
- b. Topical antibiotic and Topical steroid.
- c. Antibiotic combine steroid ointment
- d. Topical atropine 1% for 1 week
- e. Cycloplegic refraction 1-2 weeks post operatively
- f. Visual rehabilitation:
 - 1. Aphakic contact lens
 - 2. Glasses/bifocal glasses
 - 3. Patching
- g. Intraocular pressure

V. COMPLICATION

- ix. Amblyopia
- x. Visual axis opacities
- xi. Glaucoma
- xii. Strabismus
- xiii. IOL complication
- xiv. Irregular pupil
- xv. Heterochromia iridis
- xvi. Endophthalmitis
- xvii. Retinal hemorrhage and detachment
- xviii. Cystoid macular edema
- xix. Corneal edema

VI. ALGORITHM



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CONGENITAL NASOLACRIMAL DUCT OBSTRUCTION

Morm Pheakdey, Sim Sreylin, Mar Amarin

I. DEFINITION:

Congenital nasolacrimal duct obstruction (CNLDO) is a common cause of epiphora in children, with an incidence ranging from 1.2% to 30%. It is thought to result from incomplete embryonic nasolacrimal duct canalization.

II. ETIOLOGY:

The etiology of congenital nasolacrimal duct obstruction is most commonly a membranous obstruction at the valve of Hasner at the distal end of the nasolacrimal duct that has delayed involution.

1. Systemic Associations:

- ☐ Down syndrome
- ☐ Ectrodactyly, ectodermal dysplasia, clefting (EEC)
- ☐ Branchio-oculo-facial syndrome
- ☐ Craniometaphyseal or craniodiaphysal dysplasia
- ☐ LADD syndrome (lacrimo-auriculo-dento-digital)
- ☐ CHARGE syndrome (coloboma of the eye, heart defects, atresia of the nasal choanae, retardation of growth and/or development, genital and/or urinary abnormalities, and ear abnormalities and deafness)
- ☐ Goldenhar syndrome

III. DIAGNOSIS:

A. Signs and symptoms:

- ☐ A history of tearing, mucous discharge and epiphora of one or both eyes is typical with concordant increased tear lake, mucous or mucopurulent discharge, and epiphora found on exam.
- ☐ When pressure is applied over the lacrimal sac, there may be reflux of mucoid or mucopurulent material from the punctum.
- ☐ The periocular skin may be chapped from continual exposure to tears and discharge.
- ☐ The signs and symptoms are usually worse with a concurrent upper respiratory infection.

B. Diagnostic procedure:

- A fluorescein dye disappearance test can be helpful in confirming the diagnosis of nasolacrimal duct obstruction.
- A drop of 1% fluorescein is instilled into the eyes or introduced on a moistened pledget.
- The disappearance of dye from the tear film after 5 minutes is observed.
- Retained dye in a thickened tear strip is diagnostic of an obstruction.
- The test is most useful if the disease is unilateral, and the findings of the affected eye can be compared to those of the normal eye.

IV. DIFFERENTIAL DIAGNOSIS:

- 1) Acquired nasolacrimal duct obstruction
- 2) Acute conjunctivitis
- 3) Congenital anomalies of the upper lacrimal drainage system
- 4) Epiblepharon
- 5) Trichiasis
- 6) Dacryoceles
- 7) Obstruction of the inferior meatus due to nasal structures.

V. MANAGEMENT:

A. Nonsurgical:

- The management of simple CNLDO is conservative, including lacrimal sac massage or compression up to the age of 1 year.
- The aim of this treatment is to increase the hydrostatic pressure within the lacrimal sac in order to overcome the usually thin perforable membranous obstruction at the distal end of NLD.
- Cleansing the lids and lashes with cooled boiled water

B. Surgical:

- 1) Probing after non-resolving with sac massage at or after age of 1 year
 - The principle of probing is to pass a rigid probe into the lacrimal drainage system directed towards the distal end of NLD with the aim of overcoming the membranous obstruction.
 - Procedure:
 - Each punctum is dilated using a Nettleship dilator.
 - A probe of appropriate size (size “0” or “1” at 1 year of age) is passed gently along the vertical (preferably upper) canaliculus and then orientated horizontally while applying lateral counter-traction on the lid.

- The probe is then directed into the lacrimal sac until it touches its medial wall, and a hard stop is felt. The probe is then rotated 90°, pointing downward and slightly
- laterally and is gently advanced down the nasolacrimal duct until the
- membranous obstruction at its distal end can be overcome with use of minimal force.
- The importance of endoscopic guidance during the entire procedure cannot be overemphasized and is preferable, where available.
- Alternatively, some surgeons like to insert a second metal instrument such as a Guibor guide along the floor of the nose, in order to feel the contact of metal on metal as the probe emerges under the inferior turbinate to touch this instrument.
- However, false passages would give a false positive result and hence not very reliable.
- Fluorescein-stained normal saline, injected via the same canaliculus can be retrieved with suction from the nose to confirm nasolacrimal duct patency.

2) What to do if probing fails

a. Intubation

- ☐ The nose has been prepared with decongestant
- ☐ Nasal endoscopic guidance is used to view the inferior meatus
- ☐ Tubes come with a metal introducer
- ☐ Each end is placed through the system via the canaliculi, into the sac, and down the nasolacrimal duct into inferior meatus. They should be retrieved under direct visualization
- ☐ Optimal time to leave tubes in place: 3-6 months

b. Ballon catheter dilatation

- ☐ Is an alternative to intubation
- ☐ It has a high success rate, similar to that of intubation, but is significantly more expensive

c. Dacryocystorhinostomy (DCR)

- ☐ Children rarely require a DCR
- ☐ Indication:
 - Persistent epiphora despite probing and intubation
 - Complex congenital abnormalities of the lacrimal outflow apparatus (particularly involving the upper nasolacrimal duct)

- Acquired disease usually caused by infection or trauma

VI. COMPLICATION:

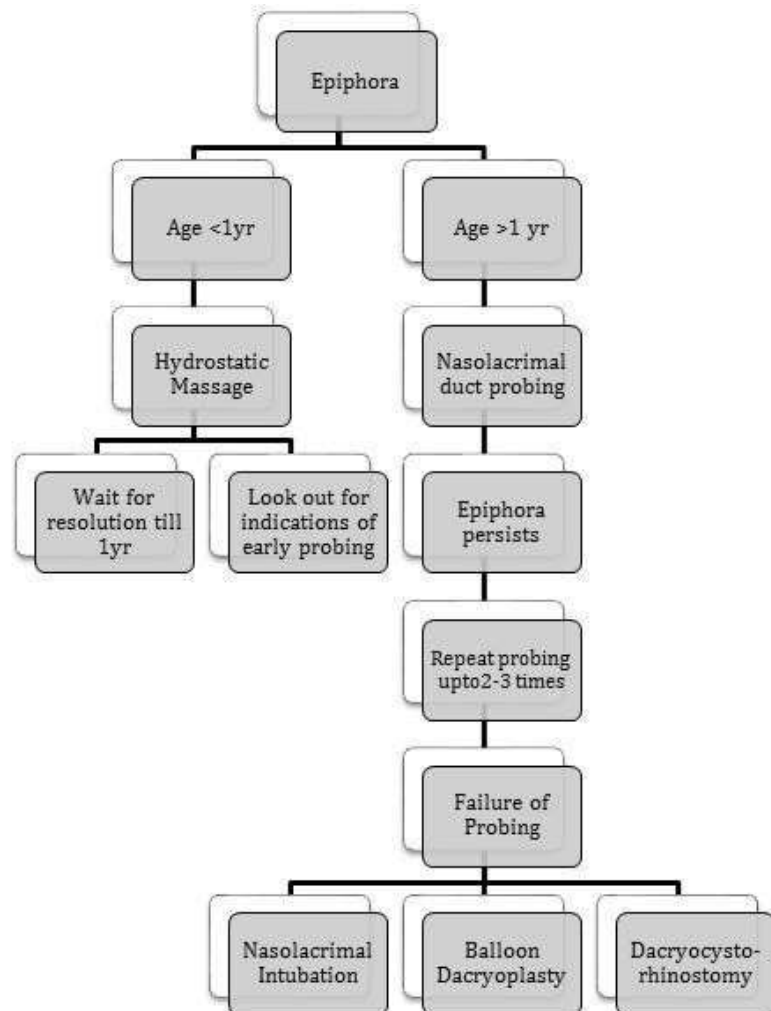
A. Probing

- 1) False passage of the duct
- 2) Nasal bleeding
- 3) Restenosis of the duct

B. Intubation

- 1) Cheese-wiring through canaliculi
- 2) Dislocation superiorly or inferiorly
- 3) Infection
- 4) Epistaxis
- 5) Sinusitis
- 6) Granulomatous reaction
- 7) Scarring for drainage system
- 8) Ocular surface can be affected by conjunctival irritation and corneal abrasion

VII. ALGORITHM:



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CORNEA WOUND LACERATION

Hun Tithsya, Tor Remy

I. CASE DEFINITION:

Corneal laceration is a common ocular injury due to direct or indirect trauma [1]. The laceration could be either partial or full thickness. Prompt management is necessary to prevent the risk of infection, restore ocular integrity, and minimize visual loss [2].

II. ETIOLOGY:

Blunt and sharp objects with high velocity are the general causes of corneal laceration [3]. Young age and male sex are the risk factors of corneal laceration [3]. The reason could be due to the increased activities in this population with blunt and sharp objects.

III. DIAGNOSIS PROCEDURE:

A. History:

- Past history of the location of accident, site of injury, objects of injury, and time of injury should be documented.
- Past medical history: immune status, diabetic mellitus, hypertension, previous tetanus vaccination.

B. Sign and Symptoms:

- Visual Acuity
- Intraocular Pressure
- Ocular Motility

C. Technical Procedure:

a. Slit Lamp Examination:

- Eyelid: swelling, laceration, foreign body
- Conjunctiva: chemosis, subconjunctival hemorrhage, laceration, foreign body
- Cornea:

- Location of laceration: central, paracentral, peripheral, total
- Size of laceration: vertical and horizontal length of laceration
- Characteristic of laceration
- Depth of laceration: partial vs full-thickness
- Leakage of aqueous: fluorescein should be used to confirm the flow of aqueous humor (Seidel's test)
- Sclera: associated laceration
- Anterior Chamber: flat / form anterior chamber, hyphema, fibrin formation.
- Iris: prolapse, incarcerated, dialysis in clock hour
- Crystallin Lens: anterior capsule integrity, presence of lens materials, cataract formation, phacodonesis in clock hour, subluxation / dislocation
- Vitreous: prolapse of vitreous into anterior chamber or extraocular, hemorrhage, opacity, foreign body
- Retina: foreign body, retinal breaks, retinal detachment, choroidal effusion
- b. Imaging Study:
 - B-scan: vitreous opacity, foreign body, retinal detachment
 - X-ray and CT-scan: foreign body, bone fracture
- c. Laboratory
 - Full blood count
 - PT / PTT
 - Renal and Liver Function
 - Infectious disease: HIV, Hepatitis B and C

IV. THERAPEUTIC APPROACH [4]:

A. Primary Wound Repair:

- a. Pre-operative consideration:
 - Corneal laceration should be promptly sutured (within 24 hours) to reduce the risk of infection, to restore ocular integrity, minimizing astigmatism and visual loss.
 - Care should be taken regarding the risk of intraocular foreign body such as trauma with small objects, iris defects, break of anterior capsular of lens, and presence of lens materials.
 - In case of dirty wound, suspects of intraocular foreign body, break of anterior capsule of lens, duration of more than 24 hours, intravitreal antibiotic prophylaxis should be initiated.

- Partial thickness laceration could be managed medically

b. Intraoperation:

- General anesthesia or regional block should be considered
- Corneal wound should be thoroughly cleaned prior to the repair
- A nylon 10.0 is preferred to prevent further scarring formation
- Suture:
 - o Suture should be placed with equal length on each side of the wound
 - o Suture depth should be at 90% of corneal thickness
 - o Suture knot should be buried as far as possible from visual axis
- Antibiotic injection:

Presence of anterior capsule rupture, wound duration of more than 24 hours, and the presence of intraocular foreign body should indicate the need for intravitreal antibiotic (Ceftriaxone 2.25mg/0.1ml)

B. Post-ops consideration:

- Topical antibiotic and topical steroid should be given hourly
- Antiglaucoma medications: in case of high intraocular pressure
- Oral antibiotic
- Oral analgesic
- Tetanus vaccination / booster should be considered in case of dirty wound or metallic origins
- Close follow-up

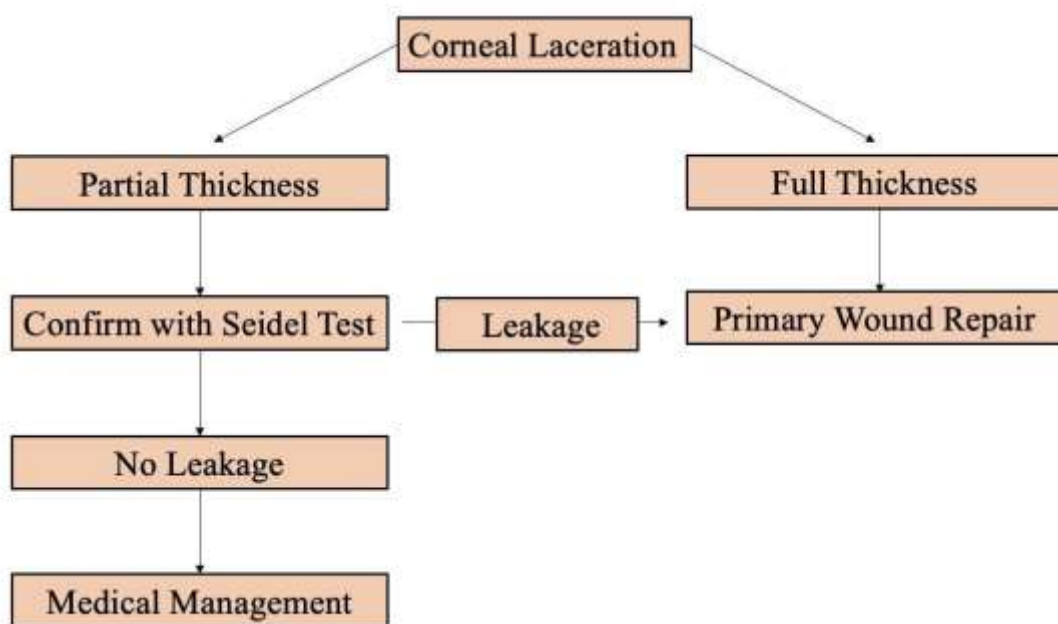
C. Follow up:

- Visual Acuity
- Intraocular Pressure
- Wound sealing
- Suture complication: loose, reaction, infection
- Anterior chamber reaction: cells, hyphema, hypopyon
- Vitreous: opacity, hemorrhage
- Retina: retinal breaks, retinal detachment

V. COMPLICATION [5]:

- Wound leakage and anterior chamber collapse
- Corneal ulcer
- Endophthalmitis
- Cataract formation
- Glaucoma: angle recess, angle-closure glaucoma, lens-related glaucoma
- Retinal detachment
- Corneal scar
- Irregular astigmatism

VI. ALGORITHM:



VII. REFERENCE:

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CORNEAL ULCER

Hun Tithsya, Tor Remy

I. CASE DEFINITION:

Corneal ulcer is defined as the epithelium defect with the loss of underlying stromal tissue [1]. Many of the cause of corneal ulcer, such as infection, autoimmune, inflammation, and trauma, were identified [2]. As one of the most devastating, sight-threatening ocular disease, corneal ulcer is regarded as an ocular emergency [1][3].

II. ETIOLOGY:

The etiology of corneal ulcer includes microbial infection, ocular trauma, inflammatory ocular surface disease, autoimmune [1][2][4]. One of the most common causes of visit is infection. Bacterial, fungal, and parasite keratitis would damage cornea tissues causing various symptoms, corneal scarring, corneal thinning and perforation [5].

III. DIAGNOSIS PROCEDURE:

A. History:

◆ Past ocular history:

- Trauma: object of injury such as vegetative matter
- Previous ocular surgery
- Previous ocular disease: ocular surface inflammation, herpes infection
- Previous medication: topical corticosteroid

◆ Past medical history:

- Systemic disease: diabetes mellitus, hypertension, rheumatoid arthritis
- Immune status: immunocompromised, immunocompetent

B. Technical Procedure:

a. Slit Lamp Examination:

- ☐ Eyelid: swelling
- ☐ Conjunctiva: degree of hyperemia, type of discharge, chemosis
- ☐ Cornea:
 - Location of lesion: central, paracentral, peripheral, total
 - Thickness: compared defect area with healthy area
 - Epithelium defect: size, edge of defect: rolled, sharp, edematous

- Characteristics of stromal infiltrate: purulent, dry, smooth or feathery border, presence of satellite lesion
- Depth of Infiltrate: superficial deep
- Descemet fold
- Endothelium: Keratic precipitate, endothelial plaque
- Presence of melting, descemetocoele, perforation
 - ☐ Anterior chamber: cells, hyphema, hypopyon
 - ☐ Iris: iris defect
 - ☐ Crystallin lens: posterior synechiae, cataract formation
 - ☐ Vitreous: vitreous cells, vitreous opacity
 - ☐ Retina: presence of retinitis
- b. Laboratory Finding:
 - ☐ Microbial culture:
 - Corneal scraping should be initiated 24 hours after antibiotic cessation
 - Blood agar, chocolate agar, Sabouraud's agar
 - ☐ Serology for infectious and autoimmune disease

IV. THERAPEUTIC APPROACH:

A. Conjunctival hooding:

Conjunctival hooding is an ocular surgical technique in which the vascularized conjunctiva is introduced on the cornea to manage various corneal diseases such as perforated corneal ulcer, trauma, and inflammation [6]. Since the 20th century conjunctival hooding has been modified from total to partial, including conjunctival bridge, monopedicle conjunctival flap, and advancement flap [7].

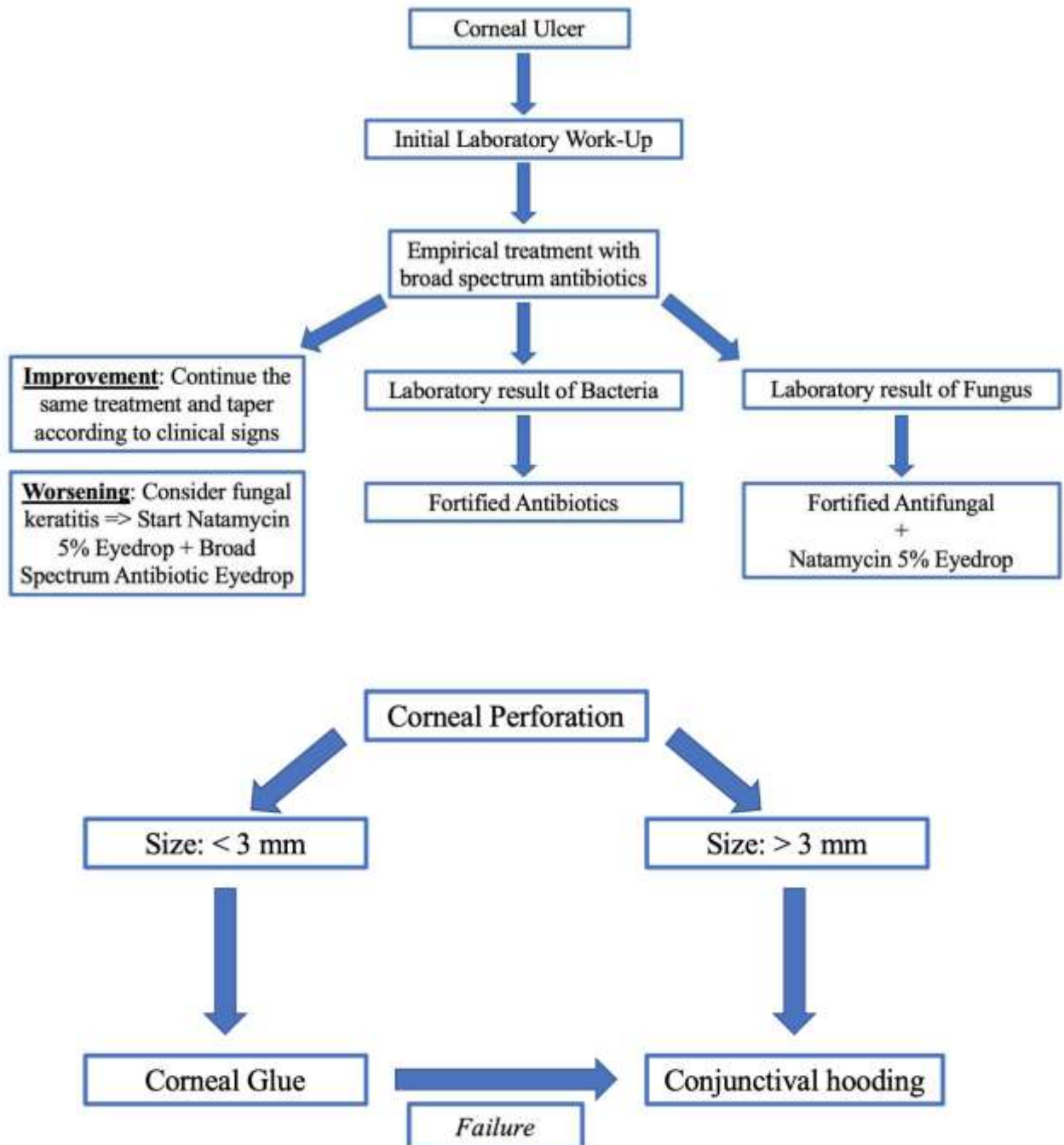
B. Indication of conjunctival hooding:

- Impending perforation of corneal ulcer
- Perforated corneal ulcer of >3 mm
- Traumatic corneal perforation of >3 mm
- Ocular surface inflammation
- Bullous keratopathy
- Corneal dystrophies

V. COMPLICATION:

- Loose suture and graft retraction
- Graft melting
- Conjunctival granuloma

VI. ALGORITHM:



VII. REFERENCE:

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DIABETIC RETINOPATHY

Kong Piseth, Un Leng and Retina Team

I. DEFINITION:

Diabetic retinopathy: chronic progressive, potentially sight-threatening disease of the retinal microvasculature associated with the prolonged hyperglycemia and other conditions linked to diabetes mellitus such as hypertension.

II. RISK FACTORS:

- Duration of diabetes (>30 years old incidence DR after 10 years is 50%, <30 years old incidence DR after 10 years is 90%)
- Poor control diabetes, pregnancy, hypertension, cardiovascular disease, renal impairment, smoking, cataract surgery, obesity and anemia

III. CLINICAL SIGNS:

Non proliferative DR		Proliferative DR	
Mild	Microaneurysm only	PDR with HRC:	1. NVD $\geq$ 1/3 of DD. 2. less extensive NVD, if vitreous or preretinal hemorrhage is present. 3. NVE $\geq$ half disc area, if vitreous or preretinal hemorrhage is present
Moderate	Microaneurysm, HEX, CWS		
Severe (4: 2: 1 Rule)	☐ Intraretinal he in 4 quadrants ☐ Venous beading in 2 quadrants ☐ IRMA 1 quadrant	Advanced PDR	1. Extensive vitreous hemorrhage precluding grading. 2. Retinal detachment involving the macula. 3. phthisis bulbi.

❖ Clinically Significant Macular Edema

1. Retinal thickening at or within 500 μ m of the fovea
2. Hard exudates at or within 500 μ m of the fovea if associated with adjacent retinal thickening.
3. Retinal thickening one disc area (1500 μ m) or larger, any part of which is within one disc diameter of the fovea.

IV. INVESTIGATION

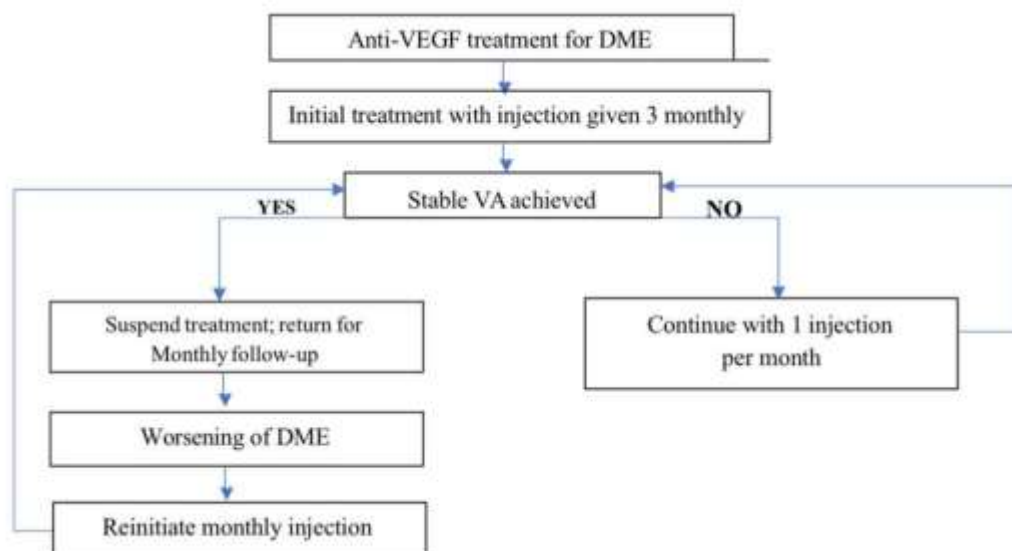
1. **Color fundus:** for diabetic retinopathy screening
2. Fluorescein angiography: guide during CSME treatment, identify macular capillary non perfusion, identify areas of NV causing recurrent VH despite full PRP.
3. Optical Coherence Tomography (OCT): treatment and follow up diabetic macular edema
4. Ultrasonography (B scan): rule out RD and TRD when media opacity

V. MANAGEMENT RECOMMENDATIONS (FROM AAO, ICO, RANZCO, ETDRS AND NHMRC)

Retinopathy	CSME	Focal/Grid laser	PRP	Follow up (month)
Normal	No	No	No	24
Mild NPDR	No	No	No	12
	Yes	Yes	No	2-4
Moderate NPDR	No	No	No	6-12
	Yes	Yes	No	2-4
Severe NPDR	No	No	Sometimes	2-4
	Yes	Yes	Sometimes	2-4
PDR	No	No	Usually	2-4
	Yes	Yes	Usually	2-4
High risk PDR	No	No	Yes	2-4
	Yes	Yes	Yes	2-4

❖ **Sometimes and Usually:** Treatment should be considered especially in patients with T2DM, poor follow-up compliance, impending cataract extraction, renal disease, pregnancy, and severe disease in the fellow eye.

❖ **Anti-VEGF treatment for DME and re-treatment schedule (Avastin)**



❖ **Indication of vitrectomy:**

- Non clearing vitreous hemorrhage > 3 months
- TRD involving or threatening macular
- TRD + RRD
- Diffuse DME associated with posterior hyaloid traction
- Dens subhyaloid or premacular hemorrhage
- Red blood cell-induced (erythroclastic) glaucoma and “ghost cell” glaucoma
- Anterior segment neovascularization with media opacities preventing PRP.

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ECTROPION

Ouk Sokhean, Pok Thorn

I. DEFINITION

Eyelid ectropion is present when the lid margin is displaced outward from the ocular surface lid margin. This eyelid malposition is commonly encountered in clinical practice and generally affects the lower eyelid.

II. ETIOLOGY

Normally, the posterior surface of the lower eyelid is closely apposed to the globe, with the edge of the lid margin positioned slightly above the inferior corneal limbus. The eyelid margin is square-shaped with the mucocutaneous junction occurring just posterior to the meibomian gland orifices. The marginal orbicularis muscle (muscle of Riolan) is responsible for tight apposition of the eyelid margin to the ocular surface. A relative shortage of skin, horizontal eyelid laxity, orbicularis muscle weakness, thinning of the normal square-shaped eyelid margin, and disinsertion of the lower eyelid retractors may all contribute to the development of ectropion.

1. Involutional Ectropion

Involutional ectropion typically progresses from mild punctal eversion secondary to medial retractor laxity, to more diffuse ectropion with associated horizontal lid laxity.

2. Cicatricial Ectropion

Cicatrization of the anterior lamella in the lower eyelid can occur following trauma, after lower eyelid blepharoplasty, due to facial actinic changes and with other dermatologic conditions. Detailed evaluation and determination of the cause is necessary as occasionally acute causes such as allergic blepharitis can be reversed with discontinuation of the offending agent or a trial of topical steroids. Cutaneous fibrosis and infiltration can be seen with certain malignancies and infiltrative processes. Suspicious lesions should be biopsied prior to eyelid surgery. Epidermal contraction can also be seen following long-standing involutional or paralytic ectropion. Cicatricial disorders can be segmental or involve the entire lower eyelid, and the treatment depends on the cause, extent, and location of the changes.

3. Paralytic Ectropion

Strong orbicularis oculi tone is required for correct function and position of the lower eyelid; however, without horizontal eyelid laxity, facial nerve paralysis seldom leads to ectropion.

4. Mechanical Ectropion

Classically, eyelid margin neoplasms can physically displace the lower eyelid margin, or through gravitational forces, weigh the eyelid margin down and away from the globe. Commonly, patients with marginal eyelid tumors have other mechanisms that lead to ectropion including cicatricial and involutional changes.

III. DIAGNOSTIC PROCEDURE

The defining cause of ectropion may be ascertained through historical clues, but the most important indication is found with careful physical examination. Specific attention should be paid to the eyelid margin, character of anterior and posterior lamella, horizontal laxity of lower eyelid, horizontal extent of the ectropion across the eyelid margin, and canthal position.

1. Eyelid Margin

The transition between nonkeratinized conjunctival epithelium and keratinized eyelid margin epithelium is abrupt and should occur just behind the orifices of the meibomian glands. Posterior migration of the mucocutaneous junction occurs in long-standing ectropion, and the level of keratinization can be assessed using rose Bengal or lissamine green if it is not easily visualized. The normal square eyelid margin takes on a rounded appearance in chronic ectropion. When severe, this marginal cicatrization can cause posterior rotation of the eyelash line, resulting in trichiasis after the ectropion is corrected surgically.

2. Eyelid Puncta

Evaluation of the punctal position and aperture is important. When observing the eyelid in its natural resting position, the punctum should not be visible; if the punctum is perceptible, eversion may be present. Chronic malposition can lead to stenosis or obliteration of the lower puncta, and surgical repair of the puncta may be required.

3. Eyelid Distraction Test

Horizontal eyelid laxity can be assessed with vertical distraction testing of the lower eyelid. Normal values are 6 mm or less. While laxity tends to increase with age, excessive laxity may need to be addressed with horizontal eyelid tightening techniques.

4. Snap Back Test

Another method of assessing horizontal eyelid laxity and eyelid tone is performing a snap back test. When the normal lower eyelid is downwardly distracted from the globe and released, it returns to position abutting the globe without blinking. The eyelid tone can be graded from 1–4 + based on the amount of time or number of blinks required for the eyelid to return to its normal position, normal being immediately without blinking. This test is most helpful in mild cases of ectropion and does not rule out paralytic or cicatricial ectropion.

5. Character of Anterior Lamella

Anterior lamellar shortening is the hallmark of cicatricial ectropion. Mild cases of cicatrization can be easily overlooked; vertical rhytids in the medial lid are an early sign of skin shortage. Limitation of superior migration of the lid margin in upgaze and eversion of the margin when opening the mouth widely also indicate inadequacy of the anterior lamella. In severe cases of cicatricial ectropion, it can be impossible to manually reposition the lid because of tethering from the tight skin.

6. Orbicularis Oculi Muscle Tone

At rest the orbicularis oculi muscle in the lower eyelid functions to keep the eyelid margin in normal position, and good muscle tone is required to keep the tarsal plate apposed to the ocular surface. In addition, normal muscle tone is required for the tear pump to function correctly. Passive and active orbicularis contracture can be assessed by asking the patient to squeeze their eyes and release.

7. Lateral Canthus Evaluation

Lengthening and poor fixation of the lateral canthal tendon is one of the most common mechanisms for horizontal eyelid laxity and can be observed in many cases of ectropion. At times, the entire canthus is dehiscient. Repair of horizontal laxity or canthal instability is mandatory for the long-term success of ectropion surgery.

8. Medial Canthus Evaluation

Poor fixation of the medial canthus can be evaluated by horizontal distraction testing. The degree of punctal displacement is evaluated while pulling the eyelid temporally. Excessive distraction may signify medial canthal tendon laxity. When mild, medial canthal tendon laxity can often be observed. In patients with marked dehiscence, surgeons may attempt to repair the tendon, a procedure that is difficult because of the close proximity of the tendon to the lacrimal drainage system.

9. Posterior Eyelid Inspection

Inspection of the conjunctival surface of the lower eyelids allows visualization of the relationship between the tarsal plate and the lower eyelid retractors. Occasionally, ectropion can be caused by excessive eyelid retractor laxity or dehiscence. If the lower eyelid retractors are dehiscient, this may require surgical repair.

IV. DIFFERENTIAL DIAGNOSIS

1. Eyelid retraction secondary to proptosis (e.g. thyroid eye disease), excessive tissue removal with lower blepharoplasty, or inferior rectus recession without disinsertion of the lower lid retractors
2. Floppy Eyelid Syndrome

V. THERAPEUTIC APPROACH

1. Lateral Tarsorrhaphy

- Indication
- Mild to moderate ectropion
- Mild or congenital eyelid retraction
- Post-operative care: Place antibiotic ointment in the lateral canthal angle and on the lid margin four times per day for one week

2. Eyelid Shortening by Lateral Tarsal Strip Fixation

- Indication
- Mild to moderate horizontal laxity of the lower or upper eyelid, especially when
- resulting from stretching of the lateral canthal ligament
- Post-operative care: Apply antibiotic ointment to the suture line 4 times per day for 7 days
- Potential surgical complications
- Persistent laxity
- Overcorrection
- Lateral canthal angle dystopia
- Trichiasis at the lateral canthus
- Redundancy of the lateral eyelid skin

3. Medial Spindle Tarsconjunctival Resection

- Indication
- Ectropion of the medial one-third to one-half of the eyelid with punctal eversion
- Post-operative care: Apply antibiotic ointment to the conjunctival fornix and skin suture twice daily until the suture dissolves
- Potential surgical complication: Canalicular injury

4. Full Thickness Marginal Wedge Resection

- Indication
- Ectropion from diffuse horizontal lid laxity
- Post-operative care
- Apply ice compress intermittently for 24 hours
- Place antibiotic ointment on the suture line and in the fornix 4 times daily for 7 days.
- Potential Surgical Complications
- Corneal abrasion
- Eyelid margin notch

5. Anterior Lamella Eyelid Lengthening with Skin Graft

- Indication

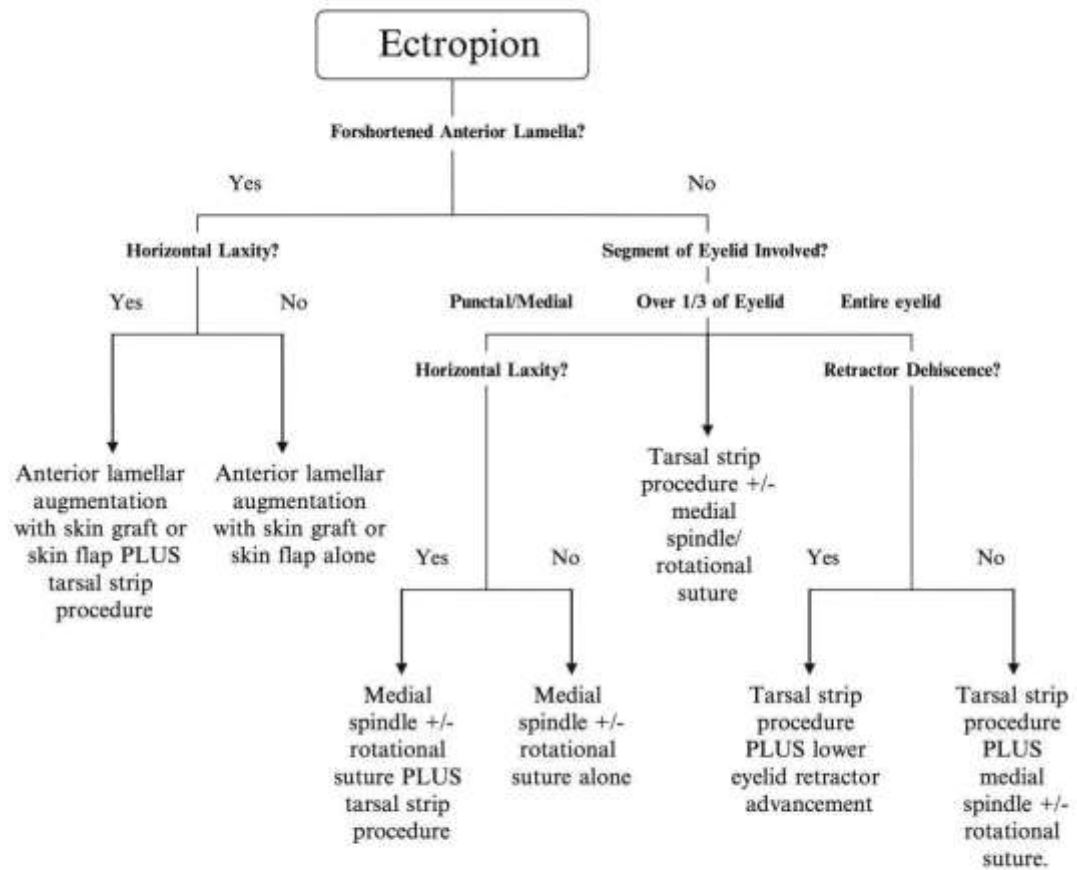
- Moderate to severe cicatricial ectropion due to anterior lamella shortening
- Post-operative care
- Apply antibiotic ointment to the suture line and place a firm dressing over the eyelid
- Change the dressing after 4 days
- Remove the dressing and frost suture after 10 days
- Potential surgical complications
- Persistent or recurrent ectropion
- Lagophthalmos the donor upper eyelid
- Graft failure

VI. COMPLICATION

Corneal infection leading to corneal ulcer is the most serious possible complication of ectropion. It can sometimes cause long-term damage. If your cornea gets scarred, light can't pass through it normally to reach the rest of your eye. Corneal infection and corneal ulcer can severely damage vision. This needs emergency evaluation and treatment. Symptoms of corneal infection and corneal ulcer include:

- Blurry vision
- More sensitivity to light
- Severe eye pain and soreness
- Increased eye redness
- Eye discharge
- Swollen eyelids
- White spot on your cornea that you may not be able to see

VII. ALGORITHM



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ENTROPION

Or Leakhena, Pok Thorn

I. DEFINITION

Entropion is an inward turning of the eyelid margin and appendages (pilosebaceous unit and mucocutaneous junction) are directed posteriorly towards the cornea and ocular surface.

II. ETIOLOGY (2)

2.1 Congenital Entropion:

- disinserted lower lid retractors, posterior lamella vertical insufficiency or kinking of the tarsal plate.
- facial paralysis

2.2 Involutional Entropion:

- horizontal and vertical lid laxity,
- Lower lid retractor weakness,
- Orbicularis oculi override.

2.3 Spastic Entropion

- local irritation, infection or recent ocular surgery

2.4 Cicatricial Entropion

- Scarring and shortening of the posterior lamella caused by chronic inflammation.
- trauma, post- surgery or radiation therapy.
- chronic antiglaucoma medications (miotic agents, prostaglandin analogues) and some chemotherapy drugs.

III. DIAGNOSTIC

3.1 Symptom:

- foreign body sensation,
- redness,
- tearing and discharge.
- Dry eye syndrome
- punctate epithelial erosions

3.2 Physical examination :

- blepharo or facial spasm

- Skin irritation or infection
- Cicatricial changes.
- Lid margin: trichiasis, distichiasis and epiblepharon.

3.3 Slit lamp examination:

- cornea with fluorescein: abrasions, corneal scarring or thinning, and neovascularization of the cornea.
- Symblepharon
- forniceal shortening
- margin abnormalities (inflammatory)

3.4 Test

- The snap back test: Horizontal lid laxity
- Resistance to traction and persistence of entropion: cicatricial.
- Eyelid eversion
- The overriding, or superior migration of the preseptal orbicularis oculi:
- lower lid retractor disinsertion
- Measure the lower lid retractor function

IV. DIFFERENTIAL DIAGNOSTIC

- Trichiasis
- Distichiasis
- Epiblepharon

V. THERAPEUTIC APPROACHES

Management of entropion should be directed at the specific etiology.

5.1 Medical management

- Ocular lubrication, artificial tears, or contact lenses.
- Botulinum toxin injection can also be considered in a patient with spastic entropion.
- Controlled the inflammation or infection by Topical and/or oral antibiotic and anti-inflammation before the surgical management. (2)

5.2 Surgical management

- Rotation or Everting suture (Quickert sutures), this technique is only temporary and often results in recurrence.

The procedure requires local anesthesia. A double-armed 4-0 chromic or vicryl sutures are placed from the deep inferior fornix below the inferior tarsal border, through the lower lid retractors, and exits through the skin superior to the level of insertion. Exiting more superiorly causes more lid eversion. This procedure indirectly tightens the lower lid retractors and forms scarring between the retractors, orbicularis and skin to

prevent override.

- Lateral tarsal strip operation or wedge resection corrects all three etiologic factors in involution entropion.

A lateral canthotomy and inferior cantholysis is performed to release the lower lid. The mucocutaneous junction is removed from the lid margin, and the anterior lamella is excised off the tarsus. Conjunctiva is debrided from the posterior tarsus. A horizontal incision through the superior tarsal plate creates a strip that is trimmed to achieve the desired amount of tightening. The strip is attached to the internal aspect of the lateral orbital rim (4)

- Tarsal fracture operation is useful in cases of upper lid mild to moderate cicatricial entropion. A posterior horizontal tarsal incision is made 2mm distal to the eyelid margin. The full thickness incision of the tarsus permits the eyelid margin to be rotated away from the globe. The margins should be stabilized with sutures. (3)

VI. COMPLICATION

Complications of entropion repair: (5)

- Hemorrhage,
- Undercorrection or overcorrection, lower lid retraction,
- Infection,
- Wound dehiscence,
- Scarring,
- Corneal injury,
- Mild ptosis, and eyelid margin necrosis.

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ESOTROPIA

Sok Kheang, Ouk Soleaphy

I. DEFINITION

Esotropia is a convergent misalignment of the visual axes. Esotropia can be categorized in a variety of ways, usually based on age of onset or underlying causes:

- Infantile esotropia
 - Acquired esotropia
- Accommodative esotropia
 - Normal accommodative convergence/accommodation (AC/A) ratio
 - High AC/A ratio

Partially accommodative esotropia

Nonaccommodative esotropia Sensory esotropia

Other (e.g., Duane syndrome, congenital fibrosis syndrome)

Infantile Esotropia

Infantile esotropia presents before the age of 6 months.⁴ Intermittent esotropia during the first 3 months of life may occur and does not necessarily predict the development of constant strabismus. The presence of esotropia is confirmed by the cover-uncover test or Hirschberg light reflect test in younger children, and it is generally measured using the prism and alternate cover test or Krimsky test. Children with infantile esotropia are at risk for amblyopia, although the presence of cross-fixation may diminish this risk prior to surgical correction. Characteristics of infantile esotropia include the following:

Onset before the age of 6 months without spontaneous resolution

Nonaccommodative or partially accommodative etiology

A constant angle of deviation that may increase with time¹² Frequent cross-fixation with the fixing eye in adduction

Additional features that may not be present at the time of diagnosis include latent or manifest latent nystagmus (see Detection of Nystagmus section under Examination), dissociated vertical deviation, oblique muscle dysfunction, A or V patterns, and optokinetic nystagmus asymmetry for nasal versus temporal pursuit.

Acquired Esotropia

Acquired forms of esotropia typically develop after age 6 months and may be accommodative, partially accommodative, or nonaccommodative. These children are at risk for amblyopia.

Accommodative Esotropia

Characteristics of accommodative esotropia include the following:

- An accommodative component that is usually associated with hyperopia
- Typical onset between the ages of 1 and 4 years, with an average age of onset of approximately 2 years. However, it may appear in infancy in older children, or develop after surgical correction of infantile esotropia
- Development that may be precipitated by illness, fever, or minor trauma
- Binocular visual function that may be normal prior to the onset of deviation

Etiology is related to excessive convergence in a child who usually has bilateral hyperopia, and correcting it eliminates esotropia (accommodative esotropia). Sometimes correction of the hyperopia results in normal alignment at distance fixation but a persistent esotropia at near (accommodative esotropia with a high AC/A ratio). Less frequently, children have normal alignment at distance fixation with no significant hyperopia, but they develop a constant or intermittent esotropia with near fixation (accommodative esotropia with a high AC/A ratio).

Partially Accommodative Esotropia

Children with acquired partially accommodative esotropia experience some improvement of their esotropia when they wear corrective lenses for hyperopia. Partially accommodative esotropia can present with normal or high AC/A.

Nonaccommodative Esotropia

Children with nonaccommodative esotropia have an acquired esotropia that is approximately equal in amount at distance and near fixation and have either no significant refractive error or no improvement in the angle of esotropia with correction of refractive error. If onset is acute and other neurological signs and symptoms are present, patients should be evaluated and treated promptly. Neuroimaging should be considered. Adults may also develop an acute acquired comitant esotropia. This is particularly common in adults who are myopic.

Sensory Esotropia and others

Sensory esotropia is associated with unilateral or bilateral vision loss on a structural basis. Consecutive esotropia occurs in some children with exotropia after surgery or, less commonly, spontaneously. Other conditions that are associated with esotropia include Duane syndrome, congenital fibrosis syndrome, craniofacial abnormalities, and ocular myasthenia gravis. Pseudoesotropia is not true strabismus but is caused either by certain facial features or a disparity between the visual and pupillary

II. DIAGNOSIS

Examination

Visual acuity and verification of eyeglass prescription with a lensometer

Binocular alignment at distance and near in primary gaze, up and down gaze, and horizontal gaze positions, if possible. If eyeglasses are worn, alignment testing should be performed with correction; alignment testing without correction may also be appropriate in some circumstances.

Extraocular muscle function (ductions and versions, including incomitance such as found in some A and V patterns)

General facial examination to note any pronounced dysmorphology or asymmetry

Detection of latent or manifest nystagmus

Assessment of head posture

Sensory testing, including fusion and stereoacuity

Cycloplegic retinoscopy/refraction

Funduscopy examination Additional testing

III. TREATMENT

Treatment should be considered for all forms of esotropia, and binocular alignment should be established as soon as possible, especially in young children, to maximize binocular potential, to prevent or facilitate treatment of amblyopia, and to restore normal appearance. Significant refractive errors should be corrected. Amblyopia treatment is usually started before surgery because it may alter the angle of strabismus and/or increase the likelihood of good postoperative binocularity.

Treatment for esotropia include the following: Correction of refractive errors

Bifocal eyeglasses Extraocular muscle surgery Botulinum toxin injection

Correction of Refractive Errors

Correction of significant refractive errors should be the initial treatment for children with esotropia. Eyeglasses are generally prescribed for +1.00 D or more of hyperopia in a child with new-onset esotropia. For patients with accommodative esotropia, realignment by cycloplegia-determined eyeglasses or contact lenses alone is successful in most cases. In general, a greater degree of hyperopia indicates a higher likelihood that the refractive error is an important etiologic factor of esotropia. The aim of treatment is to correct hyperopia sufficiently to restore alignment, and in most cases a prescription is given to correct the full refractive error as determined after cycloplegia.

Bifocals

An esodeviation greater for near than for distant targets is found in some cases. Convergence excess is defined clinically as a near esodeviation 10 prism diopters D or greater than the distance deviation (high clinical AC/A ratio) with the use of full hyperopic correction. Bifocal treatment can be considered in patients with potential for sensory fusion who maintain essentially aligned eyes at distance but have a manifest esotropia at near while wearing their full hyperopic correction.

Amblyopia Treatment

Amblyopia treatment is typically initiated before surgical treatment of strabismus. The esotropia may increase or decrease with the treatment of amblyopia. Surgical treatment of esotropia in the presence of moderate to severe amblyopia has a lower success rate than in the presence of mild or no amblyopia.

Extraocular Muscle Surgery

Although two-muscle surgery is most frequently performed, three- or four-horizontal muscle surgery may be required for large-angle deviations. Two-muscle surgery is the better initial choice even for large-angle deviations, regardless of magnitude, to reduce the risk of consecutive exotropia. In patients with deviations more than 60 prism diopters, additional extraocular muscle surgery is more often required to achieve ocular alignment.

Bilateral medial rectus-muscle recessions are commonly performed as the initial surgical procedure. Most surgeons prefer unilateral surgery (single-muscle recession or recession with resection or plication) for patients with irreversible amblyopia or other causes of substantially reduced vision in one eye. Operating on both eyes is preferable in specific clinical circumstances, such as V-pattern esotropia with inferior oblique-muscle overaction or nullpoint nystagmus with compensatory face turn.

IV. POST TREATMENT EVALUATION

Even when initial treatment results in good binocular alignment, follow-up is essential, because the child remains at high risk for developing amblyopia, losing binocular vision, and having a recurrence of strabismus. Children who are well aligned and do not have amblyopia may be followed every 6 to 12 months.

Children with esotropia should be assessed for hyperopia at least annually and more frequently if visual acuity decreases or the esotropia increases. Detection of uncorrected or undercorrected hyperopia is essential in the child with a recurrence of esotropia after successful initial treatment. Cyclopentolate 1% is effective in most patients for obtaining cycloplegia for refraction. In some patients, more hyperopia may be documented after regular eyeglass wear. If the esotropia appears to be accommodative in etiology but is not controlled with the current eyeglasses, repeat cycloplegic refraction should be performed before concluding that the esotropia has a nonaccommodative component. Recurrence of esotropia or consecutive exotropia that is not responsive to

eyeglasses, patching, or medical treatment may indicate the need for repeat strabismus surgery if the magnitude of the strabismus is sufficient.

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EXOTROPIA

Ouk Soleaphy, Sok Kheng

I. OVERVIEW

This guideline is designed for ophthalmologists managing children with strabismus (syn. squint), which is defined as a pathological misalignment of the visual axes. This is a broad subject and the reader is referred to comprehensive texts, for further information see bibliography.

The guidelines are intended to give general principles of management. It is assumed throughout this document, professionals deal with common and uncommon cases of strabismus will have had adequate training and experience to manage children with these conditions. This document represents the current view of best practice endorsed by the College. Please also refer to the Royal College Quality standards document¹ and Ophthalmic services for children².

The management of strabismus in childhood is multidisciplinary matter and usually involves parents and children, ophthalmologists, orthoptists and optometrists. General practitioners, health visitors, pediatricians and pediatric neurologists may also become involved in the management of children with strabismus. It is desirable that these parties are committed to locally agreed care pathways, covering visual screening, referral, assessment, treatment and monitoring of progress of children identified with strabismus. The latter is particularly important, as it is common that information will change with development, and multiple follow up visits will be required. Written information about strabismus should be available. Adequate time should be made available by the clinicians involved, in order to explain the terminology, the possible treatments, what they involve and for the consenting procedures that are required for surgical intervention.

II. INTRODUCTION

The classification of strabismus may be based on a number of features

Classification of Strabismus

- Intermittent
- Constant
- Infantile Acquired
- Accommodative Non-accommodative
- Comitant Incomitant
- Horizontal (eso/exo) – deviation Cyclotorsional and vertical

Strabismus is a common condition in childhood affecting 2.1% of the population³, with an increased prevalence associated with assisted delivery (forceps or caesarean section), low birth weight (including premature infants), neuro-developmental disorders. Neuro-developmental strabismus (associated with a neuro-developmental

problem) is independently associated with maternal smoking later in pregnancy, maternal illnesses in pregnancy and low birth weight for gestational age.

Approximately 60% have eso-deviations and 20% exo-deviations⁴. Strabismus may lead to a failure to develop binocular vision, and amblyopia, either of which may prevent an individual pursuing certain occupation. The appearance of ocular misalignment may interfere with social and psychological development with potentially serious effects for all patients with strabismus^{5,6,7,8,9}.

Timely diagnosis and appropriate treatment of children with strabismus can reduce the prevalence of amblyopia and ocular misalignment in later childhood and adult life. Correction of strabismus has been reported to improvement in motor coordination¹⁰.

Strabismus may be the present symptom in children with a serious eye or brain condition (e.g. retinoblastoma, hydrocephalus or brain tumour). All professionals involved with the management of strabismus need to be able to recognise this, and either initiate onward referral or arrange for appropriate investigation and management.

III. AIMS OF MANAGEMENT OF STRABISMUS

The aim of strabismus management is to achieve good visual acuity in each eye, restore normal ocular alignment (as near as possible, which may be a small under or over correction) and maximize the potential for sensory cooperation between the two eyes (the development of binocular single vision, which includes 3D vision, or stereopsis). While normal binocular single vision is the goal, sub-normal levels may be useful and may prevent later recurrences.

IV. HISTORY

It is important to elicit a detailed medical history including details of the onset of symptoms and other visual problems or general health issues. Prematurity, birth history and developmental milestones should be noted. The onset of strabismus is generally, before the age of 5 years. Cases may be intermittent. A family history may be present.

Most cases of strabismus occur in children who are otherwise healthy and often appear asymptomatic. The presence of behaviour change, or headache should be considered suspicious of serious neurological disease.

Parents and/or carers may report the presence of strabismus. However, small angle strabismus (as well as high refractive error and anisometropia causing amblyopia) may not be recognised unless the child is screened. Therefore, screening of visual acuity at 4 years of age either by an orthoptist, or by an orthoptically managed service, with onward referral to an orthoptist as appropriate is recommended. Strabismus is more common in children who have a positive family history and in those who have developmental ocular abnormalities or some specific systemic disorders. In such children a higher level of surveillance is recommended.

V. VISION TESTS AGE GUIDE

- Forced choice preferential looking grating: 0-1 year
- Cardiff cards : 0-2 years
- 3 m logMAR (uncrowded) Kay pictures :18 months to 3 years
- 3 m logMAR crowded Kay pictures : 2 – 4 years
- 3m logMAR crowded letters: Above 3 years

VI. OCULAR MOVEMENTS

The versions and ductions need to be assessed. In small children it may not be possible to bring the eyes into all areas adequately. Occlusion of an eye for a short period (up to an hour) or use of the vestibulo-ocular system may help

VII. OCULAR EXAMINATION

The pupils must be examined. An afferent defect is particularly important in cases of constant unilateral squint with reduced vision.

This includes the optic nerve and quality of the nerve fiber layer, macula and retinal periphery. The use of the indirect ophthalmoscope is recommended as standard in small children (less than 5 years), however in older children, the direct or 90 diopter/slit lamp examination can add information.

Aims of management of Strabismus:

- To detect/exclude serious underlying ocular or neurological disease
- To maintain or restore optimal visual acuity in each eye
- To maintain or restore normal (or subnormal) binocular single vision
- To restore appropriate ocular alignment
- To eliminate double vision, or other induced symptoms (e.g. asthenopia)
- To correct significant abnormal (compensatory) head posture
- To improve binocular visual field (in the case of esotropia correction)

VIII. MANAGEMENT

Sensory XT

Vision loss after 2 years of age will often cause exotropia and this is termed sensory exotropia. Surgery is for cosmetic indications. Monocular surgery on the poor vision eye is suggested to protect the “good eye”: recess the lateral rectus muscle and tighten the medial rectus by a resection or plication.



Congenital XT

Congenital exotropia is rare and occurs in the first few weeks of life. Like congenital esotropia, congenital exotropia patients have poor fusion potential. There is often an associated neurological or systemic disorder such as arthrogryposis, albinism, or craniosynostosis.



Clinical Features

- Rare
- Onset by 6 months of age, usually present at birth
- Constant exotropia
- Large angle >40 PD

Etiology

Unknown

Treatment

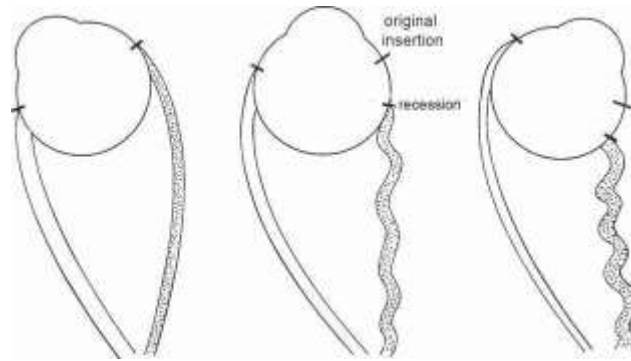
- Correct amblyopia if present.
- Surgery for XT after 6 months of age, do bilateral lateral rectus recessions.

Muscle Recession

A muscle recession moves the muscle insertion to a new location closer to the muscle's origin creating muscle slack. Muscle slack created by a recession reduces muscle strength as per Starling's length tension curve. The initial slackening of muscle fibers is taken up by fiber reorganization but there is probably a persistent change in both the recessed muscle and the antagonist. Surgical charts on the amount of recession for a specific deviation reflect the exponential character of the length tension curve. For example, each 0.5 mm of a bilateral medial rectus recession will correct approximately 5 prism diopters (PD) of esotropia up to a recession of 5.5 mm. However, after 5.5 mm of recession, each additional 0.5 mm of recession results in 10 PD of correction. Clinically, this is important, as we must be extremely careful when measuring large recessions because relatively small errors in measurement will result in large errors in eye alignment.

An inadvertent over recession of only 1.0 mm on a planned 6.0 mm bilateral medial rectus recession could result in a 20 PD overcorrection.

A unilateral rectus muscle recession will induce incomitance, as rectus muscle recessions have more of an effect in the field of action of the muscle. Note that when the eye rotates towards the recessed muscle (*right drawing*), the moment arm shortens and muscle slack increases. This results in progressive weakening of the rotational force as the eye turns towards the recessed muscle. In contrast, on eye rotation away from the recessed

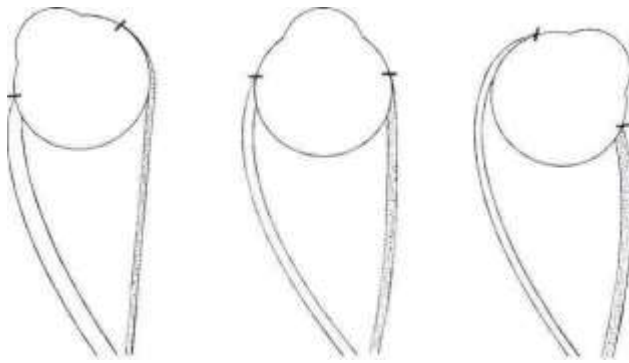


Drawing shows a rectus muscle recession. Note that in primary position (center drawing) the length of the moment arm remains unchanged and is equal to the radius. Thus, the effect of a standard rectus muscle recession on a strabismus in primary position is not due to a change in leverage, but is based on the amount of

muscle slack created. This is of practical importance because when measuring a recession, measure from the muscle insertion to quantify the amount of slack produced.

Muscle Tightening Procedures (Resection)

Rectus muscle tightening procedures include resections, tucks, and plications. These procedures tighten by shortening muscle. Increased muscle tension will result in muscle fiber hypertrophy, and this has been described as a temporary effect after muscle resection in the animal model. After a few weeks the muscle fiber diameter returned to baseline. Thus, muscle tightening procedures have a relatively small effect on the length tension curve and no effect on the moment arm. For the most part, muscle tightening procedures correct strabismus by creating a tether, or a leash, not by increasing muscle function. Often a resection is touted as a muscle strengthening procedure, but in reality, removing a section of the muscle tightens the muscle but does not improve muscle strength. Clinically, a resection produces incomitance as the tightened muscle restricts rotation away from the resected muscle. For example, a medial rectus resection limits abduction. It should be noted that tightening both medial rectus muscles restricts divergence and creates an esoshift which is greater in the distance than near. Tightening the medial rectus muscles does not improve convergence and is not effective in correcting convergence insufficiency.



The effect of a rectus muscle tightening procedure, such as a resection.

Note that the tightened muscle (the shaded muscle) causes restriction and limits eye rotation away from the tight muscle.

Recession and Resection

Resections (and Wright plication) can be combined with recessions of the same eye, and this is called a **recession–resection**, or an “**R & R**” procedure.

The effect of the recession and resection is additive in regard to limiting eye rotation. The recession reduces rotational force towards the recessed muscle while the resection restricts rotation away from the resected muscle. Thus, the recession–resection procedure causes significant incomitance. For example, a recession of the left lateral rectus muscle and resection of the left medial rectus muscle to treat a comitant exotropia will correct the exotropia in primary position, but would create an esotropia in left gaze. This is because the left eye would have limitation of abduction as compared to the normal adduction of the unoperated right eye. Limited rotations after an R & R procedure usually improve over several months to years, but some residual incomitance often persists. Since the R & R procedure induces incomitance, it is best used to treat incomitant strabismus. Monocular R & R is also the procedure of choice for the treatment of sensory strabismus so surgery is performed only on the blind eye.

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GLAUCOMA

CHUKMOL Kossama and Glaucoma team

I. NATURE OF THE DISEASE

1- Angle Closure

- ☐ Close angle
- ☐ ± Elevated IOP
- ☐ ± Glaucomatous optic nerve damage
- ☐ ± Visual field change

2- Open Angle

- ☐ Angle wide open
- ☐ Glaucomatous optic nerve damage
- ☐ ± Elevated IOP
- ☐ ± Visual field change

II. CLINICAL ASSESSMENT OF GLAUCOMA

Assessment	Equipment	Detail
History	Checklist	CC, Family History, Past ocular and medical history, History of drug use,
VA + Refraction	Auto refract	DVA, NVA, UCVA, BCVA, VA with and without glasses, refractive status
IOP	GAT/Air-puff/I-care	Before gonioscopy and Dilation, In case of measuring diurnal variation Record time 30 min interval*
Anterior Segment	Pachymetry + Slit-lamp	Pachymetry: Central cornea thickness, normal range 540 um SL: Cornea change, Pupil, lens position
Angle Structure	Gonioscopy/ A- OCT	Open, Close, NV, PAS, Recess after trauma, always record grade
Optic nerve	Slit-lamp and OCT- A/ OCT	Cupping, Color, notching, vessels around the disc, RNFL thickness + GCL map
Fundus	Slit-Lamp and Fundus photo	Abnormalities beside Glaucoma
Visual Field	Perimetry	☐ General Case: SITA-Standard 24-2/ 10-2 in case of tunnel vision ☐ Early Detection: · Change illumination from white to blue yellow · Use 24-2C ☐ Patient not cooperate: SITA FAST

III. ANGLE CLOSURE GLAUCOMA

Diagnosis	Finding	Treatment	Surgical Option (When pressure uncontrolled with medical)
Acute angle close	- At least 2 of the following complaints/symptoms: ocular pain, a history of intermittent visual blurring which may include the complaint of seeing halos, and nausea or vomiting. - At least 3 of the following signs are required: IOP greater than 21 mm Hg, corneal edema, conjunctival injection, a mid-dilated minimally reactive pupil, and a shallow anterior chamber (Iris Trabecular Contact/ Iris bowing)	1- Lower IOP Oral CAI 250 mg 3-4 times/day for 3 days Topical Steroid 2 hourly 2- Constrict Pupil Topical Pilocarpine 2% 3-4 times/day 3- Laser Iridotomy when cornea clear enough (Affected eye and Fellow eye)	Surgical PI/ Lens extraction ± GSL
Chronic Angle closure Glaucoma (Pupil Block)	Irido-Trabecular Contact/ Iris bowing	1- Lower IOP Topical CAI ± Topical Beta blocker ± Topical Alpha-Agonist ± PG 2- Constrict Pupil Pilocarpine 2% 3- Laser Iridotomy (Both eye)	Lens extraction + GSL ± Trabeculectomy + MMC
Plateau Iris	Irido-Trabecular Contact, Flare Iris	1- Lower IOP Topical CAI ± Topical Beta blocker ± Topical Alpha-Agonist ± PG 2- Constrict Pupil Pilocarpine 2% 3- Laser Iridotomy 4- Identify syndrome or configuration	- Surgical PI - Lens extraction ± GSL ± Trabeculectomy + MMC

Progression	Finding	Treatment	Follow up
Stable	No Change to Angle, optic nerve and VF IOP reach Target	Same	3-6 months
Unstable	Persistent Angle closure and increase ± Increase optic nerve damage ± Increase VF Damage	Additional IOP Lowering needed 30%	Every months till it's Stable

b. Unstable Close Angle Glaucoma (International Recommendation)

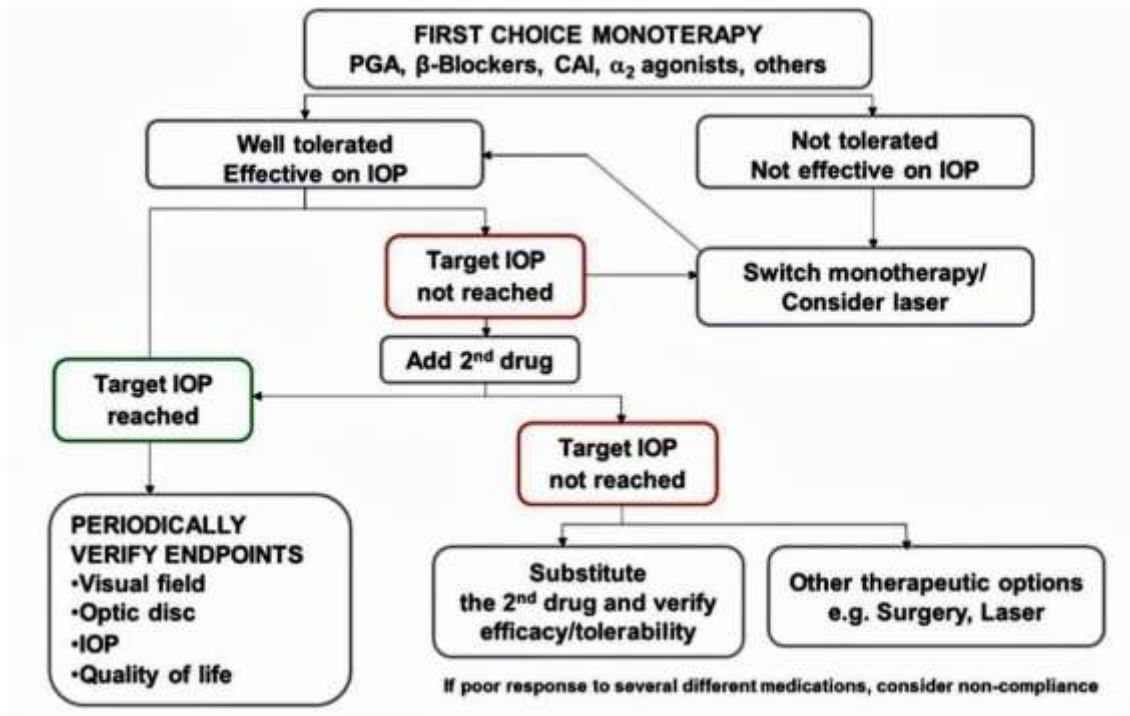
Glaucoma Severity	Finding	Suggest IOP Reduction	Treatment Consideration
Early	Persistent Angle closure + Optic nerve Damage ± Visual fields loss	Lower IOP $\geq$ 25%	Medication/ Lens extraction + IOL ± GSL
Moderate/Advanced	Persistent Angle closure + Optic nerve Damage +	Lower IOP 25%-50%	Medication + Cataract/Lens removal + IOL + GSL Trabeculectomy or tube (Valve/ Non- Valve)
End Stage (Refractory Glaucoma)	Blind eye ± Pain	Lower IOP 25%-50% (If Painful)	Medication + revise previous surgery and or CCC/CPC

IV. OPEN ANGLE GLAUCOMA

Glaucoma Severity	Finding	Suggest IOP Reduction	Treatment Consideration
Early	Optic nerve Damage ±	Lower IOP $\geq$ 25%	Medication Or Laser Trabeculoplasty/SLT/ Laser MP3
Moderate/Advanced	Optic nerve Damage + Visual fields loss	Lower IOP 25%-50%	1-Medication + Laser 2- MIGS 3- Trabeculectomy with MMC or Tube (with or without cataract removal, and IOL)
End Stage (Refractory Glaucoma)	Blind eye ± Pain	Lower IOP 25%- 50% (If Painful)	Medication + revise previous surgery and or CCC/CPC

b. Follow up

Progression	Finding	Treatment	Follow up
Stable	No Change to Angle, optic nerve and VF IOP reach Target	Same	3-6 months
Unstable	Increase IOP ± Increase optic nerve damage ± Increase VF Damage	Additional IOP needed by 30%	Every till it's Stable (If still not control Management depend on Stage)



5. Laser Setting

Laser Parameter	Selective Laser Trabeculoplasty (SLT)	Argon Laser Trabeculoplasty	Laser Iridotomy	Laser Iridoplasty
Laser Type	YAG	Argon	YAG	Argon
Spot Size	400 microns	50 microns	-	200 – 500 microns
Power	0.4-2 mJ	300 – 1000 mW	2 – 8 mJ	200 – 400 mW
Application Site	Trabecular Meshwork	TM junction (PT-NPT)	Peripheral Iris	Peripheral iris
Handheld Lens	SLT lens	Goldmann Gonioscopy lens	Laser Iridotomy lens	Goldmann Gonioscopy lens
Treatment Circumference	180 – 360 degrees	180 – 360 degrees	-	180 – 360 degrees
Number of Burn	50 spots per 180 degrees	50 spots per 180 degrees	-	20-40 spots per 180 degrees
Number of Setting	1 or 2	1 or 2	1	1 or 2
End Point	Bubble formation	Blanching at the junction TM	Fullthickness iris opening	Contraction burn

Pregnancy Status	1 st Line	2 nd Line	3 rd Line
1 st Trimester	Alpha-Agonist		
2 nd Trimester	Alpha-Agonist	Beta-Blocker (Monitor Fetal heart rate and growth)	Topical and Oral CAI (Monitor Fetal growth retardation)
Early Trimester	3 rd Alpha-Agonist	Beta-Blocker	Topical CAI
Late Trimester	3 rd Topical CAI		
Breastfeed	Topical CAI	Beta-Blocker	PGanalogue

Note

- **Laser:** ALT and SLT can be performed in all trimesters
- **Surgery:** Glaucoma surgery can be performed with caution in the second and third trimester (note anesthetic, sedative agents and antimetabolite may cause a potential risk to fetus)
- **DrugCategory**
 - Beta-blockers, CAI, PG: category C
 - Alpha agonist: category B

INTRAOCULAR FOREIGN BODIES (IOFB)

Sorn Bottomalen, Un Leng, Pro. Kong Piseth

I. CASE DEFINITION

- Intraocular foreign bodies (IOFBs) are defined as intraocularly retained unintentional projectiles that require urgent diagnosis and treatment to prevent blindness or globe loss.
- IOFBs are typically caused by striking metal on metal such as hammering on a chisel.

II. ETIOLOGY

- IOFBs have been described from glass from car windscreens
- Plastic fireworks and trimmers
- Organic material in rural settings
- Shotgun pellets
- Graphite pencil lead
- Fragments from lawnmower blades

III. DIAGNOSTIC PROCEDURE

i. Clinical argument

- Ocular retention of a small FB depends on careful attention details and circumstances of injury, and scrutiny for evidence of ocular penetration, such as a small entry site in anterior sclera and signs of vitreous disturbance.
- Mechanical injury
 - Globe: penetration, perforation, or rupture
 - Anterior segment: angle recession, ciliary body cleft, hyphema; lens capsule injury, cataract formation, zonular dehiscence, subluxation.
 - Posterior segment: vitreous liquefaction, vitreous hemorrhage, abnormal, vitreoretinal traction, retinal hemorrhage, retinal tear/dialysis, retinal detachment.
- Introduction of infection
 - endophthalmitis, panophthalmitis
- Toxicity
 - Siderosis, Chalcosis
- Siderosis (ferrous FB)
 - Dissociated iron has a predilection for deposition in epithelial tissue (lens, RPE), causing toxicity and cell death
 - RPE toxicity causes photoreceptor loss, decrease Visual acuity, constricted

Visual field, and RAPD

- Clinical: injection, heterochromia (iris reddish brown), increase IOP (2° glaucoma), anterior capsular cataract, reddish ferrous deposits at lens epithelium, coarse degenerative pigment dispersion, and retinal detachment. ERG may show generalized depression with a-wave reduction or more selective b-wave attenuation and loss of oscillatory potentials.
- Chalcosis (copper FB)
 - Pure copper IOFBs result in rapid fulminant endophthalmitis
 - Chalcosis results from FB of alloys of copper (brass, bronze) and mirror the ocular signs of Wilson's disease: Kayser–Fleischer ring, anterior 'sunflower' cataract, yellow retinal plaques.

ii. Technical procedure

a. Imaging study

- CT scanning: with thin sections, is the investigation of choice if the IOFB cannot be seen on fundoscopy
- X-ray: Screening for a suspected foreign body can be performed by plain X-ray. Orbital CT with 2mm slices in all cases, even if an IOFB is clearly visible.
- B-scan: detecting non-radiolucent foreign bodies but is a relatively inefficient method for detecting small metallic foreign bodies, especially if they are embedded in the ocular coats. Foreign bodies can give rise to high amplitude echoes provided they are appropriately orientated to the sound beam; a variety of artefacts arising from metallic particles aid in their identification and localisation, but the main value of ultrasound is in determining the vitreoretinal complications of foreign body impaction
- MRI absolute contraindication to a ferromagnetic particle rotates or moves in the magnetic field
- ERG: chronic retained IOFB => low b-wave

IV. DIFFERENTIAL DIAGNOSIS

- Superficial cornea foreign body
 - Cornea abrasion
 - Acute conjunctivitis
 - Uveitis
 - Congenital glaucoma
- Intraocular foreign body
 - Uveitis
 - Retinal detachment
 - Vitreous hemorrhage
- Intraorbital foreign body
 - Orbital cellulitis

V. THERAPEUTIC APPROACH

i. **Stabilize patient:** choice between local anesthesia and general anesthesia depends on several factors:

- a. Age of patient
- b. Preference of surgeon
- c. Difficulty of the case
- d. Duration of surgery

Advantages:

- General Anesthesia:
 - Complex surgery
 - Young patient
- Local Anesthesia:
 - Ability to position patient immediately after surgery
 - Reduce the cardiopulmonary risk to patients
 - Shorter turnover time
 - Cost-effectiveness

ii. **Directed treatment:**

- Surgical planning should address the following issues:
 - location of the foreign body in the eye
 - surgeon's ability to visualize and localize the foreign body
 - size and shape of the foreign body
 - composition of the foreign body (ferromagnetic vs nonferromagnetic)
 - encapsulation of the foreign body
- Removal of IOFB
 - Place infusion cannula for PPV
 - Use a straight/nonbeveled entry to reduce chance of suprachoroidal infusion
 - If poor visualization, use a 6-mm cannula for infusion line
 - Place additional cannulas
 - Use a similar nonbeveled entry
 - Orient the incision parallel to the limbus to allow for enlargement to remove IOFB if necessary.
 - Perform PPV
 - Create PVD if not already present: stain vitreous with triamcinolone
 - Carefully shave vitreous base to visualize possible additional foreign material and decrease vitreous snag when removing IOFB.

- Remove IOFB
 - Consider placing PFCL to protect macula from inadvertent contact with IOFB
 - Try not to remove IOFB through the entry wound. Remove IOFB through an enlarged, strategically placed sclerotomy instead. The larger the wound, the easier the removal. Preop CT measurements may guide wound size
 - ❖ Remove one cannula and enlarge the wound with MVR blade.
 - ❖ If RD repair needed after IOFB removal, make sclerotomy for IOFB removal at 12 o'clock position if possible.
 - ❖ Consider a limbal wound to remove IOFB in an aphakic eye.
 - ❖ Ensure the corneal endothelium is coated with viscoelastic prior to removal.
 - Forceps
 - ❖ If IOFB is small, then grasp with intraocular forceps of choice
 - ❖ Consider larger IOFB forceps if available
 - ❖ Grasp oblong IOFB at the smallest end.
 - Intraocular magnet
 - ❖ Will not hold IOFB strongly enough for removal through a sclerotomy. A bimanual approach is necessary.
 - ❖ Place chandelier for illumination.
 - ❖ Enlarge sclerotomy to adequate size.
 - ❖ Lift IOFB off retina and transfer to forceps for removal
 - Snare
 - ❖ Unusually shaped, round, large foreign bodies often cannot be grasped with forceps and will require a snare. Commercially available snares can be purchased, or the Finesse Flex Loop (Alcon) can be used or a snare can be fashioned from Prolene suture .
 - ❖ Straight cannula, 25G or 27G anterior segment cannula.
 - ❖ 9-0 Prolene suture looped through the cannula.
 - ❖ Suture can be retracted by assistant or by capturing in the plunger of BSS-filled syringe.
 - ❖ Suture can also be tied at apex with 2 or more loops to create a basket effect.
 - ❖ Remove cannula and enlarge sclerotomy to appropriate size.
 - ❖ Ensnare the IOFB, tighten, and remove.
 - ❖ If IOFB is oblong, then grasp it nearest to the end as possible to ease removal through sclerotomy.
 - Consider external magnet (this approach carries risk of choroidal

hemorrhage) if metallic IOFB is embedded in sclera.

- ❖ Use transillumination through external sclera to identify IOFB location.
- ❖ Create sclerotomy at site of IOFB to expose embedded IOFB.
- ❖ Use external magnet to remove.
- ❖ Consider repositing or excising any expelled tissue.
- ❖ Close sclerotomy with 8-0 nylon suture.

□ After IOFB removal:

- Reattach retina if RD present.
- Endolaser to retinal break(s) and possibly endolaser 360 degrees peripherally.
- Consider silicone oil tamponade vs gas.
- Tamponade and Silicone Oil Tamponade.
- Administer 0.1 mL of intravitreal vancomycin 1 mg/mL and 0.1 mL of ceftazidime 2.25 mg/mL after eye watertight. Check allergies.

□ Inject 0.3 mL of subconjunctival vancomycin (50 mg/1 mL) and 0.3 mL of dexamethasone 4 mg/1 mL.

iii. **Monitoring**

a. Clinical monitoring

- Have a high index of suspicion for endophthalmitis.
- With that in mind, control postop inflammation with topical steroid.
- Consider course of systemic steroids in nondiabetics.
- Use topical antibiotic and cycloplegic drops.
- Oral fluoroquinolone for 10 days. Check allergies.
- Monitor closely in the postop period.

b. Laboratory monitoring

- Blood pressure
- Any history of using anticoagulants medication
- Systemic diseases: DM & HTA
- History of strokes or severe cardiomyopathy
- Blood test related to coagulation factor, glycemia level, HbA1c%, cholesterol level, triglyceride, etc.

VI. COMPLICATION

Frequent post-operative examinations are performed in ensuing weeks and months for evaluation of any post-operative complications such as the following:

- Endophthalmitis: 2% - 48% of IOFB cases
- Anterior segment injuries including corneal scar, traumatic cataract, or

hyphema 60-75% of cases with IOFBs

- Retinal tear or detachment is 5%-40% of eyes with IOFBs
- Proliferative vitreoretinopathy (PVR)
- Sympathetic ophthalmia is 0.2% - 2.0 % of cases
- Optic neuropathy

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LENS REMOVAL WITH IOL IMPLANTATION AND TRABECULECTOMY IN LENS-INDUCED GLAUCOMA

Heng Sophal, Ngy Meng

I. DEFINITION

Lens-induced glaucoma (LIG) is a secondary glaucoma caused by lens-related conditions that lead to elevated intraocular pressure (IOP). Two primary types of LIG are phacolytic glaucoma and phacomorphic glaucoma. These conditions result from the disruption of the lens, either through leakage of lens proteins or lens swelling, respectively, both of which obstruct aqueous humor outflow. The combination of lens removal with intraocular lens (IOL) implantation and trabeculectomy is a surgical approach designed to address both the cataract and elevated IOP in these patients.

II. ETIOLOGY

i. Phacolytic Glaucoma

- a. **Pathophysiology:** Phacolytic glaucoma occurs when lens proteins leak through an intact but hypermature lens capsule. The leakage of these high-molecular-weight proteins leads to an inflammatory response, causing blockage of the trabecular meshwork and subsequent increase in IOP.
- b. **Risk Factors:** Advanced age, hypermature cataract, and a history of gradual vision loss are common risk factors. It typically occurs in patients with longstanding, untreated cataracts.

ii. Phacomorphic Glaucoma

- a. **Pathophysiology:** Phacomorphic glaucoma is caused by the physical enlargement and forward displacement of the lens due to intumescence (swelling) of a mature cataract. This results in secondary angle-closure glaucoma as the swollen lens narrows the anterior chamber angle, impeding aqueous outflow.
- b. **Risk Factors:** Risk factors include mature or hypermature cataracts, which increase the lens volume, leading to mechanical blockage of the anterior chamber angle.

III. DIAGNOSIS

i. History

- a. **Phacolytic Glaucoma:** Patients often present with a history of gradual vision loss, ocular pain, and redness. The onset of symptoms is typically insidious, with a history of untreated, longstanding cataracts.

- b. **Phacomorphic Glaucoma:** Patients may report sudden or subacute onset of severe eye pain, headache, blurred vision, and halos around lights. These symptoms are typically associated with a history of progressive vision decline due to mature cataract.

ii. **Ocular Examination**

- a. **Visual Acuity:** Significantly reduced in the affected eye, often due to the dense cataract.
- b. **Intraocular Pressure:** Elevated in both conditions, with IOP often higher in phacomorphic glaucoma due to angle closure.
- c. **Slit-lamp Examination:**
 - 1. **Phacolytic Glaucoma:** A hypermature cataract with a deep anterior chamber and presence of inflammatory cells and flare in the anterior chamber.
 - 2. **Phacomorphic Glaucoma:** A swollen, intumescent lens with shallow anterior chamber, often associated with angle closure.
- d. **Gonioscopy:** Essential for assessing the angle status. In phacomorphic glaucoma, the **angle** is often closed or narrowed, while in phacolytic glaucoma, the angle may be open but blocked by inflammatory material.

iii. **Imaging**

- a. **B-scan Ultrasound:** Useful in phacolytic glaucoma to confirm the presence of a hypermature cataract when the view is obscured.
- b. **Anterior Segment OCT/UBM:** Can be helpful in assessing the anterior chamber angle and lens position in phacomorphic glaucoma.

IV. **MANAGEMENT**

- i. **Non-Surgical a. Medical Management:** Immediate medical therapy is aimed at reducing IOP and controlling inflammation.
 - 1. **Phacolytic Glaucoma:** Topical steroids, IOP-lowering medications (beta-blockers, prostaglandin analogs), and hyperosmotic agents if IOP is significantly elevated.
 - 2. **Phacomorphic Glaucoma:** IOP-lowering medications, hyperosmotic agents, and possibly cycloplegics to relax the lens-iris diaphragm.
- ii. **Surgical**
 - a. **Lens Removal with IOL Implantation:**
 - 1. **Phacolytic Glaucoma:** The primary treatment is cataract extraction, which involves removing the hypermature lens and implanting an IOL to restore visual acuity and relieve IOP.
 - 2. **Phacomorphic Glaucoma:** Cataract extraction is also the definitive treatment. It relieves the pupillary block and restores the anterior chamber anatomy, thereby lowering IOP.

b. **Trabeculectomy:**

1. **Indications:** Trabeculectomy is often performed if IOP remains uncontrolled following cataract extraction or if there is pre-existing significant glaucomatous damage.
2. **Technique:** Involves creating a partial thickness scleral flap, a sclerostomy to allow aqueous outflow, and suturing the flap to regulate IOP.
3. **Combined Surgery:** In some cases, trabeculectomy may be performed concurrently with lens extraction, particularly in phacomorphic glaucoma with severe angle closure.

iii. **Post-operative Care**

- a. **Topical Steroids and Antibiotics:** To manage inflammation and prevent infection.
- b. **Cycloplegics:** To prevent synechiae formation and reduce discomfort.
- c. **Monitoring:** Close follow-up is required to monitor IOP and ensure the resolution of glaucoma.

V. COMPLICATIONS

- i. **Phacolytic Glaucoma:** Persistent inflammation, residual glaucoma, posterior capsule opacification.
- ii. **Phacomorphic Glaucoma:** Persistent angle closure, risk of chronic angle- closure glaucoma, cystoid macular edema, corneal edema, and hypotony post-trabeculectomy.
- iii. **Glaucoma Progression:** Despite surgery, some patients may experience ongoing glaucomatous damage.
- iv. **Endophthalmitis:** A rare but serious infection that can occur postoperatively.
- v. **Hypotony:** Over-filtration from trabeculectomy can result in low IOP, potentially leading to choroidal effusion or hemorrhage.

VI. ALGORITHM

An algorithm would include initial medical management to control IOP and inflammation, followed by timely cataract extraction and IOL implantation. If IOP remains elevated or glaucomatous damage is significant, trabeculectomy or other glaucoma surgeries may be indicated.

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NASOLACRIMAL DUCT OBSTRUCTION

Prak Kimsreng, Pok Thorn

I. DEFINITION

NLDO is a blockage of the lacrimal drainage system leading to stagnation of tears flow and causes local complications such as infection (dacryocystitis). Local complications could be sight-threatening, as well as life-threatening for immuno-compromised patients.

II. ETIOLOGY

- Acquired: Primary acquired obstruction (mostly aged related), Secondary acquired obstruction: infectious, inflammatory, neoplastic, traumatic and mechanical.
- Congenital (5% of newborn): blockage at the valve of Hasner or maldevelopment of the punctum and canaliculus. Spontaneous resolution around 90% within the first year of life.

III. CLINICAL FEATURES

- Increased tear lake (>1mm)
- Mucous or mucopurulent discharge
- Epiphora
- Others: tender sac, reflux from punctum (when pressure applied on sac).

IV. CLINICAL DIAGNOSIS

- Eliminate secondary tearing such as allergy and conjunctivitis is the main point for diagnosis (by history and examination)
- Dye disappearance test: fluorescein drops (DDT) in children, if the dye persists after 5 min it equals to obstruction
- Probing and irrigation test (SWO): cannula passing into canaliculus and irrigated with saline. If saline passes into the nose, it means patent. If there is reflux or sac distention, it means obstruction.

V. PARACLINICAL DIAGNOSIS (BEING USED IN THE COUNTRY)

- For non-complicated cases, no specific paraclinic examination is being used
- CT or MRI or Ultrasound in case of history of severe trauma, neoplastic suspected

VI. MANAGEMENT

It is focused on reestablishing the nasolacrimal duct or creating new channel surgically and it depends on causes.

- Congenital: start from observation with or without lacrimal massage to surgical procedure (probing after 1st year of life) and being more discussed in another pediatric topic.
- Acquired: Depend on the causes. Mostly frequent practice we meet Primary acquired NLDO. Dacryocystorhinostomy with silicone tube intubation (DCR +tube) is the gold standard management.
- In case of dacryocystitis, systemic antibiotherapy is administered followed by incision and drainage (Medication is listed below the diagram of surgery management)

VII. COMPLICATION

Since dacryocystorhinostomy with silicone tube intubation is a definitive treatment of NLDO, recurrence, bleeding, infection, silicone tube extrusion or scar are complications seen post operatively.

VIII. PROGNOSIS

Probing in congenital cases and DCR + tubing in primary acquired NLDO in adult have high successful rate (90 to 98%).

IX. FOLLOW UP

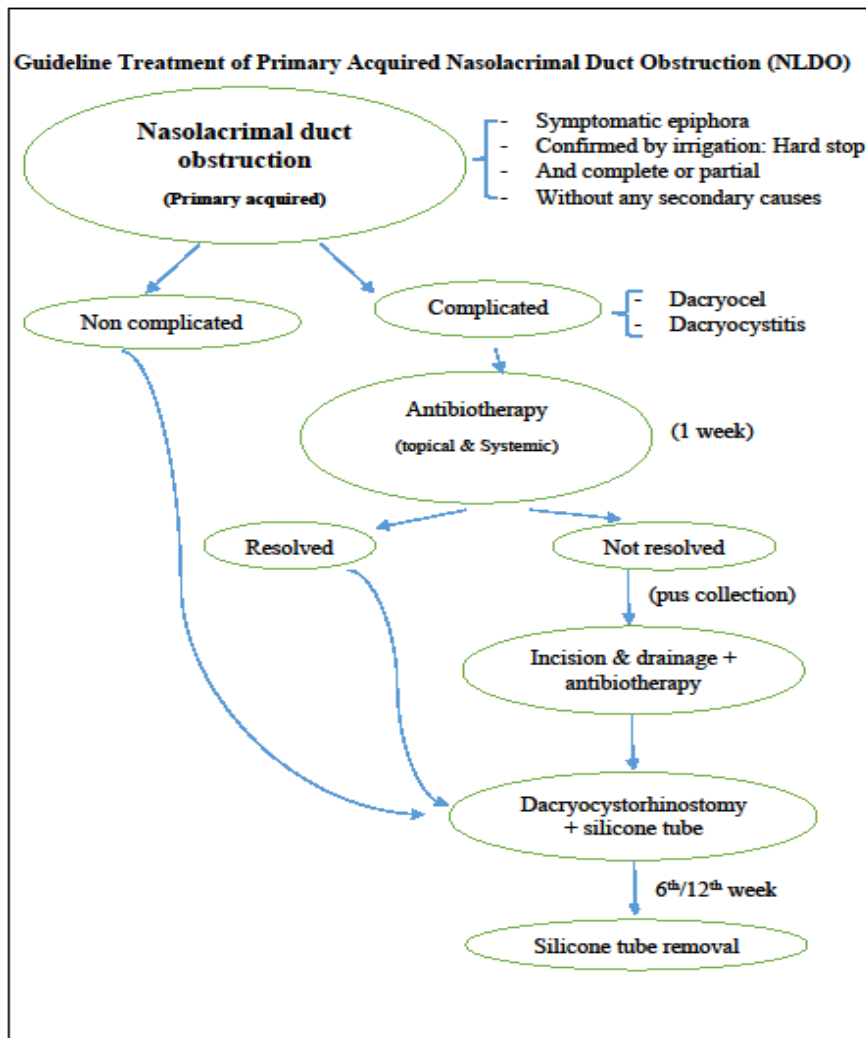
For post-operative DCR with silicone intubation, we follow up the patient 1st day, 1st week, 6th week and 12th week consecutively.

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Medication for Primary Acquired NLDO

1. Systemic antibiotic:
 - Quinolone or Beta-lactamine
2. Systemic anti-inflammatory:
 - NSAID is being used for severe inflammatory cases
3. In case of immuno-compromised and severe dacryocystitis:
 - IV Antibiotic is used combine oral/IV analgesic and Anti-inflammatory
 - Blood test and other investigations to find the immunocompromised causes
4. Post op DCR medication:
 - Topical antibiotic combined with steroid eyedrop: 4times/day (in eye for 1 week), after 1 week change to antibiotic eyedrop till tube removal then back to antibiotic combined steroid for another 1 week.
 - Topical antibiotic ointment: 3times/day (on wound for 2weeks)
 - Systemic antibiotic and anti-inflammatory as mentioned above (Duration: accordingly)



ANGLE CLOSURE GLAUCOMA

Sea Bunseng, Chukmol Kossama

I. DEFINITION

- ☐ Primary angle-closure glaucoma indicates loss of visual function and structural damage to the optic nerve occurring in the setting of an occluded drainage angle, with an assumed underlying mechanism mediated by elevated intraocular pressure.
- ☐ The acute presentation of angle-closure is a well-known ophthalmic emergency.

II. ETIOLOGY

A widely used classification identifies four potential levels of obstruction each progressively more posterior.

- Obstruction at the level of the pupil (i.e. pupil-block)
- At the level of the ciliary body (i.e. plateau iris and/or peripheral iris crowding)
- At the level of the lens or resulting from lens intumescence (lens induced angle- closure)
- Behind the lens, within the vitreous or the retina

Natural History Staging of Angle-closure Disease

- *Primary angle closure suspect (PACS)* – this is defined by iridotrabecular contact in the presence of normal intraocular pressure and no evidence of glaucomatous optic neuropathy.
- *Primary angle closure (PAC)* – In this second stage, intraocular pressure is elevated. There is no evidence of glaucomatous optic neuropathy. Other forms of ocular pathology, such as lens opacities or retinal vascular disease, may be present.
- *Acute angle closure (AAC) crisis* – The physical signs that follow the dramatic symptoms of an acute angle closure crisis do not always have an impact on the long-term visual prognosis of patients, but they are relevant because of the potential (often reversible) impact on current visual function. Symptoms are typically attributable to lens opacities or a dilated, unresponsive pupil resulting from iris ischemia. Glaucomatous optic neuropathy may be seen in the setting of an acute symptomatic presentation. It is likely this was a pre-existing phenomenon and not the result of an acute pressure rise spanning a matter of hours or days. In AAC, there is a need for expeditious management which differs from routine care, which makes the acute crisis an important variant in disease natural history.
- *Primary angle closure glaucoma (PACG)* – In the final stage, glaucomatous optic neuropathy has developed in the setting of an occluded

drainage angle. Glaucoma is defined by structural abnormalities in the pre-laminar optic nerve head, together with reproducible visual field loss.

III. DIAGNOSIT PROCEDURE

i. Clinical argument

a. Symptoms

- Painful red eye
- Headache
- Blurred vision
- Rainbow-colored haloes
- Nausea, vomiting and abdominal pain (sometimes misdiagnosed as gastroenteritis).
- Precipitating factors include dim light and certain drugs (e.g., bronchodilators, cough mixtures, cold and flu medication, antidepressants, antihistamines and anticonvulsants).

b. Slit lamp Examination: (2) (3)

- Refractive status – hyperopic eyes tend to have shallower anterior chamber angles
- which places them at risk for angle closure
- Conjunctival injection around the cornea (red eye)
- Mid-dilated nonreactive pupil
- Corneal haze
- Diminished red reflex and a hard globe, with intraocular pressure (IOP) of between 50 and 80 mmHg.
- Limbal anterior chamber depth (LACD or the “van Herick” test) showing shallow peripheral and central anterior chamber:
- Mild aqueous flare and cells: Examination of the anterior chamber may reveal a slight amount of aqueous flare (haziness) and cells, indicating inflammation.
- Sectoral iris atrophy may occur, releasing pigment into the anterior chamber that
- dusts the corneal endothelium and anterior lens capsule,
- Swollen optic nerve: The optic nerve may appear swollen during an acute attack, indicating optic nerve head edema.
- Central retinal artery and central retinal vein occlusion may also occur

ii. Technical procedure

a. Gonioscopy

- Gonioscopy remains the cardinal technique for diagnosis and monitoring of primary angle-closure disease.
- Gonioscopy will reveal angle closure for 360 degrees, confirming the

diagnosis of AAC.

- The ability to perform dynamic gonioscopy provides clinicians with unique insights into anterior segment drainage angle anatomy that is not possible with other examination techniques.

Grading System Scheie

The Scheie system is based upon visibility of the anatomical structures of the angle. They are graded Wide followed by I through IV, with Wide being open with all structures visible, and IV being closed with no visible structures

Shaffer-Kanski

Grade	Visibility	Interpretation
Wide	Wide	Open, all structures visible
I	Slightly Narrowed	Ciliary body visible, but recess obscured by the last roll of the iris
II	Apex not visible	Ciliary body not visible
III	Posterior half of trabecular meshwork not visible	Ciliary body, scleral spur, and posterior half of the trabecular meshwork not visible
IV	None of the angle structures visible	Ciliary body, scleral spur, and trabecular meshwork not visible

The Shaffer-Kanski system is a numbers-based grading system using degrees of angularity.

Grade Angle (°)	Visibility of structures	Risk of angle closure
0	0 No structures visible	Closed angle
1	10 Schwalbe's line, possibly anterior trabecular meshwork visible	Closure possible
2	20 Schwalbe's line and trabecular meshwork visible	Narrow, closure unlikely
3	20-35 Schwalbe's line, trabecular meshwork and scleral spur visible	Closure impossible
4	35-45 All structures visible from Schwalbe's line to ciliary band	Closure impossible

b. Imaging study

Anterior Segment Optical Coherence Topography (AS-OCT)

- ☐ AS OCT is particularly useful because it offers a quick, non-contact method of creating a documentary record of angle anatomy which can be reviewed remotely in virtual clinics, and used for audit.
- ☐ AS OCT can be performed in lower light conditions that may reveal unrecognized angle- closure disease.
- ☐ Ultrasound Biomicroscopy (UBM)

It is high-frequency B scan ultrasound provides high-resolution cross-sectional images of the anterior segment of the eye to the anterior vitreous. Because it uses sound, it can pass through opaque structures to visualize structures hidden from direct clinical examination, such as the ciliary body (5). It is particularly helpful for evaluating plateau iris and other ciliary body pathology

IV. DIFFERENTIAL DIAGNOSIS

Main differential diagnosis is versus secondary angle closure glaucoma

- Neovascularization of the angle – blood vessels from iris that cross scleral spur to arborize along trabecular meshwork, Peripheral Anterior Synechiae (PAS)
- Anterior chamber inflammation – keratic precipitates, posterior synechiae, iris bombé, inferior PAS (as opposed to primary angle closure where PAS tend to develop superiorly)
- Iridocorneal endothelial (ICE) syndrome – beaten-bronze corneal endothelium, corneal edema, high PAS that can extend anterior to the Schwalbe line, iris atrophy, corectopia.
- Drug induced (e.g. systemic topiramate) – acute bilateral disease, acute myopic shift, uniformly shallow chamber with anterior iris and lens displacement, ciliochoroidal effusion or detachment
- Aqueous misdirection (malignant glaucoma) – flattening of central and peripheral anterior chamber, anterior displacement of lens (crystalline or intraocular lens) or vitreous face, clear “aqueous” zones in vitreous
- Lens-induced angle closure – thick cataract, unstable or subluxed lens
- Nanophthalmos – small but normal eye with short axial length, microcornea, large lens, and thickened sclera
- Iris or ciliary body mass lesions or cysts – irregular contour of iris or neovascularization, hyphema or vitreous hemorrhage, episcleral sentinel vessel, ciliary body mass through dilated pupil or on ultrasound biomicroscopy
- Epithelial and/or fibrous downgrowth – wound dehiscence or gape, epithelial cysts in anterior chamber adjacent to wound, gray sheetlike membrane covering anterior segment structures that whitens with argon laser.
- Pseudophakic or aphakic pupillary block – pupillary obstruction or synechiae to anterior hyaloid surface, the intraocular lens, or posterior capsule
- Ciliary body engorgement associated with retinal vascular occlusion or panretinal photocoagulation – anterior rotation of iris and lens; retinal signs of each underlying etiology, e.g. retinal hemorrhages for vein occlusion and extensive retinal laser scars from photocoagulation.
- Retinal surgery
 - Encircling scleral buckle – relatively deep central anterior chamber with peripheral iris flattening, choroidal effusion
 - Pars plana vitrectomy - pupillary block from expansile gases pushing iris and lens forward, silicone oil in the anterior chamber, non-patent iridotomy

V. THERAPEUTIC APPROACH

The overall goals for management are

- To reverse or prevent the angle closure process
- To control intraocular pressure elevation

- To prevent damage to the optic nerve

a. Medical therapy

The role of medical therapy in acute angle closure attacks is to lower IOP, reduce pain, and clear corneal edema in preparation for iridotomy.

Topical

- Beta blockers (Timolol 0.5% b.i.d, Betaxolol 0.5% b.i.d)
- Selective alpha agonists (Brimonidine 0.2% b.i.d)
- Carbonic anhydrase inhibitors (Brinzolamide 1% t.i.d, Dorzolamide 2% t.i.d)
- Miotics (e.g., pilocarpine 2%) may help break an early angle-closure attack, but may be ineffective if the iris is already ischemic. High-concentration miotics (e.g., pilocarpine 4%) should be avoided because of the potential for forward displacement iris-lens diaphragm.
- Prostaglandin analogues (Tafluprost 0.0015%, Latanoprost 0.005%, Travoprost 0.004%, Bimatoprost 0.03%) – unreliable effect in acute attack because of slow onset of action
- Hyperosmolar agent (e.g. 5% sodium chloride) – assists in clearing corneal edema
- Prednisolone 1% - decreases inflammation Systemic
- Carbonic anhydrase inhibitors – oral acetazolamide's maximum IOP reduction is reached in 2-4 hours and lasts for 6-8 hours. Intravenous acetazolamide drops the
- IOP within 2 minutes with a peak effect noted by 10-15 minutes. In acute situations,
- a single dose of 500 mg acetazolamide should be given orally if the patient is not vomiting. Regular acetazolamide is preferred over the sustained-release sequel form because of quicker onset of action. If the patient is vomiting, acetazolamide can be given intravenously.
- Osmotic agents
 - Mannitol can decrease the IOP 30 mm Hg or more within 30 minutes of administration. The recommended intravenous dose is 0.5-1.5 g/kg body weight as a 15% or 20% solution, delivered at 3 to 5 mL/minute. Frail patients with cardiac or conditions may develop circulatory overload, pulmonary edema, congestive heart failure, and electrolyte imbalance. A rapid reduction in cerebral volume may result in subdural hematomas from vein rupture between the sagittal sinus and cortical surface. Therefore, patients receiving IV mannitol should be monitored in a hospital setting.
 - Oral osmotic agents:
 - *Glycerin*: 1 to 1.5 g/kg body weight of a 50% solution. Onset of pressure reduction is typically 10 to 30 minutes. Avoid in diabetics because the increased caloric load can cause ketoacidosis.
 - Isosorbide is commercially available as a 45% (45 g/100 mL) solution (Ismotic; Alcon Surgical). The recommended dose is 1 to 1.5 g/kg body weight. Its effect is similar to glycerin's but is safe for use in diabetics

because it is not metabolized.

- Although less common, oral agents can also cause subdural hematomas. Headache and gastrointestinal upset are common adverse reactions.

b. Surgical Therapy

- *Paracentesis* can be performed in an acute setting.
 - Technically, it can be difficult to perform on a phakic eye in pain with a shallow chamber, and there is a risk of permanent damage to the cornea, lens, and iris.
 - Devastating complications such as endophthalmitis and choroidal hemorrhage from a rapid pressure drop may occur.
 - The effects are typically short-term, because, as the ciliary body begins to form aqueous again, the IOP will inevitably rise.
 - This procedure can be used in cases of extreme IOP elevation to “buy time” until medications take effect or iridotomy can be performed.
- Laser Iridotomy Should be performed as soon as possible in the affected eye and in the contralateral eye to avoid an attack of acute angle closure glaucoma in the future. In case of conventional therapy (medical therapy and LPI) not working, surgery is another option
 1. Lens Extraction
 - Extra capsular cataract extraction
 - Manual Small Incision Cataract surgery
 - Phacoemulsification of lens with intraocular lens (IOL) implantation
 2. Trabeculectomy
 - Can be considered when surgical risks are outweighed by the potential benefits.
 - Indicated if glaucoma damage is moderate to advanced, if the rate of progression has been rapid, if prior laser has been unsuccessful, or if there is significant risk of future progression that will result in a symptomatic disability
 - Surgical Technique
 - Trabeculectomy surgery can be performed under local or general anesthesia.
 - A traction suture (placed partial thickness through the peripheral clear cornea) is often used for adequate exposure of the surgical site (superior quadrants of the eye).
 - A conjunctival peritomy is created to expose the superior bare sclera.
 - This peritomy may be limbal based or fornix based.
 - Gentle cautery is used to achieve hemostasis.

- A partial thickness scleral flap is created in the superior sclera, hinged at the limbus. There are multiple variations in shapes (rectangular, triangular, trapezoidal) and size of this flap. The scleral flap is dissected forward until the bluish gray zone at the limbus is exposed.
- Dissection of the subconjunctival space to provide a pathway for aqueous flow is then performed.
- A paracentesis is created in the peripheral clear cornea prior to creating an ostomy into the anterior chamber from underneath the scleral flap.
- Viscoelastic material may be injected to stabilize the intraocular pressure as the ostomy is created to avoid anterior chamber shallowing.
- There are several ways to create the ostomy underneath the scleral flap.
- A blade is first used to enter the anterior chamber from just behind the hinge of the scleral flap. Entrance into the anterior chamber is then enlarged using the blade, vannas scissors, and/or a Kelly punch device.
- Often, a peripheral iridectomy is created at this point, using forceps to pull the peripheral iris through the ostomy site, and scissors to create the iridectomy.
- In pseudophakic eyes, some surgeons omit this surgical step.
- Sutures (usually 9-0 or 10-0 nylon) are then used to close the scleral flap. The sutures may be placed in an interrupted manner, or releasable sutures may be placed. The flap should be closed tightly enough to allow the anterior chamber to remain formed, but loosely enough to allow for aqueous drainage at the edges of the scleral flap.
- The conjunctival peritomy is then closed using absorbable or nylon sutures.
- A subconjunctival injection of corticosteroids may be given at the end of the surgical procedure.

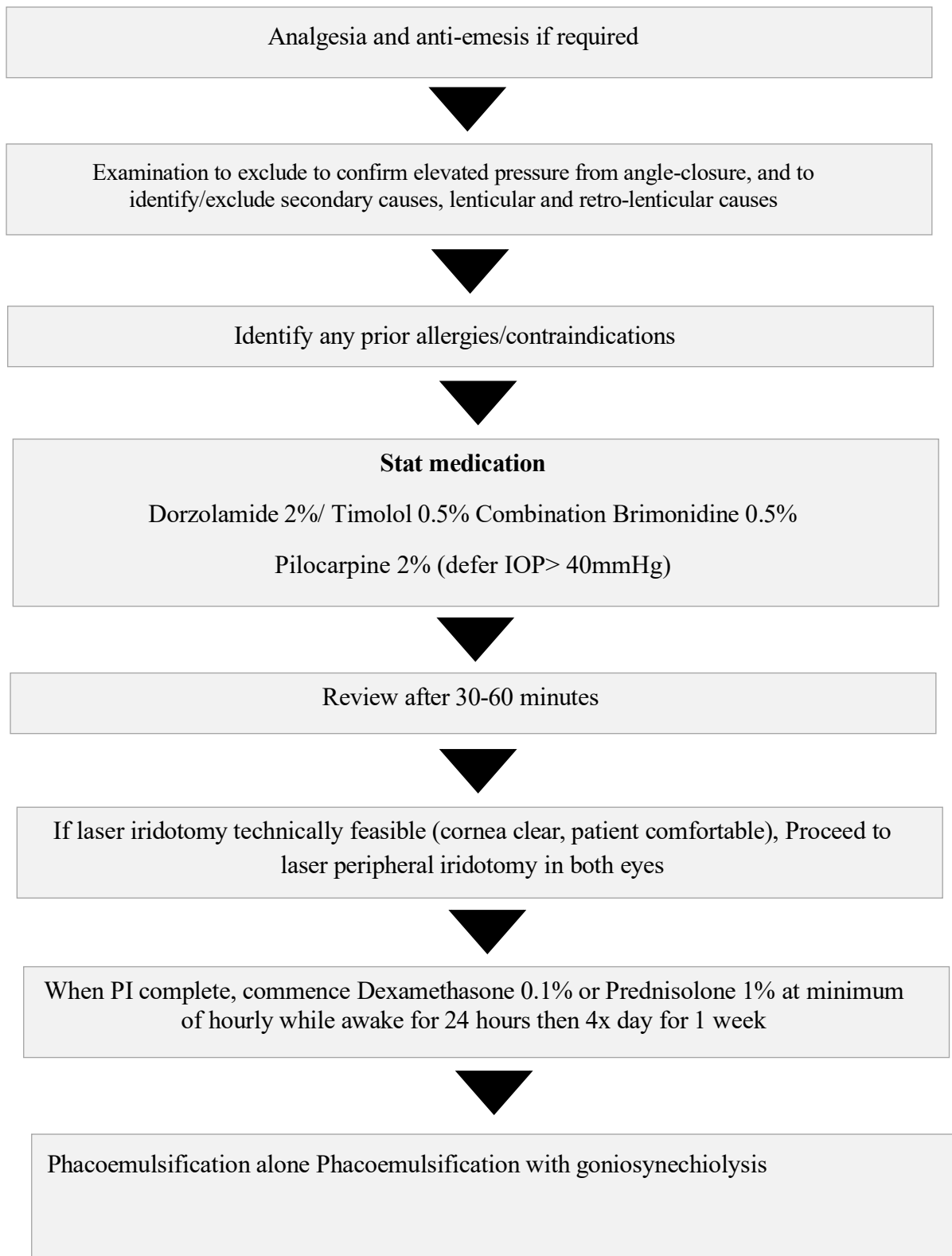
Additional long term effect with anti-metabolic agent- (Mitomycin C or 5-Fluorouracil)

VI. COMPLICATION

- a. Laser Peripheral Iridotomy
 - Transient Increase in IOP
 - Anterior Chamber Bleeding (Hyphema)
 - Visual Dysphotopsia
 - Cataract Progression
 - Iridotomy closure and necessity for react LPI
 - Aqueous misdirection syndrome
- b. Trabeculectomy (6)
 - Intraoperative

- Conjunctival button hole
- Hyphema
- Vitreous Prolapse
- Aqueous misdirection
- Post Operative
 - Choroidal effusion
 - Shallow or flat anterior chamber
 - Wound Leak
 - Hyphema
 - Persistent corneal edema
 - Encapsulated bleb
 - Dysesthesia
 - Cystoid macular edema
 - Suprachoroidal hemorrhage
 - Aqueous misdirection
 - Hypotony maculopathy
 - Blebitis or endophthalmitis
 - Bleb leak

VII. ALGORITHM



“PACS PLUS”

Criteria for Referral of People with Suspected Occluded Angles to the Hospital Eye Service

Angle Criteria

Either – a limbal chamber depth grade $< \frac{1}{4}$

Or – an anterior segment OCT showing irido-trabecular contact (ITC)

PLUS: one of the following criteria

- People with only one “good eye” in which deterioration of vision may threaten independent living or livelihood
- Vulnerable adults who may not report ocular or vision symptoms
- Family history of significant angle closure disease
- High hypermetropia ($> +6.00$ diopters)
- Diabetes or another condition necessitating regular pupil dilation
- Those using antidepressants or medication with an anticholinergic action
- People either living in remote locations (such as foreign aid workers, armed forces stationed overseas or oil rig workers etc.) where rapid access to emergency ophthalmic care is not possible.

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RETINOBLASTOMA

Soung Mengsreang, Pok Thorn

I. CASE DEFINITION

- Retinoblastoma (RB) is an intraocular embryonal tumor, typically occurring during early childhood.
- The condition is the most common eye cancer in children, affecting approximately 1 in 20,000 individuals.
- 40% of cases are bilateral and 60% of cases are unilateral.
- 10% of cases have an identifiable family history. In these situations, the condition is inherited in an autosomal dominant manner.
- Early identification and treatment while the tumor is confined to the eye has a very high cure rate (approximately 99%).

II. ETIOLOGY

- Heritable: Almost all cases of retinoblastoma occur due to mutation in one of the alleles of the retinoblastoma gene (*RBI*).
- Non-heritable

III. DIAGNOSIS PROCEDURE

1. Clinical features

- Retinoblastoma is an intraocular tumor, typically diagnosed before the age of five.
- Leukocoria
- Strabismus
- Red painful eye
- Poor vision
- Asymptomatic
- Orbital cellulitis
- Heterochromia iridis
- Hyphema
- The diagnosis should always be considered if there is a family history of retinoblastoma, until this risk is excluded with molecular investigations.

Three Tumor grow patterns are usually seen:

- Endophytic
- Exophytic

- Diffuse infiltration tumor

2. Technical Procedure

Ultrasound of eyes: confirm intraocular origin, possible calcifications.

CT scan of brain and orbit:

- To detect intraocular calcifications
- Optic nerve for possible extension of retinoblastoma

MRI:

- optic nerve invasion or intracranial extension is suspected

Genetic testing Ophthalmology review:

- Fundoscopy and visual acuity
- EUA with tonometry and indirect ophthalmoscope
- Bone marrow aspirate – metastasis
- Lumbar puncture – CSF for cell count and cytospin.

The International Intraocular Retinoblastoma Classification (ICRB)

Group	Subgroup	Reference	Features
A	Very low risk	Small tumor	□ RB ≤ 3 mm (in basal dimension) □ At least 3 mm away from the foveola and 1.5 mm from the optic nerve □ No vitreous or subretinal seeding is present
B	Low risk	Larger tumor Macula Juxtapapillary Subretinal fluid	□ RB > 3 mm (in basal dimension) □ Macular location (≤ 3 mm to foveola) □ Juxtapapillary location (≤ 1.5 mm to optic nerve) □ Additional subretinal fluid (≤ 3 mm from margin) □ No vitreous or subretinal seeding is present
C	Moderate risk	Focal seeds	Focal subretinal and/or vitreous seeds ≤ 3 mm from the tumor

D	High risk	Diffuse seeds	☐ Diffuse subretinal and/or vitreous seeds >3 mm from the tumor
E	Very high risk	Extensive retinoblastoma	☐ Extensive retinoblastoma ☐ occupying >50% of globe or any of the following: ☐ Secondary neovascular glaucoma ☐ Tumor anterior to anterior vitreous face or touching the lens ☐ Diffuse infiltrating retinoblastoma ☐ Massive intraocular hemorrhage ☐ Aseptic orbital cellulitis ☐ Phthisis bulbi

IV. DIFFERENTIAL DIAGNOSIS

- Cataract
- Coats' disease
- retinopathy of prematurity
- toxocariasis, choroidal coloboma
- vitreous hemorrhage
- myelinated retinal nerve fibers
- astrocytic hamartoma
- Corneal opacities.

V. THERAPEUTIC APPROACH

TREATMENT OPTIONS

Enucleation:

Indication Enucleation:

- children presenting with advance tumor (group E eyes), especially if unilateral failure of all possible effective therapies
- active tumor in an eye with no visual potential
- anterior segment invasion
- phthisical eyes after-high-dose chemotherapy
- when the visualization of the tumor is compromised due to corneal opacity, cataract, or vitreous hemorrhage.

Procedure:

Avoid perforating the globe and obtain an adequate length of optic nerve (>15 mm).

- Gently place a wire speculum.
- Use conjunctival scissors to perform a perilimbal conjunctival peritomy around the whole eye.
- Perform a tenotomy in all four quadrants using curved tenotomy scissors.
- Gently place a muscle hook under each of the four rectus muscles. Place muscle traction sutures 2 to 3 mm from the muscle insertion.
- The order of tagging and cutting the rectus muscles is based on the distance to the limbus: first medial, then inferior, then lateral, and finally the superior rectus. Cut the muscles in between the traction suture with conjunctival scissors.
- The superior oblique and inferior oblique muscles are now identified and cut.
- After all, six of the extraocular muscles have been severed, use the four traction sutures to exert gentle traction on the globe and facilitate globe prolapse.
- Curved tenotomy scissors are then inserted by the lateral approach and the optic nerve is identified near the orbital apex.
- The optic nerve is then cut just above the orbital apex to avoid damage to the important structures there. This ensures adequate length of the optic nerve (>15 mm).
- Pack the socket immediately with gauze and keep in place for 5 minutes to stop bleeding
- Inspect the enucleated globe for any evidence of extra-scleral extension of the tumor. Measure the length of optic nerve using calipers.
- Send the globe for detailed histopathology analysis.
- After stopping the bleeding, identify the posterior Tenon's capsule. Place an adequate-sized implant in the intraconal space.
- The anterior Tenon's capsule and the conjunctiva are then closed with absorbable sutures in two layers.
- plain conformer with a draining pore is then placed in the socket. The conformer can be secured in place with central suture tarsorrhaphy.
- A pressure patch is applied for 24 hours.

Chemotherapy:

This may be given systemically as first line treatment.

Indication chemotherapy:

- patients with extraocular disease, in the subgroup of patients with intraocular disease with high-risk histologic features, and
- patients with bilateral disease in conjunction with aggressive focal therapies

Chemotherapy for intra ocular retinoblastoma. Drugs: Carboplatin, Etoposide, Vincristine

(CEV) Dose

1. Carboplatin 500mg/m² Day 1
2. Etoposide 100mg/m² Day 1 and Day 2
3. Vincristine 1.5mg/m² on Day 1.

- * Each cycle is given every 28 days for 6 cycles.
- * If giving pre – enucleation treatment, ONLY 2 cycles are to be given prior to enucleation. This is to prevent resistance to chemotherapy.
- * Chemotherapy to start 5/52 post enucleation, provided WCC > 4.0, Neutrophil count > 0.1, Platelets >100, haemoglobin 8g/dl and child are clinically well.
- * Vincristine to be given first, then carboplatin then etoposide

VI. COMPLICATION

If retinoblastoma is left untreated patient is likely to develop the following complications:

- ☐ Retinal detachment
- ☐ Retinal necrosis
- ☐ Orbital invasion
- ☐ Optic nerve invasion
- ☐ Blindness
- ☐ Intracranial extension
- ☐ Secondary neoplasms
- ☐ Metastasis
- ☐ Tumor recurrence
- ☐ Temporal bone hypoplasia
- ☐ Cataract
- ☐ Radiation neuropathy
- ☐ Radiation retinopathy

VII. ALGORITHM

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TRACTIONAL RETINAL DETACHMENT

LY MARINA, KONG PISETH

I. DEFINITION

Tractional retinal detachment (TRD) is defined as the separation of the neurosensory retina from the retinal pigment epithelium (RPE) due to the traction force caused by proliferative membranes present over the retinal surface or vitreous which result from different etiologies.

II. ETIOLOGY

Retinal tractional detachment can result from proliferative retinopathies.

- Anterio-Posterior Traction
- Tangential Traction

The causes of tractional retinal detachment are divided into 2 types

- Affecting adult population
 - Proliferative diabetic retinopathy
 - Retinal vein occlusion (RVO)
 - Retinal vasculitis
 - Trauma
 - Sick cell retinopathy
 - Posterior uveitis
 - Proliferative vitreoretinopathy (PVR)
 - Penetrating Eye Trauma
- Affecting pediatric population
 - Retinopathy of prematurity (ROP) and its complications
 - Familial exudative vitreoretinopathy (FEVR)
 - Persistent fetal vasculature (PFV)
 - Toxoplasma retinitis
 - Toxocara retinitis
 - Sick cell retinopathy
 - Trauma
 - PVR

- Incontinentia pigmenti

III. DIAGNOSIS PROCEDURE

a. Clinical Argument

i. Symptoms

- Distorted vision
- Straight line or suddenly appear curved vision
- Visual field defect
- Floater and flash of light
- Central vision loss (in association with MD)
- Sudden drop in vision (in association with VH)

ii. Signs

- Smooth, concave, localized fibrosis
- Fibrovascular proliferation
 - Along Arcade vessel
 - Proliferation to vitreous
 - Not extend to the ora serrata;
 - May have pseudoholes or true holes in a combination
- RRD can be associated in case of SRF present with RB
- Relative clinical sign depends on etiology

b. Imaging Study

i. Optical coherence tomography

- Assessing the level and severity of membranes
- Localized potential space for surgery
- Fovea status: ERM and VMT could be presented
- Ellipsoid zone: for prognosis

ii. Fundus fluorescein angiography

- Useful in doubtful cases
- Define cause of Traction retinal detachment
- Define macular and peripheral ischemia

iii. B scan

- In case of media is hazy or obscured
- 2 types of TRD describe in Ultrasound
 - X pattern: narrow vitreo retinal adherent
 - H pattern: broad vitreo retinal adherent
- Identification of traction and retinal detachment area
- State of PVD

iv. Blood test: depend on cause of disease

IV. THERAPEUTIC APPROACH

- a. Stabilized patient: the management of TRD is divided into two types:
 - Systemic management: Controlling patient underlying disease
 - Control Diabetes mellitus: Glycemia and HbA1c Level
 - Control Hypertension: Systolic and Diastolic Pressure
 - Control Renal disease: Nephropathies, Renal Failure, etc.
 - Control Hemoglobinopathy: Anemia, Cancer, etc.
 - Ocular management: the decision for TRD surgery depends on many factors
 - Systemic factors needed to be stabilized
 - Patient understanding before surgery
 - Nature of disease after surgery needed to be informed to patients
 - Visual and anatomical outcome
- b. Empirical treatment
 - Controlling inflammation: Topical Steroid or NSAID
 - Controlling IOP: Anti-glaucoma medication
 - Prevent co-infection: Topical antibiotic
 - Controlling hemorrhage and soften fibrous tissue (TRD in DR): Anti VEGF 1 week before surgery
 - Controlling with PRP laser:
 - SSStabilized the eye in case of PDR with TRD
 - Potential risk of tear during surgery
 - Help induction of PVD during surgery
 - Reduced time needed to perform endolaser during surgery
- c. Direct Treatment
 - Indication for TRD
 - Duration less than 6 months
 - TRD threatening Macular and optic disc
 - TRD in combination with RRD
 - Surgery indication: PPV + MPC + EDL + Tamponade

- PPV:
 - ☐ Three port sclerotomies
 - ☐ Peripheral vitrectomy (outside-in technique: release traction and avoid tension to break)
 - ☐ Core vitrectomy
 - ☐ Manage membrane and PVD
- MPC:
 - ☐ Segmentation
 - ☐ Delamination
 - ☐ En-bloc dissection
 - ☐ Combination
- EDL:
 - ☐ Quadrant or full EDL depends on underlying disease
 - ☐ Avoiding 3 and 9 o'clock region
 - ☐ Curve probe is recommended in phakic eye
 - ☐ Setting of the laser vary between cases
 - ☐ Additional Anti VEGF can be considered
- Tamponade: Depend on cases
 - ☐ Air or
 - ☐ Gas SF6 or C3F8 or
 - ☐ Silicon oil 1500 or 5000 cst

d. Monitoring

- Clinical Monitoring
 - Dissection area of TRD
 - Level of Vitreous hemorrhage
 - Retina thickness
 - Level of tamponade
 - Laser reaction of EDL
 - Intra ocular pressure
 - Status of Lens
 - Sclerotomies wound
- Laboratory monitoring
 - CBC
 - Coagulation factor
 - Glycemia
 - HbA1c level
 -etc.

”

V. DIFFERENTIAL DIAGNOSIS

- Taut Posterior hyaloid
- Vitreo macular traction syndrome

VI. COMPLICATION

- Intraoperative complication:
 - Iatrogenic retinal break
 - Iatrogenic cataract
 - Vitreous hemorrhage
 - Choroidal effusion
- Postoperative complication
 - Vitreous hemorrhage
 - Secondary IOP increase
 - Retinal Detachment (RRD)
 - Epiretinal Membrane
 - Cataract
 - NVG
 - Chronic inflammation: CME
 - Endophthalmitis and SO

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VITREOUS HEMORRHAGE

Khoy Sothearith, Prof. Kong Piseth

I. CASE DEFINITION

Bleeding into vitreous humor is a vitreous hemorrhage. Depending on the location of the blood, it is described as:

- Preretinal: The blood lies in front of the retina. The vitreous humor is contained in a membrane. This membrane gets detached, and blood collects between it and the retina. The blood does not clot and may have a distinct upper level.
- Intravitreal: The blood is mixed in the substance of vitreous humor. Though dispersed in the gel, it may settle down because of gravity. The blood in the vitreous clots quickly, as the gel promotes platelet collection. The blood is red at first and changes to brown or yellow with time. The red blood cells (RBCs) remain intact for months and are broken down without being consumed by macrophages. The iron released by such RBC breakdown can damage the retinal cells.

II. ETIOLOGY

Common causes of vitreous hemorrhage include:

- Vitreoretinal: PVD with/without retinal tear
- Retinovascular – With Neo-vascularization
 - Proliferative diabetic retinopathy
 - Retinal vein occlusion
 - Sick cell retinopathy
 - Radiation Retinopathy
 - Retinopathy of Pre-maturity
 - Occlusive vasculitis (e.g., Behcet's disease)
- Retinovascular – Without Neo-vascularization
 - Macro-aneurysm
 - Retinal Angioma Arterio-venous malformation of the retina
 - Persistent hyaloid artery
- Choroidal – Choroidal neo-vascularization
 - Age-related macular degeneration
 - Pathologic myopia
 - Peripheral exudative hemorrhagic chorioretinopathy (PEHCR)

- Inflammatory/Infectious choroidopathies
- Ocular Trauma
- Closed globe injury
 - Open globe injury
 - Valsalva Retinopathy
 - Shaken baby syndrome
- Systemic causes
 - Blood disorders
 - Terson's syndrome
 - Familial retinal arteriolar tortuosity

Infrequent causes of vitreous hemorrhage:

- Complicated surgical procedures
 - Complicated cataract surgery with or without IOL Trabeculectomy
 - Globe perforation secondary to peri-bulbar/retro-bulbar anesthesia
 - Penetrating keratoplasty
 - Traumatic secondary IOL insertion
- Retinal and Choroidal tumors
 - Retinoblastoma
 - Vasoproliferative tumors
 - Melanoma and melanocytoma
 - Cavernous hemangioma
- Vascular
 - Retinopathy of pre-maturity
 - Coats disease
 - Ocular ischemic syndrome
- Inflammation
 - Pars planitis
 - Behcet's disease
 - Syphilitic retinitis
 - Multiple sclerosis with retinal vasculitis
 - Systemic lupus erythematosus
- Blood disorders
 - Thrombocytopenia Thrombotic

- thrombocytopenic purpura
- Protein C deficiency
- Anti-coagulant therapy
- Miscellaneous
 - Shaken baby syndrome
 - Newborn after vaginal delivery
 - Retinoschisis (Senile and juvenile)

III. DIAGNOSIS PROCEDURE

History

- Suddenly, painless visual loss or haze is a common presentation
- Patients may describe a red hue to their vision
- Patients may describe new onset floaters, shadows, or "cobwebs"
- Symptoms may be worse in the morning if blood settles on the macula during the night
- History of diabetes, hypertension, sickle cell disease, trauma, previous retinal conditions or ocular surgery may help lead to the diagnosis
- Visual acuity is variable based on the location, size, and degree of vitreous hemorrhage. In severe cases, patients may have dramatically reduced visual acuity and/or visual field defects.
- Slit lamp examination usually reveals red blood cells in the anterior vitreous. Presence of iris rubeosis and intraocular pressure should be noted
- Gonioscopy should be used to detect neovascularization of the angle
- Dilated fundus exam may reveal hemorrhage diffusely spread throughout the vitreous cavity, or the blood may conform to the anatomy of the vitreous. For example, hemorrhage in the subhyaloid space may result in a scaphoid (boat-shaped) hemorrhage. Occasionally, a detached posterior hyaloid face will have blood adherent to its posterior surface
- If vitreous hemorrhage is associated with acute PVD, retinal tear or detachment should be ruled out using scleral depression
- It is important to thoroughly examine the fellow eye, as it will often reveal clues as to the etiology
- In chronic vitreous hemorrhage, the red blood cells become dehemoglobinized and the hemorrhage takes on a khaki color or even yellow or white color
- B-scan ultrasonography can detect vitreous hemorrhage, PVD, retinal tears, retinal detachment, tractional membranes, intraocular tumors, and foreign bodies

- Fluorescein angiography may be useful in the setting of mild to moderate vitreous hemorrhage to help identify neovascularization
- If an open globe injury is suspected, orbital CT scan is indicated to characterize the orbital bony structures, assess the integrity of posterior the eyewall (though clinically its presence is denoted by hypotony and other features), and rule out intraocular foreign body
- Blood pressure should also be checked
- If familial exudative vitreoretinopathy is suspected, examination of family members is helpful Laboratory test
- Laboratory testing is used according to clinical suspicion to diagnose diabetes, sickle cell disease, leukemia, thrombocytopenia, and other hematologic abnormalities

IV. DIFFERENTIAL DIAGNOSIS

- Vitreoretinal
 - Posterior vitreous detachment with or without retinal tear, with or without rhegmatogenous retinal detachment
- Retinovascular
 - With neovascularization
 - Proliferative diabetic retinopathy
 - Retinal vein occlusion
 - Hypertensive retinopathy
 - Sickle retinopathy
 - Radiation retinopathy
 - Retinopathy of prematurity
 - Familial exudative vitreoretinopathy
 - Nonperfusion from retinal vasculitis (e.g. sarcoidosis)
 - Without neovascularization
 - Retinal vein occlusion
 - Macroaneurysm
 - Familial retinal arteriolar tortuosity
- Choroidal
 - Choroidal neovascularization secondary to:
 - Age-related macular degeneration
 - Peripheral exudative hemorrhagic chorioretinopathy
 - Pathologic myopia

- Inflammatory/infectious choroidopathies (e.g. sarcoidosis, punctate inner choroidopathy, presumed ocular histoplasmosis syndrome, etc)
- Choroidal tumors (e.g. uveal melanoma)
- Trauma
 - Shaken baby syndrome
 - Closed globe injury
 - Open globe injury
 - Avulsion of prepapillary loop
 - Valsalva maneuver
- Blood disorders (thrombocytopenia, leukemia, hemoglobinopathies, etc.)
- Terson syndrome
- Inflammatory/infectious conditions causing vitritis mimicking chronic vitreous hemorrhage (multifocal choroiditis with panuveitis, endophthalmitis, etc.)
- Primary central nervous system lymphoma masquerading as chronic vitreous hemorrhage

V. THERAPEUTIC APPROACH

- Treat the underlying etiology as soon as possible General treatment
- If the retina can be adequately visualized and the etiology of the vitreous hemorrhage can be determined, treatment of the underlying cause may be attempted, if needed
- If the retina can be adequately visualized and the etiology of the vitreous hemorrhage can be determined, but the vitreous hemorrhage does not permit safe treatment of the underlying etiology
 - pars plana vitrectomy is indicated
 - A short period of cautious observation for vitreous clearing may be reasonable.
 - for example, occasionally a retinal tear associated with vitreous hemorrhage can be seen with a bright indirect ophthalmoscope, but adequate uptake of laser spots to the posterior margin of the break cannot be achieved
 - Red laser may be tried in such cases as it has better penetration.
 - Cryotherapy can be considered in these cases, but the dual risk of cryotherapy and vitreous hemorrhage potentially leading to proliferative vitreoretinopathy should be considered compared to the risk of pars plana vitrectomy with endolaser
- If the retina cannot be adequately visualized in 360 degrees and the etiology of the vitreous hemorrhage is unknown
 - prompt pars plana vitrectomy is indicated

- a short period of cautious observation for vitreous clearing may be reasonable.
- For example, in the case of a superior retinal tear with dense vitreous hemorrhage, adequate treatment of the superior retinal tear may be achieved. However, undetected inferior retinal breaks may be present
 - Between office visits, retinal detachment may progress underneath vitreous hemorrhage without a change in the patient's symptoms.

Therefore, if the view is not clearing briskly, pars plana vitrectomy is indicated

 - Neovascularization of the iris or angle in the setting of new dense vitreous hemorrhage would prompt earlier surgical intervention
- Diabetics frequently present with vitreous hemorrhage, and in general the same principles apply as outlined above.
 - Of note, new vitreous hemorrhage in diabetics cannot always be assumed to be secondary to diabetic retinopathy if there is an inadequate view; diabetics are susceptible to retinal tears and detachments like the general population.
 - The severity of the fellow eye may give a clue as to the etiology of the vitreous hemorrhage, but asymmetric levels of retinopathy are relatively common
 - In any case, diabetic patients with proliferative diabetic retinopathy, new vitreous hemorrhage prohibiting adequate panretinal photocoagulation, and no history of panretinal photocoagulation benefit from pars plana vitrectomy with intra-operative panretinal photocoagulation or intravitreal anti-vascular endothelial growth factor (anti-VEGF) injection. In this scenario, if the vitreous hemorrhage does not clear in about one month, many surgeons will perform pars plana vitrectomy
 - In the case of an established patient with known proliferative diabetic retinopathy and known panretinal photocoagulation, a new or recurrent vitreous hemorrhage is far more likely to be secondary to diabetes than a retinal tear. Longer periods of observation (3 to 6 months) before considering pars plana vitrectomy may be reasonable in these situations

Medical therapy

- Patients are instructed to minimize strenuous activity, as an increase in blood pressure may disrupt the newly formed clot and cause new active bleeding.
- Patients are also instructed to keep their head of bed elevated to allow settling of the blood, improving their vision and permitting more complete fundoscopic examination
- Bilateral patching and bedrest may facilitate settling of blood
- Neovascularization from proliferative retinopathy, associated with diabetes or otherwise, is often treated with panretinal photocoagulation if the view is adequate. This will cause regression of neovascularization and help reduce the risk of further hemorrhage
- Intravitreal injection of anti-VEGF agents may be used to cause regression of

neovascularization in proliferative retinopathies, particularly if there is no view to perform panretinal photocoagulation

- Many surgeons use pre-operative anti-VEGF agents 1 to 7 days before pars plana vitrectomy for vitreous hemorrhage in diabetics, as regression of neovascular membranes reduces intra- and post-operative bleeding and dissection of tissue may become easier
- Intravitreal injection of an anti-VEGF agent is usually indicated when the cause of vitreous hemorrhage is neovascular age-related macular degeneration Medical follow up
- Patients with systemic causes of vitreous hemorrhage are followed closely by an internist or endocrinologist in addition to close ophthalmology follow

- Pars plana vitrectomy is indicated for
 - vitreous hemorrhage accompanied by retinal detachment or break seen on B-scan
 - nonclearing vitreous hemorrhage
 - many cases of intraocular foreign body
 - vitreous hemorrhage with iris neovascularization or associated with hemolytic or ghost-cell glaucoma
 - dense vitreous hemorrhage of unknown etiology
- Depending on the etiology of vitreous hemorrhage, endolaser may also be placed intraoperatively. Some surgeons will also perform a concurrent air-fluid exchange after the vitrectomy or may even inject intravitreal anti-VEGF at the conclusion of case
- In cases of open globe injury with vitreous hemorrhage but without intraocular foreign body, most surgeons close the eyewall first and address the vitreous hemorrhage as a second stage procedure

Surgical follow up

- Frequency of follow up is specific to the surgical indication

VI. COMPLICATION

- *Hemosiderosis bulbi*
- *Proliferative vitreoretinopathy*
- *Glaucoma: ghost cell glaucoma, hemolytic glaucoma, hemosiderotic glaucoma*

VII. ALGORITHM

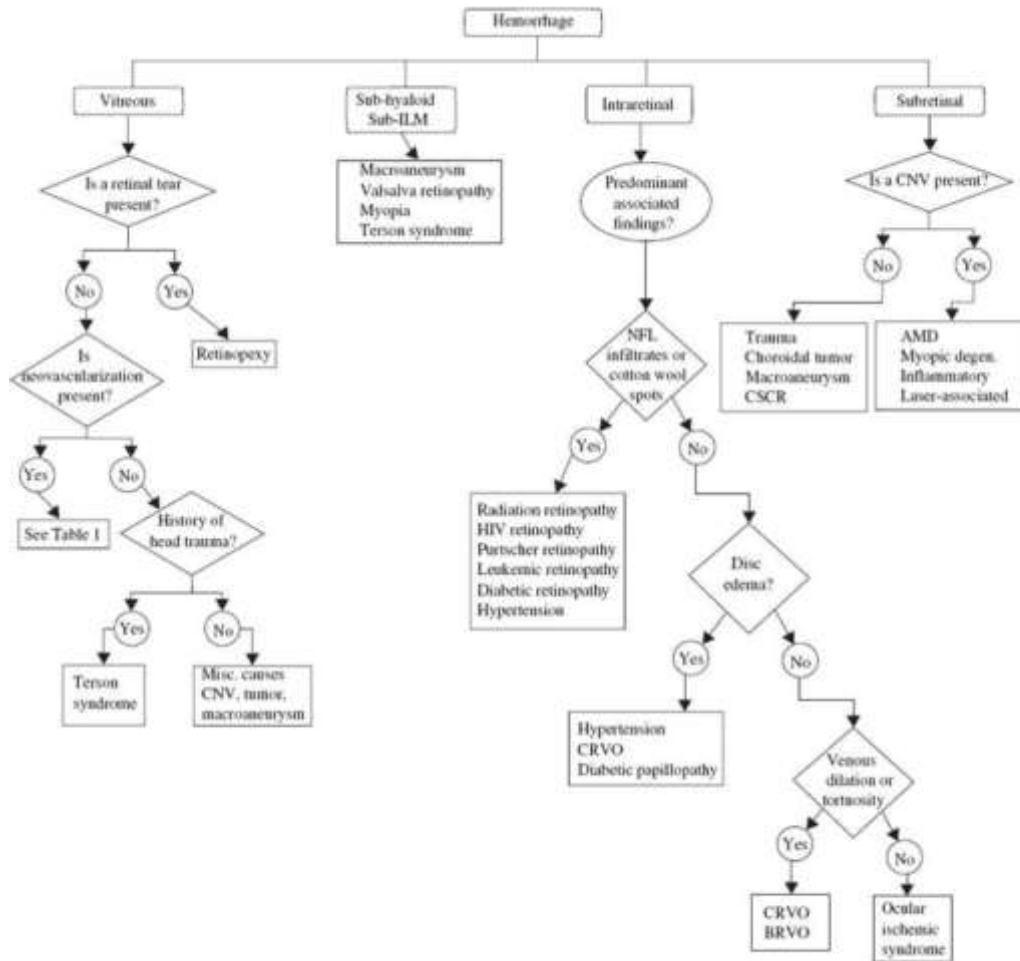


FIGURE 12-2. Diagnostic evaluation of hemorrhage of the posterior segment. ILM, internal limiting membrane; CNV, choroidal neovascularization; AMD, age-related macular degeneration; CSCR, central serous chorioretinopathy; CRVO/BRVO, central branch retinal vein occlusion; NFL, nerve fiber layer.

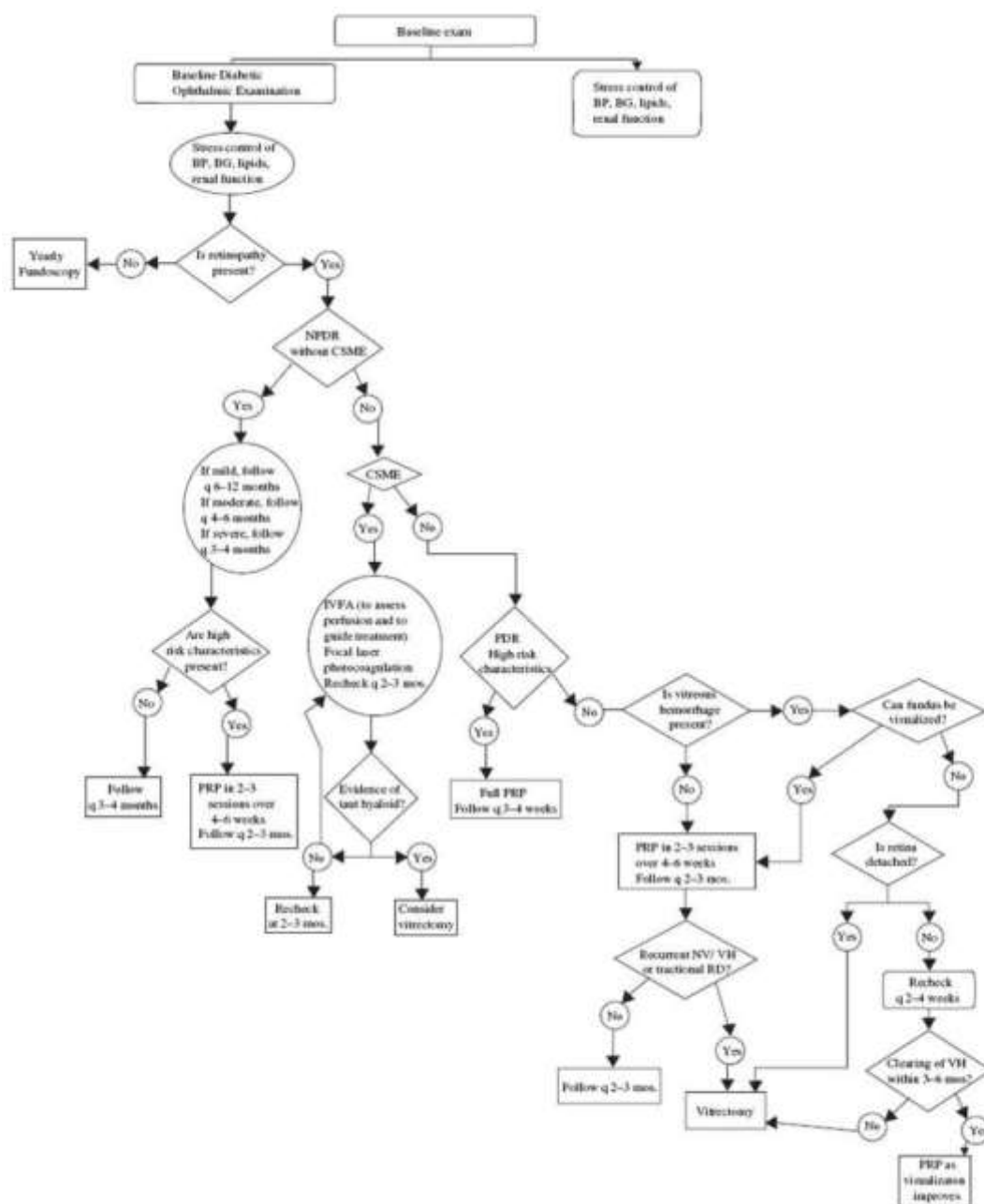


FIGURE 12-4 Ophthalmic management guidelines of the patient with diabetes mellitus. BP, blood pressure; BG, blood glucose; NPDR, nonproliferative diabetic retinopathy; CSME, clinically significant macular edema; PDR, proliferative diabetic retinopathy; PRP, panretinal photocoagulation; NVL, neovascularization of the iris; NV, neovascularization; VH, vitreous hemorrhage; RD, retinal detachment.

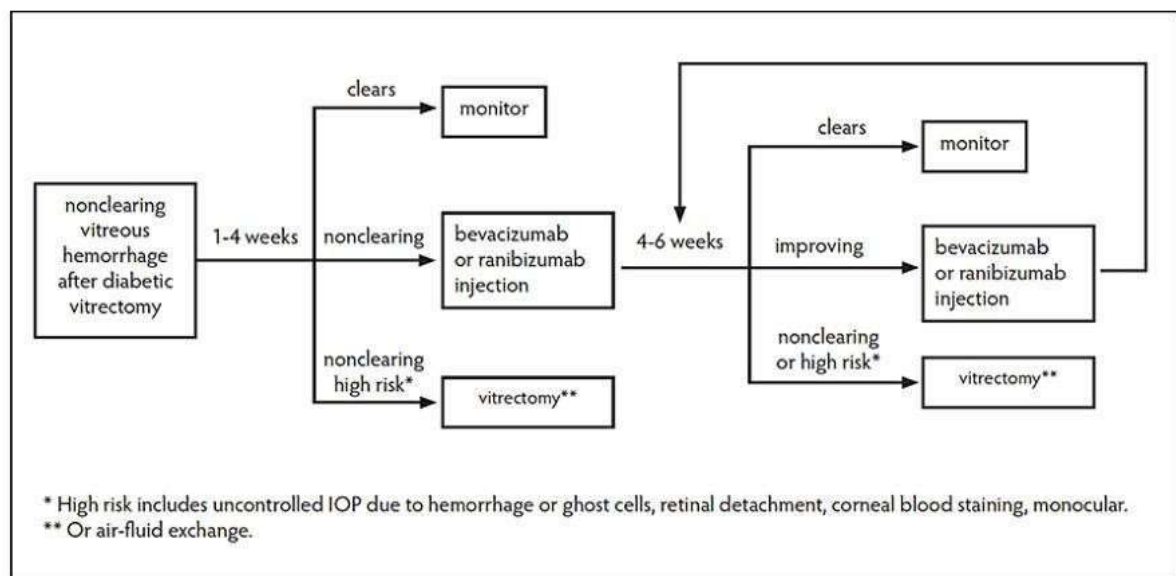
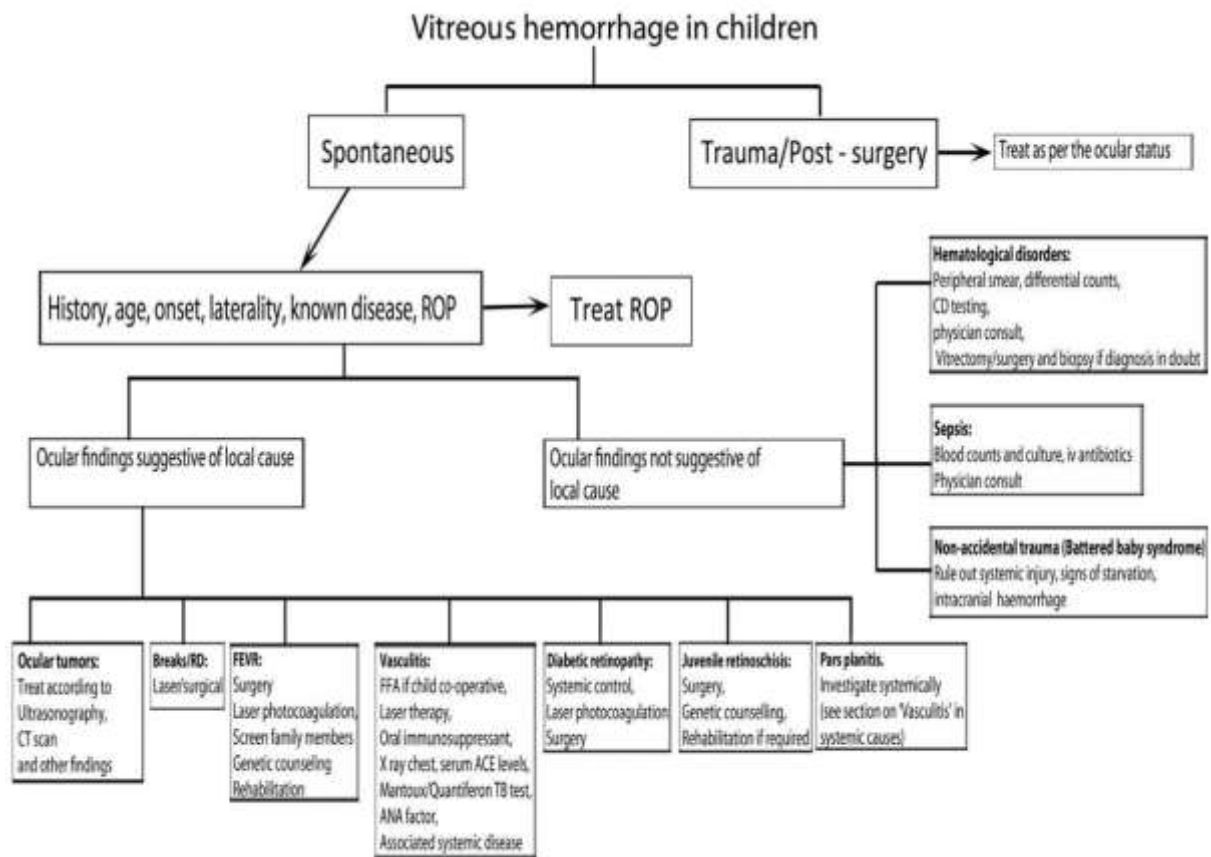


Figure 1. Algorithm for management of postoperative recurrent diabetic vitreous hemorrhage.

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ZONULAR DEHISCENCE WITH LENS SUBLUXATION OR DISLOCATION CATARACT MANAGEMENT GUIDELINES

CHHOUR Long, CHANTHAN David, NGY Meng

I. DEFINITION

Rupture of the fibrous strands connecting the ciliary body and the crystalline lens' capsule that may cause lens shaking, subluxation or dislocation into the vitreous cavity or anterior chamber.

II. ETIOLOGY

- Ocular trauma.
- Exfoliation syndrome.
- High myopia.
- Marfan syndrome.
- Inborn errors of metabolism.
- Complicated intra ocular surgery.

III. DIAGNOSTIC PROCEDURE

3.1. Clinical argument

- Symptoms.
- Ocular history.
- Systemic history.
- Assessment of visual function status.

3.2. Technical procedure

- Measurement of visual acuity.
- Ocular alignment and motility.
- Pupil examination.
- Measurement of intra ocular pressure.
- External eye examination including lids and lashes.
- Full slit lamp examination with dilated pupil.

- Evaluation of fundus.
- Ultrasonography (B-scan) if the view of fundus is obscured.

IV. THERAPEUTIC APPROACH

1. Surgical indication

- The treatment is indicated when the visual function no longer meets the patient's needs and surgery provides reasonable of vision improvement.
- There is evidence of lens induce complications.
- Lens opacity limited view for posterior segment treatment.
- The surgery should not be performed if it will not improve visual function and there is no sign of lens induce complications.
- The surgery should not be performed if the patient can't safely undergo the procedure because of coexisting medical or ocular conditions.

2. Preoperative care

- The ophthalmologist examines the patient preoperatively to ensure that the patient is fit for surgery.
- After history taking and examination, accurately documents symptoms findings and indication for treatment.
- Discuss with the patient about risk and benefit of the procedure.
- General health evaluation including note of current medication, record of allergies.
- Assessment ability of patient to cooperate with the procedure
- Formulate surgical plan including type of anesthesia, IOL type and power and postoperative refractive target.
- Informed consent for surgery written detailing the reason for, benefit of and possible complication of the surgery.

2. Biometry

- To allow the selection of correct IOL power to achieve the desired refractive result after cataract surgery.
- Should compare between both eyes and repeat it if needed
- Axial length measurement by A-Scan ultrasound or laser interferometry.
- Central cornea curvature measure by keratometry or topography.
- White to white measurement of cornea diameter.
- Anterior chamber depth.
- Selection of formula based on eye's condition: Barrette Universal II, SRK/T, Holladay I&II. Haigis and Hoffer Q slightly better with short axial lengths < 22mm.

3. Anesthesia

- The plan of anesthesia should be decided after discussion between patient and doctor.
- Retro bulbar or peri bulbar anesthesia is needed, as the procedure may take longer than normal phacoemulsification.
- General anesthesia is required for children or patient who have mental, physical or medical condition not suitable for surgery under local anesthesia

4. Infection prophylaxis

- A thorough face washes before surgery is recommended.
- Skin preparing with 10% Povidone iodine.
- Povidone iodine 5% irrigated into conjunctival sac immediately pre operative.
- A watertight wound closure.
- Intracameral or topical antibiotic.

5. Post operative care

- Topical antibiotic and corticosteroid.
- Topical NSIAD or others if necessary.
- High-risk patient should be seen within 24 hours.
- Routine patients should be seen within 48 hours.
- Assessment of visual function, intra ocular pressure.
- Frequency and timing of subsequence visit depend on refraction, visual function and medical condition of the eye.
- A written instruction about eye drop, hygiene and symptom that require to come back early.

6. Surgical technique

- 5 1. Lens in place with zonular defect less than 2 quadrant phacoemulsification with capsular tension ring (CTR) should be performed.
 - Side port incision.
 - Clear cornea incision 2.2- 2.75 mm.
 - Continue curvilinear capsulorhexis ideally a bit larger than regular to reduce stress on zonule.
 - CTR insertion.
 - Hydrodissection.
 - Phacoemulsification.
 - Irrigation and aspiration removal of epinucleus and cortex.
 - IOL implant in the bag.
 - Viscoelastic removal.

- Wound sealing.

5.2 Lens subluxation

- Subluxated soft lens should be removed by Pars Plana Lensectomy technique (PPL).
- Subluxated hard lens should be removed by Intra Capsular Cataract Extraction (ICCE) anterior vitrectomy technique. In these cases, the IOL implantation should be done either Tran scleral suture fixated lens or sutureless Tran scleral haptic fixate lens which will be detailed below.

5.3 Lens dislocation in the vitreous

Dislocated lens in the vitreous should be removed by Pars Plana Vitrectomy and Phacofragmatome. Then the IOL implantation should be done by transscleral suture fixated lens or sutureless Tran scleral haptics fixated lens.

Intra Ocular Lens Implantation technique in case there are not enough capsules to support IOL:

Tran scleral Suture Fixated IOL

- After cataract removal and vitrectomy done.
- Create scleral tunnel wound superior 7mm wide.
- Make 2 scleral flaps or groove at 3- 9 o'clock position.
- Pass through 10/0 or 9/0 prolene suture 2mm from limbus under scleral flap or groove.
- Pull out prolene suture through main wound superior and secure each side of haptic PMMA lens preferably 6.5- 7mm optic diameter.
- Insert IOL into posterior chamber and pull both side of prolene suture to center IOL then tighten the notch under scleral flap or groove to avoid conjunctival erosion.
- Wound closure.
- Apply antibiotic and steroid.

Tran scleral haptic fixated IOL (Yamane)

- Less invasive no need to do conjunctival peritomy.
- Wound smaller compared to suture lens technique.
- After removal of cataract and vitrectomy has been done.
- A 3 pieces IOL insert into anterior chamber with trailing haptic externalized.
- A thin wall 30 G needle made an angle sclerotomy through

conjunctiva 90 degree from lens incision, 2mm posterior to limbus. As it enters the sulcus the needle goes toward the leading haptic which threaded through it with intra ocular forceps.

- Another sclerotomy is made 180 degrees from the first one with 30G needle through which the trailing haptic is threaded.
- After confirming of centration both haptics is externalized into conjunctiva and then heated with low temperature cautery to create flange. The flanges are tucked through conjunctiva into scleral tunnel.

Iris-fixated IOL

- Anterior Chamber Approach
 - Paracentesis creation
 - Injection of viscoelastic to protect corneal endothelium and stabilize the chamber.
 - Insertion of IOL through the main incision.
 - Fixation to the iris using enclavation needles.
- Posterior Chamber Approach
 - Pars plana vitrectomy (if necessary).
 - Sclera's incision for IOL insertion.
 - IOL enclavation to posterior iris surface.

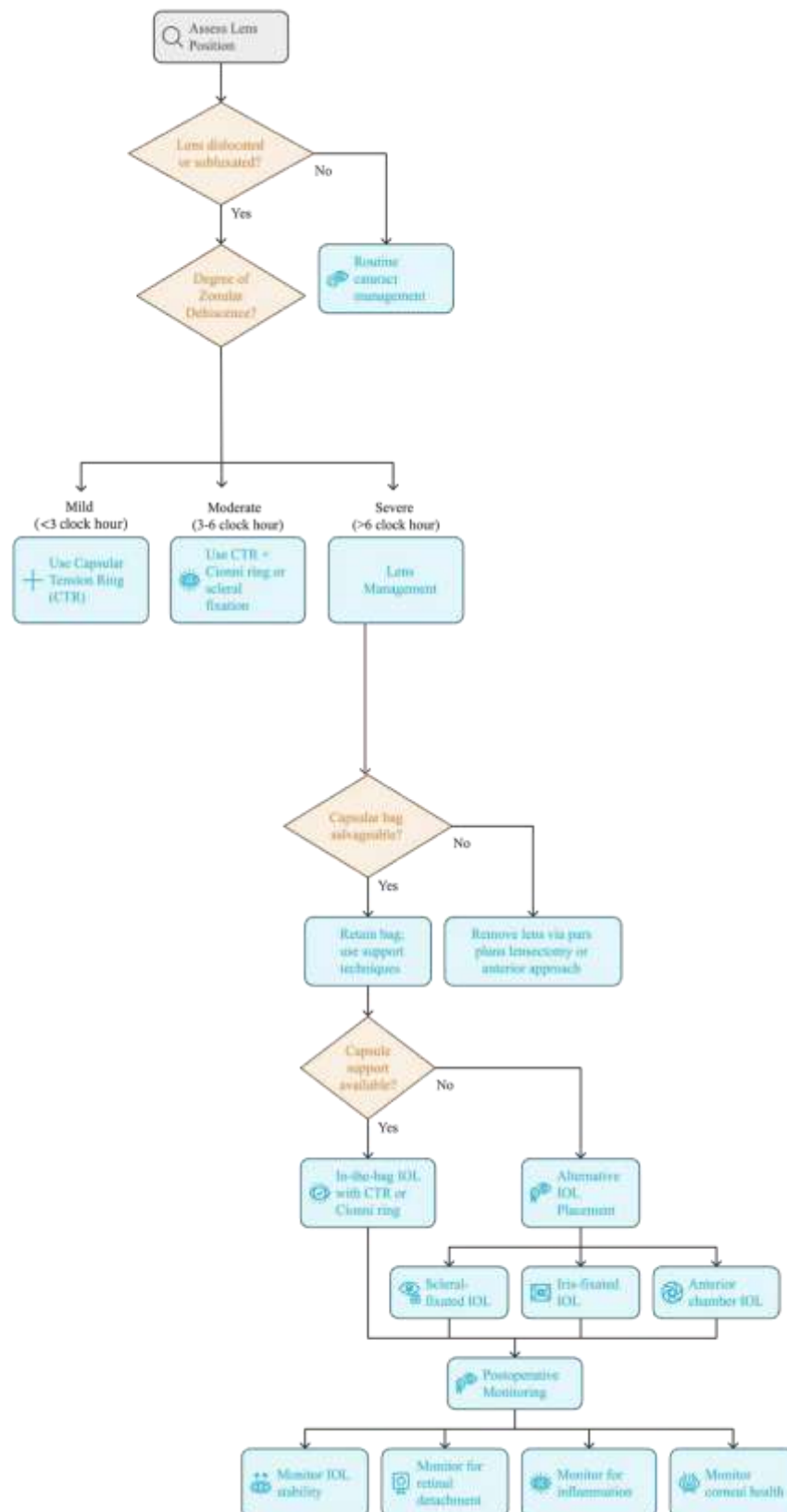
Anterior chamber IOL

- Considered if the corneal endothelium is healthy, anterior chamber is not shallow, or risk of glaucoma

V. COMPLICATIONS

- IOL off center, tilted.
- Hyphema, vitreous hemorrhage.
- Endophthalmitis.
- Cystoid macular edema.
- Retinal detachment.
- Glaucoma.
- Off target post operative refraction.
- Uveitis.
- Bullous keratopathy.

VI. ALGORITHM



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